INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Manufacturing

What the industry is, how it runs, and how to model it — for architects who have never worked in it

A working guide for an enterprise architect walking into a factory for the first time. Discrete, process and batch, and why a solution built for one fails in another. The bill of material, the routing and the order that decides everything downstream. What a plant costs to run and how its margin is really made. The layer no other industry has — the control systems below the ERP, and the boundary where IT meets OT. Then the part that matters most: how to turn all of it into a model a production manager would recognise as true.

South African lens, with North American practice flagged where it differs. Sourced throughout. October 2026.

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CHAPTER 1

The industry in one chapter

A manufacturer buys inputs and physically changes them, and everything difficult about the industry follows from that one fact. This chapter is the book compressed: enough to follow the conversation in a factory this afternoon, and a map of where the rest of the detail lives.

Two ordinary mornings

At twenty to six on a Tuesday in Paarl, a beverage line is already forty minutes into its run. The pasteuriser reached temperature at four. Nobody will turn it off before Thursday night, because bringing it down and back up again costs most of a shift and a quantity of product that is neither raw material nor finished goods while the temperature profile settles. The shift supervisor is looking at two numbers: litres filled against the plan, and the lab's clearance on the batch that ran overnight. The second number has not arrived. Until it does, 14 pallets stand in the warehouse, physically present, counted in the stock system, and not available to sell.

Six hundred kilometres away in Ekurhuleni, a press shop is running its third order of the morning. The first was 4 000 brackets against a vehicle maker's weekly electronic release, sequenced to arrive at an assembly line in a four-hour window. The second was 60 of a superseded part for the replacement-parts trade, made once a quarter, priced at four times the original margin. The third is a trial run for an engineering change the customer issued eleven days ago, which has not yet reached the costing system, so the plant is building a part whose standard cost does not exist. A setter is on his second changeover of the shift and he is the reason the schedule holds.

Same group, same general ledger, same month-end. Almost nothing else in common.

That is the first thing to take from this book. Manufacturing is not one business. The word covers a beverage line that must never stop and a press shop that stops between every order; a plant whose product is defined years before the order arrives and a plant whose product does not exist until the order defines it. The differences are not matters of degree. They disagree about what the fundamental objects are, and a design that is correct in one is wrong in the other.

Throughout this book the worked example is **Karoo Industrial Group**, an invented South African manufacturing group with two operating businesses. **Karoo Foods** runs a beverage line in Paarl and a dry-goods and canning site in Gqeberha: batch and process work, made to stock, with recipes, co-products, allergen segregation and shelf life. **Highveld Precision Components** runs one plant in Ekurhuleni: discrete work, made and assembled to order, tier-two automotive, with sequenced delivery to one vehicle maker and a make-to-stock replacement-parts business alongside it. Nothing about Karoo Industrial Group is real. It exists because almost every real South African plant is several archetypes at once, and an example that carries only one teaches the wrong lesson.

One convention before anything else. South African official statistics use the decimal comma, so manufacturing production fell 4,3% and not 4.3%. This book follows that usage for South African figures. Mixing the two in one table has caused real errors in real reports.

Physical transformation, and everything that follows from it

A manufacturer acquires inputs and changes their physical form, composition or assembly state, then sells the result. That is the whole definition, and it is the only one you need. A trader moves and holds goods without transforming them. A service business produces no physical thing with a parts list and a route. A manufacturer does both.

Three consequences follow immediately, and between them they generate most of manufacturing's architecture.

It owns three inventories at once, in different states, valued differently

Raw material is purchased and not yet consumed. It is valued at purchase cost plus landed cost, and it belongs to procurement's world. Work in progress has been released to the floor, partially transformed, and is not yet a sellable thing. It is valued at material consumed plus labour and overhead absorbed **to the operation reached** — which is a shop-floor fact, not a financial one. Finished goods are complete and sellable, at full standard cost.

Work in progress is the hardest inventory in business to value correctly, for exactly that reason. Its value depends on how far through the route a thing has travelled, and the only system that knows how far is the one on the floor.

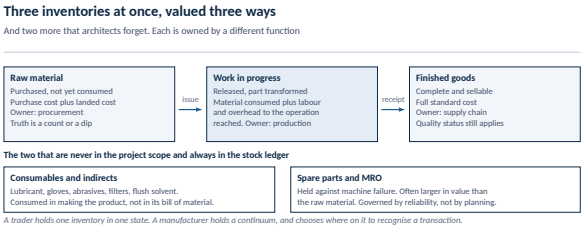


Figure 1.1 The three inventories a manufacturer owns at once, and the two it forgets. Work in progress is the hard one: its value depends on how far through the route the material has travelled, which only the shop floor knows.

Two more inventories exist and architects routinely forget both. **Consumables and indirects** — lubricant, gloves, abrasives, filters, the solvent used to flush a line between colours — are consumed in making the product but are not in its parts list, so they are charged to a cost centre rather than to an order. **Spare parts and maintenance stock** are held against machine failure, are frequently larger in value than the raw material, and are governed by

reliability engineering rather than by production planning. Their stock turns are deliberately terrible: you hold a R2 million gearbox for five years precisely so that you never need it urgently.

A trader has one inventory in one state. A manufacturer's stock ledger has to represent a continuum, and the points on that continuum at which it recognises a stock transaction are a design decision with direct financial consequences. Recognise too few and you cannot see where value is sitting. Recognise too many and the floor spends its shift confirming transactions instead of making things.

Owning all three at once is what makes the data model hard, and it is worth being precise about why, because the reason is not volume.

The three inventories have **different units of truth**. Raw material in a silo is a level reading from an instrument with a tolerance. Work in progress on a press line is a count of pieces at an operation. Finished goods in a warehouse is a pallet scan. The same physical material passes through all three and changes measurement regime twice, and the tolerance of the instrument is frequently wider than the plant's margin.

They have **different valuation rules and different owners**. Raw material value changes when a buyer negotiates. Work-in-progress value changes when an operation is confirmed. Finished-goods value changes when the annual standard cost roll happens. Three functions, three cadences, one ledger that must balance.

They have **different availability rules**. Stock is not available merely because it is present. A batch sitting in the warehouse at Paarl is counted, valued and unsellable until the laboratory releases it, so quality status is a first-class attribute of a stock position and not a note on a certificate. A planning engine that promises quarantined material has not got a configuration defect; it has a data-model defect.

And they are **joined by transformation, not by movement**. In a trading business, stock moves from A to B and the same thing arrives. In manufacturing, five things go in and one different thing comes out, usually alongside some scrap and sometimes alongside a co-product. Every stock transaction is therefore a statement about a conversion, and the conversion has to reconcile: inputs equal outputs plus losses plus the change in inventory, within tolerance. When it does not, somebody must decide whether the error is in the meter, the laboratory, the recipe or the gate. Chapter 9 builds the model; chapter 10 builds the reconciliation.

It owns capacity, which costs money whether or not it runs

A press line, a kiln, a reactor, a filling line, a furnace: these were bought, are depreciated, are maintained, are insured, are staffed on a shift pattern, and consume energy at idle. The cost exists before a single unit is made.

This is the defining economic difference from a business that buys and resells. A retailer's cost base is roughly proportional to the shops it operates and the hours they open. A manufacturer's cost base is fixed against a capacity envelope, and only becomes a cost per unit when it is divided by volume.

Unit cost is therefore a function of volume

If fixed manufacturing overhead is R10 million a month and the plant makes 100 000 units, overhead per unit is R100. At 50 000 units it is R200. The product did not change.

That single arithmetic fact drives absorption costing, variance analysis, the decision to make or to buy, the temptation to overproduce so as to absorb overhead, and most of the arguments between a plant manager and a financial controller. There is nothing structurally equivalent in a business that does not transform anything.

> The integrity question in manufacturing architecture is one sentence: **for this physical thing, at this moment, what is its state, its location, its genealogy and its accumulated cost — and do the shop floor and the ledger agree?** Manufacturing IT failures are overwhelmingly failures of that one answer. They surface as inventory variance at stock count and as unexplained production variances at month-end. Keep the sentence. You will use it in every workshop.

What that means for the shape of the estate

There is no single core system in manufacturing in the way that a bank has a deposit core. There are four things that must exist and must agree, and in most real plants they are in different systems with different owners and different clocks.

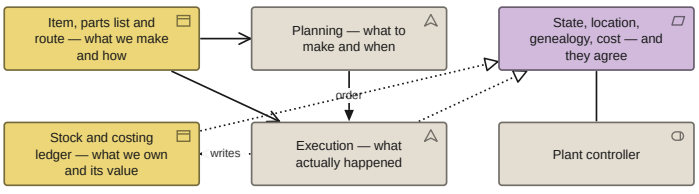


Figure 1.2 The four things that must exist and must agree. There is no single core system in manufacturing; there is a master, a ledger, a planner and an execution layer, usually under different owners and on different clocks. Chapters 11 and 12 turn this into applications.

An **item, parts-list and routing master** says what we make and how. A **stock and costing ledger** says what we own, where, in what state, at what value. A **planning engine** says what we should make and when. An **execution layer** says what actually happened on the floor, by machine, by shift, by batch. Chapter 11 turns this into a conceptual application model and chapter 12 into a real landscape, but the shape is worth holding from the first page, because every integration argument you will have is an argument about which of those four is authoritative for a particular fact.

"What did this cost to make?"

This is the question you will be asked in week one, and it is the first question you should refuse to answer in a single number. Not out of pedantry. There are at least nine defensible answers for one physical unit, all simultaneously true, and the person asking usually wants a different one from the person who will quote it back at them.

Take four of them.

Material cost is the parts list at standard price. It is the dominant cost in most assembly and most food manufacture, it is what a buyer thinks about, and it ignores everything else.

Variable or marginal cost is every cost that varies with the unit. It is the right number for a short-run decision — accept this export order at a discount, run the line a sixth day — and it is the wrong number for anything else, because a plant that prices everything on variable cost never recovers its fixed costs or funds the replacement of its assets.

Standard full absorbed cost is material plus labour plus variable and fixed overhead at the standard rate. This is the basis on which inventory is valued and reported cost of sales is struck. It is also stale by construction and depends entirely on a volume assumption made months ago.

Fully loaded cost adds selling, general and administrative cost, distribution, warranty and the cost of capital. It is the only basis on which anyone can say a product is truly profitable, and the allocation of the overhead in it is arbitrary.

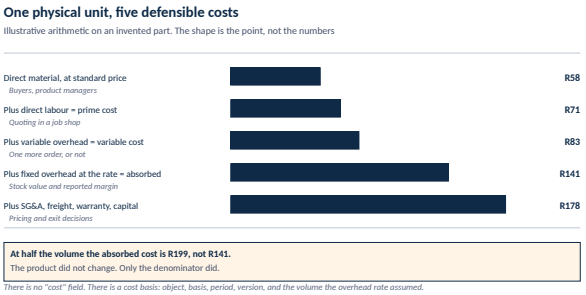


Figure 1.3 Five of the nine defensible answers to "what did this cost to make?", on an invented part. Every figure here is illustrative. The last panel is the one that causes the arguments: absorbed cost moves with the volume assumption and not with the process.

For completeness, here are all nine, because sooner or later somebody will quote one of the other five at you.

Answer	What it includes	Who uses it, and what it is blind to
Material cost	Structure material at standard price	Buyers. Ignores everything else
Prime cost	Direct material plus direct labour	Quoting in a job shop. Ignores overhead entirely
Variable or marginal cost	Everything that varies with the unit	Short-run decisions. Never recovers fixed cost
Standard full absorbed cost	Material, labour, variable and fixed overhead at the standard rate	Stock valuation and reported margin. Depends on the volume assumption and is stale by construction
Actual full cost for the period	Standard plus or minus all variances, in total	The finance team. Cannot be allocated to one unit without a convention
Actual job cost	Everything booked to one production order	Engineer-to-order and job shops. Does not exist for repetitive flow
Fully loaded cost	Absorbed cost plus selling and administration, distribution, warranty, capital	Pricing and exit. The allocation of the overhead is arbitrary
Cash cost	Cash out of the door, excluding depreciation	Shutdown and survival decisions. Pretends the asset was free
Replacement-basis cost	Current replacement cost of inputs and current-cost depreciation	Reinvestment decisions in inflation. Not a reportable basis

Why do they differ? Three mechanisms, and each one is a thing an architect has to build.

The volume assumption. Full absorbed cost contains an overhead rate, and that rate was set by dividing budgeted overhead by a budgeted activity level. Produce less than that assumption and the same physical product carries a higher cost per unit with no change in the process. This is not an artefact to be routed around. It is the single most common cause of "the margin report and the plant report disagree".

Standard against actual. A manufacturer values stock and cost of sales at a pre-set, period-frozen standard cost, and books the difference from actual as a variance. Two entirely legitimate answers therefore exist in the same ledger in the same month and they differ by the variance. Worse, standards are usually re-set once a year, so a time series of unit cost contains a discontinuity at every standard-cost roll that has nothing to do with the factory.

Which cost object. For a product made repeatedly on a line, the cost object is the product. For a one-off fabrication, the cost object is the job, and there is no meaningful unit cost of the product at all — there is the cost of job 47821. For a contract that runs three years, the cost object is the contract.

> Never build a field called `unit_cost`. Build a **cost basis dimension**: cost object (unit, batch, order, product, family), basis (standard, actual, variable, fully loaded), period, version, and the volume assumption where one applies. A manufacturing data model with a single unit-cost column has already failed, and the failure surfaces about eighteen months later as a reporting layer that nobody trusts.

One further complication, and it recurs through the whole book. Costs accumulate **by level**. A finished good's cost contains the rolled-up standard cost of its subassemblies, each of which was itself a cost object, so the cost at the top hides variances that occurred three levels down. An analyst asking why a product's cost rose needs the **cost component split** — material, labour, overhead, subcontract, freight — preserved through the roll-up, not just the total. Preserving component split through a multi-level roll-up is a specific and commonly skipped requirement, and in South African automotive it is not optional: the production incentive described later in this chapter cannot be substantiated without it.

Absorption costing in one page, honestly

You will meet the consequences of absorption costing in every reporting conversation you have in a manufacturing business, so it is worth one page now rather than a surprise in month four.

Absorption costing assigns all manufacturing costs — direct material, direct labour, and both variable and fixed manufacturing overhead — to the units produced. It is the basis required for external financial reporting under IFRS, because fixed and variable production overheads form part of the conversion cost of inventory. The alternative, **variable costing**, assigns only variable costs to units and treats fixed overhead as a period expense. Variable costing is used for internal decisions and is not a permitted basis for reported inventory.

The mechanism, because you will have to build it:

1. Pick a cost centre structure, usually one per work centre, department or line.
2. Budget each cost centre's overhead for the period: depreciation, maintenance labour, supervision, indirect labour, energy, consumables, insurance, rates, site services.
3. Pick an activity base to absorb it on — machine hours, labour hours, units, kilograms, batches. The choice determines which products carry which share of the overhead. A plant that absorbs on labour hours and then automates will systematically under-cost its automated products.
4. Set the absorption rate: budgeted overhead divided by budgeted activity. **This fixes the denominator volume assumption for the whole period.**
5. Absorb overhead onto each unit as it passes through, at rate times actual activity consumed, per the route.
6. At period end, measure under- or over-absorption: overhead absorbed less overhead actually incurred. That is the overhead absorption variance, and it goes to the profit and loss account.

Steps 3 and 4 are master data with effective dates, held per cost centre per period, and they are inputs to the standard cost roll-up. A costing architecture that does not version them cannot explain why unit cost changed. Step 5 requires the route to be accurate and the shop floor to confess actual activity, which it does imperfectly, which is why step 6 is never zero.

Now the consequence that matters. Because fixed overhead is absorbed into units **produced**, producing more units reduces reported unit cost and increases reported profit, even if nothing is sold.

Work it through on an invented plant. Fixed manufacturing overhead R10 million a month, variable cost R60 a unit, price R150, normal volume 100 000 units, so the absorption rate is R100 and standard full cost is R160.

- Make 100 000 and sell 100 000: absorbed R10m, incurred R10m, no volume variance.
- Make 120 000 and sell 100 000: absorbed R12m against R10m incurred, a R2m favourable volume variance, and 20 000 units of finished goods now carry R2m of fixed overhead into the balance sheet. **Reported profit rises by R2 million** because fixed cost was capitalised into inventory instead of expensed.
- Make 80 000 and sell 100 000: a R2m adverse variance expensed, reported profit falls by R2m — while the business actually generated more cash, because it emptied the warehouse.

Three things follow, and all three show up in real plants. A plant manager measured on unit cost or on absorption has an incentive to run the line and convert cash into finished goods; this is created by the accounting system, not by bad people. Profit is negatively correlated with cash in the short run, so a destocking quarter looks bad on the profit and loss account and good in the cash flow statement. And IFRS pushes back, partly: the allocation of fixed overhead is to be based on normal capacity, unallocated overhead in a low-production period is expensed, and the amount allocated per unit is not to be increased because production was low or the plant was idle. That is precisely the rule that stops a plant hiding idle capacity in inventory, and it is the rule most often configured wrongly.

> Any manufacturing dashboard that shows profit without inventory movement beside it is actively misleading. If you build one, you will be the person who built it. The research behind this book did not retrieve the text of IAS 2; before any of the accounting positions in this chapter reach a client deliverable, read the standard rather than this paragraph. One thing worth knowing now: standard costing is expressly permitted under IAS 2 where the results approximate cost and the standards are regularly reviewed, so "standard costing is not IFRS-compliant" is wrong. LIFO, on the other hand, is not available under IFRS at all, which is the largest costing divergence with United States practice and matters whenever a South African plant reports to a North American parent.

South African manufacturing: the actual size and shape

Now the local ground. If you are working in this market, four facts will reorder your assumptions, and the first is about size.

Manufacturing is roughly **11,5% to 12,1% of gross value added**, not the twenty-plus per cent that older material and most dinner-party conversation assume. On the Q2 2026 national accounts, manufacturing gross value added was R123 433 million for the quarter, about 11,5% of total gross value

added on a quarterly basis and about 11,7% annualised; for the full year 2025 it was R516 986 million, about 12,1% [11]. Those share percentages are arithmetic from the release's own value-added tables, because the statistics agency does not publish an industry-share table.

And the figure is about to move for reasons that have nothing to do with factories. Statistics South Africa has stated its intention to rebase the national accounts to 2022 and benchmark the estimates, with publication expected in October 2026, and that the figures may change after that [11]. Any manufacturing share-of-economy figure quoted in a document shipping after that date has to be re-checked against the rebased series. Say so in the document.

Manufacturing is also about 9,5% of total employment — 1,587 million people in the first quarter of 2026 against total employment of 16,754 million [12] — which is a *lower* share than its share of value added. That tells you South African manufacturing is more capital-intensive than the economy average, which is exactly what you would expect from the three consequences at the start of this chapter.

> Name the survey whenever you give a manufacturing employment number. The household survey and the enterprise survey do not agree and both are official: in the same quarter the household survey showed manufacturing up 39 000 on the prior quarter and down 90 000 year on year [12], while the enterprise survey showed manufacturing up 7 000 on the quarter [23]. They measure different universes. Two widely circulated claims — that manufacturing "bled 100 000 jobs", and that it "lost 45 995 jobs in two years" — are irreconcilable with each other and with both official series. Do not use either.

Where the industry actually is

The division weights of the official production index are the most useful single table in South African manufacturing, because they tell you where the industry is as opposed to where the press coverage is.

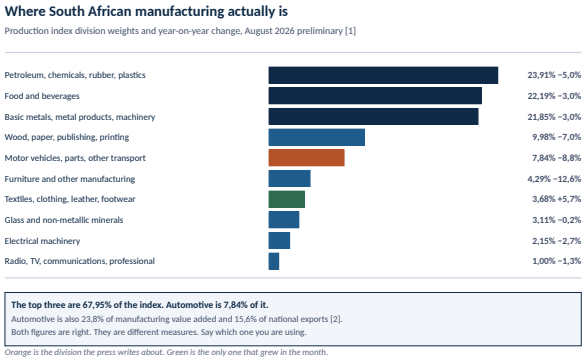


Figure 1.4 South African manufacturing by production-index division weight, August 2026 preliminary [1], with automotive's value-added and export shares from naamsa's 2025 data [2]. The three largest divisions are 67,95% of the index — my arithmetic on the published weights.

Three divisions are 67,95% of the production index [1]: petroleum, chemicals, rubber and plastics at 23,91%; food and beverages at 22,19%; and basic iron and steel, non-ferrous metals, metal products and machinery at 21,85%. An architect who understands those three understands most of the industry.

Automotive is 7,84% of the index [1]. Say that out loud, because it contradicts almost every manufacturing conference agenda you will attend. A book about manufacturing written around automotive has written about 7,84% of the industry, and the archetypes it teaches — sequenced delivery, a tiered supplier base, a model-year change — do not transfer to a cement kiln or a canning line.

So why does automotive get the policy attention? Because index weight is not the only measure, and on the other measures it is large. Vehicle and component manufacturing was **23,8% of domestic manufacturing value added** in 2025, and automotive exports were R291,0 billion, **15,6% of total South African exports** [2]. Both figures are correct. They are different measures — a volume index weight against a value-added share — and conflating them is a common error. Cite one, and say which.

Two further numbers from the same source reorder a roadmap. Light vehicle **imports are now 69,1% of light vehicle sales**, up from 62,7%, with India at 56,2% and China at 23,3% of import origin and passenger-car imports at 82,8% of passenger-car sales [2]. The domestic assembly base is an export business that happens also to serve a largely import-supplied home market. And **component exports fell while vehicle exports rose** [2] — the component base is losing ground even as assembly grows, which is the problem the local-content policy in this chapter is aimed at.

Capacity, and what it tells you about the margin reporting

Large manufacturers used **76,6% of production capacity** in the May 2026 reference month, down from 77,8% a year earlier, with food and beverages highest at 81,3% [13]. Read that alongside the absorption mechanism above: an industry running at about three-quarters of its assumed volume carries a structurally higher unit cost than its standard costs assume and reports persistent adverse volume variances. That is a structural feature of South African manufacturing margin reporting, not a cyclical one.

Treat the figure carefully. It covers large enterprises only, it is self-reported against each respondent's own notion of capacity, and it is not comparable to a client's internal number. And note the combination that food and beverages shows: the highest utilisation of any division, and a largest negative contributor to the quarter's manufacturing decline [15]. High utilisation with falling output means demand-constrained, not capacity-constrained.

One habit to form now: production or sales?

These are different series, measured differently, and they diverge for months at a time. In the August 2026 release, manufacturing **production was down 4,3% year on year** while **sales were R307 094 million, up 2,8%** in current prices [1]. Both are correct. Production is a volume index; sales is a rand value

including price inflation and including inventory drawdown. A plant can sell more than it makes, for a while, by emptying the warehouse.

Every statement that manufacturing grew or shrank has to declare whether it is volume or value, and so does every dashboard you build for a client. The same trap recurs at plant level with the same mechanism, and it is the first reconciliation defect to look for in a group reporting layer.

The structural constraints, in their current state

This is the section most likely to be wrong in any manufacturing material you pick up, because the constraints moved and most writing has not.

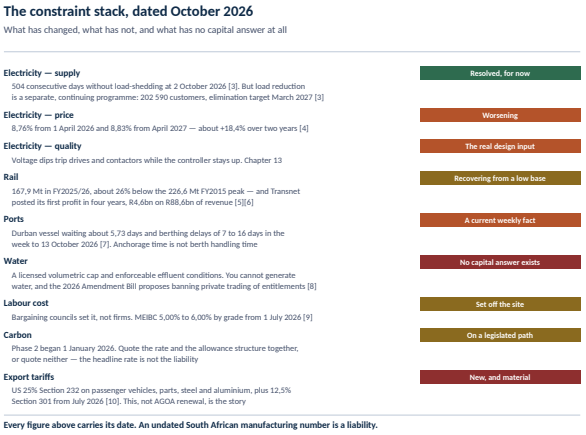


Figure 1.5 The structural constraints as at October 2026, each with its date. Most published manufacturing material still has the first row wrong, and almost none of it carries the last three.

Load-shedding has ended, and writing it in the present tense dates a document instantly

Eskom recorded **504 consecutive days without load-shedding as at 2 October 2026**, counting from 16 May 2025, with the energy availability factor at 68,11% for the financial year to date against 62,59% in the prior year and diesel spend down 80,49% to R1,16 billion [3]. The September 2026 purchasing managers' survey **did not list electricity among the cited constraints at all**; respondents named shipping delays at Durban, fuel prices and short-term borrowing costs [14].

Two qualifications, both load-bearing. **Load reduction is a different programme and it continues** — localised curtailment at peak in areas with network overloading, affecting 202 590 customers as at 2 October 2026, with elimination targeted for March 2027 [3]. A plant on an affected feeder is still interrupted. And **the electricity story is now a price story**: 8,76% approved for direct customers from 1 April 2026, 9,01% for municipal bulk purchasers from 1 July 2026, and 8,83% from April 2027 in the same determination [4], which compounds to about **+18,4% over two years** against general inflation well below that. For an arc furnace, a ferroalloy smelter, a cement kiln or a chlor-alkali plant, that is a margin event, not a utility-bill item.

There is a third change that matters more to you than either. The real electricity problem for the plant floor is now **voltage quality, not availability** — dips that trip drives and contactors while the controller itself stays up on its own supply, leaving a system that believes a pump is running when it is not. South Africa has a national standard for this and it is the best-evidenced design input in the whole research set. Chapter 13 handles it properly.

A consequence for your first roadmap: the standby generation, inverters, batteries and solar that manufacturers installed in 2022 and 2023 are now assets whose original justification has gone. The energy-management system, the generator run-hour data, the solar generation data and the tariff model have become a **cost-optimisation problem** — time-of-use arbitrage, peak shaving, wheeling — rather than a continuity problem. That is a completely different set of requirements against the same instrumentation, and nobody has rewritten them.

Logistics: rail has not recovered, and both halves of that are true

Rail volumes were **167,9 Mt in FY2025/26, up 4,9%**, against a peak of **226,6 Mt in FY2015** and a 2030 target of 250 Mt — about 26% below peak, with a gap of roughly 80 Mt to the target [5][6]. In the same year Transnet posted its **first annual profit in four years**, R4,6 billion on R88,6 billion of revenue, with container volumes up about 7% to 4,58 million containers and rail access agreements concluded with 11 private train operating companies [5]. Both facts are true and a document that carries only one of them is propaganda in one direction or the other.

Road carries roughly 86% of tonnage at about 94c per tonne-kilometre against rail's 66,5c on direct cost, and 117c against 68,5c once external costs are included [6]. Ports are a current weekly fact rather than a 2023 memory: in the week of 7 to 13 October 2026, Durban's seven-day average vessel waiting time was about 5,73 days with berthing delays of 7 to 10 days at Pier 1 and 10 to 16 days at the Durban Gateway Terminal [7].

What that does to a manufacturer is specific, and generic sources miss it. Inbound raw material lead time is long and, more damagingly, **variable** — a 10-day transit with a plus or minus 12-day berthing distribution requires safety stock sized to the variability, not to the mean. So an import-dependent South African plant carries port risk on its balance sheet as inventory, and a group inventory-reduction target set in Europe and applied here is a service-level decision disguised as a working-capital decision. Bulk exporters are rail-bound and therefore capacity-rationed: the volume you can sell is the volume you can get onto a train, which is a commercial constraint on production planning that standard planning engines have no concept of.

Water: the constraint with no capital answer

Electricity scarcity was answerable with capital — you can buy a generator. **You cannot generate water.** Industrial abstraction is a licensed volumetric cap and effluent quality is an enforceable licence condition, so growth beyond the licence is a licence amendment with a lead time measured in years. The

2026 National Water Amendment Bill proposes banning private water trading [8], which removes the mechanism by which a plant acquires an existing entitlement rather than applying for a new one.

The responses available are to reduce consumption per unit, to recycle and reuse, to secure an alternative supply, or to move. All four are multi-year and three are regulated. And a client that has not modelled water per unit of output at a product and process level has no basis for choosing between them — which makes it a data-architecture problem, because it needs metering at process level and most plants meter at site level.

Labour, carbon and trade, in one paragraph each

Wages are set off the site. Bargaining councils, not firms, set manufacturing wages in South Africa: a statutory body of employer organisations and unions whose collective agreement binds the parties and may be extended by the Minister to non-parties across a whole industry. The metal and engineering council's increases effective 1 July 2026 run from 5,00% at grade A to 6,00% at grade H — an inverted differential, which compresses upward rather than narrowing the gap — and the circular names several employer bodies as **not** covered and does not mention extension to non-parties [9]. "All metal and engineering employers must pay it" contradicts the source. A related trap for anyone reading a motor industry agreement: component manufacturers' increases are on **actual** rates of pay, not on the prescribed minimum, which is a materially more expensive formulation because it lifts everyone already above scale.

One more labour fact that breaks imported models: **direct labour is substantially not variable here**, because retrenchment is a consultation procedure under the Labour Relations Act and shift arrangements are collectively agreed. A North American contribution-margin or breakeven analysis overstates flexibility. And a 24/7 continuous pattern needs four or five crews, not three — five is the honest number once leave, training and absence are covered.

Carbon is a scheduled cost, not an environmental footnote. The tax entered Phase 2 on 1 January 2026 and the rate rises on a legislated path toward 2030, with a punitive rate above a mandatory carbon budget [17]. But the headline rate is not the liability: Phase 1 allowances reached a combined 95% maximum, so the effective rate was a fraction of the headline. The Phase 2 allowance percentages could not be verified in the research for this book and the available sources contradict each other. **Quote the rate and the allowance structure together, or quote neither.** What is not in doubt is the architectural consequence: emissions have to be computed per process, per fuel, per period, with activity data and emission factors, auditable, and reconciled both to a tax return and to a budget allocation. That is a measurement and reporting system, and it is now statutory.

Trade: the tariffs are the story, not the trade preference. The African Growth and Opportunity Act was extended to 2028 and South African eligibility is unresolved, but the live fact is a **25% Section 232 tariff** on South African passenger vehicles, relevant parts, steel and aluminium, plus a **12,5% Section 301 tariff** from July 2026 with Section 232 goods exempt from it [10]. A 25% tariff on vehicles and steel is not offset by duty-free access elsewhere in the schedule. Any statement about South African access to the United States market needs to be dated to the month.

Finally, two tax facts that will save you an embarrassment. **Section 12I is closed and has been since 31 March 2020** — writing that manufacturers can claim the industrial-policy investment allowance is wrong by six years. What is open is Section 12L energy-efficiency savings at R0,95 per kWh saved to 31 December 2030, the special economic zone 15% rate, and the research and development deduction at 150% [16]. Section 12L requires a measured and verified baseline, which is a metering requirement before it is a tax one.

And one incentive that is a data requirement in disguise. The automotive production incentive works through **customs duty rebate certificates earned on local value addition**, not through a tax credit. To claim it, a component maker must compute, per product and per period, the value of locally originating material, local labour and local factory overhead as a percentage of the ex-factory selling price, auditable to source documents, with the imported content of every purchased input evidenced by a declaration from its supplier; and if that declaration is not obtained, the entire purchase price is treated as imported content [18]. That is parts-list-level country-of-origin and cost-component split, propagated through several tiers of supply. **A plant whose costing does not preserve component split through a multi-level roll-up physically cannot substantiate the claim.** It is the most consequential and most commonly under-scoped data requirement in South African automotive manufacturing, and it is the reason chapter 9 spends as long as it does on the bill of material.

What the rest of the book does, and in what order

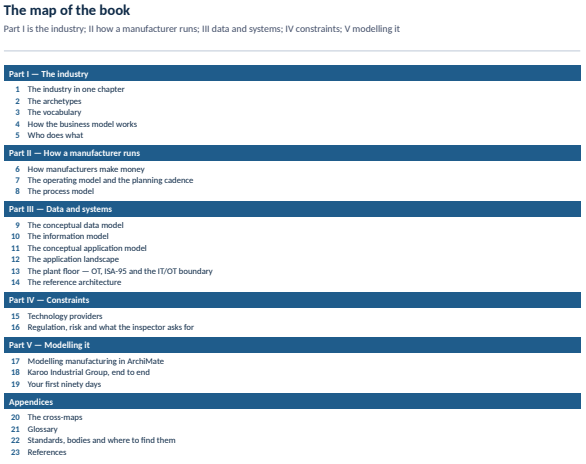


Figure 1.6 The twenty-three items, in five parts and four appendices.

Part I finishes the orientation. Chapter 2 is the one that earns the book: the archetypes, on three independent axes, because a solution designed for one archetype fails in another and you will not find out until week two of your first engagement. Chapter 3 is the vocabulary — not a glossary, but the thirty or forty words whose everyday meaning is wrong, each taught as a mechanism. Chapter 4 covers the business model, chapter 5 the people and the jobs, with what each will and will not tell you in an interview.

Part II is how a manufacturer runs. Chapter 6 takes the money seriously: margin by family, mix effects, working capital through three inventories, capital intensity, channel economics, and tolling, where the manufacturer never owns the material. Chapter 7 is the operating model and the planning cadence, from the monthly sales and operations cycle down to the detailed schedule and the fence. Chapter 8 is the process model, and it is where this book states its one notation exception.

Part III is data and systems, and it is the longest part. Chapter 9 is the conceptual data model, with the bill of material taken properly. Chapter 10 is the information model, 11 the conceptual application model, 12 the real landscape. **Chapter 13 is the one that makes this volume different from the others in this series** — the plant floor, the control systems below the business systems, and the boundary where the two meet. No other industry in this set has an operational-technology layer. Chapter 14 builds a reference architecture, and it has to do so honestly, because manufacturing has no single maintained vendor-neutral industry reference model covering the business end to end. That absence is a finding, and chapter 14 says what to do about it.

Part IV is the constraints you cannot design away: the provider market in chapter 15, and in chapter 16 the regulation, the inspections and the retention clocks, with the plain warning that it is architecture material and not legal advice.

Part V is modelling it. Chapter 17 is ArchiMate for manufacturing, chapter 18 walks Karoo Industrial Group end to end, and chapter 19 is your first ninety days. The appendices carry the cross-maps, a glossary of more than 150 terms, the standards bodies and where to find them, and the references.

Read chapter 2 next, even if you read nothing else. The three axes in it are the difference between an architect who can place a client on sight and one who spends a month discovering that the client is three businesses.

THE CHAPTER IN TEN LINES

- Manufacturing is physical transformation, and three things follow: three inventories at once, capacity that costs money whether or not it runs, and a unit cost that is a function of volume.
- There is no single core system. There is an item, parts-list and route master, a stock and costing ledger, a planning engine and an execution layer, and the integrity question is whether the floor and the ledger agree.
- "What did this cost to make?" has at least nine defensible answers. Build a cost basis dimension, never a unit-cost field, and preserve the cost component split through a multi-level roll-up.
- Absorption costing capitalises fixed overhead into units produced, so producing more raises reported profit with nothing sold. Any dashboard showing profit without inventory movement beside it is misleading.
- Manufacturing is about 11,5 to 12,1 per cent of South African gross value added, and the national accounts rebase to 2022 may move it. Hedge every share figure accordingly.
- Three divisions are 67,95 per cent of the production index. Automotive is 7,84 per cent of it, and also 23,8 per cent of manufacturing value added and 15,6 per cent of national exports. Different measures; say which.
- Name the survey whenever you quote a manufacturing employment number. The household and enterprise surveys disagree and both are official.
- Load-shedding ended: 504 consecutive days at 2 October 2026. Load reduction is a separate, continuing programme. The electricity story is now tariffs, about plus 18,4 per cent over two years, and voltage quality.
- Rail is about 26 per cent below its 2015 peak and Transnet posted its first profit in four years. Both are true. Durban berthing delays make inbound lead-time variability a balance-sheet item.
- Water has no capital answer. Abstraction is a licensed cap and effluent quality a licence condition, so growth beyond the licence takes years.
- Bargaining councils set wages, not firms, and direct labour is substantially not variable in South Africa. A 24/7 pattern needs four or five crews, not three.
- The automotive production incentive is a customs duty rebate regime, and claiming it requires country-of-origin and cost-component split at bill-of-material level, propagated through several supply tiers.

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CHAPTER 2

The archetypes

A solution designed for one kind of manufacturer fails in another, because the kinds disagree about what the fundamental objects are. Three independent axes tell you which kind you are looking at — and the answer is usually more than one, in the same plant, on the same shift.

The chapter that earns the book

A solution designed for one archetype does not merely perform worse in another. It fails, because the archetypes disagree about what the fundamental objects are. In one, the customer order is created months after the product already exists. In another, the product does not exist until the order defines it. A data model that assumes the first will not survive contact with the second, and no amount of configuration will rescue it.

You will not discover this in week one. You will discover it in week two, in a workshop, when somebody from the other plant says something that cannot be true in the model you have just drawn. This chapter exists so that the sentence lands as recognition rather than as a surprise.

Three independent axes are in play, and practitioners blur them constantly.

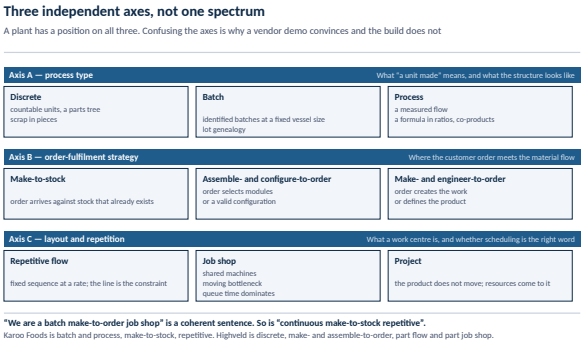


Figure 2.1 The three axes, with the positions on each. They are independent: knowing a plant's process type tells you nothing about where its customer order enters, and neither tells you what its layout is.

A plant has a position on all three, and the three do not predict each other. "We are a batch make-to-order job shop" is a coherent sentence; so is "continuous make-to-stock repetitive". What is not coherent is treating one axis as though it were the other two, which is exactly what happens when someone says "we're a process manufacturer" and the room hears "so forecasting matters and configuration does not".

Axis A — discrete, batch and process

Discrete manufacturing

Output is countable, separable units: a vehicle, a pump, a circuit board, a chair, a bottle. Units have identity and can be serialised. Material is **assembled**, and the assembly is in principle reversible — you can take a pump apart.

The **order** is for a part number or an item code and a quantity. The **parts list** is a tree: parent, components, subcomponents, with quantities per parent that are integers or simple decimals. The **route** is a sequence of operations on work centres, each with a setup time and a run time per piece. **Inventory** is counted in pieces, and stock accuracy is a counting problem. A **unit made** is a unit that passed the last operation, so the question "how many did we make" has an answer you can point at.

The dominant architectural problem is **engineering change control**: which revision was used, when it became effective, which orders were already in flight when it changed, and what happens to the stock of the superseded component. Add serialisation and as-built genealogy wherever traceability is regulated.

Process, or continuous, manufacturing

Output is a flowing, measured substance: fuel, chemicals, cement clinker, paper, steel, liquid milk. Material is **transformed irreversibly** — you cannot un-react a reaction or un-pulp pulp. The plant typically runs for weeks or months between shutdowns.

The **order** is often a tonnage of a grade against a long-term contract with a price formula, not a line on a sales order. The structure is not a parts list at all: it is a **formula in ratios or percentages**, frequently with allowed substitution ranges, and frequently with **co-products and by-products** — one input stream yields several saleable outputs simultaneously, in proportions you only partly control. Costing those co-products requires an allocation convention (by sales value, by physical volume, or by net realisable value) and the convention chosen *determines* each co-product's reported margin. That is a costing decision disguised as a modelling decision, and it is made once, usually by accident.

Work the mechanism once, on an invented refinery stream, because it comes up constantly in food, metals and chemicals and nobody explains it. One process run consumes R100 000 of input and produces 80 tonnes of a main product worth R1 500 a tonne and 20 tonnes of a co-product worth R500 a tonne. Allocate the R100 000 by physical volume and every tonne carries R1 000, so the main product shows a margin of R500 a tonne and the co-product shows a loss of R500 a tonne. Allocate by sales value instead and the main product carries R92 308 and the co-product R7 692, so both show the same

margin percentage and neither looks like a problem. **Nothing physical changed. The plant did not run differently.** Somebody chose an allocation convention, probably at implementation, probably without a meeting.

The consequence is that in a co-product plant, **product profitability is partly an accounting artefact**, and a rationalisation decision taken on reported co-product margin is taken on a number the convention created. The architect's duty here is small and important: make the convention explicit, version it, and make sure anybody reading a co-product margin can see which convention produced it.

The **route** is a flow through units of operation with residence times, not a sequence of discrete operations, and capacity is a rate: tonnes an hour, cubic metres an hour, metres a minute. **Inventory** is measured by mass, volume or level — by instrument, with instrument error. Stock accuracy is a **measurement** problem, not a counting problem, and a tank dip, a belt scale, a flow meter and a silo level sensor each have a tolerance that is frequently wider than the plant's margin.

"A unit made" is the hardest question here. The saleable product is often determined **after** production by the laboratory: the same physical tank can be sold as several grades depending on the assay. The dominant architectural problems are therefore **mass balance and reconciliation** — inputs equal outputs plus losses plus inventory change, within tolerance — and **grade and specification management**.

Batch manufacturing

The middle case, and the most common in South African food, beverage, pharmaceutical, cosmetics, paint, adhesive and speciality-chemical plants. Material is processed in discrete, identified batches through a sequence of process steps, then usually packed into discrete units.

The structure is a **recipe scaled to a batch size**, and the batch size is itself a constrained quantity: the vessel holds 5 000 litres, so you make one or two or three batches, not 1,37. That **batch-size granularity** is a planning constraint discrete manufacturing does not have at all, and planning engines configured for discrete work produce order quantities no vessel can run.

The **route** is a sequence of phases and operations against equipment, with cleaning between batches and frequently equipment-dependent parameters — the same product made on reactor 2 has different times from reactor 1. **Inventory** is lot-controlled with expiry dates and a quality status: quarantine, released, rejected, on hold. **Stock is not available merely because it is present.** The dominant architectural problem is **lot genealogy and release**: full forward and backward traceability from raw-material lot through every intermediate to every pack shipped, retrievable fast enough to execute a recall.

There is a second structure in batch and process plants that discrete people never meet: the **campaign**, or product wheel. The schedule is not a list of orders. It is a repeating cycle through grades in a fixed sequence — a 14-day wheel, say — chosen so that the changeovers between consecutive grades are the cheap ones. In paint you go light to dark; in food you run allergen-free before allergen-containing; in steel and paper the grade sequence minimises transition loss. Changeover is therefore not a number on an item. **It is a matrix, from product to product, it is large, it is maintained by process engineering, and most planning engines cannot consume it.**

The wheel also sets the inventory. A product made once every 14 days needs at least 14 days of cover plus safety stock, by arithmetic, regardless of how good the forecast is. Shortening the wheel is the single most effective inventory intervention available in a process plant, and it is bought entirely with changeover reduction. If a client's standing inventory-reduction proposal is blocked, this is usually why, and the blockage is in engineering rather than in planning. Chapter 6 puts money on it and chapter 7 schedules it.

The object model differs, and here is the table that proves it

	Discrete	Batch	Process
Structure	tree, integer quantities	recipe at a fixed batch size	formula in ratios, with co-products
The thing made	a unit	a batch	a flow, split into lots by convention
Capacity	pieces per hour per work centre	batches per shift per vessel, including cleaning	rate per hour, continuous
Changeover	a setup on a machine	cleaning and validation between batches	a grade transition, producing off-spec material
Quantity truth	a count	a weigh	a meter with a tolerance
Yield loss	scrap units	batch rejection, part-batch loss	continuous loss, evaporation, flare, purge
Traceability unit	serial or batch	batch	lot by time window or by tank
Dominant problem	change control	lot genealogy and release	mass balance and grade

> Read that table as a data-model specification, because that is what it is. Nothing in the discrete column transfers to the process column by configuration. The grain of the central transaction is different, the unit of measure is different, and the definition of a completed thing is different.

The non-stoppable asset, and why power interruption hit manufacturing hardest

This deserves its own section, because it is the structural explanation for something you will be asked about in every South African engagement, and because almost everyone gets the mechanism wrong.

A cracker, a kiln, a blast furnace, a paper machine, a glass tank, a continuous caster, a chlor-alkali cell line, a ferroalloy smelter: these **cannot be stopped and started at will.** Stopping a cement kiln means a cool-down and a re-heat measured in days and costing millions. Stopping a blast furnace risks freezing the hearth, which can mean the permanent loss of the asset. A glass tank can crack. These are not inconveniences scaled up from a retailer closing a shop for two hours; they are a different category of event.

Four mechanisms compound it.

The cost of an interruption is not proportional to its length. It is dominated by the fixed cost of the restart: each restart produces off-spec material and start-up scrap, so a plant interrupted six times for ten minutes is far worse off than a plant interrupted once for an hour. Any availability measure that counts minutes lost is measuring the wrong thing for this class of asset.

A batch interrupted mid-process frequently cannot be quality-released. In a pharmaceutical or food plant the process parameters are part of the specification; if the hold time or the temperature profile was broken, the batch is scrapped, at full standard cost including everything already consumed. The loss is not deferred output. It is destroyed material.

Cold chain is loss, not delay. Dairy, meat, poultry, frozen vegetables, ice cream, vaccines: the product is gone.

And the services restart on a sequence. Compressed air, vacuum, cooling water, effluent handling: losing any of them trips everything downstream even where the main drive had backup power. Scoping an uninterruptible supply around the controllers and not around the services is a classic and expensive mistake.

Now the part that matters most to you as an architect.

> **A grade transition produces material that is neither raw material nor product.** When a paper machine changes from one grade to another, when a pipeline switches product, when a reactor comes back up after an interruption, there is a physical period during which the output is neither the thing you were making nor the thing you are about to make. That material is real. It has a mass, it has a cost, it occupies a tank, and it has a disposition — sell it off-grade, rework it, or dump it.
>
> **No discrete-manufacturing system models transition material**, because in a discrete world a thing is either part A or part B. A system that knows only "product A" and "product B" cannot account for it, and **the mass balance will not close.** The usual symptom is a plant whose physical stock and ledger stock diverge by a steady amount every month that nobody can explain, which is then written off as a counting problem. It is not a counting problem.

Load-shedding itself has ended — 504 consecutive days without it as at 2 October 2026 [4] — but the mechanism has not, because a localised load-reduction event, a voltage dip or a municipal supply failure does the same thing to a kiln. The decision a process manufacturer faced during those years was never "generate power or lose some sales". It was "generate, or lose the batch, the asset, the quality release and the customer's delivery scorecard". That is why the response looked like over-engineering to a reviewer from a country with a reliable grid, and why it was not.

Where the axes meet: the mixed-mode plant

One more case before axis B, because it is common in South Africa and it is the one that defeats a tidy design.

A **mixed-mode** plant does process or batch work and then discrete work, in the same building, on the same order. A food group makes a bulk emulsion in a cook-and-fill plant, then fills, labels and cases it on a packing line. A steel works casts and rolls, then cuts, bends and fabricates. A pharmaceutical plant granulates and compresses, then blisters and cartons.

The handover between the two halves is a named integration problem, and it has four parts.

The unit of measure changes. Upstream the quantity is kilograms or litres; downstream it is units and cases. The conversion is item-specific, sometimes batch-specific because density varies by grade, and occasionally not constant at all. Every quantity crossing the boundary has to carry its unit and whether it was converted.

The identity changes. Upstream the identified thing is a batch or a lot; downstream it is a pack, a case and a pallet. Genealogy has to survive that change of identity, which means the packing line must record which bulk lot it drew from and when, precisely enough to resolve a tank changeover that happened mid-shift.

The constraint moves. The vessel constrains the upstream half and the packing line constrains the downstream half, and which of them binds depends on the pack format being run. A planning engine that models the whole thing as one resource gets the answer wrong in both directions.

And the quality rules differ. The bulk is released on a laboratory result; the pack is released on a line check and a label verification. Two release gates, two owners, two systems, one physical thing.

Karoo Foods is exactly this shape at Paarl, and chapter 18 walks the handover in detail.

Axis B — where the customer order enters the flow

This axis is about one thing: the point at which the customer order meets the material flow. It has several names — the order-penetration point, the decoupling point, the postponement point — and everything upstream of it is driven by forecast while everything downstream is driven by the order.

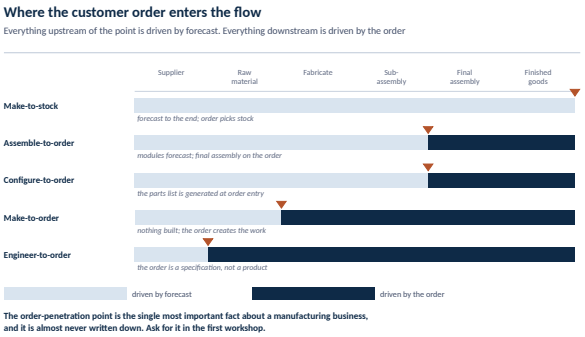


Figure 2.2 The order-penetration point for each fulfilment strategy. The marker is where the customer order meets the flow; pale is forecast-driven and navy is order-driven. Ask for this in the first workshop, because it is almost never written down.

Make-to-stock. The order arrives against finished goods that already exist. Fulfilment is a picking and despatch problem. The parts list is fixed and the customer never sees it. Inventory sits in finished goods, in a distribution network, close to the customer, and finished goods is the largest inventory. Planning is forecast-driven: typically a rolling 12 to 18 months at family level, three to six months at item level, frozen two to four weeks. The dominant architectural problem is **forecasting and finished-goods deployment** — how much of what, in which warehouse — plus shelf life, because make-to-stock finished goods age. Most South African food, beverage, packaging, cement and basic steel long products sit here.

Make-to-order. The order arrives for a known product at a known specification, but nothing is built. The order creates a production order and the customer waits a manufacturing lead time. Inventory sits in raw material and purchased components; work in progress exists only while an order is live. The dominant problem is **promising a date**. Available-to-promise is meaningless because there is no stock to promise from; you need **capable-to-promise**, a check against material *and* capacity *and* the existing order book. That is a different engine with different data, and it is where most make-to-order implementations fail. Structural steel fabrication, industrial pumps and valves, and speciality chemicals in customer-specific grades sit here.

Assemble-to-order. The order selects a combination of pre-made modules. Modules are made to forecast and held; final assembly waits for the order. The structure is two-tier: module parts lists are fixed and forecast, and the top level is created per order from the selected modules. Inventory sits deliberately at the module level — you hold variety's common ancestors, not its leaves. The dominant problem is **forecasting at module level and choosing the postponement point**, plus a final-assembly scheduling process that is separate from the material-planning run for modules.

The extreme form of this archetype, and the one you will meet in South African automotive, is **sequenced just-in-time supply**, which is how a large part of South Africa's component sector works: seven domestic light-vehicle makers buy original-equipment components locally alongside R151,0 billion of imported ones, on 2025 figures [3]. The customer does not place an order in any sense a sales system would recognise. It publishes a schedule: a rolling electronic release, usually weekly, with a firm window of a few days, a forecast tail of several weeks, and a separate sequence message naming the exact parts for the next few hours of its assembly line in the order they will be fitted. The supplier delivers to a line-side location inside a window measured in minutes, in sequence, and is measured on it.

Three things follow and all three are architecture. **The demand signal is a document you receive, not an order you create**, so the order object in your system is derived and the customer's schedule version is the authority. **The schedule changes under you**, weekly or more often, which means every version has to be retained or you cannot explain what you were asked to do. And **the delivery measure belongs to the customer**, computed on the customer's data, with penalties attached, so the supplier must run its own delivery measure and the customer's, reconciled — permanently, as a function and not as a project.

Configure-to-order. The customer specifies a configuration from a defined option set with rules, and the configuration is valid or invalid — validity is computed. The parts list is **generated at order entry** by a configurator from a generic structure plus a rule set, and the price is computed the same way. There may be millions of theoretically valid configurations, so there is no pre-existing parts list for the thing ordered. This is the archetype where "where is the product master?" has the most surprising answer: **it is in the rule engine.** Rules version, options are introduced and withdrawn with effective dates, and an order placed under last month's rules must remain buildable and costable. Truck and bus bodies, commercial vehicle specification, industrial machinery and conveyor systems sit here.

Engineer-to-order. The order arrives as a specification, a drawing or a performance requirement — not a product. The product is designed after the order, often with the customer. Each order is a project with a contract, milestones and progress claims. The parts list is created *during execution*, incrementally, and it changes while manufacturing is already under way; engineering releases it in packages and "structure maturity" is a tracked measure. Material is often not fungible: it was bought for job 47821 and cannot be used elsewhere without a transfer. The dominant problem is that **the project is the cost object, the parts list is a deliverable rather than a master, and change is the business model** — variation orders are how these businesses make money. Mining and minerals-processing plant, materials-handling systems, pressure vessels and bespoke capital equipment sit here.

What moves as you go from make-to-stock to engineer-to-order

- Customer lead time rises; inventory moves upstream and falls in total.
- Forecast importance falls and order-book importance rises.
- The product master becomes less pre-defined and more generated.
- The cost object moves from the product to the job.
- Capacity planning moves from rate-based to resource-and-schedule-based.
- The dominant risk moves from obsolete finished goods to cost overrun on a fixed price.

Axis C — repetitive, job shop, cellular, project

This axis describes layout and flow, and it determines what a "work centre" means and whether scheduling is even the right word.

Repetitive flow. Product moves through a fixed sequence at a fixed rate; layout follows the product. Capacity is a rate and the constraint is the line, so the planning activity is **sequencing and levelling**, not routing. Material is fed to the line side by a pull signal or a sequenced delivery, work in progress is deliberately tiny, and reporting is by line and shift, usually with backflushing rather than per-unit confirmation. The architectural problem is line-side material supply, and the fact that a stoppage anywhere stops everything.

Job shop. Many different products in small quantities, each with its own route through shared general-purpose machines; layout follows the process, so all the lathes are together. Capacity is machine-hours by machine group and the constraint moves with the mix. Each job queues at each machine, and **queue time dominates total lead time** — the large majority of elapsed time is waiting, not cutting. That is why quoted lead times in a job shop bear almost no relation to the sum of the operation times, and why an architect who computes a lead time from a route will be wrong by a factor.

Cellular. Machines grouped by product family into cells, so a family flows within a cell. A compromise, associated with lean. Capacity is the cell.

Project. The product is too large or too fixed to move — a plant, a ship, a mining installation — so resources move to the product. Capacity is crews and skills, not machines, and the schedule is a critical-path network. The architectural problem is **progress measurement**: how complete is it, really, and what will it cost to finish.

The full matrix

This is the table to photograph and keep. Six common combinations, and what each one implies.

	Make-to-stock repetitive discrete	Make-to-stock continuous process	Batch make-to-stock or make-to-order
The order	item code, quantity, date, against stock	tonnage of a grade on a contract with a price formula	item code and quantity, maybe customer artwork and a shelf-life rule
The structure	fixed, shallow tree, versioned by change order	a formula in ratios, with co-products and by-products	a recipe at a fixed batch size, plus a packaging structure
Where inventory sits	finished goods in a distribution network	bulk in tanks, silos and the stockyard; little variety	lot-controlled finished goods with expiry, plus bulk intermediates
Planning horizon	12 to 18 months at family level, 2 to 4 weeks frozen	annual contract, monthly campaign, shutdown cycle of 1 to 5 years	3 to 6 months forecast, a weekly batch wheel
Capacity is	units an hour on a line	tonnes an hour, and availability of the train	batches a shift per vessel, including cleaning
If you build the wrong thing	a configurator nobody needs	you count pieces where there are none	you promise quarantined stock

And the three where the order comes first.

	Assemble- or configure-to-order discrete	Make-to-order job shop	Engineer-to-order project
The order	a configuration of options and rules, priced at entry	a part number or drawing, quantity, required date	a specification, a scope of work, a contract with milestones
The structure	a generic structure resolved by a configurator, or module lists plus a per-order top level	fixed per part, but thousands of them	created during execution, released in packages, changing while building
Where inventory sits	modules at the postponement point	raw material, purchased parts, work in progress in queues	project-allocated material; work in progress is the asset
Planning horizon	modules 4 to 12 weeks, final assembly in days	the order book, weeks	the contract, 6 months to 5 years
Capacity is	assembly rate plus module supply	machine-hours by group	crew-weeks by discipline
If you build the wrong thing	an item code per order, and master data drowns you	dates promised from infinite capacity, and every one missed	the structure frozen as a master, and a business that lives on change stops

Three worked failure modes

These are real in shape. Each is a design that was correct where it came from.

One: available-to-promise deployed into a make-to-order plant

A group standardises on a template built for its make-to-stock food business. The sales team gets available-to-promise, computed from finished-goods stock less committed orders.

It is deployed at a fabrication plant that builds nothing until an order arrives. There is no finished-goods stock, so available-to-promise returns zero on every line. The sales team does the only thing available to them: they override it, by hand, using judgement and a phone call to the plant. Within a quarter the order book contains dates that no system generated and nobody can defend, so the master schedule built from that order book is fiction, so the material plan built from the master schedule orders the wrong steel, so the plant expedites, so lead times lengthen, so the next promise is worse.

The plant needed **capable-to-promise against finite capacity**: a different engine, needing routes, work-centre calendars and the existing load. Nobody costed that, because on the template it looked like the same box.

How to spot it in week one. Ask what proportion of confirmed dates were entered by a person rather than returned by the system, and ask to see the distribution of the gap between the first promised date and the delivered date. If the second is wide and the first is high, the promise is being manufactured by hand and the whole plan downstream of it is decoration.

What it costs. Not the implementation. The cost is that the plant loses the ability to measure delivery performance at all, because there is no system-generated promise to measure against. Chapter 3 shows why that is a permanent loss rather than a reporting gap.

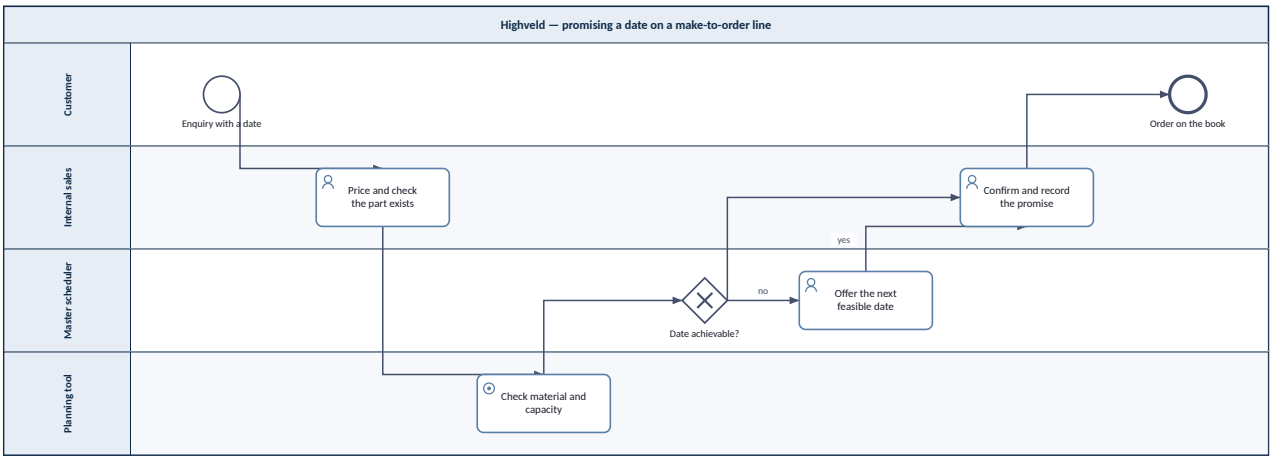


Figure 2.3 Promising a date where nothing is built yet. The check in the planning-tool lane is the one the make-to-stock template does not contain, and substituting an availability check for it is failure mode one. Modelled in BPMN rather than ArchiMate because the hand-off between lanes is the point; chapter 8 states the exception properly.

Two: one parts list per product, deployed into a configure-to-order business

A discrete data model with a clean `product → structure → components` relationship is taken into a business that sells configured machines. Every configuration the sales team sells becomes a new product code with its own structure.

The item master grows by thousands of codes a month. Planning becomes impossible, because every code has a demand history of exactly one and no statistical method works on a history of one. Costing degrades, because each new code needs a cost roll. And the configurator's rules — which are the actual product master — end up re-implemented in three places: in the quoting spreadsheet, in the structure-generation script, and in the heads of two sales engineers.

The fix is not more master data. It is to stop creating item codes for configurations, hold a generic structure plus a rule set as the master, and plan against a **planning bill** whose quantities are option take-rates rather than counts. That is a different data model, not a cleanup.

A planning bill is worth a sentence now because it is the mechanism that makes this archetype plannable. It is a pseudo-structure whose components are real items and whose quantities are percentages representing expected option take-rates: 60% diesel and 40% petrol, 70% manual and 30% automatic. Explode a forecast of 1 000 machines against it and you get 600 of one engine and 400 of the other. It is correct practice, and it fails in three predictable ways: the percentages go stale because option take-rates drift continuously and the bill is updated annually; the options are not independent, because high trim correlates with automatic and the correlation matrix is the data nobody holds; and it is owned by nobody, because marketing owns the mix, planning owns the bill and engineering owns the options. **Assign the owner explicitly.** It is the cheapest recommendation in this chapter and one of the most effective.

Three: counting pieces in a process plant

A discrete execution system is deployed on a continuous line because the group already owns the licences. It demands a good-unit count per order. The plant has a flow meter and a laboratory result, and no count of anything.

So the integration is built to derive a count: mass divided by nominal unit mass. The nominal is wrong by 0,8%, which nobody notices because 0,8% is inside everyone's tolerance for everything. Within a quarter the stock ledger and the tanks disagree by more than the plant's profit for the period, and the disagreement grows monotonically because the error is a bias, not noise.

The lesson is the one in the warning above, stated the other way round: **mass balance is not a report you add later. It is the transaction model.** If the plant's quantity truth is a meter, the system of record has to be built on a measured quantity with a tolerance, and the reconciliation has to be a first-class process with an owner, not a month-end spreadsheet.

The archetype is a property of the order, not only of the plant

Here is the sentence that saves the most rework in this book.

A single South African plant is frequently several archetypes at once, and this is not untidiness — it is how the business makes money. A food manufacturer runs make-to-stock for its branded lines and make-to-order for retailer private label with customer-specific artwork, using the same bulk blend and different packaging. A steel fabricator runs make-to-stock for standard sections and engineer-to-order for structures. A component supplier runs make-to-stock for the replacement-parts trade and sequenced delivery to a vehicle maker's assembly line, off the same presses, in the same shift.

Any solution that assumes one archetype per legal entity meets a counter-example immediately. Not eventually. The second plant you visit.

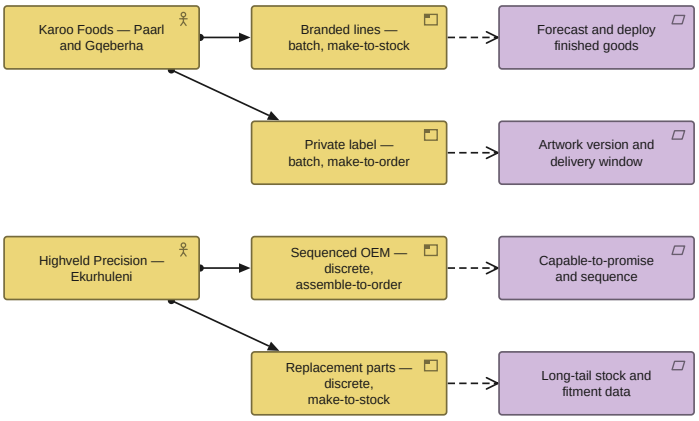


Figure 2.4 Four archetypes across two businesses in one invented group, and the different thing each one demands of the architecture. Nothing about Karoo Industrial Group is real, but a group with four archetypes and one ledger is entirely ordinary.

Karoo Industrial Group is deliberately built this way. Karoo Foods is batch and process and make-to-stock, except for the private-label lines that are made to order against a retailer's artwork and delivery window. Highveld Precision Components is discrete and make-to-order, except for the replacement-parts range that is made to stock and the sequenced programme that behaves like assemble-to-order with a four-hour window. Four archetypes, two businesses, one group ledger. Nothing about Karoo Industrial Group is real, but that shape is.

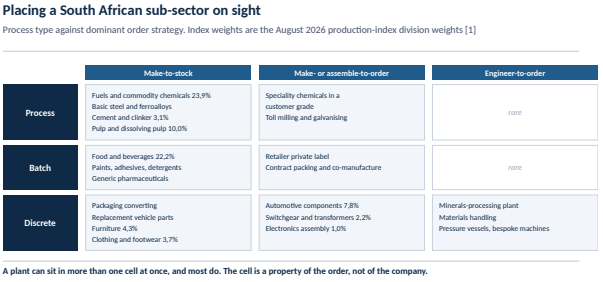
What follows for the architecture is specific and testable:

- **The control mode is an attribute of a material segment, not of a plant.** The deliverable is a map: for each material and each part of the flow, which control mode governs it, where the decoupling points are, and therefore which data must be accurate to what latency. A plant with no such map has three control modes fighting over the same item master.
- **The order type has to carry the archetype**, because the promise rule, the planning rule, the costing rule and the delivery measure all differ by archetype and all hang off the order.
- **One reporting hierarchy will not serve.** A flow measured across cost centres and a cost measured by cost centre are two different trees over the same transactions.
- **Ask the question at order level in the first workshop.** Not "what kind of manufacturer are you" — the answer is always the flattering one. Ask instead: of last month's order lines, how many were shipped from stock, how many started a production order, and how many started a design?

There is a second-order effect worth knowing about, because it surprises people. Two archetypes competing for one physical resource is not a planning inconvenience; it is a permanent commercial argument with a technical shape. At Highveld, the sequenced programme has a four-hour delivery window and a customer scorecard with penalties attached, while the replacement-parts range is made to stock against a forecast and has no window at all. Both run on the same presses. Every week somebody decides which one gives way, and the decision is taken by whoever shouts loudest unless the system makes the trade-off visible. Making it visible means costing the changeover, holding the customer's measure alongside your own, and showing the inventory consequence of each choice. That is three requirements, and none of them appear in a standard scope.

The eleven South African sub-sectors, placed

Now put the axes to work. The table below is the orientation list: what each sub-sector makes, who it sells to, and the constraint that binds it. Weights are from the August 2026 production index [1].



Placing a South African sub-sector on sight

Process type against dominant order strategy. Index weights are the August 2026 production-index division weights [1]

	Make-to-stock	Make- or assemble-to-order	Engineer-to-order
Process	Fuels and commodity chemicals 23,9% Basic steel and ferroalloys Cement and clinker 3,1% Pulp and dissolving pulp 10,0%	Speciality chemicals in a customer grade Toll milling and galvanising	*rare*
Batch	Food and beverages 22,2% Paints, adhesives, detergents Generic pharmaceuticals	Retailer private label Contract packing and co-manufacture	*rare*
Discrete	Packaging converting Replacement vehicle parts Furniture 4,3% Clothing and footwear 3,7%	Automotive components 7,8% Switchgear and transformers 2,2% Electronics assembly 1,0%	Minerals-processing plant Materials handling Pressure vessels, bespoke machines

A plant can sit in more than one cell at once, and most do. The cell is a property of the order, not of the company.

Figure 2.5 The South African sub-sectors placed on two of the three axes, with production-index division weights where the sub-sector maps to a division [1]. Use it to place a client on sight, then check the placement against their order lines.

Sub-sector	What it makes and who buys it	Binding constraint
Petroleum, chemicals, rubber, plastics 23,91%	Fuels, commodity and speciality chemicals, polymers, fertiliser, paints, tyres, converted plastics. Sells to everyone, including the rest of manufacturing	Feedstock and energy cost; carbon exposure on the most emissions-intensive processes in the country; import-parity pricing against the rand; very large single assets, so one shutdown is a macroeconomic event
Food and beverages 22,19%	Milling, baking, sugar, oils, dairy, poultry, canning, snacks, beer, wine, soft drinks. Sells to concentrated grocery retail and wholesale, food service and regional export	Agricultural input volatility, so margin is a spread and not a mark-up; retailer buying power over price, terms and delivery measures; cold chain; water; recall exposure, which makes lot traceability non-negotiable
Basic metals, metal products, machinery 21,85%	Crude and rolled steel, ferroalloys, non-ferrous semis, fabrication, pipe, castings, pressure vessels, mining and agricultural equipment. Sells to mining, construction, automotive, agriculture	Electricity cost and the fact that a smelter cannot be interrupted without risking the furnace; scrap availability and price; import competition; a 25% US tariff on steel and aluminium [2]; weak mining and construction demand
Wood, paper, publishing, printing 9,98%	Timber and board, pulp including dissolving pulp for export, packaging grades, tissue, print. Sells to global viscose markets, domestic packaging and a declining print market	Plantation fibre on an 8 to 20-year rotation, so supply is a decade-lagged constraint; dollar pulp prices; water, as the most water-intensive large process industry; large scheduled shutdowns that dominate quarterly results
Motor vehicles, parts, other transport 7,84%	Light and commercial vehicles; catalytic converters, engine parts, harnesses, pressings, seats. Sells to export markets first and the domestic market second	Local-content incentive mechanics; a 25% US tariff on vehicles and parts [2]; port and rail performance on both inbound components and outbound vehicles; model-cycle decisions taken offshore; bargaining-council labour cost
Furniture and other manufacturing 4,29%	Household and office furniture, mattresses, and a residual category including jewellery, sporting goods and medical devices. Sells to concentrated furniture retail and hospitality	Consumer credit conditions and durable-goods demand; import competition; board and timber input cost. Weakest division in the index in August 2026 at −12,6% year on year [1]
Textiles, clothing, leather, footwear 3,68%	Yarn, fabric, garments, technical and home textiles, leather, footwear. Sells to concentrated clothing retail and to uniform and workwear contracts	Import competition including direct-to-consumer e-commerce that bypasses the retail channel; retailer sourcing policy, so demand is a procurement decision rather than a market; lead time and flexibility, which is an architectural problem. The only division growing in August 2026, at +5,7% [1]
Glass and non-metallic minerals 3,11%	Cement, clinker, aggregate, ready-mix, bricks, lime, glass containers and flat glass, refractories. Sells to construction, infrastructure, beverage and automotive	Construction demand, which is the whole story; cement imports; electricity, coal and carbon on a very carbon-intensive process; the non-stoppable kiln; and an economic radius set by freight cost per tonne-kilometre, which makes the market regional by physics
Electrical machinery 2,15%	Transformers, switchgear, cable, motors, batteries, accessories. Sells to the utility, municipalities, mining and construction	State and municipal procurement, including local-content requirements in public tenders, which makes demand partly a creation of policy; the grid build programme; import competition at the commodity end
Radio, television, communications, professional equipment 1,00%	Instrumentation, telecoms equipment, a small medical-device base. Sells to telecoms operators, industry and health	A very small scale base, which means most electronic assembly here is low-volume and high-mix — a job shop, not a line
Pharmaceuticals (inside the chemicals division)	Generic solid-dose and liquid medicines, sterile injectables, anaesthetics. Sells to state tenders, private wholesale and retail pharmacy	Regulated pricing from above and input cost from below; registration and good-manufacturing-practice compliance, so a process change is a regulatory event; very high fixed cost in sterile capacity, which makes it brutally utilisation-sensitive; active-ingredient import dependence

One cross-cutting note. **Packaging** is spread across the chemicals, metals, paper and glass divisions but behaves as a sector in its own right: derived demand, so its constraint is its customers' volume; input commodity prices passed through with a lag, so margin is a timing position; very high capital intensity on single lines, so one line is a large fraction of national capacity; and an economic radius, because empty packaging is expensive to move, which is why packaging plants sit next to fillers.

What to do with this in week one

Walk the floor before you interview anybody senior. Then ask four questions, in this order.

1. **Of last month's order lines, how many shipped from stock, how many started a production order, and how many started a design?** That gives you axis B, by volume, as a fact rather than a self-description.
2. **What is the quantity truth here — a count, a weigh or a meter?** That gives you axis A, and it tells you immediately whether you are going to be arguing about mass balance.
3. **Does the product move through the machines, or the machines through the product, and does the constraint move with the mix?** That gives you axis C.
4. **Which of those answers changes if I ask about the other plant, or the other customer?** That tells you how many archetypes you are actually designing for, which is the number that determines the size of the programme.

Chapter 3 gives you the words to understand the answers. Chapter 7 turns axis B into a planning cadence, chapter 9 turns axis A into a data model, and chapter 18 walks Karoo Industrial Group through all three.

THE CHAPTER IN TEN LINES

- Three independent axes govern a manufacturing business: process type, order-fulfilment strategy, and layout and repetition. They do not predict one another.
- Discrete counts pieces, batch makes identified batches at a constrained vessel size, and process measures a flow with a formula, co-products and a meter tolerance. The data model differs in grain, unit and definition of a completed thing.
- Process manufacturing's assets cannot be stopped at will, which is why power interruption hit manufacturing categorically harder than other sectors. The cost of an interruption is dominated by the restart, not by its length.
- A grade transition produces material that is neither raw material nor product. No discrete system models it, and without it the mass balance will not close.
- The order-penetration point is the single most important fact about a manufacturing business and is almost never written down. Everything upstream of it is forecast; everything downstream is order.
- Available-to-promise is meaningless in a make-to-order plant. It needs capable-to-promise against finite capacity, which is a different engine with different data.
- In configure-to-order, the product master is the rule engine. Creating an item code per configuration drowns the item master and makes planning impossible.
- In a job shop, queue time dominates lead time, so a lead time computed from the route will be wrong by a factor.
- The archetype is a property of the order, not only of the plant. Almost every real South African plant is several at once, and any solution assuming one archetype per legal entity meets a counter-example immediately.
- The architect's deliverable is a control-mode map per material segment, not a tool choice.
- Three divisions are two-thirds of South African manufacturing, and each carries a different binding constraint. Place a client on the matrix, then verify the placement against last month's order lines.

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CHAPTER 3

The vocabulary

Manufacturing's barrier to entry is not its complexity. It is forty words you already know that mean something else here. This chapter teaches each one as a mechanism, so that you can follow the conversation and, more usefully, hear when two people in it are using the same word differently.

What this chapter is, and what it is not

Chapter 21 is the glossary and it runs to more than 150 terms. This is not that.

This chapter is the thirty or forty words whose everyday meaning is **wrong** — words you already know, used by manufacturing people to mean something else, precisely enough that using them loosely marks you out in the first five minutes of a workshop. Each one is taught here as a mechanism rather than as a definition, because the definition is not the point. The point is what the word does to a number, a data model or a decision.

It is organised by domain rather than alphabetically, because the traps cluster. Read it once through, then come back to a section the morning before the workshop it belongs to.

One habit to form while you read. When somebody in a plant gives you a measure, the useful response is almost never "what is it?" It is **"measured how, from when, and compared against what?"** Most of the entries below exist because that question has more than one defensible answer.

Yield, scrap and rework

Yield, sometimes called final or overall yield, is good units out divided by units started. It permits rework: a unit that failed, was fixed and then passed counts as good.

First-pass yield, also called first-time-through or right-first-time, is units passing **without any rework** divided by units started. It is always less than or equal to yield.

Rolled throughput yield is the product of the first-pass yields of every step. This is the measure that exposes what practitioners call the hidden factory, because a ten-step process at 98% first-pass yield per step has a rolled throughput yield of 0,98 to the tenth power, about **81,7%** — a number most plants would not recognise as their own.

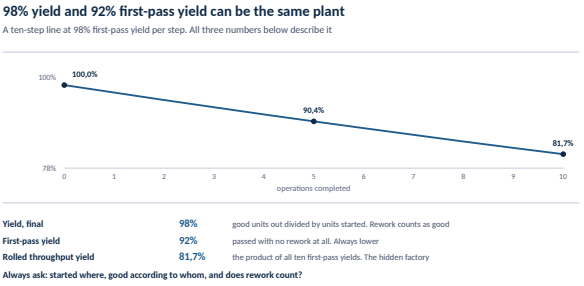


Figure 3.1 Three true statements about one ten-step line at 98% first-pass yield per step. Rolled throughput yield is my arithmetic on the stated step yield. A yield number without the stage it was measured at is not a number.

So a plant quoting 98% yield and a plant quoting 92% first-pass yield may be the same plant on the same day. Neither is lying. They measured at different stages, with different treatment of rework.

Scrap is material irrecoverably lost. Three distinctions matter and all three are modelling decisions. **Planned scrap** is carried in the structure as a scrap or shrinkage factor, so it is inside the standard cost; **unplanned scrap** appears as a usage variance. **Start-up scrap** is the predictable loss at the beginning of a run or after a changeover, so it is a function of the *number of runs*, not of the volume — which means a plant that halves its batch size doubles its start-up scrap and nobody attributes it correctly. **Running scrap** is proportional to volume.

Rework is material recovered by additional processing. It consumes capacity and labour that were planned for new production, so a plant with high rework has a capacity problem it cannot see in its production numbers.

Yield loss in a process plant is a different animal. It is output mass divided by input mass, where the loss is evaporation, purge, flare, dust, slag, trim, or straightforward conversion inefficiency. It is bounded by chemistry or physics, not by workmanship, and there is a **theoretical yield** against which the plant's distance is the real performance measure.

The scrap arithmetic, and the five years of unexplained variance it causes.

If a component has a 10% scrap rate, you do not multiply the requirement by 1,10. You divide by 0,90. To get 100 good pieces you must start 100 ÷ 0,90 = **111,1**, not 110. The difference is 1,1 pieces per hundred, every time, for ever.

Compounded through a structure it gets worse, because scrap applies at each level:

Levels at 3% scrap each	Correct uplift (divide by 0,97 each time)	The wrong arithmetic (add 3% each time)	Gap
1	3,09%	3,00%	0,09 pp
3	9,55%	9,00%	0,55 pp
5	**15,93%**	**15,00%**	**0,93 pp**

Five levels at 3% is about **15,9%**, not 15%. Just under one percentage point of material, on every order, for the life of the configuration.

It never appears as a defect, because nothing fails. It appears as a **permanent, unexplainable adverse material usage variance** that survives three improvement programmes and two system replacements, and that everybody eventually treats as a fact of nature. If a client has one, check the scrap uplift formula before you check anything else. It is a twenty-minute test.

Time, rate and the four words that are confused most often

Takt time is available production time divided by customer demand in that time. If a line has 27 000 productive seconds in a shift and demand is 450 units a shift, takt is 60 seconds. Note what that means: **takt is a demand-derived target rhythm, not a measurement of the plant.** It changes when demand changes, without anything in the plant changing. It is not a performance measure and should never appear in a performance report.

Cycle time has at least three uses and they are not the same. *Operation cycle time* is the time for one operation on one unit. *Line cycle time* is the interval at which completed units leave the line, which is one divided by the output rate, and is the figure you compare against takt. *Total cycle time* or throughput time is the elapsed time for one unit to traverse the whole process **including waiting**, and it is frequently used interchangeably with lead time, which is why the term is dangerous.

Lead time also has several uses. *Manufacturing lead time* runs from order release to order completion. *Customer or quoted lead time* runs from order placement to customer receipt. *Cumulative lead time* is the longest path through the structure from the lowest-level purchased part to the finished good, and it is the real constraint on how responsive a make-to-order business can be — and the number almost nobody calculates. *Replenishment lead time* for a purchased item runs from order to goods-receipted-and- available, which is longer than the supplier's quoted transit because it includes receiving, inspection and put-away. In South Africa, add the port.

Throughput is where two professions collide in one room. To a planner it is good units or value produced per unit of time. In theory of constraints it is specifically **sales revenue minus totally variable cost** — the rate at which the system generates money, not units. Both usages are correct within their tradition and both will be used in the same meeting. If a client says "throughput", establish which one before you write anything down.

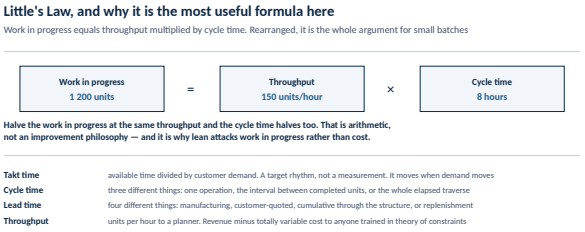


Figure 3.2 Little's Law with illustrative quantities, and the four time words beneath it. The law is an identity, not a theory — which is why work in progress and lead time cannot be managed separately.

Little's Law ties them together: work in progress equals throughput multiplied by cycle time. Rearranged, cycle time equals work in progress divided by throughput. That is why reducing work in progress shortens lead time at constant throughput, and it is the arithmetic behind lean's preoccupation with work in progress. It is the single most useful formula in this chapter and it is not a theory; it is an identity.

Capacity, which is four numbers wearing one word

A single capacity figure is meaningless. Ask which one.

Term	What it is	Who uses it
Theoretical, design or nameplate	Output at design rate, 24 hours a day, no stoppages, no changeovers, no quality loss	Equipment vendors and press releases
Rated, nominal or installed	Theoretical, adjusted for a declared operating pattern and a maintenance allowance	Investment announcements. Meaningless without its assumed shift pattern, which is usually omitted
Effective	Rated, less planned maintenance, changeovers and planned non-productive time	Planners, as a starting point
Demonstrated or proven	The highest sustained output actually achieved over a representative period, at acceptable quality, **in the current product mix**	**The only one that should enter a plan**
Protective	Deliberate slack at non-constraint resources, so they can recover from disruption without starving the constraint	Anyone running to the theory of constraints

Two consequences. **Demonstrated capacity is mix-specific**: change the mix and it changes, because changeovers and run rates change. And **utilisation is only interpretable once you say which capacity**. Utilisation against theoretical and utilisation against demonstrated, for the same plant in the same month, can differ by 25 percentage points.

South Africa publishes an official series — large manufacturers used 76,6% of production capacity in the May 2026 reference month [1] — and you will be asked to compare a client's internal number to it. Do not. It covers large enterprises only and it is self-reported against each respondent's own notion of capacity.

Work centre, and the two standards that disagree

Work centre is where work happens: a machine, a group of machines, a cell, a line, a bench. In an enterprise resource planning system it carries available capacity by shift, formulas that turn standard values and quantity into a capacity requirement and a cost, a cost centre and activity types — which is how time becomes money — and, in a well-configured estate, a link to the equipment record in the maintenance system, which is the join that lets a maintenance outage become a capacity reduction and which almost nobody populates.

But the word also appears in the control-systems standard that the plant floor argues in, and there it means something related but different. In the ISA-95 equipment hierarchy, `WORKCENTER` is the **generic** term for a level of the hierarchy whose specific name depends on what kind of manufacturing you do — a process cell, a production line, a production unit or a storage zone — with `WORKUNIT` the generic term below it. The two models do not line up with the plant and storage-location model of an enterprise system, and reconciling them is a real project rather than a mapping exercise. Chapter 13 does this properly, including the related fact that the asset hierarchy and the production hierarchy are two different trees over the same steel.

> Two further corrections about ISA-95 worth carrying from the start, because they are everywhere in circulating material. The equipment hierarchy is **not** a seven-level stack ending in a control module — the enumeration has fifteen values, and several of them are alternatives selected by process type rather than levels stacked on one another. And equipment module and control module are **ISA-88** concepts that ISA-95 references, not ISA-95's own. Chapter 13 and chapter 22 give the versions and the part structure, which have moved recently.

On time, and why 96% and 78% can be the same performance

On-time-in-full is the customer-facing measure and it is the most frequently mis-specified measure in manufacturing architecture. Every one of the following is a live design choice, and two plants in one group that chose differently are not comparable.

Which date is the clock measured against?

Date basis	What it means	What it hides
Customer request date	What they originally asked for	Nothing. It is the hardest, and the only one the customer recognises
First confirmed date	What we said when we accepted the order	A bad promise counts as a success
Latest confirmed date	The most recent re-promise	Everything. You can reach 100% by re-promising until you are right, while the customer is furious
Schedule of record	The date on a customer-managed schedule	That the date itself changes weekly, upstream of you

Then three more choices. **What does on time tolerate** — exactly on the day, a window of minus zero plus two days, a delivery slot measured in minutes for sequenced automotive supply, or a booking window at a customer's warehouse? Note that **early is frequently a failure**, not a success. **What does in full mean** — order level, where one short line fails the whole order, or line level, where each is scored separately? Order-level is arithmetically lower and gets worse as orders get larger, so the measure can decline purely because order sizes grew. **And where does the clock stop** — at goods issue, at carrier collection, at the customer's gate, or when the customer books the receipt into their own system, which adds their receiving delays to your measure?

> Systems that **overwrite the promised date on reschedule** destroy the ability to measure delivery performance at all, and this is extremely common. An order line needs, at minimum: request date, first promise date, current promise date, despatch date, delivery date, quantity ordered, confirmed, despatched and received, and a reason code for any variance — all retained, none overwritten. Check this in your first fortnight. It is a one-query test and it decides whether the client can measure anything at all.

A South African footnote with teeth: the large customers publish their own supplier scorecards, with their own definitions and their own data, and they will not accept yours. A manufacturer therefore needs its own measure *and* each major customer's measure, reconciled. That reconciliation is a

permanent function, not a project.

OEE, and the three traps inside it

Overall equipment effectiveness is availability multiplied by performance multiplied by quality [2]. Availability is run time over planned production time; performance is ideal cycle time multiplied by total parts, over run time; quality is good parts over total parts. Put more intuitively, it is fully productive time divided by planned production time: of the time we intended to make good product at full rate, what fraction did we achieve?

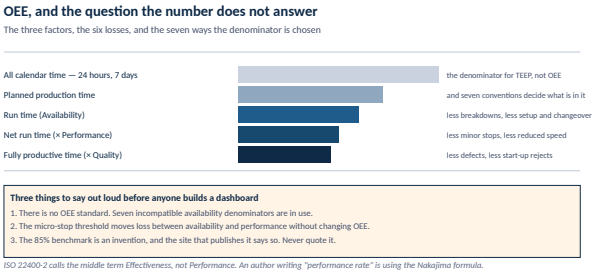


Figure 3.3 The loss cascade behind overall equipment effectiveness, with illustrative proportions, and the three things to say before anybody builds a dashboard on it. The second bar is where the seven denominator conventions do their damage.

It is a genuinely useful measure on a single constraint asset. It is also the single most abused number in manufacturing reporting, for three reasons.

Trap one: there is no OEE standard. The calculation is a de facto convention, not a published one [2][3]. The research behind this book found **seven incompatible availability denominators** in use: all calendar time; scheduled or shift time; scheduled time less planned maintenance; scheduled time less maintenance and less breaks; scheduled time less periods of no demand; scheduled time less external causes such as a supply interruption or a strike; and the variant that counts all unscheduled time, which is a different measure called overall operations effectiveness. A plant that excludes planned maintenance from the denominator looks better than a maintenance-light plant that does not, for no operational reason.

Trap two: the micro-stop threshold moves loss between the factors without changing the product. Downtime shorter than a company-set threshold — typically one to five minutes — is conventionally counted under performance rather than availability. Two plants with identical physical behaviour and different thresholds report different availability and different performance, and the **same** OEE. So OEE can be comparable in a group where its components are not, which is the opposite of what everyone assumes, and it means a drill-down from a comparable headline into incomparable parts is a trap waiting in every dashboard.

Trap three: the benchmark is an invention.

> **The 85% OEE benchmark — the one everybody quotes as the mark of an excellent plant — is made up, and the site that publishes it says so**, describing it as a convenient, compelling and completely artificial benchmark [2], and a second source states that treating 85% as the mark of excellence is in many cases incorrect [3]. **Never put 85% in a client deliverable as a benchmark.** It is usually attributed to Nakajima's work on total productive maintenance by way of multiplying 90% availability by 95% performance by 99% quality, but that attribution could not be sourced in the research for this book and should not be repeated as fact. Set incremental targets against the plant's own history instead.

Two further points that will save you an argument. OEE is explicitly documented as **unsuitable for comparing machines, for benchmarking and for aggregating to department or plant level** [3] — a line's OEE is not the average of its stations'. Compute it on the constraint, or on the line treated as one resource. And **ISO 22400-2's middle term is Effectiveness, not Performance**, so an author writing "performance rate" while citing the ISO standard is actually using the Nakajima formula. If you are specifying measures against the standard, use its words.

> When a client asks for a group manufacturing dashboard, the first deliverable is **not a dashboard**. It is a measure definition register: one row per measure, with an owner, a formula, a denominator convention, a reason-code hierarchy where one applies, and a declared comparability scope. Ship the dashboard first and it will be rebuilt within a year.

Quality words, where the distinction has legal force

QC, quality control, is **inspection and testing**: measuring product and process against specification, in the laboratory and on the line. It is reactive and product-focused, and it detects defects. **QA**, quality assurance, is **system and prevention**: procedures, specifications, validations, training records, supplier approvals, change control, investigation of deviations, audits — and the authority to **release or reject**.

Get this right, because practitioners will correct you, and in a regulated plant the distinction has legal force: **the release decision belongs to QA, not to QC and not to production.** QA holds a veto over despatch, which means it must be modelled in the system as a **status on the stock**, not as a note on a certificate. An execution system that lets production despatch unreleased material in a pharmaceutical or food plant is not misconfigured. It is a compliance failure.

Four more words that go with it. A **specification** is the set of attributes and limits a material must meet, and it is versioned and dated like any other master data. **Quarantine** is a stock status, not a place, although it is usually also a place. A **non-conformance** is a recorded instance of something failing its specification, and a **deviation** is a recorded, approved departure from a procedure before the fact; confusing the two will get you corrected in a pharmaceutical plant within a minute.

And one timing fact that belongs in every planning conversation. **The laboratory's turnaround time is a planning constraint.** If a release test takes 48 hours, finished goods sit in quarantine for 48 hours, and that is inventory, lead time and cash. Plants rarely model laboratory turnaround in their lead times. They should, and it is a cheap change.

The words for what gets made and how

Bill of material, usually just "the BOM", is the structure of components that go into a parent. The thing to know now, and the reason chapter 9 spends so long on it, is that **there are at least four of them over the same parts, all simultaneously true**: the engineering structure as the product was designed, the manufacturing structure as the plant actually builds it with its consumables, phantoms, scrap factors and operation assignments, the planning structure with percentage quantities for options, and the service structure as a technician decomposes the product in the field. They are not views of one structure. They have their own identity, lifecycle, approval and effectivity.

Routing is the sequence of operations needed to make the parent, each assigned to a work centre, each with standard times. **Operation** is one step in it. In a process plant the equivalent object is a **recipe** with phases, which is a different shape, not a renamed one.

Standard time is the planned time for an operation, split into setup time, which is per run, and run time, which is per piece. It is the foundation of both the capacity model and the labour element of the cost. **It is also always wrong**, and for a reason worth understanding rather than complaining about: standard times are set by an industrial engineer at commissioning and are rarely re-timed, so a labour efficiency variance is often measuring the age of the standards rather than the performance of the people. Ask when the standards were last re-timed. If the answer is "when the line was commissioned", you now know what the variance report means.

Phantom, or transient assembly, is an item that exists in the structure to group components but is never stocked and never gets a work order of its own; the planning run blows straight through it to its components. The correction to carry: **phantom is a property of the structure line, not of the item.** The same item is legitimately a phantom in one parent and a stocked assembly in another, and an item-level flag makes that impossible.

Production order, work order, process order, batch are near-synonyms with real differences. An order is the authority to make a quantity of something and the object that collects its cost. In batch and process plants the equivalent carries the recipe and its parameters. In a pure pull system there may be no order at all, which has a consequence described below under backflush.

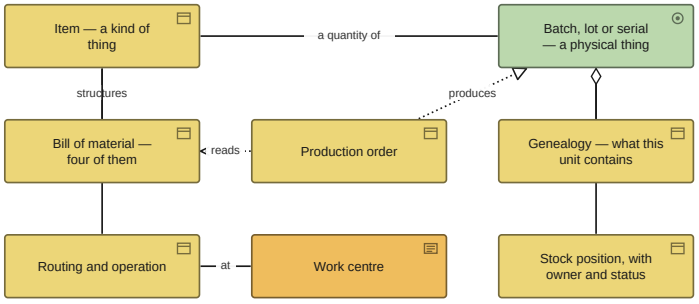


Figure 3.4 The core objects the vocabulary in this chapter describes, and the line between the product model on the left and the physical model on the right. Chapter 9 adds attributes, cardinality and the parts this view deliberately leaves out.

Identity and traceability

Batch and **lot** are used interchangeably in most plants and mean a quantity of material produced or received together, treated as homogeneous, and given one identifier. **Serial number** identifies one individual unit. Which of the two a plant uses is driven by how fine its traceability obligation is and by whether the product has individual identity at all.

Where-used reads the structure upwards: given a component, which parents consume it, at every level, as at a date. **Genealogy** reads actual consumption: given an actual lot, which actual units consumed it.

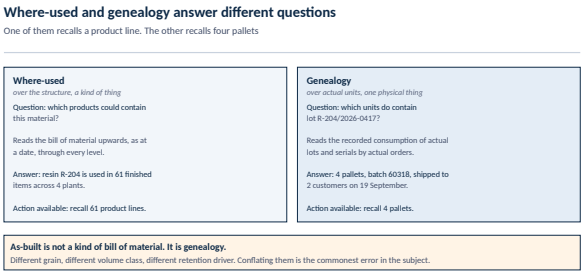


Figure 3.5 The two questions traceability is asked, with invented identifiers. Where-used reads the structure; genealogy reads what was actually consumed. Only the second one bounds a recall.

> **Where-used tells you which products *could* contain the contaminated lot. Only genealogy tells you which *do*.** That is the difference between recalling a product line and recalling four pallets, and it is a difference measured in millions of rand and in whether the brand survives the week. If a client's traceability design can only answer the first question, they do not have traceability. They have a parts list.

As-built is not a kind of bill of material. It is genealogy: a statement about *one* physical unit, not about a kind of thing. Different grain, different volume class — millions of rows against thousands — and a different retention driver.

Backflush is the practice of consuming the standard quantities of components automatically when the finished quantity is reported, instead of issuing each component explicitly. It is cheap, it reduces the data-entry burden on operators, and it is the right answer in many repetitive plants.

> **Backflushing systematically hides material usage variance**, because it consumes the *standard* quantity by definition. An over-issue or an unreported scrap event cannot show up as a variance; it surfaces months later as a stock-count difference instead, where nobody can attribute it to anything. If a client has unexplained inventory shrinkage and backflushes, you have found the mechanism.
>
> Worse, where substitution happens on the floor, backflush produces a **confidently false genealogy** — it records the component the structure named, not the one actually consumed. A gap in traceability is a problem. A wrong answer that looks complete is a different and larger problem, and it is the one that fails a recall.

Planning words, distinguished properly

These six are used loosely and mean specific things.

Term	What it actually is
S&OP	Sales and operations planning: a monthly, cross-functional, management-level process that reconciles demand and supply over a rolling 18 to 24 months at product-family level, producing one agreed plan. Its purpose is not a better forecast; it is to force a single set of numbers and surface the decisions the gap requires
MPS	The master production schedule: what the plant will build, by item and period, over the medium horizon. It is a **decision**, not a calculation — a human chooses it, and an MPS that exceeds demonstrated capacity propagates infeasibility into everything downstream
MRP	Material requirements planning: the calculation. Explode the schedule through the structure, net against on-hand and on-order, apply lot sizing, offset by lead time, produce planned orders and exception messages. It assumes infinite capacity and known, constant lead times
MRP II	Manufacturing resource planning: MRP plus capacity, shop-floor control, purchasing and a financial feedback loop. Its contribution over MRP is the **closed loop** — the plan is tested for feasibility and the result feeds back
RCCP	Rough-cut capacity planning: tests the master schedule against a handful of critical resources using a bill of resources per family. Coarse, fast, and the right tool for a monthly conversation. The wrong tool for a daily one
APS	Advanced planning and scheduling, also finite capacity scheduling: sequences work against finite resources, honouring calendars, sequence-dependent changeovers, tooling, skills and material. Produces a feasible sequenced schedule — when the constraint model is complete, which is the condition that usually fails

Two related terms. **Available-to-promise** is computed from the master schedule plus inventory less committed orders; **capable-to-promise** asks whether we can actually do this by then given everything already committed, and needs finite capacity. Chapter 2's first failure mode is what happens when the first is substituted for the second.

The fence. Time fences are dates in the planning horizon at which the rules about who may change the plan, and how, change. Conventionally there is a **frozen zone** from now to one or two weeks where nothing changes except by escalation, because material is committed, labour is rostered and setups are sequenced; a **firm zone** out to roughly eight weeks where the master scheduler may change sequence and quantity within capacity; and a **free zone** beyond it where the planning run plans as it likes. Two distinct fences exist: the **demand time fence**, inside which the forecast is ignored and only actual orders count, and the **planning time fence**, inside which the system will not create new planned orders on its own.

> **A time fence shorter than the cumulative material lead time is arithmetically unenforceable.** If the longest path through the structure to a purchased part is nine weeks and the firm fence is four, then anything inside the fence depends on material that was ordered outside it, on a forecast, and the fence cannot protect what it claims to protect. This is the commonest configuration error in this area and it is checkable in an afternoon: compute cumulative lead time per end item, compare it to the configured fence, and count the items where the fence loses.

There is a second mechanism in the same area that is almost never diagnosed. **Updating planned lead times from observed performance creates a positive feedback loop**: a longer observed lead time produces a longer planned lead time, which releases work earlier, which puts more work in progress on the floor, which by Little's Law lengthens the observed lead time. Plants that "tune" lead times annually from actuals are often tuning themselves into a slower factory.

Ownership: tolling and contract manufacturing

Tolling, also toll processing, is where the customer supplies the material and the manufacturer processes it for a **conversion fee**. Title to the material never passes to the processor. **Contract manufacturing**, also co-manufacturing or co-packing, is where the manufacturer buys the material *and* makes the product to the customer's brand and specification. The two are routinely confused and they have opposite balance sheets.

In tolling the material is not the processor's inventory, so the balance sheet is light, the revenue is the fee rather than the value of the goods, and every ratio to revenue is incomparable to an owning manufacturer's. The risk is **yield**: a tolling contract specifies an agreed yield and material lost above the allowance is at the processor's cost, at the *material's* value, which may dwarf the fee. A one per cent yield miss on an expensive input can wipe out the entire conversion margin.

> **Ownership is a first-class attribute of inventory, and omitting it is one of the very few genuinely unrecoverable design errors in manufacturing.** In a tolling plant the material belongs to the customer. Stock in one silo may be owned by three parties. The same tonne of flour may be the plant's asset on Monday and a customer's on Tuesday. You cannot retrofit ownership onto historic stock movements, because the information was never captured — so the cost of getting this wrong is not a migration, it is a permanent hole in the history.

The lean words, defined so you are not bluffed by them

You will be told a plant is "doing lean". These are the words that will follow, and knowing what they actually claim is the difference between a useful conversation and a nodding one.

Word	What it is	What it demands of the systems
Muda, mura, muri	Waste, unevenness and overburden. Muda is the famous one — overproduction, waiting, transport, over-processing, inventory, motion, defects — and the other two are the ones usually dropped and the ones that matter operationally	Measurement aligned to a flow rather than to cost centres
Kaizen	Continuous improvement by many small changes made by the people doing the work	Almost nothing, which is why it is popular
Gemba	The actual place where the work happens. "Go to the gemba" means go and look rather than read a report	Nothing. Do it anyway, in your first week
Kanban	A signal that authorises replenishment of what was consumed. A card, a container, an empty square. The number of cards times the container quantity **is** the maximum work in progress — that is the whole control system	Container quantity, card count and loop lead time as governed master data; signal latency in minutes, not overnight
Heijunka	Levelling: producing the same quantity and mix every period, absorbing demand variability into a small finished-goods buffer rather than transmitting it into the plant	Mixed-model sequencing at master-schedule level, and a demand-smoothing policy with a named owner
SMED	Single-minute exchange of die: the systematic reduction of changeover time by separating work that needs the machine stopped from work that does not, and converting the first into the second	Changeover as a measured, reason-coded event, and the from-to changeover matrix as master data. Most planning engines model setup as a constant per item and therefore cannot see the benefit
Jidoka	Automation with a human quality: equipment and people that stop automatically when something abnormal happens, so defects are not passed on	The authority and the mechanism to stop, which is organisational before it is technical
Andon	The visual signal and board showing line status and calls for help	Low-latency status. This is the one place lean asks for more technology, not less
Poka-yoke	Error-proofing: designing the operation so the mistake is physically impossible rather than merely discouraged	Nothing, usually. It is a fixture, not a feature
5S	Sort, set in order, shine, standardise, sustain. Workplace organisation so that abnormality becomes visible	Nothing, which is why "we're doing lean" frequently means "we painted the floor"

> **"Lean means no inventory" is wrong**, and repeating it will cost you credibility with anyone who has actually run a levelled line. Heijunka **deliberately holds finished-goods inventory** to absorb demand variability instead of transmitting it into the plant and its suppliers. Lean's claim is not that inventory is always bad; it is that **lead time, not cost, is the primary variable**, and that inventory conceals the problems that lengthen it. Those are different claims and only the second one is lean's.

One structural point to carry into chapter 7. A pull system and a material-requirements- planning push system make genuinely **incompatible** demands on the same data: one decides in advance from a forecast and needs an order to carry plan, cost and confirmation; the other decides after the fact from consumption and has no order at all. Most real plants run both, legitimately, on different material segments. The architect's deliverable is a control-mode map, not a tool choice.

The control-system words you will meet in chapter 13

Chapter 13 is the chapter no other volume in this series has, and it will go faster if these eight words are already familiar. One line each.

- **PLC** — programmable logic controller. An industrial computer that reads its inputs, runs its program and writes its outputs on a fixed cycle, typically every 1 to 50 milliseconds. Determinism is the property that matters: one that *sometimes* takes 80 milliseconds is broken, not slow.
- **DCS** — distributed control system. A single engineered control system covering a whole process plant, from one vendor, configured from one environment and sharing one configuration database. What you are buying is engineering coherence, and what it costs you is the ability to change vendor.
- **SCADA** — supervisory control and data acquisition. A software layer that polls or subscribes to many independent controllers, presents a consolidated view, logs data and alarms, and lets an operator issue commands. It adds an integration problem and removes a lock-in; the distributed control system does the reverse. That trade is the actual decision.
- **HMI** — human-machine interface. The graphic the operator looks at. **It is a view and it owns nothing** — except, very often, undocumented business logic in a script behind a button, which is invisible to every system inventory and is lost when the panel is replaced.
- **Historian** — a store of tag values over time, at high rate and long retention, built for one query shape: give me these forty tags between these two timestamps. It is not a database in the sense you mean: no joins, no transactions, no referential integrity.
- **Tag** — a named value in a controller, with a data type, an address, engineering units and a scan class. A mid-size plant has tens of thousands. **A tag is not a data entity**: it has no identity over time, no relationships, no business meaning and no transactional context, and its value is a sample rather than a fact. Chapter 13 gives the six reasons in full, because this is the deepest conceptual gap for an architect arriving from the business side.

- **Setpoint** — the value a control loop is trying to hold, as distinct from the process variable, which is what it is actually reading. The gap between them is the control problem, and both are tags.
- **Interlock** — logic that prevents an action while a condition is unsafe or unready. The distinction that matters: an interlock on a run *command* believes the motor is turning because something told it to start. An interlock on run *feedback* knows. South African voltage dips make that distinction expensive.

How to use a list like this

Three things, and then chapter 4.

Do not deploy the vocabulary to sound fluent. Use it to ask better questions. "Is that first-pass or final?" is a better sentence than any definition in this chapter, and it costs nothing if you are wrong.

Write the register before the report. Most of the traps above become harmless the moment somebody writes down, for each measure the business uses, what it means here, who owns it, and what it may be compared with. That document is cheap, nobody has it, and it is the single most useful thing you can produce in your first month.

And keep asking the question. Measured how, from when, and compared against what? Every entry in this chapter is an answer to that question having more than one answer.

THE CHAPTER IN TEN LINES

- A plant quoting 98 per cent yield and one quoting 92 per cent first-pass yield may be the same plant. Always ask: started where, good according to whom, and does rework count?
- A 10 per cent scrap factor means dividing by 0,90, so 111,1 and not 110. Five levels at 3 per cent is 15,9 per cent, not 15, and the gap becomes a permanent unexplainable usage variance.
- Throughput means units per hour to a planner and revenue minus totally variable cost in theory of constraints. Both will be used in the same meeting.
- Takt is a demand-derived target rhythm, not a measurement of the plant, and it changes when demand changes without anything in the plant changing.
- Little's Law is an identity: work in progress equals throughput times cycle time. Halve work in progress at constant throughput and lead time halves.
- Capacity is four numbers. Only demonstrated capacity should enter a plan, and it is specific to the current product mix.
- Work centre means related but different things in an enterprise system and in ISA-95, where WORKCENTER is the generic term for a hierarchy level whose name depends on the process type.
- On time has four or more defensible date bases, plus choices about tolerance, in-full grain and where the clock stops. Two plants reporting 96 and 78 per cent may be performing identically.
- There is no OEE standard, seven incompatible availability denominators are in use, and the micro-stop threshold shifts loss between availability and performance without changing OEE.
- The 85 per cent OEE benchmark that everybody quotes as a mark of excellence is an invention, and the site that publishes it says so. Never put it in a client deliverable.
- Where-used tells you which products could contain the contaminated lot. Only genealogy tells you which do. That is the difference between recalling a product line and recalling four pallets.
- Backflush consumes standard quantities by definition, so it hides usage variance and, where substitution happened, produces a confidently false genealogy.
- A time fence shorter than the cumulative material lead time is arithmetically unenforceable, and it is checkable in an afternoon.
- Ownership is a first-class attribute of inventory. In a tolling plant the material belongs to the customer, and ownership cannot be retrofitted onto historic stock movements.
- Lean does not mean no inventory. Heijunka deliberately holds finished goods to absorb demand variability; lean's claim is that lead time, not cost, is the primary variable.

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- **2.** Vorne Industries, OEE FAQ (oee.com) — formula, the six big losses, the denominator conventions, and the statement that the 85% benchmark is artificial https://www.oee.com/faq/
- **3.** Wikipedia, “Overall equipment effectiveness” — used for the catalogue of calculation conventions, the micro-stop threshold and the documented criticisms, not for benchmark values https://en.wikipedia.org/wiki/Overall_equipment_effectiveness

CHAPTER 4

How the business model works

Before you can model how a manufacturer makes money, you have to know who its customer is, what it actually sells, and how it has been configured to deliver it. Most of that was decided by other people, some of them in a gazette.

Three questions, not one

Chapter 6 does the arithmetic — margin, absorption, variance, cash. This chapter comes first because the arithmetic only makes sense once you know what is being sold and to whom.

A manufacturer's business model answers three questions.

1. **Who is the customer, and what can that customer compel?** In manufacturing this is not a marketing question. An automotive assembler can compel the quality scheme you certify to, the message format your orders arrive in, the label on your box and the order in which your lorry is loaded. A retailer can compel your pack artwork and delivery window; a government department, an audited local-content calculation. Each is a system requirement written by someone who does not work for you.
2. **What is actually sold?** Sometimes a product. Sometimes only the conversion — the customer's material arrives, you change its state, you invoice a fee, and the material was never yours. The cases need different inventory models, and one difference cannot be fixed later.
3. **How is the firm configured to deliver it?** Where it stopped making and started buying. How much capital is sunk, and how old the decision was. Which route the product takes to the person who uses it, and where along that route the price is set.

Then a fourth question, which is really the first one again: **what data obligation comes attached?** Sequenced supply drags a cumulative shipped balance and a customer label standard. Tolling drags an ownership attribute on every stock movement. A local-content incentive drags country of origin and a cost-component split down to the bill-of-material line. None of these are reporting niceties. Each is the difference between a business that can substantiate what it says and one that cannot.

> **BEFORE YOU CLASSIFY ANYTHING**
>
> The business model is a property of the **order**, not of the company or even the plant. Highveld Precision Components — one of two invented firms this book uses, and nothing about it is real — runs sequenced supply to an assembler, make-to-stock aftermarket parts and quoted make-to-order industrial work through the same presses. Three models, one asset base, one item master. Any design that assumes one model per legal entity meets its counter-example in week two.

The customer writes your architecture

OEM supply

Selling components or assemblies into another manufacturer's production line. This is the sharpest case, and the one where an architect most often mistakes a customer requirement for a design choice.

How the order arrives. Not as a purchase order. As a schedule. The customer transmits a planning schedule over weeks or months and a shipping or call-off schedule over days, refreshed weekly or daily, in a message format it specifies. In Europe and South Africa those are the EDIFACT delivery-forecast and delivery-just-in-time messages; North America uses a different pair under a different standard family, and a supplier shipping to both markets runs both. The quantities are frequently **cumulative**: the message does not say "send 400", it says "your cumulative shipped total against this part should be 128,400 by Friday". Reconciling cumulative-received against cumulative-shipped is how discrepancies surface, and a system that models each release as an independent order cannot do that reconciliation at all.

What the contract looks like. Multi-year, for the life of a vehicle or machine model. Annual price reductions written in, on the assumption that your cost falls as volume accumulates. Tooling often owned by the customer and standing in your plant — an asset on your floor that is not on your asset register.

What the customer can compel. This is the list that surprises people.

- **The quality scheme.** In automotive, certification to IATF 16949 — still its 2016 first edition, with a second edition in development and publication planned for mid-2027 on dates the oversight body itself calls indicative [1].
- **The message set and the transport.** Which messages, which version, which protocol, which acknowledgement model.
- **The label.** Format, symbology, data content, placement. Not negotiable, and different per customer, so label specification is master data with a customer dimension.
- **The quality performance regime.** Defects measured in parts per million, with formal escalation: a problem triggers containment, then a controlled-shipping level where you inspect and certify every box at your own cost, sometimes with a third party doing it in your plant.
- **The launch process.** Participation in the customer's product-approval and part-submission process, with specified documents at specified gates. Standard practice in automotive rather than something I could source here, so get the requirement from the customer's manual.
- **The delivery sequence.** In sequenced supply the customer says not merely how many and when, but in what order the parts must sit on the trolley so that they come off in build sequence at its station.

The data obligation. A cumulative balance per part per customer. A label standard per customer. Traceability to whatever depth the customer's scheme demands, which may be the serial number of a safety-relevant component. The customer's own definition of on-time-in-full, which will not be yours,

computed on its clock and published on its scorecard. And engineering change arriving from outside your business, on the customer's timing, against parts you have already built and shipped.

The architectural consequence, stated plainly. A tier-two supplier's application architecture is substantially written by its customers. A useful question in week one is: *which of our interface standards are ours?* The answer is usually "the ones nobody outside cares about". Standardising the inside while accepting that the edges are heterogeneous by contract is the only design that survives.

Where it hurts. Total dependence — a model change can remove the product entirely — and competition from a large import stream inside your own customers: original-equipment component imports by the South African assemblers were **R151.0 billion in 2025** [2].

Aftermarket and replacement parts

The same part, or an equivalent, sold for repair rather than original fitment.

How the order arrives. As a discrete order against stock, from a distributor, a workshop chain or a franchise network — and the expectation is availability, not lead time. Nobody waits six weeks for a water pump.

What the customer can compel. Little, individually. Collectively, the catalogue: which part fits which model, year, variant and engine. Fitment data is a genuine and underestimated data problem, with its own change rhythm and no natural owner inside a manufacturing business.

The data obligation. An enormous item count, because you must support every model year you ever supplied; slow-moving stock with a long tail; packaging and branding distinct from the original-equipment version of the same physical part, which means one manufactured item and several sellable ones; and that fitment catalogue. Replacement-parts imports were **R107.5 billion in 2025** [2], so this is also an import-competing business.

Retail private label

Manufacturing to a retailer's brand and specification.

Orders arrive against a listing won in a tender, called off weekly or daily into a delivery window. The retailer compels the specification and any change to it; the audit regime, usually a scheme audit plus its own; the pack artwork and its revision cycle; the delivery window and the on-time measure, computed its way; trading terms; and in South Africa, expectations about your empowerment level and local sourcing.

The data obligation. Artwork and label version control as a production input rather than a marketing asset, because printed packaging obsoletes the moment the customer changes the design, and the write-off is real money. A product master for products you cannot sell to anyone else. The retailer's item identifiers alongside your own. And when the listing is re-tendered and lost, the packaging in your store becomes scrap on the day of the decision.

Branded consumer

Selling under your own brand, through distributors, wholesalers or retailers. Orders arrive against a price list you publish and discounts you negotiate, which looks like control and is less than it appears, because what the customer compels is listing fees, promotional allowances, rebates and settlement terms — which is to say, the gap between your invoice price and what you realise.

The data obligation. Gross-to-net. The accrual for promotions your customer will claim months after the volume moved, against volumes you do not control. This is routinely the largest unmanaged item in a branded manufacturer's accounts and almost never in a manufacturing programme's scope, because it looks like a sales problem. It is not: trade spend has to be accrued per customer per promotion per period and settled against deductions that arrive as unexplained short payments. The gross margin here is the best in the business and funds the most cost below the line.

Industrial and B2B distribution

Selling components, equipment or materials to other businesses, on a discrete purchase order or a call-off against a framework agreement, frequently with a quoted lead time rather than an expectation of stock. The customer compels technical conformity — a standard, a test certificate, a material certificate, a data sheet — and the data obligation is certificates and material test reports tied to the **batch or heat**, not to the order, plus published technical data that matches what you actually make.

Export

How the order arrives. As an order, a schedule or a letter of credit — and the commercial terms decide what business you are in. Ex-works, free on board, cost-insurance-freight and delivered-duty-paid are four different companies with four different balance sheets, four different revenue-recognition points and four different sets of liabilities.

The data obligation. Country of origin and the computation behind it. Customs classification per item. Declarations tied to the underlying commercial and movement documents. And a sequencing trap worth knowing before it costs money: a **manufacturing drawback claim must be lodged within six months of export, only after export with proof attached, and the export declaration must already carry the correct procedure code and drawback item before the goods leave the country** [12]. Get the declaration wrong and the claim is irrecoverable. Nothing in the warehouse system knows that.

Where it hurts. Logistics and currency. In the week of 7–13 October 2026, vessel waiting at Durban averaged about 5.73 days, with berthing delays of 7–10 days at Pier 1 and 10–16 days at the Durban Gateway Terminal [3]. That is not a footnote to an export plan. It is inventory.

Government and utility

Through a tender, then a contract, then call-offs. The customer compels the specification, usually by reference to a standard; your empowerment level as an award criterion; penalties for late delivery; and an audited **local-content declaration**. For goods in a designated sector a bidder must meet a stipulated threshold of local production and content, declared on the prescribed bidding document and calculated according to the national local-content standard [4][5].

The data obligation is the same origin and cost-component data an automotive incentive claim needs, computed by a different formula against a different standard. A plant serving both assemblers and state tenders **computes local content two different ways from the same bill of material** — which is the strongest argument in this chapter for holding origin and cost split as governed master data rather than deriving it per claim. The designated-sector list and its thresholds change, and I could not verify either here.

Archetype	How the order arrives	What the customer compels	The data obligation
OEM supply	Electronic schedule, cumulative quantities, daily or weekly	Quality scheme, message set, label, defect regime, often the sequence	Cumulative balance per part, their label standard, their on-time definition, inbound change
Aftermarket	Discrete order against stock	Availability; collectively, the fitment catalogue	Very large item count, fitment data, separate pack for the same part
Retail private label	Call-off into a delivery window against a won listing	Specification, audit regime, artwork cycle, delivery window, terms	Artwork version control, their item identifiers, products you cannot sell elsewhere
Branded consumer	Trade order against a published price list	Listing fees, rebates, promotional allowances	Gross-to-net, trade-spend accrual per customer per promotion
Industrial and B2B	Purchase order or framework call-off, quoted lead time	Technical conformity and certificates	Certificates tied to batch or heat; agent and distributor recorded consistently
Export	Order, schedule or letter of credit on stated terms	Destination-market rules, origin evidence	Origin and classification per item; declarations sequenced before despatch
Government and utility	Tender, contract, call-off	Specification by standard, local content, empowerment level	Audited local-content calculation on a second formula from one bill of material

> A manufacturer does not have *a* customer archetype. It has a mix, and the mix is a dimension on every transaction. Channel must drive pricing basis, terms, the on-time definition applied, the quality regime, the labelling specification, the electronic message profile and the margin basis. A data model that carries customer but not channel cannot answer the only question that matters: which of these are we actually making money in?

What is actually sold

Product, conversion, or capacity

Tolling — also toll processing or toll manufacturing. The customer supplies the material, you process it, you invoice a conversion fee, and **title to the material never passes to you.** Grain milling for traders, metal rolling and galvanising, beverage co-packing and chemical blending all work this way, often on lines that also run owned material.

Contract manufacturing — co-manufacturing, co-packing, and in some sectors contract development and manufacturing. You buy the material *and* make the product to the customer's brand and specification. The material is yours. The brand is not.

These two are constantly confused, including by the people doing them. They are not variants of one thing: the accounting, the risk and the data model all differ.

In tolling the input is the customer's, so it is not on your balance sheet at all — quantity, location and condition only — revenue is the conversion fee, and the dominant risk is **yield**. The stock ledger must hold non-owned stock segregated by owner, with quantity accounting and no inventory value, and reconcile yield per contract per batch. In contract manufacturing the input is yours, revenue is the full product price, the risk is obsolescence of brand-specific material and single-customer dependence, and the systems hold a product master for products you may not sell to anyone else.

The tolling risk inverts the usual intuition about where exposure sits. Your revenue is the fee; your liability is the material. **A one per cent yield miss on an expensive input can wipe out the entire conversion margin on a contract**, and nothing in your revenue line hints at that asymmetry. Every ratio computed against revenue is meaningless for a tolling plant.

Ownership is an attribute of inventory, and this one you cannot fix later

In a plant that tolls, the same silo, tank, rack or yard holds material belonging to several parties at once. The same physical tonne of flour may be the plant's asset on Monday and a customer's on Tuesday, because the customer bought it under a supply agreement and handed it straight back for conversion.

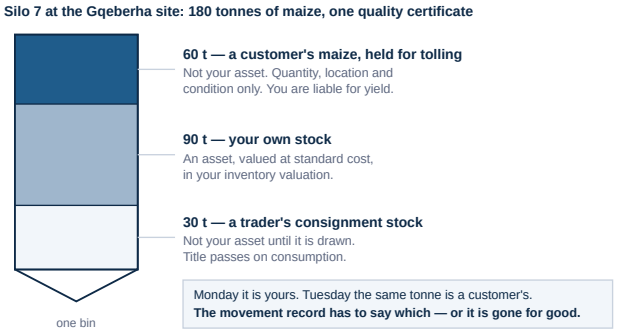


Figure 4.1 Ownership in a tolling plant. One silo, one certificate of analysis, three balance sheets. Illustrative, using this book's invented food business.

Most enterprise systems can be made to do this, through a subcontracting or consignment construct built for the opposite direction. That is a nuisance. The unrecoverable part is different.

> THE ONE DESIGN ERROR WITH NO REMEDY
>
> **Ownership is a first-class attribute of inventory, and omitting it is one of the few genuinely unrecoverable errors in a manufacturing data model.** If the stock-movement history does not record whose material moved, you cannot reconstruct it. The quantities are right, the batches are right, the genealogy is right, and the ownership is gone — because it was never a fact in the record, only an understanding in somebody's head and a clause in a contract. You cannot retrofit ownership onto historic movements. Every yield reconciliation, every disclosure of third-party stock, every dispute about what was lost and whose it was, depends on a field that was not there.
>
> Ask the client on day two whether a stock movement carries an owner. Not in design.

The architectural statement of it: inventory identity is not *item plus location plus batch*, it is **item, location, batch, status and owner**. Status carries the quality veto; owner carries the balance sheet; and those are the two that get dropped.

Where the firm stops: vertical integration and make-or-buy

Vertical integration is how much of the chain from raw input to finished product sits inside the company. A beverage business that blow-moulds its own bottles is more integrated than one that buys them.

The decision presents itself as a one-off capital question and it is nothing of the kind. **It is reopened every time an input price, an exchange rate or a capacity position moves**, which in South Africa is continuously. A supplier's quote reopens it per component; an exchange-rate move for every imported input; falling utilisation in the other direction, as a reason to insource work to fill an asset; a duty or anti-dumping determination by moving a landed price double digits; and a local-content or empowerment consequence for reasons that appear in no cost column.

Three consequences for an architect.

First, the item master churns at the make-or-buy boundary. A manufactured part becomes a purchased one. Its bill of material becomes irrelevant but must be retained, because the parts you already built are traced through it. Its costing basis changes from a roll-up to a purchase price. A system where procurement type is a flag with no history and no effective date cannot explain last year's cost.

Second, almost every make-or-buy analysis is done on the wrong number. Compare the *avoidable* cost of making — incremental material, labour and variable overhead, avoidable fixed cost, and the contribution forgone from whatever else that capacity could have made — against the *full* cost of buying, including freight, duty, inspection, supplier management and the inventory a longer lead time requires. What is actually compared is full absorbed standard cost against the supplier's quote, which overstates making and understates buying every time, in the same direction, producing a structural bias toward outsourcing that nobody intends.

Third, in South Africa the right answer includes items in neither cost column. Local value addition drives an incentive claim, so buying in can forfeit money a landed-cost comparison never sees. Spend with a black-owned local supplier earns scorecard recognition worth real money in tenders; internal manufacture earns none. And an imported component's effective lead time, and therefore its safety stock, is materially worse than the lead time its supplier quotes [3].

In practice that engine is a spreadsheet owned by one person. **That spreadsheet is a system. Treat it as one, model it, and find out what it reads.**

The asset decides what you can propose

Capital intensity, and the twenty-year shadow

Manufacturing's defining balance-sheet feature is a long-lived, specific, immobile asset base: roughly 20 to 50 years for process plant, 10 to 20 for machine tools, 5 to 10 for automation and information technology. Three things follow.

Capacity comes in lumps. You cannot buy seven per cent more kiln. You buy a kiln. This is why capacity additions in process industries overshoot demand and depress prices for years.

Replacement is a cliff, not a slope. A kiln relining, a furnace reline or a paper-machine rebuild is one large event on a multi-year cycle. It is a capital plan, not a maintenance budget, and the plant has been arguing about its date for a decade.

Exit is expensive and slow. Closing a plant means retrenchment cost, site rehabilitation and contractual commitments on inputs you agreed to take whether you need them or not. ArcelorMittal South Africa confirmed the wind-down of its Longs Business, and production of long steel at Newcastle Works ceased in November 2025 [6] — a headline-sourced fact to re-source before it goes into a client document. The lesson stands regardless: **in a capital-intensive business the decision to stop is taken years before the asset stops, and is reversed several times on the way.**

The consequence for you is uncomfortable and worth saying out loud. **A plant's economics are dominated by decisions taken ten or twenty years before you arrived** — where the site is, how material flows through it, what the asset can and cannot do, what shift pattern the agreement permits. Your room is the information layer and the decisions above it. An architecture that requires the physical flow to change is not an architecture; it is a capital proposal wearing one, and it will be judged as one by people who have been judging capital proposals for thirty years.

Maintenance capital versus growth capital

Both are capitalised, so this is not an accounting distinction. It is a management one, and it is the most-manipulated number in manufacturing disclosure.

- **Sustaining or maintenance capital** keeps existing capacity available at existing cost: replacements, relines, overhauls, statutory compliance, safety.
- **Growth capital** adds capacity, adds products or reduces unit cost.

The incentive to misclassify runs one way. Sustaining capital is a proxy for the cash cost of staying in business, so calling spend "growth" flatters free cash flow. There is no auditable boundary: a rebuild that replaces a worn asset *and* adds five per cent capacity is honestly both.

One honest test is available to you. **Over a full asset cycle, sustaining capital should approximate depreciation on a replacement-cost basis.** Persistently below it means the asset base is being consumed — which presents first as falling availability, then as unplanned outages, then as a catch-up programme arriving at the same time as your implementation.

Why depreciation policy changes reported margin

For a manufacturer, depreciation on production assets is a **conversion cost**. It is absorbed into inventory and it lands in cost of sales, not below the operating line. Therefore the depreciation policy is a gross-margin policy. Six ways it moves the number, with no operational change whatsoever:

1. **Useful life.** Extend a plant's life from 20 years to 30 and annual depreciation falls by a third.
2. **Method.** Straight line charges by time; units-of-production charges by output. Units-of-production makes depreciation per unit constant and **removes the volume effect from unit cost** — a materially different margin profile in a cyclical business.
3. **Componentisation.** Where significant components of one asset have different lives — a kiln shell, its refractory lining, its drives — they are depreciated separately, and finer componentisation front-loads the charge onto the short-life parts.
4. **Residual value and impairment.** A revised residual value changes the depreciable amount. An impairment reduces the carrying amount and therefore *reduces future depreciation and raises future reported margin* — so a plant that has been impaired looks more profitable afterwards.
5. **Leases.** Lease accounting that brings right-of-use assets onto the balance sheet puts depreciation inside cost of sales and interest below the line, for equipment that used to be a rental charge. Margin comparisons spanning that change are not like-for-like.
6. **Revaluation.** A revalued asset base carries higher depreciation and lower reported margin.

Items 3, 5 and 6 rest on accounting-standard requirements I did not retrieve here; let the controller cite the standard. What *is* sourced is the ground under the whole section: fixed production overheads must be allocated on the basis of **normal capacity**, the per-unit allocation may **not** be increased because output was low or the plant idle, and unallocated overhead is expensed in the period [7].

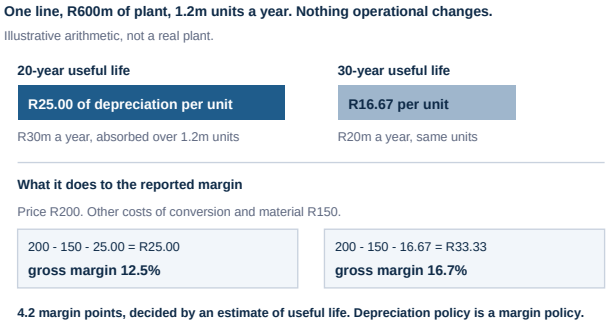


Figure 4.2 Depreciation is a conversion cost, so it is absorbed into inventory and lands in cost of sales. An estimate of useful life therefore moves reported gross margin with no operational change at all.

> **THE ASSET REGISTER IS A COSTING MASTER**
>
> Asset to cost centre, cost centre to absorption rate, absorption rate to standard cost, standard cost to inventory value, inventory value to reported margin. That is a live dependency chain, and a capitalisation decision, a life revision or an impairment propagates all the way down it. Very few manufacturing data models represent the chain, which is why nobody can explain why unit cost moved after a revaluation. The asset register is not a finance back-office artefact. Model it as master data with effective dates.

Channel, route to market and where the price is actually set

Four routes, with different consequences. **Direct**: you hold the stock, take the credit risk and own the relationship. **Distributor**: they buy and resell, the sale completes at your gate, and stock, credit risk, the end-customer relationship and visibility all move to them — you stop seeing demand and start seeing replenishment, which is a different and more volatile signal. **Agent**: they introduce and service, you keep the stock, the risk and the invoice, and

pay commission. **Own retail or own distribution**: you carry the channel's cost structure and gain its data. The distributor and agent cases are frequently **recorded both ways in the same system for the same customer**, so reported revenue and margin depend on which record a transaction happened to hit.



Figure 4.3 Pricing power and gross margin are not the same axis. The best gross margin in the building usually has the least control over the price finally realised.

Two things about pricing power are worth understanding rather than merely noting.

Where it sits is not where the margin percentage is highest. A branded consumer product has the best gross margin in the building and the least control over what it realises, because the realised price is net of fees, rebates and allowances claimed after the volume moved. A sequenced automotive part has the worst gross margin and a contractually certain price. Certainty and level are different things, and their planning consequences are opposite.

Where the customer is a retailer, part of your result is trading terms rather than price. Supplier income — settlement discounts, volume rebates, promotional contributions, listing payments — is negotiated annually, earned conditionally and claimed retrospectively, often as a deduction from a payment rather than an invoice. It sits outside the systems a manufacturing programme touches, and it is routinely larger than the improvement the programme is funded to deliver.

The South African constraints that are business-model inputs

What follows is not a chapter on policy. Each item changes the model, the data or the programme, and each is dated because every one of them will move.

Trade and industrial policy is an input, not a footnote

South Africa's characteristic industrial-policy instrument is the **master plan**: a negotiated compact between government, business and labour in a sector, with localisation, investment and employment commitments. The most recent consolidated status document I could retrieve is a briefing to a parliamentary committee **dated 27 October 2020** [8], covering plans for clothing and textiles, poultry, sugar, automotive, steel and furniture.

Read that date twice. Those figures are **2020 targets with a 2020 status, not achievements**, and the current status of every one is something I could not verify. Quote a master-plan target with its date and status, or do not quote it. For scale: the automotive target is 60% local content, and the highest reported local content of any locally made vehicle is "more than 50%" [10].

The automotive incentive, and the data requirement nobody scopes

The Automotive Production and Development Programme is the one sector-specific production incentive of real consequence, and an architect working anywhere near automotive components has to understand it — not because of the money, but because of what it demands the business be able to produce.

Start with the correction. APDP is **not a tax credit**. It is a **customs-duty-rebate-certificate regime** [11]. The production rebate certificate is a duty credit issued at duty value, claimable **within 12 calendar months of the eligible product's invoice date**, valid for 12 months from the first day of the quarter in which the claim was submitted, usable only by the original holder, and transferable **once** to another South African importer at the administering commission's sole discretion [11]. A North American reader hears "incentive" and thinks of a credit against tax. This is a tradeable customs instrument with an expiry date, and it belongs in a register.

Now the data requirement. To claim the production incentive, a component manufacturer must compute, per product and per period, the value of local-origin material, local labour and local factory overhead as a percentage of the ex-factory selling price, auditable to source documents, with the imported content of every purchased input evidenced by a declaration from its supplier [11]. That declaration is Form C2, and the sanction for its absence makes the requirement concrete: **if no Form C2 is obtained, the entire purchase price is treated as imported content** [11].

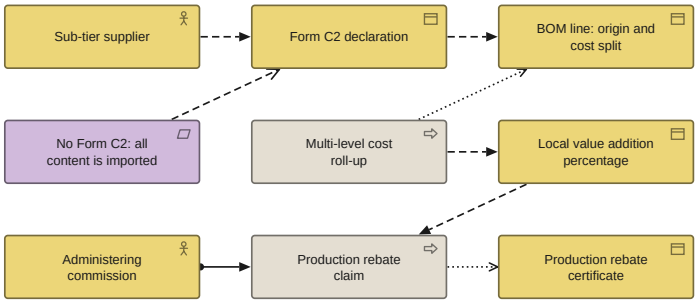


Figure 4.4 The automotive incentive's data chain, in ArchiMate. The requirement element carries the sanction: absent a supplier declaration, the whole purchase price counts as imported content [11].

Put those together and you have a **bill-of-material-level country-of-origin and cost-component split, propagated through multiple tiers of supply**, with document retention and a 12-month claim window. Two common failures follow. A bill of material carrying no origin per component cannot support the computation at all. And a costing model that rolls up multi-level standard cost as a **single total** — rather than preserving the split into material, labour, overhead, subcontract and freight through every level — physically cannot substantiate a claim, because the local-content percentage is a function of the components of cost, not of the cost. Preserving component split through multi-level roll-up is cheap at design time, expensive later, and skipped constantly.

Phase-two enhancements to the programme have been discussed publicly. I could not verify their content or status, and neither should you from secondary commentary. **Print nothing about phase two that does not come from the gazetted regulations.**

Tariffs, not AGOA, are the story

The African Growth and Opportunity Act gives qualifying sub-Saharan African exports duty-free access to the United States, and has mattered most to apparel, automotive and agro-processing. Its renewal is the less important half of the position [9]:

- AGOA expired in September 2025 and has since been extended **to 2028** — two years against the 15 South Africa sought — with the country's continued eligibility **unresolved** and the next annual determination taking effect 1 January 2027.
- Separately and more damagingly, a **25% Section 232 tariff** applies to South African passenger vehicles, relevant automotive parts, steel and aluminium, plus a **12.5% Section 301 tariff** from July 2026, with Section 232 goods exempt from the latter.

A 25% tariff on vehicles and steel is not offset by duty-free access elsewhere in the schedule. **Any statement about South African access to the United States market must be dated to the month**, and a roadmap built on an export assumption needs the tariff position as an explicit, dated assumption with an owner.

The incentives that are open, and the big one that is closed

From a tax summary last reviewed 29 May 2026 [13]:

Incentive	Position	Terms
Section 12I industrial policy projects	**Closed.** Only projects approved before **31 March 2020** retain benefits	Investment allowance of 35% or 55% of qualifying asset cost
Section 12L energy-efficiency savings	**Open to 31 December 2030**	**R0.95 per kWh** saved against a measured baseline
Special economic zone regime	Open; ceases for years of assessment from 1 January 2031	**15% corporate rate**, plus building, employment, customs and value-added tax relief
Research and development deduction	Open for ten years from 1 January 2024	Qualifying current costs **150% deductible**, pre-approval required

Three notes. **Writing that a manufacturer can claim Section 12I is wrong by six years. Section 12L is a data requirement in disguise**: it pays against a measured and verified baseline, meaning metering, a verification process and an evidence trail, which no plant has lying around. And the accelerated depreciation allowance under Section 12C is **not** covered by that source; the rates commonly quoted from memory could not be verified here, so **do not write them**.

Carbon tax: quote the rate and the allowances together, or neither

Phase 1 of the carbon tax ran to 31 December 2025. **Phase 2 began on 1 January 2026, and the rate is R308 per tonne of carbon-dioxide equivalent, up 31% from R236** [14]. The legislated path rises toward **R462 by 2030**, and emissions above a mandatory carbon budget would face **R640** [15].

> **CHECK THE RATE BEFORE YOU PRINT IT**
>
> The two research passes behind this book disagree. A February 2026 commentary gives R308 from 1 January 2026; a July 2026 commentary describes R236 as the then-current rate while agreeing that Phase 2 began on 1 January 2026. They cannot both be right, and a search to settle it was not available. Read the rate off the Carbon Tax Act as amended and the current SARS carbon tax guide, and write down the date you read it. The structural point — that the headline rate is not the liability, because the allowances do the work — holds on either figure.

And now the part that makes the headline rate nearly useless on its own. The liability is computed after **allowances** — a basic tax-free allowance plus trade-exposure, performance, fugitive, carbon-budget and offset allowances — and in Phase 1 the combined maximum reached **95% for process and fugitive emissions and 85% for combustion** [15]. At those maxima the effective rate is a small fraction of the headline.

> **Phase 2 allowance percentages could not be verified, and the two available sources contradict each other** on the offset allowance — one describing an enacted increase, the other a larger proposal [14][15]. Until the enacted position is read from the Act as amended, **quote the rate together with the allowance structure, or quote neither**. A business case that applies the headline rate to gross emissions is wrong by an order of magnitude.

The architectural consequence is the same either way, so you can design before the numbers settle: emissions per process, per fuel, per period, from activity data and versioned emission factors, auditable, reconciled to the tax computation and the budget position. **Every allowance is a dated, per-category, per-taxpayer parameter, not a constant.**

The home market is supplied by imports

Light-vehicle imports were **69.1% of light-vehicle sales in 2025**, up from 62.7% in 2024, with passenger-car imports at 82.8% of passenger-car sales. India was the largest origin at 56.2% of imported units and China second at 23.3% [2].

Read that beside the export numbers. Total automotive exports were R291.0 billion in 2025 and 15.6% of all South African exports, with vehicle exports rising to R229.8 billion — while **component exports fell**, from R63.4 billion to R61.2 billion [2]. The assembly base is an export business serving a largely import-supplied home market, and the component base is losing ground even as assembly grows. That is the structural fact behind every conversation about volume with a component manufacturer, and the reason the local-content data requirements above will not relax.

Labour is substantially not variable

This is the correction most likely to be imported from elsewhere and most expensive to get wrong.

Wages in South African manufacturing are set largely by **bargaining councils** — statutory bodies of employer organisations and unions in a defined industry, whose collective agreements bind the parties and may be extended by the Minister to non-parties across the whole industry. **A manufacturer's wage bill is therefore partly set by a negotiation it may not have attended.** The metal and engineering council's main agreement runs 1 July 2024 to 30 June 2027, with increases effective 1 July 2026 of **5.00% to 6.00% by grade** — on an inverted differential, 5.00% at the lowest grade and 6.00% at the highest [16]. Read the circular carefully: it applies to the parties only, names several employer bodies as not covered, and does not mention extension to non-parties [16] — so "all metal and engineering employers must pay it" contradicts the source. In the motor industry council, component manufacturers' increases are on **actual rates of pay**, not on the prescribed minimum [17] — a two-word distinction that lifts everyone including those already above scale, and the most expensive technical detail in South African wage agreements.

Then the structural point. Dismissal for operational requirements under the Labour Relations Act is a **procedure**: consultation, disclosure, consideration of alternatives, facilitation for large-scale retrenchments, severance. Shift arrangements and manning levels are collectively agreed. Taken together, **direct**

labour is not variable in the short run, and a contribution-margin or breakeven model imported from a North American template overstates flexibility in a way that makes every marginal-order decision look better than it is.

One piece of arithmetic to carry into any conversation that begins "just add a shift". A seven-day, three-shift pattern is **21 shift-slots a week**. One crew cannot cover more than about five of them within ordinary hours. So a 24/7 continuous pattern needs **four or five crews, not three** — and **five is the honest number** once annual leave, sick leave, training and absence are covered. Plants running four fill the gap with structural overtime, which costs more than the fifth crew and never appears as a crewing decision.

The fully loaded cost of an hour on a continuous pattern is a substantial multiple of the basic rate once crewing ratio, allowances, overtime, standby, council and statutory contributions, leave accrual and training are counted. **I am deliberately not giving you a multiplier**: any number I printed would be an invention. Get it from the client's payroll, before you build a capacity model that trades labour against capital.

Empowerment is a procurement data model

Broad-Based Black Economic Empowerment is a scorecard regime that assigns a measured entity a level, and that level determines how much of the entity's invoice value its *customers* may recognise on their own scorecards. For a manufacturer the dominant element is **enterprise and supplier development, at 40 points — the largest single element on the generic scorecard** [18]: preferential procurement at 25 points, supplier development at 10 and enterprise development at 5, each with a **sub-minimum of 40% of its points**, and an entity that misses a sub-minimum **risks dropping a level** overall [19].

The correction commonly got wrong: enterprise and supplier development contribution targets are expressed as a percentage of net profit after tax, not of procurement spend [19]. The spend-based part of the element is the preferential-procurement score, a different computation on a different base.

Two caveats. The scorecard point totals in the source available to me are internally inconsistent — the weightings do not sum to the stated total — so **read the authoritative weightings from the current Codes** [18]. And as at a February 2026 alert, the Code amendments affecting this element were **still in draft** [20].

What is stable is the architecture:

1. **Every supplier carries an empowerment level, a verification basis (agency certificate or sworn affidavit), an expiry date as a structured field, black-ownership and black-woman-ownership percentages, and an enterprise-size classification** [19][20].
2. **Those attributes are a dated series, not a current value.** Each spend transaction is recognised according to the supplier's status *as at the transaction date*. A supplier whose certificate lapsed in March contributes differently before and after it, and one current-value column cannot compute that.
3. **Certificate expiry monitoring is an operational control with a financial consequence**, and **the scorecard reconciles to the accounts-payable ledger**, less the exclusions the Codes allow — so spend classification is itself a scorecard input, and misclassification is the commonest verification finding.
4. **Contributions are a beneficiary register with a lifecycle**, not a ledger account: loans, grants, early payment, discounts, mentoring and shared capacity, per beneficiary, per year, against a profit-based target and a recognition limit [19].

Put that beside the origin requirement from the automotive incentive and the local-content declaration from public procurement, and the conclusion is blunt. **In a South African manufacturer the vendor master is not a finance master. It is a compliance master**, and a plant whose vendor master lacks these attributes cannot produce its own scorecard, cannot substantiate an incentive claim and cannot bid into designated-sector public procurement — three business-stopping consequences from one data gap.

What to take into the model

The business model is not a slide. It is a set of constraints already decided — by customers who do not work for you, by capital decisions taken before you arrived, and by instruments published in a gazette. Your contribution is to write them down as what they are: dimensions, attributes, dated parameters and named assumptions, so the next person can see which of them moved.

Chapter 5 introduces the people who hold all of this in their heads, and which of them to interview first.

THE CHAPTER IN TEN LINES

- The business model is a property of the order, not of the company; most plants run several at once.
- An OEM customer can compel the quality scheme, the message set, the label, the defect regime and often the delivery sequence, so a tier-two supplier's architecture is substantially written by its customers.
- Sequenced schedules are usually cumulative quantities, so a system that models each release as an independent order cannot reconcile what was shipped.
- Tolling sells conversion, not product: revenue is a fee and the liability is the customer's material, so every ratio to revenue is incomparable.
- Ownership is a first-class attribute of inventory, and omitting it is unrecoverable because you cannot retrofit it onto historic stock movements.
- Make-or-buy is reopened whenever input prices, exchange rates or capacity move, and each revision changes the bill of material, the supplier master and the costing basis.
- Depreciation is a conversion cost, so useful life, method, componentisation, impairment, leases and revaluation all move reported gross margin with no operational change.
- Sustaining capital persistently below depreciation on a replacement-cost basis means the asset base is being consumed.
- The automotive incentive is a customs-duty-rebate-certificate regime, and it needs bill-of-material-level origin plus cost-component split preserved through multi-level roll-up.
- Section 12I closed on 31 March 2020; Section 12L at R0.95 per kWh saved runs to 31 December 2030 and requires a measured baseline.
- Carbon tax Phase 2 is R308 per tonne from 1 January 2026, but the liability is after allowances, and the Phase 2 allowance percentages could not be verified.
- Direct labour is not variable in the short run in South Africa, and a 24/7 pattern needs four or five crews, with five the honest number.
- Enterprise and supplier development is 40 points, the largest single element, and its contribution targets are a percentage of net profit after tax, not of procurement spend.
- In a South African manufacturer the vendor master is a compliance master, carrying dated empowerment credentials, origin and local-content attributes.

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CHAPTER 5

Who does what

Fifty-odd jobs, most of them with a title you have not met. This chapter is the cast list: what each person does all day, what they will tell you, what they will not say unless you ask, and which system they live in.

Why you need the cast list before you need the model

An architect's first three weeks in a manufacturing business are interviews. You will not be given a process model. You will be given forty people, most of whom have a title you have not met, and about half of whom hold something in their heads that is not written down anywhere.

This chapter is the cast list. For each role: what the job actually is, what the person does all day, what they will tell you, **what they will not say unless you ask**, and which system they live in. The gap between the third and the fourth is where the requirements are.

Three working rules before the list.

Name the job as the plant names it. An architect who says "the production scheduling resource" has already identified themselves as someone who has not been in a factory. The job is called a scheduler, or a planner, or sometimes just Pieter.

Assume no title means what you think it means. The same word does different jobs in different plants, and the difference is not cosmetic — it changes who you interview and what you ask them.

Count the ones who are not on the organisation chart. The shop steward, the bargaining-council agent, the inspector and the customer's quality engineer all have the power to stop your programme, and none of them reports to your sponsor.

One technique, because it changes what you get out of all of these conversations. **Do not ask people to describe the process. Ask them for the last three exceptions.** "When did this last not work, and what did you do?" produces the workaround, the spreadsheet, the phone call and the override — which is the real process. "Talk me through how an order flows" produces the induction pack, which somebody wrote in 2016 and nobody has followed since.

Title	In one plant	In another	So ask
Planner	Forecasts demand	Sequences machines for tomorrow	"What horizon do you work on, and in what tool?"
Scheduler	Runs the master schedule in the ERP	Runs a spreadsheet against die availability	"Who decides the sequence, and who can change it?"
Engineer	Designs the product	Keeps the plant running	"Do you own the drawing or the machine?"
Superintendent	Senior person on site at night	A rank above supervisor, day shift only	"Who can stop the plant at 02:00?"
Quality manager	Owns the system and the release	Runs the laboratory	"Who signs the release, and can production overrule it?"
Controller	Plant accountant	Instrumentation and control engineer	Nothing. Work out which one from the context, then ask

The plant

Plant manager, or works manager. Accountable for everything inside the fence: output, cost, safety, quality, people, and the site's relationship with its community and its regulator. Usually an engineer by training, 15 to 30 years in, and frequently the most powerful individual in the company below the executive. *All day:* the three things that went wrong, a capital argument, a customer visit, a conversation with the union. *Will tell you:* the real constraints, which assets are reliable and which are nursed, which numbers in the corporate pack they do not believe, and the political history of the last three improvement programmes. *Will not say unprompted:* the workarounds that make the plant work — the spreadsheet that actually schedules the line, the group chat that runs shift handover, the material borrowed between orders. *Lives in:* the monthly pack, and email. **The most important interview on the site, and it should not be your first.**

Production manager. Runs output day to day across shifts: manning, the handover with planning, and the decision about what runs next when the schedule becomes impossible. *Will tell you:* where the schedule breaks, how often it is overridden and by whom, and what "schedule adherence" actually means here. *Lives in:* the ERP's production orders and a daily meeting.

Shift superintendent. On a continuous plant, the senior person on site at 02:00, with operational command across all areas for that shift and the authority to stop the plant. Not every plant has the role; where it exists it is the single point through which a night-time incident, decision or data correction passes. *Ask:* who holds it, and what they are allowed to decide without waking anyone.

Shift supervisor, or shift manager. Owns one shift's output, safety and discipline in one area, typically managing 10 to 60 people. The level at which the plan meets reality. *Will tell you:* the real reasons for downtime, as distinct from the reason codes entered against it, and which numbers are written up at the end of the shift from memory. *Lives in:* a paper logbook, a shift handover sheet, and whatever terminal the plant gave them.

> CHECK THIS IN WEEK ONE, BECAUSE IT IS NEARLY ALWAYS PARTLY TRUE
>
> **If shift supervisors are entering production data retrospectively from paper, the data carries a systematic bias and every metric built on it inherits that bias.** Downtime gets rounded to the nearest convenient cause, short stops vanish, and the shift's output is reconciled to what the counter says rather than recorded as it happened. You cannot fix this in the reporting layer. You can only find out how big it is.

Line leader, team leader, or cell leader. The first-line working leader of a line or cell. In a plant running a Toyota-derived system, this is the pivotal role: owner of standard work, trainer, and first responder to an abnormality. **How much time and authority the line leader has been given is the best single indicator of whether a lean programme is real.** *Lives in:* the line's hour-by- hour board, which is usually a whiteboard.

Operator, machine operator, or process operator. Runs the machine or the process. In discrete assembly, performs standard work to a rhythm; in process, monitors a control system and intervenes. **Operators are the primary data-entry population of any execution system**, which means every additional field you ask for costs cycle time, and will be degraded or circumvented if it costs too much. *Will tell you, if asked properly:* exactly which defects recur and why, where the process is unstable, and which screen is slow. *Lives in:* the terminal, the control screen, and a handwritten sheet.

Setter, set-up technician, or tool setter. Changes the machine over between products — dies, tooling, moulds, formats, recipes — and brings it back to a quality standard. Scarce, high-skill, and the acknowledged bottleneck in a high-mix plant: the schedule is frequently sequenced around setter availability rather than around the customer. **Knows more about changeover loss than anyone in the building.** *Lives in:* nothing. Their knowledge is not in a system, which is the finding.

Materials handler, feeder, or forklift driver. Moves material to and from the line, stages the next order, returns what was not used. Almost every movement they make is supposed to be a stock transaction, and a proportion of them are not. **This is where inventory accuracy is actually won or lost**, long before anyone counts anything. *Lives in:* a scanner, when there is one.

Storeman, or stores controller. Controls the raw material store or the maintenance store: receipts, issues, binning, counts. In the maintenance store the inventory is often large in value and deliberately terrible in turns, because its purpose is insurance against a machine failure rather than throughput. *Will tell you:* which issues are recorded after the fact, and which bins nobody trusts.

A 24/7 pattern is 21 shift-slots a week. Every cell has to be filled.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
06:00	A	A	D	D	C	C	B
14:00	B	B	A	A	D	D	C
22:00	C	C	B	B	A	A	D

Four crews, A to D, cover all 21 slots — if nobody is on leave, in training, sick or absent.
Crew A works six shifts in this week. Somebody always does.

Five crews is the honest number. Plants running four fill the gap with structural overtime.

Figure 5.1 A four-crew continuous pattern, illustrative. The arithmetic is the point: 21 slots a week, and the fifth crew is what covers leave, training and absence.

Engineering

The word "engineer" does at least six jobs in a manufacturing business, and conflating two of them produces a model that cannot represent an engineering change.

Process engineer. Owns the process: the recipe or the method, the parameters, process capability, troubleshooting and improvement. Decides what the machine should be set to and why. **Owns the data that should be master data and usually is not** — parameter set-points by product by machine, which in many plants exist as a laminated sheet beside the machine. *Will not say unprompted:* that the official parameters and the ones actually used have diverged, and which product has a "trick" that only works on one machine. *Lives in:* the control system, a spreadsheet, and their own notebook.

Industrial engineer. Owns the work rather than the process: time study, method study, line balancing, layout, standard times, labour standards, capacity models. **The source of the routing data that the entire costing and planning system depends on.** *Ask one question:* when were the standard times last re-timed? If the answer is "when the line was commissioned", then the labour efficiency variance is measuring the age of the standards, not the performance of the people. *Lives in:* a time-study file and the ERP's routing.

Manufacturing engineer, or production engineer. Owns the industrialisation of a product: tooling, fixtures, gauges, equipment specification, and the launch. In automotive, runs the product-approval process with the customer. **This is the person who turns a design into something a plant can build**, and therefore the person who owns the manufacturing bill of material.

Design engineer, product engineer, or technical. Owns the product definition: drawings, specifications, the engineering bill of material, and engineering change. **The engineering change process is the most important business process you will meet in discrete manufacturing**, and the one most likely to be undocumented. *Lives in:* the product-lifecycle system — Thorn, in this book's invented group — and the drawing vault.

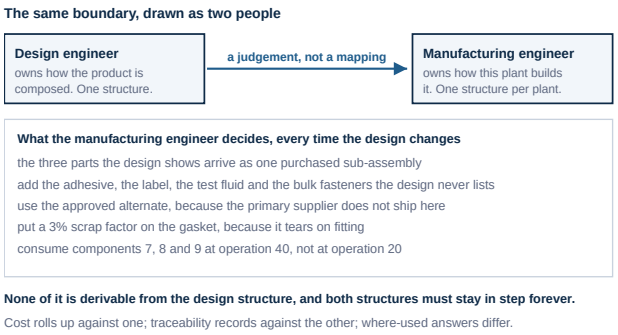


Figure 5.2 The engineering-to-manufacturing bill-of-material boundary as two jobs. The market-leading vendor's own answer to keeping the two structures aligned is a person doing a structured comparison [1].

The boundary between those last two roles is the most consequential human boundary in discrete manufacturing, because it is the engineering-to-manufacturing bill-of-material boundary in person. The design engineer's structure describes how the product is composed. The manufacturing engineer's structure describes how this plant builds it: the purchased sub-assembly collapsed to one line, the adhesives and labels and bulk fasteners the design never mentions, the alternate part this site can actually buy, a scrap factor on the gasket because it tears on fitting, and components allocated to the operation that consumes them. **None of that is derivable from the design structure. It is a judgement, made by a named person, and it has to be re-made every time the design changes** [1].

Automation and controls engineer. Owns the programmable controllers, the drives, the instrumentation and the supervisory layer: the code that makes the machine move. Frequently the only person who can change a machine's behaviour, and frequently not in the information technology reporting line. *Will tell you:* what the machine can and cannot be asked for, and what happens to the plant when the network above it disappears. *Lives in:* the controller programs, the engineering workstation, and an archive that may or may not match what is running.

Project engineer. Owns capital projects: new lines, upgrades, relocations. The person who will be commissioning a new asset while you are implementing a system on it. *Ask:* what is in the capital plan for the next 24 months, because it will collide with your roadmap.

Maintenance

Maintenance manager. Owns asset availability and the maintenance budget, in permanent tension with production over access to equipment for planned work. *Will tell you:* the backlog, and whether it is a plan or a wish.

Maintenance planner, or maintenance scheduler. Converts maintenance requirements into executable work: scopes the job, secures the spares, books the window with production, and closes out the history. The role with the most influence per hour worked in the whole maintenance function, usually the most under-resourced, and the primary user of the maintenance system. *Will tell you:* whether the asset hierarchy is usable, whether failure history is coded or free text — it is usually free text, which means it is unanalysable — and what spares availability really is. *Lives in:* the maintenance management system, Baobab in this book's example.

> THE FIRST INTERVIEW IN MOST PLANTS
>
> **The maintenance planner is the person who knows where everything actually is.** Every asset, its location, its condition, its history, what it is called on the drawing, what it is called by the fitters, and which of those two names the system uses. In most plants this is the single most useful first interview available to you — it costs an hour and it tells you whether an asset register, a functional location hierarchy and a failure taxonomy exist in usable form. Nothing else you can do in week one is worth as much.

Reliability engineer. Analyses failure history, sets the maintenance strategy per failure mode, and owns condition monitoring. **If this role exists and has data, the plant has a maintenance strategy; if it does not, the plant has a work-order backlog.** *Lives in:* the maintenance system's history, a vibration database, and a spreadsheet.

Millwright. The mechanical-and-electrical trade that keeps production equipment running: fault-finding, repair, installation, alignment. In South Africa an artisan qualification with a formal apprenticeship, and on the national scarce-skills list [2]. *Will tell you:* which machines fail the same way every month, and which standard job has not matched reality in years.

Electrician. Power, switchgear, motors, protection and the electrical side of every machine. Also the person who sees the consequences of a voltage dip before anybody else does. Also scarce [2].

Instrument technician. Calibrates and maintains the instruments the control system reads — transmitters, sensors, analysers, final elements. **Every number your historian stores comes from something this person calibrated**, which makes the calibration record part of the provenance of your reported data, not a maintenance detail. *Lives in:* the calibration register, sometimes a wall chart.

Quality

Quality manager. Owns the quality management system and its certifications, the audit programme, customer complaints, and product release. In automotive, the sector scheme; in food, a food-safety scheme; in pharmaceuticals, the licensing regime. *Will tell you:* what the last external audit found, which is the fastest route to a list of the plant's real data gaps. *Will not say unprompted:* how many customer complaints are closed without a root cause, and which specification has not been reviewed since the product launched.

Quality assurance, and quality control — get this right, because practitioners will correct you. Quality control is inspection and testing: measuring product and process against specification, in the laboratory and on the line. It is reactive, product-focused, and it detects. **Quality assurance is system and prevention**: the procedures, specifications, validations, training records, supplier approvals, change control, deviation and corrective-action management, audits — and the authority to **release or reject**. It is proactive, process-focused, and it prevents. In a regulated plant the distinction is formal and the **release decision belongs to assurance**, not to control and not to production. **That veto has to exist in the system as a status on the stock**, because an execution system that lets production despatch unreleased material in a food or pharmaceutical plant is not a configuration error, it is a compliance failure.

Laboratory analyst or technician. Runs the tests, records the results, flags the out-of-specification ones. **The laboratory's turnaround time is a planning constraint**: if a release test takes 48 hours then finished goods sit in quarantine for 48 hours, and that is inventory, lead time and cash. Plants rarely model laboratory turnaround in their lead times and should. *Lives in:* the laboratory information system, Clearwater in this book's example, or a spreadsheet.

Metrologist. Owns the measurement system itself: traceability to national standards, gauge capability studies, and the question of how much of the observed variation is the gauge rather than the process. **A plant that cannot state its measurement error cannot interpret its capability indices**, which means half its quality reporting is unfalsifiable.

Calibration technician. Executes the calibration schedule on gauges, instruments and test equipment, and keeps the certificates. Where measuring instruments are used for a prescribed purpose — a weighbridge, a trade scale, a filling line's checkweigher — the verification record is a regulatory artefact and the reading generally may not be adjusted or recomputed in software. *Lives in:* a calibration register, which is startlingly often a wall chart.

Planning and supply chain

Systems vendors conflate the planner and the scheduler. They are different people doing different work on different horizons, and the handover between them is a canonical integration problem.

Demand planner. Owns the forecast: a statistical baseline, plus what sales and marketing know, plus promotions and launches, at the grain and horizon the business plans on. Accountable for forecast accuracy and bias. **Bias matters more than accuracy and is measured less often.** *Will tell you:* that sales will not commit to a number, that the forecast is overridden upward every month, and that the promotional plan arrives too late to produce for.

Supply planner. Converts the demand plan into a feasible material and production plan: runs the requirements calculation, manages inventory targets, raises the supply proposals. The person who absorbs the gap between what sales wants and what the plant can do — and therefore the person who knows where every workaround is.

Master scheduler. Owns the master production schedule over the medium horizon and the integrity of the planning fences: what is frozen, what is firm, what is still free. *Will tell you:* how often the frozen period is violated, and by whom. The answer is "weekly" and "sales". *Will not say unprompted:* that the time fence is shorter than the cumulative material lead time, which makes it arithmetically unenforceable — chapter 7 explains why that is checkable in an afternoon.

Production planner. Releases and maintains production orders for a plant or area: order quantities, dates, material availability checks, and the daily argument with production about what is actually possible. In many plants this role and the supply planner's are one person.

Scheduler, or detailed scheduler. Sequences the next hours to days on specific equipment, honouring changeovers, tooling, labour and material. **Almost always working in a spreadsheet or a specialist tool rather than in the ERP**, because the ERP cannot honour the constraint that actually binds — the allergen changeover matrix, the die, the campaign wheel, the sequence-dependent setup. *Find this person's file in week one.* It is a system, and it is undocumented.

Materials manager, or materials controller. Owns raw-material availability and the physical stores. The person who expedites, and the person who knows which supplier's lead time is a fiction.

Buyer, or procurement officer. Places and manages purchase orders, chases deliveries, manages day-to-day supplier performance. Distinguish from the category manager or strategic sourcing role, which negotiates contracts and selects suppliers. The purchase price variance is the buyer's measure, and the buyer will tell you it is not a fair measure of their performance. They are usually right. *Will not say unprompted:* which suppliers are single-sourced because nobody has qualified a second one, and which supplier certificates on file have expired.

Expediter. Where the role exists as its own job, it exists because the plan does not work: a full-time person chasing late material and late orders through the system and the supply base. **An expediter's workload is a direct measurement of planning quality**, and nobody reports it.

Logistics coordinator. Books transport, plans loads, manages the yard and the documents. In South Africa, owns the rail-versus-road decision and the relationship with the port, and therefore owns an assumption about lead time that the planning system has probably hard-coded as a constant.

Data and systems

Master data steward, or master data analyst. Where the role exists, maintains item master, bill of material, routing, vendor and customer master against a governance standard. **Where it does not exist, master data is maintained by whoever needs it changed**, and the first finding of any manufacturing architecture assessment will be master data quality. *Ask:* who can create an item code, how many ways are there to do it, and what is the approval? The answer is usually "several people, three ways, none".

The item-master gatekeeper. Not a job title. This is the person — a planner, a costing analyst, somebody's long-serving assistant — who in practice decides whether a new item code is created and what it looks like. They are not in the governance document and they are the governance. Find them by asking three people how to get a code created and noticing whose name comes up twice.

ERP functional lead. Knows how the enterprise system is actually configured, as opposed to how it was designed: which fields are repurposed, which validation was switched off in 2019, which report everyone uses and nobody owns. Usually one or two people for an estate, and usually the constraint on your programme's pace.

Execution-system administrator. Owns the manufacturing execution or operations-management layer: the work instructions, the routings as executed, the data-collection screens, the interfaces up to the ERP and down to the controllers. Sits culturally between information technology and the plant and is frequently trusted by neither.

The historian tag-list owner. The person who maintains the tag list, the tag naming, the compression settings and the mapping between a tag and the thing it measures. **This person is usually not in information technology** — they are a control engineer, or a process engineer, or a technician who inherited it. They are also the only person who can tell you whether a tag means what its name says. *Lives in:* the historian, Steenbras in this book's example, and a spreadsheet of tag names that is the de facto data dictionary of the plant floor.

Safety, improvement, and the industrial-relations layer

Safety, health and environment officer. Owns statutory compliance inside the fence: appointments and committees, incident investigation and reporting, hazardous substance handling, and the inspection clocks on pressure equipment and lifting equipment. **In South Africa this role has become a significant data consumer and producer, and it is almost never in the scope of a manufacturing systems programme. It should be.**

Environmental officer. Owns the authorisations and their conditions: air emission licence reporting, water-use licence volumes and effluent quality, waste classification and consignment records, and now greenhouse-gas activity data for the carbon tax computation. Every one of those is a measurement obligation with a frequency, an evidence requirement and a regulator who may ask.

Occupational health practitioner. Runs medical surveillance: baseline and periodic examinations, audiometry, exposure-related screening, fitness-to-work certificates. **This role holds the longest-retained personal information in the enterprise** — chemical exposure and audiometry records run for decades — and it sits in a system nobody thinks of as a privacy system. Chapter 16 has the retention clocks.

Continuous improvement lead, or black belt. Runs the improvement programme. *Ask two questions:* which tradition — Toyota-derived lean, Six Sigma, theory of constraints, total productive maintenance — and what does the plant scorecard reward? If the scorecard is unit cost, the improvement programme is fighting the accounting system and it will lose. *Will not say unprompted:* how many of the programme's claimed savings were ever verified in the ledger.

The works council or workplace forum. The statutory consultative structure, uncommon in practice but real where it exists. Where there is none, consultation happens through the recognised unions instead, which is not the same forum and not the same agenda.

Shop steward. An elected worker representative with organisational rights, not a stakeholder to be managed. **Changes to work organisation — shift patterns, manning levels, measurement of individual performance, the introduction of monitoring technology — are consultation matters.** A project that introduces operator-level productivity measurement without consultation will be stopped, and stopped late. Involve the stewards early, in their own forum, and be straightforward about what the system will and will not measure.

Bargaining-council agent. Not an employee at all. The council's compliance officer inspects against the collective agreement: the employee's grade under the agreement's own grading structure, the rate paid against the gazetted schedule, hours by category, allowances, and fund contributions [3]. **The agreement's grade is a separate attribute from the company's job grade, and the minimum is a dated rate**, so an annual wage schedule is a mass dated-rate event in your payroll model.

Above the plant

Operations director, or group manufacturing director. Owns several plants and the comparisons between them, which is why they will ask you for a single definition of output, yield and on-time delivery, and will be disappointed by the answer. **Take the disappointment seriously and give them a measure definition register rather than a dashboard**; two plants reporting 96% and 78% on-time may be performing identically.

Group supply chain director. Owns planning, procurement and logistics across sites, and therefore owns the question of whether the group plans centrally or the plants plan themselves. *Lives in:* the planning tool — Vrystaat in this book's example — and a monthly cycle.

Finance business partner. The controller's forward-looking counterpart: business cases, capital submissions, scenario analysis. Your most useful ally in getting a programme funded, and the person who will ask what the benefit case is measured on.

Cost accountant, or plant controller. Owns standard costs, variance reporting, inventory valuation and the month-end close. **Simultaneously your most important ally and your most important constraint**, because the costing model is both an accounting requirement and the thing that distorts plant behaviour. *Will tell you:* which variances are explainable and which are a plug.

Chief information officer. Owns the estate, the budget and the risk, and in a manufacturing group usually owns the information-technology half of a boundary whose operational-technology half reports to engineering. That boundary is chapter 13's subject and it is an organisational fact before it is a technical one.

Enterprise architect. Sometimes you. Often there is already one, and they have a view, a model and a history with the plant managers. Read their material before you meet them, and ask what they have already tried.

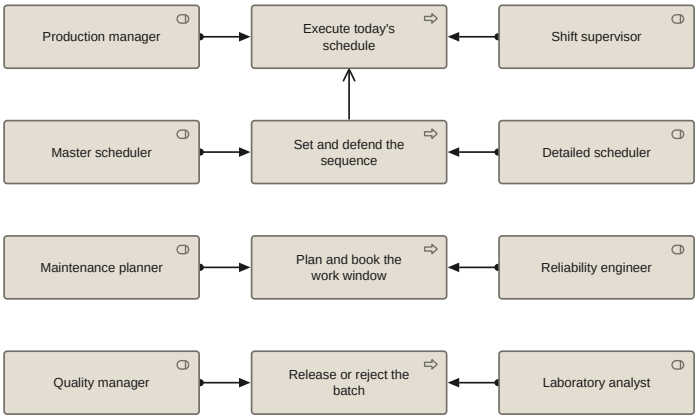


Figure 5.3 Four decisions and the pairs of people either side of them, drawn with the ArchiMate Role element. Note what the view does not show: a reporting line. Assignment to work is more useful to you than hierarchy.

Where these people live

An architect's instinct is to map roles to systems. Do it, and then notice how much of the column on the right is not a system.

Group	System class	What is actually there as well
Plant line management	ERP production orders, execution system	A paper logbook, a handover sheet, a whiteboard
Engineering	Product-lifecycle and drawing vault, control-system projects	A laminated parameter sheet beside the machine
Maintenance	Maintenance management system	Free-text failure history, a wall chart, the planner's memory
Quality	Laboratory and quality systems	Email approvals, a sample register, a gauge wall chart
Planning and supply chain	ERP planning, a specialist scheduling tool	The scheduler's spreadsheet, which is the real scheduler
Data, safety and above the plant	ERP master data, historian, reporting	The tag-name spreadsheet, the compliance file, the monthly pack

The right-hand column is the finding. Every item in it is a system without an owner, a backup or a successor, and each one is a requirement you now know about.

Who to see first

Order matters more than coverage. The sequence below is built on the one in the research for this book, and it is optimised for a specific outcome: by the end of it you should be able to say what is true about this plant that nobody has written down.

Order	Who	What you are actually getting
1	Shift supervisor, plus a floor walk on shift — including night shift	The real process, the real data capture, the workarounds
2	Maintenance planner	Asset hierarchy quality, failure coding quality, where everything is
3	Production planner and the detailed scheduler	Where the plan breaks, and which spreadsheet is the real system
4	Setter or line leader, watching one changeover	What a changeover costs, and what is not in the routing
5	Industrial engineer	Routing and standard-time currency — the foundation of costing and capacity
6	Plant controller	The costing model, the absorption base, the variance population
7	Quality manager and the laboratory	Release process, status-of-stock model, laboratory turnaround as lead time
8	Demand planner	Forecast grain, horizon, bias, and the override culture
9	Historian tag-list owner and the controls engineer	Whether a tag means what its name says
10	Master data owners, plural, because there is no single one	The governance gap
11	Safety, health and environment, and the environmental officer	The statutory reporting obligations nobody put in scope
12	Plant manager	Validation, priorities and sponsorship
13	Shop steward forum	What will stop the project

Two notes on using it. **Do the floor walk before the plant manager**, so that you can ask specific questions instead of inviting a presentation. And **watch one changeover end to end**; nothing else converts an abstract understanding of setup time into a usable one as quickly.

The skills reality

Now the uncomfortable part, and it belongs in this chapter rather than in a risk register.

The preliminary results of the 2026 Xpatweb critical-skills survey, as at 22 September 2026 with the survey closing on 30 September, put **engineers first among hard-to-fill skills at 29% of respondents, with information and communications technology specialists and artisans joint second at 25% each** [2]. The engineering category explicitly includes mechanical, industrial, automation and energy engineers plus engineering technologists and technicians. The artisan shortage covers electricians, millwrights, refrigeration and air-conditioning mechanics and lift mechanics [2].

Read that against the two sections above. **The scarcest people in the country are precisely the ones who execute everything an architect designs below the control layer** — the automation engineer who changes the machine's behaviour, the instrument technician whose calibration is the provenance of your data, the millwright who installs the device, the industrial engineer who re-times the standards your costing depends on.

Four consequences, which are design consequences and not staffing ones:

1. **Design for fewer, less specialised, more remote hands.** Standardise ruthlessly on fewer device types and fewer vendors, because each one is a competency somebody has to maintain.
2. **Prefer configuration to code at the plant**, because code requires the scarce person, and document for a technician rather than for an engineer.
3. **Institutionalise what is in people's heads** — asset registers, tag taxonomies, program archives, documented restart sequences. Every "it is in Karel's head" dependency is a single point of failure with a resignation date.
4. **Treat the integrator as part of the architecture.** The scarce skills are concentrated in the automation houses, so a plant dependent on one of them has a single point of failure in a supplier. The mitigation is program archives, documented standards and open interfaces, not a second supplier.

There is also a retention problem, partly emigration-driven for engineers. I could not source a rate for that and will not invent one, but the design response does not depend on the number: it is point three, applied harder.

Architecture that assumes those people are available is architecture that does not get built.

One more stakeholder, who does not work here

Every role above sits inside the company. One set does not, and it has a data demand as specific as any customer's: the **regulator**. The safety inspector asks for appointments, risk assessments with their clocks, and exposure monitoring by an approved authority. The air-quality authority asks for monitoring records and the share of valid hourly values behind them. The metrology regulator asks for verification history on the instrument that weighed the pack. The council agent asks for grades and rates. The empowerment verification agency asks for your suppliers' status *as at each transaction date*.

One more external party behaves the same way: **the customer's own quality engineer**, who in automotive may hold a resident position at your plant, raises problem reports against your parts and decides when containment ends. They have inspection rights and an escalation ladder, and they are not in your organisation chart either.

Treat each of them as a stakeholder with a scheduled information requirement, because that is exactly what they are. Chapter 16 sets out who inspects, what they ask to see, and where the evidence has to live — and it is the chapter that most often changes a programme's scope.

THE CHAPTER IN TEN LINES

- Name each job as the plant names it, and assume no title means what you think it means until you have asked.
- A planner may forecast, may sequence machines, or may do neither; the planner and the scheduler are different people on different horizons.
- If shift supervisors enter production data retrospectively from paper, every metric built on it carries the same bias.
- The maintenance planner knows where everything actually is, and is the single most useful first interview in most plants.
- Quality assurance builds the system and holds the release decision; quality control inspects and tests. The release veto must exist as a status on the stock.
- The design engineer and the manufacturing engineer are the engineering-to-manufacturing bill-of-material boundary in human form, and the transformation between them is a judgement.
- The setter knows more about changeover loss than anyone, and none of it is in a system.
- The historian tag-list owner is usually not in information technology, and is the only person who can say whether a tag means what its name says.
- The item-master gatekeeper is not a job title and is nonetheless the actual governance.
- Shop stewards, bargaining-council agents and inspectors can stop a programme and do not report to your sponsor.
- A 24/7 pattern needs four or five crews, not three, and five is the honest number once leave, training and absence are covered.
- Engineers were the hardest skill to fill in South Africa at 29% of respondents in the preliminary 2026 Xpatweb results, with artisans joint second at 25%.
- The scarce people are the ones who execute everything designed below the control layer, so standardise on fewer device types and document for a technician.
- The regulator is a stakeholder with a scheduled information requirement; chapter 16 sets out who asks for what.

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- **3.** MEIBC, Industry Circular 2026/06: wage increases, Metals Main Collective Agreement, 1 July 2026 to 30 June 2027 https://www.meibc.co.za/circulars/hvVZ1HqrX09C7nAGJdBx

CHAPTER 6

How manufacturers make money

Margin conversations in a factory are arithmetic, not opinion — and some of the arithmetic is designed, quite legally, to mislead you. This chapter is the arithmetic: what a thing costs, which of the several right answers you need, and why the profit number and the cash number move in opposite directions.

Walk into a plant performance meeting and you will hear three people argue about the profitability of one product, using three different numbers, all of them correct. That is not dysfunction. It is the structure of manufacturing accounting, and an architect who does not understand the structure will build reports that nobody trusts.

This chapter is the arithmetic. It is the least glamorous chapter in the book and the one you will use most. You will sit in margin conversations for the rest of your career in this industry, and your usefulness in those conversations depends entirely on whether you can follow the numbers to their source.

Three facts that generate everything else

A manufacturer buys inputs, changes their physical form, and sells the result. Three consequences follow, and the rest of the chapter is the working out of them.

It owns three inventories at once, in different states, valued differently. Raw material, valued at purchase cost plus landed cost. Work in progress, valued at material consumed plus the labour and overhead absorbed *to the operation it has reached* — which is a shop-floor fact, not a financial one, and which makes work in progress the hardest inventory in business to value. And finished goods, at full standard cost. Plus two that architects forget: consumables and indirects, which are consumed but are not in the bill of material, and spare parts, which are often worth more than the raw material and are governed by reliability engineering rather than by production planning.

It owns capacity, which costs money whether or not it runs. A press line, a pasteuriser, a reactor, a kiln: bought, depreciated, maintained, insured, staffed on a shift pattern, drawing energy at idle. That cost exists before a single unit is made.

Therefore unit cost is a function of volume. If fixed manufacturing overhead is R10 million a month and the plant makes 100,000 units, overhead per unit is R100. At 50,000 units it is R200. The product did not change. Nothing physical changed. Every argument in this chapter descends from that one line of arithmetic.

> A retailer's cost per unit sold is not materially a function of how many units it sold this month. A manufacturer's is. That single difference generates absorption costing, variance analysis, the make-or-buy decision, and most of the friction between a plant manager and a financial controller.

Cost, properly

The standard cost build-up

A manufacturer values stock and cost of sales at **standard cost**: a pre-determined, period-frozen cost per unit, built by rolling up three things.

- **Standard material quantities** from the costed bill of material, at **standard prices** from the purchasing master.
- **Standard times** from the routing — setup time and run time per operation — at **standard labour rates** per work centre.
- **Standard overhead**, applied at an absorption rate against an activity driver.

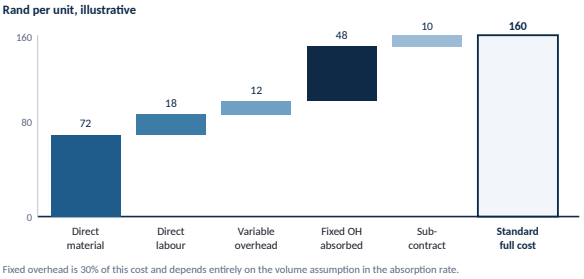


Figure 6.1 Standard cost build-up for one unit, illustrative. The component split on the left is what lets you explain a cost movement; the total on the right is what the ledger carries. Figures invented.

The roll-up is recursive. Level-three components are costed, then level two, then the finished item. A multi-level bill of material therefore has a **cost roll-up sequence**, and a change at level three only reaches the finished item when the roll-up is re-run. This matters more than it sounds: a costing engine that rolls up only the top two levels will show a stable finished-goods cost while a component cost moves 30%.

The part architects skip, and must not, is the **cost component split**. A standard cost is not one number; it is material, labour, variable overhead, fixed overhead, subcontract and inbound freight, each separately accumulated and each separately rolled up. A finished item's cost at level zero *hides* variances that happened at level three. When a margin analyst asks why a product's cost rose, the answer lives in the component split, not in the total — and preserving the split through multi-level roll-up is a specific requirement that implementations routinely skip.

It is also, in South Africa, a legal requirement for anyone in the automotive value chain. The Automotive Production and Development Programme is a customs-duty-rebate-certificate regime, and substantiating a production rebate claim requires country-of-origin and cost-component data at bill-of-material line level, propagated through multiple supply tiers [6]. A plant whose costing does not preserve component split through multi-level roll-up physically cannot substantiate its claim. At Highveld Precision Components, the invented tier-two component maker this book uses as a worked example, that dataset is maintained by one costing analyst and is the single most fragile thing in the company.

Why standards exist at all. Three reasons, and they pull in different directions.

1. **Transaction economy.** Valuing every material issue at its actual purchase lot cost requires perfect lot traceability through every operation. A standard lets the plant value a movement instantly at a known rate and reconcile later.
2. **Performance measurement.** Variance against standard is the only mechanism that separates "the process behaved differently" from "prices changed".
3. **Quoting.** You cannot quote a customer from a cost you will know only after you have built the thing.

And why they are permanently unsatisfactory. A standard is stale by construction: it is set annually at budget time, from a view of next year's prices, and by month six material prices, electricity tariffs and bargaining-council wage rates have all moved. Setting a standard is a political act — a tight standard generates permanent adverse variances and demoralises the plant, a loose one hides losses. And the annual roll creates a step change in reported margin with no operational cause whatsoever, plus an inventory revaluation that hits the profit and loss account. Any margin trend report that spans a standard-cost roll must flag the roll, or it is lying by omission.

> At Highveld the standard cost roll is effective 1 July, to align with the metal and engineering bargaining council's wage date. So the cost roll and the wage increase land in the same month and the resulting margin step cannot be attributed to either without care. This is a common and avoidable reporting defect: never let two step changes share a month if you can move one.

Absorption and the overhead rate

Absorption costing assigns all manufacturing costs — direct material, direct labour, and both variable and fixed manufacturing overhead — to the units produced. It is the basis required for reported inventory under IFRS, because IAS 2 requires conversion costs, including a systematic allocation of production overheads, to be included in the cost of inventories [1]. The alternative, **variable costing**, assigns only variable costs to units and expenses fixed manufacturing overhead in the period. Variable costing is used for internal decisions and is not permitted as the basis for reported inventory.

The mechanism, step by step, because you will have to build it:

1. **Choose a cost centre structure.** Usually one per work centre, department or line; sometimes one per machine.
2. **Budget the overhead** per cost centre per period: depreciation, maintenance labour, supervision, indirect labour, energy, consumables, insurance, factory rates, allocated site services.
3. **Choose an activity base** to absorb it on: machine hours, labour hours, units, kilograms, batches, or a blend. This choice determines which products carry which share of overhead. A plant that absorbs on labour hours and then automates will systematically under-cost its automated products — and will go on doing so for years, because nobody revisits the driver.
4. **Set the absorption rate** = budgeted overhead ÷ budgeted activity. *This fixes the denominator volume assumption for the whole period.*
5. **Absorb** overhead onto each unit as it passes through: rate × actual activity consumed, per the routing.
6. **Measure under- or over-absorption** at period end: overhead absorbed minus overhead actually incurred. That difference goes to the profit and loss account.

Steps 3 and 4 are master data with effective dates, held per cost centre per period, and they are inputs to the standard cost roll-up. A costing architecture that does not version them cannot explain why a unit cost changed. Step 5 requires an accurate routing and requires the shop floor to confess its actual activity, which it does imperfectly — which is why step 6 is never zero.

Activity-based costing replaces one or two plant-wide drivers with many activity-level drivers: number of setups, number of purchase orders, number of inspections, number of material moves. It reliably reveals that low-volume, high-complexity products are far less profitable than standard costing says, because they consume setups and expediting out of all proportion to their volume. It is intellectually dominant and operationally rare, because it needs activity data most plants do not capture. If a client asks for activity-based costing, the real project is instrumentation, not costing. Say so in week one.

Actual costing and the material ledger

Not every manufacturer runs a pure standard. The alternatives you will meet:

- **Weighted average per material per period.** Common in food, chemicals and metals, where lots are homogeneous and a monthly average is defensible.
- **Moving average price**, which re-averages the stock value on every receipt. Simple, and it back-propagates a price shock into the value of stock already on hand, which makes period-over-period unit cost comparison unreliable.
- **First-in-first-out or specific identification**, where lots are genuinely distinguishable and traceability is regulated.
- **Actual cost by production order**, used in engineer-to-order and job shops, where each job *is* the cost object. Here there is no meaningful unit cost of the product at all. There is the cost of job 47821.

Large ERP systems add a **material ledger** or periodic actual-costing run: the system holds the standard for day-to-day valuation, collects the differences, and then at period end rolls the actual prices *up through the bill of material levels* so that a finished item carries the actual cost of its components rather than their standard. This is the single most expensive month-end process in a manufacturing ERP and the one most often switched on without anyone understanding what it does to the numbers. Two things to establish before you touch it: whether the roll-up is single-level or multi-level, and what happens to the difference on stock that has already been sold.

The five variances

Variance analysis decomposes the gap between what the standard said the period's production should have cost and what it actually cost. Each variance isolates one cause and each is owned by a different person, which is precisely why they exist.

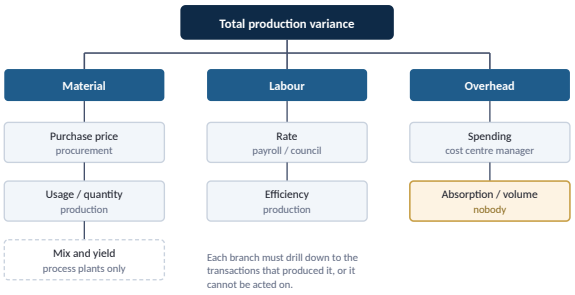


Figure 6.2 The variance tree, with the owner of each branch. The absorption variance is the one nobody in the plant owns, and the one most often read as a manufacturing failure.

Purchase price variance = (actual price − standard price) × actual quantity purchased. Owned by procurement. It tells you whether buying beat the standard. It is gamed, or accidentally corrupted, in four ways: by timing receipts across a period boundary; by negotiating a rebate that lands in a different account below the line; by buying forward in volume to book a favourable variance while creating an inventory problem; and most often by definitional drift, where "price" is ex-works in the transaction and landed in the standard. That last one produces a permanently and spuriously favourable variance, and it is worth checking in your first week.

Material usage variance = (actual quantity consumed − standard quantity for actual output) × standard price. Owned by production. This is the yield and scrap story expressed in money. It is computable only if output is reported accurately and if you know whether the standard quantity is net or gross of planned scrap. Both conventions exist. **Ask which.** A bill of material carrying a 3% scrap factor on a gasket has already absorbed that loss into standard, so planned scrap does not appear as a variance; a bill of material without the factor shows the same physical loss as an adverse variance every month, forever. Two plants with identical physical behaviour, opposite reports.

Where several inputs are substitutable within a recipe — milling, blending, smelting, rendering, refining — usage splits further into a **mix variance** (we used a different proportion of inputs than standard) and a **yield variance** (total output from total input differed from standard). This decomposition is essential in process manufacture and absent in discrete assembly. Karoo Foods, the invented food business in this book, cannot manage its wet plant without it.

Labour efficiency variance = (actual hours − standard hours for actual output) × standard rate. Owned by production. It tells you about downtime, speed loss, training and rework. It is gamed by loose standard times and by booking time to an indirect cost centre, and it is corrupted by the allocation rule: it requires hours booked against production orders, which requires either labour confirmation at a terminal or a time-and-attendance allocation. Most plants do the second badly, so labour efficiency variance is frequently an artefact of the allocation rule rather than a fact about work. Before you report it, find out how an hour gets attached to an order.

Labour rate variance = (actual rate − standard rate) × actual hours. Owned by human resources and payroll. It is driven by overtime, shift allowances, grade drift and bargaining-council increases. In South Africa this variance is substantially outside the plant's control, because the metal and engineering council's main agreement sets scheduled-rate increases by grade — 5.00% at grade A rising to 6.00% at grade H effective 1 July 2026 [4] — and in the motor industry council's component-manufacturing chapter the increases are on **actual** rates of pay rather than on the prescribed minimum [5]. That two-word distinction is the most expensive detail in South African wage agreements, because an increase on actual rates lifts everyone, including those already above scale.

Overhead absorption variance, also called the volume variance = overhead absorbed minus budgeted overhead at budget volume. It is the money consequence of running at a volume other than the one the absorption rate assumed. It is **owned by nobody in the plant, and that is the point**. It is a consequence of sales volume and of the master scheduler's decisions, and it is the variance most often misread as a manufacturing failure. Sitting beside it is the **overhead spending variance** — actual overhead incurred against budget for the activity level achieved — which the cost centre manager does own.

Period-end settlement

The variances are not a report. They are postings. At period end:

1. The cut-off is enforced: every production confirmation, goods movement and time booking for the period must be posted. This is the hard part. An order half-confirmed at month end carries work in progress, and a confirmation posted on the 2nd for work done on the 31st is in the wrong period.
2. Work in progress is determined, by order or by a period-based method.
3. Activity rates are actualised: the cost centre's actual costs divided by its actual activity, giving the actual rate against which the planned rate was charged. The difference is a price variance on the activity itself.
4. Order variances are calculated and categorised.
5. Over- or under-absorption is computed.
6. Orders are **settled**: variances post to the general ledger, to cost of sales, to an inventory revaluation, or are capitalised into inventory, depending on policy.
7. Where actual costing runs, the material ledger rolls actual prices up the levels.
8. Inventory is valued and provisions taken — obsolescence, slow-moving, net realisable value.

9. The stock ledger is reconciled to the general ledger and the physical position to the book position.

> The commonest costing configuration failure in manufacturing is step 4 done badly: variance categories not configured meaningfully, so almost everything lands in a residual bucket called something like "remaining input variance". That single misconfiguration makes the whole exercise useless to operations, and it is why the variance report in so many plants is printed, distributed and read by nobody. Check it. It is a half-day investigation with a large payoff.

And the structural problem underneath the whole close: operations measures the month in tonnes, cases and equipment effectiveness; finance measures the same month in rands of variance; and in most plants no mapping exists between the two. The variance report is therefore never used to change anything. Building that mapping is a genuine architecture deliverable and almost nobody asks for it.

The absorption trap, worked

Here is the arithmetic that explains behaviour you will otherwise find inexplicable.

Take an illustrative plant — not a real company. Fixed manufacturing overhead R10 million a month. Variable cost R60 a unit. Price R150 a unit. Normal volume 100,000 units a month, so the absorption rate is R100 a unit and standard full cost is R160 a unit.

Same sales. Three production decisions. Three reported profits.

Units produced	80 000	100 000	120 000
Overhead absorbed	R8,0m	R10,0m	R12,0m
Overhead incurred	R10,0m	R10,0m	R10,0m
Volume variance	R2,0m adverse	nil	R2,0m favourable
Fixed cost in closing stock	released	unchanged	+R2,0m added
Reported profit, versus the middle column	R2,0m lower	base	R2,0m higher
Cash generated, versus the middle column	higher	base	lower

Profit and cash move in opposite directions, driven only by how much was made.
IAS 2 counters this: allocate fixed overhead on normal capacity, expense the unallocated part of a low-production period, and never raise the per-unit allocation because the plant was idle.

Figure 6.3 The absorption trap, worked. Fixed overhead R10m a month, normal volume 100,000 units, absorption rate R100 a unit. Sales are 100,000 units in all three cases. Illustrative figures; no real company.

- **Make 100,000, sell 100,000.** Overhead absorbed R10m against R10m incurred. No volume variance.
- **Make 120,000, sell 100,000.** Absorbed R12m against R10m incurred: a **R2m favourable volume variance**. Twenty thousand units of finished goods now carry R2m of fixed overhead onto the balance sheet. **Reported profit rises by R2m** because fixed cost was capitalised into inventory instead of being expensed. Nothing extra was sold.
- **Make 80,000, sell 100,000.** Absorbed R8m: a **R2m adverse volume variance**, expensed. Reported profit falls by R2m — while the business actually generated *more* cash, because it emptied the warehouse.

Three consequences, all of which you will meet.

One: the incentive to overproduce is real and is created by the accounting system. A plant manager measured on unit cost or on absorption will run the line to absorb overhead, converting cash into finished goods. This is the single most destructive incentive in manufacturing management accounting, and it is not the product of bad managers. It is the product of the measurement. Change the measurement or expect the behaviour.

Two: profit and cash are negatively correlated in the short run. A destocking quarter looks bad on the profit and loss account and good in the cash flow statement. A stocking quarter does the reverse.

> Any manufacturing dashboard that shows profit without inventory movement alongside it is actively misleading. Not incomplete — misleading, because the two move in opposite directions for a reason that has nothing to do with operational performance. If you design one such dashboard in your career, design it with the inventory movement on the same page. This is the most consequential single design rule in this chapter.

Three: IFRS pushes back, and it is worth knowing how. IAS 2 requires the allocation of fixed production overheads to conversion cost to be based on **normal capacity**, requires unallocated overhead in a low-production period to be recognised as an expense of the period, and requires that the amount allocated per unit **not be increased** as a consequence of low production or an idle plant [1]. That is precisely the rule that stops a plant hiding idle capacity in inventory. It is also the rule most often configured wrongly, because configuring it properly means someone has to define "normal capacity" and own the number.

> Honest caveat on sourcing. The paragraph-level text of IAS 2 on normal-capacity allocation was not retrieved in the research pass behind this book; the requirement as stated above is universal practice and is what every technical summary says, but read the standard's own words before this sentence goes into a client deliverable. The two specific paragraph references in the next section were checked and are reliable.

Two corrections that will save you an argument

"Standard costing isn't IFRS-compliant" is wrong. You will hear it, often from someone confident. IAS 2 paragraph 21 expressly permits standard cost for convenience, where the results approximate cost and the standards are regularly reviewed and revised in the light of current conditions [1]. The condition matters — a standard frozen for three years is not a permitted approximation of cost — but the method is not the problem. If a client has been told it must abandon standard costing to be compliant, somebody has sold them a project they do not need.

LIFO is not an option under IFRS. IAS 2 paragraph 25 lists first-in-first-out and weighted average cost only [1]. This is the largest single costing divergence between IFRS and United States practice, and both South Africa and Canada use IFRS.

[NA] Last-in-first-out is permitted under US GAAP, and a North American group may consolidate on a LIFO basis while its South African subsidiary reports first-in-first-out or weighted average. Two inventory values then exist for the same physical stock, and the LIFO reserve is the reconciling item. The architectural consequence is concrete: the group costing method and the statutory costing method may differ, and **both must be derivable from the same transactions.** If you design a costing layer that can produce only one of them, the group reporting pack becomes a spreadsheet maintained by one person.

Margin and mix

Gross margin is not a property of the company

A manufacturer's gross margin is a weighted average of product families with structurally different margins, and the weights move every month. Where do the differences come from?

- **Material content as a share of price.** A product that is 70% material by cost has almost no room to improve and passes input-price moves straight to margin. Beverage cans, basic steel, commodity plastics and milling all sit here.
- **Value added by processing.** More transformation steps means higher gross margin and higher fixed-cost intensity. Speciality chemicals and precision components sit here.
- **Volume per item.** High-volume items absorb setup and changeover across many units; low-volume items carry setup per unit. Within one plant, the same gross-margin percentage can conceal a tenfold difference in setup cost per unit.
- **Channel**, which is covered below and which frequently matters more than the product.
- **Capacity position.** A product made on a near-full asset carries an opportunity cost; one made on an idle asset carries almost none.

Price, volume, mix and cost

A margin movement between two periods decomposes into four effects: **price** (we charged more or less), **cost** (inputs cost more or less), **volume** (we sold more or less at the same margin), and **mix** (we sold a different *composition*, so the weighted average moved without any single product's margin changing).

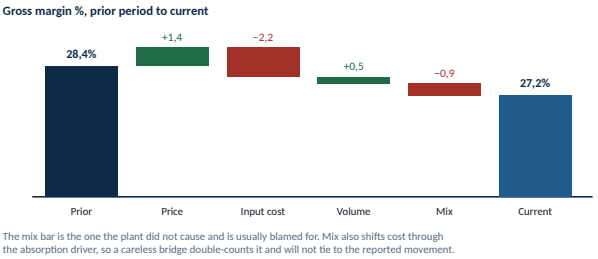


Figure 6.4 A margin bridge between two periods. Until a client has one of these, every margin conversation restarts from zero. Illustrative.

The standard argument runs like this. Group reports margin down 90 basis points. The plant insists nothing changed. Both are right: the mix shifted toward a lower-margin family. Without a price-volume-mix-cost decomposition the conversation is unresolvable, and it will recur monthly until someone builds one.

Building it requires margin at a grain of product by customer by period, on a consistent cost basis. That is a canonical reporting-layer requirement and you should expect to specify it.

A trap specific to manufacturing: mix also changes *cost*, through the absorption mechanism, because a different mix consumes different amounts of the absorption driver. A mix shift therefore moves both the numerator and the denominator, and a naive decomposition double-counts. If your bridge does not reconcile exactly to the reported movement, this is usually why.

Contribution margin, and the South African qualification

Contribution margin = selling price − variable cost per unit. It answers one question: does one more of these cover the cash it consumes and contribute something toward the fixed block? **Breakeven volume** = fixed costs ÷ contribution per unit.

It matters more in manufacturing than in most industries because the fixed block is large and lumpy, so decisions at the margin — accept this export order at a discount, run the line a sixth day, keep this low-volume item — are contribution decisions and not full-cost decisions. A product that loses money on full absorbed cost may be strongly contribution-positive, and dropping it makes the plant *worse* off, because the overhead it was absorbing does not go away. It redistributes onto everything else.

That is the **death spiral**, and it is a real failure mode of full-absorption reporting used for product rationalisation: drop the loss-making product, overhead redistributes, the next product becomes loss-making, drop it, repeat.

The counter-trap is equally real. Contribution thinking justifies every marginal order forever, and a plant that prices everything on contribution never recovers its fixed costs or funds replacement capital. Contribution is the right frame for a short-run decision inside a fixed capacity envelope. Full cost is the right frame for pricing, capital allocation and exit.

> What is actually variable is contested, and in South Africa the answer differs from the textbook. Direct material, yes. Energy, partly — a furnace or a pasteuriser has a large idle draw. **Direct labour is substantially not variable in the short run**, because operational-requirements dismissal under sections 189 and 189A of the Labour Relations Act is a procedure with consultation, disclosure and severance, and because minimum hours and shift arrangements are collectively agreed at bargaining-council level [4][5]. An imported contribution-margin or breakeven model therefore overstates this plant's flexibility, sometimes badly. **[NA]** At-will employment and more flexible shift scheduling make North American direct labour closer to genuinely variable, so US contribution margins and US breakeven volumes are both lower. Do not copy the model across.

The architectural consequence is small and specific: contribution margin requires a **variable or fixed flag on every cost element**, maintained per cost centre, with the ability to override it per decision context. Almost no ERP ships this usefully configured, so it gets rebuilt in the reporting layer, inconsistently, by three different teams. Finding and governing that flag is a cheap win.

Working capital

Cash conversion cycle = days inventory outstanding + days sales outstanding − days payables outstanding. A manufacturer's is structurally positive and long. Three reasons.

Days inventory outstanding is the sum of three stages, not one. Raw material days are driven by supplier lead time, minimum order quantity, import transit and port risk. Work-in-progress days are driven by manufacturing lead time and batch size — and by Little's Law, work-in-progress days *are* the manufacturing lead time, in disguise. Finished-goods days are driven by forecast error, service target, campaign length and shelf life.

Days sales outstanding is real. A manufacturer sells to businesses on terms: 30, 45, 60, 90 days, sometimes longer. Nothing is paid at despatch.

Days payables outstanding is constrained by supplier power. A manufacturer buying commodity inputs from large upstream producers — steel, resin, energy, pulp, sugar — often has less negotiating power than its own customers, who are large retailers and vehicle assemblers. It is squeezed from both ends.

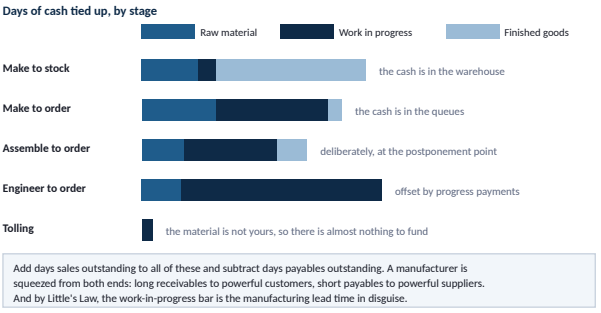


Figure 6.5 The cash conversion cycle by archetype, schematic. Identical profit and loss accounts, completely different balance sheets. Day counts are illustrative shapes, not benchmarks.

Every day of lead time added upstream appears in working capital twice: once as the inventory itself, once as the extra safety stock needed to cover the variability of a longer lead time. That is why South African import lead-time variability has a working-capital cost larger than the transit extension alone. In the week of 7 to 13 October 2026, vessel waiting time at Durban averaged about 5.73 days, with berthing delays of 7 to 10 days at Pier 1 and 10 to 16 days at the Durban Gateway Terminal [3]. Those are not transit days; they are variability, and variability is bought with safety stock.

Where the cash sits, by archetype

- **Make-to-stock**, discrete or process: in **finished goods** and the distribution network.
- **Make-to-order** job shop: in **raw material and work in progress sitting in queues**.
- **Assemble-to-order**: in **modules at the postponement point**, deliberately.
- **Engineer-to-order**: in **work in progress against contracts**, offset by advance payments and progress claims. An engineer-to-order business's working capital can be negative if its contract payment profile is front-loaded and catastrophic if it is back-loaded. The contract payment schedule *is* the balance sheet.
- **Tolling**: almost nothing, because the material belongs to someone else.

> A make-to-order business and a make-to-stock business with identical profit and loss accounts have completely different balance sheets. Same revenue, same margin, same cost base — and one holds its cash in finished goods it chose to build, the other in work in progress it cannot avoid. Any reporting design that treats inventory as one number has thrown away the only information that distinguishes them.

And total inventory turns are not comparable between companies at all. A plant that tolls holds almost no material it owns. A plant that imports holds months. A vertically integrated plant holds its own intermediates as work in progress where a non-integrated one holds them as purchased raw material. Turns comparison without a stage split and a note on vertical integration is noise.

The levers, honestly assessed: reducing finished goods means shortening the campaign cycle, which requires changeover reduction, or improving forecast accuracy, which has sharply diminishing returns. Reducing work in progress means smaller batches and shorter queues, which directly reduces lead time — the highest-value and least-funded lever in the industry. Reducing raw material through consignment stock or vendor-managed inventory shifts cost to the supplier, who prices it back in. Extending payables and shortening receivables are mostly a function of who has the power.

Theory of constraints as a rival accounting

Most of the improvement traditions make claims about how to run a plant. Theory of constraints makes a claim about how to *account* for one, and that makes it an architecture problem rather than a philosophy.

Its measurement set has three terms. **Throughput** is sales revenue minus totally variable cost — the rate at which the system generates money. **Investment** is the money tied up in the system. **Operating expense** is the money spent to turn investment into throughput. Decisions are ranked by **throughput per constraint-unit**: contribution per hour of the bottleneck resource. That is the single most useful product-ranking rule in a capacity-constrained plant and it is almost never what the margin report ranks by.

Two collisions follow, and both are real.

First, "throughput" now means two things in the same room. In operations it means good units per unit time. In theory-of-constraints usage it means revenue minus totally variable cost — a money rate, not a unit rate. When a client says throughput, establish which one before you write it into a specification. This is not pedantry; a reporting requirement written against the wrong definition produces a measure nobody can use.

Second, and much more serious: throughput accounting refuses to allocate fixed cost to units at all. That is its central methodological commitment, and it is flatly incompatible with absorption costing as a reporting basis. Lean's position is softer but lands in the same place: it measures a flow across cost centres while absorption costing measures cost centres, so the two reporting hierarchies are different trees over the same transactions.

> A plant that adopts theory of constraints or lean as its operating philosophy runs **two measurement systems in parallel, permanently.** Neither throughput accounting nor a lean flow view is permitted for statutory reporting, so absorption costing does not go away. The same production transactions must serve an absorption-costing ledger and a throughput or flow view, and reconciling them is a standing reporting function rather than a project. That is an architecture requirement. Write it down as one, in the principles, at the start — because if you do not, it arrives in year two as an unbudgeted reporting rebuild.

Customer and channel economics

A manufacturer typically serves several channels with incompatible economics out of one plant, and the channel — not the product — often determines profitability.

Channel	Economics	What it demands	The risk
Vehicle or equipment assembler supply	Lowest price, highest volume, longest contract, lowest margin; multi-year price-down agreements are normal	Delivery in hours or minutes; sequenced or just-in-time; quality in parts per million; mandatory certification; EDI as the only order channel	Total dependence; a model change removes the product; the tooling is often the customer's and sits in your plant
Aftermarket and replacement	Much higher margin, lumpier volume, enormous item count, long slow-moving tail	Availability from stock, not lead time; fitment data; different packaging and branding	Obsolescence, and a catalogue nobody can regenerate
Retailer private label	Volume, no marketing cost, no brand equity, no pricing power	The retailer's audit regime, artwork change cycle, delivery windows and on-time definition	Tenders are re-run and lost wholesale; printed packaging written off on every artwork change
Branded and direct	Highest gross margin, which then funds marketing, trade spend and listing fees	Trade spend accruals against volumes you do not control	Gross-to-net is large and routinely the biggest unmanaged line in the accounts
Export	Priced in foreign currency, so margin is a currency position; logistics cost and risk rise sharply	Incoterms discipline — ex-works, FOB, CIF and DDP are four different businesses	Tariff and trade-policy shocks arrive without notice

Two South African specifics worth carrying. The domestic component sector competes against a very large import stream *inside its own customers*: original-equipment component imports by the domestic vehicle assemblers were R151.0 billion in 2025, up from R147.4 billion in 2024, and replacement-parts imports were R107.5 billion against R104.8 billion [2]. And export economics changed sharply in 2026: a 25% United States Section 232 tariff now applies to South African passenger vehicles, relevant parts, steel and aluminium, with a 12.5% Section 301 tariff from July 2026 on goods not covered by Section 232, while AGOA was extended only to 2028 with South African eligibility unresolved [7]. For a plant whose export margin was already a currency position, that is a second exposure stacked on the first.

Distributor or agent. A distributor buys and resells, takes the credit risk and the inventory, and takes a margin. An agent sells on commission and the manufacturer carries the stock and the risk. The revenue, margin and working-capital consequences are entirely different, and the same customer relationship is sometimes recorded both ways in the same system.

> Channel is a dimension on *every* transaction, and it must drive pricing basis, payment terms, the on-time definition applied, the quality regime, the labelling and packaging specification, the EDI profile and the margin basis. A manufacturer whose data model has customer but not channel cannot answer the only question that matters: which channel are we actually making money in?

Tolling, where you never own the material

Two models are routinely confused and they have completely different accounting and system requirements.

Tolling, also toll processing or toll manufacturing. The customer supplies the material; you process it and charge a **conversion fee**. Title never passes to you. So the material is not your inventory — it is held on behalf of a third party, disclosed rather than capitalised. Your balance sheet is light and your inventory turns are meaningless. Revenue is the toll fee only, not the value of the goods, so **revenue is a fraction of the value flowing through the plant** and every ratio to revenue — margin percentage, revenue per employee, capital expenditure to revenue — is incomparable to an owning manufacturer's.

The commercial risk is **yield**. A tolling contract specifies an agreed yield; material lost above that allowance is at your cost, at the *material's* value, which may dwarf the toll fee. A one per cent yield miss on an expensive input can wipe out the entire conversion margin for the contract. So the yield reconciliation per contract per batch is not a report. It is the core commercial process.

Contract manufacturing, also co-manufacturing or co-packing, is the other model: you buy the material *and* make the product to the customer's brand and specification. The material is yours, the working-capital exposure is full, revenue is the whole product price and the margin is thin. The specific risks are obsolescence of brand-specific material — printed packaging and labels are the classic write-off — and single-customer dependence.

> **Ownership is a first-class attribute of inventory, and omitting it is one of the few genuinely unrecoverable design errors in manufacturing.** A single plant may toll and contract-manufacture on adjacent lines for different customers. The same tonne of flour may be the plant's asset on Monday and a customer's on Tuesday. Stock in one silo may be owned by three parties. You can add an ownership field to a data model at any time; you cannot retrofit ownership onto historic stock movements, which means you cannot restate the past and cannot answer an audit question about it. Most ERP systems support non-owned stock through a subcontracting or consignment construct designed for the opposite direction, and configuring it is a recurring source of pain. Find out about tolling in the first week, not the fifth month.

What this means for the model

Pull the chapter together into the things you will actually specify.

Never build a "cost" field. Build a **cost basis dimension**: cost object (unit, batch, order, product, family), cost basis (standard, actual, variable, fully loaded), period, version, and the volume assumption where one applies. A manufacturing data model with a single unit-cost column has already failed. The defensible answers to "what did this cost to make?" include material cost, prime cost, variable cost, standard full absorbed cost, actual full cost for the period, actual job cost, fully loaded cost including selling and distribution, cash cost, and replacement-basis cost. All are simultaneously true for one physical unit. Which one you need depends on the decision.

Every variance needs a drill-down to a transaction. This purchase price variance to these goods receipts; this usage variance to these material issues against these orders. The failure mode is a variance report that balances to the general ledger at the top and cannot be explained at the bottom, which happens whenever the variance is computed in finance from ledger postings while the explanation lives in the execution system, and the two have different keys.

> The join key between a production order in execution and a cost object in finance is the single most important integration in a manufacturing landscape. Everything in this chapter depends on it. Find it on day three, draw it, and find out who maintains it.

Preserve cost component split through multi-level roll-up. It is the only way to explain a cost movement, and in the automotive value chain it is the only way to substantiate a rebate claim [6].

Treat the asset register as a costing master. Asset to cost centre to absorption rate to standard cost to inventory value to reported margin is a live dependency chain. A capitalisation decision, a useful-life revision or an impairment propagates into product cost. Depreciation sits in cost of sales for a manufacturer, so **the depreciation policy is a gross-margin policy** — extending a plant's useful life from 20 to 30 years cuts annual depreciation by a third and raises reported gross margin with no operational change at all. Very few manufacturing data models represent that chain, which is why nobody can explain why unit cost moved after a revaluation.

And put the inventory movement on the profit dashboard. If you remember one thing from this chapter, remember that.

THE CHAPTER IN TEN LINES

- Unit cost is a function of volume, and almost every argument about manufacturing margin descends from that one fact.
- Standard cost is a build-up of material, labour and absorbed overhead, rolled up level by level, and the component split must survive the roll-up or no cost movement can be explained.
- The absorption rate fixes a volume assumption for the whole period, so producing more lowers reported unit cost and raises reported profit with nothing sold.
- Profit and cash move in opposite directions in the short run, so a profit dashboard without inventory movement beside it is actively misleading.
- Standard costing is permitted under IFRS; IAS 2 paragraph 21 says so, provided the standards are regularly reviewed. LIFO is not permitted at all.
- There are five variances an architect will be asked to source, each owned by a different function, and the absorption variance is owned by nobody in the plant.
- Direct labour is substantially not variable in South Africa, so an imported contribution-margin or breakeven model overstates the plant's flexibility.
- A make-to-order business and a make-to-stock business with identical profit and loss accounts have completely different balance sheets.
- A plant that adopts theory of constraints or lean runs two measurement systems permanently, because throughput accounting refuses to absorb fixed cost into units.
- Ownership of inventory is a first-class attribute, and in a tolling plant omitting it is an unrecoverable design error.
- Channel is a dimension on every transaction, because channel often determines profitability more than product does.

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CHAPTER 7

The operating model and the planning cadence

A plant is a physical thing with a timetable. This chapter is both: how the sites, lines and shifts fit together, which operating measures mean anything, and the monthly-to-daily cascade that turns a forecast into a released order — including the two configuration errors you can find in an afternoon.

A plant is a physical thing with a timetable. The physical thing is sites, lines, cells and stations, crewed on a shift pattern. The timetable is a cascade from an eighteen-month view down to what runs next on press four. This chapter is both — the operating model, the measures people argue about, and the planning cadence that turns a forecast into a released order — and it ends with the most useful thing you can produce in your first month, which is not a dashboard.

The operating model

Sites, plants, lines, cells and stations

A **site** is a piece of land with a gate, a licence and an address. A **plant** is a manufacturing entity inside it — and note at once that "plant" in an ERP is a legal and organisational construct, not a building, so one site can hold three plants and one plant can span two buildings. Below the plant: an **area**, then a **line** or **work centre**, then a **cell** in cellular layouts, then a **station** or machine, then the individual **tool**, **mould** or **fixture**, which is often the real constraint and is usually untracked.

The object the systems turn on is the **work centre**: where work happens. In an ERP it carries available capacity from shifts and operating time, formulas that turn the routing's standard times into a capacity requirement and a cost, a cost centre and activity types, queue and move times, and — in principle — a link to the equipment and functional location in the maintenance module [7]. That link is what lets a maintenance outage become a capacity reduction in the plan, and almost nobody populates it.

> Be careful with "work centre" as a word. ISA-95 uses work centre and work unit as generic terms in an equipment model that is not a stack of levels, and whose familiar lower terms are ISA-88's rather than its own [8]. So the phrase means related but different things in the control-systems standard and in the ERP. Chapter 13 does this properly; for now, record which sense each document is using.

Two trees over the same steel

The one structural point to take from the operating model is this: there are **two hierarchies over the same physical equipment, and they are different trees.**

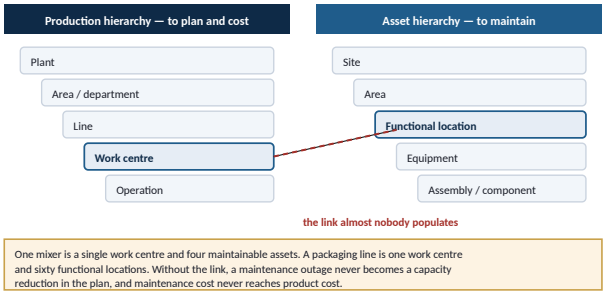


Figure 7.1 Two hierarchies over the same physical equipment. Reconciling them is a project, and until it is done maintenance cost cannot be attributed to product cost.

The **production hierarchy** exists to plan and cost work — plant, area, line, work centre, operation — and is organised by where capacity is consumed. The **asset hierarchy** exists to maintain equipment — site, area, functional location, equipment, assembly, component — and is organised by where failure happens. A mixer is one work centre and four maintainable assets.

Reconciling the two is a project, not a mapping exercise, and the reason is concrete: every work order, every downtime event, every spare part and every maintenance cost attaches to a node in the asset hierarchy, and unless that hierarchy aligns to the cost-centre structure, maintenance cost cannot reach product cost. The asset hierarchy is the most valuable and most neglected master data object in a manufacturing estate. If you find it unstable — equipment hanging off the wrong parent, functional locations invented per project — you have found a multi-year data problem on day one, and you should say so before anybody scopes a reliability programme on top of it.

Shifts, crews, and what a pattern costs

When somebody says "just add a shift", this is the arithmetic.

Pattern	Crews needed	Asset availability	What it costs you
Single day shift, Monday to Friday	1	about 2,000 hours a year	Baseline. No shift allowance.
Two-shift, double day	2	roughly double	A shift allowance on the second shift.
Three-shift discontinuous, five days	3	plant down at weekends	Cheaper than continuous, but a weekly shut-down and start-up, which is fatal for process plant.
Continuous or continental, 24/7	**4 or 5**	the asset runs	The honest answer is five once leave, training, sick leave and absence are covered.

That last row is the one most often got wrong. A 24/7 pattern needs four or five crews, not three, because each crew must get rest days, annual leave, training and absence cover. Five is the honest number. Many plants run four and fill the gap with overtime, which costs more than the fifth crew — and that is how a plant ends up with *structural* overtime: not a response to a peak but a permanent feature of the crewing model, invisible as a staffing decision because it arrives as a labour rate variance.

Over and above the basic rate, the stack per employee includes shift and night-shift allowances, Sunday and public-holiday rates, overtime, standby and call-out for trades, bargaining-council provident fund and medical contributions, unemployment insurance, the skills development levy at 1% of payroll, Compensation Fund contributions and leave accrual.

> You will be asked for a fully loaded labour cost per hour. **Do not supply a multiplier.** Any general number would be an invention; this one comes from the client's payroll, read against their own bargaining-council agreement. What you can supply is the structure — crewing ratio, allowances, statutory contributions, leave and absence cover — and let payroll fill it in. An architect who hands over a borrowed multiplier has created a number that will end up in a board paper.

Two South African facts shape all of that. **Wages are set by bargaining councils, not by the firm.** The metal and engineering council's main agreement runs to 30 June 2027, with increases from 1 July 2026 of 5.00% at grade A rising to 6.00% at grade H — an inverted differential, compressing upward rather than narrowing the gap — and the circular as issued applies to the parties only [5]. In the motor industry council, component manufacturers' increases are on *actual* rates of pay rather than on the prescribed minimum [6], which is materially more expensive because it lifts everybody already above scale too.

And **retrenchment is a procedure, not a decision**: dismissal for operational requirements under sections 189 and 189A of the Labour Relations Act requires consultation, disclosure, consideration of alternatives and, at scale, facilitation, with severance. That is why direct labour is not variable in the short run here, and why a manning model imported from North America is wrong in both directions.

> Changes to **work organisation** are consultation matters: shift patterns, manning levels, measurement of individual performance, and the introduction of monitoring technology. A project that introduces operator-level productivity measurement without consultation will be stopped, and stopped late — after the hardware is installed. Put the consultation on the plan as a dependency with a duration, not as a communications work stream.

Centralised, or plant-autonomous

Every manufacturing group sits somewhere on a spectrum, and where it sits determines what your master data problem actually is.

In a **centralised** model, group functions own the item master, the costing method, the planning parameters, the measure definitions and the reporting hierarchy. Plants execute. The master data is consistent and frequently wrong in the specifics, because a group steward who has never seen the Gqeberha canning line sets its queue times from a template.

In a **plant-autonomous** model each plant owns its own parameters, and often its own instance: the parameters are right and nothing is comparable. Karoo Industrial Group, the invented group this book uses throughout, is at that end — two ERP instances with no shared item master, so the same ingredient exists twice at two standard costs.

The position worth arguing for is **governed where it must be comparable, local where it must be accurate**, decided attribute by attribute. A workable split: item identity, classification, unit-of-measure conversions, costing method, measure definitions and the reporting hierarchy are group-governed; planning parameters, lead times, safety stock, lot sizes, routings, standard times and work-centre capacity are plant-owned. That split works because the item master in manufacturing is not one record but a matrix, with a plant-specific view of planning, scheduling and valuation for every plant that uses the item [7].

Multi-plant, multi-company and the transfer price

Most manufacturing groups move material between plants, and often between legal entities. Three things follow.

Intercompany transfer is a sale. When plant A in company X ships a semi-finished item to plant B in company Y, A books revenue and B books a purchase at a **transfer price**. That price is neither a market price nor a cost; it is set, and whoever sets it decides where the group's margin appears. Shift it and plant A looks profitable while plant B looks marginal, with no physical change.

So consolidated manufacturing margin is not the sum of plant margins. Group figures net out intercompany transfers; plant figures do not. Any reporting layer that aggregates plant margin upward without an elimination step is wrong by the whole intercompany flow, and wrong in a way that looks plausible.

And the transfer price reaches beyond reporting. It is a tax position in a cross-border group, a customs valuation on an import, and the basis on which a plant manager is judged. Ask early and in writing who sets it, on what basis, how often it is reviewed, and whether the plant scorecard uses it. If nobody knows, you have found why the plants argue about profitability.

The operating measures

Overall equipment effectiveness

The canonical formula is **Availability × Performance × Quality** [1]: availability is run time over planned production time, performance is ideal cycle time times total parts over run time, and quality is good parts over total parts. In process industries performance is restated as output rate over ideal run rate and quality as good material over all material [1]. Equivalently and more usefully, OEE is **fully productive time divided by planned production time**: of the time we intended to make good product at full rate, what fraction did we achieve?

The losses map to the three factors as the **six big losses** [1]: equipment failure, and setup and adjustment, hit availability; idling, minor stops and reduced speed hit performance; process defects and start-up rejects hit quality.

You can learn that in five minutes. The useful part is why the number cannot be compared.

Why OEE is not comparable, stated hard

There is no OEE standard. The site publishing the most-used definition states plainly that there is "not really" an international standard, though the calculation is a de facto one [1]; the general reference treatment cites a measurement-definitions standard and a German engineering guideline but notes they are not standardised across industries [2].

The numerator is uncontested; the denominator is not. What you divide run time by is a choice, and the research behind this book found seven conventions in use.

What do you divide run time by?	Effect on the number
1. All calendar time, 24 × 7	lowest; this is TEEP, not OEE
2. Scheduled or shift time	the most common convention
3. Scheduled time less planned maintenance	flatters maintenance-heavy assets
4. Also less breaks and meal times	worth 5 to 10% of a shift
5. Also less time with no customer demand	makes a loss a commercial judgement
6. Also less externally caused stoppages	supply, strike, upstream trip
7. The micro-stop threshold, 1 to 5 minutes a stop below it counts under effectiveness, not availability	**moves loss between A and P without changing OEE**

No OEE standard exists. The measure is also documented as unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level. Compute it on the constraint.

Figure 7.2 Seven availability conventions found in use, with no standard to choose between them. The last one changes the component split without changing the product. Treat any cross-plant availability comparison as unsafe until the convention is declared.

Read that figure as a design problem. Two physically identical plants in one group can report availability thirty points apart, neither of them lying. And **the seventh divergence is the subtle one**: downtime shorter than a company-set threshold, typically one to five minutes, is conventionally counted under performance rather than availability [2], so two plants with identical behaviour and different thresholds report different availability and different performance **with the same OEE.**

> OEE can be comparable where its components are not. The consequence is direct: if a group dashboard shows availability, performance and quality separately by plant and the micro-stop thresholds have not been harmonised, those columns are not a comparison and should not be drawn side by side. Harmonise the threshold or stop publishing the split.

OEE is explicitly documented as unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level [2]. That is not a cautious practitioner's opinion; it is in the documented criticism of the measure. The reason is arithmetic: a line is a serial system, so its output is governed by its constraint, and a line's OEE is therefore not the average of its stations'. Compute OEE **on the constraint**, or **on the line treated as one resource**, and say which you did.

Aggregating further is worse: whether plant OEE should be weighted by time or by value added is an open practitioner question the published guidance does not settle [1]. So any group OEE dashboard — and you will be asked to build one — is on contested ground before you write a line of code. You can still build it. You cannot build it without declaring the weighting and owning the choice.

> **The 85% benchmark is an invention, and the site that publishes it says so**, calling it "a convenient, compelling and completely artificial benchmark" and recommending incremental targets instead [1]. The general reference treatment says treating 85% as the mark of excellence is in many cases incorrect [2]. The figure is usually traced to Nakajima's total productive maintenance work via component values of 90% availability, 95% performance and 99% quality, but neither available source carries those values, so treat the attribution as commonly reported and not something I could source. **Never put 85% in a client deliverable as a benchmark.** It will be quoted back as a target, and a plant will then be measured against a number somebody made up.

> One terminology correction: the relevant international standard's middle term is **effectiveness**, not performance, so an author writing "performance rate" is implementing the Nakajima formula while citing the standard. Secondary summaries routinely use "performance rate" for it, so confirm the wording against the licensed text before you put it in a measure register. Its current published edition dates from 2014; a second edition was still at draft-international-standard stage as at October 2026 [3].

Yield, first-pass yield and rolled throughput yield

Three measures, routinely conflated, and the difference is where the hidden factory lives. **Yield** is good units out over units started, and it permits rework: a unit that failed, was fixed and passed counts as good. **First-pass yield** is units passing *without any rework* over units started, so it is always less than or equal to yield. **Rolled throughput yield** is the product of every step's first-pass yield: a ten-step process at 98% per step has a rolled throughput yield of 0.98 to the tenth power, about **81.7%** — a number most plants would not recognise as their own, and the honest measure of how much rework capacity is being silently consumed.

Then distinguish **planned scrap** — in the bill of material as a scrap or shrinkage factor, so already in the standard cost — from **unplanned scrap**, which is a usage variance; and **start-up scrap**, the predictable loss at the beginning of a run or after a changeover, from **running scrap**. Start-up scrap is a function of the *number of runs*, not of volume, so shortening batches increases it. That is the counterweight to every batch-size reduction proposal and it belongs in the business case.

Process yield is a different animal: output mass over input mass, where the loss is evaporation, purge, flare, dust, slag or conversion inefficiency. It is bounded by chemistry, not workmanship, and a **theoretical yield** exists. Distance from theoretical yield is the real performance measure, and one of the few genuinely comparable numbers in manufacturing.

> A plant quoting 98% yield and a plant quoting 92% first-pass yield may be the same plant. A yield number without the stage it was measured at, the defect definition applied, and whether rework counts, is not a measurement. Always ask: started where, good according to whom, and does rework count?

Takt, cycle, lead time, and Little's Law

These four are confused more often than any other set of terms in manufacturing, including by practitioners.

Takt time is available production time divided by customer demand in that time. It is a demand-derived target rhythm and not a measurement of the plant at all. If a line has 27,000 productive seconds in a shift and demand is 450 units a shift, takt is 60 seconds. **Takt changes when demand changes, without the plant changing** — which is why takt is not a performance measure and should never appear on a performance dashboard.

Cycle time has three uses. Operation cycle time is the time for one operation on one unit. Line cycle time is the interval at which completed units leave the line, which is one over the output rate and the figure you compare against takt. Total cycle time, or throughput time, is the elapsed time for one unit to traverse the whole process including waiting — which is why the term is dangerous, being used interchangeably with lead time.

Lead time has four uses. Manufacturing lead time is order release to order completion. Quoted lead time is order placement to customer receipt. **Cumulative lead time** is the longest path through the bill of material from the lowest-level purchased part to the finished item — the real constraint on responsiveness in a make-to-order business, and the number nobody calculates. Replenishment lead time for a purchased item is order to goods-receipt-and-available, longer than the supplier's quoted transit because it includes receiving, inspection and put-away, and here it includes the port.

Throughput is good units or value per unit time — except in theory-of-constraints usage, where it means revenue minus totally variable cost. Both live in the same room; establish which is meant before you specify anything.

Little's Law ties the set together: work in progress equals throughput times cycle time, so cycle time equals work in progress divided by throughput. That is the arithmetic behind lean's obsession with reducing work in progress and the most useful formula in this chapter — at constant throughput, halving work in progress halves lead time — and it is why work-in-progress days on a balance sheet *are* the manufacturing lead time in disguise.

The four capacities

Four numbers, one word, and only one belongs in a plan.

- **Theoretical or nameplate capacity.** Output at design rate, around the clock, with no stoppages, changeovers or quality loss. A fiction, used by equipment vendors and in press releases.
- **Rated or installed capacity.** Theoretical, adjusted for a declared operating pattern and maintenance allowance. "Rated at 140,000 units a year" — as Toyota's Prospecton plant is stated to reach under its ninth-generation Hilux programme, due for completion June 2027 [4]. **A rated figure is meaningless without its assumed shift pattern, and the assumption is almost always omitted.**
- **Effective capacity.** Rated, less planned maintenance, changeovers and planned non-productive time.
- **Demonstrated or proven capacity.** The highest sustained output actually achieved over a representative period, at acceptable quality, **in the current product mix.** The only capacity figure that should enter a plan, and it changes when the mix changes.

Utilisation is actual output over capacity, interpretable only if you say which capacity: the same plant in the same month can report figures 25 percentage points apart depending on the denominator.

> South Africa's official capacity series is a specific trap. Stats SA's release P3043 put large manufacturers' use of production capacity at **76,6% for the May 2026 reference month**, down from 77,8% a year earlier, food and beverages highest at 81,3% [9]. It covers **large enterprises only**, it is **self-reported** against each respondent's own notion of capacity, and it is therefore **not comparable to a company's internal figure.** And high utilisation with falling output means demand-constrained, not capacity-constrained — which is what food and beverages showed in 2026.

On-time-in-full, and its four date bases

On-time-in-full is the customer-facing measure and the most frequently mis-specified metric in manufacturing architecture. Each of the following is a live design choice.

Which date? Four defensible answers. The **customer's request date**, hardest and the only one the customer recognises. The **first confirmed date**, fairer to the plant, and it lets a bad promise count as a success. The **latest confirmed date**, which can reach 100% while the customer is furious, because you re-promise until you are right. Or the **schedule-of-record date** in a customer-managed schedule — a vehicle assembler's release, a retailer's order window — which changes weekly, upstream, outside your control.

What does on time tolerate? Exactly on the day; a window, commonly minus nothing plus two days; a delivery slot, measured in minutes in sequenced automotive supply; or the customer's booking window at a distribution centre. Note that **early is frequently a failure**: an early delivery to a retailer's distribution centre or to an assembly line is rejected.

What does in full mean? Order-level, where all lines must be complete or the whole order fails, or line-level, where each line is scored separately. Order-level is arithmetically lower and falls as orders get bigger, so the trend can decline purely because order sizes grew. **And where does the clock stop?** Goods issue, carrier collection, arrival at the customer's gate, or receipt booked into the customer's system — which adds their receiving delays to your measure.

And whose measurement wins? The customer's. Large retailers and vehicle assemblers publish supplier scorecards with their own definition, their own data and penalties attached, and will not accept yours. So a manufacturer needs its own measure *and* each major customer's, reconciled — a permanent function, not a project. At Highveld two definitions are in use for the same deliveries and nobody owns the difference.

> **Systems that overwrite the promised date on reschedule destroy the ability to measure on-time delivery at all.** This is extremely common and checkable in an hour: take an order line rescheduled twice and ask the system what was promised the first time. If the answer is gone, the measure is unrecoverable for every historic order, and no reporting project can fix it — only a data model change, applied forward.
>
> An order line needs, retained and never overwritten: request date, first promise date, current promise date, despatch date, delivery date, quantity ordered, confirmed, despatched and received, and a reason code for any variance.

So the first deliverable is a measure definition register

Pull the section together. OEE is not comparable across plants and its components are less comparable than it is. Yield depends on a stage boundary and a defect definition, cycle time on stage boundaries, lead time on what is included, inventory turns on vertical integration and tolling, on-time-in-full on four separate conventions, and cost per unit on the absorption base and the volume assumption. The only genuinely reliable cross-plant measures are physical output and energy per tonne — the two that dashboards neglect.

> When you are asked to build a group manufacturing dashboard, **the first deliverable is not a dashboard. It is a measure definition register**: one row per measure, with an owner, a formula, a denominator convention, the reason-code hierarchy it depends on, the systems it is computed from, and a **declared comparability scope** — within a line, within a plant, across plants, or not at all.
>
> That is the central claim of this chapter. Ship the dashboard first and it will be rebuilt, because the first time two plants' numbers meet in a board pack somebody will find that the definitions differ, and the dashboard will lose its credibility in one meeting and never recover it. The register takes two workshops per measure family and is the cheapest insurance in manufacturing architecture.

The planning cadence

Sales and operations planning

Sales and operations planning is a monthly, cross-functional, management-level process that reconciles demand and supply over a rolling horizon — typically 18 to 24 months — at the level of product family rather than individual item, producing **one agreed plan** that sales, operations and finance all work to.

Its purpose is not a better forecast. Its purpose is to force a single set of numbers and surface the decisions the gap between demand and capability requires, early enough to act. **Integrated business planning** is the extended version, including the financial plan, new product introduction and strategic initiatives; in practice the distinction is whether finance participates or observes.

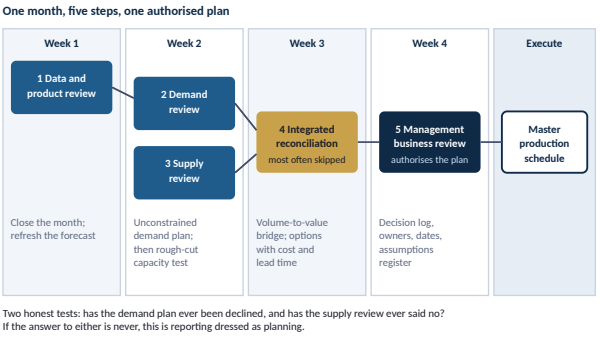


Figure 7.3 The monthly sales and operations planning cycle, with the step most often skipped marked. Skip step four and the process becomes two parallel meetings that never meet.

The five steps, and where each number actually comes from.

Step one, data gathering and product review, week one. Close the prior month: actuals by family for sales, production, inventory, forecast error and bias, and service level. Refresh the statistical forecast. Review introductions and discontinuations with their ramp-up and run-out plans. *The numbers come from* the ERP and the data warehouse.

Step two, demand review, week two. Owned by the commercial organisation. Produces an **unconstrained demand plan** — what the market will take if supply is not a limitation — deliberately unconstrained, because constraining it here hides the gap. *The numbers come from* the statistical baseline, sales intelligence, customer plans, promotions and pricing, plus customer schedules arriving by EDI from vehicle assemblers and retailers. *The failure mode:* the review becomes a negotiation about the budget. **If the demand plan is required to equal the budget, it is not a demand plan.**

Step three, supply review, weeks two to three. Owned by operations. Tests the demand plan against capacity, material, labour and logistics across the horizon using rough-cut capacity planning, and produces a **constrained supply plan**, the gaps, and the options to close them with cost and lead time: overtime, an additional shift, outsourcing, tolling, capital, inventory pre-build. *The numbers come from* demonstrated capacity by resource, routings, the maintenance plan, the shutdown calendar, the manning plan, supplier capability and inbound lead times. *The failure mode:* the supply review says yes to everything, because saying no is career-limiting. **A supply review that has never declined a demand plan is not functioning.**

Step four, integrated reconciliation, week three. The working session where the gaps close: demand against supply, volume against value, plan against financial commitment. **This is the step most often skipped, and skipping it is why the process decays into two parallel meetings.** The hard part is the **volume-to-value bridge**, which needs a consistent costing basis and a price assumption per family.

Step five, management business review, week four. The executive meeting reviews performance and the reconciled plan, takes the escalated decisions and **authorises** the plan. One numbered plan, and a decision log.

What makes it real rather than theatre is checkable in an afternoon. One set of numbers, published and versioned, with last month's plan alongside. An **assumptions register**, because when the plan misses you need to know whether the assumption failed or the execution did. Performance measured against the plan that was authorised rather than a restated one. And decisions recorded with owners and dates.

> Sales and operations planning is a **plan-versioning and scenario problem**, not a reporting problem. The requirements: multiple plan versions coexisting — statistical, demand, constrained supply, authorised, budget — at a family and period grain; a volume-to-value bridge; scenario comparison; the assumptions register as structured data rather than a slide; and an audit trail of who changed what. Almost no ERP does this, which is why the process lives in a planning tool or a spreadsheet and why the integration back to execution is where it breaks. At Karoo the cycle runs in spreadsheets and the authorised plan is re-keyed into the master schedule by hand, so the plan that executes is not provably the plan that was authorised.

From the plan to a released order

The master production schedule states what the plant will build, by item and period, over the medium horizon. It sits between the aggregate plan and material requirements planning, and it is where a family-level plan is disaggregated into specific items — finished goods, or in assemble-to-order the module or planning-bill level — by week and sometimes by day.

It is a decision, not a calculation. Material requirements planning calculates; the master schedule is chosen. A master scheduler can build ahead, level, shorten a campaign or decline. And **available-to-promise** is the master schedule plus inventory minus committed orders — so master scheduling is the mechanism by which the sales promise is made, which the name does not suggest.

> An unvalidated master production schedule causes more manufacturing-system failure than any technical defect. A schedule exceeding demonstrated capacity propagates infeasibility into material requirements planning, which orders material for production that will not happen, which produces excess inventory and a planner who no longer trusts the system. The validation step is rough-cut capacity planning, it takes a day a month, and it is the first thing dropped when the master scheduler is doing three other jobs.

Material requirements planning is the calculation: explode the master schedule through the bill of material, net against on-hand stock and scheduled receipts, apply lot-sizing, offset by lead time, and generate planned orders plus exception messages — expedite, de-expedite, reschedule, cancel. This is where the distinction between **independent demand** (a forecast or an order for a finished item, arising outside the system) and **dependent demand** (a component requirement, derived arithmetically from its parents) does its work. Forecast independent demand; never forecast dependent demand, calculate it. A plant forecasting at component level is a plant whose bills of material are not trusted, and that is the problem to fix.

What it needs: bill-of-material accuracy, because a 1% component error compounds through a multi-level structure; lead-time, on-hand and open-order accuracy; and a feasible master schedule. Industry lore puts the required inventory-record accuracy in the high nineties, but the specific thresholds usually quoted are not something I could source — treat them as folklore with a sound instinct behind them.

What it produces when the inputs are wrong: thousands of exception messages that nobody actions, which planners then ignore wholesale. **That is the universal observed failure state of material requirements planning**, and you will walk into it. The tell is a planner who works from a spreadsheet and runs the planning job only because month-end requires it.

Lot-sizing rules you will meet: lot-for-lot, fixed order quantity, economic order quantity, period order quantity, fixed period requirements, least unit cost. The rule chosen per item drives inventory, and the rules are usually set once at implementation and never revisited. Reviewing them is one of the better early wins available to you.

Material requirements planning II adds capacity, shop-floor control, purchasing and a financial feedback loop. Its contribution is the **closed loop**: the plan is tested for feasibility and the result feeds back. Every modern manufacturing ERP descends from it.

The planning bill is invisible until it fails. It is a pseudo bill of material whose components are real items and whose quantities are **percentages representing expected option take-rates**, summing to 100% or deliberately more. A family forecast of 1,000 vehicles exploded against a bill specifying 60% diesel and 40% petrol plans 600 diesel engines and 400 petrol. It exists because in assemble-to-order you cannot forecast a configuration — there may be millions — but you can forecast family volume and option mix.

It goes wrong in four ways. The percentages go stale, because take-rates drift continuously and the bill is updated annually, so you are systematically short of the option that is winning. Over-planning above 100% is correct practice and must be a conscious decision rather than an accident. Options are not independent — high trim correlates with automatic transmission, diesel with the towing pack — so a bill treating each option independently mis-plans the combinations, and the correlation matrix is the missing data nobody holds. And it is master data owned by nobody: marketing owns the mix forecast, planning owns the bill, engineering owns the options. Assigning the owner is a one-line recommendation with a large effect.

A checkable test: compare the last twelve months' actual take-rates against the current planning bill. The gap answers the question everybody asks, which is why they always run out of that one variant.

Rough-cut, detailed, and finite

Four capacity tools at four cadences, and the commonest error is using one at the wrong cadence.

- **Rough-cut capacity planning** tests the master schedule against a handful of **critical resources** — the bottleneck, the constrained skill, a key supplier — using a bill of resources per family. Coarse, fast, right for a monthly conversation and wrong for a daily one.
- **Capacity requirements planning** takes the planned orders, applies routings and computes load by work centre by period. It is **infinite capacity**: it shows the overload rather than resolving it. A diagnosis, not a schedule.
- **Infinite scheduling** plans against lead times and ignores capacity. All classical material requirements planning is infinite-capacity, which is why it will plan 400 hours of work into a 120-hour week.
- **Finite capacity scheduling**, sold as advanced planning and scheduling, sequences work against finite capacity, honouring resource calendars, sequence-dependent changeovers, tooling and mould availability, skills, material and several objectives at once. **Capable-to-promise** is the order-entry use of the same engine.

Why finite-scheduling projects fail, consistently: the constraint model is incomplete — the real constraint was the one tool, or the one setter, or the effluent permit. The optimiser then produces schedules infeasible for reasons it cannot see, the scheduler reverts to the spreadsheet within a week, and the spreadsheet becomes the system of record.

> Before specifying a scheduling tool, **enumerate the constraints the current scheduler actually honours — by watching them schedule, not by asking them.** Every constraint they apply from memory is a requirement, and the ones they cannot articulate are the ones that will kill the implementation.

There is also an alternative vendors will not propose: control material *release* at the constraint's rate rather than sequencing everything. Theory of constraints calls it drum-buffer-rope. It is release control rather than dispatching, far cheaper than finite scheduling, and often more dependable. Put it on the options list.

Time fences, and why nobody respects them

Time fences are dates in the planning horizon at which the rules about who may change the plan, and how, change.

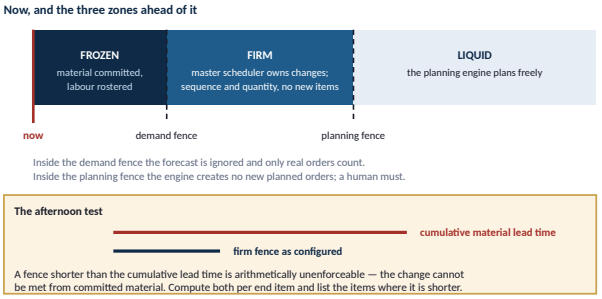


Figure 7.4 The planning horizon and its fences. The arithmetic test is at the bottom: a fence shorter than the cumulative material lead time cannot be enforced, whatever the policy says.

Two distinct fences matter and are routinely confused. Inside the **demand time fence** the forecast is ignored and only actual orders count, because the forecast period has passed. Inside the **planning time fence** the engine creates no new planned orders automatically; a human must.

Six reasons nobody respects them, and the list matters because a project that assumes they will be respected will fail.

1. **A customer with power does not recognise your fence.** An assembler's release, a retailer's promotional uplift or an export vessel booking will change the frozen week, and the commercial answer is yes.
2. **The fence protects the plant from sales, so sales do not own it and it is not defended.** It needs someone senior enough to say no to revenue, and usually nobody is.
3. **Material and quality reality break it from inside.** A failed batch, a late container, a machine failure or a quality hold forces a resequence regardless.
4. **The fence lengths came from an implementation template** and have never been validated against actual lead times and changeover constraints.
5. **Changing the plan is free in the system and expensive in the plant.** The cost of a frozen-zone change — lost changeover, scrapped setup material, overtime, expedited freight — is almost never attributed to the change or to the person who made it.
6. **The schedule is re-promised rather than defended**, which makes on-time-in-full against the latest promise look fine while the customer experiences chaos.

Three of those are architecture findings rather than observations, and they are the ones to take into a client.

> **A time fence shorter than the cumulative material lead time is arithmetically unenforceable.** If the longest path through the bill of material to a purchased component is eleven weeks and the firm fence is four, a change inside the fence cannot be met from committed material, so the fence is decorative. This is the commonest configuration error in the area and it is **checkable in an afternoon**: compute cumulative lead time per end item, compare it to the fence settings, and list the items where the fence is shorter. That list is a finding with arithmetic behind it, which no planner will argue with.

> **Updating planning lead times from observed performance creates a positive feedback loop.** A longer observed lead time gives a longer planned lead time, which releases work earlier, which raises work in progress, which by Little's Law lengthens the actual lead time, which is observed and fed back. The loop is real, common, and almost never diagnosed — partly because every step in it looks like good practice. A client with a scheduled job that refreshes planned lead times from actuals has a mechanism quietly degrading their responsiveness. Break it or damp it; do not let it run unattended.

And the third: **make the cost of a frozen-zone change visible and fence violations fall.** That is a reporting requirement with a behavioural purpose, and one of the highest-value, lowest-cost things an architect can specify. It needs a reason code on the change, the person who requested it, and a standing cost per changeover from the cost centre.

The data requirement that follows: **retain every version of the schedule and every promise date**, stamp every change with who, when and why, and report **schedule adherence** — did we build what we said, in the sequence we said — separately from **schedule attainment**, which is whether we built the volume. Plants conflate them constantly. A plant can hit 100% attainment while adhering to nothing, by building the easy items and deferring the hard ones.

Push and pull

Two control logics

Material requirements planning is push. Explode independent demand through the bill of material, net against on-hand and on-order, offset by lead time, generate planned orders — all in advance, from a forecast. Its characteristic failure is **nervousness**: a small forecast change at the top explodes into large oscillating changes in planned orders four levels down. And its core assumption, that lead times are known constants, is circular, because lead times are queue-dependent and therefore load-dependent.

Kanban is pull. Downstream consumption authorises upstream replenishment: a card, a container, an empty square or an electronic signal is released when material is consumed, the upstream process makes exactly that, and nothing is made without a signal. It needs four things — container quantity, cards in the loop, replenishment lead time, and reasonably level demand — and cards times container quantity is the maximum work in progress. **That is the whole control system**, which is why it is cheap and why it is brittle. Its failure is a silent stock-out on a demand spike, and a card population nobody resizes as demand drifts.

Five genuine incompatibilities

They are not two styles of the same thing. They make conflicting demands on the same data.

1. **Direction of causality.** Push decides in advance from forecast; pull decides after the fact from consumption. A material cannot be both planned-and-ordered and consumption-triggered without double-counting demand.
2. **Order existence.** Push needs a production order as the carrier of plan, cost and confirmation. Pure pull has no order: material is made against a signal and consumption is backflushed. The **cost object differs** — order-based against period-and-rate-based.
3. **Inventory as control against inventory as waste.** Push's safety stock is a parameter to be optimised; pull's work-in-progress cap is a limit to be lowered. The same number, opposite intent, and two teams correcting it in opposite directions indefinitely.
4. **Lead time.** Push treats it as input master data; pull treats it as an *output* of work in progress and throughput. That is the feedback loop above in its underlying form.
5. **Latency tolerance.** Push runs nightly and tolerates stale data; pull needs the signal in minutes. **Running a pull loop off an overnight extract does not work, and is attempted constantly.**

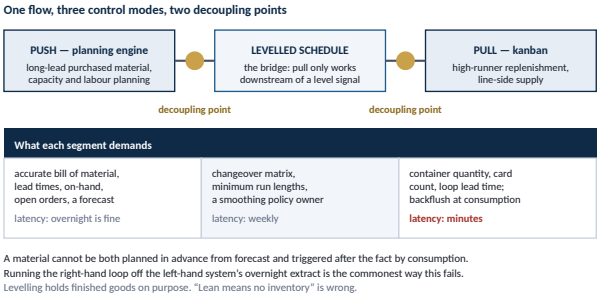


Figure 7.5 The control-mode map, which is the deliverable. Decoupling points are decisions, and each segment has its own data accuracy and latency requirement.

What plants actually do, and what to deliver

Almost every real plant is a hybrid, and the hybrid is legitimate: push for purchased long-lead material and for planning capacity and labour over the medium horizon; pull for high-runner internal replenishment and line-side supply; and a **levelled master schedule** as the bridge, because pull only works downstream of a level signal. Highveld runs three control modes against the same press capacity — sequenced releases from one assembler, statistical make-to-stock for aftermarket, quoted make-to-order for industrial customers — and the conflict between them is its central planning problem.

> **The deliverable here is not a tool choice. It is a control-mode map per material segment**: for each material and each segment of the flow, which control mode governs it, where the decoupling points are, and therefore which data must be accurate to what latency. A plant without one has three control modes fighting over the same item master, and every parameter change made to serve one degrades the other two.

And lean does not mean no inventory

Heijunka — levelling — means producing the same quantity and mix every period, absorbing demand variability into a small finished-goods buffer instead of transmitting it into the plant and its supply chain. In the limit it becomes mixed-model sequencing: rather than 300 A then 200 B, the repeating sequence A-A-A-B-B.

Note what that means. **Heijunka deliberately holds finished-goods inventory.** "Lean means no inventory" is wrong, and wrong in a way that matters, because a client who believes it will reject the one intervention most likely to stabilise their plant. Lean's claim is that unevenness causes overburden which causes waste, and that the finished goods needed to level are *less* than the capacity and work in progress needed to chase demand.

Levelling demands two things: a **sequencing capability at master-schedule level** rather than a dispatch list, able to produce a repeating mixed-model pattern honouring the changeover matrix, minimum run lengths and the downstream pull loops; and a **demand-smoothing policy with an explicit owner**, because somebody must be allowed to say "sales asked for 1,400 this week, we are making 1,000 and drawing 400 from stock." Levelling breaks on seasonal demand, engineer-to-order, and promotional volumes of five times base for two weeks. There, level the base and chase the peak, and say in the plan which is which.

Confirmation and backflush

Between the schedule and the ledger sits the confirmation: the report that work happened. Four options.

Per-operation confirmation has the operator confirm quantity good, quantity scrapped and time at every operation: most accurate, highest burden, and it degrades under production pressure, so the data is worst exactly when the plant is busiest. **Milestone confirmation** confirms only key operations and backflushes the rest — the usual answer. **Backflushing** consumes standard bill-of-material quantities and standard times automatically on reporting the finished quantity. **Automatic confirmation from a machine counter** is most accurate for quantity, blind to reason, and needs reconciling against the manual scrap record.

Backflushing systematically hides material usage variance, and the reason is definitional rather than accidental: it consumes *standard* quantities by construction. So an over-issue, a substitution or unreported scrap cannot appear as a usage variance, because the system has already decided what was consumed. It surfaces months later as a **stock-count difference** instead.

If a client has unexplained inventory shrinkage and backflushes, you have found the mechanism, and you can confirm it in a day by checking whether the stock adjustment accounts are dominated by backflushed components. And where substitution occurs, backflush produces a **confidently false genealogy** — the system records component X consumed when component Y physically was — which is worse than a gap, because a gap is visible and a false record is not.

The architectural point is larger than the defect. **The confirmation strategy determines which variances are visible, and therefore what the business can manage.** It is always presented as a usability decision — how much typing do we make the operator do — and it is actually a controllership decision. Make sure whoever signs it off knows that.

The improvement traditions, as system requirements

Clients adopt these as philosophies. Read them as requirements documents: each makes specific demands on the data, and some of those demands conflict.

Tradition	Primary claim	Primary measure	What it demands of the systems	Where it conflicts
Lean	Lead time, not cost, is the primary variable	lead time, first-pass yield, inventory by stage	low-latency status; value-stream rather than cost-centre measurement; pull master data	with absorption costing and unit-cost scorecards
Six Sigma	Variation is the loss, not the average	capability indices, defects per million opportunities	individual measurements with time order preserved; gauge studies; long retention; deep traceability	with lean's line-led, visual, low-system posture
Theory of constraints	Only the constraint matters	throughput per constraint hour, buffer status	capacity load by resource; material release control; buffer colour status	with absorption costing, flatly
Total productive maintenance	Operator care prevents most failure	equipment effectiveness, mean time between failures	a stable asset hierarchy aligned to cost centres; reason-coded downtime; a maintenance system; sensor time series	with production's reluctance to release operator time
SMED	Setup is a variable, not a constant	changeover time, batch size, campaign length	the from-to changeover matrix as master data; sequence-dependent setup in the scheduler	with planning engines that model setup as a constant per item

Two notes on that table, because they are the parts that bite. **Six Sigma asks for something no reporting layer provides**: individual measurements with their time order preserved, plus the measurement system's own error, retained longer than a transactional archive policy allows. If the improvement function asks for a data warehouse while production asks for a line display, both traditions are present and nobody has reconciled them.

And SMED exposes a checkable architectural defect. Changeover time is a matrix — from product by to product — because white to black is fast and black to white needs a full flush, and in food a changeover from an allergen-containing line to an allergen-free one needs validated cleaning. That matrix is master data, large, maintained by process engineering, and **most planning engines model setup as a constant per item and therefore cannot see the benefit a changeover-reduction programme creates.** The plant cuts changeover by 40%, the scheduler plans as though nothing happened, and the improvement never reaches inventory. At Karoo the allergen rules live on a laminated sheet on the wall, which is why the dry plant is scheduled in a spreadsheet.

A tradition's philosophy is not your problem. Its data requirements are.

THE CHAPTER IN TEN LINES

- There are two hierarchies over the same equipment, a production one and an asset one, and reconciling them is a project rather than a mapping.
- A 24/7 continuous pattern needs four or five crews, not three, and a plant running four fills the gap with structural overtime that costs more.
- Never supply a fully loaded labour multiplier; supply the structure and let the client's payroll fill it in.
- There is no OEE standard, seven availability denominators are in use, and the micro-stop threshold moves loss between availability and effectiveness without changing OEE.
- OEE is documented as unsuitable for comparing machines, for benchmarking and for aggregating to plant level; compute it on the constraint or on the line as one resource.
- The 85% benchmark is an invention and the site that publishes it says so, so it must never appear in a client deliverable.
- The first deliverable for a group manufacturing dashboard is a measure definition register with an owner, a formula, a denominator convention and a declared comparability scope.
- Sales and operations planning is a plan-versioning and scenario problem, not a reporting problem, and the reconciliation step is the one most often skipped.
- A time fence shorter than the cumulative material lead time is arithmetically unenforceable, and the check takes an afternoon.
- Updating planned lead times from observed performance creates a positive feedback loop that lengthens lead time, and it is almost never diagnosed.
- Push and pull make genuinely incompatible demands on the same data, so the deliverable is a control-mode map per material segment, not a tool choice.
- Backflushing consumes standard quantities by definition, so it hides usage variance and surfaces it later as inventory shrinkage.

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CHAPTER 8

The process model

Twelve processes run a manufacturer, and the value of modelling them is almost entirely in the seams rather than the boxes. This chapter states the notation exception for the whole volume, teaches the four levels, and draws eight level-2 flows with the systems named and the hand-off failures called out.

Everything in this book that is architecture is modelled in ArchiMate 4. This chapter is the one exception, and it is worth stating the exception once, here, for the whole volume.

The notation exception

Level-2 process models in this book are BPMN 2.0, not ArchiMate [1][2].

The reason is specific rather than stylistic. ArchiMate has no swimlanes. It can express that a role is assigned to a process, and it can show one process triggering another, but it cannot lay a flow out so that the *hand-off* is the visible thing on the page. In a manufacturing process model the hand-off is the whole point. Almost every failure in this chapter happens at a boundary: between engineering and manufacturing engineering, between the execution system and the ledger, between the laboratory and the warehouse, between the maintenance planner and the production manager. A notation that cannot draw the boundary cannot draw the failure.

So: ArchiMate for the architecture — capabilities, application components, services, data objects, technology. BPMN for the modelled flow of work, at level 2 only. And one rule to keep them from drifting apart: **every BPMN lane must correspond to an ArchiMate Role or an application component that exists in the model.** If a lane has no counterpart, either the lane is wrong or your application model is incomplete. That one constraint is what stops a process library and an architecture repository becoming two unrelated collections of pictures.

> Chapter 17 covers modelling manufacturing in ArchiMate properly, including how to attach these process models to the architecture without duplicating them. Do not model the same flow twice in two notations. Model it in BPMN, and reference it.

The four levels, and who reads each

Before you draw anything, decide which level you are drawing. Most wasted modelling effort in manufacturing comes from drawing at the wrong one — usually level 3 when level 1 was asked for.

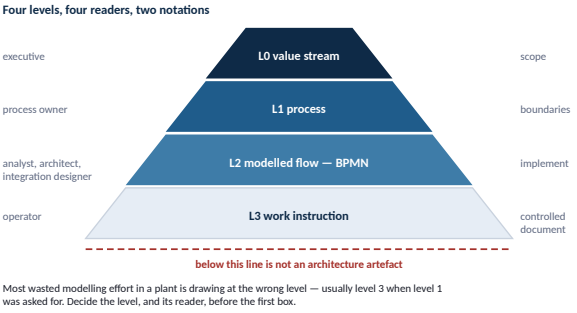


Figure 8.1 The process level pyramid. Decide which level you are drawing, and for whom, before you draw a box.

Level 0, the value stream. One page. Plan, source, make, deliver, service, with the major material and information flows between them. No roles, no systems, no decisions. Its reader is an executive, and its purpose is to agree scope and ownership. Written on a wall, it should let the chief operating officer point and say "that box is where our problem is".

Level 1, the process. One page per process, five to twelve steps, still no swimlanes. Its reader is a process owner. Its purpose is to agree the steps and their sequence, and — critically — to agree the *boundaries*: where this process starts, where it ends, and which other process picks up. Most arguments that look like level-2 arguments are unresolved level-1 boundaries.

Level 2, the modelled flow. BPMN with lanes, with the systems named, with the decision gateways drawn. Its readers are the business analyst, the solution architect and the integration designer, and its purpose is to be *implementable*: this is the artefact from which you derive interfaces, data requirements and test cases. Everything else in this chapter is level 2.

Level 3, the work instruction. How one person does one task at one station, with the screen, the button, the torque setting, the sample size. Its reader is an operator, its owner is manufacturing engineering or quality, and it is a controlled document under the quality scheme with a revision and an approval. **It is not an architecture artefact and you should not be writing it.** Say so clearly when asked, because clients routinely ask for level 2 and then grade it against level 3.

Level	What it shows	Who reads it	Where it lives
0	the value stream on one page	executives	the architecture repository
1	steps and boundaries, no lanes	process owners	the architecture repository
2	lanes, systems, gateways	analysts, architects, integrators	the architecture repository
3	one task at one station	operators	the quality management system, controlled

The conventions this book uses at level 2

A level-2 model is useful only if it is readable, and readability is a matter of restraint. Five conventions, which are worth adopting as a house standard whether or not you adopt these particular ones.

Five lanes, seven columns. If a flow needs more, it is two processes, or it has level-3 detail in it. Split it and draw the boundary between the halves as an explicit start and end event, so that the two models join without ambiguity.

A lane is a role or a system, never a department. "Operations" is not a lane. "Shift supervisor" is. "Production" as a lane name hides the fact that three different people touch the flow, and it is exactly the ambiguity you are modelling to remove.

Name the system in the lane where work happens in a system. A lane called "Meridiem / WMS" is honest about the fact that two systems share a step and that the sharing is unresolved; a lane called "the ERP" when four systems are involved is not.

Model only the decisions that change the path. Every gateway should correspond to a real rule somebody applies, with a data item behind it. A gateway labelled "approved?" with no named approver and no recorded decision is decoration, and it will be implemented as a status field nobody sets.

Draw the exception that happens weekly, not the one that happens never. A model that shows only the happy path is a model of the documented process, not the real one. A model that shows every conceivable exception is unreadable. The test is frequency: if it happens in a normal month, draw it.

Testing a level-2 model for completeness

There is a standards-derived device that makes this easy, and it is the single most useful structuring tool in this part of the subject. ISA-95's third part decomposes the operations layer into four **operations areas** — production, maintenance, quality and inventory — and gives each the same eight-activity pattern: definition management, resource management, detailed scheduling, dispatching, execution management, data collection, tracking, and performance analysis [7]. The earlier functional models published by the manufacturing-execution industry association cover similar ground from a product perspective and are worth knowing for vocabulary, though their own published history records conflicting dates for one of them [8].

Use the eight activities as a checklist against any level-2 model you have drawn. Does the flow say where the *definition* comes from — the recipe, the routing, the control plan — and who maintains it? Does it say how *resources* are confirmed as available and qualified? Does it include *tracking*, so that someone can answer "where is this now"? Does it include *performance analysis*, or does the process simply stop when the work does?

> A capability that does execution and data collection but has no definition management is not a whole process, and the gap will be filled by a spreadsheet. That is the most reliable way to predict where a shadow system will appear, and you can predict it from the model before anyone has bought anything.

The twelve processes a manufacturer runs

Here is the whole set. Eight are drawn in this chapter; the rest are described in prose at the end, because they are either covered elsewhere or are variants of something already drawn.

Process	Grain	Primary systems	Drawn here
Ideate to launch	a product	PLM, CAD, QMS, ERP	in prose
Plan to produce	an order	planning tool, ERP, APS, MES	yes
Make-to-stock replenishment	an item at a location	demand planning, ERP, MES	yes
Order to cash, make to order	an order line	CRM, ERP, APS, WMS, TMS	yes
Procure to pay	a purchase order line	ERP, EDI, WMS, LIMS	yes
Quality hold to disposition	a lot or batch	LIMS, QMS, ERP, MES, WMS	yes
Maintenance work management	a work order	CMMS or EAM, ERP	yes
Engineering change	a change order	PLM, ERP, MES	yes
Recall	a lot and its descendants	MES, ERP, WMS, TMS	yes
Physical inventory and cycle counting	a stock location	ERP, WMS	in prose
Sales and operations planning	a product family	planning tool, ERP	chapter 7
Month-end costing close	a period	ERP	in prose

A note on the worked examples throughout. The lanes and system names below belong to Karoo Industrial Group, an invented South African manufacturing group used across this book: **Meridiem** the group ERP, **Loomhouse** the execution system at the Highveld component plant, **Clearwater** the laboratory system, **Thorn** the product lifecycle system, **Vrystaat** the planning tool, **Baobab** the maintenance system. **None of it is real.** The processes are typical; the company is not a case study.

Plan to produce

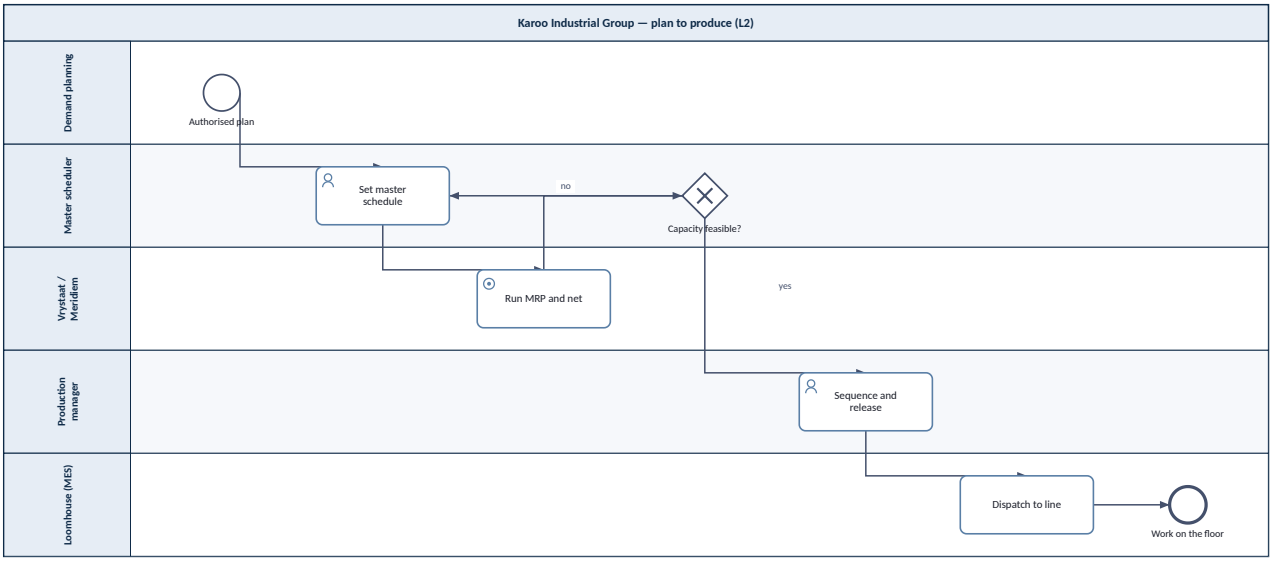


Figure 8.2 Plan to produce, level 2. Release is the moment the plan becomes a commitment with a cost object attached. The seam that fails most often is order change after release.

The flow from an authorised plan to work on the floor. The master scheduler disaggregates the authorised family plan into items and periods. Material requirements planning explodes it through the bills of material, nets against stock and open orders, applies lot sizing, offsets by lead time, and produces planned orders. Capacity is evaluated — either coarsely in the ERP or properly in a scheduling tool — and then planned orders are converted to production orders, availability-checked, and released. Release is the moment that matters: it is the point at which the plan becomes a commitment with a cost object attached.

Two steps between release and execution are routinely left out of models and are both real. **Dispatch** is how the released order reaches the line — as a printed dispatch list, a work queue on a screen, or a sequence that the supervisor re-orders by hand. **Material staging** is the picking and movement of components to line side, which in most plants is a warehouse transaction and in some is an execution-system one, and the boundary between those two is a design decision nobody has written down. Then execution: setup, run, operator interaction, parameter capture and in-process checks, touching the execution system, the control system, the historian and the laboratory. Then confirmation of quantities good and scrapped, labour and machine time, and component consumption. Then goods receipt of finished product into stock at a quality status, disposition, and finally order settlement, where actual costs are collected and the variances of chapter 6 are calculated.

The hand-off failures, in the order you will find them.

- **Scheduling on stale routings.** The scheduling tool consumes routings and standard times from the ERP. If those were last re-timed at commissioning, the schedule is wrong by whatever the plant has learnt since, and the scheduler has already stopped believing it.
- **Order changes after release that do not propagate.** The commonest single defect at this seam. The order is released and downloaded to the execution system; then the quantity changes, or a component is substituted, or the order is cancelled. If the interface was built as a one-way push with no change or cancellation handling, the execution system is running a superseded order and nobody knows.
- **Staged material consumed against the wrong order.** Components are picked to line side for order A and physically used on order B because B was running when the material arrived. The stock ledger and the genealogy are both wrong from that moment.
- **Confirmation aggregated so coarsely that genealogy is lost.** One confirmation for a shift, when the lot changed twice during it.
- **Timing.** The confirmation posts in the next ERP batch window, so stock is wrong overnight — and the night-shift planner works from the wrong numbers.
- **Product shipped before disposition**, because the quality hold is advisory rather than blocking.

Make-to-stock replenishment

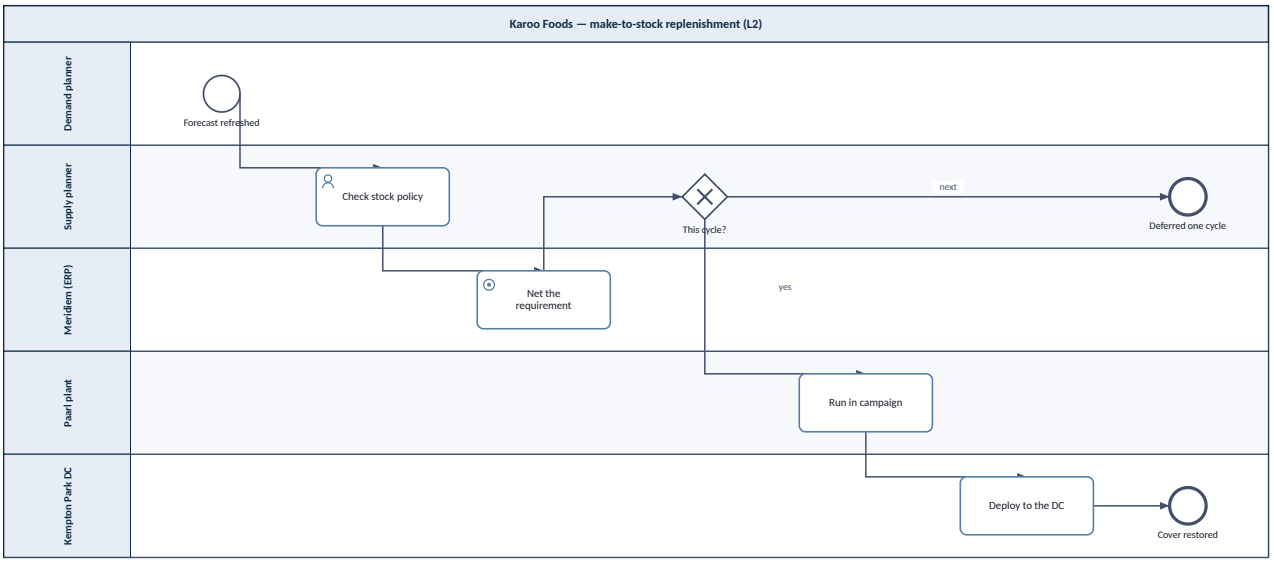


Figure 8.3 Make-to-stock replenishment, level 2. In a process plant the question is not whether to make it but whether to make it this cycle, and the cycle is usually in no system at all.

Shorter, and the trap is elsewhere. Forecast refreshes, inventory policy is checked, the net requirement is computed, and the decision is made to produce. In a process plant the decision is not "do we make it" but "**do we make it this cycle or next**", because the plant runs a repeating sequence of grades, and that changes the inventory arithmetic entirely. A product made once every fourteen days needs at least fourteen days of cover plus safety stock, whatever the forecast says.

Hand-off failures. Forecast produced at a different granularity or on a different calendar from the planning run, so the netting is wrong at the edges — a defect that is invisible in the middle of the horizon and systematic at its boundaries. Inventory policy parameters loaded once at implementation and never reviewed, which is the dominant failure in this process and is worth a week of anybody's time. And the big one: **the campaign cycle is known to the planner and represented in no system**, so the planning engine's proposals are routinely overridden, and once a planner overrides a system weekly they stop reading it at all.

Order to cash in a make-to-order business

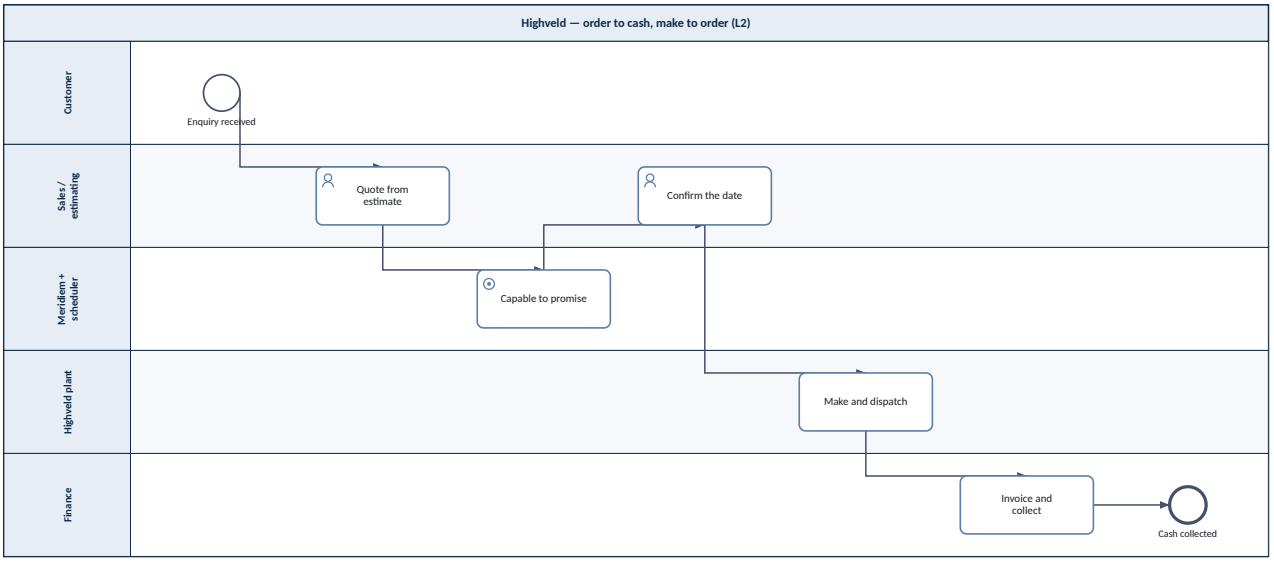


Figure 8.4 Order to cash in a make-to-order business, level 2. Everything turns on where the promise date comes from — a plan, or a capacity check.

The defining property of make-to-order is that the promise is made before the capability to keep it has been checked. So the architecture question is concentrated in one step: what does the promise date come from?

Available-to-promise is computed from the master schedule plus inventory minus committed orders. **Capable-to-promise** asks a finite-capacity engine whether the work can actually be done by then, given everything already committed. The first is arithmetic on a plan; the second is a question about reality. Clients will describe the first and mean the second.

Two steps before the promise deserve attention. **Configuration** resolves the customer's chosen variant to a specific product definition — in assemble-to-order through a configurator and a variant model, in engineer-to-order through design work. And in engineer-to-order proper, **the order's bill of material is created per order**, which is the defining architecture property of the archetype: the product does not exist until the order defines it, so the item master, the bill of material and the routing are all order-scoped objects rather than standing master data. A system designed on the assumption that a product definition precedes an order cannot represent this at all, which is why engineer-to-order businesses so often run their real product structure in the design tool and keep the ERP for invoicing.

Downstream, two more. **Progress visibility to the customer** is the most common unmet requirement in make-to-order, and it is unmet because it needs execution-level status exposed outside the plant. And the shipment carries documentation — packing list, certificate of analysis, certificate of conformity, export paperwork — which is a data requirement, not an administrative one.

Hand-off failures. Quoted against an estimate and built against a different structure, so the margin on the job is unknown until settlement — and by then the quote has been repeated eleven times. Availability checked against a plan rather than a capacity position, so promises are undeliverable at the moment they are made. In engineer-to-order, engineering release late while the plan is never updated, so the whole downstream plan is fiction. And the as-built configuration not recorded against the serial number, so when a warranty call arrives nobody can establish what was actually shipped.

Procure to pay

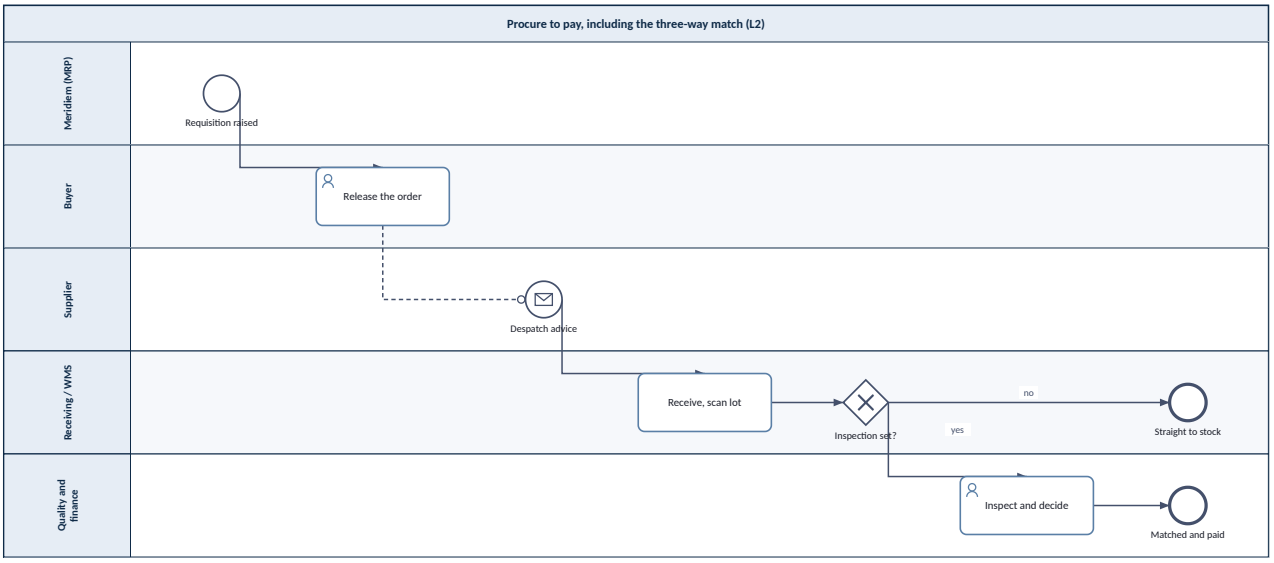


Figure 8.5 Procure to pay, level 2, with the goods-receipt and three-way-match step. The gateway depends on one field on the item master, and when it is unset uninspected material reaches unrestricted stock.

The manufacturing-specific wrinkles are what make this worth drawing. In high-volume supply the purchase order is a frame and the actual call-off is a repeated **delivery schedule** sent by EDI. **Subcontracting** sends your components to a supplier who returns a processed part, which requires a stock-at-supplier model. **Consignment stock** is supplier-owned, on your site, settled on consumption. And **tooling you own but keep at the supplier** is a fixed asset at a third-party location that nobody counts.

The step to look at hardest is **goods receipt and the three-way match**: purchase order, receipt and invoice must agree before payment. It sounds mechanical. It is where units of measure diverge.

Hand-off failures. No despatch advice, so receiving is blind and the lot number is keyed by hand from a printed label. **No inspection lot created, because the material master's inspection type was never set** — so uninspected material goes straight to unrestricted stock and is consumed, which is both a quality failure and an audit finding. A certificate of analysis arriving as an email attachment and filed nowhere, so the receipt has no supporting evidence when an inspector asks. And receipt quantity in a different unit of measure from the invoice, which is the single most common reason a three-way match fails on volume: the order is in tonnes, the delivery note is in bags, the invoice is in kilograms, and the conversion factor is grade-specific. Where the operation was externally processed, add **subcontract settlement** as a separate step, because the supplier is paid for conversion while the material was always yours — and the stock ledger has to have been tracking material at a third party throughout, which at Highveld it does in a spreadsheet.

Quality hold to disposition

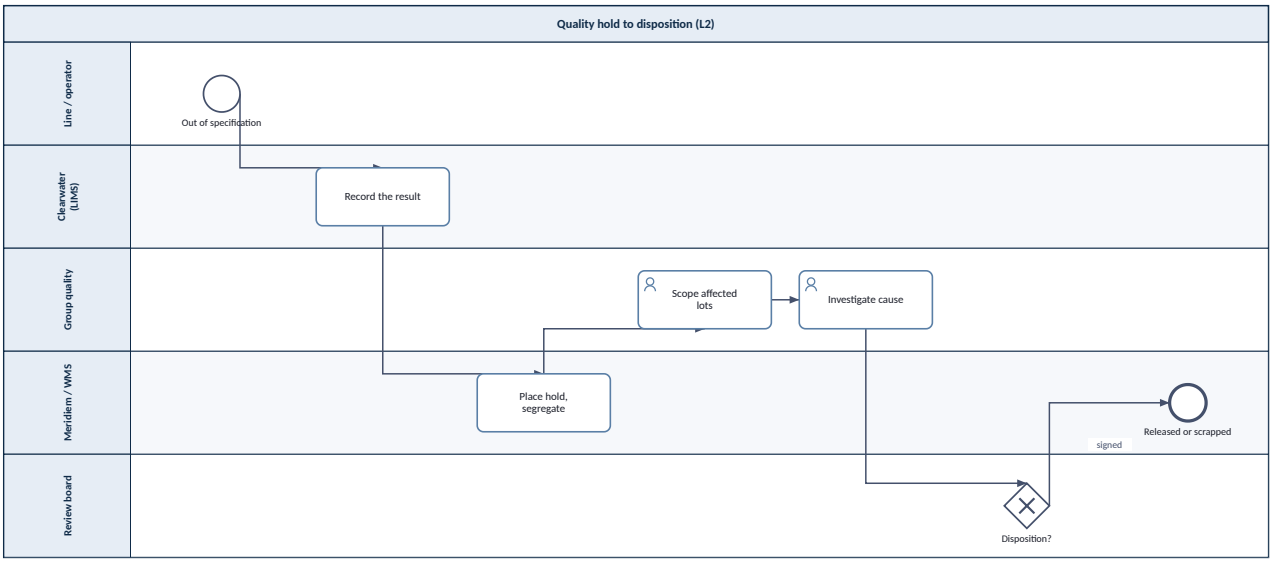


Figure 8.6 Quality hold to disposition, level 2 — the most systems and the least clear ownership of any process in a plant. Five systems can hold a status; only one can be the authority.

Model this one carefully: it is the process with the most systems and the least clear ownership. Draw it on your second day and you will have found most of what is wrong with the client's data architecture.

A hold is triggered by an out-of-specification in-process result, a final test failure, a deviation, a customer complaint, a supplier notification, an operator's observation, a failed in-line inspection or an environmental excursion. Then the material's status changes so it cannot be consumed or shipped — and the first architecture question arrives immediately.

> **Where is the hold the truth?** The candidates are the ERP's batch status, the warehouse system's stock status, the execution system's lot status, the quality system's event status, and the laboratory system's sample disposition. All five can hold a status. **Only one can be the authority that stops material moving, and the other four must read it.** If nobody has decided, the answer in practice is that the hold is set in one system, not propagated, and the pick happens anyway. That is not a hypothetical; it is the normal state.

The step that tests the architecture is **scope determination**. If the root cause is a process parameter excursion between 02:40 and 04:15, the scope is every unit produced in that window — which requires the historian and the execution system to agree on what time it was. They frequently do not. A plant whose unit-level timestamps come from a controller that has drifted since commissioning cannot resolve a time window to a set of units at all.

The disposition itself is a decision with a fixed and surprisingly short set of outcomes, and it is worth learning the list because each one has a different system consequence. **Use as is**, with a recorded justification. **Rework**, which creates a new order with its own genealogy edge. **Repair**, which is rework that does not restore full specification. **Regrade or downgrade**, which posts as a material-to-material conversion and therefore changes what the stock ledger thinks it owns. **Return to supplier**, which reopens the purchase. Or **destroy**, which posts scrap with a reason code and a cost. The decision is taken by a named authority with a recorded rationale and signature, and then the material's status changes back — at which point somebody must reconcile the released quantity against the held quantity. If the cause is systemic, a corrective action follows, and that action may itself become an engineering change or a document revision.

Hand-off failures. The hold set in one system and not propagated. The time window that cannot be resolved to units. Disposition recorded in an email rather than as a signed decision by a named authority. Rework executed without a genealogy link, which breaks the trace graph at exactly the point a recall will need it. **Release without quantity reconciliation** — the released quantity may legitimately be less than the held quantity, and if nobody reconciles the difference the stock ledger silently gains or loses material. And a corrective action never linked to the change that resolved it, so its effectiveness cannot be verified.

Engineering change

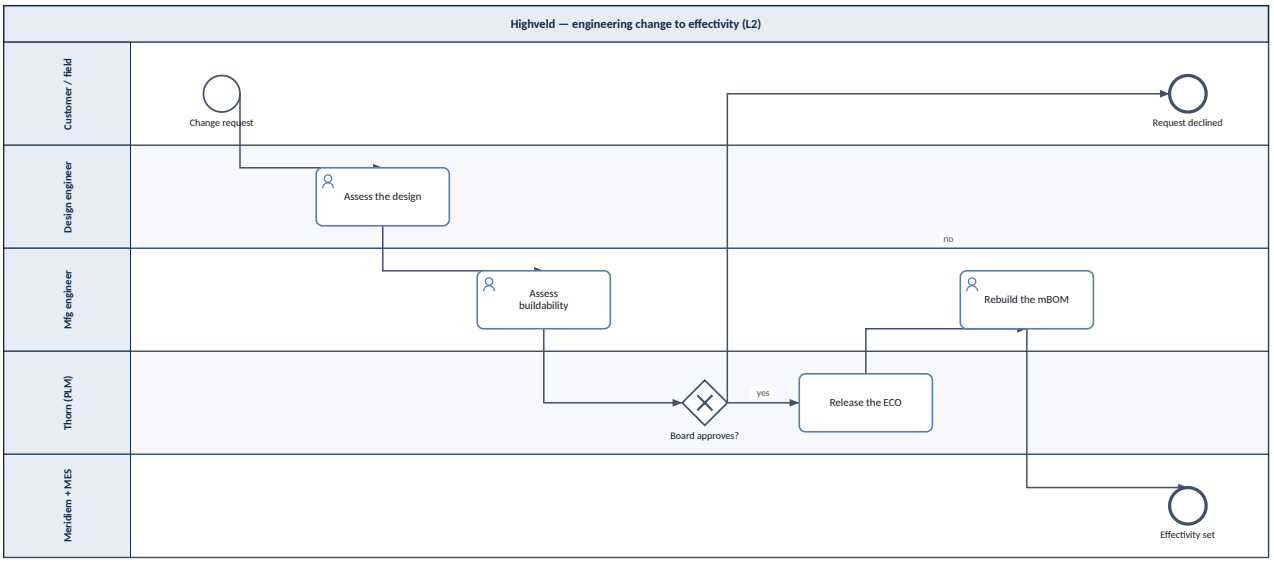


Figure 8.7 Engineering change, level 2. It crosses three system classes, and asking whether a change is implemented requires checking five of them — which is why the chain has no single status.

This is the flow that crosses product lifecycle management, the ERP and the execution system, and it is where the engineering-versus-manufacturing bill-of-material boundary actually hurts.

The engineering bill of material is how the product is designed. The manufacturing bill of material is how it is built: flattened or restructured to match stations, with consumables, adhesives, fasteners, packaging and labels added, with scrap and yield factors applied, with alternates resolved to the one this plant actually uses, with phantom groupings, and with every component allocated to an operation. **The transformation between them is a judgement, not a mapping.** A manufacturing engineer decides that the four fasteners in the design come from a bulk bin, that the sub-assembly the design shows as three parts is bought complete, that a 3% factor applies to the gasket because it tears on fitting. None of that is derivable from the design. The market-leading vendor's own answer to keeping the two structures aligned is a structured human side-by-side comparison with a change summary and checklists [3] — which is the honest state of the art, and worth knowing before anybody promises a client automation.

Then **effectivity**, which is the mechanism that makes a past design reproducible. Valid from a date; valid from unit 1,247 onward; valid for lots made from this material batch; valid for this configured variant; or carried on a change number that every affected bill-of-material line references, so one change activates or reverses atomically across thousands of lines. **Revision without effectivity is the most common implementation defect in this area**, and it is invisible until the first field failure or the first audit, when "what was the approved design when we built unit 1,100?" turns out to have no answer.

Hand-off failures. The design-to-manufacturing gap above. Manufacturing engineering to the ERP, usually a manual re-entry or a partial interface that carries the bill of material and not the routing. The ERP to the execution system, where work instructions go stale because the instruction is a document with no effectivity link, so the operator is shown last year's version. The supplier notification, which is almost always the late one, so the supplier ships the old part after the effectivity date. And the structural one: **the chain has no single status.** Asking "is change order 4471 implemented?" requires checking five systems, which is why mature manufacturers build a change dashboard and immature ones hold a weekly meeting.

> Establish the client's own vocabulary before designing anything here. Change request, change order and change notice are used inconsistently between vendors and companies — one major vendor's material uses engineering, manufacturing and active change notices rather than the request-and-order pairing in general use [3]. The words are frequently the whole argument, and you will waste two workshops discovering that two departments mean different things by the same three letters.

Maintenance work management

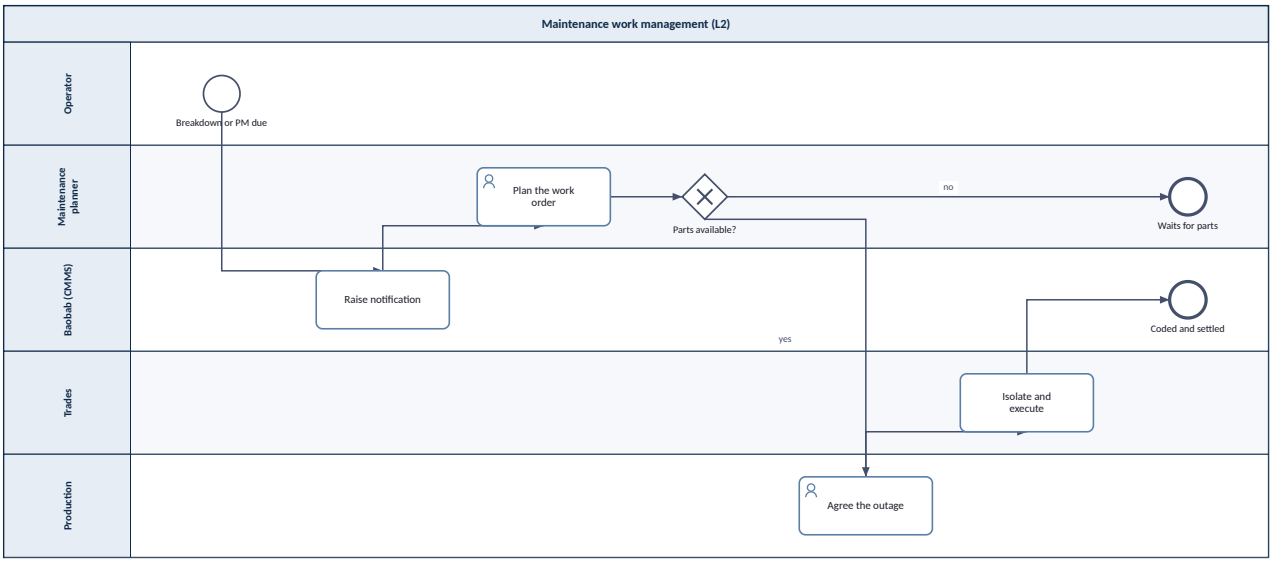


Figure 8.8 Maintenance work management, level 2. The hand-off that causes the most friction in any plant is the one with production, and the integration that would fix it is rarely built.

Demand arrives from a preventive plan reaching its due date or counter, a condition-monitoring alarm, an operator's notification, a breakdown, a statutory inspection falling due, or a project. A notification is raised against a functional location with a symptom and a priority. The planner converts it to a work order and adds the task list, the trades, the duration, the parts and the permits. Parts are reserved. The work is scheduled. Isolation and permits are issued. The work is done. Completion records the **failure coding** — object part, damage, cause — plus as-found and as-left condition and any measurement readings. Costs settle to the equipment and the cost centre. Failure data aggregates against an equipment taxonomy so that strategy can be revised; the reference taxonomy most often cited for this is ISO 14224 [4].

Two steps in that chain are worth drawing attention to because they are safety-critical and almost always analogue. **Permit to work and isolation** — lockout and tagout, energy isolation, confined-space or hot-work permits, gas testing — is usually paper, and where it is digital it is a separate system that nothing else reads. And **scheduling the work into the production schedule as a capacity outage** is the integration almost nobody builds, which is the subject of the box below.

Hand-off failures. Notifications raised against the wrong functional location, or none at all, so history accumulates nowhere. Parts reserved in the maintenance system and not in the ERP, so the reservation is invisible to material requirements planning and the part is consumed by production. Permits on paper, so there is no digital record of isolation state. Failure codes captured as free text, which makes the whole reliability dataset unusable — and this is close to universal. Reliability analysis never performed at all.

> And the one that causes the most day-to-day friction in any plant: **the maintenance schedule and the production schedule are never reconciled.** Planned maintenance is cancelled at the last minute by production, or production is stopped at the last minute by maintenance. The integration that would fix it is the link from the equipment record to the work centre, so that a maintenance outage appears in the capacity plan as a capacity reduction. Almost nobody builds it. It is cheap, it is checkable, and it is the single most useful integration you can recommend in a maintenance review.

Recall

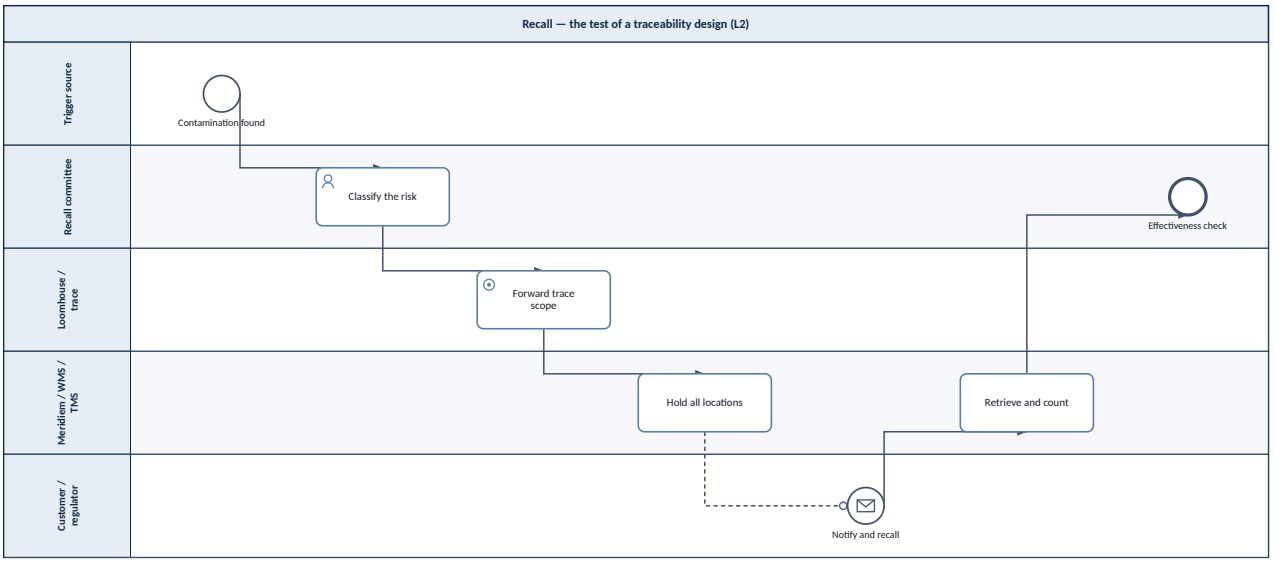


Figure 8.9 Recall, level 2. Where-used says which products could contain the lot; only genealogy says which do. That difference is four pallets against a product line.

Recall is where a traceability design is finally tested, and it is the reason the rest of the data model matters.

The distinction that decides the cost of the event: **where-used tells you which products *could* contain the contaminated lot; only genealogy tells you which ones *do*.** That is the difference between recalling a product line and recalling four pallets. Where-used is a structural query against the bill of material. Genealogy is a recorded graph of what was actually consumed into what, at the grain of the unit or the lot, captured at execution time or not at all.

The scope query is a forward trace from a raw-material lot to every finished lot it entered, then to every despatch, then to every customer. It is a graph traversal, it is many-to-many in process manufacture, and target response times are measured in hours. The event-level vocabulary for exchanging these records between companies is standardised — the global identification body publishes event and business-step vocabularies for exactly this [5] — but the internal capture is yours.

In South Africa the food traceability obligation is **Regulation R638**, which replaced the earlier instrument, and it requires records kept for six months beyond shelf life, a traceability system, a recall procedure, and notification of a recall incident to the local inspector and to the National Directorate: Food Control [6].

> Two corrections worth carrying into any traceability conversation. **South Africa has no equivalent of the United States food traceability rule**: there is no prescribed food traceability list, no prescribed critical tracking events and no prescribed key data elements. Importing that structure by default over-specifies a domestic manufacturer, sometimes by a large margin. And the familiar "two-dimensional barcodes will be required by 2027" claim does not survive contact with the source — the identification body's own material uses no mandate language and frames adoption as incremental. Write it as a voluntary industry capability ambition or do not write it.

Hand-off failures. The trace cannot narrow the scope, so the recall is wider and more expensive than it needed to be — this is where every earlier traceability decision is paid for, in cash. Holds not propagated to all systems, so product keeps moving. Shipped quantities by lot not available at customer granularity, so recall effectiveness — quantity recovered against quantity shipped, which is a reported number — cannot be computed. And mock recalls run on a favourable scenario, a single-lot finished item, rather than a hard one: a raw-material lot that went into eleven products.

> If you want one test of a client's traceability architecture, this is it. Ask them to run a mock recall from a **raw-material lot**, not a finished item, and to tell you which customers received product containing it. Time the answer. A plant that cannot do it in a day cannot do it in a crisis.

The four not drawn here

Ideate to launch. Opportunity, concept and feasibility, design, engineering bill-of-material release, design verification, manufacturing engineering, process risk analysis, control plan, supplier qualification, ERP master data creation, tooling, trial run, first-article inspection, customer approval, release to production. Two things about it. The step that takes longest and is always underestimated is **ERP master data creation** — item master with every plant view, bill of material, routing, production version, costing — and getting it wrong means material requirements planning misbehaves for the first six months of the product's life. And the control plan, which is the authoritative statement of what gets measured where, is supposed to configure the execution system and the laboratory sampling plans; in most plants it is a spreadsheet, so the in-process checks happen on paper. Across the whole flow, the characteristic failure is that **the launch project's data is created in project tools and never migrated into the operational systems**, so the control plan an auditor asks for in year three is a file on a departed engineer's laptop.

Physical inventory and cycle counting. Policy — full annual count, perpetual cycle counting, or value-and-movement-stratified counting. Count documents created with stock frozen or a cut-off recorded. Physical count, blind or not: **blind counting, where the counter is not shown the book quantity, is correct and is resisted.** Recount above a threshold, variance analysis, adjustment with a reason code, root cause. The manufacturing-specific complications are real: work-in-progress counting is genuinely hard, because material is in a vessel, in a pipe, on a conveyor, half-processed. Tank and silo quantities are *measurements* with an error band rather than counts, so reconciling a measured level against a book quantity derived from transactions is a different problem from counting boxes. And bulk solids settle and bridge, so level is not volume is not mass. Note too that in process industries evaporation, moisture gain and settling are **real physical variance, not error.**

Sales and operations planning is chapter 7. The only point to repeat here is that its characteristic failure is a modelling failure: the whole cycle runs in spreadsheets with no link to the systems that execute it, so publishing the plan is a re-keying exercise and the plan that executes is not provably the plan that was authorised.

Month-end costing close. The process that forces every data quality problem in the plant into visibility, and the one architects most often leave out of scope. Chapter 6 works through its mechanics. The modelling point is the cut-off: an order half-confirmed at month end carries work in progress, and a confirmation posted on the 2nd for work done on the 31st is in the wrong period. Everything downstream of a weak cut-off is noise.

Running the workshop

You will model these processes in a plant, with the people who do them. Four things make the difference between a model that is true and a model that is tidy.

Do it on the floor, not in a meeting room. Walk the flow in the direction the material moves, with the supervisor, before you draw anything. This is not a courtesy; it is the only way to see the steps nobody mentions — the bench where rework accumulates, the pallet of material "borrowed" from another order, the clipboard beside the machine that holds the real downtime record, the laminated sheet that holds the changeover rules. None of those appear in a documented process and all of them are requirements.

Model the real process, not the documented one. This is the trap, and it is the main reason process-modelling exercises in plants produce models nobody recognises. The documented process exists to satisfy a certification audit. The real process exists to get product out of the door. They differ, and the difference is almost always where the architecture work is. Two questions find it reliably: *"when was the last time it did not go like that?"* and *"what do you do when the system is down?"* The second answer is a process, usually undocumented, frequently validated nowhere, and often the one that runs for three days a year.

Get the right people in the room, and be careful about the order. The plant manager is the most important interview and should not be the first — do the floor walk first so you can ask specific questions. Shift supervisors will tell you the real reasons for downtime as opposed to the reason codes entered, and which data is keyed at the end of a shift from memory. **If shift supervisors are entering production data retrospectively from paper, every metric built on that data inherits a systematic bias**, and you should establish this in week one, because it is nearly always true to some degree.

Name the systems at every step, and the data object that moves. A lane with no system named is a lane you have not finished. A step whose output you cannot name is a step you have not understood. This is what turns a process model from a description into an input to an integration design.

Sequence the workshops so that each one has the previous one's output. A workable order for a first engagement: walk the floor; then level 0 with the operations director, half a day, to agree scope; then level 1 per process with the process owner, two hours each, to agree boundaries; then level 2 per process with the people who do the work and somebody who knows the systems, half a day each. Twelve processes at that rate is roughly three weeks of workshops and three weeks of writing up, and quoting less than that is how process-modelling exercises end up delivering level-1 diagrams labelled as level 2.

Two practical warnings. **Do not model during the workshop.** Draw on a wall, photograph it, and model afterwards; a modelling tool on a projector slows the conversation to the speed of the slowest typist and the room stops talking. And **do not let the quality manager's documented process into the room first.** It will anchor everybody, and the differences you are looking for will go unmentioned because nobody contradicts a controlled document in front of its owner. Model the real process, then compare it to the documented one afterwards, and treat the gaps as findings for both.

> And one discipline that costs nothing. **Record the hand-off failures on the model itself**, as annotations at the boundaries, in the client's own words. That list — not the boxes — is the value of this chapter and of the workshop. The boxes will be more or less the same at every manufacturer you ever visit. The seams are where the engagement is.

THE CHAPTER IN TEN LINES

- Everything that is architecture in this book is ArchiMate 4; level-2 process models are BPMN 2.0, because ArchiMate has no swimlanes and in a process model the hand-off is the point.
- Every BPMN lane must correspond to a Role or an application component that exists in the architecture model, or the lane is wrong.
- Decide the level before you draw: level 0 agrees scope, level 1 agrees boundaries, level 2 is implementable, level 3 is a controlled work instruction and not your job.
- Most arguments that look like level-2 disputes are unresolved level-1 boundaries.
- The commonest defect in plan to produce is an order change after release that the interface does not propagate.
- In make-to-stock process plants the campaign cycle is known to the planner and represented in no system, which is why the planning output is overridden weekly.
- Quality hold to disposition has five candidate systems for the hold status and usually no decision about which one is the authority.
- The engineering-to-manufacturing bill-of-material transformation is a judgement, not a mapping, and the vendor state of the art is a structured human comparison.
- Revision without effectivity is the commonest product lifecycle implementation defect and is invisible until the first audit.
- Maintenance and production schedules are rarely reconciled, and the cheap fix is the link from equipment to work centre.
- Where-used tells you which products could contain a contaminated lot; only genealogy tells you which do.
- South Africa has no equivalent of the United States food traceability rule, so importing its data structure over-specifies a domestic manufacturer.
- Model the real process, not the documented one, and run the workshop on the floor rather than in a meeting room.
- Record the hand-off failures on the model itself; that list, not the boxes, is the value of the exercise.

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CHAPTER 9

The conceptual data model

Twenty-odd things a manufacturer talks about, each with its grain, its identity, its lifecycle, what owns it — and what breaks when the model is wrong. The last of those is where the value is, because the reader will walk into an estate where several of them already are.

A conceptual data model is a list of the things a business talks about and the relationships between them. No attributes, no keys, no system names. For a manufacturer it runs to about twenty entries, and an architect who can name them, say what one row of each is, and say what breaks when the model is wrong can hold a credible conversation in a plant.

That last clause is the work of this chapter. Each object gets the same treatment: what it is as distinct from what it is confused with, what one row is, and the consequence when the model is wrong. The consequence is why the object is here.

Three rules run underneath all of it. **A manufacturer's model is three models wearing one coat** — a *product* model, a *process* model and a *physical* model — and nearly every serious failure is two of the three collapsed into one table. **Nothing is a flag**: most attributes you want on a master record are really dated, plant-specific, function-owned assertions, and "is this item active?" has no yes or no. And **quantity and value are different facts about the same event**, kept by different people on different clocks.

The item master, and why it is the hardest master data object in any industry

An **item** is the identity of a *kind of thing* the enterprise designs, buys, makes, stores, moves, inspects, costs, sells or disposes of. SAP calls it a material, discrete engineering a part, process people an ingredient. It is not a physical thing: the physical thing is a batch, a serial number or a stock position.

One row is one kind of thing, globally, independent of plant. That is the only grain that survives a second site. Everything plant-specific belongs on a *plant extension*. One flat table with a plant column forces a choice between one row per item-plant, destroying global identity, and one row per item, destroying plant-specific planning — and neither is recoverable without a migration. The grain is four levels deep — global item, item-plant, item-storage-location, item-valuation — and on top sit dated relationships that are not attributes at all: item to supplier, to structure, to routing, to specification.

One row, six owners who disagree

A customer record is owned by one function and disputed by nobody. An item record is **co-owned by six or seven functions who each need attributes the others consider noise**, and each needs the record to exist before it can supply its part of it.

Function	What it needs on the item	What breaks if denied
Engineering	Part number, revision, drawing, specification	Cannot release a change
Planning	Procurement type, planning type, lot size, lead time, fence	Every planned order is wrong
Costing	Valuation class, price control, standard cost, component split	Cannot explain a variance
Quality	Inspection type, specification, sampling plan, shelf life	Cannot raise an inspection lot
Purchasing	Purchasing group, order unit, tariff code, country of origin	Cannot clear customs
Warehousing	Storage conditions, weight, volume, dangerous-goods class	Cannot store it legally
Sales	Sales unit, division, pricing reference, delivering plant	Cannot take an order

A single item in a large ERP carries several hundred fields across those views. So: **the item master is not a table, it is a workflow with a table at the end of it.** Chapter 10 builds the workflow.

Item type, procurement type, status

Item type is a behavioural contract, not a label: it decides which views exist, which transactions are legal, whether stock is kept, whether a structure may exist, and how the thing is valued. Raw material, semi-finished, finished good, packaging material, operating supply, spare part, non-stock item, service item, phantom, configurable, co-product, by-product, tool, trading item.

Procurement type — made, bought, both, or subcontracted — is the cleanest proof of the grain rule. An item made at Ekurhuleni and bought at Gqeberha cannot have one global procurement type. Planning parameters sit beside it on item-plant: planning type, lot-sizing rule, rounding profile, lot-size limits, safety stock, lead times, the time fence, the planning strategy, and the planning controller, who is a *person* and therefore the accountability route.

Status is not one field. There are at least five, each with its own transitions and owner: engineering status, moved only by the change process; plant item status, owned by planning per plant; sales status per sales organisation; quality status, qualified for a given use or supplier; and deletion flags, which are administrative. Collapse them and you get the two symmetrical failures every manufacturer has met — an engineering-obsolete part still orderable, and a sellable part blocked because a drawing was superseded. Note also what the lifecycle does not contain: **deletion is not a state a manufacturer reaches**, because an item consumed in a batch made ten years ago must stay resolvable for as long as that batch's genealogy must be.

> The item-master rule with no exceptions: **base unit of measure is immutable once a single transaction has posted against the item.** Change it afterwards and the historical quantity series cannot be reconstructed, by anyone, ever.

Units of measure, and the error that never announces itself

This looks like plumbing. It is one of the places a manufacturing model is most likely to be *silently* wrong, because after it breaks every quantity still has a number in it. Four unit roles need four fields: **base**, what stock is held in and what conversions anchor to; **order unit**, what the supplier invoices in, which is a dated property of the item-supplier relationship and not of the item; **issue unit**, what the floor works in — base kilograms, issue bags; and **sales unit**, which belongs to the item-customer relationship. A conversion is a *dated, three-part fact*: item, from-unit, to-unit, numerator, denominator, valid-from. Conversions are **item-specific** — one bag is 25 kg for this item and 50 kg for that one. And they **change**: a supplier moves to 20 kg bags, and if the conversion is overwritten rather than dated, every historical receipt silently restates. Rounding belongs here too, as a *profile* on item-plant, because rounding applied at presentation time produces plans that do not add up.

Catch weight is the hard case, and it is normal in meat, poultry, fish, dairy, cheese and some chemicals and metals. A catch-weight item is transacted in two units at once, **neither derivable from the other**: a carton of boneless beef is one carton and 27.4 kg, and the next is one carton and 26.1 kg. There is no conversion factor, there is a measured second quantity per physical unit. So stock is a *pair* per position and every inventory table widens; buying may be per kilogram and selling per carton, which makes margin a function of a measured weight. One Karoo Foods line at Gqeberha runs catch weight and the Paarl lines do not, which is exactly the split that defeats one group-wide inventory design.

> Get units wrong and the failure is invisible for months. Planning orders twelve times or a twelfth of what is needed. The plant believes it has 1,000 kg and has 1,000 bags. **Every quantity in every table needs its unit stored beside it. A bare number is a defect.**

The bill of material

A bill of material looks like a tree and is not. It is **a set of dated, qualified, versioned, context-specific structural assertions over a shared set of items**, and different functions make different, simultaneously true, contradictory assertions about the same parts.

Several structures over the same parts

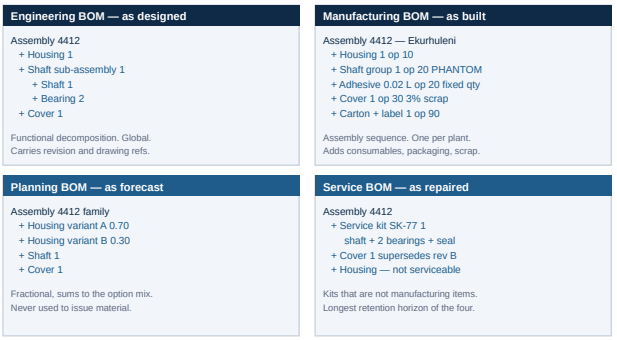


Figure 9.1 Four structures over the same parts. They are not views of one bill of material — they have their own identity, lifecycle, approval and effectivity, and they happen to reference the same items. Illustrative.

Three sentences make this section worth writing.

1. **These are not views of one bill of material. They are separate structures with their own identity, lifecycle, approval and effectivity, which happen to reference the same items.**
2. The engineering-to-manufacturing transformation is a *judgement*, not a mapping: that the four fasteners in the design come from a bulk bin and become a consumable line; that the three parts the drawing shows are bought as one assembly; that this plant uses the alternate because the primary supplier will not ship here; that 3% scrap applies to the gasket because it tears on fitting; that components seven, eight and nine are consumed at operation 40. None of it is derivable from the engineering structure. The market-leading product lifecycle vendor's own published answer to keeping the two aligned is a side-by-side comparison with a change-summary panel and checklists — a human doing a disciplined diff [4]. That is the honest state of the art, and worth knowing before you promise a client automation.
3. **As-built is not a bill of material.** A bill of material is a statement about a *kind* of thing; as-built is a statement about *one* thing. Different grain, different volume class — as-built is transactional and runs to millions of rows where a structure is master data in the thousands — different retention driver, different consumers. As-built is genealogy.

Header and line

The header identifies a structure: parent item, usage, plant or global, alternative number, status, base quantity and unit, validity, approval, change number. **Usage is part of the key**, not a filter and not a flag, because engineering, manufacturing, costing and planning structures coexist for one parent. **The alternative number** is how several simultaneously valid structures for one parent, usage and plant coexist, and selection among them is a *rule* — never a preference. And **base quantity** is why a structure is not a list of quantities: "to make 1,000 kg of this, use…" lets a component be 0.3 kg without rounding error.

The line is where the interesting data lives, and it is routinely modelled as parent, child, quantity — three columns where thirty are needed.

Line attribute	Why it exists
Position number	Stable identity of the *position*, independent of what sits in it
Item category	Stock, non-stock, variable-size, document, text, class — decides whether a requirement is generated at all
Operation assignment	*Where* it is consumed. Drives staging, backflush timing, work-in-progress value
Fixed-quantity indicator	One test certificate, one litre of flush per run, whatever the run length
Change number in and out	What authorised this line, and what removed it
Reference designator	Which of the 48 identical capacitors this is. In electronics "C47" is a real identity and the part number is not

One row is one position in one structure version — not one parent-child pair. Two lines may legitimately carry the same component: the same screw at two operations with different quantities. Treating parent-and-child as the key destroys reference designators, operation assignment and fixed-quantity lines in one move. The three-column model cannot answer the questions actually asked on a shop floor, in a recall and in a variance investigation: when did this change and on whose authority, which operation consumes it, does this quantity scale, is this the primary or the approved alternate, and where is it on the drawing.

Explosion and where-used

Explosion descends recursively from a parent to its ultimate components, multiplying quantities and applying scrap and yield at each level. **Where-used** is the reverse, and it is the question that justifies the structure: a supplier has failed, and somebody must say what cannot be built. Both are parameterised in ways a naive recursive query ignores — *as at what date*, since every line is dated and "the BOM" with no date is not a question with an answer, and *which usage*, since engineering and manufacturing explosions give different part lists.

Phantom assemblies

A **phantom** — transient assembly, blow-through item — exists in a structure *to group components*, is never stocked and never has an order of its own. Requirements planning blows through it. Good reasons exist: it is a design unit built and consumed in one operation; it collapses a group of eight components used in forty products from 320 lines to forty; it preserves engineering identity inside a flattened manufacturing structure; or it *is* the kit.

They hurt because a phantom is a thing that is sometimes real. The moment the line stops and half-built harnesses sit in a bin, the phantom has stock and the model says it cannot. Mature designs express that as "phantom with occasional stock" — blow through when there is none, consume stock when there is.

> **Phantom is a property of the BOM line, not of the item.** The same item is legitimately a phantom in one parent and a stocked assembly in another. Good systems express it as an explosion type on the component line with an item-level default. An item flag alone makes the legitimate dual use impossible — and that design error ships in real systems. If a client says a sub-assembly "cannot be stocked because it is a phantom", you have found it.

Alternates, substitutes, and the direction of interchangeability

Four distinct mechanisms get called the same thing.

Mechanism	Meaning	Where it lives
Alternative BOM	A whole different structure for the same parent	Header, alternative number
Alternative item group	At *this position*, any of these parts is acceptable	Lines sharing a group
Approved substitute	B may be used wherever A is called for	A *part-to-part* relationship, no BOM involved
Ad-hoc substitution	We used B today because A was out	A fact on the order and in genealogy

Alternative item groups carry three numbers doing different work: **priority**, the preference order; **usage probability**, the *planning* split — 70/30 — which generates dependent demand for all the alternatives in proportion so that both are actually available; and **strategy**, whether selection is automatic or manual. Usage probability is the subtle one: an item at 100% with a 0% sibling means the sibling is never planned, so never available, so the alternate exists only on paper.

> **Interchangeability is not symmetric.** "B may replace A" does not imply "A may replace B". One-way interchange is the *normal* case on a supersession: the new part fits everywhere the old one did and not the reverse. Model it as a symmetric link and you have authorised the warehouse to fit the wrong part. Conditions matter too — interchangeable for manufacture but not for service, or only above a serial number.

Scrap, yield, and the arithmetic done properly

Four numbers, four meanings, four places they belong. This is a reliable interview question for a manufacturing data modeller, because almost nobody gets all four right.

Number	Meaning	Where it belongs	Effect
Component scrap	Part of *this component* is lost in use — offcuts, spillage, test	BOM line	Uplifts that component only
Assembly scrap	Part of *the parent* is lost in assembly	Item-plant, or header	Uplifts every component and the order quantity
Operation scrap	Part is lost at *a specific operation*	Routing operation	Uplifts only components consumed before that operation
Yield	Proportion of input that becomes good output	Recipe or item-plant	The positive form, and the primary number in process plants

Operation scrap is the only one that needs the structure and the routing read together, which is why it is the one most often missing.

Now the arithmetic. **Scrap of 10% is not an uplift of 10%.** To ship 100 good units when a tenth of what you start is lost, start $100 \div (1 - 0.10) = \mathbf{111.1}$, not 110.

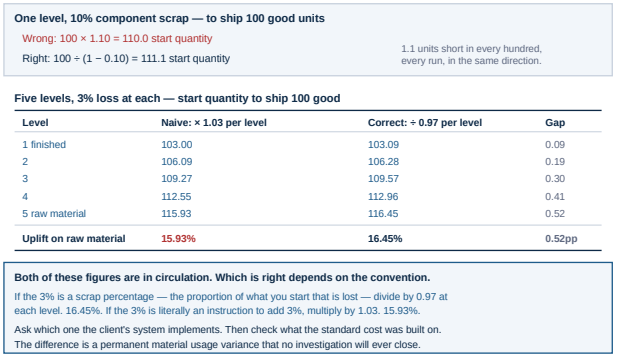


Figure 9.2 The scrap arithmetic, done properly and done the way it is usually done. The five-level figure that circulates — 15.9% — is itself the one-level error repeated: it multiplies by 1.03 instead of dividing by 0.97.

Over a multi-level structure the error compounds — and the usual statement of the compounding is itself worth checking. A five-level structure losing 3% at each level does *not* need $1.03^5 - 1 \approx 15.9\%$ more raw material than the naive explosion. Multiplying by 1.03 per level is the same mistake as multiplying by 1.10 for 10% scrap, one level at a time. If the 3% is a scrap percentage, each level divides by 0.97 and the true requirement is $1 \div 0.97^5 \approx 1.1645$, an uplift of about **16.4%**. The half a percentage point between them is the difference between two wrong answers, both in circulation; 1.03^5 is right only where each level's number is literally an instruction to add 3%, which is a rarer convention.

> Build the standard cost on the naive explosion while the plant consumes the compounded quantity and you have created a permanent, structural, unexplainable material usage variance. It will be investigated every month for years and never closed, because the cause is not in the plant, it is in a formula. How common a cause this is of persistent usage variance cannot be said from public sources — but the arithmetic is checkable in an afternoon.

Process industries invert the emphasis: yield there is a function of process conditions rather than a constant, so modelling it as a fixed scrap percentage surfaces as inventory which will not balance. And **theoretical and actual must both exist** — the structure holds the standard, the confirmation holds the actual, and if the model stores one number the variance cannot exist as a fact.

Effectivity

Effectivity is the rule that says *when, or to what, a structural assertion applies*. Four regimes, not interchangeable, chosen by the industry rather than by preference.

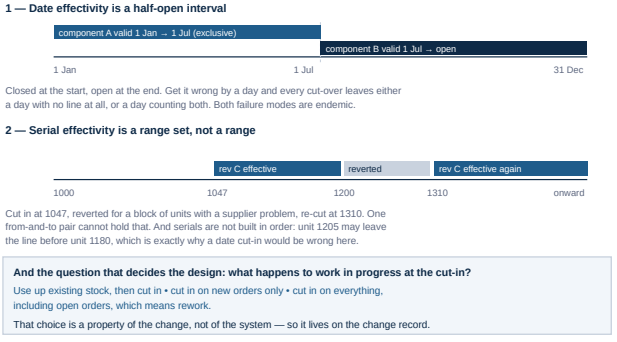


Figure 9.3 Effectivity on a timeline and on a serial axis. Date effectivity is a half-open interval; serial effectivity is a range set, because changes are cut in, reverted and re-cut.

Regime	Where it belongs	The hard part
Date	Most discrete and process manufacturing	*Which* date — order creation, planned start, actual start, release? Systems default differently and the answer changes the exploded structure
Serial	Aerospace, heavy equipment, rail, defence	Serials are not built in order: unit 1050 may be built before 1047, so a date cut-in is *wrong*
Lot	Pharmaceutical, food, chemical	Must bind to the released formula version at batch creation and then freeze, because a batch record must reflect what was authorised then
Unit	Engineered make-to-order, capital equipment	Effectivity becomes an object with its own identity and ranges, separate from date and serial

Three mechanics matter. Date effectivity is a **half-open interval**, closed at the start and open at the end, or every cut-over leaves a one-day gap or a one-day double-count. **The "no line" case must be representable**: a component removed by a change is a row whose validity ended, not a deleted row. And **serial effectivity is a range set, not a range**. And what happens to work in progress at the cut-in — use up stock, cut in on new orders, or cut in on everything, which means rework — is a *property of the change*, so it lives on the change record.

Revision, and engineering change as a data event

Two things are called revision. **Item revision** is the design state of a *part*; **structure version** is the state of a *structure*; either can change without the other. The governing rule of part numbering in discrete manufacturing is the **form, fit and function rule**: if a change affects form, fit, function or interchangeability it gets a new part number, and if not, a new revision of the same number. So a revision is an assertion that old and new stock are mutually substitutable, and warehousing, service and quality rely on it.

Engineering change is an entity, not metadata. The change record carries description, reason code, originator, affected items, structures, routings and documents, the effectivity rule, the disposition of existing stock and work in progress, approvals, implementation status, cost impact and regulatory impact. Every changed line carries the change number that brought it in and the one that took it out — and that pair is what makes "why is this component here" answerable years later. **Disposition of existing stock** is the field architects omit and the plant cares about most: use as is, rework, scrap, return to supplier, segregate for one customer.

One change also fans out further than most change processes reach — a structure version, a routing, a tool, a sampling plan, a standard cost, a supplier re-qualification, a label, a homologation — and the change record is the only place that fan-out can be tracked. A process that stops at the structure guarantees one is missed, usually the regulatory one. The vocabulary is not standard either [4], so **establish the client's own words first.**

Configurable and variant structures

A family with fourteen independent options has more variants than it is worth modelling as items, so the structure is parameterised: characteristics with typed value domains; a **super-BOM**, the union of every component that could be needed, each line carrying a selection condition; a **super-routing** on the same pattern; the **configuration**, a valid set of values belonging on the order; and the **variant**, a configuration recurring often enough to earn its own item number. Then two things decide whether it works: **where the rule set lives**, because three copies is the normal state and the normal source of quotes that cannot be built, and that **constraint satisfaction is a solver, not a filter**.

Recipes, co-products, by-products — and transition material

Discrete structures assemble. Process recipes transform, and the differences are not cosmetic.

- **Structure and process conditions are inseparable.** A master recipe carries temperature, pressure, pH, hold times and addition rates as part of the specification. A structure with a routing beside it cannot express "add B over 20 minutes while holding 65 °C". Force the recipe into structure-plus-routing and the parameters move into free text or into the controller and out of the governed record — which in a regulated plant means the authorised process is not the documented process.
- **Quantities do not scale linearly.** A catalyst may be fixed-quantity; a cleaning solvent may scale with vessel surface area rather than batch mass.
- **Co-products are outputs with their own value**, carried as *negative* component quantities with an apportionment rule splitting the batch cost. That rule is an accounting policy with a data representation, and it is the thing an auditor questions.
- **By-products** are outputs of low, zero or *negative* value — one with a disposal cost makes the batch cost higher than the input cost, which breaks naive costing arithmetic.
- **Recycle and rework streams** feed output back as input, so the structure is cyclic and an explosion over it will not terminate without a convergence rule. And **potency** means an active ingredient is dosed by *activity*, not mass, so the consumption quantity is a function of the *lot*, computed at issue.

Then the object no discrete system has. **Transition material** is what comes out of a continuous or semi-continuous asset between grades, or after an interruption: neither the raw material it was nor the product it was meant to be. It has a mass, it occupies a vessel, it has a disposition — reprocess, downgrade, sell as off-spec, dispose — and a value that is usually neither of its neighbours'. **No discrete-manufacturing system models it, and without it the mass balance will not close.** When the Paarl pasteuriser is interrupted, what is left in the line is transition material; with nowhere to put it, the only way to make the numbers add up is an adjustment with no cause attached, which becomes permanent, and the plant's yield figures become a negotiation.

Ten questions that settle a bill-of-material design

Run these in a workshop, in order.

1. **Whose structure is this?** Two answers means two structures with two usages.
2. **Does this assertion have a date, a serial, a lot or a unit?** That names the effectivity regime, and it is not negotiable later.
3. **Could this position hold a different part legitimately?** Then it is an alternative item group.

4. **Does this quantity scale with order size?** No means a fixed-quantity line.
5. **Is this thing ever physically put down?** No means a candidate phantom; sometimes means phantom with occasional stock.
6. **Where is this consumed?** If the answer is an operation, the routing must exist first.
7. **Is the loss specific to this component, to the assembly, or to an operation?** Three fields.
8. **What authorised this line, and what will authorise its removal?** No answer means there is no change process and everything above is decoration.
9. **Will anyone need to explode this as it stood on a past date?** Almost always — so no in-place updates, anywhere.
10. **Must any of it be reproducible to an outside party?** Then the retention horizon is the regulatory one, not the product's commercial life.

Routing, operation, work centre, resource, capacity

A **routing** is the ordered set of operations that turn components into the parent, with the resources and times each needs. It is the process model to the structure's product model, joined to it at the component-to-operation assignment. Process industries call it a master recipe and treat it as one object with the formula. **One row is one operation in one routing version for one parent at one plant** — plant-specific always, because the same product on different equipment is a different process. The header's **lot-size range** is how a 100-unit run and a 10,000-unit run use different processes, and also how the costing run picks a routing.

Two operation attributes are underrated. **The operation number is the stable identity of a step and the structure's lines point at it**, so renumbering silently re-points material staging — which is why sequences go 0010, 0020, 0030. And **the control key is the behavioural contract**: does this operation schedule, cost, confirm, subcontract, raise an inspection lot? Usually copied from a default.

Work centre, capacity and equipment are three objects, not one. A work centre — resource, in process plants — is the *place and means* where an operation happens. Capacity is its *available time*: a dated, shift-driven quantity with a utilisation rate and a number of individual capacities, against a **factory calendar** of working days, public holidays and shutdown windows with dated exceptions. Equipment is the *physical machine*, with a serial number, a maintenance history and a book value. One work centre may hold three machines; one machine may serve two work centres; a work centre may be people with no machine at all. Merge the scheduling construct with the physical one and the maintenance tree and the scheduling tree fight.

> `WORKCENTER` is also an ISA-95 term, where it is the *generic* name for the grouping of equipment typically scheduled by level 3 or 4 functions, with `WORKUNIT` below it [1] — which is why "work centre" means related but non-identical things in a control-systems conversation and in an ERP one.

Standard times, and why they are always wrong

Setup, run, queue, move and wait are the numbers on which capacity planning, scheduling, lead time, product cost and plant performance all rest. They are also the least reliable numbers in the estate. Queue feeds lead time only and is usually its largest single component. Move is usually ignored, which breaks the schedule on a long shop floor. And **wait is a mandated post-processing delay** — cure, cool, rest, test turnaround — which gets modelled as queue, so it is "optimised" away and product is moved before it has cured.

They are wrong for structural reasons, not for want of effort. They are set once, under project pressure, often by copying a vendor default, and then they are load-bearing for a decade, encoding an operating condition that has changed. Setup is sequence-dependent and one number per operation cannot express a changeover *matrix*. And the standard is political, because it is simultaneously a planning input and a performance yardstick.

The architectural response is not "make the standards right". Date them on the routing version, so a change is a change and not an overwrite. Take actuals from machine data where you can. Make the **variance between standard and actual a first-class reported fact** by operation, resource, product and period, not a by-product of a cost run. And let scheduling use actual-derived distributions while costing uses the authorised standard — **these are allowed to be different numbers**.

Alternate routings, parallel sequences, subcontract operations

A parent may have several routings: a primary and an alternate on different equipment, a rework routing, one for a lot-size band, one valid only while the primary line is down. Selection is by **production version** — a named triple of structure alternative, routing and lot-size range, with validity. Without it, nobody can say which structure and routing a given order used.

A routing is not always a line. Real processes branch: two sub-assembly paths converge at a final operation, an inspection runs in parallel with a cure. That needs sequences with explicit finish-start relationships and offsets — a directed acyclic graph with lags. **A strictly sequential model forces the planner to serialise genuinely concurrent work, inflating lead time by the full duration of every parallel branch**, which then inflates safety stock downstream.

Subcontract operations carry consequences larger than they look. The operation raises a requisition on release, so a production order spawns a procurement document. Components may be *provided* by us — stock owned by us and physically at their site, a stock type that must exist and is frequently missing. And their process is in *our* genealogy.

The order: production, work, process, batch

The order is the authorisation and record of *one instance of making something*: what, how much, by when, using which structure and routing version, against which demand. A **production order** is discrete ERP. A **work order** is the shop-floor term — and in maintenance systems it means a *maintenance* order, so in an estate with both the word is ambiguous and the model must disambiguate by type. A **process order** is driven by a master recipe with phases rather than operations. A **batch** is the execution of a recipe on a specific equipment train, with the batch record as the legal artefact; the batch-control standard's part 1 defines that vocabulary and, unlike ISA-95, carries conformance criteria [3].

Two patterns an order-centric model cannot express: **repetitive manufacturing**, which is period-and-rate based and deliberately orderless, and **kanban**, which signals replenishment by consumption and has no order object at all. Plants running either will create one dummy order per shift to satisfy a model that insists on orders, destroying the cost and traceability the model was meant to provide.

The states run planned, which is a planning output and not an order; firm planned; created; scheduled; released; partially confirmed; technically complete, meaning no more activity but cost still open; closed and settled; archived. Cancellation is possible only before any posting, because **deletion must be impossible once a posting exists**. **Release** is where the order becomes a legal and financial object; **settlement** is the point after which cost cannot change, and it is tied to the financial period rather than to the plant's sense of being finished. The gap between them is where plant and finance disagree about whether an order is done.

> **Fix the versions at release, by copying rather than by pointing.** The order must store which structure alternative, which routing, which production version and which change-number state it was released against. Point at the master instead and a later master change rewrites history, corrupting every downstream reconstruction — cost, genealogy and regulatory record together.

Confirmation and backflush

Confirmation is the act of telling the system what happened: quantities produced and scrapped, activities consumed, times taken, by whom, at which operation, when. It is the only mechanism by which the plan becomes actual, and in most plants it is the weakest data in the estate. One row is **one reported event against one operation of one order** — not one per order and not one per shift — and confirmations are reversible by cancellation documents, never by update. Two fields are usually missing and both matter: the **scrap reason code**, without which there is no loss analysis worth doing, and the **variance reason**, without which the cost variance has no narrative.

Backflush is automatic component consumption triggered by a confirmation: confirm 100 good units and the system issues the structure quantity for 100 units of every backflush-flagged component. It assumes the structure is right, so where scrap or yield is wrong it bakes that error into the inventory balance every time. It assumes substitution did not happen. And it defers error detection to the physical count, where the discrepancy appears as unexplained loss with no cause attached. **If a client has unexplained shrinkage and backflushes, you have found the mechanism.**

> Backflush where substitution occurs produces a **confidently false genealogy** — a record saying A was consumed when B was — and a confident falsehood is worse than a gap, because a gap prompts a question and a falsehood ends one. Backflush is appropriate where consumption is deterministic and lot identity need not be exact. Where either fails — pharmaceutical, allergen-controlled food, aerospace, anything serialised — **lot-specific explicit issue is the only defensible design**, and the extra scanning is the price of a traceability claim that will survive a regulator.

Batch, lot, serial, and genealogy as a graph

A **batch** or **lot** is a quantity produced or received under uniform conditions and treated as homogeneous for quality purposes. **One row is one batch of one item**, keyed by item plus batch number, because batch numbers are not globally unique and a supplier's batch number is theirs. The distinction a batch design turns on: **a batch is homogeneous by assumption, not by fact.** The assumption is what lets one certificate of analysis cover it, so batch size is a *risk* decision. South African law sharpens that: under the compulsory-specifications regime a test result is generally valid for the whole batch unless shown otherwise [6], which makes batch size an exposure decision rather than a production convenience.

A **serial number** is the identity of one physical unit. Serials and batches are not alternatives: a serialised unit is normally also in a batch, and both must be recorded, because the batch carries the test results and the serial carries the field history. There are four distinct uses of a serial number, routinely conflated: manufacturing identity for genealogy, warranty identity for the technician, the regulatory identity a recall uses — which may be prescribed rather than chosen — and asset identity. These can legitimately be four different numbers on one object.

Genealogy is a graph

Genealogy is the recorded relationship between material consumed and material produced, at batch or serial grain, through every production and movement event. **It is a directed acyclic graph, not a tree**, and every property that makes traceability hard follows from that.

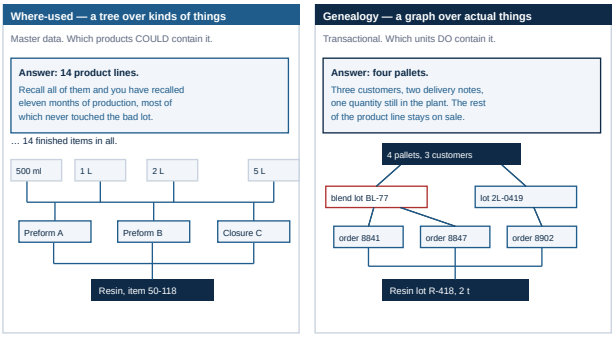


Figure 9.4 The same contaminated resin lot, asked twice. Where-used is a tree over kinds of things and answers which products could contain it. Genealogy is a graph over actual things and answers which ones do — and the blended lot on the right has two parents, in proportion. Illustrative.

One input batch fans out to many outputs: one resin lot feeds 400 orders over three weeks, producing 400 output lots that reach 90 customers. One output has many inputs. Blending is the hard case, because afterwards the output lot's parentage is a *set with proportions*. Rework reintroduces output as input, creating a cycle unless each pass is a distinct node. The graph spans organisations, and those links are the weakest in the chain. And **the graph spans systems**: the execution system holds production links, the warehouse system movement links, the ERP procurement and sales links. **The graph exists in no single system**, which is the most important architectural fact here.

Forward trace asks *where did this go?* — the recall question. Backward trace asks *where did this come from?* — the root-cause question. Both must be answerable to a stated depth, as at a stated date, with a stated completeness, and both must include *movement* history as well as *transformation* history.

> **Where-used tells you which products *could* contain the contaminated lot. Only genealogy tells you which *do*.** That is the difference between recalling a product line and recalling four pallets. Where-used is master data over kinds of things; genealogy is transactional data over actual things. A client who has one and believes they have the other finds out during a recall.

Recall simulation is the real test

A traceability design never tested by simulation is a claim rather than a capability. Pick a real input lot at random and ask for a forward trace. **Time it**, against the window the business or the customer's certification scheme requires. **Measure the spread**: a design returning "everything we made in March" has technically answered and practically failed. The metric is the ratio of product implicated to product actually affected — the **amplification factor** — and reducing it is what traceability engineering *is*. Then check the edges, the past dates, and whether the system can state what it does **not** know, because a trace that silently omits is worse than one that flags. Four levers reduce amplification: smaller batches, lot-specific consumption, first-in-first-out enforced physically, and not blending a quality-critical input.

Shelf life, retest, quarantine

Shelf life is a dated property of a batch, and there are **two periods, not one**: total shelf life until expiry, and minimum remaining shelf life, the proportion that must remain at despatch to a given customer. The second is contractual, different per customer, and almost always missing — without it you despatch product the customer rejects at the gate. A **retest date is not an expiry date**: material past retest is unapproved pending retest, not scrap, so modelling retest as expiry destroys usable inventory every month. And **quarantine is a stock type, not a batch attribute**.

Quality

Quality is a data domain, not a department. Ten objects carry it.

Object	What one row is	Note
Specification	One item, one version, one validity	Versioned, never edited
Characteristic	One property in one specification version	Method, unit, target, limits
Sampling plan	One plan, referenced by many specifications	Sample size, severity, accept criteria
Inspection lot	One trigger event	The *instance* of inspection
Result	One measured value, one characteristic, one sample	Append-only
Certificate of analysis	One batch, one recipient context	Issued, superseded, reissued
Non-conformance	One recorded failure	A fact about an *outcome*
Deviation	One authorised departure from the authorised process	A decision about a *process*
CAPA	One corrective and preventive action programme	A programme of *work*
Usage decision	One disposition per inspection lot	Irreversible, and it moves stock

Four points carry the section. **Specifications are versioned and inspection lots bind to a version**, so a batch released in March against version 3 shows version 3 for ever — because "which version was this released against" is the first question of any investigation. **Results are append-only**, and that original out-of-specification result is the most important record in the file. **Disposition moves stock** — and downgrade moves it to *a different item number* with its own cost and price, which is the one architects miss: reclassification is a transfer from item A to item B with a valuation difference, not a status change. And **non-conformance, deviation and CAPA are three objects with three lifecycles and three owners**; collapsing them is the commonest quality-data failure.

Inventory, and ownership as a first-class attribute

Ask a plant how much of a material it has and the honest answer is a set of questions: which plant, which location, which batch, which status, whose, valued at what, as at when? Every one of those is part of the key of an inventory position.

Stock type is the orthogonal axis — neither a location nor a batch status, but the usability *and ownership* state of a quantity at a place: unrestricted; quality inspection; blocked; in transit; consignment, physically ours to use and legally the supplier's until consumed; subcontracting, ours and at their site; customer-owned, which is tolling material; project stock; and work in progress. Work in progress is the position that breaks the model: once a component is issued to an order it has left material inventory and has not arrived as the parent, so in the *quantity* world it has ceased to exist, while in the *value* world it sits on the balance sheet. So:

> *Quantity and value genuinely live in different systems, with different grains, different owners and different period logic — and that is not a defect to be fixed, it is a fact to be reconciled.*

They meet at exactly three places: the goods movement document, the inventory valuation at period end, and the plant-to-finance reconciliation that chapter 10 builds.

One mechanic underneath all of that: **the movement document is the only truth.** On-hand is the sum of movements, and each movement carries three dates — posting, document and entry — which differ whenever a Friday night shift is posted on Monday morning.

Ownership is a first-class attribute of inventory, and omitting it is one of the few genuinely unrecoverable design errors in this chapter. In a tolling plant the material belongs to the customer. Stock in one silo may be owned by three parties. The same tonne may be the plant's asset on Monday and a customer's on Tuesday. Leave ownership out and the balance sheet is overstated, the customer's books show an unexplained loss, and consignment stock sits on the wrong accounts. **You cannot retrofit ownership onto historic stock movements**, because the information was never captured — so this is the one item on the list that has to be right in version one.

Equipment, functional location, and the two trees over the same steel

A **functional location** is a place in the plant where equipment performs a function — "line 3, filler position, bearing housing". It is structural and permanent. **Equipment** is a physical, individually identified object that is maintained, has a serial number, a history and often a book value, and it is mobile between functional locations. A pump is overhauled and swapped out; a spare goes into the same functional location. The **location** accumulates the history of the *function* — this position fails every eight months, a process problem. The **equipment** accumulates the history of the *object* — this pump has been rebuilt four times, an asset problem. So **install and dismantle is a dated relationship**, not an attribute, or this year's failures are attributed to whatever happens to be installed today.

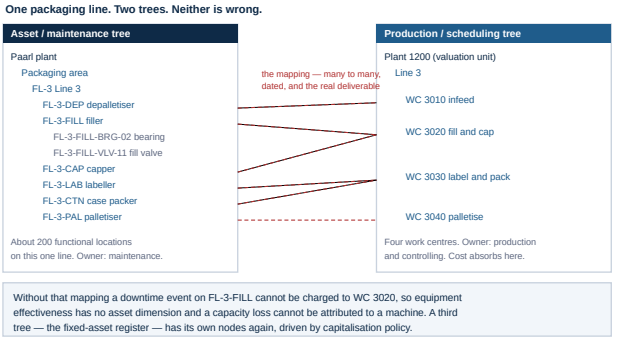


Figure 9.5 The asset hierarchy and the production hierarchy over one packaging line. Both are correct, they do not match, and the dated many-to-many mapping between their nodes is the thing an architect actually has to deliver.

The asset hierarchy and the production hierarchy are two different trees over the same steel. The maintenance tree answers what we maintain, where cost accumulates, what is critical, what is depreciating; its nodes run functional location to equipment to assembly to component, and one packaging line may be 200 of them. The production tree answers what we schedule, where capacity exists, where cost is absorbed; the same line may be four work centres. **The correct design is both trees, with an explicit dated many-to-many mapping between their nodes, and no attempt to make one the parent of the other.** The mapping is the deliverable: it is what lets a downtime event recorded against a functional location be charged against the right work centre.

Three more objects complete the maintenance side. A **task list** is the standard set of operations for a maintenance activity, structurally the same object as a production routing. A **maintenance plan** is the schedule — time-based, performance-based from a counter, or condition-based from a measured threshold, the last two requiring measurement data. And a **measurement point** is a named place where a value is taken, which is where this model meets the plant floor: the same bearing temperature is also a **tag** in the historian.

Supplier, approved manufacturer, source list, call-off

Manufacturing supplier data is not procurement supplier data. Three distinctions do the work.

First, the supplier is not the manufacturer. A distributor sells a part made by a manufacturer in a factory, and in quality-critical manufacturing the identity that matters is the *manufacturer and the site*. So the model needs four objects where most estates have one: the supplier party, which contracts and invoices; the manufacturer; the manufacturing site, with its own certifications and audit history; and the **approved manufacturer entry** — item, manufacturer, site, manufacturer part number, qualification status and evidence, validity. Beside it sits the **source list**. Those two answer different questions. *Is this part, from this maker, at this site, acceptable?* is a quality answer. *May planning buy from this supplier for this plant right now?* is a procurement answer. A part can be on the source list after the site changed and was never re-qualified — a quality incident waiting to happen, which modelling the two as one list makes undetectable. **Manufacturer part number is a first-class identifier**, not free text in a description, because it is how the entry is checked at receipt.

Second, qualification status is dated and evidence-backed: the site or supplier, the scope, the status, the evidence, valid-from, valid-to, re-audit due. Scope matters, because qualified for machining is not qualified for heat treatment. And a lapsed qualification must be *detectable by query*, which is only possible if the dates are fields rather than a scanned certificate.

Third, scheduling agreements and call-offs are a different shape from purchase orders. A purchase order is a discrete commitment. A scheduling agreement fixes terms, total quantity, validity and price conditions **without fixing delivery dates**, and schedule lines are issued against it. The part architects miss is that schedule lines carry a **commitment classification** — firm, where the supplier may produce and will be paid; semi-firm, where they may buy material; and forecast, which carries no commitment. Automotive release schedules do this formally, and Highveld Precision Components lives on it. So schedule *versions* must be retained, because a supply dispute is about what was communicated when; the firm-and-forecast boundary must be a field; and cumulative received quantity must be tracked, because a cumulative mismatch is the commonest electronic-data-interchange reconciliation failure in the sector.

The plant hierarchy, and the reconciliation nobody budgets for

Two models describe the same physical plant, they do not match, and reconciling them is a real project nobody puts in a plan.

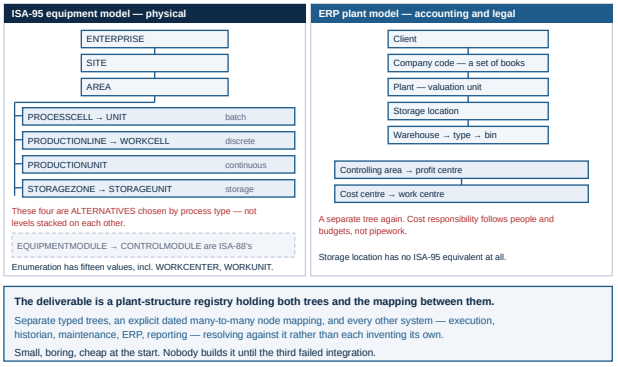


Figure 9.6 The ISA-95 equipment model against the ERP's plant model. The middle of the ISA-95 structure is a set of alternatives selected by process type, not a stack of levels, and equipment module and control module come from ISA-88.

The **ISA-95 equipment model** is the standard reference decomposition of a manufacturing enterprise, and two things about it are routinely misreported. **First, it is not a seven-level stack ending in control module.** The published equipment-element level enumeration carries **fifteen** values, and the middle of the structure is a set of **alternatives chosen by process type, not levels stacked on each other**: a batch plant uses process cell and unit, a discrete plant production line and work cell, a continuous plant production unit, a warehouse storage zone and storage unit [1]. `WORKCENTER` and `WORKUNIT` are the *generic* terms [1]. **Second, equipment module and control module are ISA-88 concepts referenced by ISA-95, not ISA-95's own** [1], and the batch-control lineage explains why they exist [3]. ISA-95's part 1 is at its 2025 edition, which most circulating material does not reflect [2].

The ERP's structure is driven by *accounting and legal* boundaries: client; company code, a legal entity with a set of books; plant, a valuation and planning unit that is also often a physical site; storage location; then warehouse number, storage type, section, bin — and separately a controlling area, profit centres, cost centres and work centres. They fail to line up in predictable ways.

Mismatch	Why it happens	What it costs
One ERP plant covers several ISA-95 areas	A plant is a valuation unit; areas are physical	Plant measures cannot be decomposed to the units that drive them
Two ERP plants on one site	Two legal entities behind one fence	A movement between buildings becomes an intercompany transfer, not a forklift trip
One ISA-95 unit maps to several work centres	Work centres are costing and scheduling constructs	A downtime event cannot be attributed without a mapping rule
Storage location has no ISA-95 equivalent	It is an accounting subdivision	A bin-level position has no accounting home
ISA-95 depth exceeds ERP depth	Control modules exist in the controller, not in the ERP	Tag-to-asset binding has no ERP anchor

The pattern that works is small, boring and high-value: **a single governed plant-structure registry** holding both hierarchies as separate typed trees, plus an explicit, dated, many-to-many mapping between their nodes, with every other system resolving against it. Almost nobody builds it until the third failed execution-system integration. The alternative is the mapping encoded in integration code and tribal memory, where it cannot be queried or corrected.

The model on one page

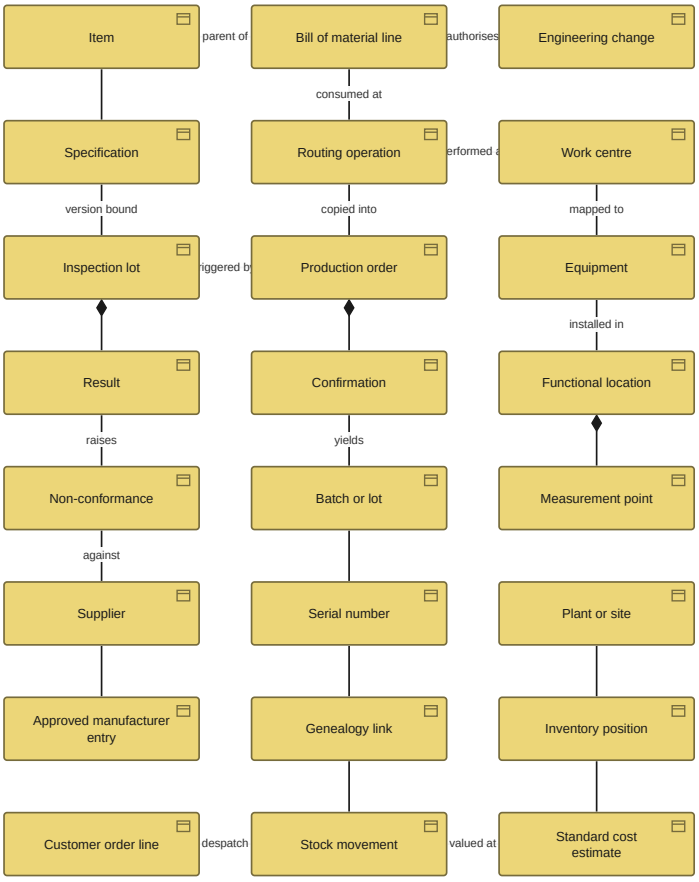


Figure 9.7 The conceptual model: twenty-four business objects and the relationships that carry the most weight. Edges from every object back to the item are omitted on purpose — drawing them would make a star and teach nothing.

Two things about that drawing. It is **abridged on purpose**: every object references the item, and those edges are left out because drawing them makes a star and teaches nothing. And it is **the beginning of a conversation** rather than a conclusion: the value of putting it on a wall is that a production manager points at it and says the thing that is wrong, which is the fastest structural review available anywhere.

Karoo Industrial Group, its people, its plants and its systems are invented, and nothing here is a case study.

THE CHAPTER IN TEN LINES

- A conceptual model is about twenty business objects with no attributes, no keys and no system names, and its value is the consequence stated for each one.
- A manufacturer's model is a product model, a process model and a physical model in one coat, and most serious failures are two of the three collapsed into one table.
- The item master is co-owned by six or seven functions who each need the record to exist before they can fill in their part, so it is a workflow with a table at the end of it.
- Base unit of measure is immutable once any transaction has posted, and a quantity stored without its unit is a defect.
- Engineering, manufacturing, planning and service bills of material are different structures over the same parts, not versions of one structure.
- Phantom is a property of the BOM line, not of the item, and an item-level flag makes legitimate dual use impossible.
- Ten per cent scrap means dividing by 0.90 and starting 111.1, not adding ten per cent, and the error compounds down every level of the structure.
- Interchangeability is directional: B replacing A does not imply A replacing B, and one-way interchange is the normal supersession case.
- Backflush where substitution happens produces a confidently false genealogy, which is worse than a gap; lot-specific issue is the only defensible design in regulated or serialised production.
- Where-used says which products could contain a contaminated lot; only genealogy says which ones do.
- Ownership is a first-class attribute of inventory and cannot be retrofitted onto historic movements.
- The ISA-95 equipment model and the ERP plant model do not match, and the asset tree and the production tree are two different trees over the same steel.

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- **5.** IFRS Foundation, IAS 2 Inventories, issued standard (2023 annotated text) https://www.ifrs.org/content/dam/ifrs/publications/html-standards/english/2023/issued/ias2.html
- **6.** National Regulator for Compulsory Specifications Act 5 of 2008, consolidated text (SAFLII), section 20 https://www.saflii.org/za/legis/consol_act/nrfcsa5o2008543/

CHAPTER 10

The information model

Chapter 9 said what the objects are. This one says how they are held, mastered, moved and reported on — including the layer no other industry in this series has, and the reason a single corporate retention period is always wrong.

Chapter 9 said what the objects are. This chapter says how they are actually held, moved, mastered and reported on — which is where a conceptual model stops being a diagram and starts being a programme. Five things carry it: who owns the creation of master data, how plans and costs are held as data, the layer no other industry in this series has, the reporting layer and the reconciliation that is always wrong, and retention, which in manufacturing is a genuine constraint on the design rather than a policy appendix.

Master data governance, and the item-creation workflow

Chapter 9 ended the item-master section with a line worth repeating: the item master is not a table, it is a workflow with a table at the end of it. Manufacturing master data governance is harder than its equivalents elsewhere for one structural reason. **The item is co-owned by six or seven functions, and each needs the record to exist before it can supply its part of it.** That circularity is the whole governance problem, and no amount of ownership policy dissolves it.

So "who owns item creation?" is the wrong question, and showing why is the useful move. Nobody owns the record. Different functions own **different fields of it**, and what is owned centrally is the process, the rules and the arbitration.

Owned centrally	Owned by a function
The decision that this is a new item and not a duplicate	Engineering attributes, revision, specification reference
The number	Plant planning parameters, per plant
The rules each view must satisfy	Valuation class, price control, cost estimate
The arbitration when two functions disagree	Inspection setup, specification binding, shelf life
The release to active, on evidence of completeness	Tariff code, country of origin, storage and hazard class

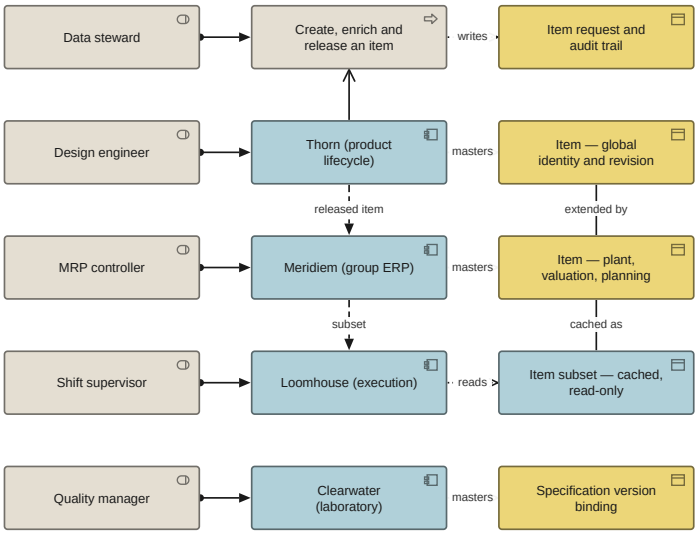


Figure 10.1 Where the item master is actually mastered, and how it flows. The global item and its revision belong to the product lifecycle system, the plant, valuation and planning extensions to the ERP, and everything below holds a cached read-only subset. The roles on the left are the people who own the fields.

The workflow is the deliverable, and it has nine steps. **Request**, with the intended use — buy, make, sell, consume, service — because intended use determines which views are mandatory. **Duplicate check**, before anything else, by description tokens, classification, manufacturer part number, specification and physical characteristics; not by description string equality, which catches nothing. **Classification** assignment, which is what makes the next duplicate check work. **Number assignment. Parallel field collection** by owning function, with a *completeness state per view* rather than one status. **Validation** against rules: a batch-managed item needs a shelf life, a sold item needs a sales unit, a hazardous item needs a dangerous-goods class and a safety-data-sheet reference. **Release per view**, so an item can be purchasable before it is sellable. **An audit trail** of who asserted which field when, retained for the life of the item. And **periodic re-validation**, because the mandatory set changes when the regulations do.

> Two properties separate a workflow that works from one that is bypassed. A **fast path** — an emergency item created with a minimum set and an enforced completion deadline, because a line stopped at 02:00 will not wait for a workflow. And a **measured cycle time**, because the real reason governance is bypassed is that it is slow. The fix is to make it fast, not to make it mandatory.

Part-number intelligence, and the argument against it

"Intelligent" part numbers encode meaning in the number: `BRG-6205-SKF-STD` tells a human it is an SKF 6205 bearing. The practice is deeply embedded, and the argument against it is strong.

In favour: humans read it, mistakes are visible, sorting groups like with like, and it works without a system. Against, and these are the reasons it loses:

- **The classification changes and the number cannot.** Restructure the commodity taxonomy and either every number is wrong or every part is renumbered — and renumbering is impossible once the number is on drawings, in customers' systems, on labels and in historic genealogy.
- **It runs out.** Two digits for a category is 99 categories until it is 100.
- **It encodes attributes that are properly *data*** — material, size, supplier — so those attributes exist twice and disagree. The number wins in human use and the field wins in queries, so there are two truths.
- **It is ambiguous under change.** A part whose material changes has a number asserting the old material. Change the number and interchangeability breaks; keep it and the number lies.
- **It leaks** a supplier's identity and sometimes a cost structure onto a part a customer holds.
- **It makes the number a schema**, so every numbering decision becomes a data-modelling decision taken by whoever happens to be creating a part that afternoon.

The position to take: numbers should be **meaningless, sequential and permanent**; meaning lives in **classification and attributes**, which can be restructured freely; and the human need the intelligent number served is met by good search over good classification, which is a solved problem. The migration is hard, so the argument has to be won on the cost of the alternative — which is the next thing.

Duplicate parts as a measurable cost

This is the most useful argument an architect has in this area, because it converts a data-quality complaint into a number. A duplicate costs in seven places: **inventory**, two safety stocks and two reorder points for one demand stream, which is the largest single component; **purchasing**, volume split so neither number reaches a price break; **obsolescence**, where both stocks are written off and the less-used duplicate at close to 100%; **engineering time** re-specifying and re-drawing a part that already exists; **quality**, two qualification records and a real risk that one is qualified and one is not; **transaction cost**, two sets of master data maintained across every view for ever; and **analytical cost**, since spend analysis, where-used and risk analysis are all wrong by the duplication rate.

> Published figures for "the average cost of a duplicate part" and for typical duplication rates in industrial item masters circulate widely. **None of them is reliably sourced, and this book will not quote them.** The method is better than the benchmark anyway: take a sample of the item master, resolve duplicates by specification and manufacturer part number, compute the duplication rate, and multiply by the cost components above using the company's own figures. That takes a few weeks and it produces a number the client cannot dismiss — which a borrowed statistic cannot.

Classification, and why an item has several

Classification is what makes duplicate detection, spend analysis and search work. Manufacturing uses several schemes at once, for different purposes.

Scheme	What it is	Where it is used
UNSPSC	Segment, family, class, commodity — four two-digit levels giving an eight-digit code, with an optional two-digit business function. Developed by UNDP and Dun & Bradstreet in 1998 [7]	Spend analysis and procurement category management. Too coarse for engineering identity
eCl@ss	A European industrial classification *and* property dictionary — classes plus standardised typed properties per class, which is the part UNSPSC lacks [8]	Engineering and maintenance item specification, supplier catalogue exchange. The property dictionary is what enables attribute-based duplicate detection
ECCMA open technical dictionary	A registry of concepts identified by concept identifiers, in the ISO 22745 and ISO 8000 family; ISO 8000 is the master-data quality standard [9]	Where master-data quality must be contractually specified and verified
GS1 identifiers and classification	Global trade item numbers and a product classification	Finished goods going into distribution; case and pallet identification
Company commodity codes	The company's own taxonomy	Nearly always exists, nearly always the one actually used, nearly always mapped to nothing

> **UNSPSC is no longer GS1-managed.** GS1 US managed the code set until 31 December 2024, and **UNDP has handled change requests, revisions and updates since 1 January 2025** [7]. Any procurement document or data-governance standard that names GS1 as the custodian is out of date, and the reference consulted gives the latest code-set version as UNv26.0801, released 14 August 2023, with 158,448 items [7].

The design rule follows: **these are not alternatives, and an item does not have "a" classification.** It carries several at once, each for a purpose, each independently maintained, each dated — a company commodity code for internal category management, UNSPSC for spend analysis, eCl@ss or a

company property set for engineering attributes, a customs tariff code for trade, a dangerous-goods and hazard classification for safety, a waste classification for disposal, and a trade-item identifier family. Force one hierarchy to serve all of them and it serves none.

Plan objects as data

Planning is a chain of progressively firmer objects, and the architectural discipline is to keep them distinct, because each has a different owner, a different authority and a different reversibility.

Object	Grain	Firmness
Forecast	Item or family × location × period	None. A prediction
Demand plan	Item × location × period × version	A commitment to a number, not to an order
Independent requirement	Item × location × date	Demand that exists on its own
Dependent requirement	Item × location × date × source	Derived. **Never editable**
Master production schedule	Item × location × period	Planner-owned, deliberately stable
Planned order	Item × plant × date × quantity	Overwritten by the next planning run
Firm planned order	As above, plus a firming indicator	Not overwritten; the run plans around it
Requisition	Item × plant × date × quantity	Firm internally, no external obligation

Independent versus dependent demand is the central distinction in all of planning, and the rule that follows is absolute and routinely broken: **you may not manually adjust dependent demand.** Doing so decouples the component plan from the parent plan, and the two drift until a shortage appears with no traceable cause. If a component genuinely needs demand of its own — service parts, samples, a planned build-ahead — that is *additional independent demand on the component*, modelled as such, and then visible as a separate, attributable source.

The **horizon and the fence are data constructs, not policies**, and they are three different things. The **planning horizon** is how far forward the plan extends, and it must be at least the cumulative lead time of the deepest structure or long-lead items are never ordered in time; an under-length horizon produces an apparently stable plan with permanent shortages at the bottom. The **planning time fence** is the near-term period inside which the planning run may not create or change orders automatically. The **forecast consumption horizon** is the period in which actual orders consume forecast rather than adding to it, and a misconfigured one is a common cause of an inflated plan.

The fence is a governance control expressed as data. Its data requirements are specific: the fence length per item-plant, the firming rule — automatic on entering the fence, or manual — an **exception log of every change made inside the fence**, and a reason code on each. Without the log the fence is advisory and the plan churns anyway. Multi-zone fences, with different authority per zone, are the mature pattern.

Two more properties are worth insisting on. **Pegging** is the recorded chain from a supply element up through every dependent requirement to the independent demand that caused it, and it matters because a planner's real question is never "what is the plan?" but "why is this order here?". Pegging is expensive to maintain and it is the difference between a planning system people use and one they override. And **plan versions must be retained**: the plan as it stood at the moment of commitment is the only basis on which forecast accuracy can be measured, and forecast accuracy is the only basis on which safety stock can be set rationally. A demand plan maintained in place has no accuracy and therefore no improvement path.

Costing as data, and the standard that constrains it

A standard cost is not a number, it is a **structure**: the bill of material and the routing exploded for a given lot size at a given date against given prices and rates. Material from the component quantities including scrap and yield, labour and machine from activity quantities times activity rates, external from subcontract prices, and absorbed overhead from a base times a rate.

The cost component split is the requirement that the result be carried decomposed and **preserved through every level** of the explosion: raw material, packaging, direct labour, machine, setup, energy, subcontract, scrap, variable and fixed overhead, inbound freight, duty. Three properties make it worth the effort. It survives roll-up, so a parent's "material" component contains its children's material and not their total cost. It makes margin analysis possible by driver rather than in aggregate. And it is what tells you whether a price increase is labour or copper.



Figure 10.2 The cost component split surviving a three-level roll-up. The parent's material component contains its children's material, not their total cost — which is the only form in which a cost change can be explained. Illustrative figures in rand, October 2026.

A standard cost stored as a single number cannot be explained, and a cost that cannot be explained will be disbelieved and worked around. In South Africa the split has a second, harder use: the Automotive Production and Development Programme is a customs-duty-rebate regime that requires country-of-origin and cost-component detail propagated through multiple supply tiers, so a costing model that does not preserve the split through a multi-level roll-up physically cannot substantiate a claim. Chapter 16 does the regulation; the point here is that the data model decides whether the claim is possible at all.

Activity rates are period-dated, because a rate is the cost centre's planned cost divided by its planned activity quantity for a period — so a cost estimate is only meaningful with its rate period stated. And the rate embeds a **planned volume assumption**, which is the mechanism by which under-utilisation becomes an unabsorbed cost.

The accounting standard constrains the data model

This is where an architect can add real value, because the constraint is specific and most designs miss it. IAS 2 is the inventories standard, both South Africa and Canada report under IFRS, and three of its paragraphs are design requirements rather than accounting niceties [6].

Paragraph	What it requires	What the data model must hold
9	Inventories measured at the **lower of cost and net realisable value**	Both numbers, per item per period — not one
13	Fixed overheads allocated on **normal capacity**; the per-unit allocation **not increased** for low output or idle plant; unallocated overheads **expensed in the period**	Normal capacity, actual activity, the rate, the absorbed amount and the under- or over-absorbed residual, as separate dated facts per cost centre per period
16(a)	**Abnormal** wasted material, labour and production cost expensed as incurred, not capitalised into inventory	Scrap separable into normal and abnormal — which needs the scrap standard, the actual scrap, and reason codes on the actual
21	**Standard cost may be used** where the result approximates cost and standards are regularly reviewed	The standard-cost version history and the variance analysis, which are the *evidence* that the condition is met
25, 27	Cost assigned using **FIFO or weighted average**, consistently within a class	The cost formula as a stored, disclosable attribute per class

> Two corrections worth making out loud with a client. **"Standard costing is not IFRS-compliant" is wrong** — IAS 2 paragraph 21 expressly permits it, subject to approximation and regular review [6]. And **LIFO is not an option under IFRS**: paragraph 25 lists first-in-first-out and weighted average only [6]. That is the largest single costing divergence from United States practice, and it catches architects who learned costing from North American material.

The material ledger is the alternative to all of that, and it is expensive. Standard costing values every movement at standard and throws the differences into variance accounts. Actual costing on the material-ledger pattern instead keeps, per material per plant per period, the standard price, the quantities and values of every receipt and issue, and the accumulated differences — then at period end computes a periodic unit price and **rolls the differences up the structure**, so a raw-material price variance lands in the cost of the finished good that actually consumed it rather than in a general variance account. Three data consequences follow. Multi-level price determination needs the full consumption graph for the period, not just balances. The close must run bottom-up by low-level code. And the result is a *second* valuation of the same inventory, which has to be reconcilable to the first. Without the full period graph, multi-level determination silently falls back to single-level and the answer is wrong in a way nobody notices.

Variance categories exist because each has a different owner: input price belongs to purchasing, input quantity to production or to a wrong bill of material, resource efficiency to production, scrap to production and quality, mix to production and planning, lot size to planning, and output price to finance. A single "production variance" number is attributable to nobody and therefore acted on by nobody. The categorisation is the deliverable.

Period-end settlement is an ordered sequence — post everything, calculate work in progress for incomplete orders, calculate variances for complete orders, settle them to stock or to a price-difference account, revalue at actual cost if the material ledger is in use, close — and the order matters, because it runs bottom-up by structural level. One property of it generates a problem later in this chapter: **once the period is closed those costs are frozen, while the plant's own quantity records may still be being corrected.**

Time-series and tag data, which no other industry in this series has

A **tag** is a named, typed, continuously sampled signal: a temperature, a pressure, a flow, a motor state, a counter, a setpoint, an alarm condition. A mid-sized plant has tens of thousands; a large process plant hundreds of thousands. A tag is not a column and not a row. It is **an object with its own master data**: name, description, engineering unit, data type, source address or node, scan rate, compression settings, archival policy, valid range, alarm limits, asset binding, process-function binding, quality handling, calibration reference.

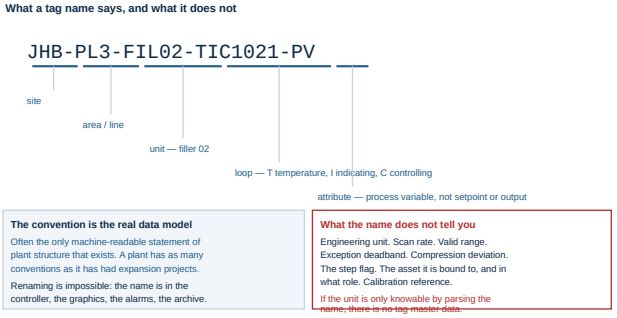


Figure 10.3 A tag name, taken apart. The convention is the only machine-readable statement of plant structure in many estates, and it is also not a substitute for tag master data.

Naming conventions are the plant's real data model, because the tag name is in practice the only metadata many estates have. A disciplined convention is a compressed composite key carrying site, area, unit, loop and attribute, and the loop letters are not arbitrary: `TIC` means temperature-indicating-controller under the instrumentation identification standard, which is why an engineer can read `FIC`, `LAH` or `PDT` without a data dictionary [10]. Four consequences follow. The convention is often the only machine-readable statement of plant structure that exists, so an asset hierarchy built by parsing tag names will be right most of the time — which is exactly why the exceptions are so damaging. A plant has as many conventions as it has had expansion projects: the 1987 units use one, the 1999 debottleneck another, the 2014 line a contractor's house style, the 2022 packaging hall the machine builder's defaults. Renaming is effectively impossible, because the name is referenced in the controller, the graphics, the alarm database, the archive, every report, every spreadsheet and the piping diagrams — **so the convention you inherit is the convention you keep.** And the name is not a substitute for metadata: if the only way to know a tag's unit is to parse its name, the estate has no tag master data.

Asset-tag binding is the highest-value piece of plant data integration there is. A tag bound to a functional location or a piece of equipment, and through that to a node in the plant-structure registry of chapter 9, can answer questions in business terms: the effectiveness of this line, the energy per tonne of this unit, the temperature excursions on the batches of this product. An unbound tag answers questions only in tag terms, which means a human translates every time, which means the data is used by three people and trusted by none. The binding is `(tag, asset node, role, valid-from, valid-to)` — dated, because instrumentation is replaced and re-pointed, and **role**-typed, because "the inlet temperature of" is a different fact from "the ambient temperature near".

Why a historian's data model is not relational, and should not be forced to be

Six structural reasons, each with a consequence.

1. **The write pattern is append-only, monotonic in time and enormous.** A hundred thousand tags at one sample a second is 8.64 billion points a day. Row-per-sample in a general-purpose relational table is the wrong physics: index maintenance on every insert, no locality, and a table that cannot be queried while it is written.
2. **Compression is semantic, not generic.** A historian stores *changes*, so the stored series is a sparse, lossy-by-design representation from which values at arbitrary times are interpolated — not a dense table of readings.
3. **Queries are range-and-aggregate, not point-and-join.** "Average of these 40 tags over these 90 days in 15-minute buckets" is the normal query.
4. **Every value has a quality.** A reading is `(timestamp, value, quality)`, where quality may be good, bad, substituted, questionable, out-of-service or stale. **Analytics that ignore quality produce confidently wrong averages**, because a frozen sensor reads a plausible constant and a failed one often reads zero. There is no equivalent in ordinary relational data, and it is the commonest failure in first-generation plant analytics.
5. **Time is contested.** Source, gateway and server timestamps differ, clocks drift, and devices buffer and replay after an outage — so **late-arriving data in the past is normal, not exceptional.** Historians handle backfill natively; a relational pipeline usually does not.
6. **Retention is tiered by nature**: raw at full rate for weeks, minute aggregates for a year, hourly for a decade, with full-rate raw kept indefinitely for regulated periods and investigated events. That is a policy per tag group, not a global one.

The rule to state plainly: **do not move the series. Move the question to the series, and move the aggregate to the warehouse.** The historian owns the raw series. The analytical layer holds decision-grain aggregates — per shift, per order, per batch, per equipment, per downtime event — computed with explicit quality handling and tagged with the calculation version. Break that and three things follow: replicating raw tags into a relational warehouse collapses on volume and cost, computing effectiveness in two places with two slightly different definitions collapses credibility, and storing aggregates without a calculation version makes them irreproducible.

Exception and compression silently change a reported number

This is the part that catches everyone, and it is concrete enough to work through.

A historian of the dominant type applies **two separate reduction mechanisms at two different points in the pipeline**, both using deadband-style tests [1]. **Exception reporting** happens at the interface, before data is sent: the interface sends only values differing enough from the last sent value, which conserves network bandwidth. Its settings are a deadband in engineering units, the same deadband as a percentage of the tag's span, and a maximum time between events so that a value is sent even when nothing changes [1]. **Compression** happens at the archive, using a swinging-door algorithm that "removes data that can be safely recreated by interpolation" [1]: it tracks upper and lower slopes from the last archived point and archives a point when a new value's slope falls outside those bounds, with a compression deviation in engineering units or as a percentage of span, and a maximum interval between archived values [1].

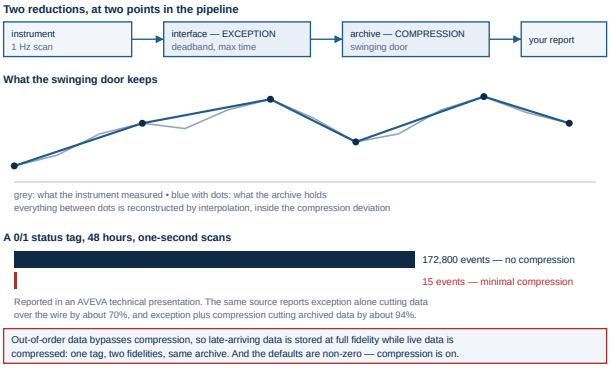


Figure 10.4 Exception at the interface, compression at the archive, and what they do to a reported number. Magnitudes from an AVEVA technical presentation; the sparkline is schematic.

Three facts about that mechanism matter more than the mechanism itself.

The magnitudes are large. A 0-or-1 status tag over 48 hours at one-second scans is **172,800 events with no compression and 15 events with minimal compression** [1]. An offshore pilot cited in the same source reports exception alone cutting data over the wire by about **70%**, and exception plus compression cutting archived data by about **94%** [1]. A downstream oil-and-gas study in the same material found reductions from 0.2% to 95%, with most tags reduced by 25% or more, while all tags had maximum deltas below measurement precision [1] — which is the honest counterweight: done properly, compression does not move the value beyond what the instrument could resolve anyway.

The fidelity is not uniform within one archive. Out-of-order data bypasses compression [1], so late-arriving data is stored at full fidelity while live data is compressed, and the same tag therefore has two fidelities in the same archive depending on how the data arrived. Collection mechanisms differ too: in the same vendor's material, adapters apply deadband filtering but ignore exception settings, and connectors perform no exception testing at all [1]. **And the defaults are non-zero, which means compression is on unless somebody turned it off** [1].

Therefore every query against a time grid is interpolating, and interpolation is a modelling choice. Ask for a value at 09:14:03 and if no point exists at that instant, an analogue tag returns an interpolated value and a stepped tag returns the last known one — and a `step` attribute on the tag decides which. A tag mis-flagged as analogue when it is really a discrete state will return physically impossible intermediate values: a valve 37% open when it has two positions. A time-weighted average over archived points is not the arithmetic mean of the original scans, and the two diverge most where the signal moved fastest, which is exactly during the transients that matter.

> So two reports of the same thing can both be right and differ: one reads archived values and one an interpolated grid; one uses a time-weighted average and one an event-weighted one; one reads the historian and one the execution system's own capture of the same signal at a different rate.
>
> The deliverable is therefore a statement, **per tag class**, of exception and compression settings, the step flag, the retention period, and the interpolation and aggregation method reports must use — with those settings **under change control**, because changing a compression deviation changes the fidelity of all future data for that tag and does not change the past. A tag's series has a discontinuity in information content at the moment somebody tuned it, and nobody records when that was.

That chains straight into data integrity. Where a record is evidence — an emission compliance record, a batch record, a pressure-vessel test — the number in the report has to be reconstructable, and "reconstructable" now means knowing the compression settings and the aggregation method that were in force. A compliance dataset built on the historian is defensible. The same dataset rebuilt as a relational extract, with the settings undocumented, is not.

The reporting layer, and the reconciliation that is always wrong

Effectiveness computed from tag data

Overall equipment effectiveness is the most-quoted and least-comparable number in manufacturing, and precision about which definition is in use is most of the value an architect adds here.

ISO 22400 is the international standard for key performance indicators for manufacturing operations management, and part 2 holds the definitions [2]. Its decomposition is availability, multiplied by effectiveness, multiplied by a quality ratio [3]. Two version facts first, because both are commonly got wrong: **ISO 22400-2's current published edition is 2014** [2], and the second edition was still at draft stage in October 2026 [4]; part 1 of 2014 was *confirmed* rather than revised.

> **The standard's middle term is Effectiveness, not Performance** [3]. That is not pedantry. The widely used total-productive-maintenance formulation calls the middle term the *performance rate* and computes it against a different denominator and a different ideal rate. An architect who writes "performance rate" while citing ISO 22400 is implementing one thing and citing another. And there is no OEE conformance scheme to appeal to: the research behind this book found at least six incompatible availability denominators in use [5], and the widely repeated 85% "benchmark" is called "a convenient, compelling and completely artificial benchmark" by the site that publishes it [5]. **Never put 85% in a client deliverable as a benchmark.**

The data requirements the formula implies are where the work actually is. **Planned busy time is a decision, not a measurement**: is a planned maintenance window excluded, a no-demand idle shift, a changeover, a tea break? Each choice moves the number by points, so two plants with identical physical performance can report figures far apart purely through the time model. The time model must therefore be an explicit, governed, **versioned** artefact — a partition of all calendar time into mutually exclusive, exhaustive, typed states — and the number must carry the time-model version. **Ideal cycle time is a master-data decision** too: nameplate rate, design rate, demonstrated best rate, or the routing standard, each giving a different middle term, and the choice recorded per item-resource and dated. **Produced and good quantities must reconcile to the goods receipt**, or the plant's effectiveness and the plant's output disagree. Everything comes from tag data with quality flags, so a machine-state tag that is bad-quality for twenty minutes is not twenty minutes of running. And **the measure does not aggregate by averaging**: a line's figure is not the mean of its stations', it is computed on the constraint or on the line as one resource.

Downtime reason trees

A downtime event is `(resource, start, end, duration, reason, sub-reason, detecting source, acknowledging user, linked order, linked notification, linked quality event, comment)` — and **the reason tree is the asset**. A flat list of reasons is useless, for reasons worth stating because each is a design decision somebody will get wrong.

- It is a **hierarchy two to four levels deep**: category, then system, then component, then failure mode. Deeper than four and nobody completes it accurately.

- It must be **mutually exclusive and exhaustive at every level**, because time must be fully accounted for. An "Other" node is mandatory and **its share is the quality metric of the tree**: an Other share above a few per cent means the tree does not fit the plant.
- It is **versioned and dated**. Reparenting a node in place silently restates last year's analysis, so trend analysis across a revision needs a mapping between versions.
- **Reason is not cause.** "Motor failure" is a reason; the cause lives in the maintenance order's failure analysis as object-part, damage and cause codes. Two vocabularies, both needed.
- **The first cause and the blocking cause are different.** When station 4 fails, stations 1 to 3 are blocked and station 5 is starved. All five are down and only one has a root cause. Recording five independent events quintuple-counts the loss, so the model needs **event propagation**: one originating event with dependent blocked and starved events linked to it.
- **Micro-stops** below a threshold are invisible to manual reporting and only capturable automatically — and the threshold, commonly one to five minutes, shifts loss between availability and the middle term **without changing the product of the three** [5]. So the composite can be comparable where its components are not.
- **Reduced-rate running** is a loss that is not a stop at all and appears only in the middle term. A plant that measures downtime well and rate loss not at all systematically under-reports its largest loss.

Plant reporting against the financial close

Two cycles, two clocks, two sets of rules. Plant reporting runs on shifts, days and weeks against a production calendar, at the grain of resource, order, operation, batch, product and crew, in physical units, within minutes, and it is **corrected continuously as better data arrives**. Financial reporting runs on accounting periods that need not align to shifts or weeks, at the grain of cost centre, profit centre and account, in currency, days after period end, and it is **frozen at close**, with prior-period corrections going to the current period.

> *The plant's numbers keep improving after the period closes and the financial numbers do not. A quantity corrected on the 4th for a movement on the 29th lands in the new period financially and in the old period operationally. That is not an error in either system; it is the designed behaviour of both.*

So the difference is permanent and irreducible, and **the architecture's job is to make it explained rather than to make it zero.**

The plant-to-finance reconciliation

The most useful single artefact in a manufacturing reporting architecture is a standing, automated, line-item reconciliation between plant output and financial output, with every difference attributed to a named, typed cause.

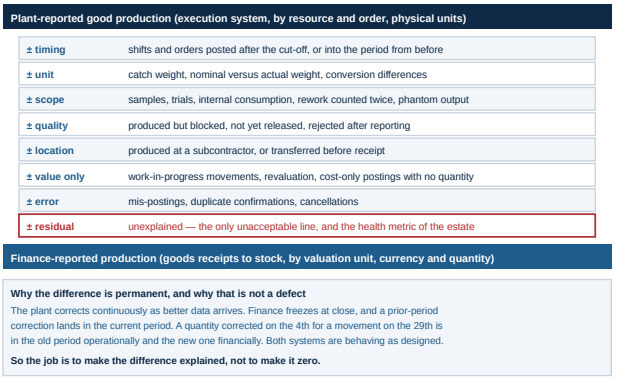


Figure 10.5 The plant-to-finance reconciliation. Every difference needs a type and an owner; the residual is the only unacceptable line, and its size is the health metric of the whole data estate.

Four rules about it. **It must be a standing report, not a year-end exercise** — run monthly it finds process defects, run annually it finds an unexplainable number and a search for somebody to blame. **Every difference needs a type and an owner**, and an unexplained residual is the only unacceptable line; its size is the health metric of the whole data estate. **It requires the plant-structure mapping of chapter 9**, because without a work-centre-to-cost-centre and plant-to-site mapping the two sides cannot even be put on the same rows. And **it is the single best diagnostic available for the quality of a manufacturing data architecture**, because every defect in chapter 9 eventually shows up in it: unit conversions, scrap factors, backflush errors, misposted stock types, unsettled orders, unmapped hierarchies.

> An architect who wants to know how bad a manufacturing data estate is should ask for this reconciliation and watch what happens. **Where it does not exist, the request itself is the finding.**

The measure definition register

Which is why the first deliverable of any group manufacturing reporting programme is not a dashboard. It is a **measure definition register**: one row per measure, with its formula, its time scope, its unit, its data sources, its owner, its version, and — the column everyone forgets — its **declared comparability scope**, saying whether this measure may be compared between lines, between sites, or only with itself over time. The architectural value of ISO 22400 is less its formulas than that discipline [2].

Six measures need the register most, because each has four or more defensible definitions: *a unit made* (started, completed, completed good first time, shipped, invoiced), *on time* (against request date, first promise, latest promise or a customer-managed schedule), *yield* (which stage boundary, and does

rework count), *capacity* (theoretical, rated, effective or demonstrated), *cost per unit* (standard or actual, at what absorption volume), and *availability* (which denominator). Two plants reporting 96% and 78% on-time-in-full may be performing identically. A plant quoting 98% yield and one quoting 92% first-pass yield may be the same plant.

Moving data between systems

Chapter 13 does the mechanics of the information-technology-to-operational-technology boundary. Three things belong here, because they are information-model problems rather than integration problems.

Unit-of-measure conversion across a system boundary needs an owner. If the execution system converts, the historian does not, and the ERP converts with a different dated factor, then three systems hold three quantities for one event. The rule: convert once, at a named boundary, with the dated factor recorded on the message, and carry both the transacted unit and the base quantity.

Time synchronisation, because timestamps disagree for structural reasons and not through carelessness. A controller clock drifts, a gateway stamps on receipt, a server stamps on write, and a buffered device replays an hour later. Every integration therefore needs a stated clock authority, a stated tolerance, and a rule for which timestamp is authoritative for which purpose — the source clock for process reconstruction, the posting clock for financial attribution.

The grain mismatch between a tag reading and a production confirmation. A tag reading is an instant; a confirmation is an interval with a quantity. Joining them is a join on *time ranges*, not on keys, and the interval boundaries — order start and end, batch phase transitions — come from the execution system and the ERP. **That join is the core of every manufacturing analytics platform, and its correctness depends entirely on clock alignment and asset-tag binding.** Both are unglamorous, and both are where these projects fail.

Retention as a data-model constraint

In most industries retention is a policy appendix. In manufacturing it is a design constraint, for one reason that is easy to demonstrate and hard to argue with.

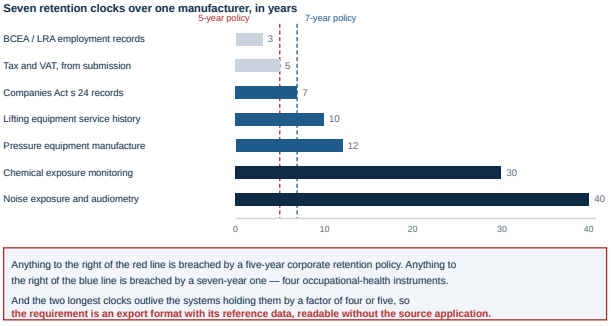


Figure 10.6 Seven retention clocks over one manufacturer, against the two retention policies a company is most likely to write. Each period needs re-checking at publication; chapter 16 carries the instruments.

Record class	Period	Instrument
Chemical exposure assessments and air monitoring	**Minimum 30 years**	Hazardous Chemical Agents Regulations 2021, reg 9(e) [11]
Noise exposure assessment, monitoring, audiometry including baseline	**40 years**	Noise-Induced Hearing Loss Regulations, reg 11(e)–(f) [12]
Pressure equipment manufacturing records, held by the manufacturer	**Minimum 12 years**	Pressure Equipment Regulations, reg 14(3) [13]
Lifting equipment full service history	**At least 10 years**	Driven Machinery Regulations 2015, reg 18(7)(c) [14]
Company records, annual financial statements, accounting records	**Seven years**, longer where another regulation says so	Companies Act 71 of 2008, s 24 [15]
Tax records where a return was submitted; VAT records	**Five years from submission**	Tax Administration Act, s 29(3)(a); VAT Act, s 55(1)(a) [15]
Employee name, occupation, hours, remuneration; collective-agreement records	**Three years**	Basic Conditions of Employment Act, s 31; Labour Relations Act, s 205 [15]

> **A single corporate retention period is always wrong.** A five-year policy breaches the Companies Act. A seven-year policy breaches four occupational-health instruments. The obligation is **per-record-class, not per-company**, and the data model has to carry a retention class on the record rather than a retention setting on the system.

Then the consequence nobody budgets for. The two longest clocks — 30 years for chemical exposure data and 40 years for noise exposure and audiometry — sit on data held in occupational-health and monitoring systems that are replaced every seven to ten years. **The retention obligation outlives the application by a factor of four or five, so a retention design that assumes the system will still exist is not a design.** What is actually required is an **export format with its reference data, readable without the source application** — which is an unglamorous, cheap thing to build at the start and an impossible thing to reconstruct afterwards. Chapter 16 works through the instruments and what an inspector asks to see; the data consequence is this one, and it belongs in the information model from the first version.

Everything in this chapter about Karoo Industrial Group, its plants, its people and its systems is invented, and the regulatory material is architecture material rather than legal advice: it says where evidence has to live, not whether anybody complies.

THE CHAPTER IN TEN LINES

- Nobody owns the item record; functions own different fields of it, and what is owned centrally is the process, the rules and the arbitration.
- An item-creation workflow without a fast path and a measured cycle time will be bypassed, and the fix is to make it fast rather than mandatory.
- Part numbers should be meaningless, sequential and permanent, because meaning belongs in classification and attributes that can be restructured.
- No published figure for the cost of a duplicate part is reliably sourced; measure the duplication rate on a sample of the client's own item master instead.
- UNSPSC is no longer GS1-managed: GS1 US ran it to 31 December 2024 and UNDP has handled it since 1 January 2025.
- An item carries several classifications at once, each dated and each for a purpose; one hierarchy serving all of them serves none.
- Dependent demand may never be manually adjusted, and a planning time fence shorter than the cumulative lead time cannot be enforced.
- IAS 2 paragraph 21 expressly permits standard costing, and paragraph 25 lists only FIFO and weighted average, so LIFO is not an IFRS option.
- IAS 2 paragraph 13 is what makes normal capacity, actual activity and the absorption residual separate stored facts rather than accounting detail.
- Historian exception and compression silently change a reported number: a status tag over 48 hours at one-second scans goes from 172,800 events to 15.
- Out-of-order data bypasses compression, so one tag can hold two fidelities in the same archive, and the defaults mean compression is on unless somebody turned it off.
- ISO 22400-2's middle term is Effectiveness, not Performance, and its current published edition is 2014.
- The plant-to-finance reconciliation is the best single diagnostic of a manufacturing data estate, and where it does not exist the request is the finding.
- A single corporate retention period is always wrong: five years breaches the Companies Act and seven years breaches four occupational-health instruments.

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- **13.** Pressure Equipment Regulations, GN R.734 of 15 July 2009, consolidated (SAFLII), regulation 14(3) https://www.saflii.org/za/legis/consol_reg/per354/
- **14.** Driven Machinery Regulations, 2015, GN R.540, consolidated (SAFLII), regulation 18(7)(c) https://www.saflii.org/za/legis/consol_reg/dmr2015283/
- **15.** SAICA, Guide on the Retention of Records, June 2023 — Companies Act s 24, Tax Administration Act s 29(3)(a), VAT Act s 55(1)(a), BCEA s 31, LRA s 205 https://saicawebprstorage.blob.core.windows.net/uploads/Guide-on-Retention-of-Records-June-2023.pdf

CHAPTER 11

The conceptual application model

The twelve applications a manufacturer needs, named by what they do rather than by product — with the roles that use each one and the data each one masters. Draw it before you know what the client owns, because its real job is to show you the boxes nobody filled.

The model you draw before you know what they own

A conceptual application model names the applications a business needs, described by what they do rather than by whose logo is on them. It is the artefact you draw in your first week on a manufacturing engagement, on a whiteboard, with a plant manager in the room, before you have the inventory of what is installed and before anybody has said the word *vendor*.

It is worth drawing first for a reason that sounds backwards. The purpose of the model is not to describe the estate. Chapter 12 does that. The purpose is to give you a list of the things a manufacturer must have an application for, so that when you walk the estate you can notice the boxes nobody filled. In manufacturing those gaps are consistent and they are expensive: detailed scheduling, quality, and maintenance. A plant will have an ERP it paid millions for, a control system that runs the physical process beautifully, and no application at all for deciding what order to run things in, no application for its specifications and test results, and no application for the equipment that stops twice a week. All three functions are being performed. They are being performed in Excel, by one person, who is on leave.

The second reason to draw it early is that it is the only diagram in your pack that a production manager can argue with on equal terms. Johan Terblanche at Highveld Precision Components will not engage with a technology reference architecture and should not be asked to. He will engage, hard, with a page that says *this is the thing that decides which job runs next*, because he does that job every morning with a printed list and a pencil, and he will tell you exactly which box is wrong.

Twelve capabilities is the right order of magnitude. A single-plant job shop needs about seven of them. A regulated pharmaceutical manufacturer needs all twelve and will carry a thirteenth for validation and change control. A group like Karoo Industrial Group — invented for this book, two operating businesses, three plants, nothing about it real — needs all twelve and has to decide whether each is one instance for the group or one per plant, which is the second-hardest question in the chapter.

Three layers, and why confusing them costs a year

Three things get drawn on one page and the result is unusable. Keep them apart.

A **business capability** is something the business can do: design a product, plan supply, make to a schedule, ship and invoice, keep the plant running, prove the product conforms. It is independent of how, it has an owner with a job title, and it changes about once a decade.

A **conceptual application capability** is a coherent lump of software function that realises part of one or more business capabilities. "Detailed scheduling" is one. It is independent of who supplies it. It changes when the industry changes, which in manufacturing means when a standard moves or a customer imposes a requirement — electronic batch records became a capability because regulators made records reviewable, and serialisation became one because two legislatures made a unique pack identifier compulsory for medicines.

A **system** is a thing with a licence key, a version, a support contract and a renewal date. Loomhouse is a system. Meridiem is a system. Pastorale, which is what Karoo Foods actually uses to run its batch records, is forty Excel workbooks and an Access database, and it is a system too, because it has a maintainer and a failure mode.

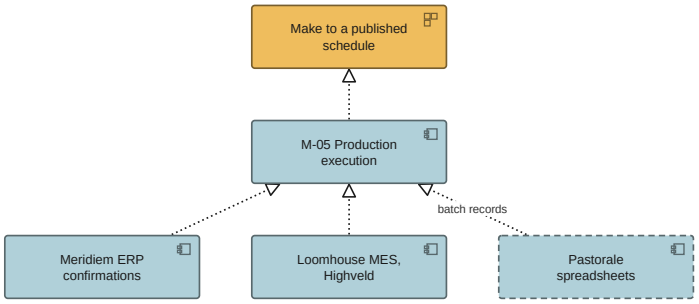


Figure 11.1 Three layers, one capability threaded through them. Realisation runs upward and is many-to-many: the spreadsheet estate realises part of production execution at Karoo Foods, which is why its batch records cannot be reviewed by exception.

The relationship between the layers is realisation and it is many-to-many in both directions. One application capability is realised by three systems; one system realises nine application capabilities, which is exactly what an ERP is. Draw the layers on separate pages and the conversation about replacement becomes tractable: you can say *we are replacing the system that realises six of these, and four of the six are realised better elsewhere already*. Draw them on one page and you have produced a picture that supports no decision.

> In manufacturing there is a fourth layer that other industries do not have, and it belongs in the technology architecture rather than here: the physical equipment. A filling head, a pasteuriser, a press. It is tempting to put it on the application model because the control logic that runs it is software. Resist. The control logic is M-06; the steel is a technology-layer `Equipment` element and chapter 17 shows how to model it.

What goes in a box

Four tests, and they settle most arguments in about a minute each.

One: is it named by what it does, not by who sells it? "MES" is not an application capability. It is a market category, and three manufacturers mean three different things by it. "Production execution" is a capability. "Historian" is a product category; "process data historisation" is a capability, and once you name it that way you can ask whether a plant's historisation is done by a historian, by the DCS's own archive, or by a technician exporting a CSV every Friday.

Two: does it master something, or decide something? A box that neither holds authoritative data nor makes a decision is a screen or an interface, and it belongs one layer down. This test alone removes "dashboards", "mobile" and "reporting portal" from the model.

Three: could a different supplier provide it tomorrow without changing the business? If not, you have described an implementation and should rename it. The test has a sharp edge in manufacturing: process control usually fails it, because the control logic is inseparable from the equipment it drives and from the vendor that engineered it. That is a real finding about the capability, not a flaw in the test, and it is why M-06 is the one box on the page you should never expect to put out to tender on its own.

Four: does it have a grain? Every manufacturing application capability has a characteristic unit of data, and the grain is the single most useful attribute you can write next to a box. M-01's grain is a financial document. M-05's grain is a unit, a container or an operation instance. M-07's grain is a tag value at a timestamp. Two capabilities with grains three orders of magnitude apart cannot be the same application, however much a vendor wants them to be, and the grain mismatch is where the integration cost lives.

Three things that are not application capabilities

"Industry 4.0." It is a programme name, a funding line and sometimes a department. It is not something the manufacturer does that another box does not do. A model with an Industry 4.0 box has a box that overlaps nine others, and the first time the model is used to scope anything, somebody will claim their project is inside it.

"Integration." Integration is how capabilities reach each other. In manufacturing it is expensive, difficult and the subject of most of chapter 13, and it belongs in the technology architecture. A box called Integration on a conceptual application model is an admission that the boundaries between the other boxes were not thought through, because the integration requirement *is* the boundary.

"The digital twin." The term covers at least five different things, and until a client says which one they mean there is nothing to put in a box. Chapter 12 separates them. On a conceptual application model, three of the five meanings are already M-07, one is M-08, and the fifth — a first-principles process simulation — belongs to process engineering and has no place in an application portfolio at all.

The twelve

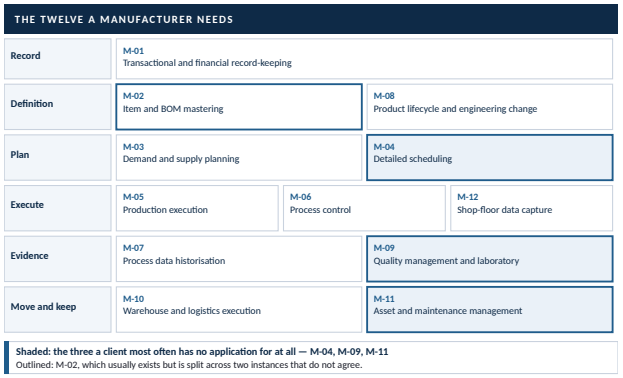


Figure 11.2 The twelve conceptual application capabilities, grouped for reading. The grouping is a convenience; several capabilities serve more than one domain. The outlined three are the ones a client most often has no application for at all.

For each capability: what it must do, what it should master, what it must not master, and who uses it. The *must not master* line is the one that does the work, because almost every manufacturing architecture failure in this book is a case of a capability mastering something that belonged to its neighbour.

M-01 Transactional and financial record-keeping

Must do: post every movement of material and money with a document trail that reconciles to the general ledger; net multi-level requirements across plants and purchased lead times; procure, receive, three-way match, value inventory, cost an order, settle it, and produce statutory reporting. **Should master:** the financial representation of production — goods movements, inventory valuation, order costs and variances, the chart of accounts, the supplier and customer records, and the organisational structure of company, plant and storage location. **Must not master:** the sequence of work on a line,

the state of a unit part-way through an operation, a machine's downtime reason, or an engineering revision. **Roles:** the plant accountant, who closes the month and spends three days explaining variances; the buyer; the materials controller; the master-data steward who nobody has appointed.

M-02 Item and bill-of-material mastering

Must do: hold the item in all its organisational scopes, with the planning, costing, batch, shelf-life and unit-of-measure parameters each plant needs; hold the manufacturing bill of material with quantity-per, scrap factor, item category and effectivity; hold the routing or recipe with its operations, work centres and standard values; and bind a particular bill to a particular routing for a quantity range. **Should master:** the item identifier, the manufacturing structure, the routing, and the work-centre definition. **Must not master:** the as-designed structure, which is M-08's, nor the as-built structure, which is M-05's. Those are three different things over the same product and chapter 9 explains why conflating them destroys both traceability and cost. **Roles:** manufacturing engineering, which builds the structure; planning, which lives off its parameters; costing, which values it.

This is the capability on which every other one depends, and the first of the chapter's two decisive questions is about it.

M-03 Demand and supply planning

Must do: build a statistical forecast and let the business override it with judgement; run a consensus demand review and a supply review against rough-cut capacity; hold inventory policy by item and location; and produce one agreed plan with the decisions recorded. **Should master:** the consensus demand plan, the supply plan, the inventory policy parameters, and the record of what was decided and declined. **Must not master:** the item, the bill of material or the stock position. **Roles:** the demand planner; the supply planner; the commercial director who wants the forecast changed; the Group Supply Chain Director who chairs the review — at Karoo that is Priya Naidoo, and her problem is that the same review covers a beverage line that plans in cases and a components plant that plans against a customer's call-off schedule.

M-04 Detailed scheduling

Must do: sequence work on finite resources against real constraints — sequence-dependent changeovers, shared tools and moulds, tank and vessel occupancy, minimum and maximum run lengths, cure and hold times, resource calendars that differ machine by machine — and publish a sequence a supervisor can act on. **Should master:** the published schedule and the sequence. **Must not master:** the actual sequence that was run, which belongs to M-05, and this distinction is the whole point of the capability. **Roles:** the scheduler or planner; the shift supervisor who reorders the list; the production manager who arbitrates.

> The schedule is a plan, the dispatch list is an instruction, and the actual sequence is a record. They are three objects with three lifecycles. Designs that make them one object fail, and the measure everybody wants — schedule adherence, did we make what we said in the order we said — requires all three to exist. Almost no manufacturer can compute it.

M-05 Production execution

Must do: receive the released order, dispatch operations, enforce sequence and operator qualification, present the correct revision of the work instruction for that order's effectivity, capture material consumption by lot at the point of use, record labour and machine time, prompt for a downtime reason within seconds of a stop, build the genealogy record, and in regulated manufacture assemble the electronic batch record so that quality can review by exception rather than read three hundred pages. **Should master:** execution state, the operation confirmation, the as-built genealogy, the downtime event and its reason, and the actual sequence. **Must not master:** the item, the bill of material, the released order's commercial terms, or the financial valuation of what it consumed. **Roles:** the operator, whose hands are the production process and for whom every second of screen interaction is lost output; the line leader; the production manager; the quality reviewer.

M-06 Process control

Must do: read the instruments, execute control logic on a deterministic scan, hold the interlocks and the sequences, present the operator graphic, raise and manage alarms, and in a batch plant run the procedural hierarchy of the recipe against allocated equipment. **Should master:** the control logic, the live process state, the alarm, the operator action log, and the master recipe's procedural content. **Must not master:** anything with a commercial meaning. A control system that is the system of record for production counts is a finding, and a common one. **Roles:** the process or control engineer, who owns it and will not let you near it; the panel operator, who runs the plant from it; the instrument technician and the millwright, who are the scarcest people in the estate — engineers were the hardest skill to fill in South Africa at 29% of surveyed employers in the preliminary 2026 Xpatweb results, with artisans joint second at 25% [6].

M-07 Process data historisation

Must do: collect tag values at rate, store them for years, answer time-range queries over tens of tags quickly, buffer locally when the link upward drops, and carry an asset model above the tags so that a report can be written against *Reactor 2, jacket temperature* rather than against `JHB-30-R02-TIC1021-PV`. **Should master:** the tag value time series and nothing else. **Must not master:** the asset hierarchy as an authoritative structure, the batch record, or any transaction. **Roles:** the process engineer doing a yield investigation; the reliability engineer; the data team, who will try to copy the whole thing into a lakehouse as triples and reproduce every problem at greater cost.

> This capability silently changes the numbers it reports, and chapter 13 shows how. A historian's archive does not hold what the instrument measured; it holds the smallest set of points from which the measured curve can be rebuilt within a configured deviation [4]. Write that on the box now, in week one, so that nobody later builds a quality argument on a number they think is a measurement.

M-08 Product lifecycle and engineering change

Must do: vault the CAD and its revisions; hold the part as a business object separate from its file; hold the engineering bill of material; manage the change chain from request through authorised order to verified closure, with the effectivity stated; hold the configuration baseline so that the approved definition of a unit shipped three years ago can be reconstructed; and carry the bill of process that manufacturing engineering builds from it. **Should master:** the as-designed part and structure, the document revision, the change object and its effectivity, and the baseline. **Must not master:** the manufacturing structure, the stock position, or the routing's standard times. **Roles:** the design engineer; the manufacturing engineer who performs the engineering-to-manufacturing transformation by hand, because it is a judgement and not a mapping; the change board, whose minutes an auditor will ask for.

M-09 Quality management and laboratory

Must do: hold the specification with its version and effective date; hold the sampling plan; register the sample and its chain of custody; capture results, ideally from the instrument; evaluate against the specification version in force at the time; issue the certificate of analysis; manage non-conformance, deviation, corrective and preventive action, change control, audit and complaint; and be the authority that holds material and releases it. **Should master:** the specification, the sample and result, the quality event, and the disposition. **Must not master:** the stock position, though it must be able to freeze it. **Roles:** the laboratory analyst; the quality manager — Sipho Khumalo at Karoo, who is accountable for releasing a batch he did not see made; the regulatory affairs officer; and in a regulated plant the person who signs.

The hold-and-release authority is worth a sentence of its own. A hold can be raised in four different capabilities. Only one of them can be the authority that stops material moving, and the other three must read it. Leave this undeclared and material will move while on hold, which in food or pharmaceutical manufacture is the defect that ends careers.

M-10 Warehouse and logistics execution

Must do: hold stock at bin, pallet or handling-unit granularity with its quality status; direct receipt, putaway, replenishment, picking and loading as tasks; respect lot and shelf-life allocation rules; build loads, select carriers, produce transport and dangerous-goods documentation; and run the dock, the yard and the weighbridge. **Should master:** stock by location and handling unit, the warehouse task, and the shipment. **Must not master:** work-in-progress, which it will claim and should not get, and the quality status, which it must read from M-09. **Roles:** the warehouse supervisor; the picker with a scanner in a glove; the logistics planner; the gatehouse clerk whose weighbridge reading is a legal measurement under the Legal Metrology Act, 2014 and therefore cannot be adjusted, rescaled or rounded in software [5].

M-11 Asset and maintenance management

Must do: hold the functional location hierarchy — the *position* in the plant, independent of what is installed in it — and the equipment installed at each position with its own history; hold criticality from a structured analysis; schedule preventive work by time or by counter; take condition readings; raise notifications and authorise work orders with permits, parts and trades; and record completion with a failure code that supports reliability analysis. **Should master:** the functional location, the equipment, the maintenance plan, the work order, the failure history and the spares relationship. **Must not master:** the production hierarchy. The asset tree and the production tree are two different trees over the same steel and reconciling them is a real project. **Roles:** the maintenance planner — Karel du Toit at Highveld, who knows where everything actually is; the artisan; the reliability engineer; the production manager who cancels the maintenance window.

The failure code deserves emphasis. Free-text failure recording destroys the data asset the capability exists to build, and three years of work orders then yield no reliability analysis at all. ISO 14224:2016 gives a published equipment taxonomy and failure coding scheme, written for oil and gas and used far beyond it, and starting from it rather than inventing one is the cheapest good decision available in this box [3].

M-12 Shop-floor data capture

Must do: get physical events into software — scans, counts, weights, label prints, badge reads, andon calls, inspection verdicts — with a reliable timestamp and a station identity, from devices that work in gloves, in wet and caustic areas, in freezers at -25°C and in explosive atmospheres. **Should master:** the scan event, the count, the weight and the print event. **Must not master:** anything derived. **Roles:** the operator; the line leader; the industrial engineer who sizes the interaction.

This is the cheapest capability on the page and the most under-used. A photo-eye on a conveyor and a counter in a controller gives an accurate production count for a fraction of an execution system's cost, and most plants either have it and do not read it, or count by hand. It is also the capability with the hardest physical constraint in the book: a data-entry screen that costs eight seconds per unit on a line running 600 units an hour is not a usability problem, it is arithmetically impossible, and the operators will defeat it by scanning one unit in ten.

The thirteenth box, where the product is regulated

Where the product is a medicine, a medical device or food for a customer who audits, add one more: **validation and controlled-document management**. It must hold the controlled procedure at its revision with read-and-understood acknowledgement, the training record that gates who may run which operation, the change-control record for any change to a validated system or process, and the validation evidence itself. It masters the controlled document and the validated state. It must not master the engineering revision, which is M-08's.

It is a separate box rather than part of M-09 for one reason: **its scope is every system in the regulated perimeter, not the quality function.** The moment you draw it you have also drawn the list of systems whose change cadence is now governed by it, which is the conversation that stops somebody promising continuous delivery into an electronic batch record.

From business capability to application capability

The mapping is many-to-many and the table below is the version to take into a workshop. Read it across: a business capability, the application capabilities that realise it, what the business will tell you it masters, and whose job it is.

Business capability	Application capabilities	What is mastered	Roles in the room
Design a product	M-08, M-02	Part, revision, eBOM, change, baseline	Design engineer, manufacturing engineer
Plan supply	M-03, M-01, M-02	Demand plan, supply plan, policy	Demand planner, supply planner
Decide the sequence	M-04, M-05	Published schedule, actual sequence	Scheduler, shift supervisor
Make to a schedule	M-05, M-06, M-12, M-02	Execution state, control logic, genealogy	Operator, production manager, control engineer
Prove it conforms	M-09, M-05, M-07	Specification, result, disposition, batch record	Analyst, quality manager
Move and store material	M-10, M-01	Stock by location, task, shipment	Warehouse supervisor, logistics planner
Keep the plant running	M-11, M-07, M-12	Functional location, work order, failure history	Maintenance planner, artisan, reliability engineer
Account for it	M-01, M-02, M-05	Document, valuation, variance	Plant accountant, costing analyst
Trace and recall	M-05, M-09, M-10, M-12	Genealogy, lot, aggregation, dispatch	Quality manager, logistics, technical director

Two things to notice. **M-02 appears in five rows and M-05 in five**, which tells you where the architectural weight sits. And **"trace and recall" is realised by four capabilities and mastered by none of them**, which is why a recall takes three weeks at a plant that believed it was traceable. Where a business capability has no single mastering application, say so on the page. That sentence is worth more than the diagram.

The two questions that decide every manufacturing application architecture

Everything else follows from two answers, and you can get both in the first week if you ask plainly.

One: what is the system of record for the item, the bill of material and the stock position?

Not "which systems hold them" — all of them hold them. Which one is authoritative, for each of the three, and who may change it.

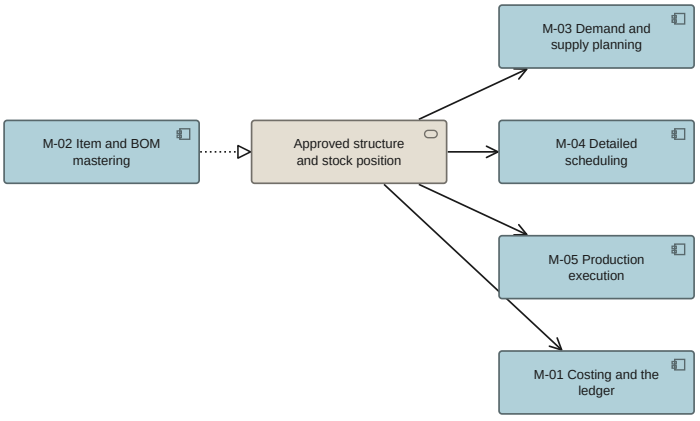


Figure 11.3 The first decisive question, drawn. Four capabilities consume an authoritative item, structure and stock position. Split the mastering capability and all four become reconciliations.

The reason this question comes first is that the three objects are the spine of the model. The item identifier is referenced by every other capability. The bill of material determines what is consumed, what is costed, what is traced and what is planned. The stock position determines whether an order can be released and whether a promise can be made. If the answer is split — and at Karoo Industrial Group it is, because Meridiem runs as two instances that do not share an item master — then every cross-plant question becomes a reconciliation, intercompany transfers need a mapping table that someone maintains by hand, and group-level planning is performed on an export.

Three ways the answer goes wrong, in descending order of frequency. **Split by instance**, as above: two ERP instances, two item masters, one group. **Split by layer**: the item exists in the engineering system with one identifier and in the transactional system with another, joined by a convention rather than by a relationship. **Split by stewardship**: one system of record, no steward, and the master-data table open to direct edit by anyone who noticed a shortage — which is how an item comes to have fourteen days of safety stock that nobody can explain.

Write the answer down as a table with four columns: object, mastering capability, who may change it, and how a consumer learns it changed. That table is the most valuable half-page you will produce in the first month.

Two: where is the boundary between transactional record-keeping and production execution?

This is the question chapter 12 fights out at length, because it is the most common integration argument in manufacturing architecture. Set it up here, because the conceptual model is where the argument can still be had cheaply.

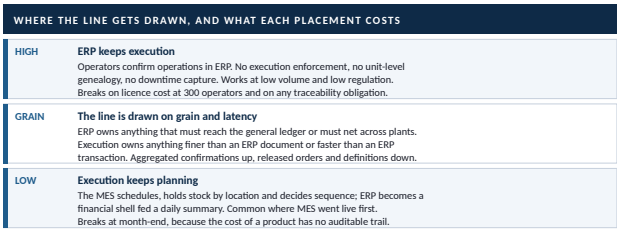


Figure 11.4 The second decisive question. The middle band is the principled answer and the one the boundary standard supports; the other two are what you will usually find, and each has a predictable bill attached.

The principled answer is the middle band: draw the line on **grain and latency**. M-01 owns anything that must reach the general ledger or must net across plants. M-05 owns anything whose grain is finer than an ERP document or whose latency requirement is tighter than an ERP transaction. The interface carries aggregated confirmations upward and released orders plus effective definitions downward. That is also the position the published boundary standard takes, and ISA-95's object names — production schedule going down, production performance coming back — are the vocabulary to use for it [1][2].

What actually decides it in practice is rarely principle. It is whichever system was implemented first, and after that it is licence cost: if shop-floor confirmation in the transactional system costs three hundred named users, the boundary moves whatever the architecture says. An architect who does not know the licence metric is not qualified to run the argument.

Three specific sub-questions always need an explicit answer, because leaving any of them implicit guarantees two systems both holding the capability and a nightly reconciliation report that nobody reads:

1. **Where is material consumed** — by backflush at order confirmation, or in real time by lot at the point of use with a summarised issue upward? The second is correct for traceability and considerably harder.
2. **Who decides the sequence**, and does a supervisor's resequencing travel back up?
3. **Who holds and releases** material, and which of the four candidates is the authority?

What the model is actually for

Three uses, and one honest limitation.

It scopes a procurement. Three vendors answering the same list of capabilities are comparable; three vendors answering their own product boundaries are not. The question to put to each is not "do you cover manufacturing execution" but "which of these twelve do you master, which do you read, and what do you assume about the ones you do not provide". The second half of that question is where integration cost hides, and it is the half vendors answer worst.

It frames the instance question. For each of the twelve: one instance for the group, or one per plant? There is a defensible answer in both directions and it differs by box. M-01 and M-08 want to be single. M-05, M-06, M-07 and M-12 must be plant-resident, because the plant has to keep producing when the link to head office is down, and chapter 13 makes that a design rule rather than an aspiration. M-04 is the interesting one: it wants to be central for cross-plant optimisation and local for responsiveness, and it is not in the execution path, so it can live centrally provided the plant holds a published dispatch list that survives the link dropping.

It finds the gaps, which is the point of drawing it before you know what the client owns.

> Run the model as a checklist, not as a target. For each of the twelve ask: which application does this, who owns it, and what happens if that person leaves? The boxes that come back with a name and no application are the finding. In South African mid-market manufacturing they are almost always M-04 detailed scheduling, M-09 quality and laboratory, and M-11 asset and maintenance management — in that order.

How to run the week-one workshop. Put the twelve on a wall, one per sheet, and ask three questions per sheet in this order: *who does this today*, *in what*, and *what happens when that person is on leave*. Do not ask "which system do you use", because the answer is the system somebody bought, not the system that does the work. The third question is the one that produces the findings: a capability with a named person and no application behind them is a risk you can price, and it is also the only part of the exercise a plant manager will find obviously worth their morning. Write the answers on the sheets in the room and photograph the wall. That photograph, not the tidy diagram you draw afterwards, is the evidence.

The limitation, stated plainly. You draw this model before you know what the client owns, and it is the artefact you will revise most in your first month. That is not a defect in the method; it is the method working. The first draft is a hypothesis built from the industry; the revision is what you learn from the plant. Expect to move two boundaries, split one box because the client genuinely has two incompatible archetypes under one roof, and delete nothing.

What you should not do is revise it quietly. Keep the first draft, date it, and keep a short note of every change and why. When somebody asks in month four why quality is a separate capability from execution, the answer is that Sipho Khumalo said in week two that he cannot be the authority for release if the authority lives inside the system the production manager owns, and he was right. A conceptual application model with a change log is an argument with evidence. Without one it is a drawing.

THE CHAPTER IN TEN LINES

- A conceptual application model names what the applications must do, with no product names on the page.
- Twelve capabilities cover a manufacturer: record-keeping, item and BOM mastering, planning, scheduling, execution, control, historisation, product lifecycle, quality, warehouse, maintenance and shop-floor capture.
- Keep business capability, application capability and system on three separate pages; realisation between them is many-to-many in both directions.
- Write the grain next to every box. Two capabilities whose grains differ by three orders of magnitude cannot be one application.
- The must-not-master line settles more arguments than the must-do line.
- Question one: what is the system of record for the item, the bill of material and the stock position, and who may change each.
- Question two: where is the boundary between transactional record-keeping and production execution — drawn on grain and latency if principle wins, on licence cost if it does not.
- Industry 4.0, integration and the digital twin are not application capabilities and each one makes the model worse.
- The model is a checklist for finding gaps, and the gaps are consistently detailed scheduling, quality and maintenance.
- Expect to revise it hardest in month one, and keep a dated change log so the revisions are an argument rather than a drawing.

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CHAPTER 12

The application landscape

What is actually installed, class by class: what each system is for, what it masters, what its data is made of, and what breaks. Including the argument you will be standing in the middle of within a month — where the line between the business system and the execution system is drawn, and who drew it.

Six questions, asked of every class

Chapter 11 drew the applications a manufacturer needs. This chapter describes what is actually installed, class by class. By the end of it you should be able to walk into a plant, hear four acronyms in a sentence, and know what each one owns, what it is lying about, and which two of them are quietly disagreeing about yesterday's production.

Ask the same six questions of every class, in this order, and write the answers in a table:

1. **What is it for?** One sentence, in the plant's own words, not the vendor's.
2. **What does it master?** The entities for which it is authoritative, and only those.
3. **What does it integrate with**, in which direction, on what trigger?
4. **What is its grain?** The characteristic unit of data, which predicts everything else — storage, retention, interface design and cost.
5. **How is it licensed and deployed?** Including the metric, because the metric moves boundaries more often than principle does.
6. **How does it fail?** Not whether. Every class here has a characteristic failure, well known to the people who live with it.

The few product names here appear so that you recognise one when a plant manager says it. They are not recommendations and not a shortlist; chapter 15 handles providers properly.

System class	System of record for	Grain	Breaks on
ERP	item, BOM, routing, order, cost, ledger	document	stale standard values; backflush at the wrong operation
MES / MOM	execution state, genealogy, downtime	unit, operation	order change not propagated; synchronous posting upward
APS	the published schedule	task on a resource	constraint model never maintained
PLM / PDM	part, eBOM, revision, change, baseline	part-revision	revision without effectivity
WMS / WES	stock at a bin, the task	handling unit	quality status not read, so a held lot is picked
TMS and yard	load, shipment, freight cost	consignment	weighbridge re-keyed by hand
EAM / CMMS	functional location, equipment, work order	work order	free-text failure codes; counters descoped
LIMS	specification, sample, result, CoA	test result	specification version not recorded against the result
QMS	non-conformance, deviation, CAPA	quality event	hold-and-release authority split four ways
Historian	the tag value time series	tag at a timestamp	compression set once and never reviewed
Shop-floor capture	scan, count, weight, print	event	unsynchronised local clock; operators defeating it
IIoT / namespace	usually nothing — ask	message	nothing durable subscribes, so the data is gone

Figure 12.1 The six questions answered in one page, for twelve classes. Read the last two columns together: the grain tells you what the system is made of, and what it breaks on is nearly always a consequence of it.

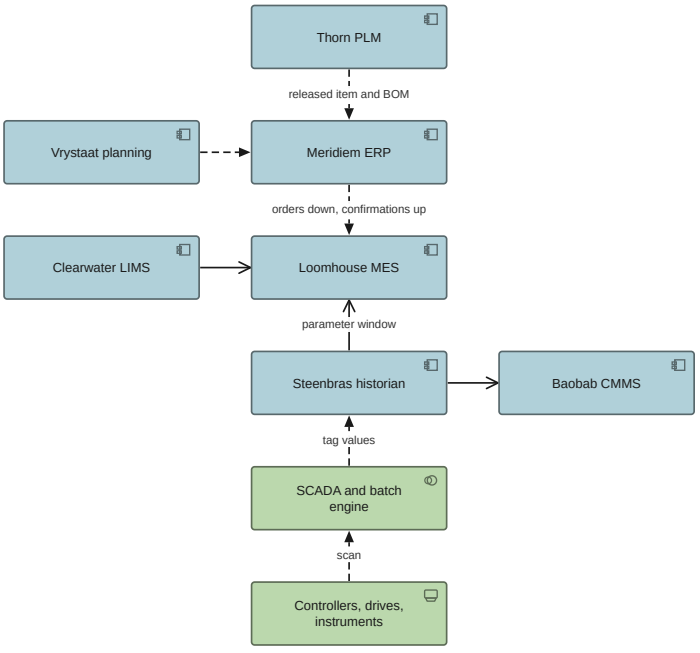


Figure 12.2 The estate as a layered view, with the two halves that meet. Orders go down and confirmations come up on the same interface; tag values go up and nothing goes down. The unlabelled flows carry the plan into the ERP, the disposition into execution and running hours into maintenance. Karoo Industrial Group is invented.

The shape of what you find

A manufacturer's estate is not one landscape but two that meet: IT above the line, and below it the controllers, distributed control systems, supervisory software and instrument loops that chapter 13 is about. Expect one instance of the transactional system and one of product lifecycle; **one set of execution, historian, supervisory, control and data-capture systems per plant**, sometimes per line, sometimes per machine builder; and shadow instances of planning and scheduling in Excel. The count of controllers in a mid-size plant runs to dozens and the count of instruments to thousands. That shape is practitioner consensus rather than a surveyed figure — no South African manufacturer publishes a portfolio or controller count — and the way to settle it at a client is a passive network discovery, which is in any case the first deliverable of most plant security programmes.

The most useful early question is not "what systems do you have" but **"who owns the OPC server"**. If the answer is the control systems engineer at the plant, the estate is led from the plant and IT is told afterwards. If the answer is corporate integration, the plant has probably built a path around it.

ERP in manufacturing

ERP is the Level 4 system of record for the commercial and financial representation of manufacturing. **It is not the system that runs the plant**, and designing as though it were is the most common architectural error in this industry.

The manufacturing modules, specifically

Item master. The anchor. A manufacturing item carries far more than a traded article record: a material type, a procurement type, a lot-sizing rule, a planning strategy, base and alternative units of measure with conversion factors, batch management, a serial-number profile, shelf life, valuation class and price control, lead times, safety stock, storage and hazard classification, inspection types — **and a separate plant-level view for every plant that uses it.** That last point is structural. The same item has different planning parameters, different costs and different status in each plant. Architects arriving from a single-site mindset design one item record and then discover the data model is a matrix.

Bill of material. Component lines with quantity-per, a scrap or loss factor, an item category and validity dating or change-number effectivity. Multi-level by reference. Alternative bills exist for the same item in the same plant, selected by lot-size range or production version. Phantom assemblies are structural groupings never stocked and blown through at order creation — and **phantom is a property of the bill-of-material line, not of the item**, which chapter 9 explains and which an item-level flag makes impossible to represent.

Routing. The sequence of operations, each at a work centre, each with standard values for setup, machine and labour time, and a formula that turns those plus quantity into a capacity requirement and a cost. Control keys determine whether an operation is costed, scheduled, confirmed or subcontracted. Component allocation binds bill-of-material lines to operations, which is what makes backflushing at the right point possible. Process plants use a master recipe with phases instead, aligning loosely with the batch standard in chapter 13.

Work centre. Available capacity by shift, scheduling and costing formulas, a cost centre and activity types — which is how time becomes money — and **a link to the equipment and functional location in maintenance**, which is the join that lets a maintenance outage become a capacity reduction and which almost nobody populates.

Material requirements planning. Explodes demand through bills, nets against stock and supply, applies lot-sizing and lead-time offsets, and produces planned orders and requisitions. **It assumes infinite capacity** and will happily plan 400 hours of work into a 120-hour week. It runs nightly in most plants, so an urgent change made at 10:00 is invisible to the floor until tomorrow.

Production order, process order and production version. The order is the execution object: a header, a component list copied from the bill, an operation list copied from the routing, a settlement rule, and **a status network**. The statuses — created, released, partially confirmed, confirmed, technically complete, closed, plus deletion and user statuses — are the integration contract. An execution interface that does not model the transitions will release work the transactional system has not released, or confirm against an order it has closed.

Product costing. Three distinct activities that get conflated: the **standard cost estimate**, frozen at planned prices for a year or a quarter; **order costing and settlement**, which collects actuals and produces variance categories including — in process industries — yield; and **actual costing**, which revalues periodically and rolls actual prices up through the levels. The cost of a product is a different number in each, and all three are correct.

Plant maintenance and quality. Functional locations, equipment, task lists, plans, notifications and orders; inspection plans and lots, results, usage decisions, batch status. Many manufacturers run these modules as their maintenance and quality systems; many run specialists.

What ERP is genuinely good at

Four things, and defend them rather than letting a point solution erode them.

- **The financial representation of production, with auditability.** Every goods movement has a document, every document an accounting document, every accounting document a ledger posting, and the chain reconciles. Nothing else in the estate does this.
- **Multi-level netting across the whole product structure and supply chain.** Its weakness is capacity; its strength is that it will correctly net a finished-goods demand against work in progress four levels down at another plant with a 90-day purchased lead time.
- **Master data with organisational scoping and effective dating.** The plant-view model and change-number effectivity are genuinely sophisticated and genuinely necessary.
- **Procurement, inventory valuation and statutory reporting.** Mundane, and the reason the system exists.

The three things it is always asked to do and always does badly

Finite-capacity scheduling and sequencing. Capacity requirements planning shows a load profile; it does not sequence. It cannot reason about a sequence-dependent changeover — light colour before dark, allergen-free before allergen-containing, low viscosity before high — about a tank that must be emptied before refill, a cure time that must elapse between operations, or a tool shared by two work centres. Asked to do it, it produces a schedule that violates physics or a dispatch list in date order that the supervisor reorders by hand.

Real-time execution and operator interaction. Its transaction model is document-oriented, latency-tolerant and designed for a trained user at a desk. The floor needs sub-second response, a touch target usable in a glove, enforcement of operation sequence, machine-state capture, downtime prompting within seconds, and continued operation when the link to the data centre is down. Confirmation transactions work for low-volume assembly; at high-rate packaging they cannot.

Anything that must be re-run on a changed assumption. Simulation, what-if, costing a design that does not exist. The data model records what has happened or is committed; it has no first-class concept of a version of the future that is not a plan.

Licensing, deployment, and how it fails

The classic metric is named users plus engine charges, and the manufacturing trap is that **shop floor operators are users**. Three hundred operators confirming operations is three hundred licences of some class, which is frequently what makes an execution layer cheaper than the simple option. Document- or metric-based models shift the calculus the other way: a high-volume confirmation stream now carries a cost per document, which pushes toward aggregated confirmation and away from unit-level traceability — a licensing decision with a traceability consequence. In South Africa the mid-market default on any shortlist is SYSPRO, founded in South Africa in 1978 by Phil Duff, headquartered in both the United Kingdom and South Africa, with Advent International's majority stake announced on 13 August 2024 [1]; "independent South African ERP" is no longer an accurate description.

How it fails: the item master open to direct edit with no stewardship, so planning parameters are tuned by whoever noticed the shortage. The bill maintained only here, so engineering's structure exists in CAD file names and change control is email. Standard costs frozen for a year while input prices move 30%, so every variance report is noise and the plant stops reading it. Backflush configured at the wrong operation, so work-in-progress is wrong all day and corrects itself at settlement. Routings with standard values accurate in 2014, so the capacity plan is out by 20% and the scheduler has learnt to ignore it. Orders left partially confirmed for months, carrying phantom value until a year-end write-off.

MES and MOM, and the boundary war

Is there a distinction?

MOM is the standards-derived and broader term; MES is the market term for a product. In a client conversation they are used interchangeably with a slight difference in status.

The provenance is worth knowing. MESA International published a sequence of models from 1992 onward, including the eleven-function model of 1996 [2]. **MOM** comes from Part 3 of the boundary standard, which decomposes Level 3 into four operations areas — production, maintenance, quality and inventory — each with the same eight activities: definition management, resource management, detailed scheduling, dispatching, execution management, data collection, tracking and performance analysis [3].

That eight-activity pattern is the most useful structuring device in this chapter. It lets you ask of any plant-level capability: **which operations area, and which activity?** A product that does production execution and tracking but not production definition management is not a whole execution system, and the gap will be filled by the transactional system or by a spreadsheet.

What it actually does

Work-order execution: receive the order, dispatch operations, enforce sequence, manage splits, merges and rework loops, hold and release, report completion. It is the system of record for execution state at a granularity ERP does not carry — which unit is at which station, how far through operation 30 it is, who is on it now.

Operator interaction: the screen the operator actually uses, with the correct revision of the work instruction for that order's effectivity, forced acknowledgements, mistake-proofing checks, electronic signatures, and qualification enforcement.

Electronic batch record: in regulated manufacture, the executed recipe with every parameter, lot consumed, signature and deviation assembled into a reviewable document. **Review by exception** — a reviewer reads only the records with exceptions rather than three hundred pages — is the single largest return case for an execution system in pharmaceutical manufacture.

Genealogy, labour and machine time capture, and downtime with a reason. The reason hierarchy is a political object, because it decides whose number looks bad: "material not available" blames planning, "changeover" blames the schedule, "mechanical failure" blames maintenance. Architects consistently underestimate how long agreeing that taxonomy takes.

Grain, licensing, deployment

The grain is the point. ERP's grain is the order and the confirmation; a day's production might be three documents. Execution's grain is the unit, the container, the lot, the operation instance, the scan, the state change and the parameter sample. A line running 600 units an hour with twelve recorded parameters per unit generates 7,200 facts an hour from one line.

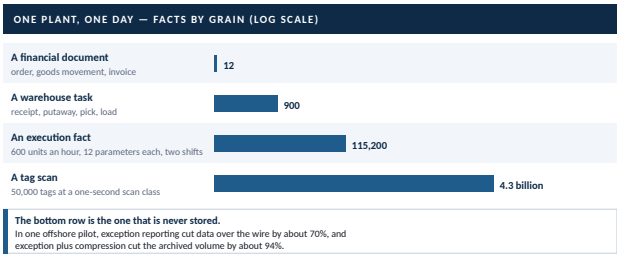


Figure 12.3 Grain decides everything downstream. Four classes of fact from one plant in one day, on a logarithmic scale. Two systems whose grains differ by three orders of magnitude cannot be one application.

That is why execution databases are large, why their retention policy is a regulatory question rather than a storage question, and why summarising upward is mandatory. Licensing is per site, per line, per concurrent client or per named user, usually with a separate charge for regulated-industry modules. Deployment is overwhelmingly on premises at plant level, because the system must survive the loss of the wide-area link. The emerging pattern is a plant-local execution tier with a cloud aggregation tier above it, and your job is to be explicit that the cloud tier is **not in the execution path**.

The boundary war, and how it actually gets settled

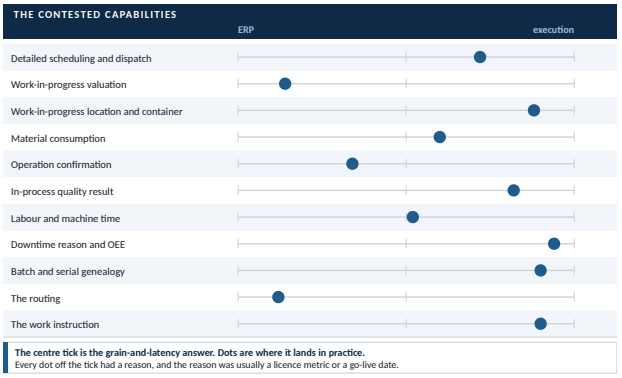


Figure 12.4 The contested line. Each capability is genuinely available in both systems and both claims are reasonable; the dot is where it usually ends up in a South African mid-market estate. Author's judgement, not a survey.

Each of those capabilities is genuinely available in both systems, and both claims are reasonable. Six patterns settle it, in descending order of how well they work.

1. **By grain**, which is the principled answer and the one the boundary standard supports. Aggregated confirmations up; released orders and effective definitions down.
2. **By a written system-of-record declaration per entity.** Entity, master system, consumers, update direction, latency. Unglamorous, two workshops per operations area, and it prevents most of the pain. If this book teaches one artefact for manufacturing architecture, it is this one.
3. **By who implemented first.** The honest and most common determinant.
4. **By licence cost.** Three hundred operators settle more arguments than any principle.
5. **By vendor consolidation.** When both come from one supplier the boundary is set by the product's own integration content: a genuine advantage and a genuine lock-in.
6. **By nobody**, which is the failure state. Both hold the capability, both are updated, a reconciliation runs nightly, the plant trusts one and finance the other, and the two figures for yesterday differ by 2% permanently.

Three sub-questions always need an explicit answer: where material is consumed; who decides the sequence and whether a supervisor's resequencing travels back up; and who holds and releases.

> The most common and most avoidable integration defect in manufacturing: confirmations posted synchronously from the floor into ERP, so the line stops when ERP is in its nightly batch window. Queue them locally and post asynchronously with idempotency keys, always.

How else it fails: implemented as a data-capture front end with no execution enforcement, so it adds a system without removing a spreadsheet. A one-way order download with no handling of change or cancellation, so it executes orders that no longer exist. A downtime reason hierarchy built from the machine builder's fault codes rather than the plant's management categories, producing 1,400 reasons of which eleven are used. Work instructions held as PDFs with no link to the engineering revision. Genealogy captured but the recall query never written. And customisation so heavy — because every plant genuinely is a special case — that the upgrade path is dead within two releases, which is why execution estates are full of unsupported versions.

APS and finite-capacity scheduling

An advanced planning and scheduling tool takes demand, supply, capacity and constraints and produces a sequenced, time-phased allocation of work to resources that respects real limits. The distinction from material requirements planning is the whole point: **capacity is finite and lead time is an output of the schedule rather than an input.** It also expresses constraints that planning cannot: sequence-dependent changeover matrices, shared tools and moulds, campaign and run-length rules, tank and vessel occupancy, maximum hold time between operations, resource-level calendars, and a weighting between objectives that conflict.

And the schedule it publishes is never the schedule the plant runs. Seven reasons, and treat every one as a requirement rather than a problem to eliminate. The routings and standard times it runs on are stale, so the schedule is arithmetically perfect and physically impossible. The constraint model is incomplete — the supervisor knows line 3 cannot run product X on a Monday because the water is different after the weekend shutdown. Reality arrives faster than the re-plan cycle. The planner optimises delivery and the supervisor optimises getting through the shift without a stop, and both are rational. Material does not arrive. Quality intervenes. And nobody tells the schedule it was wrong, because actuals flow back on a different cadence.

> The schedule is a plan, the dispatch list is an instruction, and the actual sequence is a record. Three objects, three lifecycles. The measure that matters is schedule adherence — did we make what we said, in the order we said — and it requires all three to exist, which is why almost nobody measures it.

Architecturally it sits between the two big systems, reads from both and writes to both, and is therefore the integration node with the worst latency characteristics in the estate. Three-way integration with a planner in the middle is the hardest thing in manufacturing integration and should be priced accordingly. It is, however, **not in the execution path**, so it can live centrally provided the plant holds a published dispatch list that survives the link dropping.

How it fails: bought as an optimiser when the planners wanted a Gantt chart they could drag, so the output is overridden and the spreadsheet continues. Constraint model built once and never maintained. Run frequency too low to matter. Integrated for master data but not for actuals, so it plans from yesterday. And the political failure — it makes the capacity shortfall visible for the first time, the number is unwelcome, and the project is reframed as a reporting problem.

PLM and PDM

PDM is the vault: CAD files, their revisions, their references, check-in and check-out, where-used and a release state. **PLM** is that plus the structure around it: the part as a business object independent of its file, the engineering bill of material, change management, configuration management, requirements, variant and option management, manufacturing process planning, supplier collaboration and substance-compliance data.

It masters the as-designed part, the document revision, the engineering bill, revision and effectivity, the change chain, the configuration baseline and increasingly the bill of process. Its grain is the part-revision and the bill line, and a single change order can touch thousands of them atomically.

The handover that nobody automates

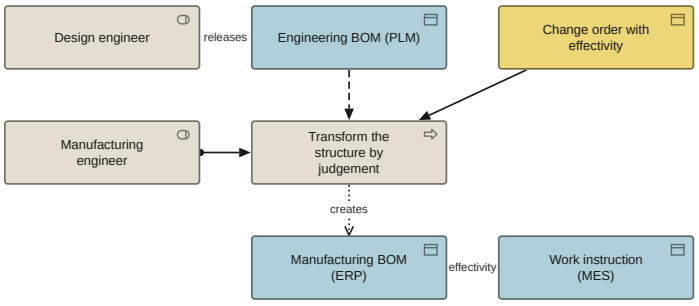


Figure 12.5 The handover nobody automates. The transformation in the middle is a judgement made by a person, which is why the market leader's own answer to keeping the two structures aligned is a structured side-by-side comparison.

The engineering bill is how the product is designed; the manufacturing bill is how it is built. They describe the same product, they are structurally different, and **the difference is intrinsic rather than a data-quality defect**. The manufacturing structure is flattened or restructured to match stations; it adds consumables, adhesives, lubricants, bulk fasteners, packaging and labels; it drops items the design shows that arrive inside a purchased sub-assembly; it carries phantoms and scrap factors; it resolves alternates to the one this plant actually uses; it exists once per plant; and it allocates components to operations.

None of that is derivable from the engineering structure. It is a judgement made by a manufacturing engineer who decides that four fasteners come from a bulk bin, that a three-part sub-assembly is purchased complete and collapses to one line, that this plant uses the alternate because the primary supplier does not ship here, and that a 3% scrap factor applies to the gasket because it tears on fitting. The market-leading product's own answer to keeping the two aligned is a structured side-by-side comparison with a change-summary panel, visual cues and checklists [4] — that is, a human doing a careful comparison, which is the honest state of the art.

Four consequences, each an architecture problem. Two structures must be kept in step across every change, forever. Cost is computed against one and quoted against the other, and nobody owns the reconciliation. Traceability is recorded against the manufacturing structure, so linking a field failure to the lots of the as-built component needs both structures and the link between them — usually an item number that happens to be the same. And where-used gives different answers in each, differing by exactly the phantom and consumable lines.

Revision, effectivity and change

Revision is *which version*. **Effectivity is when or where that version applies**, and it comes in kinds you must distinguish: date, serial number, lot, configuration, and change-number effectivity, where one change object carries the validity and every affected line references it, so a change can be activated or reversed atomically across thousands of lines.

Revision without effectivity is the most common implementation defect in this class, and it is invisible until the first field failure or the first audit. Without it, "what was the approved design when we built unit 1,100?" has no answer.

The change chain runs request, evaluation, a board decision whose minutes an auditor will ask for, then an authorised order carrying the effectivity and the disposition of existing stock, then implementation in five places: the vault, manufacturing engineering, the transactional system, the execution system and the supplier. The hand-offs fail predictably, and the whole chain has no single status, which is why asking "is change order 4471 implemented?" means checking five systems.

> One question to put to any client running a long-installed engineering system: what is its support horizon? Oracle's Agile PLM has a large installed base, particularly in electronics and life sciences, and the vendor's newer direction is cloud product lifecycle management inside its applications suite. **What Agile's current support position and roadmap actually are could not be established from a public source in the research for this book**, and it is settled only by the vendor's own lifetime support policy document. Treat it as a live architecture question for anyone running it, not as a settled fact in either direction.

WMS, warehouse execution and TMS

A manufacturer's warehouse is not a distributor's warehouse. It holds three stock populations with different behaviours: raw material and components, received and held in a quality status until released; work in progress, which may be in a staging area, in a tank, on a rack or on the line; and finished goods, produced and held until disposition.

Work in progress is the stock class that the warehouse, execution and transactional systems all claim and none owns completely. That is the same argument as the boundary war with different nouns, and it needs the same written answer.

The warehouse system masters stock by bin, pallet or handling unit with its status; the task and its state; the receipt, putaway, replenishment, pick, pack and load transactions at bin granularity; and the physical layout. Its grain is the handling unit at a bin. The boundary with the transactional system is whether that system's storage location is the sum of warehouse bins maintained by interface, and the right answer depends on whether anything outside the warehouse needs bin-level visibility. Usually nothing does.

Below it, where there is automation, warehouse execution does short-interval work release and balancing, and **warehouse control talks to the controllers on the conveyor and the sorter in real time. The control layer is operational technology** — it has a scan rate, it must not stop, and its supplier is a materials-handling integrator rather than a software company. Patch it on a Tuesday and the conveyor stops.

Transport management handles load building, carrier selection, rating, tendering and freight settlement; the manufacturing-specific parts are inbound transport, bulk and tanker movements with prior-load restrictions, dangerous goods, and dock and yard management. **The weighbridge deserves its own line**: it is a legal-metrology instrument under the Legal Metrology Act, 2014, administered by the National Regulator for Compulsory Specifications, which requires instruments used for trade to be of an approved type and verified [5]. Software may therefore not adjust, rescale or round the reading — and it is almost always integrated through a serial connection to a gatehouse PC with a manual re-key.

How it fails: stock accuracy diverges because the three populations are reconciled on three cycles. Staging and consumption modelled as warehouse tasks when they are production events. **Quality status held in one system and not the other, so a picker picks a lot that is on hold — the most consequential integration defect in food and pharmaceutical manufacture.** And handling-unit identifiers generated locally rather than globally, so two plants' pallets collide the first time one ships to the other.

EAM and CMMS

Maintenance management is work orders, preventive schedules and spares; enterprise asset management is that plus asset lifecycle, capital projects, condition monitoring, reliability analysis and whole-life cost. The distinction is real and the words are used loosely.

The data model here is more standardised than most of manufacturing, and that is an opportunity. Learn five objects.

Functional location is a *position* in the plant where equipment is installed, independent of what is installed there. It is the most important and most frequently misunderstood concept in maintenance data: it exists so that failure history accumulates against the position — where the process stress is — and not only against the serial-numbered pump that gets swapped out and rebuilt. **Equipment** is the individually identified physical thing, installed at and dismantled from locations, carrying its history with it. **Criticality** drives strategy and spares policy and should come from a structured analysis; in most plants it comes from an engineer's opinion in

1. **Maintenance plan and task list** schedule standardised work by time, by counter or by condition. **Measuring point and counter** are the link to the plant data: counters fed from the historian or the controller, which is the single most valuable plant-to-enterprise integration in this domain, because counter-based preventive maintenance is far better than calendar-based and requires only a tag.

Strategies run from reactive, through preventive by time or counter, to condition-based with a measurement and a threshold, to predictive with a model of remaining useful life. **Be sceptical of the last.** Predictive maintenance needs labelled failure history, and the whole point of a well-maintained plant is

that it does not have many failures, so the training data for the failure modes that matter is thin. No credible local study of realised benefit was found in the research for this book; what you can reasonably do is ask a supplier for a before-and-after comparison of mean time between failures for a named asset class at a named plant.

For the taxonomy, start from **ISO 14224:2016**, which provides a standardised equipment taxonomy and a standardised failure-mode, failure-cause and failure-mechanism coding scheme [6]. It was written for oil and gas, it is used far beyond it, and adopting it means the client's reliability data is comparable with something.

How it fails, and the first one is expensive: **failure codes captured as free text, so three years of work orders yield no reliability analysis at all** — which destroys the data asset the system exists to build. The location hierarchy built once from the drawings and never updated, so new equipment has no location. Criticality unpopulated, so a safety-critical valve and a car-park light get the same programme. Counter integration descoped, so the plant services machines that have not run. Work orders closed without labour or parts, so cost per asset is unknown. And two systems — the ERP module for cost, a plant tool for execution — reconciled never.

LIMS and QMS

The laboratory system masters the specification with its version and effective date; the sampling plan; the sample with its source, time and chain of custody; the test and result with its method, instrument, analyst and evaluation; out-of-specification and out-of-trend investigations; the certificate of analysis; and stability studies, which run for months or years and break most workflow tools.

Two points matter architecturally. **A specification is versioned and effective-dated, and the version used must be recorded against the result**, because a change does not retrospectively re-judge old batches — and without the version, a tightened limit fails two years of released product in any report that re-evaluates. And results should come from the instrument: where they are hand-typed from a printout you have both a data-integrity finding and the source of most transposition errors.

The quality management system holds the non-conformance and its disposition; the deviation, planned or unplanned; corrective and preventive action; change control; audit; complaint and field action; training records that gate execution; and controlled documents. Corrective action is **the object linking a quality event to a change in the product or process**, and therefore to an engineering change or a document revision. That link is almost never systemised.

Where product is pharmaceutical, medical device or food, an overlay changes the architecture. Every system in scope needs a validated state, which means **the change cadence of a validated execution or laboratory system is structurally slower than IT's** — propose continuous delivery into one and you need a very specific answer about how validation keeps pace. Audit trails are a functional requirement with a data-volume consequence. Data integrity under the ALCOA principles is assessed across the whole data path, which includes the historian's compression settings and the clock. And supplier assessment extends validation obligations to the software vendor and any cloud host, which is one real reason these systems stay on premises.

How it fails: specifications maintained in both the laboratory system and the ERP inspection plan, drifting, so a lot passes in one and fails in the other. Hold-and-release authority split across four systems with no declared owner. Corrective action used as a ticket queue with no root-cause discipline. And document control as a shared folder, so the operator is reading an uncontrolled printout taped to the machine.

Shop-floor data collection

The thin layer that gets physical events into software: fixed terminals and rugged tablets, handheld scanners, badge readers, andon pull-cords and stack lights, counting sensors and the machine's own digital outputs. It masters the scan event, the count, the weight and the print.

The design constraints are not ones an IT architect expects. Gloves, so no small touch targets and no typed passwords. Noise, so no audio feedback. Wet and caustic areas, so sealed enclosures. Explosive atmospheres, so certified devices at five times the price and a fraction of the choice. A freezer store at -25°C, which kills consumer tablets and their batteries. And the operator's hands are the production process.

Identification. The manufacturing two-dimensional standard is **Data Matrix**: small, tolerant of damage through error correction, and capable of being marked directly onto metal or plastic by laser or dot-peen. Direct part marking matters architecturally because it makes the identifier permanent and physical, so **the serial number must be allocated before the mark is made and cannot be changed afterwards.** Radio-frequency identification works where barcodes cannot — returnable containers, processes that destroy a label, tool and die identification — and does not pay for item-level tagging of low-value product.

Label printing is under-appreciated and a recurring source of incidents. A label is a regulated artefact carrying allergen declarations, net content, origin, batch code, best-before date and hazard pictograms, and its template is an application with business logic in it — date arithmetic, unit conversion, check digits, conditional content — sitting in a file managed by whoever bought the printing software. Print is also a physical act with no transaction: a command that succeeds in software and fails at the printer produces a gap, a reprint produces a duplicate, and serialised printing therefore requires print-and-verify with an inline scanner reading what was actually printed. Label printers are plant devices with firmware and default credentials, and among the most commonly found unmanaged items in an asset inventory.

Weighing appears in four roles that must not be conflated: dispensing against a recipe target, where the weight is a recipe actual; checkweighing in line, which governs net-content compliance and controls giveaway; trade weighing, which is legal metrology [5]; and loss-in-weight dosing, where the signal is a control input and not a record. Alongside it sits in-line inspection: metal detection and X-ray for foreign bodies, vision for fill level, cap presence and date-code legibility, leak testing, temperature verification.

The pattern common to all of them is the reject. The reject must be confirmed — did the mechanism actually divert it — counted, reconciled, and recorded against the batch. An unconfirmed reject means a non-conforming pack may be in the shipment, and counting is the only way to know.

Two warnings on vision. **Deep-learning inspection changes the validation question**: a rule-based check is deterministic and testable, but a trained classifier's behaviour depends on its training set, so re-training is a change to a quality-critical control and the model is a validated artefact with a version. Most deployments do not treat it that way. And **vision is the most common place where a machine builder's black box sits in the data path**: the

vendor's software on the vendor's PC with a local database and one digital output that says pass or fail. Always ask what data exists inside the box and what it will cost to get out.

Traceability and serialisation

The vocabulary, precisely

A **batch** or **lot** is a quantity produced under uniform conditions; a **serial number** identifies one unit. **Genealogy** is the recorded relationship between what went in and what came out, at every operation. **Forward trace** — given an input lot, which outputs contain it and where did they go — answers *what must we recall*; **backward trace** answers *what caused this*. **Aggregation** is the containment hierarchy: item in carton, carton in case, case on pallet, because nobody opens a pallet to read the items. And **recall simulation** is running a forward trace against a hypothetical lot to test whether the data and the process can answer in time. That drill is an architecture test and the best way to prove or disprove a traceability claim.

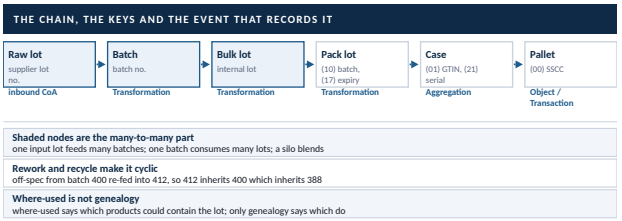


Figure 12.6 The chain from a supplier lot to a pallet, the GS1 key that identifies each stage, and the EPCIS event type that records the transition into it. The shaded nodes are where the tree becomes a graph.

In discrete assembly, genealogy is a tree and traversal is bounded. In process manufacture it is a directed graph with many-to-many edges and possible cycles: one raw lot feeds many batches, one batch consumes many lots, a silo blends, rework creates cycles, repack creates new lot numbers for the same material. A single forward trace in a food or chemical plant commonly expands to thousands of downstream lots within three or four hops.

> The decision that determines whether a recall is narrow or wide is made years earlier, when somebody decides the batch boundary and the silo model. A plant that records which silo compartment a load went into can narrow a recall. A plant that records "received into Silo 3" cannot.

The second consequence is performance. A four-level traversal over three years of transaction data in a system built for transactional writes takes hours. Mature implementations **materialise the genealogy** into a purpose-built structure maintained as production happens. Immature ones discover the problem during the recall.

GS1 keys, upstream of retail

Learn the application identifiers, because those numbers are what is encoded on the label and what the receiving system parses. **GTIN** identifies a trade item at each packaging level, and the manufacturer as brand owner allocates them, which makes allocation a governance process the manufacturer owns. **SSCC** identifies one logistics unit, and is the join between the physical pallet, the despatch message and the receiving scan; its absence makes shipment-level traceability impossible. **GLN** identifies a party or a location. The identifiers you will meet most are (01) GTIN, (10) batch, (17) expiry, (21) serial, (00) SSCC and (3103) net weight.

EPCIS standardises the event: what happened, when, where and why, with a controlled vocabulary supplying the values; the current vocabulary version is 2.0, ratified June 2022 [7]. Five event types matter. ObjectEvent is something happening to objects; AggregationEvent is objects put into or taken out of a container; TransactionEvent associates objects with a business transaction; **TransformationEvent is the manufacturing event**, and it carries a transformation identifier precisely so that a continuous process can be represented by several events sharing one; and AssociationEvent records one object fitted to another [7].

There is a nice detail in the vocabulary. The business step **assembling** is distinguished from transformation because the original objects remain recognisable or the process is reversible [7]. That is the discrete-versus-process distinction appearing inside the standard itself, because GS1 had to model both.

> **Do not write that GS1 requires a two-dimensional carrier by 2027.** GS1 US's own page uses no mandate language, calls it a milestone in the transition to two-dimensional barcodes, states that it is not necessary to implement all two-dimensional capabilities by that date, names one retailer-side capability — scanners able to read the code and extract the identifier at point of sale — and frames adoption as crawl, walk, run [8]. It is a voluntary industry capability ambition with no day and no month. The manufacturer-side consequence is nonetheless real: putting batch, expiry and serial into a two-dimensional carrier on a production line is a printing, verification and data-availability project with a capital component, and the data has to come from the execution system and the item master.

The regulatory drivers, and the South African position

Pharmaceutical serialisation is the strongest driver and it is extraterritorial. The European Falsified Medicines Directive and the United States Drug Supply Chain Security Act are **different architectures, not different parameters** — one a centralised repository verified at dispense, the other a distributed trading-partner tracing model — so a manufacturer exporting to both implements both. Note too that serialisation uses its own level numbering, L1 to L5, which cuts across the control-system levels in chapter 13: "Level 4" in a serialisation conversation and "Level 4" in a plant architecture conversation are different things.

For food, the North American position is prescriptive and the South African one is not, and importing the former by default over-specifies a domestic manufacturer.

- **United States.** The food traceability rule requires additional records of critical tracking events for foods on a designated Food Traceability List. The compliance date most commonly quoted, 20 July 2028, **comes from a proposed rule whose own text says "if finalized"** [9]. Secondary sources treat it as settled. It is not.
- **South Africa.** The obligation is **Regulation R638**, gazetted June 2018 and replacing R962 of

1. It requires records of processing, production and distribution kept for **at least six months beyond the product's shelf life**, a traceability system, a recall procedure, and notification of a recall incident to the local inspector and to the National Directorate: Food Control [10]. **There is no Food Traceability List, no prescribed critical tracking events and no prescribed key data elements.**

The practical driver in South Africa is therefore the customer, not the regulator: a manufacturer supplying a major grocer or exporting to the European Union implements to the customer's standard and to a benchmarked food-safety scheme, not to R638's minimum. Design to R638 for a purely domestic manufacturer and to the destination regime for an exporter — and say in the architecture principles which one you assumed.

IIoT platforms, unified namespace, edge and the digital twin

This section is deliberately sceptical, because the spend in this category is large, the realised value is uneven, and telling the parts apart is a substantial share of your professional value to a manufacturing client.

IIoT platforms. Three things are substantive. Connectivity breadth: a product with two hundred maintained protocol drivers saves a genuine integration cost where the plant has a long tail of old and odd equipment, and that unglamorous thing is what the money buys. Edge-to-cloud buffering done properly, because store-and-forward with guaranteed delivery and ordered replay is hard to build. And a modelling layer above tags, if and only if somebody populates it. **What is a slide is the claim of replacement.** A platform does not do work-order execution, batch records, genealogy, qualification enforcement or financial posting. **The question to ask is: which of my existing systems does this retire?** If the answer is none, it is an addition, and additions justify themselves on their own.

Unified namespace. Every system publishes its current state to a central broker under a hierarchical topic structure mirroring the plant, and every consumer subscribes to what it needs.

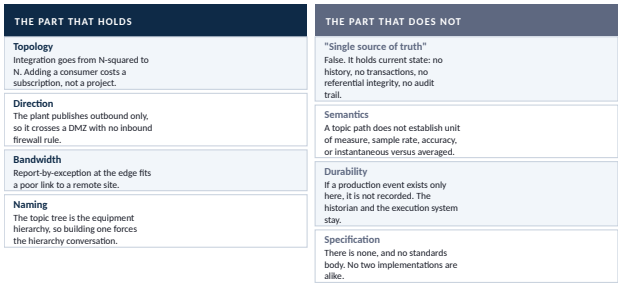


Figure 12.7 Unified namespace, split honestly. The left column is a genuinely strong architectural argument. The right column is what the marketing says and the mechanism cannot support.

The topology argument is genuinely strong. What you must not repeat is the "single source of truth" claim, and the word that makes it false is *current*. There is also **no specification and no standards body** — the concept is promoted through integrator and vendor channels, which does not make it wrong but does mean there is nothing to conform to. The defensible position: adopt the topology; specify the payload convention, for which Sparkplug is an Eclipse Foundation specification and **Sparkplug 3.0 is published as ISO/IEC 20237:2023**, so it can be named with a standard number in a procurement document [11]; insist the namespace is a transport and contextualisation layer rather than a system of record; and insist that something durable subscribes to everything and persists it.

Edge computing. The three real reasons are not the ones usually given. Resilience is the actual reason: the plant must keep producing when the wide-area link is down, so anything in the execution path is local. Latency for control is a hard boundary set by physics — though genuine control at the edge is rarer than claimed, and most "edge analytics" is monitoring, which tolerates latency. Bandwidth is real at remote sites. What is a slide is "edge AI" as a category: running inference on a gateway is straightforward, and the hard parts are the training data, the model lifecycle, the validation, and the fact that a model on forty gateways across six plants is 240 deployments to version and monitor. **The edge is an operations problem disguised as an architecture opportunity.**

Digital twin. The term covers at least five different things and you must insist on disambiguation before discussing it.

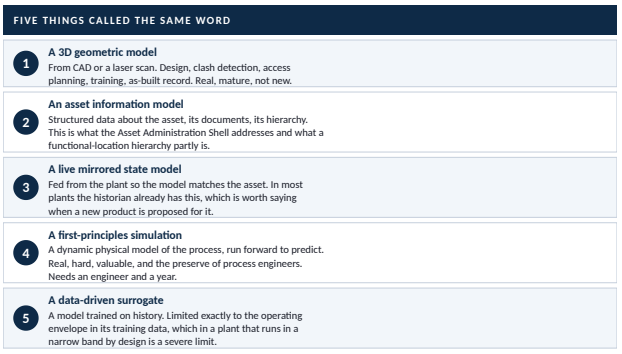


Figure 12.8 Five different things called a digital twin. The one-page taxonomy, the client's stated use case mapped to one of them, and the data that case needs, is the deliverable that kills the most bad projects.

The commercial use of the term blurs the first, third and fourth so that a three-dimensional viewer with live tag values is sold with the benefit case of a first-principles simulator. The genuine value cases are narrow: operator training simulators, which are mature and demonstrably valuable; commissioning and control-logic testing against a simulated plant, which shortens start-up; and design-stage process simulation, which has been standard engineering practice for forty years. A "digital twin of our factory" with no defined scope, no defined data requirement and no defined decision it supports should be refused until it has all three.

> **On adoption figures for digital twin and the Asset Administration Shell, the finding is the absence.** The research behind this book found no credible independent survey of adoption or realised benefit, in any geography. Every statistics source returned was search-optimised content aggregating other such content, with no primary research behind the numbers, and several were evidently machine-generated. **Assume any specific percentage you are shown is fabricated until somebody names the survey instrument and the sample.**

The standards themselves are real, which is a separate question from adoption. The Asset Administration Shell exists as two version lines that must not be conflated: **IEC 63278-1 Edition 1.0 of 2023-12 for the structure**, and the Industrial Digital Twin Association's **IDTA-01001 metamodel at version 3.x** [12][13]. For a South African manufacturer the realistic 2026 position is that you will meet the Shell as a file your German machine supplier sends you and have nowhere to put it. The question is not whether to adopt it but which system is its home when a package arrives — usually product lifecycle or asset management, neither of which has an importer today.

The shape of a South African estate

A closing characterisation, offered as judgement rather than measurement: the research behind this book flags the typical-estate description as unverified, because no South African manufacturer publishes a portfolio inventory and the way to settle it is a walkdown.

What you will usually find: **one transactional system, often eighteen years old, frequently in more than one instance** that do not share an item master. A product lifecycle system in the discrete businesses and nothing of the kind in the process ones, where the recipe lives in the batch engine and a second copy lives as a bill of material for costing. Planning in the ERP's own engine plus three spreadsheets of real sophistication maintained by one person. Scheduling on paper. **One execution system per plant at best, and more often execution split between the ERP's confirmation screens and a local application somebody built.** A historian at the process sites and nothing at the discrete ones. A maintenance system bought in the last five years and populated to about 40% of its intended depth. Laboratory and quality in Excel unless the product is regulated or a customer audited it.

And it is overwhelmingly on premises, for reasons that are physical rather than cultural, and mostly correct ones.

1. **The execution path cannot cross a wide-area link.** Universal rather than South African, and it alone keeps execution, supervisory, historian, gateway and batch systems in the plant.
2. **Connectivity to the plant, not to the cloud, is the constraint.** Cloud regions now exist in Johannesburg and Cape Town, which materially improved the position. A plant on an industrial estate in Gauteng with dual fibre is fine. A mill, a smelter or a food-processing site in a rural area frequently has one path, with restoration measured in days after a cable is cut or stolen.
3. **Validated-state and supplier-assessment obligations** make cloud hosting of batch-record systems a documentation exercise many mid-market manufacturers decline.
4. **The capital cycle.** Plant IT is bought with the plant and replaced at a turnaround. A 2014 supervisory system on a 2014 server has a 2029 replacement date and no case before then.

The pattern that works is two tiers: a plant-resident execution and acquisition tier that survives isolation, and a central or cloud tier for aggregation, analytics, planning, product lifecycle and the transactional system. The boundary is the execution path, and the test of any proposed migration is one sentence: *can the plant run for 72 hours without this?* If not, it stays local.

THE CHAPTER IN TEN LINES

- Ask six questions of every system class: what it is for, what it masters, what it integrates with, its grain, its licence metric, and how it fails.
- ERP is the system of record for the financial representation of production, and it is not the system that runs the plant.
- The three things ERP is always asked to do and does badly are finite scheduling, real-time execution and anything re-run on a changed assumption.
- MOM is the capability space from the boundary standard; MES is the market term for a product; use the four operations areas and eight activities to find the gaps.
- The ERP to execution boundary is settled on grain and latency if principle wins, and on go-live dates and licence metrics if it does not.
- The engineering and manufacturing bills of material differ intrinsically, and the transformation between them is a human judgement that nobody has automated.
- Revision without effectivity is the commonest product-lifecycle defect and stays invisible until the first field failure or audit.
- Free-text failure codes destroy the data asset a maintenance system exists to build; start from ISO 14224 instead.
- Whether a recall is narrow or wide was decided years earlier, by whoever set the batch boundary and the silo model.
- GS1 requires no two-dimensional carrier by any date, and South Africa has no FSMA-204 equivalent: R638 asks for records six months beyond shelf life, a traceability system, a recall procedure and notification.
- A unified namespace holds current state only, so it is a transport and contextualisation layer and never a system of record.
- No credible independent adoption data exists for digital twin or the Asset Administration Shell; treat any percentage as fabricated until the survey is named.

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CHAPTER 13

The plant floor — OT, ISA-95 and the IT/OT boundary

The layer no other industry in this series has. ISA-95 and Purdue as they really are, why a tag is not a data entity, what a historian quietly does to a number before you read it, and why a South African plant must be designed to keep producing in the dark and off the network.

Below the ERP

No other industry in this series has this chapter, because no other has an operational technology layer. A bank's estate ends at a database. A manufacturer's carries on downward through supervisory software, controllers, instrument loops, drives and safety systems, until it reaches a valve that either opened or did not. The two halves differ in almost every dimension you assume is settled.

- IT measures availability in nines and change in sprints; the plant measures availability in years of uninterrupted running and change in shutdown windows that come twice a decade, and patches only when the builder has validated the patch against that firmware revision.
- IT's unit of data is a record; the plant's is a **tag**, sampled on a scan, and its clock may be a controller that has drifted four minutes since commissioning.
- IT's security priority runs confidentiality, integrity, availability; the plant's runs safety, availability, integrity, confidentiality, and that inversion is rational, not primitive.

Everything hard in manufacturing architecture happens at that seam. And one fact generates most of the non-obvious requirements you will meet: *the physical process does not stop because the software stopped.* A plant whose execution system is down keeps making product, because the material is already in the vessel and the reaction is already running. From that come store-and-forward queues, local buffering at the line, validated paper fall-back procedures, and an execution system that must come back and reconcile hours it did not see.

ISA-95, honestly

ISA-95 is the vocabulary of manufacturing architecture. Learn it, use it, and understand what it does not give you. It is maintained by the International Society of Automation as ANSI/ISA-95 and published internationally as IEC 62264 [1].



Figure 13.1 The levels, and what each actually contains. The timescales are a practitioner ordering, not a statement from the standard, which describes activity scopes rather than scan rates.

The figure's right-hand column is a practitioner ordering, not anything the standard states.

The version facts, which almost all circulating material gets wrong

Part 1 is the 2025 edition. ANSI/ISA-95.00.01-2025 replaces the 2010 edition, announced on 10 April 2025 [1][2]. Almost every slide and textbook in circulation cites 2010. It is not a rewrite: the functional enterprise-control figure was updated to mark the enterprise-to-manufacturing boundary, and clauses moved to an annex [2].

There are eight parts plus a technical report — not five. Most secondary material, including the body text of the most-read encyclopaedia article on the subject, says five, while that same article's reference list contradicts it [4]. Parts 6, 7 and 8 exist and matter: a messaging service model, an alias service for reconciling the different names one thing has in different systems, and information exchange profiles [1][4].

> **Do not print a table of ISA-95's parts with their designations and dates.** The research behind this book could verify Part 1's 2025 designation and little else; the part years in circulation come from a community-edited source that is internally inconsistent about them. Write the shape — eight parts and a technical report — and buy the current list from ISA.

The ISA and IEC lineages are out of step and are different documents. The 2025 ANSI part was worked from IEC 62264-1 Edition 2.0, dated 2013-05 [2][3]. "(IEC 62264-1 Mod)" in the designation means the ANSI text is based on the IEC text with modifications, and ISA's release does not say what they are. Citing the two interchangeably with one year is wrong, and a contract clause must name which document it means. Note too that **IEC 62264-6 is a Publicly Available Specification — a pre-standard** [3].

B2MML is MESA's, not ISA's, and it is older than the current Part 1. It is an XML implementation of the ISA-95 data models published by MESA International, with BatchML as its ISA-88 sibling, royalty-free with attribution [5]. **Its current release is V7.00.00, of 24 November 2020, built to the**

2018/19 editions — five years older than the 2025 Part 1. Do not write that B2MML implements the current ISA-95; write that it implements the 2018/19 editions and that the gap is a fact to design around.

"ISA-95 compliant" means almost nothing

There is no conformance certification scheme for ISA-95. The contrast is instructive, because the same body is perfectly capable of writing one: IEC 62443 has ISASecure, with named programmes for the development lifecycle, for components and for systems [6], and ISA-88 Part 1 carries conformance criteria in its Clause 9 [7]. ISA-95 has neither — and the absence cannot be cited to a document, which is the honest way to state it [1].

The equipment hierarchy, and the correction that matters most

The pyramid is the famous picture; the useful content is the **equipment hierarchy model**: enterprise, site, area, and then — depending on the kind of manufacturing — process cell and unit, or production line and work cell, or storage zone and storage unit [1].

Read that again, because it is the correction almost every secondary treatment gets wrong. **This is not a seven-level stack ending in a control module.** The equipment-element enumeration, as published verbatim in the OPC Foundation's ISA-95 companion specification, carries fifteen values [15], and process cell, production line, production unit and storage zone are **alternatives selected by the kind of process**, not levels stacked on each other. The generic terms are work centre and work unit — which is why "work centre" means related but different things in ISA-95 and in an ERP. And **equipment module and control module are ISA-88 concepts that ISA-95 references**, not ISA-95's own.

Two consequences. **This is the only standard hierarchy you can hold a client to**, and it is the backbone of tag naming, of an asset model, of the maintenance hierarchy and of a namespace topic tree. And **it does not match an ERP's plant-and-storage-location model** — reconciling them is a real project, as is reconciling the asset and production hierarchies, two trees over one set of steel.

Three limits to hold alongside it [8]: it is a reference model and not an implementation; it assumes a tree, so it breaks where lines branch; and **it does not solve naming** — Parts 6 and 7 give a mechanism, not content, and the practitioner advice, agree your naming convention before you integrate anything, is the most valuable sentence in the ISA-95 literature.

The Purdue model as it is actually used

PERA is not the Purdue security model

The Purdue Enterprise Reference Architecture was developed by Theodore J. Williams and the Industry-Purdue University consortium; the canonical reference is Williams' 1992 technical guide, and the levels everyone draws come from the Purdue Reference Model of 1995, where they are **layers for integrating enterprise control applications** [9].

PERA says nothing about cybersecurity. Neither the 1992 guide nor the reference model contains any security content at all [9]. The Level 3.5 industrial demilitarised zone, the enterprise zone, the cell and area zones and the whole zone language are a later overlay from the ISA-99 and IEC 62443 community and from vendor reference architectures. Attribute the industrial DMZ to Williams in a workshop and the OT engineer in the room will correct you, and will be right.

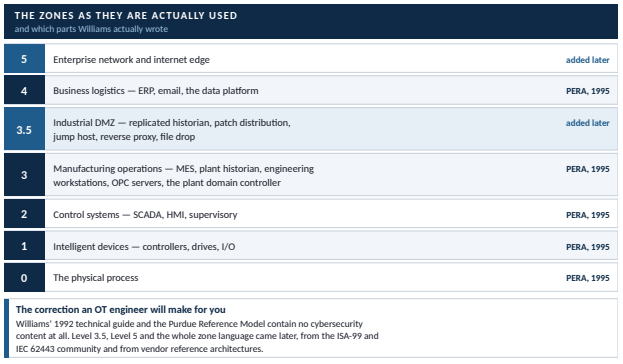


Figure 13.2 The zones as a security assessment actually uses them, with the provenance of each marked. Levels 3.5 and 5 and the whole zone language are later additions from the ISA-99 and IEC 62443 community, not from Williams.

Where the diagram and the plant diverge

Five divergences to expect on almost every assessment.

1. **Flat networks.** A plant built in 1998 and extended five times has one broadcast domain, so the levels exist on the drawing and not in the switch configuration.
2. **The dual-homed engineering workstation**, one card on the plant and one on corporate.
3. **The machine builder's remote-access box**, wired into the cell zone at commissioning, unknown to IT and in no asset register.
4. **Level 2 talking to Level 4 directly**, over a database link somebody built in 2011, and **historian collection crossing the DMZ in the wrong direction**.
5. **Shared infrastructure that nobody zoned**: time, name and directory services, antivirus consoles, backup agents, the wireless controller. Each is a conduit and each is undocumented.

And the characteristic failure: zones drawn on Purdue levels rather than on **consequence**, so a safety-critical cell and a packing line sit at the same security level.

ISA-88 and why recipes reorganise everything

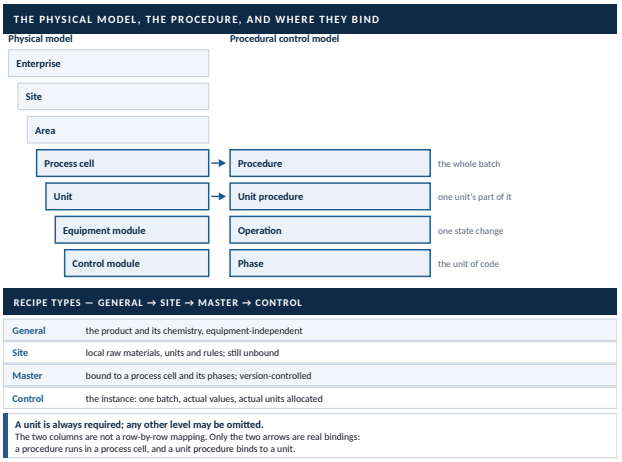


Figure 13.3 The batch-control models and the recipe types. Enterprise, site and area are shared with the ISA-95 equipment hierarchy; below that, ISA-88 refines. The phase is the unit of code a control engineer writes.

Part 1's current US edition is ANSI/ISA-88.00.01-2010, approved 6 December 2010, replacing the 1995 original, with conformance criteria in Clause 9 [7]; internationally **IEC 61512-1 is 1997** [10]. **The dates of Parts 2 to 5 are unverified and should not be stated from memory**: ISA's committee page lists no parts at all, and the most detailed public status account is a 2006 conference paper whose status columns are two decades stale [11]. That is itself the finding.

Three concepts architects miss [7]. **Basic, procedural and coordination control are three different things**: basic control is the regulatory and interlocking layer that runs whether or not a batch is executing; procedural control is the recipe-driven sequence; and **coordination control arbitrates** — which batch gets the shared transfer line, which unit is allocated — which is where the hard engineering is. **"Held" is not "stopped"**: held means suspended with intent to resume while the material is kept viable, so an interface must carry a batch held for six hours without the order looking complete or failed. And **the procedural state model in the standard is an example, not a mandate** — so an execution system that assumes one plant's state set will be wrong at the next.

Why process manufacturers' systems are organised around recipes

In discrete manufacturing the production definition is a **bill of material plus a routing** — two objects joined at the order, with integer quantities and scrap as an exception. In process manufacturing it is a **recipe**, which *contains* the formula and the procedure as two parts of one versioned object [10]. Four consequences ripple out.

1. **Material and sequence cannot be changed independently.** Raising the catalyst dose may require a longer hold at temperature, so change control must version them together, and a discrete pattern that versions the bill and the routing separately produces invalid combinations the moment it meets a process plant.
2. **Quantities are continuous and scaled**, some ingredients linearly and some not, so conversion with density, concentration and moisture is intrinsic rather than an edge case.
3. **Yield is variable by physics.** A 94% yield is the process, not an error, and the batch is the unit of quality disposition.
4. **Equipment allocation is a first-class scheduling object** — a train of units with transfer constraints, not a work centre for a duration.

This bites hardest in a mixed-mode manufacturer, and Karoo Foods — invented, like the rest of the running example — is one. A bulk emulsion is made in a recipe-driven process plant, then filled and cased on a discrete line with a bill of material and a routing. The handover carries a lot number and kilograms; the downstream a count of cases. **Name that interface explicitly**, because unit conversion, lot-to-serial linkage and yield reconciliation land on it at once — and it is usually a figure written on a clipboard.

The control layer

PLC and PAC

A **programmable logic controller** is an industrial computer with deterministic cyclic execution: read inputs into an image, execute the program, write outputs, repeat. The **scan cycle** is the fundamental unit — typically one to fifty milliseconds — and the program is written to complete inside it. Determinism is the property that matters: a controller that sometimes takes 80 ms is broken, not slow. A **programmable automation controller** adds memory, concurrent tasks at different scan rates, structured data types and often an embedded protocol server, so it can run its own northbound interface where a classic controller needs a gateway.

Programming is standardised by IEC 61131-3, and here is a version trap. IEC 61131-3:2013, Edition 3.0, **was withdrawn on 22 May 2025 and superseded by IEC 61131-3:2025, Edition 4.0** [12][13]. Nearly every source still cites 2013. The "five languages" claim is contestable too: the 2013 edition specifies two textual languages, instruction list and structured text, and two graphical ones, ladder diagram and function block diagram, with sequential function chart defined as a structuring element rather than a fifth language in the same sense [12].

Safety is separate on purpose. Safety functions run on a safety instrumented system certified to a safety integrity level, with changes under functional-safety management of change. Propose reading from it and expect resistance; accept it, and take a read-only feed through a hardware-enforced one-way path. The Triton intrusion of 2017, whose first symptom was an unexplained trip, is why.

DCS, SCADA, and the difference that survives

A **distributed control system** is a single engineered control system covering a whole process plant — controllers, input and output, operator and engineering stations, historian, alarm management and a database, all from one supplier, configured from one environment, sharing one tag database. **The defining property is not "distributed" — everything is distributed. It is engineering coherence.** A loop configured once appears automatically on the operator graphic, in the historian, in the alarm list and in the reports, because they are views of one database.

Supervisory control and data acquisition is a software layer that polls or subscribes to many independent controllers, presents a consolidated view, logs data and alarms, and lets an operator issue setpoints.

> **An authoritative, citable statement of the SCADA-versus-DCS difference could not be found.** Every public comparison located in the research for this book was vendor marketing or search-optimised content with no primary source, and **ISA-95 itself places both at Level 2 and makes no distinction** [1]. What follows is practitioner judgement, offered as such.

The functional overlap is now almost total; what survives is commercial and organisational. With a distributed control system you buy a coherent system and configure it; with supervisory software you buy a visualisation platform and integrate it to controllers chosen separately, so the tag database, the alarm configuration, the graphic and the historian point are four configuration acts somebody must keep consistent. **The distributed control system removes an integration problem and adds a lock-in; supervisory software adds an integration problem and removes a lock-in**, and that trade is the actual decision. Ownership differs too: process control engineering owns the first and it is untouchable, while the second is often owned by an automation engineer or by IT, and is therefore more exposed to well-meaning change.

Why a tag is not a data entity

This is the deepest conceptual gap for an architect arriving from IT. A tag is a named value in a controller or supervisory system, with a data type, an address, engineering units, a range, alarm limits and a scan class. A mid-size plant has tens of thousands.

1. **A tag has no identity over time, only a name.** It is a *position* whose value is whatever the instrument reads now. Replace the instrument with a different model and range and the name is unchanged while the series silently changes meaning. No version, no effective date, no key.
2. **A tag has no relationships.** Nothing says this flow is the feed into Reactor 2, that Reactor 2 is in Area 30, or that the flow and the temperature on one vessel belong together. **The relationships live in the tag name**, and in the operator graphic, which is a picture.
3. **Two tags that look alike are not comparable.** A level in metres against a level in percent of span, with only the engineering-units attribute to say which, and it is often blank.
4. **A tag has no transactional context** — no order, batch, lot, product or shift. **Attaching that context is the entire job of the Level 2 to Level 3 boundary**, done by correlating tag time ranges against order start and end times, which is why clock disagreement corrupts product data and not just reports.
5. **A tag's value is a sample, not a fact** — true at a scan instant, to the instrument's accuracy, subject to the filter configured in the transmitter.
6. **A tag is written, and writing is dangerous.** A write can come from a faceplate, a protocol client, a script or an engineering station, and a write to the wrong tag moves a physical thing. This is why read-only is a design stance rather than a limitation.

A design that treats tags as the manufacturing data model produces an unusable data platform. The work is an asset model above them — equipment instances, each of a type, each with named attributes bound to tags — which is what historian asset frameworks, OPC UA information models and the Asset Administration Shell are all independently attempting.

Historians, and the number that changes itself

A process historian stores tag values at high rate with long retention and answers queries of the form *give me these forty tags between these two timestamps, interpolated to a ten-second grid*. **It is not a database in the sense an IT architect means**: the model is tag, timestamp, value, status; writes are appends at the current time; reads are time-range scans; and there are no joins, no transactions and no referential integrity.

Tag naming conventions are the plant's real data model. A tag name is a compressed composite key carrying plant, area, unit, equipment, loop and measurement type, which is why an engineer reads `FIC`, `LAH` or `PDT` without a dictionary. **A plant has as many conventions as it has had expansion projects.** And **renaming is effectively impossible**, because the name is referenced in the controller, the graphic, the alarm database, the archive, every report and the piping drawings — so the convention you inherit is the one you keep, and the asset model is how you add meaning without renaming.

Exception and compression, and how they silently change a reported number

The reference implementation applies **two separate reduction mechanisms at two different points in the pipeline**, both using deadband-style tests [14].

Exception reporting happens at the interface, before data is sent: only values differing enough from the last sent value are transmitted, with a deviation in engineering units or as a percentage of span, and a maximum time between events [14]. Note the inconsistency across collection mechanisms — adapters apply deadband filtering but ignore exception settings, and connectors perform no exception testing at all [14] — **so the same tag collected two ways has two fidelities**.

Compression happens at the archive, using a swinging-door algorithm that removes data which can be safely recreated by interpolation [14]. Two details matter more than the algorithm. **Out-of-order data bypasses compression** [14], so late data is stored at full fidelity while live data is compressed, and one tag has two fidelities in the same archive. And **the deviation defaults are non-zero** [14] — compression is on unless somebody turned it off.

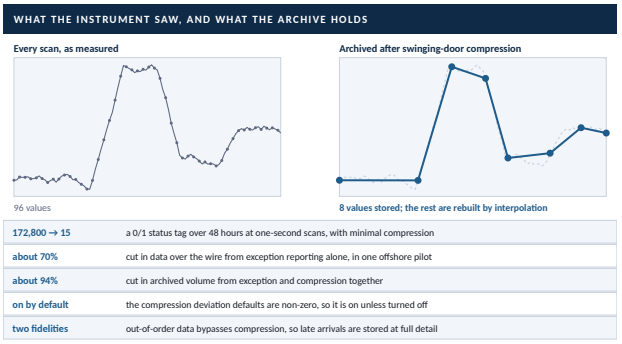


Figure 13.4 The set piece of this chapter. The same two days of one analogue tag: every scan on the left, and on the right what the archive keeps after swinging-door compression, with the discarded values shown dashed. Every query against a time grid is interpolating between the blue points.

Now the part you must explain to a data scientist, slowly, more than once.

- **The archive does not contain what the instrument measured.** It holds the smallest set of points from which the curve can be rebuilt to within the configured deviation [14].
- **Therefore every query against a time grid is interpolating, and interpolation is a modelling choice.** Ask for a value at 09:14:03 and if no point exists the historian returns an interpolated value for an analogue tag and a stepped, last-known value for a discrete one. **A `step` attribute decides which**, and a tag wrongly flagged returns physically impossible intermediates — a valve 37% open when it has two positions.
- **Therefore averages are not what they look like**, and two reports can both be right and differ. A time-weighted average over archived points is not the arithmetic mean of the original scans, and they diverge most during the transients that matter.
- **Therefore compression settings are a governance object.** Changing the deviation changes all future data for that tag and not the past, so the series has a discontinuity in information content at the moment somebody tuned it, and nobody records when.

> **Your deliverable here** is a statement, per tag class, of the exception and compression settings, the step flag, the retention period and the aggregation method reports must use, all under change control. It is the difference between a data platform that can support a yield investigation and one that cannot.

In the regulated case this stops being an engineering nicety. Data integrity under the ALCOA principles is assessed **on the whole data path**. A regulator can ask why the archived value differs from the instrument reading, and "compression" is an answer only if the settings are documented, justified and controlled. The same applies to the clock, which is next.

The protocol layer

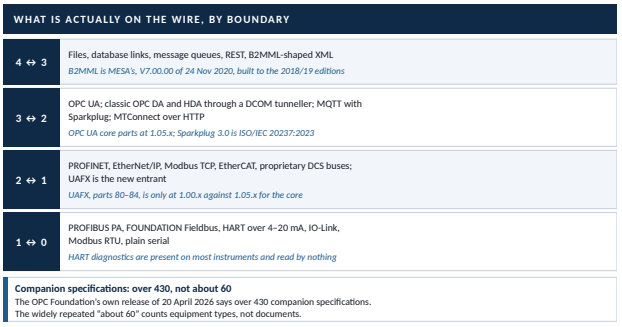


Figure 13.5 What is actually on the wire at each boundary, with the version fact that most often goes stale beside it. The prevalence ordering is practitioner consensus; a passive protocol discovery settles it at any given plant.

OPC UA is standardised as IEC 62541. It is not "OPC with better security": classic OPC was built on a Microsoft component technology that made it Windows-only and firewall-hostile, while OPC UA is platform-independent with its own secure channel and — critically — **an information model rather than a flat list of tags**. It has two communication patterns: client-server, and a publish-subscribe part that makes it usable for high-rate telemetry and for crossing a DMZ outbound only. **Companion specifications** turn it from a transport into interoperability: each is a standardised information model for a domain, so every machine of a kind exposes the same nodes with the same meaning — the difference between *I can read tags from this machine* and *I can read tags from any machine of this class without a mapping project*.

> **The count is over 430, not about 60.** The OPC Foundation's own release of 20 April 2026 states there are over 430 companion specifications [16]. The widely repeated "about 60" counts *equipment types*, not documents, and is an order-of-magnitude error.

Maturity is not uniform, and the version numbers say so. Core parts sit at 1.05.x; the field-exchange family, parts 80 to 84, is only at 1.00.x [15]. Several companion specifications carried unfilled template text in their titles at retrieval [15]. **Published does not mean implemented**: at a machine purchase, name the specification, the version and a certification requirement, and ask whether *this firmware revision* exposes the nodes.

MTConnect defines a shared vocabulary for machine data through three components: the machine, an adapter translating native signals into a simple line protocol, and an agent exposing them over HTTP [17]. Its centre of gravity is North American discrete metalworking. Its advantage is that it is **read-only and HTTP-based**, essentially impossible to use to write to a machine — worth saying out loud when an OT engineer objects to connectivity. Its limit is machine monitoring.

> **MTConnect's ANSI designation is a full major version behind its documentation**, and the documentation site was internally inconsistent about its own current version at retrieval [17]. **Which version holds ANSI approval is unverified.** Do not write "ANSI/MTC1.4-2018 is the current MTConnect standard" as though that were the whole truth.

MQTT is a lightweight publish-subscribe transport with a topic string and an opaque payload. It specifies nothing about what is in the payload, what the topics are called, or how a subscriber learns the current value of something it has just subscribed to — and in a plant those three gaps are the whole problem. **Sparkplug fills them** with a topic namespace, a payload encoding, a birth-and-death certificate mechanism so a subscriber gets a full tag definition and current state on connect, and a state convention distinguishing *the value has not changed* from *the publisher is gone*. **Sparkplug is Eclipse's, not the OPC Foundation's**, and **Sparkplug 3.0 is published as ISO/IEC 20237:2023** [18] — so it can be named with a standard number in a procurement document, which matters because it is commonly described as informal. It fails predictably: the retain flag set on data messages, creating stale values read as current; sequence numbers ignored; and no replay, because the broker is a live bus, not a log.

The Asset Administration Shell is two version lines and citing one misrepresents the landscape: **IEC 63278-1 Edition 1.0, 2023-12, for the structure** [19], and the Industrial Digital Twin Association's **IDTA-01001 metamodel at version 3.x** for the implementable specification [20].

RAMI 4.0 is a pre-standard, not an IEC or ISO standard: DIN SPEC 91345 of April 2016 and internationally IEC PAS 63088:2017 [21][22], and both a publicly available specification and a DIN SPEC are pre-standards. It is a three-axis classification scheme with no reference data model, no interface specification, no protocol and no conformance criteria. You cannot build anything from it.

Integration, as it really is

The protocol discontinuity

There is no shared primitive. An IT system asks a question and gets an answer; a fieldbus master polls a slave register on a fixed cycle and gets sixteen bits with no semantics. The discontinuity is bridged in one place — a protocol gateway — and **that single component is the most architecturally significant and least well-managed item in the estate**: a Windows server under a desk, two versions behind, configured by a contractor in 2017, with the configuration existing only on that machine.

Two discontinuities beyond protocol. **Typing**: whether a pair of bytes is a float, two integers or a bitfield is gateway configuration, so getting it wrong produces plausible wrong numbers rather than errors. **And the semantics of absence**: a plant read returns the last value, or zero, or a stale value with a good quality flag, and **quality flags are almost universally discarded at the boundary**, so a consumer cannot distinguish *the pump is stopped* from *we lost communication with the pump*.

The DMZ breaks request-response

Correct zoning says nothing in the enterprise zone connects directly to anything below the DMZ. That is good security, and it breaks the pattern most IT architects default to: a synchronous call from an enterprise system into a plant system needs an inbound rule the plant will refuse.

The patterns that work, in order of preference: **outbound publish from the plant** to a broker in the DMZ with the enterprise subscribing from there, so the plant initiates every connection — which is why publish-subscribe is accepted where an HTTP interface is not; a **replication node in the DMZ** written from below and read from above, of which a historian replica is the canonical example; **file exchange through a DMZ drop**, ugly, universal and devoid of every real-time property; and a **data diode** where the consequence of compromise is extreme.

The rule: design every plant integration as plant-initiated and asynchronous. Treat a requirement for a synchronous enterprise-to-plant call as something to renegotiate. Where it genuinely cannot be — a line that must read a customer order's configuration before building a unit — cache the data in the plant rather than opening the firewall.

Polling, events and the queue that must not fill

Polling is the plant default: stateless and self-healing, so a missed cycle corrects itself. Events are efficient and introduce state. **A missed event is invisible** — polling degrades to staleness, events degrade to wrongness — so any event-based plant integration needs sequence numbers with gap detection or a periodic full-state refresh, and you should design both. **Deadbands make events lie**: a drift smaller than the deadband produces no events for hours while the consumer's last value looks live. And **polling at the wrong rate corrupts data silently** — poll a counter that wraps at 65,535 every 30 seconds on a line making 3,000 units a minute and you lose wraps. Nyquist applies to plant data and nobody checks it.

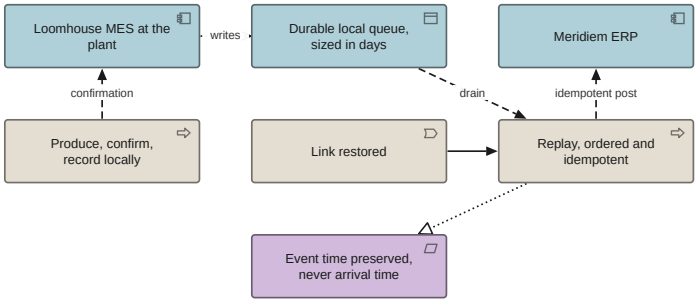


Figure 13.6 Store and forward, as a pattern to specify rather than discover. The requirement at the bottom is the one most often missed: replayed data carries the event time, never the time it arrived.

Store and forward is mandatory, not optional, because the plant keeps producing. Five questions must each have an answer. *How long must the buffer hold?* Not "a few hours" — the longest credible outage, which on a single-path link to a remote site is a multi-day event, so size it in days of production at full rate and alert on its depth. *What happens when it fills?* Drop oldest, drop newest, overwrite or stop producing — **a business decision with a regulatory dimension**, because in regulated manufacture losing records is not an option and the answer may be "stop the line". *Is the replay ordered and idempotent?* Forty thousand replayed confirmations must not double-post, and they carry event time, never arrival time. *Will the downstream system accept backdated transactions?* An ERP refuses a posting into a closed period. *Must the plant operate on a stale plan?* Yes — the dispatch list, recipe, specification and work instruction are resident locally, which means they can be stale, which means an effectivity change during an outage will not take.

Why timestamps disagree

This is the integration problem most likely to produce wrong answers that look right, and it deserves a paragraph in every manufacturing architecture document you write.

Six sources of disagreement. **Unsynchronised clocks** in the controller, the supervisory server, the historian, the execution system, the handheld, the label printer and the ERP, many with no time client at all. **Drift**, at seconds per week, so a controller commissioned in 2019 is minutes out. **The timestamp assignment point**: a value can be stamped by the device, by the gateway, by the interface when it sent, or by the server when it received — four times for one measurement, which is why the protocol distinguishes source from server timestamp and why most integrations use whichever the interface returned by default. **Timezone and daylight saving**, which South Africa does not observe, so the problem reappears only for a group with plants

abroad. **Resolution**, from millisecond controller events to a date-only ERP document, so joining them requires a window and the window is an assumption. And **ordering across sources**, so causality inverts and nobody notices.

What it corrupts: batch windows, so parameters land on the wrong batch; downtime, so a stop lands on the wrong shift; genealogy, so a component scanned at 06:00:02 joins the wrong unit; sequence-based root cause, so the investigation concludes the alarm followed the trip when it preceded it; and regulated batch records, which makes it a data-integrity finding.

What to specify. One authoritative time source for the site with a documented distribution path; network time at minimum and precision time where sub-millisecond ordering matters; every device recorded with its time source; **UTC stored everywhere, local time as presentation only**; source timestamps preserved end to end; and a periodic drift audit. It is cheap, it is boring, and it is the most valuable line you can put in a manufacturing integration standard.

Units, clocks and the message sets

Unit-of-measure conversion is a correctness problem, not a rounding problem. The conversion factor is master data, and master data is wrong. Worse, several conversions are **per lot rather than per item**: litres to kilograms needs density, which varies with temperature; wet kilograms to dry needs a moisture result. Genuine dual-unit support with lot-specific conversion is a hard selection criterion. And at the plant boundary a tag in bar read as kPa is wrong by a factor of 100 and looks entirely plausible, while **the engineering-unit attribute is frequently blank** — so make it mandatory on every tag that crosses.

Three clocks, none aligned. The plant clock runs shifts, so a production day is not a calendar day. The ERP batch window runs the nightly cycle, during which the plant must keep producing and queuing. The financial calendar runs periods with a cut-off that must be enforced at the plant and is not. A batch starting at 23:40 and finishing at 02:15 crosses all three, and **a reported figure for "yesterday" depends on which clock defined yesterday.**

Manufacturing EDI is older, deeper and more mandatory than in most industries, because the automotive and aerospace supply chains run on it. South Africa and Europe run **UN/EDIFACT** — purchase order, delivery forecast, just-in-time call-off, despatch advice, invoice. North America runs **ANSI ASC X12**, where 830 and 862 are the automotive planning and shipping-schedule pair. German automotive adds **VDA**, whose fixed-length formats are giving way to EDIFACT-based replacements, and European automotive adds **Odette** with its own peer-to-peer file transfer protocol, which must be planned for specifically.

Three things are hard beyond format. **The schedule, not the order, is the instruction**: rolling forecasts and firm releases make the order a continuously revised quantity-by-date grid expressed as a cumulative received quantity, and a cumulative discrepancy between the two parties is the most common dispute in the sector. **Labels are part of the message set**, and a wrong label is a rejected delivery. And **penalties are real**: a line stoppage at an assembler is charged to the supplier, which is why automotive suppliers over-invest in integration resilience.

OT security, concretely

Why patching a controller is not like patching a server

Six reasons, all true at once. **Availability is the top of the triad, not the bottom**, so a patch that risks an unplanned stop is a worse outcome than the vulnerability. **A patch usually requires a controller stop**, so the window is a shutdown that may be annual. **The supplier must validate the patch** against that firmware and configuration, or support is void and, in regulated industry, so is the validated state — there is a technical report in the IEC 62443 family devoted to patch management because the industry treats it as a distinct discipline [23]. **Re-validation costs more than the patch**, and **a change to a safety system is a functional-safety activity. Some equipment is out of support entirely** — a plant has controllers from 1996, so the vulnerability is permanent. And **the asset inventory does not exist**, which is the real reason most programmes stall.

A vulnerability programme with a 30-day critical-patch target is inapplicable below Level 3, and proposing it destroys your credibility in one meeting. The OT equivalent is **compensating control**: segmentation, allow-listing on fixed-function hosts, removable-media control, brokered and monitored remote access, and a patch cadence tied to the shutdown calendar with documented risk acceptance in between.

The air gap, and where the real path is

"The plant network is air-gapped" is almost never true, and the bridges are predictable: the dual-homed workstation, the builder's remote-access path, removable media — of which Stuxnet is the canonical proof — shared directory and time services, the historian feed. **Air gap is not a control; it is an assertion.** The useful question is not "are you air-gapped" but **"show me the asset inventory and the firewall rule base"**.

Remote vendor access is the real attack path, and there is data. Dragos's 2026 OT Cybersecurity Year in Review reports that **73% of all-time incident-response cases involved compromised VPN or jump-host credentials** [24], and puts **manufacturing at more than two-thirds of ransomware victims** in 2025, with 119 groups affecting 3,300 industrial organisations, up 49% from 80 the year before [24].

> That is a **vendor source** — an OT security firm reporting its own incident-response caseload, with an obvious selection bias toward organisations large enough to call one. Quote it with the attribution and the bias attached. The conclusion stands: manufacturing is not a low-risk sector that happens to have interesting technology. It is the primary target, because it has high downtime cost, a large Windows estate, poor segmentation and a willingness to pay.

Every machine came with a support contract, and support means remote access installed at commissioning with the builder's preferred tool, terminating **inside the cell zone, below every control the enterprise built**. There are commonly dozens in one plant. Specify instead: a single brokered path terminating in the DMZ and never below it; per-person named identities with multi-factor authentication; just-in-time enablement against a work order; session recording for changes to control logic; an inventory of existing paths with a decommissioning plan; and **a clause in the machine purchase specification** that future equipment uses that path.

The standards, with their uneven dates

IEC 62443 is what you design and procure to. Zones group assets that share security requirements, conduits group the communications between them, and **security levels are evaluated per technical requirement, not used to classify whole products** [23]. Hold three distinctions: **SL-T**, the target set by risk assessment; **SL-A**, the level achieved as built; and **SL-C**, what a supplier can demonstrate. The gap between the first two is the remediation programme.

> **The editions are uneven, and the famous parts are the oldest.** 62443-2-1 is Edition 2.0 of 2024-08 and added a maturity model; 62443-2-4 is Edition 2.0 of 2023-12. But **62443-3-3 is still Edition 1.0 of 2013-08** and 62443-4-2 Edition 1.0 of 2019-02 [23]. The technical requirement sets everybody quotes are the oldest live documents in the family.

NIST SP 800-82 is what you reason with and write policy from. Revision 3, September 2023, is the current final publication [25]; Revision 4 exists only as an Initial Public Draft dated 21 September 2026 [26], and much secondary material still calls Revision 2 current. Revision 3 is where the title changed from industrial control systems security to **operational technology security**. Its overlay mapping general security controls into an OT context is the valuable part, and the part nobody opens.

There is no South African manufacturing OT security mandate. The applicable law is general: privacy law, the Cybercrimes Act 19 of 2020, whose reporting duties fall on communications providers and financial institutions rather than manufacturers, and occupational health and safety law as it bears on safety systems. The Critical Infrastructure Protection Act 8 of 2019 creates a designation regime, and a manufacturer operating a designated facility has obligations a general manufacturer does not — but designations are not public, so the client's own legal function is the only source. An absence cannot be proven from a document, so write it as what it is: no sector mandate was found, and the drivers are customer assessment and insurance.

What you are actually expected to produce

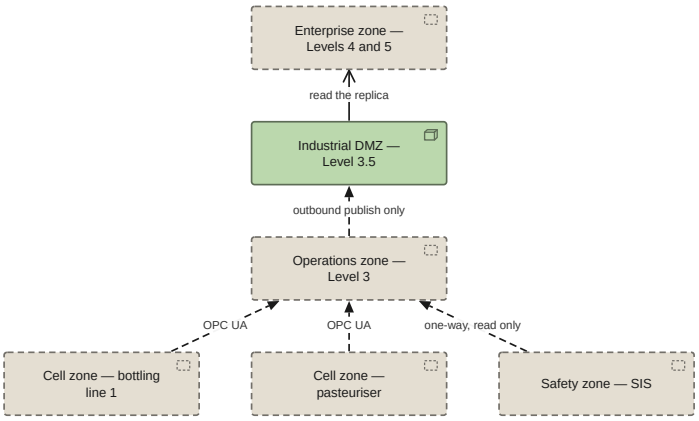


Figure 13.7 A zone and conduit model for the invented Paarl site. Every conduit is plant-initiated and outbound; the safety zone is read-only through a one-way path. The deliverable is only real once it is reconciled against the firewall rule base.

Five deliverables. The first two answer the most frequently asked and least well answered question in the field.

1. **An OT asset inventory.** Every device below the DMZ: make, model, firmware, addresses, location, function, criticality, owner, support status, and *how it is reachable*. Built by passive discovery, physical walkdown and configuration extraction — **not by an IT scanner, because active scanning can crash old controllers and will not be permitted.**
2. **A zone and conduit model** from a consequence-based risk assessment, with a target security level per zone and every conduit listed with its protocol, direction and enforcing control — **reconciled against the actual firewall rule base**, which is the step that makes it real.
3. **A segmentation plan**, a **remote-access architecture** with the inventory of existing paths and their decommissioning plan, and a **control baseline per zone** with compensating controls named where patching is impossible.
4. **Requirements in the procurement specification for new plant**: component capability level, no shared accounts, no inbound remote access, patchability and supported-life commitments, and an open northbound interface. **This is the deliverable with the most effect per hour spent, because it stops the estate getting worse**, and the one architects most often omit.
5. **An OT incident response plan distinct from the IT one**, with the plant's degraded-mode procedures as its core, the authority to stop production named, and controller restoration from a known-good archive that has been restore-tested.

And one thing to insist is **not** your deliverable: the risk acceptance. You state the exposure; the business accepts the residual in writing, per zone. In OT that matters more than elsewhere, because the residual is permanent.

The South African OT reality

Voltage quality is the design input, and it is written down

This is the best-evidenced South African design input in the whole research set, and most architects have never heard of it. NRS 048-2:2025, Edition 5, approved by NERSA on 6 May 2025 and superseding the 2015 fourth edition [27], defines a dip as a sudden reduction in RMS voltage lasting more than 20 ms and up to 3 s, characterises it by depth and residual voltage, and classifies dips into windows.

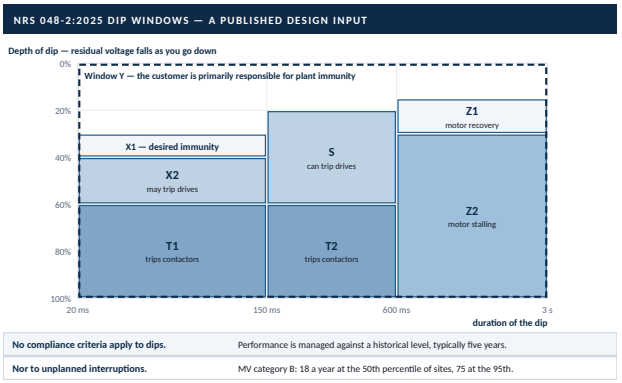


Figure 13.8 NRS 048-2:2025 Edition 5 dip windows, drawn from the published characteristics. Window Y is explicitly the customer's immunity obligation. There are no compliance criteria for dips or for unplanned interruptions.

Read that from an architect's seat. The national quality-of-supply standard says, in effect: **dips in window Y are the customer's problem to be immune to**, dips in windows X2 and S will trip your drives, dips in window T will drop your contactors, and a rural overhead-fed site should expect tens of sustained interruptions a year with no recourse. The standard publishes characteristic counts — for a medium-voltage network of mostly overhead lines, **18 unplanned sustained interruptions a year at the 50th percentile of sites and 75 at the 95th** — with **no compliance criteria** [27].

What breaks, in order

1. **Variable-speed drives trip** on undervoltage or direct-current bus collapse, typically within a dip in window X2 or S [27]. A tripped drive is a stopped pump, fan, conveyor or agitator.
2. **Contactors drop out** on a deeper dip in window T [27], de-energising motors directly, and a contactor does not re-close by itself.
3. **Controllers usually survive**, because they are small loads on an uninterruptible supply — **and that is precisely the problem.** The controller stays up, retains its state, and now believes a pump is running that has stopped. Interlocks based on a run *command* rather than a run *feedback* do not notice.
4. **Operator stations reboot** if unprotected, losing visibility for the two to five minutes of a boot, during which the process is unattended.
5. **Instruments with slow start-up** take minutes to hours to read validly again.
6. **The process itself is damaged.** A batch that lost agitation sets. An extruder that stopped with polymer in the barrel solidifies and needs stripping. A furnace locks out and needs a purge. **These are hours-to-days recovery events, not seconds.**

The restart sequence is the thing architects do not know exists. A process plant cannot be re-energised in one act. It has a start-up sequence: energise in an order that avoids inrush tripping the incomer; purge combustion equipment before re-light; re-establish instrument air, cooling water, steam and nitrogen before process; bring loops from manual to automatic in order; and, in a batch plant, decide for each in-progress batch whether it can resume, must be held, reprocessed or discarded. **That sequence may take hours and is often partly in an operator's head** — which is why the procedure-automation standard for continuous operations is more relevant in South Africa than in most markets.

Uninterruptible supply and generator scope is an architecture decision, not a facilities one. The uninterruptible supply bridges seconds to tens of minutes, and the question is *which* loads: controllers, input and output, operator stations, network switches, the protocol server, the execution server, the historian. **Every item not on it fails independently**, and the most common finding is a protected controller behind an unprotected switch, so the plant has a running controller nobody can see. The generator bridges hours to days, and the question is whether it carries enough to *run* the process or only to *shut it down safely* — radically different sizes, and "safe shutdown only" is legitimate but must be chosen explicitly, because it means every interruption is a production loss.

Load-shedding ended; the discipline did not

Writing load-shedding in the present tense will date this book instantly. Eskom reported **504 consecutive days without load-shedding as at 2 October 2026**, from 16 May 2025, with the energy availability factor at 68.11% year-to-date against 62.59% a year earlier [28].

> **"Load reduction" is a separate, continuing programme and must not be conflated with load-shedding.** The same Eskom statement reports load reduction affecting **202,590 customers as at 2 October 2026**, down from 1.69 million, with elimination targeted for March 2027 [28]. It targets network overloading and is predominantly residential, so a manufacturer is less likely to be directly affected — but its workforce is, and a municipal site may be.

The architectural consequence persists, for four reasons. **The installed base was built for it**: from roughly 2019 to 2025 South African plants invested in uninterruptible supplies, generators, inverters, solar, transfer switching and procedural and software changes to tolerate power loss, and that estate is what an incoming architect inherits. **Power loss did not become rare, only less scheduled**: municipal faults, transformer failures, cable theft and storms continue to interrupt individual sites, and **an unscheduled interruption is architecturally worse than a scheduled one**, because the scheduled one

allowed a controlled shutdown. **Voltage quality is a separate, continuing problem. And the discipline is correct anyway** — which is the point to carry out of this chapter. *Load-shedding taught South African manufacturers a design discipline most of the rest of the world lacks.* Five rules follow, good anywhere:

1. **Assume power loss as normal, not exceptional.** Every state machine needs a defined behaviour on unexpected de-energisation and a defined recovery.
2. **Interlock on feedback, not on command.** A controller must know whether the motor is turning, not whether it told it to.
3. **Persist state before you need it**, on local disk at the line, power-fail-safe.
4. **Make the restart sequence a designed artefact**, documented and tested.
5. **Specify uninterruptible-supply and generator scope as part of the application architecture**, and design the degraded mode for a dark plant, not just a disconnected one.

The test

The requirement, as a design rule: *the plant must continue producing conforming, traceable product for a defined period with no connectivity to any system above Level 2, and must reconcile on recovery without loss of record.* The period is agreed with the business — "indefinitely" is not an answer, "72 hours at full rate" is, and it sizes the local storage and the serial-number block.

The test is to pull the uplink on a Tuesday and see what happens. Most manufacturers have never done this deliberately. All of them have done it accidentally, and the incident report from that accident is the best requirements document available to you. Ask for it in week one.

One last constraint, and it is a people constraint. Engineers were the hardest skill to fill in South Africa at 29% of surveyed employers in the preliminary 2026 Xpatweb results, with artisans joint second at 25%, and that category includes the millwrights and electricians who execute everything you design below Level 2 [29]. **Design for fewer, more remote, less specialised hands**, and remember that every *it is in Johan's head* dependency is a single point of failure with a resignation date.

THE CHAPTER IN TEN LINES

- The physical process does not stop because the software stopped, and most non-obvious manufacturing requirements follow from that one fact.
- ANSI/ISA-95.00.01 is the 2025 edition, there are eight parts plus a technical report, and the ISA and IEC lineages are different documents with different dates.
- B2MML is MESA's, at V7.00.00 of 24 November 2020, built to the 2018/19 editions, so it predates the current Part 1.
- ISA-95 compliant is uncertifiable: there is no conformance scheme, where IEC 62443 has ISASecure and ISA-88 Part 1 has conformance criteria in Clause 9.
- The ISA-95 equipment hierarchy is not a seven-level stack: several values are alternatives chosen by process type, and equipment module and control module are ISA-88 concepts.
- Williams put no cybersecurity in PERA. Level 3.5, Level 5 and the zone language came later from the ISA-99 and IEC 62443 community.
- IEC 61131-3:2013 was withdrawn on 22 May 2025 and superseded by IEC 61131-3:2025.
- A tag has no identity, no relationships, no business meaning and no transactional context, so an asset model above the tags is the actual deliverable.
- Exception and compression silently change a reported number: 172,800 events become 15 on one status tag, and out-of-order data bypasses compression entirely.
- The OPC Foundation reports over 430 companion specifications as at April 2026; the familiar sixty counts equipment types, not documents.
- Sparkplug is Eclipse's and Sparkplug 3.0 is ISO/IEC 20237:2023, so it can be named with a standard number in a procurement document.
- Design every plant integration as plant-initiated and asynchronous, with a durable local queue sized in days and event timestamps preserved on replay.
- Remote vendor access is the realistic attack path, and the famous IEC 62443 requirement parts are the oldest documents in the family.
- NRS 048-2:2025 assigns window-Y dip immunity to the customer, names the windows that trip drives and contactors, and publishes interruption counts with no compliance criteria.
- Load-shedding ended at 504 consecutive days as at 2 October 2026, load reduction is a separate continuing programme, and the design discipline of assuming power loss remains correct.

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- **26.** NIST, SP 800-82 Revision 4 Initial Public Draft, released 21 September 2026 https://csrc.nist.gov/pubs/sp/800/82/r4/iprd
- **27.** NERSA, NRS 048-2:2025 Edition 5, Electricity supply: Quality of supply, Part 2 (approved by NERSA 6 May 2025, superseding the 2015 fourth edition) https://www.nersa.org.za/files/files/2025/05/NRS-048-2-Edition-5.pdf
- **28.** Eskom, "More than 500 days without loadshedding, EAF improves to 68.11%...", media statement, 2 October 2026 https://www.eskom.co.za/more-than-500-days-without-loadshedding-eaf-improves-to-68-11-diesel-costs-fall-by-r4-79-billion-sustained-emissions-performance-below-target-levels-and-load-reduction-reduced-to-202-590-customers/
- **29.** Polity, "AI specialists and senior project managers now among South Africa's top 10 hard-to-fill skills", 22 September 2026, reporting preliminary Xpatweb 2026 Critical Skills Survey results https://www.polity.org.za/article/ai-specialists-and-senior-project-managers-now-among-south-africas-top-10-hard-to-fill-skills-2026-09-22

CHAPTER 14

The reference architecture

Banking has one. Insurance has one. Manufacturing does not — and the evidence for that negative, properly assembled, is the most useful thing you can tell a manufacturing board. Then this chapter shows you what to build instead.

In her second week at Karoo Industrial Group, Ingrid Lourens is asked the question every architect gets asked. Nomthandazo Mahlangu, the group COO, wants to know which industry reference model the programme will work to. A consultancy has told her that banking has one and insurance has one, and she would like to know what manufacturing's is called so she can stop paying people to argue about words.

The honest answer is that there is not one — not a weak one, not a contested one. No body publishes a maintained capability model, process taxonomy, canonical data model and service catalogue for a manufacturing business as a whole enterprise, the way industry bodies do in financial services, where BIAN sits under banking and ACORD under insurance.

A negative conclusion delivered badly sounds like an excuse. Delivered with evidence it changes what the architecture team is for. If a governing model existed, the team's job would be to adopt and tailor it. Since none does, the job is to assemble one from sourced parts and get it endorsed — more work, more defensible, and it produces a house standard the business owns. So this chapter works through what manufacturing does have, states the negative conclusion with the nine pieces of evidence behind it, reports the two honest counter-indications, and then gives you the constructive answer.

> **NEVER STATE A STANDARD'S VERSION FROM MEMORY**
>
> Four of the editions in this chapter moved inside a single research window: ISA-95 Part 1 to 2025, ANSI/ISA-62443-2-1 to 2024, ISO 9001 to 2026 and ISO 14001 to 2026. Most circulating material cites the superseded editions. Every version here carries a source and a date. Re-check each before you repeat it, because the next four will move inside your programme.



Figure 14.1 What manufacturing has, by layer and by body. Seven layers, six or seven organisations, no published reconciliation between any two of them. Versions as at October 2026 — re-check every one.

What manufacturing has instead of a reference model

The inventory, drawn in the figure above: a control-systems hierarchy and boundary standard, a batch-control standard, a measure-definition standard, an operational-technology security series, three standardised interoperability stacks from three bodies, an identification system, a digital-twin representation standard in two version lines, engineering exchange formats from five publishers, and a stack of sector quality schemes that function as de-facto information architectures because they dictate records. Six or seven organisations, different layers, **not reconciled with each other.** The joins are yours to make.

A second warning runs through everything below. **None of these is a design.** ISA-95 models what information crosses a boundary; it is not an application architecture. ISO 22400 is a set of measure definitions, not a reporting design. IATF 16949 is a conformance obligation, not a data model. An architect who hands a plant a diagram of ISA-95's levels and calls it a target state has produced a document that satisfies nobody, and the plant engineering team will say so.

ISA-95 and IEC 62264 — the one everybody has heard of

Who ISA is, and why that matters commercially

The International Society of Automation is a United States nonprofit professional association, founded in 1945 and headquartered at Research Triangle Park in North Carolina [2]. It develops standards through sector committees — ISA95, ISA88, ISA99, ISA101, ISA106 and others — publishes them as American National Standards under ANSI, and sells them through its own store [1][2][5]. Several series are developed with simultaneous IEC review and adoption, which is where the parallel ANSI/ISA and IEC numbering comes from [10].

Three properties of ISA as a body bear on your work. **The standards are sold, not published freely**: a full ISA-95 part set plus the 62443 series is a real budget line [2][5]. **The committee pages are poor documentation of current status** — the ISA95 page gives scope but no part numbers or dates, and the ISA88 page lists no parts at all [2][5], so the authoritative body's own page will not tell you the current edition. And **ANSI maintenance rules bite**: an ANSI/ISA standard must be reaffirmed or revised every five years [6].

What ISA-95 actually scopes

ISA-95 is *Enterprise-Control System Integration*. Its committee scope is the interface between manufacturing control and other enterprise functions, the terminology for it, and the information exchange across it [2]. The word that matters is *interface*. **ISA-95 is scoped to a boundary, not to an enterprise.**

It has two famous artefacts. The first is the functional hierarchy — the levels 0 to 4 taxonomy every slide deck reproduces. Chapter 13 dealt with it; the honest framing is that the levels classify activity, and are not a deployment topology or a segmentation model. The second artefact earns its keep: the **object model set**, covering personnel, equipment, physical assets, material, process segments, production capability, product definition, and production schedule and performance [2]. Those are the nearest thing manufacturing has to a canonical information model at the boundary between the enterprise systems and the plant.

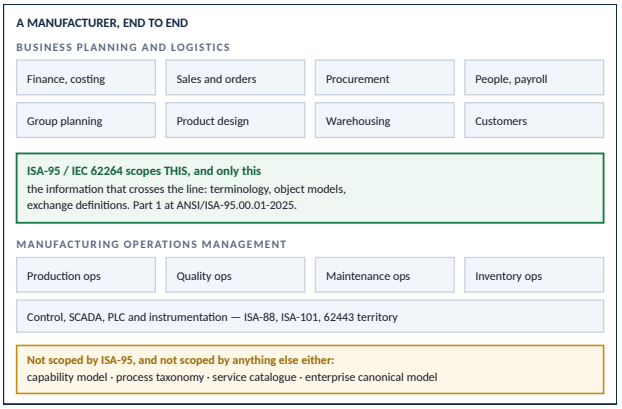


Figure 14.2 What ISA-95 scopes, against a whole manufacturer. The standard governs the information crossing one line. The four artefacts an architect most wants — capability model, process taxonomy, service catalogue, canonical model — are scoped by nothing.

The version fact that dates most published material

Part 1 is now **ANSI/ISA-95.00.01-2025, (IEC 62264-1 Mod), *Enterprise-Control System Integration — Part 1: Models and Terminology***, announced on 10 April 2025, superseding the 2010 edition [1][3]. ISA describes the revision as clarifying the boundary between the enterprise domain and the manufacturing and control domains, improving internal consistency, and responding to containerised workloads, data-centric architectures and standards-based metadata [1]. Almost everything in circulation cites the 2010 edition: a procurement specification or a vendor conformance claim saying "ISA-95.00.01-2010" was written before April 2025, or copied from something that was.

The parenthetical **"(IEC 62264-1 Mod)"** means the ANSI/ISA document is based on IEC 62264-1 with modifications — and ISA's release does not say what the modifications are [1]. The consequence is contractual: the two texts are closely related but not guaranteed identical, so a conformance clause must name which document it means. Citing "ISA-95 Part 1" and "IEC 62264-1" interchangeably with one year on them is wrong.

Eight parts and a technical report, with the part table withheld

The series runs to **eight parts plus ISA-TR95.01-2018**, a master-data profile technical report — not the five commonly asserted. The parts cover terminology, object-model attributes, activity and object models for manufacturing operations management, business-to-manufacturing transactions, messaging, aliasing and information exchange profiles [4].

What this chapter will not print is a table of part designations and dates: every edition year for Parts 2 to 8 in the available secondary source is unverified, and that source is internally inconsistent [4]. Buy the current list from ISA. Note also that **IEC 62264-6 is a Publicly Available Specification** — a pre-standard, not a full IEC standard, which matters if you are about to write it into a contract.

There is no ISA-95 conformance scheme

This is the single most useful sentence to carry into a vendor meeting. **"ISA-95 compliant" is uncertifiable.** No conformance scheme exists, no certification body issues certificates, and no register lists conforming products.

Draw the contrast. ISA/IEC 62443 *does* have one — ISASecure, with separate certifications for the development lifecycle, for components and for systems [10]. ISA-88 Part 1 carries conformance criteria in its own text. ISA-95 has neither. So never ask an MES or ERP vendor "do you support ISA-95?", to which the answer is always yes. Ask: **which parts, which editions, which objects, in which binding, and show me the schema.**

Licensing, and the clause nobody checks

ISA standards are copyrighted and sold, with no free normative text [2][5]. A conformance clause citing ISA-95 therefore obliges both parties to hold the document — so check that your client owns a licensed copy before you write one. More than one architecture contract has been signed by an organisation that cannot read the standard it agreed to be measured against.

B2MML — MESA's, not ISA's, and six years behind the standard it binds

Somebody has to turn ISA-95's object models into usable schema, and it is not ISA. **MESA International** maintains **B2MML — Business To Manufacturing Markup Language — and BatchML, through its XML Committee** [7]. Architects routinely assume B2MML is an ISA deliverable. It is not, and the licensing follows: any company may use it royalty free provided credit is given to MESA, with the download behind a MESA account and the schema set also on GitHub [7][8].

The dates are the point. The current release is **Version 7 (V7.00.00), announced on 24 November 2020**, still the latest in MESA's own repository [7][8], and it was built to the **2018/19 editions** of ISA-95 and IEC 62264 [7]. It therefore **predates the 2025 Part 1** by five years.

> **"OUR PRODUCT SUPPORTS B2MML"**
>
> That sentence tells you almost nothing. Ask which version — the repository carries an archived legacy release as well as V7 [8]. Then ask which ISA-95 edition the vendor's own object model follows, because v7 binds the 2018/19 editions and Part 1 is at 2025 [1][7]. A canonical-model strategy built on B2MML is reasonable, but you own the gap as a design decision. Discovering it in integration testing is worse.

Nothing here suggests abandonment. The repository is live, the licence is permissive, and the schema set ships an extensions schema for the local additions every manufacturer needs [8]. What is not available is a published mapping from v7 to the 2025 Part 1, which is the thing you would most like to have.

The control-side standards: ISA-88, ISA-101 and ISA-106

ISA-88 and IEC 61512 are the batch-control standards: recipe-driven production, the separation of recipe from equipment, the procedural control model, the physical model, and the general/site/master/control recipe distinction. Chapter 8 used them to explain why a food or pharmaceutical plant can run a new product without a controls project.

Three things about ISA-88 matter regardless of edition. **Part 4, the Batch Production Record, is the architecture-relevant part** — it defines what a batch record is as a data object, and in regulated process manufacturing that record is the artefact the whole system estate exists to produce. **The recipe-and-equipment separation** maps onto an application-architecture question: which system owns the master recipe, which the control recipe, and where does the site recipe get localised? Most process-plant integration disputes reduce to that. And **the physical model is where ISA-88 and ISA-95 touch** — ISA-95's equipment hierarchy is recognisably ISA-88's, and the two are compatible by intent, not by any published reconciliation.

> **THE PUBLIC RECORD ON ISA-88 IS TWENTY YEARS OLD**
>
> ISA's own ISA88 page gives scope and a store link and lists no parts, no designations and no dates [5]. The fullest public status account located is a 2006 ISA Expo paper whose status columns are, necessarily, from 2006 [6]. Part 1's current American edition is ANSI/ISA-88.00.01-2010, not the 1995 original, and IEC 61512-1 is 1997. Parts 2 to 5 are unverified and this chapter does not state them — which is itself the finding.

ISA-101 is a lifecycle standard for human-machine interfaces in process automation — philosophy, style guide, toolkits, design, implementation and maintenance, rather than prescribed screens. The product listing located is ISA-101.01-2015, with a technical report on usability [9]; whether that edition has been revised is unverified. Its value is that **it makes "we will replace the HMI" a governable project rather than a taste argument**, because it forces an HMI philosophy document and a style guide into existence as deliverables — in a repository, a Principle and a Standard, in the one layer where architects usually have no standing.

ISA-106 addresses procedure automation for continuous processes — the gap ISA-88 leaves. Continuous plants have proceduralised activities (start-up, shutdown, grade change, abnormal-situation response) that live in operators' heads or on paper. ISA's own coverage, dated August 2023, describes it as a new standard [11]; its part structure and dates are unverified. Treat it as emergent, but know the name.

ISO 22400 — the measure definitions, and why they are a data artefact

Automation systems and integration — Key performance indicators for manufacturing operations management. Part 1 is **ISO 22400-1:2014, edition 1, published October 2014**, and ISO's catalogue records that it was last reviewed and confirmed in 2025 — a 2023 review stage marked "to be revised" ended in confirmation in August 2025 instead [12]. Part 2 defines the individual measures and is the part practitioners buy; **its current published edition is 2014, with a second edition still at draft-international-standard stage in October 2026** [12]. So the standard most often cited for operating-measure definitions is twelve years old, deliberately. That is stability, not neglect.

ISO 22400 does not stop the argument about what availability means; it makes it *resolvable*, because both sides can be asked which definition they use. The consequence is a reference-model decision: **measure definitions are data-architecture artefacts, not reporting artefacts.** If overall equipment effectiveness is computed in the historian, the MES and the group reporting layer with three denominators, no amount of dashboard work fixes it. The standard's real use is as the authority behind one Measure definition in the repository, owned by one function, computed in one place.

What goes stale here is not the standard but vendor claims of ISO 22400-compliant measures. Ask which measures, from which part, at which edition.

ISA/IEC 62443 — the one item here that becomes contractually binding

The operational-technology security series is developed by ISA's ISA99 committee as American National Standards with simultaneous IEC review and adoption, and maintained on the IEC side by TC65 WG10 [13][10]. First published from 2007, the documents were originally ANSI/ISA-99 and were renamed around 2010 [13]. Four groups: 1-x general concepts and profiles, 2-x policies and procedures, 3-x system, 4-x component [13].

Two facts reorder an OT security target state.

The editions are uneven, and the famous parts are the oldest. ANSI/ISA-62443-2-1-2024 was announced on 28 January 2025, updating the 2010 version, reorganising requirements into security programme elements and **adding a maturity model** [10]; the IEC side shows 62443-2-1 Edition 2.0 (2024) and 62443-2-4 Edition 2.0 (December 2023) [13]. But **62443-3-3 — the system security requirements and security levels part everyone quotes — is still Edition 1.0 (August 2013)**, and 62443-4-2 is Edition 1.0 (February 2019) [13]. Use the 2024 edition's maturity model to state where the programme is, while knowing that the security-level requirements have not moved in over a decade.

There is a real conformance scheme, which makes this series different from everything else here. Its zones-and-conduits concept is the legitimate home of the segmentation discussion people wrongly attach to ISA-95's levels; its security levels give a target-setting vocabulary that maps onto a Requirement in a repository; and its 4-1 and 4-2 pair lets a manufacturer push security obligations onto equipment vendors — the most useful lever you have in a plant with a twenty-year installed base.

> **NEVER CITE A 62443 STATUS SUMMARY WITHOUT ITS DATE**
>
> The most-cited public article on the status of the 62443 standards is dated 25 May 2021 and describes as "in development" several documents that have since published [14][13][10]. Date it or drop it.

Three interoperability stacks, three bodies, no reconciliation

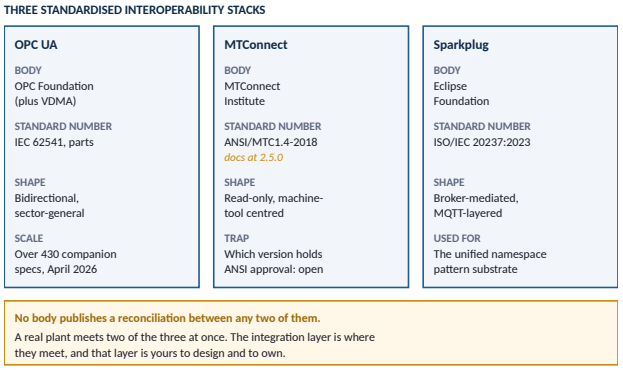


Figure 14.3 Three stacks, three bodies. All three are standardised; none is reconciled with the others. Which MTConnect version holds ANSI approval could not be verified and is a trap, not a fact.

OPC UA is a service-oriented industrial interoperability architecture — address space, information model, services, mappings, profiles, domain facets — standardised as **IEC 62541** in parts [15]. Three facts matter architecturally. **PubSub includes an MQTT mapping** [15], which makes "OPC UA or MQTT?" a false choice; the real question is whether a flow wants a client-server request model or a broker-mediated publish model. **The specifications are free of charge but not free of terms** — the document portal presents a licences-and-terms gate, and while reference implementations are open source and free to members, non-members use them under GPL 2.0, a genuine constraint for anyone shipping a product [15][17]. And **the maturity is not uniform**: core parts sit at 1.05.x while field exchange, the UAFX parts, is only at 1.00.x [15].

Then the number usually wrong by an order of magnitude. The OPC Foundation's own release of **20 April 2026 states there are "over 430" OPC UA companion specifications** [16]. The widely repeated figure of about sixty is a different unit of count — models for over sixty *types of industrial equipment* [15], which is not sixty documents. Quote the Foundation's figure with its date and explain both countings, because somebody in the room will have the other number. Note also that many come from joint working groups — the best-documented being **VDMA**, whose specifications carry the OPC Foundation mark but whose licensing terms are not the Foundation's and are unverified [18] — and that **umati is a marketing brand, not a specification**, whose published figures are from a 2019 trade-fair showcase [19].

MTConnect is a read-only, HTTP and XML standard for getting data out of machine tools, narrower than OPC UA and originating in the American machine-tool industry. Its version position is a trap: the site's own designation is **ANSI/MTC1.4-2018** [20], the current documentation set is at **version 2.5.0** [21], and the association hosting committee activity announced a **version 2.0** release [22]. **Which version holds ANSI approval is unverified.** Do not write "ANSI/MTC1.4-2018 is the current MTConnect standard" as though that were the whole truth.

Sparkplug puts a defined topic namespace, payload encoding and state-management convention onto MQTT, so MQTT becomes a plant-floor interoperability layer rather than just a transport. Two attributions to get right. **It is the Eclipse Foundation's, not the OPC Foundation's** [24] — architects file it under OPC because both solve plant-floor interoperability. And **Sparkplug 3.0 was published as an international standard, ISO/IEC 20237:2023** [23][24], so it can be named with a standard number in a procurement document, which very few people know. It is the substrate of the unified namespace pattern — a pattern with standards backing, but **no standards body publishes "the unified namespace"**, so claims about it are practitioner claims, not standards claims.

The Industrie 4.0 conceptual layer — a pre-standard and two assessment instruments

RAMI 4.0, the Reference Architecture Model Industrie 4.0, is the three-axis model — hierarchy levels, lifecycle and value stream, six architecture layers — that most people meet first. Its standards status is the thing to get right: it is published as **DIN SPEC 91345**, a DIN *specification*, which is a pre-standard produced outside the full consensus process, not a DIN, EN, IEC or ISO standard [25]; its origin document is a ZVEI and VDI-GMA status report of July 2015 [26]. It also exists as IEC PAS 63088:2017 — again a Publicly Available Specification. **Do not describe RAMI 4.0 as an IEC or ISO standard.**

My assessment, offered as assessment: RAMI 4.0 is pedagogically influential and operationally rare. It is excellent for explaining why "digitise the factory" is several different projects. This research found no evidence of a maintained conformance programme, certification or tooling comparable to

62443's. I invite contradiction, but I would not build a governance model on it.

The Asset Administration Shell is RAMI 4.0's one idea that became implementable: a standardised digital representation of an asset carrying submodels — nameplate, technical data, documentation, carbon footprint — with resolvable identifiers, so asset information travels with the asset. It is **two version lines, and citing one misrepresents the position**. IEC 63278-1 Edition 1.0 (2023-12) is the structural international standard [28], adopted in Germany as DIN EN IEC 63278-1 (2024-09) [29]; the **Industrial Digital Twin Association publishes the implementable metamodel, IDTA-01001, at version 3.0.2, posted March 2025** [27].

The standardisation is real, recent and pulled along by regulation rather than advocacy — where a rule demands that product data travel, the shell is the available mechanism. My assessment for a South African manufacturer: you will **receive** shell data from imported European equipment long before any local plant publishes it. Either way the capability to ingest and store a submodel set belongs in the asset-data design now.

> **THE INDUSTRIAL INTERNET CONSORTIUM DOES NOT EXIST UNDER THAT NAME**
>
> The body founded as the Industrial Internet Consortium renamed itself the **Industry IoT Consortium**, and on **18 January 2024 the Object Management Group announced it was integrating the consortium within the Digital Twin Consortium** [30]. The **Industrial Internet Reference Architecture is at version 1.9** on the consortium's own page, which **states no publication date and mentions neither body at all** [31]. That last detail is the most useful signal available about the reference architecture's maintenance intensity. As an artefact it is a viewpoint framework — closer to a method than to a reference model, and not to be listed alongside ISA-95 as though the two were comparable.

GS1 above the factory floor

GS1's identification system is usually taught as a retail matter. In manufacturing it does different and arguably more important work, and an architect who meets it only as barcodes will mis-specify traceability. The keys sit in the **GS1 General Specifications**, current release **26.0 (2026)** [32][33].

Key	What it identifies	Why a manufacturer cares
GTIN	A trade item at a packaging level	A consumer unit, an inner pack and a case are three GTINs with a hierarchy; an item master with one row per "product" cannot reconcile a despatch against a receipt
SSCC	One physical logistic unit, once	The hinge between the plant's production lot and the customer's received shipment — and the key most often reused in error
GLN	A party or a location	Plant, warehouse, legal entity, ship-to, bill-to and remit-to are distinct; conflating them surfaces at the first invoice dispute
GRAI, GIAI	A returnable asset; an individual asset	Kegs, cylinders, pallets, totes; then individual plant assets

Application Identifiers are the grammar [32]. **Batch or lot, expiry date, production date, serial number and variable weight all arrive as identifier-prefixed elements inside one data carrier.** A plant system that parses the GTIN and discards the rest has thrown away the traceability data the business is about to be audited on. The carrier-specification and placement sections are the most escalated integration problem in manufacturing logistics, because they decide whether a customer's receiving scanner can read the label your printer applied, at line speed, through shrink wrap.

Two further pieces are worth knowing by name. **EPCIS** is the traceability event standard, adopted as ISO/IEC 19987:2024 [34], and **its transformation event is the one that matters in manufacturing and not in retail**: the event type that records inputs consumed and outputs produced, which is a production step. You need not invent a vocabulary for "this lot became that lot". **GS1 Digital Link** turns an identifier into a resolvable web URI so one carrier serves machine identification and human-facing information; the reference site publishes version **1.1.3** [35], and the line is 1.x, not 2.x.

> **THE GS1 2027 SENTENCE, WRITTEN CORRECTLY**
>
> **GS1 does not require a 2D barcode by 2027.** GS1 US's own page calls Sunrise 2027 a milestone in the transition to 2D barcodes, states in terms that it is not necessary to implement all the capabilities of 2D by that date, names one retailer-side capability — checkout scanners able to read 2D and extract the GTIN — and frames adoption as crawl, walk, run [36]. The correct sentence: *Sunrise 2027 is a voluntary industry capability ambition coordinated by GS1, not a regulation and not a mandate; the practical question for a manufacturer is whether its customers will require 2D-carried data, and when.*
>
> The manufacturer-side consequence is real even though the mandate is not. Adding batch, expiry and serial data to a 2D carrier is a labelling, printing, verification and data-availability project with a capital component — and the data has to come from somewhere, which makes it an MES and item-master question, not a packaging one.

Engineering and plant-lifecycle data exchange

This cluster is usually left out of enterprise-architecture treatments because it looks like an engineering concern. It decides whether a plant's as-built data is usable for the next thirty years.

ISO 10303, "STEP", is the long-running product-data exchange standard, structured as application protocols. The one that matters now is **AP242: ISO 10303-242:2022, Managed model-based 3D engineering** [37], with a newer architectural direction in ISO/TS 10303-4000:2024, a core model [38]. You meet AP242 as a **contractual deliverable format** in aerospace, automotive and defence supply chains, where a customer requires geometry plus product and manufacturing information in a neutral rather than a native CAD format. Three consequences: the PLM system must export *and validate* AP242; validation is where projects fail; and a long-life programme needs a reading capability for files produced decades earlier.

QIF, the Quality Information Framework, is an XML schema library for metrology and quality data — characteristics, measurement plan, results, statistics — in a form machines can act on. Published by the Digital Metrology Standards Consortium, **its latest version is ANSI QIF 3.0 of December**

2018 [39]: it has not moved in eight years. Adoption is narrow and deep — most manufacturers never meet it, and those in model-based supply chains cannot avoid it. Its value is that it makes quality data a structured asset rather than a PDF of a measurement printout.

Two more to know by name. **JT** is a lightweight 3D visualisation format standardised as ISO 14306, whose version line has moved — ISO 14306-3:2025 and ISO 14306-4:2026 are in the catalogue [40]; you meet it where a large customer standardises on it for viewing assemblies, and **in a PLM estate it is a derived artefact**, so the governance question is whether derivation is automated and versioned against the master. **ISO 15926** is the process-plant lifecycle data standard — a data model plus a published reference data library of plant object classes, so plant information survives handover between contractors, owners and operators [41]. It is hard, rigorous and sparsely implemented outside oil, gas, chemicals and large capital projects, where it is unavoidable and usually met through CFIHOS rather than directly.

CFIHOS is the most useful item in this section, and almost nobody outside capital projects has heard of it. It specifies **what information must be handed over at the end of a capital project, and in what structure**: an information requirements list, a tag, document and equipment data model, and a reference data library. It exists because handover from contractor to owner-operator was, and largely still is, a catastrophe of spreadsheets.

The current position, dated: **version 2.0, including an extended reference data library, released as a major update in November 2025; managed by IOGP; and all documents available free of charge** [42]. A manufacturer commissioning a new line or a capital upgrade needs exactly this document — what data the contractor must deliver, in what form, before retention is released. The value is large and the cost is zero. Using it outside its home sector needs pruning, because its reference data is process-plant-shaped.

> **"CFIHOS IS BUILT ON ISO 15926" IS AT BEST INCOMPLETE AT VERSION 2.0**
>
> The CFIHOS standards page retrieved **does not mention ISO 15926 at all.** It names alignment with ISO 14224, the Energistics unit-of-measure standard and JIP33's reference data library instead [42]. The familiar claim may be true of the lineage, but it is not what the current page says, and the formal relationship at version 2.0 is unverified. Settle it from the specification's normative references before repeating it in a handover specification.

The sector quality schemes are the real reference architecture

A quality scheme is not an architecture. But a scheme that compels a manufacturer to produce specific records, on demand, to an auditor, with traceable lineage, is an architecture constraint of the same force as a regulation. That is what turns IATF 16949, the aerospace series, ISO 13485, the food schemes and good manufacturing practice into architecture documents: **they determine what must be recorded, by whom, with what approval, retained for how long, and retrievable in what time.**

In the absence of an industry reference model, **the record obligations of the applicable quality scheme are the nearest thing a manufacturer has to a mandated information architecture.** The pattern to teach: for each scheme, ask *what records does this compel, and which system is the authoritative source of each?* Answer that and the scheme has generated a data-ownership matrix, which is a real architecture artefact.

Two editions that just moved, and the one most likely to be stale in print

Scheme	Current position, dated	Note
ISO 9001	**Sixth edition, ISO 9001:2026, published 16 September 2026** [43][44]	**ISO 9001:2015 and its 2024 amendment are withdrawn** [43]. The certification transition period is set by accreditation-forum resolution, not by ISO, and is unverified
ISO 14001	**ISO 14001:2026, published 15 April 2026** [45]	Stronger alignment with climate change, biodiversity and resource efficiency [45]
ISO 45001	**ISO 45001:2018 remains the published edition** [46]	A draft circulated in 2026 pointing at a possible 2027 edition; stage and date unverified
IATF 16949	**Still the 2016 first edition** [47]	A second edition is planned for mid-2027. The churn is in the Auditor Guide (5th edition, January 2025), the Rules (6th edition, 1 January 2025) and the Sanctioned Interpretations (November 2025) [47][48]
AS9100	**Being renamed, not just revised** [49]	Sector designations are moving to a single IA designation, with the **IA 9100 series expected early 2027**; "Rev E" is third-party vocabulary and unconfirmed
ISO 13485	Current edition unverified [50]	Deliberately not aligned to the common high-level structure, and referenced by device regulators
FSSC 22000	**Version 6: published 31 March 2023, mandatory 1 April 2024, recognised by GFSI 23 August 2024** [51]	A version 7 page exists [52]; v7's own dates are unverified — do not print one

ISO 9001:2026 is the single most likely stale claim in any quality-architecture text. A document written before October 2026 that names ISO 9001:2015 as current names a withdrawn standard. Check your own templates.

What each scheme compels, which is what makes it architecture

Scheme	What records it compels, which is what makes it architecture
Scheme	What records it compels
---	---
The common ISO layer	Documented information under control, with version, approval, distribution and retention; competence records per person per role; audit programmes with findings and closure; nonconformity and corrective action with root cause; measurement results with calibration evidence
IATF 16949	All of the above, plus the automotive core tools: quality planning, the part approval package, process failure-mode analysis, measurement system analysis, statistical process control — each evidenced per part and per process
The aerospace series	Configuration management of what was built to which drawing revision with which deviations; material and process traceability including special processes with qualified sources; counterfeit-part prevention; first-article inspection; **retention for the life of the product**
ISO 13485	A device master and device history record per device or batch; the design history file; risk management; unique device identification; complaint and post-market surveillance with a route back to a batch; **validation of every computerised system in the quality chain**
Food schemes	One-up and one-down traceability; control-point monitoring carrying measurement, limit, person, time and corrective action; allergen and changeover verification; cleaning and environmental monitoring; supplier approval; label approval; calibration of every control-point device
GMP, inspected not certified	The batch record; computer-generated audit trails independent of the operator; electronic signatures with identity and meaning; validation documentation per system; data-integrity principles; deviation, change-control and corrective-action records linked to batches

Three rows deserve a sentence, because that is where the work lands.

The automotive part approval package is an assembly problem, and the assembly problem is the architecture problem. One submission covers design records and authorised changes, process flow diagrams, the failure-mode analysis, the control plan, measurement studies, dimensional and material results, capability studies, laboratory documentation, sample and master parts, checking aids and a warrant — all tied to one part, one process, one date, one approver. **No manufacturer holds that in one system.** It spans PLM, MES, LIMS, the quality system, metrology and document management, which is why automotive manufacturers end up with a quality-document integration layer nobody designed.

The medical-device validation obligation has the largest architectural consequence here: it determines release cadence, environment strategy and the cost of a cloud upgrade the manufacturer does not control.

The food layer's parts are different kinds of thing, and newcomers conflate them. Hazard analysis and critical control points is a *method*, not a certifiable standard in most schemes. ISO 22000 is a management-system standard; FSSC 22000 is a certification *scheme* built on it. And **GFSI certifies nobody** — it benchmarks schemes against its Benchmarking Requirements, version 2024, in two handbook parts, and customers then accept recognised schemes [53].

> **THE FOOD SCHEME GIVES YOU SOMETHING PRECIOUS**
>
> A mock recall is a timed exercise, so the retrieval requirement arrives as **a non-functional requirement with a number attached** — rare and precious in architecture work. Karoo Foods has four hours in its customer agreements. Use the number: it is the only thing that will get the genealogy design funded.

The live item to watch in the last row is the **computerised systems annex of the European good-manufacturing-practice guide, which is under revision** — a consultation draft was published with a reasons-for-changes document [54], and the international inspection co-operation scheme circulated a concept paper [55]. That annex governs validation, audit-trail, access-control and record-retention obligations for every computerised system in the chain, which is to say the architecture. Its final text and adoption date are unverified.

Stated hard: in pharmaceutical manufacturing the record obligations *are* the architecture requirements. A target state for such a plant without a validation strategy, an audit-trail design and a change-control model is not an architecture.

Maturity models, and what they are actually for

None of the Industry 4.0 maturity models is a reference architecture, and treating one as a reference architecture is a category error that will cost you a quarter.

SIRI, the Smart Industry Readiness Index, is INCIT's — the International Centre for Industrial Transformation publishes and governs it, certifies assessors and runs the portal [56]; it began as a Singapore economic-development initiative and was taken up in a World Economic Forum programme [57]. Its structure is well made — **three building blocks** (process, technology, organisation), **eight pillars**, **sixteen dimensions**, **maturity bands 0 to 5**, assessed on site by a certified assessor over two to three days — and it **prescribes no target systems and no technical architecture** [56]. So it tells you where you are, not what to build. The sixteen dimensions are still a defensible externally validated scoping frame, and the focus-dimension output is a prioritisation artefact a board will accept.

The German national academy's maturity index is the other instrument you will hear named. No primary source for its current edition, structure or licensing terms could be located, so this chapter says only that it is a German academy-published maturity model with development stages.

The conclusion, with the evidence

Now the claim, with the nine items that support it rather than an assertion to take on trust.

1. **The closest candidate is scoped to a boundary, not an enterprise** — the interface between manufacturing control and other enterprise functions [2], with the 2025 revision described as clarifying *that boundary* [1].
2. **Its canonical binding is stale relative to the standard it binds.** B2MML v7 (November 2020) implements the 2018/19 editions; Part 1 is at 2025 [1][7][8].
3. **The batch-control standard's own publisher does not publish its current part list** [5]; the fullest public status account is from 2006 [6].
4. **The protocol layer is maintained, large and vendor-governed rather than industry-governed** — an IEC standard [15] with over 430 companion specifications as at April 2026 [16], many published by a national engineering association rather than a neutral body [18][19]. A healthy interoperability ecosystem, not a reference architecture.
5. **A second and a third standardised protocol lineage exist alongside it.** Sparkplug 3.0 is ISO/IEC 20237:2023 [23][24]; MTConnect carries an ANSI designation [20] with its documentation a major version ahead of it [21][22].
6. **The Industrie 4.0 conceptual layer is a pre-standard plus assessment instruments.** RAMI 4.0 is a DIN SPEC [25][26]; SIRI prescribes no architecture [56]; the Industrial Internet Reference Architecture is a viewpoint method whose home has moved twice and whose page reflects neither move [30][31].
7. **The identification layer belongs to a different body with a different scope** — identification and carriers, not enterprise capability or process [32][33].
8. **The engineering-data layer belongs to yet more bodies** — ISO 10303 [37][38], ISO 14306 [40], QIF [39], ISO 15926 [41], IOGP's CFIHOS [42]: five publishers, one an industry association.
9. **The sector quality schemes, which between them specify more of a manufacturer's information architecture than anything above, come from six different bodies and none is reconciled with ISA-95 or with each other** [43][45][47][49][51][53][54].

The two honest counter-indications

An argument that reports no evidence against itself is not an argument. Two things pull the other way, and you should know both.

First, ISA-95's 2025 revision uses language more ambitious than a boundary standard's. ISA's own release frames the new edition as providing **shared ontologies and semantic models for manufacturing operations information**, responding to data-centric architectures and standards-based metadata [1]. That is not how a body describes an interface specification. **Whether the 2025 text actually broadens scope beyond the boundary is unverified**, and it is settleable only by reading the licensed document — an afternoon's work if your client owns a copy. If it does broaden scope, the conclusion above needs softening, and this chapter says so in advance rather than being corrected later.

Second, and more consequentially: Catena-X is the nearest thing to a maintained, multi-company, vendor-neutral manufacturing data architecture that exists. The **Catena-X Automotive Network Association** is the governance body whose members define business areas, develop standards and drive adoption; **Cofinity-X** is the operating company handling data-space registration and onboarding [58][59]. It describes itself as fully operational and reports in a **September 2026 network report that it grew by more than a factor of five to over 1,000 companies**, with business areas covering supply-chain resilience, sustainability, quality and master data. A battery-passport scenario is tied to an **EU requirement effective 18 February 2027** [58].

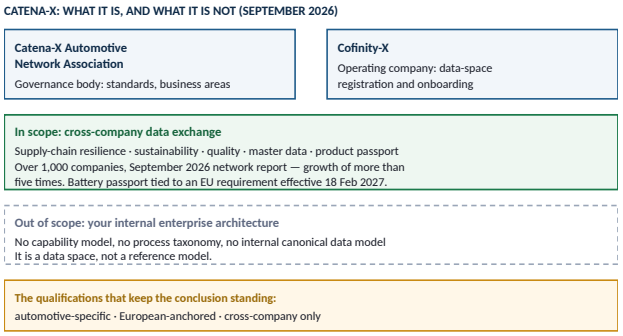


Figure 14.4 Catena-X, the honest counter-indication. The nearest thing to a maintained, multi-company, vendor-neutral manufacturing data architecture — and the single most likely way this chapter dates.

But it is automotive-specific, European-anchored, and scoped to **cross-company data exchange**, not to a manufacturer's internal enterprise architecture. It is a data space, not a reference model. Those three qualifications keep the conclusion standing.

> **THIS IS THE SINGLE MOST LIKELY WAY THIS CHAPTER DATES**
>
> For a South African automotive component manufacturer selling into German supply chains, Catena-X is the most likely of everything here to arrive as a **non-negotiable customer requirement**, because it is driven by the vehicle makers and by European product-passport regulation rather than by a standards body's advocacy [58]. Johan Terblanche at Highveld will hear about it from his customer's purchasing organisation, not from a standards committee. If it generalises through the wider family of European manufacturing data spaces, this chapter's conclusion needs rewriting. Watch it, and date whatever you write about it.

The constructive answer: build the house reference model

You cannot adopt a manufacturing reference model, so you compose one from four sourced inputs and get it endorsed as the house standard. It is defensible precisely because each input is somebody else's published, maintained work rather than your opinion.

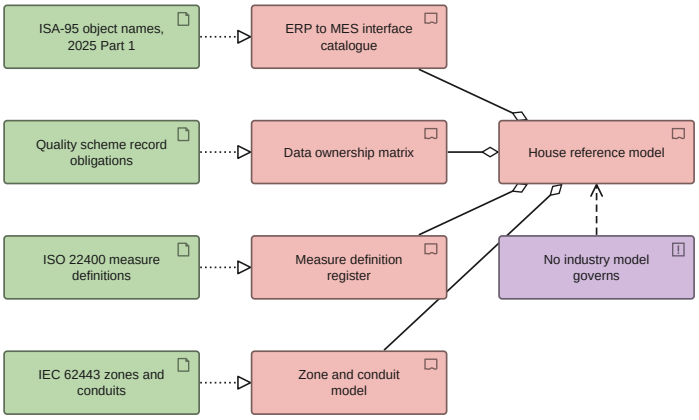


Figure 14.5 The house reference model from four sourced inputs. Each input is somebody else's published, maintained work, which is what makes the composite defensible. The principle is the artefact that converts four tables into an architecture.

Input one: ISA-95 object names, for the ERP and MES boundary. Take the object vocabulary — production schedule, production performance, material definition, material lot, equipment, personnel, process segment, product definition — and use it to classify every payload crossing the line between the enterprise systems and the plant. Not as a deployment architecture, not as a security model: as the **names on the interfaces**.

The steps. List every existing interface between the enterprise layer and the plant layer, including the files on shares and the scheduled jobs nobody owns, and classify each against an ISA-95 object. Where one interface carries three objects, record that — it is a finding, not an inconvenience; where an object has three interfaces, record that too. Then adopt B2MML v7 as the starting schema for new payloads where both vendors accept it, with your own extension schema beside it, **and write down the gap between v7 and the 2025 Part 1 as a known, owned design decision** [7][8].

Input two: the quality scheme's record obligations, for the data-ownership matrix. This input does the most work and is most often skipped. For each legal entity and each plant, identify which scheme applies — and an entity may have several. Karoo Foods runs a food scheme; Highveld runs the automotive one; the group therefore has two information architectures, not one.

Then build the two-column table per scheme: **record the scheme compels**, and **candidate system of record**, one row per record type. Fill the second column with the system that is authoritative today, not the one that should be. Where two systems claim a record, that is the ownership dispute you were hired to settle — and it now has an external obligation behind it rather than being a preference argument between two directors. Where no system holds it, you have found a gap an auditor will find later and more expensively.

Input three: ISO 22400, for the measure definitions. Create a measure definition register as a first-class repository artefact: one row per measure, with the ISO 22400 definition it follows, the denominator stated explicitly, the computing system, the owner and the version. **This register is the only place the business will ever agree what a number means**, and it belongs inside the house standard rather than in a reporting project's appendix.

Input four: ISA/IEC 62443 zones and conduits, for the technology and security layers. Use the zone model as the technology-layer structure of the reference model, not as a separate security deliverable. Each zone gets a target security level; each conduit a stated purpose, direction and protocol; each asset sits in exactly one zone. State the programme's position using the maturity model from the 2024 edition of 62443-2-1 [10]. Segmentation then lands where it belongs instead of being attached to ISA-95's levels, and you have a vocabulary to push obligations onto equipment vendors at the next capital purchase.

Then declare it

The last step converts four tables into an architecture. **Write an architecture principle saying that no single industry reference model governs manufacturing, name the four sourced inputs, and have it endorsed by the governance body.** The wording matters less than the endorsement: what you need is a document an incoming consultancy, a vendor and a new architect all have to work to.

> **WHAT THE PRINCIPLE BUYS YOU**
>
> It stops the next consultancy arriving with a proprietary "manufacturing reference architecture" and presenting it as the industry standard. It gives every vocabulary argument a place to be settled rather than relitigated. And it tells a board honestly why the architecture team exists — not to adopt somebody else's model, but to maintain the one the business owns.

A closing caution, offered as judgement. The composite is not elegant: it joins four documents from three organisations across two layers and leaves visible seams. Every architect who meets it wants to tidy it into one unified model, and every attempt produces a document nobody outside the architecture team recognises. **Leave the seams visible.** The seams are where the external sources are, and those are the whole reason the model is defensible.

THE CHAPTER IN TEN LINES

- Manufacturing has no single maintained, vendor-neutral industry reference model covering the business end to end, and nine pieces of evidence support that rather than one assertion.
- ISA-95 is scoped to a boundary, not to an enterprise, and Part 1 is now the 2025 edition; almost all circulating material cites 2010.
- There is no ISA-95 conformance scheme, so ISA-95 compliant is uncertifiable; ask which parts, which editions, which objects, in which binding.
- B2MML is MESA's, royalty-free with attribution, and its Version 7 of November 2020 binds the 2018 and 2019 editions, so it predates the 2025 Part 1 by five years.
- Three standardised interoperability stacks exist from three bodies with no reconciliation: OPC UA, MTConnect and Sparkplug, the last of which is Eclipse's and is ISO/IEC 20237:2023.
- The OPC Foundation's own April 2026 figure is over 430 companion specifications; the widely repeated sixty counts equipment types, not documents.
- RAMI 4.0 is a DIN specification and an IEC publicly available specification, which means a pre-standard, and SIRI is INCIT's assessment instrument, not a reference architecture.
- GS1 does not require a 2D carrier by 2027; Sunrise 2027 is a voluntary industry capability ambition, and GS1 US's own page says it is not necessary to implement all 2D capabilities by that date.
- ISO 9001:2015 is withdrawn and ISO 9001:2026 published on 16 September 2026, which is the single most likely stale claim in any quality-architecture document.
- The applicable sector quality scheme specifies more of a manufacturer's information architecture than any reference model does, because it dictates records.
- Catena-X is the nearest thing to a maintained multi-company manufacturing data architecture, but it is automotive-specific, European-anchored and scoped to cross-company exchange.
- Build the house reference model from four sourced inputs: ISA-95 object names, the quality scheme's record obligations, ISO 22400 measure definitions, and 62443 zones and conduits, then declare in the principles that no single industry model governs.

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CHAPTER 15

Technology providers

Who plays in each category, what each category exists to answer, and what it costs architecturally to say yes. These are examples, never recommendations — and every claim here carries a date, because seven ownership and naming changes landed inside two years.

Everything in this chapter is a map of **who plays in each category**. None of it is a recommendation, a ranking, an endorsement or a shortlist. Products are named because they are recognisable, not because they are good; omission is not a judgement any more than inclusion is a blessing. Every claim carries a date. Re-verify before you repeat it.

That is not a disclaimer bolted on for form. It is the only honest way to write this chapter, because the facts move faster than books print. Inside a single research window ending in October 2026, in this sector alone: one conglomerate completed a spin-off with a second separation in flight; a measurement group spun its whole software division out as a new listed company; one vendor bought out a minority it already controlled; a private-equity house took majority control of a South African ERP; two enterprise-architecture products were sold with the buyers undisclosed; and two products were renamed. Seven changes, two years.

So the chapter's job is not to tell you what to buy. It is to let you walk into a vendor conversation knowing **what kind of thing is being sold**, **which question the category exists to answer**, and **what it costs architecturally to say yes**.

What goes stale first, in order

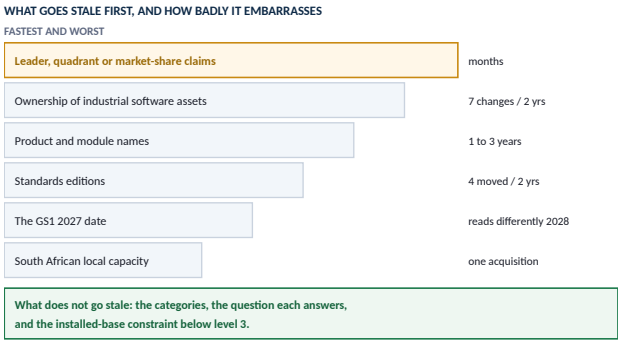


Figure 15.1 What goes stale first. The bar length is how long a claim of that kind survives. Leader and market-share claims go first and embarrass hardest, which is why none appears in this chapter.

The ordering matters because it tells you which sentences to date and which to leave out.

Any "leader", quadrant or market-share statement goes stale in months, and embarrasses hardest. This chapter contains none, deliberately. If you need one for a steering committee, cite the analyst and the year and present it as a datum about market *perception*. An analyst placement is a statement about a vendor's execution and vision as assessed on a date; it is not a statement about whether the product fits a batch plant in Paarl.

Ownership of industrial software assets goes stale next, then **product names**: a reader quoting a product name from a three-year-old deck is often quoting a name that no longer exists — MindSphere is Insights Hub [20][21], OSIsoft is AVEVA PI [22][23], Timescale the company is TigerData [24][25], EOH is iOCO [26], and AS9100 becomes IA 9100 on revision. Then standards versions, then the GS1 2027 date, and last the South African local-capacity position, which changes with one acquisition or one emigration wave.

The ownership corrections, all dated

These are the ones an author gets wrong, and several appear in material published this year.

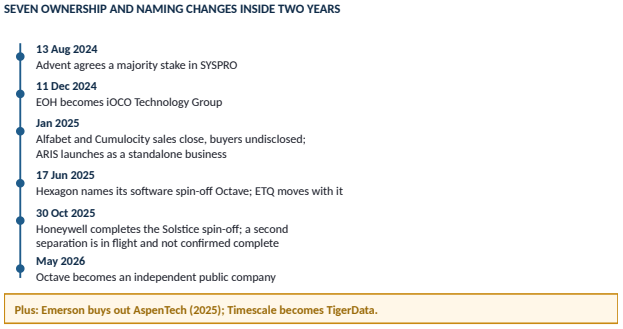


Figure 15.2 Seven changes, two years. Each is dated and sourced in the table that follows. This is why an undated ownership sentence is a liability rather than a fact.

Claim you will meet	The position, dated
"SYSPRO is an independent South African ERP"	**Wrong on both counts.** Founded in South Africa in 1978 by Phil Duff; **Advent International announced a definitive agreement for a majority ownership stake on 13 August 2024**; headquartered in **the UK and South Africa**, about 400 employees, claiming more than 15,000 customers in over 60 countries [1]
"Hexagon's EAM and quality products"	**Hexagon named the spin-off Octave on 17 June 2025** and its own page states Octave became an independent public company in **May 2026**, moving Asset Lifecycle Intelligence, Safety Infrastructure and Geospatial, **ETQ** and Bricsys [2][3][4]. **Whether HxGN EAM — the former Infor EAM — moved is UNVERIFIED** [2]
"Honeywell owns X"	Needs a date. The **Solstice Advanced Materials spin-off completed 30 October 2025**; the **Automation and Aerospace separation was "on track for completion in the second half of 2026" and is not confirmed complete** [5][6]
"STARLIMS is Abbott's"	**Francisco Partners acquired STARLIMS from Abbott Laboratories** [7][8]. Completion date and consideration UNVERIFIED
"Plex competes with Rockwell"	**Plex and Fiix are both Rockwell Automation's** [9][10]. That is one vendor's two products, not a comparison
"Software AG ARIS/Alfabet"	**Two separate things now.** Software GmbH announced in early January 2025 that **ARIS would launch as a standalone business** and that the **sales of Alfabet and Cumulocity had closed with buyers undisclosed**; Silver Lake has owned Software AG since May 2023 [11][12]. **Who owns Alfabet is UNVERIFIED**

Also settled and dated: **AspenTech is wholly Emerson's** since the 2025 buy-out of the remaining shares [13][14]. **AVEVA, including the former OSIsoft PI System, is Schneider Electric's** [22][23] [27]. **Sparta Systems TrackWise is Honeywell's**, agreed at $1.3bn [15] — which Honeywell depends on the separation above. **Critical Manufacturing has been an ASMPT subsidiary since 2018** [16]. **SAP completed the LeanIX acquisition in November 2023** [17]. **Infor is Koch Industries'** [18]. **IFS completed a CAN$1bn Copperleaf acquisition** and was valued at EUR 15bn in an April 2025 minority-stake sale, with EQT, Hg and TA named as existing owners [19][28]. **MindSphere is Insights Hub** [20][21]. **Timescale is TigerData** [24][25]. **EOH is iOCO, effective 11 December 2024** [26].

> **OWNERSHIP THIS CHAPTER DELIBERATELY DOES NOT ASSERT**
>
> **Epicor. QAD. Anaplan. Tulip. iBase-t. 42Q. Canary. Braincube. TwinThread. Alfabet. Bizzdesign. And the entire OT security category.** Each is named below as a player in its category with no owner stated, because the ownership could not be confirmed from a primary source at October 2026. If you need one for a contract, get it from the vendor's own corporate page on the day. An ownership claim is the cheapest thing in a document to check and the most expensive to get wrong.

How to read a category, and how to read a price

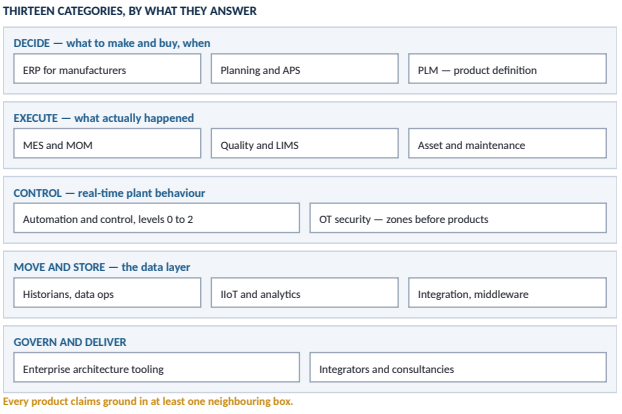


Figure 15.3 The thirteen categories, grouped by the question each exists to answer. Group the market this way and a vendor's claim to neighbouring ground becomes visible as soon as it is made.

Two habits make vendor conversations shorter. The first is to **name the category a product belongs to, and the adjacent categories it will claim to cover.** Every product here claims ground in at least one neighbouring category, because that is where the next sale is: a historian vendor shows you dashboards, an MES vendor shows you maintenance, an ERP vendor shows you scheduling. The question is not whether the feature exists but whether the vendor's customers *use* it as their system of record for that thing.

The second is to ask for **the pricing metric rather than the price.** Industrial software pricing is negotiated, bundled and almost never published, and any number is wrong within weeks. What is stable is the metric, and the ones that recur, as practitioner observation, are: named or concurrent users; plants, sites or lines; tags or data points; devices or assets; orders or documents; revenue bands for mid-market ERP; and consumption for anything cloud-native. **The metric tells you where the cost goes when the business grows**, which is the only part of a price that matters in a five-year architecture.

ERP for manufacturers

The system of record for the commercial and financial transaction set — customers, suppliers, orders, inventory valuation, costing, the general ledger — plus, in manufacturing, the bill of material, the routing, the work order and the planning run. The question it answers is "what should we make, from what, when, and what did it cost?"

ERP — examples of who plays, October 2026

SAP S/4HANA — large and group-structured manufacturers, where a global template, statutory consolidation and integration with SAP's own planning, PLM and warehousing are the point. SAP also owns LeanIX [17] and Signavio, which matters when an architecture-tool decision collides with an ERP decision. **Oracle** Fusion Cloud for the finance-led estate. **Infor** — three distinct manufacturing products and the distinction matters: CloudSuite Industrial (the former SyteLine) for discrete and mixed-mode mid-market; Infor LN (the former Baan) for project-based and engineer-to-order; M3 for process, food and distribution-intensive. Infor is Koch Industries' [18]. **Microsoft Dynamics 365** Finance and Supply Chain Management, where the Microsoft estate is the strategic platform. **Epicor**, flagship Kinetic, mid-market discrete — ownership UNVERIFIED. **IFS Cloud**, where maintenance and manufacturing are the same conversation [19][28]. **QAD**, automotive and life-sciences-leaning — ownership UNVERIFIED. **SYSPRO**, South African-developed, now Advent-controlled [1]. **Sage X3** for mid-market process and distribution. **Odoo**, cheap and open-source-cored, where the risk is not vendor lock-in but implementation-partner lock-in.

Examples, not recommendations. Ownership and cloud-edition names go stale first here.

MES and MOM — manufacturing execution and operations management

Executes and records what actually happened: dispatching work, enforcing the route or recipe, collecting counts and downtime reasons, enforcing inline quality checks, tracking genealogy, managing labour and tooling. **MOM** is the broader framing — MES plus quality, maintenance and inventory operations.

MES and MOM — examples of who plays, October 2026

Siemens Opcenter — the former Camstar, Preactor and SIMATIC IT lineage consolidated inside Siemens Xcelerator, with Execution, Quality, APS and Intelligence modules; Siemens also markets Industrial Operations X as the operations layer [29]. Differentiator: the PLM and automation estate on both sides of it. **Rockwell Automation — Plex and FactoryTalk.** Plex is multi-tenant software as a service, FactoryTalk the on-premise operations family; Rockwell announced the Plex acquisition on 25 June 2021 and announced completion [9][10]. Differentiator: the Allen-Bradley installed base. **AVEVA** — the former Wonderware and Citect lineage plus OSIsoft's PI System; Schneider Electric's since 2023 [22][23]. **Honeywell** — the former Process Solutions MES lineage plus Sparta Systems TrackWise on the quality side [15]; which Honeywell owns it after the separation is the open question [5][6]. **AspenTech** — process optimisation and scheduling rather than discrete execution; wholly Emerson's since 2025 [13][14]. **Dassault Systèmes DELMIA Apriso** [30][31]. **Critical Manufacturing** — founded 2009 in Maia, Portugal, an ASMPT subsidiary since 2018 [16]; semiconductor, electronics and medical-device lineage. **Tulip** — a low-code frontline-operations platform bought by operations rather than by IT, which is also its architectural risk; ownership UNVERIFIED. **iBase-t Solumina** — aerospace and defence configuration rigour; ownership UNVERIFIED. **42Q** — cloud MES with a Sanmina lineage [32]; current corporate status UNVERIFIED.

Examples, not recommendations. Module names inside these suites have each been rebranded within the research window, and three of the independents' owners are deliberately not stated.

The option the category does not sell you is **assembling an MES** from a broker, an industrial data-operations layer, a low-code application layer and a time-series store. It is increasingly real, and the honest assessment is that fit and cost improve and **you have just become an MES vendor with one customer**, carrying the record obligations of chapter 14 as your own validation burden. Say that before the business case is signed, not after.

Automation and control, and what installed base means

PLCs and programmable automation controllers, distributed control systems, safety systems, drives, HMI and SCADA, industrial networks, and the engineering software that configures all of it. Levels 0 to 2 of chapter 13, where the plant's real-time behaviour lives.

Automation and control — examples of who plays, October 2026

Siemens (SIMATIC, TIA Portal, PCS 7 and PCS neo); **Rockwell Automation** (Allen-Bradley Logix, Studio 5000, FactoryTalk View); **Schneider Electric** (Modicon, EcoStruxure, and AVEVA on the software side [22]); **ABB** (800xA and the Ability portfolio); **Emerson** (DeltaV, Ovation, and now AspenTech [13]); **Honeywell** (Experion); **Yokogawa** (CENTUM); **Mitsubishi Electric** (MELSEC); **Omron** (Sysmac); **Beckhoff** (TwinCAT, PC-based control and EtherCAT).

The split that changes an architecture, as practitioner analysis: **distributed-control vendors** dominate continuous and hybrid process plants — one engineering database, one operator environment, strong batch and alarm management — while **PLC and PAC vendors** dominate discrete manufacturing and machine-level control, with distributed engineering and far more heterogeneity.

Examples, not recommendations. Published product lifecycle dates move by design, and the umbrella platform brands are all relatively recent.

Now the most important paragraphs in this chapter, because they govern every shortlist you will write below level 3.

The controls vendor is chosen once and inherited forever. Control equipment lifetimes run 15 to 25 years and a changeover means re-engineering, re-validating and re-training. You do not choose the controls vendor; you choose what sits above it.

Installed base therefore determines the realistic MES, historian and IIoT shortlist, because every controls vendor sells software upward from its own hardware and prices it attractively to its own base. "We are a Rockwell plant" predicts a FactoryTalk and Plex conversation; "we are a Siemens plant" predicts Opcenter and Insights Hub; "we are an Emerson DeltaV plant" predicts AspenTech [9][13][29]. **This is not a conspiracy, it is the economics of**

the category — and an architect who ignores it produces a shortlist the plant engineering team will veto, correctly. **Heterogeneity is the norm**: two or three controls vendors, a dozen machine-builder-supplied control systems with their own interfaces, and at least one system whose vendor no longer exists. The job is an abstraction layer, not standardisation of the controls estate.

> **THE CHEAPEST HIGH-VALUE THING YOU CAN DO IN A MANUFACTURING CLIENT**
>
> Controls vendors publish lifecycle status per product line — active, mature, limited, obsolete — with dates. **Harvesting those dates into the repository as a technology-currency view is one of the highest-value, lowest-effort deliverables available to an architect in manufacturing, and almost nobody does it.** The installed base also defines the security problem: what can be patched, what can be segmented, and what cannot be touched without re-validation.

Historians and industrial data infrastructure

Captures high-frequency time-series data reliably, stores it compressed for years, and serves it back fast enough for trending and reporting. Then the newer half: reshaping plant data into something an IT data platform can consume — contextualised, modelled, named consistently across sites.

Historians and industrial data — examples of who plays, October 2026

AVEVA PI System — the former OSIsoft PI, the de-facto process historian in oil, gas, chemicals, mining, power and large-scale process manufacturing; AVEVA is Schneider Electric's [22][23][27]. Name it AVEVA PI and explain the OSIsoft lineage once, because readers meet both names in the same plant. **Canary** — lighter-weight, per-tag simplicity, where PI-class reliability is wanted without PI-class cost; ownership and licensing UNVERIFIED. **InfluxDB**, now at generation 3 [33], and **TimescaleDB** from **TigerData**, the renamed company [24][25] — general-purpose time-series stores, not process historians. **Cognite Data Fusion** — industrial data operations and contextualisation. **HighByte Intelligence Hub** — models, contextualises and routes OT data to IT targets at the edge; Siemens publishes a HighByte product page and the two announced a partnership [34][35], and **the commercial form of that arrangement is UNVERIFIED** — a reader meeting "Siemens HighByte Intelligence Hub" is meeting a partnership, not an acquisition. **Litmus Edge** — edge collection and normalisation with a large driver library.

Examples, not recommendations. Product names churn harder here than anywhere else.

The distinction to hold: **a historian stores; a data-operations layer models and moves; a time-series database is a component, not an answer.** The commonest error is buying a time-series database and discovering the hard part was never storage. It was naming, modelling and contextualisation — deciding that `TIC_1042.PV` is the outlet temperature of reactor 2 at Gqeberha, and keeping that true across 40,000 tags and three refurbishments. On price, interrogate **what the per-tag licence counts, and what happens when a line is added.**

PLM — product lifecycle management

Owns the product definition: design data management, the engineering bill of material, revisions and effectivity, engineering change, configuration management, document and specification control, requirements traceability, and the handover of the engineering bill into the manufacturing bill. In regulated manufacturing it is also the home of the design history file.

PLM — examples of who plays, October 2026

Siemens Teamcenter — broad enterprise PLM, design-tool-agnostic in practice, tied into NX and Opcenter. **Dassault Systèmes ENOVIA on 3DEXPERIENCE** — where CATIA is the design authority [31]. **PTC Windchill** — discrete manufacturing, coupled to Creo and PTC's own IIoT. **Autodesk** — Fusion, Vault and Upchain, lighter-weight, mid-market. **Aras** — upgrade-safe customisation where configuration demands are unusual. **SAP PLM** — where the ERP decision is made and the document scope is modest. **Oracle Agile PLM** — a long-installed base in electronics and life sciences; **Agile's current support position and roadmap are UNVERIFIED, and that is a real architecture question for anyone running it.**

Examples, not recommendations. Cloud-edition naming and the support horizon of the long-installed products go stale first.

The point that matters more than the vendor list: **PLM and ERP both think they own the bill of material.** Governing the transformation between them — who may change what, where effectivity is set, how an engineering change reaches an open work order — is the highest-value piece of design work in a discrete manufacturer's application architecture. Every vendor has an answer. **The answer is a process decision, not a product feature.**

Supply chain planning and advanced planning and scheduling

Decides what to make and buy, where and when, over horizons from a shift to several years: demand planning, supply and inventory planning, integrated business planning, distribution planning, and — the specifically manufacturing part — finite-capacity scheduling on constrained resources.

Planning and APS — examples of who plays, October 2026

Kinaxis — concurrent planning, bought for scenario speed across a multi-tier network. **o9 Solutions** — a graph-based platform spanning demand, supply and revenue planning. **Blue Yonder** — planning plus warehouse and transport execution; **Blue Yonder is Panasonic's, and "JDA" is a former name of the same company, not a sister product** [36][37]; the acquisition was reported at $7.1bn and completed in 2021 [37], with the exact completion date and the full lineage UNVERIFIED. **SAP IBP** — where S/4HANA is the ERP and the planning data model should not be duplicated. **OMP** — Belgian, process-industry-leaning, bought for campaigns, grades and co-products. **Anaplan** — connected planning as a modelling platform rather than a supply-chain application; ownership UNVERIFIED. **Dassault Systèmes DELMIA Quintiq** — hard combinatorial scheduling and resource optimisation [30][31]. **John Galt Solutions Atlas Planning** — mid-market demand planning where a full suite is unaffordable.

Examples, not recommendations. Artificial-intelligence feature claims are the fastest-moving and least verifiable statements any vendor in this category makes, and this chapter makes none of them.

Two things to say in every planning conversation. **Planning and scheduling are different problems with different time constants and different owners.** "Make 40,000 units in week 32" and "run product A, then C, then B on line 3 on Tuesday to minimise changeover" are not substitutes, and most failed planning programmes failed because one tool was bought for both jobs. And **both depend on master data the manufacturer almost certainly does not have at the required quality** — routings with real cycle times, changeover matrices, yield by product and line. That is a precondition, not an implementation task: if the vendor says "we will sort that out in implementation", the programme is already late.

Asset management and maintenance

The asset register and hierarchy, preventive and condition-based maintenance plans, work orders, spares and storerooms, permits and isolation, contractor management, and the cost and reliability history that makes replace-or-refurbish decisions arguable.

Asset management — examples of who plays, October 2026

IBM Maximo, delivered as Maximo Application Suite — the heavyweight in asset-intensive industry. **SAP plant maintenance** — bought because it is in the ERP and the cost integration is native, criticised for usability at the technician's hands, which is why mobility add-ons are a market in themselves (practitioner analysis). **Infor EAM, now HxGN EAM** — the former Datastream product, sold to Hexagon; and Hexagon has since spun its software out as Octave [2][3][4]. **Whether HxGN EAM moved into Octave is UNVERIFIED, and it is the likeliest vendor-attribution error in this category.** Do not write "Infor EAM" as a current Infor product without checking. **Fiix** — cloud maintenance management, Rockwell Automation's, sold alongside the controls estate [9]. **UpKeep** — mobile-first, where the alternative is a spreadsheet. **Pragma and On Key** — South African: the Pragma group owns the On Key system, hosted from data centres in **South Africa and northern Europe** (**Ireland**), with products EAMS, Insights, Action, Integrate and Connect, ISO/IEC 27001:2013 certification, a client history stated "since 1992", and a 2024 analyst recognition [38][39]. Pragma is also a **maintenance services business, not only a software vendor**, which is the differentiator and often the thing a South African manufacturer is actually buying.

Examples, not recommendations. The Hexagon and Octave product allocation is the live uncertainty.

Asset management is where **the asset hierarchy fight happens.** Finance wants a fixed-asset register, maintenance a functional-location hierarchy, operations a line and equipment hierarchy, and the historian holds a third. **Four hierarchies over the same steel is the normal starting state**, and resolving it — one physical asset identity, several classification views — is a genuine data-architecture deliverable.

Quality management and laboratory information management

Two categories that are routinely conflated. **Quality management** handles the quality *system*: controlled documents with revision and training acknowledgement, nonconformances, deviations, corrective and preventive action, audits, change control, supplier quality, complaints. **Laboratory information management** handles the *laboratory*: samples, tests, specifications, instrument interfacing, results, out-of-specification handling, stability studies, certificates of analysis.

Quality and LIMS — examples of who plays, October 2026

Laboratory: **LabWare** (independent, very large regulated-industry base); **LabVantage**; **Thermo Fisher SampleManager**; **STARLIMS** — **acquired by Francisco Partners from Abbott Laboratories** [7][8], so not an Abbott product. Quality: **Sparta Systems TrackWise** — Honeywell agreed to acquire it for $1.3bn [15], with the open question of which Honeywell it lands in [5][6]; **ETQ** — moved with Hexagon's spin-off into Octave [2][3][4]; **Veeva** Vault quality; **MasterControl**; **Siemens Opcenter Quality**, the former IBS product, where inline statistical control at the line is the requirement.

Examples, not recommendations. Three of these changed hands inside the research window.

Quality data lives in at least five places: inline checks in the MES, lab results in the laboratory system, nonconformances in the quality system, dimensional results in metrology, supplier certificates in a document store or the ERP. **The record obligations of chapter 14 do not care where it lives. They care that it can be assembled per batch, per part, per date, with signatures.** That assembly requirement *is* the integration architecture, and in regulated sectors it is also the validation burden.

IIoT platforms and industrial analytics

Connects equipment, moves telemetry somewhere it can be analysed, and provides modelling, visualisation, anomaly detection and machine-learning tooling on top. The boundary with historians and data operations is porous, and the vendors fight over it.

IIoT and analytics — examples of who plays, October 2026

The hyperscalers — the three large public-cloud providers sell ingestion, edge runtime, time-series storage, digital-twin modelling and machine learning as building blocks rather than as applications. Differentiator and risk in one sentence: unlimited flexibility and no manufacturing semantics — whatever ISA-95 or asset-shell structure you want, you build. **This chapter prints no hyperscaler service names at all**, because naming changes faster than anything else here and per-service availability by South African region could not be confirmed. Ask for the region-and-service table instead. **PTC ThingWorx** — an application-platform approach with strong design and service adjacency. **Siemens Insights Hub** — the former MindSphere [20][21], inside Xcelerator, with Industrial Operations X as the broader framing [29]. **Seeq** — self-service time-series analytics aimed at process engineers rather than data scientists; raised $50m in 2024 and appears independent at this date [40]. Bought by the people who know the process, which is why it gets used. **Braincube** and **TwinThread** — process analytics and a predictive-analytics-led platform; ownership and product sets UNVERIFIED.

Examples, not recommendations. Everything commercial here goes stale inside a year.

The honest point: **most disappointment with an IIoT platform is a data-quality and contextualisation failure, not a platform failure.** The platform works. The tags are unnamed, the asset model is absent, the downtime reasons are free text, and the timestamps come from three unsynchronised clocks. Naming and asset model first, then connectivity, then analytics. Vendors sell it in reverse because the demonstration is better that way.

Enterprise architecture tooling, and the honest treatment of ArchiMate 4

The repository: object types and relationships, diagrams, application and technology portfolios, the capability model, roadmaps, governance artefacts — and in a manufacturing client, the plant-level technology-currency and obsolescence views above.

EA tooling — examples of who plays, October 2026

OrbusInfinity (Orbus Software) — software-as-a-service platform; its own capability page describes a central repository, document library, diagramming, workflow automation, dashboards, views, permissions and collaboration; modules for enterprise architecture, business architecture, business process management and IT portfolio management; AI features; **TOGAF Version 10 and ArchiMate Version 3.2 as preconfigured content**; BPMN and ITIL support; a configurable metamodel and a CoreEA metamodel; ISO 27001, SOC 2 Type II, FedRAMP and IRAP certifications; a government edition; and integrations including ServiceNow, Jira, AWS, Power BI, Teams, SharePoint, Visio, Okta and Microsoft Entra ID, with "150+ connectors" claimed [41]. **Every item in that sentence is Orbus's own claim at a date. SAP LeanIX** — SAP completed the acquisition in November 2023 [17]; a LeanIX-versus-SAP framing is obsolete. **Bizzdesign** — modelling-led, long ArchiMate heritage; ownership and version support UNVERIFIED. **ARIS** — now a standalone business [11][12], process-modelling-led. **Alfabet** — no longer Software AG's, buyer undisclosed at announcement [11][12]; **who owns it is UNVERIFIED**. **Sparx Systems Enterprise Architect** — deep UML, SysML and ArchiMate modelling, cheap per seat; relevant here because SysML is the systems-engineering language and a manufacturer with a product-engineering function may already own it. **Ardoq** — data-driven and graph-based rather than diagram-led. **Avolution Abacus** — analysis and metrics led, with scenario and cost modelling. **MEGA HOPEX** — governance-led, strong in risk and compliance adjacency, a real advantage in regulated manufacturing. **Archi** — free and open source, the reference ArchiMate modelling tool; the right answer for a small team, a training exercise or a client with no budget, and the right place for a reader of this book to start. Its limitation is repository-grade collaboration and reporting, not modelling fidelity.

Examples, not recommendations. ArchiMate version support across every vendor here will move within months of the ArchiMate 4 release, and three of these changed hands or corporate structure inside the research window.

Three facts about ArchiMate 4 support have to be handled honestly, because this is where a consultancy's own commercial relationships make it easiest to be sloppy. **The firm publishing this book is a preferred partner of Orbus Software. No preference is expressed and none should be inferred.**

One. ArchiMate 4 is new. The Open Group announced the ArchiMate 4 Specification, published as document C260, describing it as an evolution of the language with strong user-level compatibility; the announcement page is datelined 27 April with no year printed, and the Open Group's own blog discussing the release is dated 20 May 2026 [42][43][44]. The exact publication date is UNVERIFIED — do not print one from a secondary source.

Two. There is no public certification evidence for ArchiMate 4 tool conformance. The Open Group's ArchiMate Tool Certification Register page **returned unrendered template placeholders** at retrieval, and its static text refers to conformance against **ArchiMate 3.0.1 or 3.1** [45][46]. **Therefore no tool — OrbusInfinity included — can be described as a certified ArchiMate 4 tool on the strength of a public register.** Which tools hold current certification, and whether such a programme exists at all, is UNVERIFIED.

Three. A vendor claim and a certification are different claims. OrbusInfinity's own page says ArchiMate 3.2 and TOGAF 10 preconfigured [41] — an accurate statement of a vendor claim at a date, neither flattering nor damning: **3.2 was the current ArchiMate version until 2026, and a tool shipping 3.2 content weeks or months after a new major version is normal and not a deficiency.** Treat any vendor's "ArchiMate 4 ready" marketing the same way, as a claim with a date on it pending a register entry.

THE FIVE QUESTIONS, PUT TO EVERY TOOL VENDOR INCLUDING ORBUS

1. **Which ArchiMate version does your shipped metamodel implement, and do you hold a current Open Group tool certification for it?** Ask for the register entry, not the marketing page.
2. **Can the metamodel be extended without losing upgradeability**, and what happens to custom object types on a major release?
3. **Does your TOGAF content mean a populated content metamodel** with the full object and relationship set a real repository needs, or TOGAF-labelled diagram templates? Very different products.
4. **Can the repository hold manufacturing-specific extensions** — plant, line, work centre, equipment, control system, historian tag namespace, recipe, quality record type — as first-class objects, or must they be application components with naming conventions?
5. **How is the repository exported if we leave?**

Then the finding, which is a gap in the whole category rather than a criticism of any vendor in it. **No enterprise-architecture tool was found to ship a manufacturing metamodel extension out of the box**, though a manufacturer's repository needs Plant, Line, Work Centre, Equipment, Control System, Historian Tag Namespace, Recipe, Quality Record Type and standard-conformance status as objects with relationships. **Whether any vendor ships such an extension is UNVERIFIED across the whole category — which is itself the finding.** Build it on a configurable metamodel, and make upgrade-safety of custom object types a selection criterion.

Integration and middleware in a plant

Moves data between ERP, MES, historian, laboratory system, quality system, warehouse system, PLM and the control layer, under plant-specific constraints: some flows are real-time and must not block the line, some are store-and-forward because the network drops, and a surprising amount is still a flat file on a share.

Integration — categories of player, October 2026

Enterprise integration platforms — the brokers, API gateways and integration platforms the IT organisation already owns. Strength: governance, monitoring, reuse. Weakness: usually no OT protocol support and no appetite for sub-second latency. **Industrial protocol gateways and OPC servers** — the layer that speaks Modbus, EtherNet/IP, Profinet, S7 and a hundred proprietary machine protocols and presents OPC UA or MQTT upward; **Kepware** and **Matrikon** sit here alongside the controls vendors' own servers, ownership and naming for both UNVERIFIED. **Industrial data operations** — HighByte, Litmus and Cognite above: protocol breadth plus modelling plus routing, the genuinely new category of the last five years. **MQTT brokers with Sparkplug** — **HiveMQ** and **EMQX** are commonly met; licensing and ownership UNVERIFIED. **ERP-vendor integration technology**, which arrives by default and is then used for things it is poor at. **And the flat file**, which is not a joke.

Examples, not recommendations. Gateway ownership and naming are the live uncertainties.

A large fraction of ERP-to-MES integration is a delimited file on a share with a scheduled job. Say so, then give the honest assessment: **a monitored, idempotent, reconciled file interface is better architecture than an unmonitored real-time one.** The question is not "is it file-based?" but "is it reconciled, and how do we know when it breaks?"

The point that outlives every product in that box: **the plant integration layer has a different availability requirement from the enterprise one.** If the ERP is down, the plant must keep making product and buffer the transactions. That single requirement — **store-and-forward with eventual reconciliation** — drives more plant integration design than any technology preference, and it is what enterprise integration architects most often fail to design for.

OT security

Discovers and inventories OT assets passively, because active scanning can disrupt a control system. Identifies vulnerabilities and insecure protocol use, detects anomalies in industrial traffic, segments and enforces policy at the OT and IT boundary, and manages secure remote access for vendors and integrators — the commonest real-world attack path into a plant.

OT security — examples of who plays, October 2026

Claroty; **Nozomi Networks**; **Dragos**; **Tenable OT Security**, the former Indegy; **Armis**; and **Microsoft Defender for IoT**, the former CyberX. Broad differentiators, as analysis rather than ranking: Dragos is strongest where control-system threat intelligence and incident response are the requirement; Claroty and Nozomi compete hardest on asset-discovery breadth, protocol coverage and sensor placement; Tenable appeals where its IT vulnerability management is already standard; Armis comes from the unmanaged-device-visibility direction and reaches beyond OT; Microsoft wins on bundling where the organisation is already Microsoft-committed.

Ownership, funding status and — for Microsoft specifically — current product status and naming are UNVERIFIED across this entire category. This chapter names the players and refuses to rank them, because here a ranking is both the most-requested and the most dangerous thing an architect can publish.

Examples, not recommendations. Everything commercial here consolidates constantly, which is why no ownership claim is made.

OT security is **an ISA/IEC 62443 problem before it is a product problem.** The sequence: define zones and conduits, set a target security level per zone, establish an asset inventory, then choose a detection and enforcement product to deliver the monitoring the zoning requires. **A plant that buys a**

detection platform without a zone model gets dashboards it cannot act on, because every alert raises the question "may I touch this device?" — a zoning and change-control answer, not a tooling one.

Systems integrators and consultancies, as categories of player

Integrators — categories, with examples, October 2026

Global strategy-and-technology consultancies with South African practices — bought for board-level programme credibility and the capacity to staff a large ERP programme; watch that the delivery team is frequently more junior and more offshore than the sales team (practitioner analysis). **The large South African listed IT services groups — iOCO Technology Group**, the JSE-listed group formerly known as EOH, rebranded effective **11 December 2024**, FY2025 revenue **R5.58bn**, with an explicitly named industrial-technologies capability [26][47][48]; **Altron** and the Bytes lineage; and the local NTT and Dimension Data operation. iOCO's own public record ties the rename to distancing the business from past procurement allegations under previous leadership [26] — stated factually with the citation, because the alternative is a rumour or a silence. **ERP-specialist implementation partners** — the SAP, Dynamics, Infor, Sage and SYSPRO channels [49][50]. **Controls and automation system integrators** — engineering businesses of 20 to 200 staff, concentrated near industrial centres, often certified by a controls vendor. The most important category here below level 3 and the least visible to enterprise architects; the current named firms are UNVERIFIED, so use the controls vendors' own partner locators. **Specialist manufacturing-operations consultancies. Asset-management and reliability specialists** — Pragma is the clear local example, as a combined software and services business [38][39]. **Independent enterprise architecture consultancies.**

Categories, not recommendations. Every firm name here can change: the largest listed group renamed itself inside the research window [26].

The South African provider landscape

The global category map above is misleading in South Africa in five specific ways, each of which belongs in a business case rather than in a footnote.

Who actually sells and supports locally

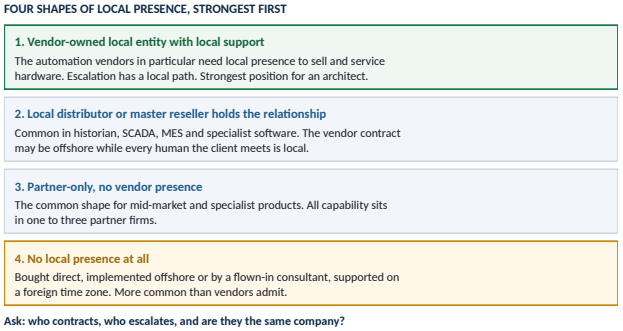


Figure 15.4 The four shapes of local presence. The shape, not the brand, predicts the support experience after go-live. The question at the foot settles it in one conversation.

For almost every product above, the local commercial reality takes one of four shapes, and the question to settle first is **who the contracting party is, who the escalation party is, and whether those are the same company.** The locally rooted names to know are **SYSPRO** [1], **Pragma and On Key** [38][39], **iOCO** [26][47][48], **GS1 South Africa**, operated by the Consumer Goods Council of South Africa and the local route to company prefixes and verification services [51], and **Sage** — not South African-owned, but with a very large local installed base in finance and payroll, which means it appears in more local manufacturing shortlists than its manufacturing capability alone would justify (practitioner analysis).

SYSPRO deserves one honest sentence, because it is the most commonly mis-described vendor in this market. **Its differentiator here is not features. It is that the implementation and support capacity is physically present, in local hours, with people who have done it in local plants.** That is a legitimate selection criterion and worth saying plainly — without turning it into an endorsement, and without repeating the claim that the company is independently South African, wrong since August 2024 [1].

Which global products have real local implementation capacity

> **THIS IS JUDGEMENT, NOT A FINDING, AND IT MUST STAY LABELLED AS SUCH**
>
> As an assessment built on the structure of the market rather than a counted survey: **deep local capacity** — many firms, many consultants, references in local plants — exists for SAP, Microsoft Dynamics 365, Sage, SYSPRO, the Siemens and Rockwell automation estates and their software, AVEVA on the SCADA and historian side, and IBM Maximo in the mining-adjacent asset-intensive sector. Mining, petrochemicals, automotive assembly and large-scale food and beverage manufacturing sustained those skill pools for decades.
>
> **Thin local capacity** — one or two firms, a handful of named individuals, references often regional rather than South African — is the assessment for the modern cloud MES and MOM products, the planning and APS category, the laboratory systems category, the industrial data-operations products, the OT security platforms and the newer architecture tools.
>
> **Do not print those lists as findings.** Print the question instead: *how many South African implementations of this exact product exist, and may I speak to two of them?*

Thin capacity has four consequences that change decisions rather than colour them. It makes **the implementation partner a single point of failure**, mitigated contractually rather than technically. It **inflates cost and duration unpredictably**, because flown-in consultants price in foreign currency plus travel and bill in time zones that halve the overlap. It **degrades support after go-live**, when a manufacturer most needs it. And it should **change the build-versus-buy calculus, not just the shortlist**.

Local-partner dependence as an architectural risk

Put it in the architecture risk register as a **delivery risk per work package**, with likelihood, impact and owner. It gets managed when it is visible and ignored when it lives in a procurement document. Five failure modes, each with a different mitigation:

Failure mode	Mitigation
Key-person dependence — knowledge sits with two or three named consultants	Named-person continuity clauses; documentation acceptance criteria; a client-side product owner at every design session
Partner acquisition or collapse [26]	Direct vendor contract where possible; assignment clauses; a tested path to a second partner
Configuration as undocumented architecture — in Odoo, Dynamics, SYSPRO and every low-code MES the real architecture *is* the configuration, and the partner holds it	Configuration documentation as a contractual deliverable with acceptance criteria; a client-held baseline; periodic independent review
Capability misrepresentation — certified in the product is not experienced in the archetype	References in the same archetype and regulatory regime, spoken to without the partner present
Escalation dead-ends — the partner is the only route to the vendor	A named vendor escalation route in the contract, tested before go-live rather than discovered during an incident

Cloud region availability and data residency

The dated facts, conservatively: **Google Cloud operates a Johannesburg region, operational since January 2024**, its first African region [52][53]; **Microsoft** has substantial South African Azure data-centre investment, reported at R20.4bn spent plus a further R5.4bn announced [52]; and **AWS** has spent, in the same report's words, billions of rand on data centres in Cape Town [52].

What this chapter will not print is a table of regions, zones or service names, because **per-service availability by region could not be confirmed and it changes monthly.** Learn the check instead: **ask the provider for the region-and-services table, by named service, on the day of the design decision.** Industrial and IoT services, managed time-series stores, digital-twin services and the newer analytical services are frequently absent from smaller regions at launch.

Three consequences that do not go stale. **"We are on a hyperscaler" does not mean "in South Africa"** — a plant-floor workload designed against a service that exists only in a European or American region has acquired a round trip of 150 to 200 milliseconds and a cross-border data flow. **Manufacturing data is mostly not personal data, except where it is** — telemetry, counts and genealogy are not; operator performance data, biometric access control, time-and-attendance, plant-floor video and quality records naming the operator are, which chapter 16 takes up. And **latency belongs in the design, not in the apology**: local execution with cloud aggregation, where the MES or edge layer continues through a link failure. That requirement is the most reliable way to tell whether a vendor has done plant work before.

Currency and licensing exposure in rands

Almost all industrial software is priced in dollars or euros, and almost all South African manufacturing revenue is in rands. A multi-year subscription is therefore a **multi-year foreign-currency liability against rand revenue**, and the exchange rate is the largest uncontrolled variable in the total cost of ownership of anything here. No exchange rate appears, deliberately: any number would be wrong within weeks.

Four things follow, all practitioner analysis. **Model the renewal at a stressed rate, not at the spot rate on the day of signature**, and take the architecture decision — more modules or fewer, more users or fewer — against the stressed case. **Per-user and per-tag metrics compound currency risk**, because growth multiplies against a moving rate; prefer metrics the client controls and can forecast. **Rand contracting is sometimes available and always worth asking for**, usually with a rate-adjustment clause, and the clause is the thing to read. And **cloud consumption is a currency exposure nobody signs for**, because consumption pricing has no fixed commitment and therefore no fixed rand cost.

> THE MOST USEFUL SOUTH-AFRICA-SPECIFIC COMMERCIAL OBSERVATION IN THIS CHAPTER
>
> **Perpetual-plus-maintenance is not obviously worse than subscription here.** The automation vendors still offer perpetual licences in some categories. In a weak-currency, high-interest-rate economy, a capitalised perpetual licence with rand-denominated local maintenance can beat a dollar subscription on total cost — which is the opposite of the global default advice, and worth modelling rather than assuming. Offered as analysis, not as a finding.

Support hours against a 24-hour plant

Many South African plants run continuously — three shifts, five to seven days, with planned shutdowns measured in days per year. Much of the software above is supported on business hours in the vendor's home time zone, so for an American vendor the South African night shift is unsupported and for a European vendor the late shift is marginal.

Five moves. **Map each application's support window against the plant's operating window explicitly**, as part of resilience tiering: if the MES is tier one for a 24-hour plant and its support window is 09:00 to 17:00 in a European time zone, the architecture has an unowned gap and somebody should sign for it. **Buy the follow-the-sun tier where it exists; where it does not, buy local first-line**, because the first hour of an incident is diagnostic and local. **Design for autonomy rather than for support**, which converts a support-hours problem into a reconciliation problem that can wait until morning. **Write recovery objectives against the shift pattern** — four hours means something different at 02:00 on a Sunday than at 10:00 on a Tuesday. And **test the**

escalation path during the window you actually fear. Support windows are contract-specific and UNVERIFIED for every product named here; they live in the schedule attached to the contract, the document nobody reads before signing.

And the framing again, because it matters more than any name above

Everything here is a map of who plays in a category. **None of it is a recommendation, a ranking, an endorsement or a shortlist.** Products were named for recognisability. Ownership was dated where it could be sourced and withheld where it could not. Analyst placements, market shares and "leader" positions were left out on purpose. And the one place where the publisher has a commercial relationship — Orbus Software — was covered on the same terms as its competitors, with the same five questions printed for a reader to put to it.

What should survive when every vendor fact here is out of date is the structure: the categories, the question each exists to answer, the installed-base constraint below level 3, the five South African distortions, and the habit of putting a date on every claim.

THE CHAPTER IN TEN LINES

- Every vendor in this chapter is an example of who plays in a category, never a recommendation or an endorsement, and every claim carries a date.
- Leader positions, analyst quadrant placements and market-share claims go stale fastest and embarrass hardest; this chapter contains none on purpose.
- SYSPRO is no longer independently owned: Advent International announced a definitive agreement for a majority stake on 13 August 2024, and the company is headquartered in both the UK and South Africa.
- Hexagon's industrial software is now Octave, named 17 June 2025 and an independent public company in May 2026, and ETQ moved with it; whether HxGN EAM moved could not be verified.
- Honeywell is becoming more than one company, so any sentence naming Honeywell as an owner needs a date on it.
- STARLIMS is Francisco Partners', TrackWise is Honeywell's, Plex and Fiix are both Rockwell's, AspenTech is wholly Emerson's, AVEVA including the former OSIsoft PI is Schneider's, and SAP owns LeanIX.
- ARIS and Alfabet are no longer one vendor offering and who owns Alfabet could not be verified, so do not write Software AG Alfabet.
- The controls vendor is chosen once and inherited forever, so installed base determines the realistic MES, historian and IIoT shortlist whether an architect likes it or not.
- No tool, OrbusInfinity included, can be described as a certified ArchiMate 4 tool on the strength of a public register, because the register returned unrendered placeholders and its static text references ArchiMate 3.0.1 and 3.1.
- OrbusInfinity's own page claims ArchiMate 3.2 and TOGAF 10 preconfigured, which is a dated vendor claim and is normal rather than a deficiency so soon after a major version.
- No enterprise architecture tool was found to ship a manufacturing metamodel extension, which is a genuine category gap and is itself the finding.
- In South Africa the shape of local presence predicts the support experience, local-partner dependence belongs in the risk register as a delivery risk per work package, and support windows live in a contract schedule nobody reads before signing.
- Price in rands against a stressed exchange rate, and consider that a capitalised perpetual licence with local maintenance can beat a dollar subscription here, which is the opposite of the global default advice.

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CHAPTER 16

Regulation, risk and what the inspector asks for

Architecture material, not legal advice. The frame is never whether you comply — it is what the record has to show. Nineteen bodies, a dozen clocks that contradict each other, and one register that answers all of them.

This chapter is architecture material. It is not legal advice. It cannot tell you whether an obligation applies to a given plant, what a provision means in a contested case, or whether anybody complies — those belong to an admitted attorney and to the regulator. What it can do is tell you **what records a manufacturing data architecture has to produce**, because an inspector will ask.

The frame throughout is **what the record has to show**, never *whether you comply*. Everything carries a date, because statutory instruments change without announcement. The deliverable is a **regulatory obligations register**: one row per obligation, with the instrument, the section, the retention period, the owner, the system of record and the repository column holding the evidence.

> **A SOUTH AFRICAN REGULATION NUMBER IS NOT AN IDENTIFIER**
>
> The same published list of food instruments carries **"R1008" against two different subjects in two different years** — colourants in 1996 and sanitary conditions for manufacture in 2004 [22]. Numbers are reused. **The identifier is number plus year plus subject**, and a register column holding only "R1008" cannot be resolved by anybody, including the person who typed it.

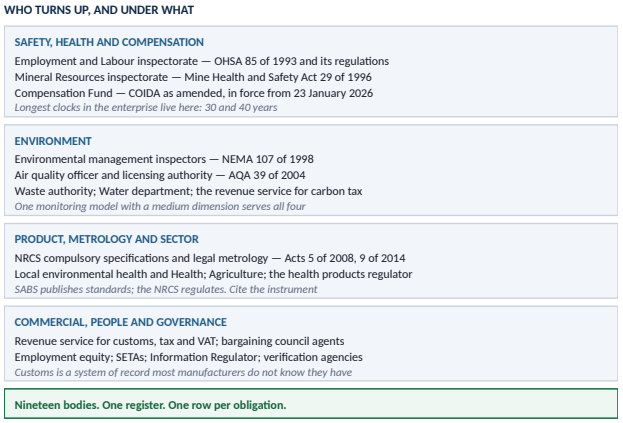


Figure 16.1 Who turns up, and under what. Nineteen bodies in four groups. Each asks for records a manufacturer either holds as data or cannot produce at all — and each group has a repository pattern that serves all of its members.

Occupational health and safety

The Act, and why the regulations are the architecture

The **Occupational Health and Safety Act 85 of 1993** works through a short set of general duties plus a large body of regulations made under **section 43** — and **it is the regulations, not the Act, that generate almost all of a manufacturer's record obligations.** The section numbers usually cited for the general duties and for written appointments could not be confirmed, so this chapter does not state them.

The Act-level obligation is one object, and it is almost always wrong. **A legal appointments register**: every appointment and delegation, by name, scope and site, with the letter reference, acceptance date, period, competence evidence and supersession chain. It is kept as files rather than as data, so it cannot answer **"who was the appointed person at this site on this date"** — the question asked after an incident.

Incidents, investigations and committees

The **General Administrative Regulations, GN R.929 of 25 June 2003** [20] set the clocks: notice to the provincial director **within seven days** of a section 24(1)(a) incident on form WCL 1 or WCL 2; **serious incidents** — death, unconsciousness, loss of a limb, or an injury likely to cause death or permanent defect — reported **immediately**, as are incidents involving **non-employees**; an **occupational disease** notified on form WCL 22 **within 14 days**; a record in the **Annexure 1** form kept **at least three years**, covering reportable incidents *and* those needing medical treatment beyond first aid; **investigation within seven days**; and the **committee examining each record at its next meeting**. **No retention period is stated for appointment records** [20].

Two consequences. **The clock fields must be data, not narrative**, or compliance with the seven-day rule is unprovable. And **one event may produce an occupational-health notification, a compensation claim and an internal non-conformance: three cross-referenced records that must not be merged.**

Hazardous chemical agents — the 1995 regime is gone

The Regulations for Hazardous Chemical Agents, GN R.280 in GG 44348 of 29 March 2021, replaced the earlier hazardous chemical *substances* regime [4]. An author working from 1995 is six years out. And **reg 2 brings manufacturers, importers, suppliers and retailers into scope, not only employers** [4] — the limb manufacturers under-read, because a chemical manufacturer is regulated both as an employer and as a supplier.

Duty	Reg	What the instrument says
Risk assessment	5(1), 5(5)	Immediately, then **at intervals not exceeding two years** — not annually — reviewed on process change
Air monitoring	6(1)–(2)	By an **approved inspection authority**, **at least every 24 months**
Medical surveillance	7(2)	Initial evaluation immediately before, or **within 14 days after**, starting work; follow-up not exceeding two years
Engineering controls	12(b)	Examination by an approved inspection authority **at intervals not exceeding 24 months**
Classification	14(a)–(c)	Manufacturers and importers **classify before supply** using Tables 4 and 5 of Annexure 1
Safety data sheets	14A	**GHS-compliant, 16 prescribed headings, reviewed at least every five years**
Retention	9	**Assessment and air monitoring: minimum 30 years.** Control examinations: at least three years. Medical surveillance is covered by 9(a) with **no period stated**

Five linked repositories follow: a **chemical inventory** bound to the item master; a **risk assessment register** with a two-year clock **and review triggers on process change** — so an engineering change affecting a hazardous agent must raise a risk-assessment review, and that link is the one architects miss; **exposure monitoring results** at exposure-group grain; **medical surveillance records**, which are special personal information; and **control examination records**.

> **THE 30-YEAR CLOCK OUTLIVES THE APPLICATION BY A FACTOR OF FOUR**
>
> Exposure records carry a **minimum 30-year retention** [4], and occupational health systems are replaced every seven to ten years. The requirement is not "keep the data in the system" but **an export format, with its reference data, readable without the source application.** Cheap at the start, impossible to reconstruct later.

Noise — 40 years, and a state-dependent test interval

The **Noise-Induced Hearing Loss Regulations, GN R.307 of 7 March 2003** [12]. The limit is **85 dBA as an eight-hour rating level**, and two things are routinely got wrong. **Noise records are 40 years, not 30** — assessments, monitoring, and medical surveillance including baseline audiograms [12]. And **audiometry is not a flat annual test**: baseline before employment or within 30 days; **annually for the first three years**; then **up to two-yearly if no referral threshold shift appears**; and **six-monthly in zones at or above 105 dBA** until no shift is evident [12]. Monitoring runs at least every 24 months by an approved noise inspection authority [12].

So **the next-due date is computed from the employee's own test history and zone** — a state machine, not a calendar; **threshold shift is derived and needs the baseline resolvable for forty years**; and **the dated employee-to-noise-zone assignment** is the evidence a compensation claim turns on, and the most commonly missing field here.

Pressure equipment — 36 months, not 12

FIVE INSPECTION-INTERVAL TRAPS

WHAT PEOPLE SAY	WHAT THE INSTRUMENT SAYS
Pressure vessels are inspected annually	Internal, external and hydraulic test at 1.25x design pressure, intervals not over 36 months
The 12-month pressure inspection	External inspection of fire-tube steam generators only
Lifting gear is tested every year	Three clocks: machine 12 months, components 6 months, tackle 3 months, signed per item
Audiometry is an annual test	Annual for 3 years, then up to 2-yearly absent a shift; six-monthly at or above 105 dBA
Scales must be verified annually	The Act fixes no interval at all. It is deferred to class-specific technical regulations

Figure 16.2 Five interval traps. Each left-hand statement is in circulation and each is wrong. The maintenance plan, the next-due calculation and the audit answer are all built on the right-hand column.

The **Pressure Equipment Regulations, GN R.734 of 15 July 2009** [11]. "Pressure vessels are inspected annually" is wrong, and maintenance plans are built on it anyway.

- **Reg 11(1)(d)**: internal and external inspection plus hydraulic test at **1.25 times design pressure, at intervals not exceeding 36 months**; for non-deteriorating equipment the internal inspection and test may be **omitted up to nine years with written approval**.
- **Reg 12**: **risk-based inspection may replace those intervals entirely**, verified by an accredited certification body.
- **The 12-month item is *external* inspection of *fire-tube steam generators* only** (reg 11(1)(c)).
- Plus **SANS 347** categorisation (reg 3), **manufacturing records kept a minimum of 12 years by the manufacturer** (reg 14(3)), and a steam-generator registration certificate that **lapses on transfer of control or removal from the premises** (reg 8) [11].

So **the next-due date is a derived field whose rule differs per equipment class and per inspection regime**, not a stored number maintained by hand — and **the 12-year manufacturing clock sits on the manufacturer, which for an in-house fabricator is the same company.**

Lifting equipment — three clocks, not one

The **Driven Machinery Regulations, 2015, GN R.540 in GG 38905**, corrected by GN R.913 in GG 39253, commencement 30 September 2015 [13].

Object	Clock	Record
Lifting machine	**12 months**, plus before first use and after each dismantling and re-erection	Performance test by a **registered lifting machinery inspector**; 110% of safe working load where no manufacturing standard applies
Ropes, chains, hooks, sheaves, brakes, safety devices	**6 months**	Competent-person examination record
Lifting tackle	**3 months**	**Signed examination record per identified item**, plus discard decisions, with traceable identification and load markings
Owner or lessor of leased gear	**10 years**	Full service history; on site, records no older than 12 months

Lifting tackle is the hard part: every sling, shackle and chain is individually identified, inspected and discardable on a three-month cycle, which is an equipment master with an inspection cycle, not a spreadsheet. **The commonest failure is tackle held as a consumable item**, after which the three-month record cannot be produced per item.

Major hazard installations — the highest-consequence open item here

> **BOTH "THE 2001 REGULATIONS APPLY" AND "THE 2022 REGULATIONS APPLY" ARE UNSAFE**
>
> The **Major Hazard Installation Regulations, 2022** were promulgated as **GN R.2989 in GG 47970 of 31 January 2023**, and **reg 25 repeals the 2001 regulations**. But **reg 26 provides for commencement on a date set by the Minister by notice in the Gazette, and this research could not confirm that the notice has been published** [3]. So **both "the 2001 regulations apply" and "the 2022 regulations apply" are unsafe. Write the uncertainty into the document**, design to the 2022 duty set because it is the more demanding, and carry commencement as an open item with a named owner. The substantive change to know: **risk assessment at intervals not exceeding five years, not three** [3].

The 2022 duty set, in clocks [3]: Form A notification **90 days** before erection or change; a registration certificate **displayed** and valid **five years**; risk assessment to **SANS 1461** by an approved inspection authority **at intervals not exceeding five years**; a prevention policy, safety report and operating licence for **high hazard** establishments within 36 months; and incident reporting at **48 hours, 7 days and 6 months** with a near-miss register. All the regulations apply to **high** hazard establishments; **low** is exempt from regs 11 to 13, **medium** from regs 12 to 13.

The substance inventory determines everything else. Whether a site is low, medium or high hazard — and therefore whether the safety report and licence apply at all — is a function of on-site quantities of named substances, aggregated under the Annexure A notes including a **2% rule** [3]. So you need a **maximum-inventory calculation per substance per site** with **the aggregation and 2% rules as a stated, versioned computation** — not a spreadsheet, because the output sets the regulatory tier — and a **tier determination record** with its date, inputs and outcome, because a change in stockholding can change the tier and trigger a 14-day notification.

Compensation for occupational injuries — the 2026 amendment

The **Compensation for Occupational Injuries and Diseases Act 130 of 1993**, as amended by the **Amendment Act 10 of 2022** [60]. Per a 27 January 2026 account, **most provisions took effect on 23 January 2026** [10]. What changed [10]: a **Chapter XA inspectorate**; **administrative penalties** replacing criminal prosecution; **claim prescription extended from 12 months to three years**; **records kept five years**; **accidents reported within seven days**; and **section 89 contractor liability** for sub-contractors' employees where the sub-contractor fails to register or pay assessments.

That last item **converts a supplier-master data-quality problem into a direct financial exposure**: a manufacturer that cannot produce a valid letter of good standing per contractor per period has assumed its sub-contractors' liability, and the evidence is **a dated attribute on the supplier record with a renewal clock**, not a filed certificate.

Environment

The frame, and why conditions are the obligation

The **National Environmental Management Act 107 of 1998** is the framework statute, and a manufacturer's exposure is usually three things: the authorisation and its conditions; the **duty of care for contamination, which is open-ended in time and attaches to land**; and the inspector, with search, seizure and directive powers.

The record is an **environmental authorisation and conditions register**, one row per authorisation with **every condition as a separate trackable obligation** carrying an owner, a frequency, an evidence type and a due date. It usually does not exist, because an authorisation arrives as a file and the conditions inside it are where the duties live. Plus a **contaminated-land record with no retention end**, because the duty of care does not expire with a tenancy.

Air quality — and the deadline that is not final

The **Air Quality Act 39 of 2004** [68] and the listed activities with minimum emission standards published as **GN 893 in GG 37054, in operation from 22 November 2013**, amended by **GN 551 in GG 38863 effective 12 June 2015** [19]. New plant to new-plant standards from 1 April 2010; existing plant to existing-plant standards by 1 April 2015, and to new-plant standards by 1 April 2020 [19].

THE 1 APRIL 2020 DEADLINE IS NOT FINAL

The regime expressly allows the **National Air Quality Officer, with the licensing authority's concurrence, to grant postponements of up to five years each** [19]. So the architecture must model **the standard that applies to this plant right now** — a function of the plant's own postponement history, not of the published date. A dataset storing "the 2020 standard" as a constant is wrong for any plant with a postponement.

Monitoring is where this becomes a data problem rather than a reporting one [19]. **Continuous monitoring** gives daily averages, must yield **at least 80% valid hourly values**, and must be **audited by a SANAS-accredited laboratory at least every two years**; **periodic monitoring** gives hourly averages measured at least annually; and licence holders report annually with **any exceedance explained and a remediation plan**.

Two things to say. **An emissions compliance dataset is a historian problem masquerading as a reporting problem** — the 80% test, bad-quality intervals and the averaging rules are time-series concerns, and an estate that builds emissions reporting on a relational extract of hourly values cannot reproduce last year's numbers. And **the national reporting obligation under the 2015 atmospheric emission reporting regulations is a different dataset from licence-condition reporting** [26][19]: same emissions, two regimes, two formats.

Waste, manifests and extended producer responsibility

The **Waste Act 59 of 2008** [28][69], its list of waste management activities [69], and the **Waste Classification and Management Regulations, 2013, GN R.634 in GG 36784** with the landfill norms R635 and R636 in the same gazette [27].

THE WASTE MANIFEST IS A GENUINE GAP IN THIS CHAPTER

The regulation imposing the manifest obligation, its prescribed contents, and the registration and reporting obligations and frequency under the national waste information system could not be verified [27][28][69] — nor the classification standard R634 references, nor the Act's duty-of-care section. "A waste manifest is required" is exactly the kind of confident, loosely cited claim this chapter exists to avoid. **Close it from the gazette before you put a manifest field list in a specification.**

The architecture is statable independent of the citation: a **waste register** at the grain of stream, plant, classification and period, plus a **movement record** at consignment grain, producing **classification per stream reconciled with the chemical classification of the input materials** — two unrelated classifications of one substance is an inconsistency an inspector finds in one question; **quantities by stream by period**, reconcilable to production and inventory; and **consignment-level chain of custody**, which has the properties of the genealogy graph of chapter 9.

Extended producer responsibility binds *producers* in identified sectors with registration, scheme participation, fees and periodic reporting of volumes placed on the market [51]. **The definition of "producer", and therefore exactly who is bound — brand owner, importer, manufacturer, converter, or more than one — is UNVERIFIED**, along with the gazette numbers and reporting fields [51]. The operative text must be read, not paraphrased.

The data requirement **cuts across the product and the packaging in a way no other obligation does**: **mass by material type per product**, so the packaging bill of material must carry composition and mass per component **at a taxonomy matching the scheme's categories, not the plant's own**, plus **units placed on the market by the obligated entity**. So **the producer determination is an attribute of the item-and-customer relationship, not of the item** — and **a contract manufacturer and a brand owner can both believe the other reports a given tonnage.** Only a per-customer **obligated-party flag** prevents that, and it is almost never present.

Water — the constraint with no capital answer

The **National Water Act 36 of 1998** [48]. Water uses requiring authorisation are enumerated in **section 21** and authorised by schedule 1 permissible use, by a **general authorisation under section 39**, or by a **licence**; the revised general authorisation for section 21(c) and (i) uses was published as **Notice 4167 in GG 49833, dated 8 December 2023** [25].

WATER HAS NO GENERATION ANALOGUE, AND THAT IS THE ARCHITECTURAL POINT

Electricity scarcity was answerable with capital expenditure. **Water scarcity is not.** Abstraction is a **licensed volumetric cap** and effluent quality an **enforceable licence condition**, so growth beyond the licence takes years rather than a purchase order. And the **2026 National Water Amendment Bill proposes banning private water trading**, removing the mechanism by which a plant acquires an existing entitlement. A capacity plan that assumes water can be bought like diesel has a hole no architecture can fill.

The repository: an **authorisation inventory** per site with conditions as trackable obligations; metered **abstraction and discharge volumes** with the aggregate as a versioned computation; and **effluent quality results** at the grain of point, date, parameter, value, method and laboratory.

The cross-cutting observation, and it saves real money: **water, air, waste and greenhouse-gas reporting all have the same shape** — a metered or sampled quantity, a versioned computation, a limit with a dated basis, an exceedance event, and a conditions register attached to an authorisation. Four separate repositories is the usual outcome and the usual waste. **One environmental monitoring model with a medium dimension** is the right design.

Carbon tax, and the allowance structure you must not hard-code

The **Carbon Tax Act 15 of 2019**, in effect from 1 June 2019 [2]. **Phase 2 began on 1 January 2026, and the rate is R308 per tonne of carbon dioxide equivalent from that date, up from R236 — a 31% increase** [1], rising toward **R462 by 2030**, with emissions **above a mandatory carbon budget facing R640 per tonne** [2]. The **carbon fuel levy from 1 April 2026 is 19 cents a litre on petrol and 23 cents on diesel** [1].

CHECK THE RATE BEFORE YOU PRINT IT

The two research passes behind this book disagree. A February 2026 commentary gives R308 from 1 January 2026; a July 2026 commentary describes R236 as the then-current rate while agreeing that Phase 2 began on 1 January 2026. They cannot both be right, and a search to settle it was not available. Read the rate off the Carbon Tax Act as amended and the current SARS carbon tax guide, and write down the date you read it. The structural point — that the headline rate is not the liability, because the allowances do the work — holds on either figure.

QUOTE THE RATE AND THE ALLOWANCE STRUCTURE TOGETHER, OR QUOTE NEITHER

The headline rate is not the liability. Phase 1 allowances reached a combined maximum of 95% for process and fugitive emissions. For Phase 2 the two available sources **contradict each other**: a National Treasury discussion paper gives the basic tax-free allowance falling to 50% in 2026 and an offset allowance of **up to 25% combustion and 20% process and fugitive** as *proposals* [2], while a February 2026 commentary states the offset allowance **increased to 15% combustion and 10% process and fugitive** as enacted [1]. **The enacted Phase 2 allowance percentages are UNVERIFIED**, and **no allowance percentage should be hard-coded from a commentary.** Every allowance is a **dated, per-category, per-taxpayer parameter**, not a constant.

Two consequences worth designing for. **The trade-exposure computation is a sales-and-purchases data problem, not an emissions one**, and that join is where it breaks. And **offset credits are inventory-like assets with serial numbers, retirement events and double-counting risk**, so they need a register, not a certificate folder.

Behind the tax sits the inventory, and this is the architecturally important sentence: companies report greenhouse gas emissions under the mandatory national reporting regulations, and **those reported emissions are the tax base** [2]. The greenhouse-gas dataset is therefore **a tax computation** needing the controls of one — a defined scope, a versioned methodology, an auditable derivation from source measurements, and the ability to reproduce a prior year's submission exactly. The **Climate Change Act 22 of 2024** [30] adds carbon budgets; draft regulations were reported in a 14 August 2025 alert [31], unverified.

Product specifications and metrology

What a standard is, and what it is not

The **South African Bureau of Standards develops and publishes SANS standards** under the Standards Act 8 of 2008 [22]. **The National Regulator for Compulsory Specifications is the regulator for compulsory specifications**, under the **NRCS Act 5 of 2008**, in operation from 1 September 2008 [15]. Two bodies, two jobs, and the distinction decides your citation.

A SANS standard is voluntary by default — a technical document, copyright, bought, and a contractual rather than a legal requirement. It becomes compulsory only when an instrument makes it so: when the regulator declares a compulsory specification incorporating it, or when a regulation references one, as the pressure equipment regulations reference SANS 347. **The instrument, not the standard, is the citation.** So the claim a competent author will make and should not is **"SABS approval is required"**: the Bureau is a standards and certification body, not a regulator [15][22].

The record is a **standards and certification register** per product and plant: which standards are claimed, which are compulsory and under which instrument, which marks are held with scheme, number, scope, body and expiry. Management-system certifications belong here too and are **contractual, not statutory**, so **keep both in one register with a `basis` attribute** — one register *without* it is how a company comes to believe a scheme requirement is a legal one.

Compulsory specifications, letters of authority, and section 20

Under **section 13** [15] the Minister may, on the regulator's recommendation, declare by *Gazette* notice that a commodity affecting public safety, health or the environment must meet a specification; **a preliminary notice inviting comment comes first, and the final notice takes effect no less than two months after publication.** It may require a mark, labelling or a **letter of authority**, and **section 14(1)–(2)** prohibits supplying a non-compliant regulated product. **There is no general list of products needing a letter of authority**: the requirement arises from the individual compulsory specification, so **the operative artefact is the specification itself** [15][56].

SECTION 20 IS THE PROVISION MOST OFTEN MISSED, AND IT MAKES BATCH SIZE A RISK DECISION

On written request a supplier must provide samples or information at its own cost, and a test result is generally valid for the whole batch unless shown otherwise [15]. Read the second half again. **The entire evidential weight sits on the manufacturer's batch definition and batch records.** If a batch is large, one failing sample implicates all of it, and the only way to narrow the exposure is a batch-identity and genealogy design that can demonstrate a smaller homogeneous population. This is the clearest regulatory illustration in this book of **why batch size is an exposure decision, not a production convenience.**

The register is at the grain of **product or family, market, specification, version, effective-from, effective-to**, carrying the letter of authority with number, scope and expiry, the mark or labelling and the artwork version carrying it, the accredited test reports, and **the link to the engineering change record — because a design change can invalidate a letter of authority. A letter of authority is product-and-version specific**, a dated certification bound to an item revision, not a field on the item. One teaching case from the published specification list [14]: **the cement specification**, 2007 version effective until 13 June 2025 and 2024 version from 14 June 2025 — the effectivity problem of chapter 9 as a regulatory event.

Legal metrology — no fixed verification interval

The **Legal Metrology Act 9 of 2014**, commenced 1 August 2014 [5]. Instruments used for a **prescribed purpose** need **type approval** (s 22) and **initial and subsequent verification** (s 24). **The Act fixes no verification interval**: section 24 defers it to the technical regulations, and the interval is instrument-

class-specific [5]. **"Scales must be verified annually" is not what the Act says**, and the gazetted maxima for the instrument classes a manufacturer uses are unverified here [5][65]. The **net mass of prepacked products sold by mass must be indicated, subject to prescribed error limits** (s 36) [5].

> THE MOST CONSEQUENTIAL OPEN ITEM FOR A PACKAGING ARCHITECTURE
>
> **Whether South African prepackages run on an average-quantity basis or a minimum-quantity basis is UNVERIFIED**, along with the error limits under s 36(1)(d) [5]. It matters more than any other gap here for a filling line, because the two bases require **completely different datasets**: an average-quantity system needs a **statistical control record per production batch** — mean, standard deviation, counts of tolerable negative error — while a minimum-quantity system needs a **per-unit minimum**. You cannot design one and discover you needed the other.

Two linked repositories. A **measuring instrument register**: type approval, verification history with the authority and the mark, next-due date derived from the applicable technical regulation, and the **prescribed purpose it is used for** — the modelling point, because **"used for a prescribed purpose" is a *use* attribute, not an instrument attribute**: the same scale is in scope on a despatch bay and out of scope in a laboratory. **Calibration records are not verification records.** Then a **net-content control record** per batch per prepackage format: declared quantity, the filling-line measurement data, the control applied, the tare determination, and the disposition of out-of-limit product.

Sector-specific obligations

Food

The parent statute is the **Foodstuffs, Cosmetics and Disinfectants Act 54 of 1972**, with the Minister of Health making regulations and enforcement principally by **environmental health practitioners of local authorities** [22]. **Hygiene: R638 replaced R962 of 2012**, requiring a **Certificate of Acceptability per food premises** [22].

> TWO FOOD CLAIMS A 2026 AUTHOR WILL GET WRONG
>
> **One. "HACCP is mandatory in South Africa" is unsafe as written. GN R.908 in GG 25123 of 27 June 2003** exists — the hazard analysis and critical control point regulations [45] — and a 2026 guide describes it as effectively mandatory for medium- and high-risk food manufacturers [22]. But **which classes of manufacturer R908 actually compels could not be verified** [45].
>
> **Two. Food labelling is still R146 of 1 March 2010.** The draft replacement, **R3337, published for comment in GG 48460 on 21 April 2023, remains a draft and is not in force as at October 2026** [49]; a July 2026 legal review records the department as still reviewing it [50], while industry commentary describes finalisation as merely "expected" in 2026 [22]. **This is the likeliest error for a 2026 author.** So: **build the nutrient-value and nutrient-profile fields now, hold warning-label status as a derived indicator, and do not commit artwork until the final regulations are gazetted.**

What **R146** requires, in the form it lands on a data model [49]: the name of the foodstuff; the manufacturer, importer or seller name and address; the **ingredient list in descending order of mass**; storage conditions; **net contents in SI units**; country of origin; **batch identification**; and **date marking** supported by a shelf-life study. Allergen declaration covers **eight categories — egg, cow's milk, crustaceans and molluscs, fish, peanuts, soybeans, tree nuts and significant gluten-containing cereals — and sulphites are not among them** [49]. And **nutrition labelling is not compulsory on all prepackaged food**: it is triggered principally by a claim, with the supporting analysis **re-verified every three years** [49].

Requirement	Where it lives in the data model
The hazard plan: hazards, control points, critical limits, monitoring, corrective action	**A critical control point is an operation with a mandatory in-process characteristic and a defined out-of-limit disposition** — the most useful mapping in this chapter
Allergen declaration against the eight categories	An item-level governed allergen list **derived from the bill-of-material explosion over ingredient allergen attributes — derived, not typed in**
Composition and contaminant limits; grade and class under the agricultural product standards	Item specification characteristics with a **category-to-limit mapping**, plus **a grade axis on the item that behaves like a variant axis**

The composition instruments a food specification must carry include R588 of 2018 on metals, **R7091 of 2026 on mycotoxins**, R692 of 1997 on microbiological standards and R2527 of 1973 on additives [22] — **each one's gazette number, date and amendment status unverified.**

Pharmaceutical

Manufacture requires a **licence under the Medicines and Related Substances Act 101 of 1965** from the **South African Health Products Regulatory Authority** [53]; the enabling section is unverified.

South Africa has no Qualified Person. The regulator's **SAHPGL-INSP-02 version 9, *Guidelines to Good Manufacturing Practice for Medicines*, dated 17 November 2025**, states that the international inspection scheme's "**Responsible Person**" is equivalent to the **Responsible Pharmacist**, who is responsible for ensuring that released batches meet the marketing authorisation's terms, the practice guidelines and the legal requirements, **supported by a release checklist** [16].

> **DO NOT CITE THE NATIONAL REGULATOR FOR GMP DETAIL**
>
> Section 2.1 of the guideline states that South Africa **adopts and adapts the PIC/S GMP Guide** [16], so the substantive requirements are that guide's. **Batch-record and documentation retention, batch numbering, deviation and change control, product quality review frequency, data-integrity expectations and the computerised-systems annex are verified absent from SAHPGL-INSP-02** [16]. Citing the national guideline for them is a sourcing error a regulatory affairs manager will catch in the first review. **Cite the PIC/S guide and its annexes**; its current revision is unverified here.

Single Exit Price is a manufacturer-side price constraint, not a retail mechanic, and two things are routinely got wrong: **it is the only price at which a manufacturer or importer may sell a medicine to any buyer, and it does not apply to the State** [17]. **The second adjustment of 2026 was 2.88%, effective 1 October 2026**, applying to all medicines in the national database as at 30 September 2026, **including all related pack sizes** [17][59].

The consequence is sharp. **The selling price is a regulated, externally set, dated attribute of a registered medicine and its pack sizes, uniform across all private-sector buyers**, so it cannot live in a customer-specific pricing structure, and **the pack size is part of the priced identity.** The adjustment applies to the whole registered portfolio on a date — **a mass dated-price event, not a repricing exercise**. For devices the regulator licenses **establishments** and requires **ISO 13485 certification as a prerequisite** [33][70].

Automotive

Vehicle categories are regulated by compulsory specifications [14]; the homologation process and type-approval record set are unverified here. The **Automotive Production and Development Programme** is administered by the International Trade Administration Commission under gazetted regulations [32], and **its phase-two benefit instruments, eligibility rules, computations and record-retention obligations are UNVERIFIED — this chapter prints none of them from secondary commentary.**

What is verified, and what reorders a costing design, is the consequence: **if no Form C2 is obtained, the entire purchase price is treated as imported content.** Substantiating local content therefore needs, per vehicle or component, a **costed bill of material decomposed by origin** — the cost component split crossed with country-of-origin and the customs classification — **propagated through multiple supply tiers.** Origin is a property of a supplied component's own supply chain, not of the buying transaction, so **a plant whose costing does not preserve the component split through multi-level roll-up physically cannot substantiate a claim.** The other unusual obligation is **homologation traceability to the build**: a unit is approved as a *type*, so the as-built configuration must be comparable against the approved specification, with a change record flagging approval-affecting changes.

Chemicals

The operative instrument is the hazardous chemical agents regulations above, and its supplier limb is the part to read twice [4]: **reg 2** brings manufacturers, importers, suppliers and retailers into scope; **reg 14(a)–(c)** requires classification before supply using Tables 4 and 5 of Annexure 1, reviewed on composition change; **reg 14A** requires GHS-compliant safety data sheets, 16 prescribed headings, reviewed at least every five years; and **reg 14B(2)** requires GHS label elements. Whether SANS 10234 is incorporated by reference is unverified [4][39].

The architecture is a **substance and mixture register** — composition with ranges, GHS classification with basis and version, label elements, data sheet version with its five-year clock, and transport and segregation classes. The integration points matter more than the register. **Composition drives classification, and composition comes from the bill of material or the recipe.** So **a recipe change is a classification review trigger, a data sheet review trigger, a label change trigger and potentially a transport-classification change** — a four-way fan-out from one engineering change, most often missed because the four outputs are owned by four different functions. And the data sheet is a **versioned document bound to a substance version, with a record of who received which version when**: the duty is to supply a compliant sheet and the evidence is a distribution record, not a file on a server.

Customs — the system of record a manufacturer does not know it has

A manufacturer that imports raw materials or exports product holds **a statutory customs record that is a system of record in its own right**: the declarations made about classification, value and origin, and the registers supporting any duty relief claimed. They are maintained by a clearing agent, a logistics provider, a trade-compliance person with a spreadsheet, or nobody — while the liability sits with the importer or exporter. **Most manufacturers' architecture diagrams do not contain this system at all.**

The frame is the **Customs and Excise Act 91 of 1964**, and duty relief runs through the tariff schedules [7]: **Schedule 3** industrial rebates for home-use processing, **Schedule 4** inward and outward processing and temporary admission, **Schedule 5** specific drawbacks on goods used in exported products, with **section 75** as the enabling provision. **The Act's own retention period is UNVERIFIED** — commonly cited as five years, but it must come from the Act [21].

> **DRAWBACK IS A SEQUENCING TRAP, NOT A PAPERWORK ONE**
>
> Three facts from the revenue service's own guide, dated 20 March 2024 and stating that it has no binding legal effect [7]. Applications may **only be submitted after export**, with proof attached; they must be lodged **within six months of the date of export**; and **the export declaration must already show the correct customs procedure code and drawback item before the goods leave South Africa.** Read the third again: **getting the declaration wrong forfeits the claim irrecoverably**, because the goods have gone. It is a **data-capture-at-despatch** problem, so the procedure code and the drawback item must be carried on the export transaction by the system that raises it.

Four datasets follow, the first two item-master extensions treated as afterthoughts. **Tariff classification per item**, dated, with the basis and its author; a free-text code with no version and no basis is the norm and indefensible under audit. **Origin per item and per supply arrangement**, dated, substantiated by a supplier declaration with its own validity, and distinguishing **non-preferential origin** for marking from **preferential origin** for duty rates; the latter needs a **rule-of-origin computation** over the bill of material, making origin a **derived, bill-dependent, periodically recomputed attribute** — the same computation as the local-content one above, to be built once. **Declaration records**, with **the link to the purchase order, goods receipt, sales order and**

despatch each corresponds to — the part missing in practice, without which a post-clearance audit cannot be answered. And **duty-relief registers** reconciling inputs imported under relief, the production that consumed them and the products exported.

A rebate claim is, in data terms, an assertion that *these imported inputs became those exported outputs*, and only the production genealogy can substantiate it. **The customs duty-relief claim and the recall trace are the same query with different filters.**

Labour, and then POPIA

Bargaining councils and employment equity

A manufacturer in a sector covered by a bargaining council is bound, where the agreement is extended to non-parties, by a collective agreement prescribing wages, grades, hours, allowances and fund contributions, and the council's agents inspect compliance. The **Metal and Engineering Industries Bargaining Council** is the largest relevant to manufacturing [57][58], and **no wage figure from a secondary source belongs in a deliverable.**

The record is a **compliance dataset at employee-period grain**: the employee's **grade under the agreement's own grading structure** — not the company's job grading, and a separate attribute — the prescribed minimum for that grade and period, the actual rate paid, hours by category and fund contributions. Two consequences: **the agreement's grade is a dated attribute and the minimum a dated rate**, so an annual schedule change is a mass dated-rate event like the medicine-price event above; and **the employee-to-agreement-scope determination is derived**, where most payroll models hold a static flag that breaks when an employee changes role or site.

Employment equity's turnover test is gone. The **Employment Equity Act 55 of 1998** as amended, the **Employment Equity Regulations, 2025, effective 15 April 2025**, and the **Determination on Sectoral Numerical Targets**, also effective 15 April 2025 and covering 18 national economic sectors, set the position [6]. **From 1 January 2025 an employer with fewer than 50 employees is not a designated employer, regardless of turnover** [6]. **Plans run from 1 September 2025 to 31 August 2030** [6], and a compliance certificate for doing business with the State is **valid 12 months** [6]. Both instruments' gazette numbers and the manufacturing sector's own targets are unverified [6].

The dataset is at employee grain with a reporting-period dimension: occupational level and category, **race, gender and disability status as self-declared**, remuneration, and **the movement events — recruitment, promotion, training, termination — that the reports require.** EEA2 reports *flows*, not only stock, so **a snapshot model cannot produce it.** This is **special personal information** collected for a statutory purpose, and that basis does not extend to anything else; and **self-declaration means the field has a provenance and a date**, with "not disclosed" a real value rather than a null.

POPIA, where it actually bites a manufacturer

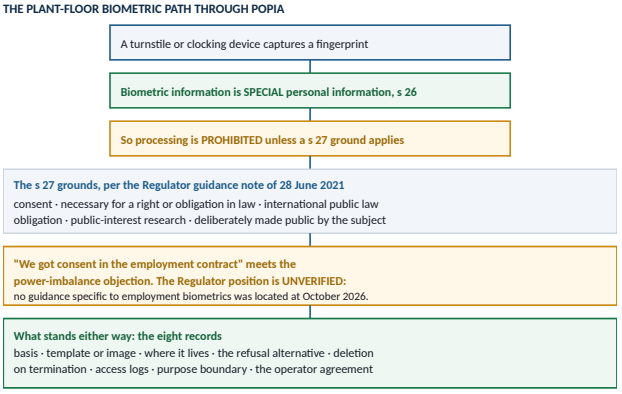


Figure 16.3 The plant-floor biometric path through POPIA. The route from a turnstile to a prohibition, and the eight records that stand whichever way the open questions resolve.

In a manufacturer the Protection of Personal Information Act 4 of 2013 bites hardest on employee and biometric data, not on customer marketing. The customer file is a few hundred corporate accounts with contact details. The employee file is thousands of people with identity numbers, bank details, medical surveillance and exposure records, race and disability declarations — and, on most plant floors, **fingerprints or facial templates used for time and attendance and access control.**

Section 26 prohibits the processing of special personal information [52], and **biometric information is in that category** [23][52]. **Section 27** sets the general exemptions, which the Information Regulator's **Guidance Note on the Processing of Special Personal Information, published 28 June 2021**, lists as consent, necessity for a right or obligation in law, an obligation of international public law, public-interest research, or information the subject deliberately made public [23].

> **"WE GOT CONSENT IN THE EMPLOYMENT CONTRACT" RUNS INTO THE POWER-IMBALANCE OBJECTION**
>
> And this chapter cannot tell you how that argument ends. **Whether the Regulator has issued guidance specific to biometric time-and-attendance in employment, its position on consent as a lawful basis in the employment relationship, whether prior authorisation under section 57 is engaged, and whether any enforcement finding on workplace biometrics has been published are all UNVERIFIED** [23]. What stands regardless is the architecture: **biometric time-and-attendance on a plant floor is a POPIA question, and most architects miss it entirely**, because it arrives as a turnstile procurement rather than as a data project.

The record	Why
The **lawful basis**, as a determination with a date and an author, naming the section 27 ground	"We have a clocking system" is not a basis
Whether a **template or an image** is stored, and **where it lives** — device, controller, cloud, which country	An image is a different exposure, and cross-border transfer is its own question
An **alternative for employees who refuse**, and **deletion on termination** with evidence	Without an alternative, consent cannot be freely given; templates outliving employment is the commonest finding
Access control and logging on the store, and the **purpose limitation boundary**	Access control reused for productivity measurement is a new purpose
The **operator agreement** with the vendor	The vendor is an operator; the manufacturer remains responsible party

Then the point that reorders the scoping exercise. Everything else POPIA requires attaches to the same employee file: exposure and surveillance records are health data about identified people, and equity declarations are race and disability data. **So the longest-retained special personal information a manufacturer holds is its occupational health file** — 30 years of chemical exposure data and 40 years of audiometry — in a system nobody thinks of as a privacy system. **A POPIA scoping exercise that starts with the customer database has started in the wrong place.**

Financial reporting as a data-model constraint

Under **Companies Act 71 of 2008, section 24**, company records are retained **seven years**, or longer where another regulation specifies, including **annual financial statements and accounting records** [21]. Then the accounting standard, because **IAS 2 *Inventories* bounds the costing design** and most architects do not know it does [18].

Para	What it requires	What it dictates in the data model
12, 13	Fixed overheads allocated **based on normal capacity**; the per-unit allocation **not increased** for low output or idle plant; **unallocated overheads expensed in the period**	**Normal capacity, actual activity and the absorption residual become mandatory, separately stored, dated facts.** A design storing only absorbed overhead cannot demonstrate compliance
21	**Standard cost** may be used **if the results approximate cost** and standards are **regularly reviewed**	**Standard costing is expressly permitted.** The version history and the variance analysis are the *evidence* the condition is met
25, 27	Costs assigned using **FIFO or weighted average**, applied consistently	**LIFO is not among the permitted formulas** — the largest costing divergence with United States practice, and both South Africa and Canada use IFRS

Two corrections out loud. **"Standard costing is not IFRS-compliant" is wrong** — paragraph 21 expressly permits it [18]. And **the normal-capacity rule is not an accounting nicety**: it is why the model must hold planned capacity, actual activity and the absorption residual separately [18].

B-BBEE and the supplier master

The **Broad-Based Black Economic Empowerment Act 53 of 2003** as amended, and the Codes of Good Practice under it. **As at February 2026 amendments to the Codes were still in draft** [36], and **whether they have been gazetted in final form, the current scorecard weightings and the enterprise-size thresholds are UNVERIFIED** [36] — the scorecard point totals in the available source are internally inconsistent, so none is printed here.

The architectural content does not depend on the open items. A manufacturer's own scorecard depends on data about *other* companies, so **the supplier master becomes a measurement instrument**: per supplier, dated — status level, the verification basis, **the expiry date as a structured field**, ownership percentages and the derived procurement recognition level; and per spend transaction, **the supplier's status as at the transaction date, not as at today.**

That is why the status must be a **dated series rather than a current value**: a supplier whose certificate lapsed in March contributes differently before and after, so **a supplier record with a single status column cannot produce a defensible scorecard.** The pattern is identical to the supplier qualification record of chapter 9.

Who inspects, and what they ask for

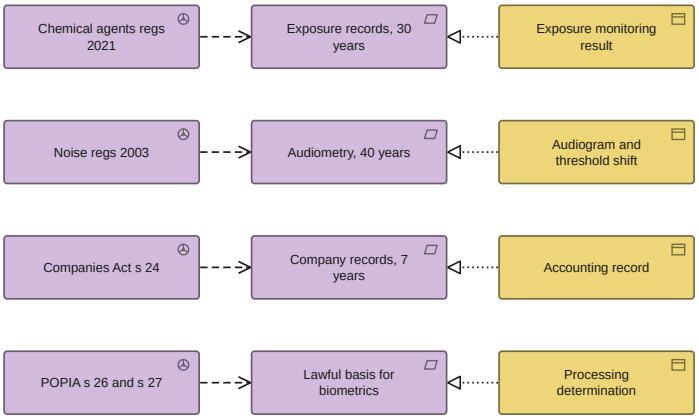


Figure 16.4 The obligation-to-evidence chain, four worked examples. Instrument, then the requirement it generates, then the object that realises it. Every row of the obligations register is this shape.

This is the most useful table in the volume, split into three only so that it fits a page. **Columns three and four are what an architect delivers.**

Safety, health and compensation

Body	What it asks to see	Where the evidence lives
Employment and Labour (health and safety)	Legal appointments; Annexure 1 records with seven-day investigations; two-year risk assessments; air and noise monitoring by approved authorities; pressure and lifting registers	Appointments register; incident repository; exposure repository at 30 and 40 years; equipment register
Employment and Labour (major hazard)	Substance inventory with data sheets; Form A; the displayed registration certificate; the SANS 1461 risk assessment; safety report and licence	Substance inventory with the Annexure A aggregation computation; tier determination record
Mineral Resources inspector	Mandatory codes of practice; risk assessments; record of hazards; medical surveillance and exit certificates	A different record set — hence **a jurisdiction attribute on the site**
Compensation Fund	Return of earnings; letter of good standing; accident register with seven-day evidence; sub-contractors' registration	Compensation repository; payroll; **contractor credential model with expiry**

Environment, product and sector

Body	What it asks to see	Where the evidence lives
Environmental management inspector	The authorisation and compliance with **every condition**; incident and contamination records	Authorisation and conditions register, per-condition owner and evidence
Air quality officer	The applicable standard with any postponement; **data availability against the 80% test**; the two-yearly accredited audit; the annual report	Emissions repository built on the historian, not a relational extract
Environment department and revenue service	Activity data with source measurements; emission factors with versions; retired offsets with serials	Inventory with calculation versions; tax computation record; offset register
Waste authority	Classification per stream with standard and date; quantities by period; consignment chain of custody	Waste register and consignment movement graph
Water department	Which section 21 uses are exercised; abstraction and discharge volumes; effluent quality with method and laboratory	Water authorisation and monitoring repository
Compulsory specifications regulator	The specification and version; letter of authority with scope and expiry; **samples at the supplier's cost, the result valid for the whole batch**	Product regulatory register bound to item **revision**; batch identity and genealogy
Legal metrology division	Type approval and verification history with marks; net-content control against the error limits	Instrument register with a **dated instrument-to-purpose assignment**
Local environmental health and Health	Certificate of Acceptability; control-point monitoring; labels against R146; recall capability	Control point as operation plus characteristic; item and label register
Agriculture and standards assignees	Class and grade; container markings; inspection records	**Grade axis on the item**; pack marking data
Health products regulator	Manufacturing licence; master and executed batch records; deviations and change control; out-of-specification results; the release decision	Versioned specifications, append-only results, copied order versions, full genealogy

Commercial, people and governance

Body	What it asks to see	Where the evidence lives
Revenue service (customs)	Tariff classification with its basis; origin with supplier declarations; declarations linked to the commercial documents; drawback within six months	Item-master trade extensions; declaration repository; duty-relief reconciliation
Revenue service (tax and VAT)	Accounting records; inventory valuation with its cost formula	Financial system plus inventory valuation record
Bargaining council agent	Grade per employee under the agreement's structure; rates against the gazetted schedule; fund contributions	Compliance dataset with **dated grades and dated minima**
Employment and Labour (equity)	EEA2 and EEA4; the 2025 to 2030 plan with targets by level; progress against sectoral targets	Equity dataset **with movement events, not a snapshot**
Sector training authority	Workplace skills plan; annual training report	Training record plus plan version history
Information Regulator	The lawful basis for biometric processing; the processing register; operator agreements; deletion evidence; the refusal alternative	Privacy register plus occupational health, time-and-attendance and access-control systems
Verification agency and the commission	Supplier status levels **as at each transaction date**, with expiries; spend classification with exclusions	**Dated supplier credential model** plus spend classification
Certification bodies (contractual, not statutory)	Documented information, internal audits, management review, corrective actions, calibration	The same repositories, with a `basis = contractual` attribute

The retention clocks, and why a single corporate period is always wrong

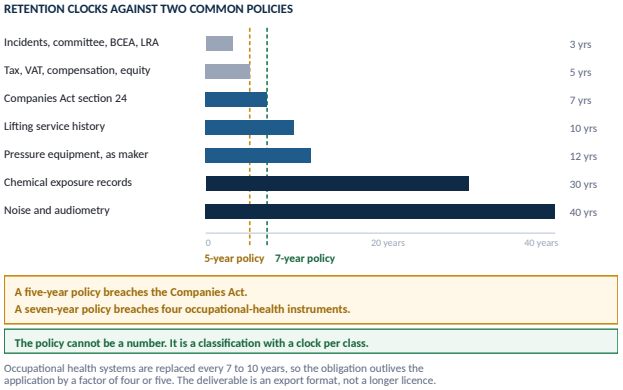


Figure 16.5 The retention clocks against two common policies. Neither line works. A five-year policy breaches the Companies Act; a seven-year policy breaches four occupational-health instruments.

Record	Period	Instrument
Noise exposure assessments and monitoring; audiometric and medical surveillance records including baseline audiograms	**40 years**	Noise regulations, reg 11(e)–(f) [12]
Chemical exposure assessments and air monitoring	**Minimum 30 years**	Hazardous chemical agents, reg 9(e) [4][21]
Pressure equipment manufacturing records, held by the manufacturer	**Minimum 12 years**	Pressure equipment, reg 14(3) [11][21]
Lifting equipment full service history, owner or lessor	**At least 10 years**	Driven machinery, reg 18(7)(c) [13]
Company records; annual financial statements and accounting records	**Seven years**, longer if another regulation specifies	Companies Act s 24 [21]
Compensation records; equity records (from expiry of the plan); tax and VAT records (from submission, extended while under audit)	**Five years**	COIDA [10]; EE Act s 26; Tax Administration Act ss 29(3)(a), 32; VAT Act s 55(1)(a) [21]
Annexure 1 incident records; committee records; control examination records; BCEA and LRA employment records	**Three years**	GAR regs 9(1), 5(1); chemical agents reg 9(g); BCEA ss 31, 29(4); LRA s 205 [20][4][21]
Medical surveillance (chemical agents); customs and excise records; skills levy records	**UNVERIFIED** — reg 9(a) covers the first with no period stated; customs commonly cited as five years	[4][21]

Now state it hard. **A single corporate retention period is always wrong. A five-year policy breaches the Companies Act; a seven-year policy breaches four occupational-health instruments.** No number satisfies both, so the policy is **a classification with a clock per class**, derived from the obligations register rather than a records-management template. And the two longest clocks are where the cost sits — **thirty years for chemical exposure data and forty for noise and audiometry, in systems replaced every seven to ten years** — so **a retention design that assumes the system will still exist is not a design.**

Last word, and the same warning as the first

This has been architecture material, not legal advice. It tells you what columns, entities, clocks and evidence trails a repository needs, because an inspector will ask for them. It does not tell you whether an obligation applies to your client or whether anybody complies: a South African admitted attorney and the relevant regulator settle those questions.

Three things to carry out. **The frame is always what the record has to show**, never whether you comply — the reframing that turns a compliance conversation into an architecture conversation. **The open items are stated as open** — the major hazard commencement notice, the waste manifest provision, the definition of a producer, the Phase 2 carbon allowances, the prepackage basis, the scoping of the hazard-analysis regulations and the customs retention period among them — each named with what would settle it, because **an unverified item written as a confident sentence is worse than a gap.** And the most reusable deliverable is the **regulatory obligations register**. Build it once and the inspector's visit becomes a query.

THE CHAPTER IN TEN LINES

- This is architecture material, not legal advice, and the frame throughout is what the record has to show rather than whether anybody complies.
- A South African regulation number is not an identifier: the same source list carries R1008 for two different instruments, so the identifier is number plus year plus subject.
- The Major Hazard Installation Regulations 2022 repeal the 2001 regulations but commencement turns on a separate Ministerial notice that could not be confirmed, so both positions are unsafe and the uncertainty belongs in the document.
- The 1995 hazardous chemical substances regime is superseded by the 2021 Hazardous Chemical Agents Regulations: risk assessment every two years, air monitoring every 24 months, safety data sheets reviewed every five years, and exposure records kept a minimum of 30 years.
- Noise records run 40 years, and audiometry is state-dependent: annual for three years, then up to two-yearly absent a referral threshold shift, and six-monthly at or above 105 dBA.
- Pressure vessels are not inspected annually: internal and external inspection plus hydraulic test at 1.25 times design pressure at intervals not exceeding 36 months, with the 12-month item being external inspection of fire-tube steam generators only.
- Lifting equipment is three clocks: the machine every 12 months by a registered inspector, components every 6 months, and tackle every 3 months signed per identified item.
- COIDA's Amendment Act took effect 23 January 2026, extending claim prescription from 12 months to three years and introducing section 89 contractor liability for sub-contractors' employees.
- The 1 April 2020 minimum-emission-standards deadline is not final, because the National Air Quality Officer may grant postponements of up to five years each, and continuous monitoring must yield at least 80% valid hourly values.
- Water has no capital answer: abstraction is a licensed volumetric cap and effluent quality an enforceable licence condition, and the 2026 amendment bill proposes banning private water trading.
- Carbon tax is R308 a tonne from 1 January 2026, but the enacted Phase 2 allowance percentages could not be verified, so quote the rate and the allowance structure together or neither.
- SABS publishes standards and the NRCS regulates compulsory specifications: a SANS standard is voluntary unless an instrument makes it compulsory, and the instrument is the citation.
- NRCS Act section 20 makes batch size an exposure decision, because a test result is generally valid for the whole batch unless shown otherwise.
- The Legal Metrology Act fixes no verification interval, and whether South African prepackages run on an average-quantity or minimum-quantity basis could not be verified, which is the most consequential open item for a filling line.
- Food labelling is still R146 of 1 March 2010 with eight allergen categories and sulphites not among them; the draft R3337 of 21 April 2023 remains a draft as at October 2026.
- South Africa has no Qualified Person: the batch-release role is the Responsible Pharmacist, and GMP detail must be cited from the PIC/S guide rather than from the national guideline.
- Single Exit Price rose 2.88% effective 1 October 2026, does not apply to the State, and is the only price at which a manufacturer or importer may sell to any buyer.
- Customs drawback is a sequencing trap: applications within six months of export, only after export, and with the correct procedure code and drawback item already on the export declaration before the goods leave.
- Employment equity's turnover test is gone, and POPIA bites a manufacturer hardest on employee and biometric data, with biometric information prohibited under section 26 unless a section 27 ground applies.
- The longest-retained special personal information a manufacturer holds is its occupational health file, not anything in a customer database.
- IAS 2 expressly permits standard cost at paragraph 21 and lists only FIFO and weighted average at paragraph 25, so LIFO is not an option under IFRS.
- A single corporate retention period is always wrong: a five-year policy breaches the Companies Act and a seven-year policy breaches four occupational-health instruments.

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CHAPTER 17

Modelling manufacturing in ArchiMate

You can already model. What you cannot yet do is model a factory, because several of manufacturing's most important objects have no honest home in the notation — and the professional answer is to say so, draw the best approximation, and name the artefact that carries the rest.

The ground rules, stated once

ArchiMate 4 gives you forty-two elements, eleven relationships and two kinds of junction. That is the whole vocabulary. Nine of the elements are shared across domains — Role, Collaboration, Path, Grouping, Location, Process, Function, Service and Event. Four are business, three are application, ten are technology, nine are motivation, four are strategy and three are implementation. The relationships fall into four families: structural (Composition, Aggregation, Assignment, Realization), dependency (Serving, Access, Influence, Association), dynamic (Triggering, Flow) and Specialization on its own.

Three corrections before anything else, because each of them will be made in front of you by somebody who knows the notation better than the person presenting.

The element is `Role`. There is no business-prefixed variant of it in ArchiMate 4. Role is one of the nine shared elements, which is exactly why it can be assigned to a business process, an application function or a technology service without changing name. An architect who puts the word "business" in front of it on a slide has told the room they learned the notation from a 2016 tutorial, and somebody will say so.

There is no Constraint element. If you need to say that a BOM position is only valid between two dates, that a conduit must meet a target security level, or that a batch may not ship before quality release, you express it as a **Requirement**. Requirement is a motivation element, it can be realized by a process or a component, and it can be associated with the thing it bears on. That is the whole mechanism. Reaching for a Constraint element is the second-commonest tell.

Never reproduce the specification. Do not paste the element table, the relationship matrix or the metamodel figure into a client deliverable. Describe and draw. Every figure in this chapter is drawn from scratch, and so should yours be. The specification is licensed; your model is yours.

One more rule, which is not about the notation but about the room. In a manufacturing engagement your audience is a production manager, a maintenance planner, a control systems engineer and a financial controller. None of them has read the specification and all of them can tell when a picture is wrong about their plant. **Correctness in the notation is cheap; correctness about the plant is the whole job.**

> The notation is a means of writing down what you found. If the model is elegant and the plant manager does not recognise the plant in it, you have produced a diagram, not an architecture.

Where manufacturing does not fit, and what to do about it

This is the part of the chapter worth reading. Manufacturing's objects were defined by control engineers, process chemists, industrial engineers and cost accountants over seventy years, in four separate traditions, before anybody wrote an enterprise modelling notation. Several of them have no honest home in ArchiMate. The professional answer is to say so, draw the best available approximation, and name the artefact that carries the real detail — not to force the object into an element that distorts it.

The object	Why it does not fit	What carries it
Effectivity on a BOM position	Dated, four regimes, not interchangeable	Logical data model; change record
Scrap factor, yield	Four numbers in three places; arithmetic	BOM line and routing operation attributes
Genealogy	A graph of instances, not of types	Traceability model; recall simulation
Control recipe	One batch, not a kind of thing	Batch production record
The historian tag	Tens of thousands; not a unit of information	Tag register; model the namespace
OEE, yield, OTIF	No metric element; Value is not a number	Measure definition register
A constraint	No Constraint element in ArchiMate 4	Requirement, in the motivation domain
Zone trust relationship	Grouping has no boundary semantics	The 62443 zone and conduit diagram

Figure 17.1 The manufacturing objects with no honest home in ArchiMate, and the artefact that carries each one instead.

The plant hierarchy

Start with the thing every engagement starts with: where is this, and what is in it.

Location is the geographic element. Use it for the site — Paarl, Gqeberha, Ekurhuleni — and for nothing else. A Location aggregates the things that are there; it does not compose them, because the same equipment can be moved and the site persists.

Facility is the technology element for physical plant: a building, a plant, a production line as a physical entity, a warehouse, a tank farm. Facility composes Facility, which is how you get depth.

Equipment is for an identified machine that does work: a filler, a press, a pasteuriser, a coating line. The test that works in practice is whether a maintenance planner would raise a work order against it. If yes, it is Equipment. If it is a space that contains machines, it is a Facility.

Now the hard part. **A plant, an area, a line and a work centre are not all the same kind of thing.** A plant is a valuation and planning unit in the ERP and usually also a physical site. An area is a physical division of a site. A line is a physical arrangement of equipment. A work centre is a *scheduling and costing construct* that may contain three machines, may be one machine shared by two work centres, or may be a group of people with no machine at all. The first three are Facility or Equipment. The fourth is not physical at all, and modelling it as Equipment is the error that makes capacity unplannable and downtime unattributable.

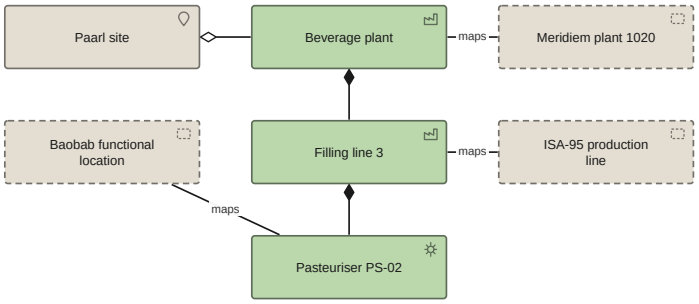


Figure 17.2 The plant hierarchy, drawn honestly: one physical tree in Location, Facility and Equipment, with the ISA-95 classification and the ERP plant as Groupings mapped onto it rather than stacked above it.

Then the mismatch that will consume a month of your engagement. **The ISA-95 equipment model and the ERP's plant hierarchy do not agree, and reconciling them is a real project.** Two facts about ISA-95's model are routinely misreported and both matter.

First, it is not a seven-level stack ending in a control module. The equipment-element level enumeration, as published in the OPC Foundation's ISA-95 companion specification, carries **fifteen values** [3]. Below enterprise, site and area, the branches — process cell and unit; production line and work cell; production unit; storage zone and storage unit — are **alternatives selected by process type, not levels stacked on each other** [3]. A batch plant uses process cell and unit; a discrete plant uses production line and work cell; a continuous plant uses production unit; a warehouse uses storage zone and storage unit. The enumeration also carries `WORKCENTER` and `WORKUNIT` as the *generic* terms for the grouping typically scheduled by Level 3 or Level 4 [3], which is precisely why "work centre" means related but non-identical things in ISA-95 and in ERP vocabulary.

Second, **equipment module and control module are ISA-88 concepts referenced by ISA-95, not ISA-95's own** [3]. They come from the batch-control lineage, where an equipment module is a functional group doing a small number of minor processing activities and a control module is the lowest grouping of sensors and actuators driven as one entity. Saying "ISA-95 goes down to control module" is wrong in attribution, and a control systems engineer will say so.

The ERP's structure, meanwhile, is driven by accounting and legal boundaries: company code, plant, storage location, and separately controlling area, profit centre, cost centre and work centre. One ERP plant can be several ISA-95 areas. Two ERP plants can sit on one physical site, which turns a forklift trip between buildings into a cross-plant transfer with intercompany pricing. A storage location has no ISA-95 equivalent at all. And ISA-95's depth exceeds the ERP's, because control modules exist in the controller and nowhere in the business system.

What to do in the model. Draw both trees, as two typed hierarchies, with an explicit dated many-to-many mapping between their nodes. Do not make one the parent of the other. The deliverable is the mapping, and the repository artefact is a **plant-structure registry** that every other system resolves against. It is small, boring and high-value, and almost nobody builds it until the third failed MES integration. Draw the ISA-95 tree with Grouping elements associated to the Facility and Equipment they correspond to; that is the honest representation, because the ISA-95 node is a *classification* of the physical thing, not the physical thing.

> There is a third tree. The asset and maintenance hierarchy — functional location, equipment, assembly, component — is deeper than the production hierarchy and built to answer different questions, and the fixed-asset register is a fourth structure again, with capitalisation units driven by depreciation policy rather than by pipework. Four trees over the same steel. A model that shows one and calls it the plant hierarchy is wrong three ways.

The bill of material

A BOM is a structure over objects. Model the objects as **Business Object** where you are talking about what the business asserts, and **Data Object** where you are talking about what a named system holds. Use **Aggregation** from the structure to its positions and **Association** from a position to the component item. That much works and reads well.

Now the honest part. **ArchiMate has no native way to express effectivity, revision or a scrap factor on a BOM line.** These are not decorations. They are the three attributes that make a BOM a set of dated, qualified, versioned assertions rather than a tree, and they are where the money is.

- **Effectivity** is the rule that says when, or to what, a structural assertion applies. There are four regimes — by date, by serial number, by lot, by configured unit — and they are not interchangeable. Serial effectivity exists because units are not built in serial order, so a date-based cut-in is wrong for aerospace and heavy equipment. Date effectivity is a half-open interval, closed at the start and open at the end, or every cut-over produces a one-day gap or a one-day double count.

- **Revision** is two things that must be kept apart: the design state of a part, and the version of a structure. A BOM can change with no part changing. A part can change with no structure changing.
- **The scrap factor** is four numbers, not one: component scrap on the line, assembly scrap on the parent, operation scrap on the routing step, and yield. Only one of the four requires the BOM and the routing to be read together.

The best available approximation in ArchiMate is a **Requirement** associated with the position it governs — "effective 1 March to 31 August 2026", "component scrap 3 per cent at this position" — and a **Data Object** realizing the position to show which system holds the authoritative line. Do that for the two or three positions that illustrate the design question, and no more. **A BOM view with forty positions on it is a data extract with boxes round it.**

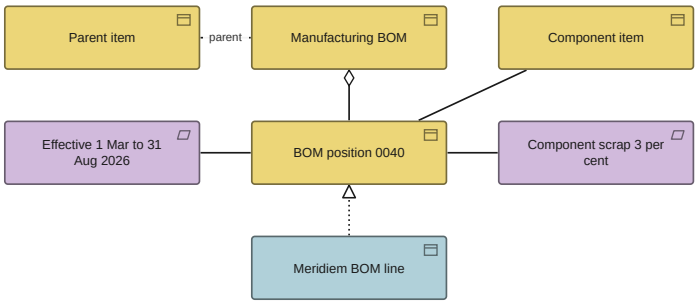


Figure 17.3 A bill of material in ArchiMate, with the two things the notation cannot express — effectivity and a scrap factor — carried as Requirements on the position they govern.

Say where the real model lives. The authoritative BOM structure lives in the ERP and the PLM, and the *design* of it lives in a logical data model with an entity for the header, an entity for the position, and about thirty attributes on the position. ArchiMate's job is to show which system asserts the structure, which caches it, who may change it, and what authorises the change. It is not to carry the attributes. Put the attribute list in a data model artefact, reference it from the view, and keep the view readable.

> And while we are here: a 10 per cent scrap factor means multiplying by 1.111, not by 1.10. To get 100 good units at 10 per cent scrap you must start $100 \div (1 - 0.10) = 111.1$. Compounded at 3 per cent over five levels it is about 15.9 per cent, not 15 per cent. If the standard cost is built on the naive explosion and the plant consumes the compounded quantity, the result is a permanent, structural, unexplainable material usage variance that will be investigated every month for years.

The routing and the work centre

A routing is the process model to the BOM's product model. Model an operation as a **Process** — it has a defined start, a defined end and a transformation in the middle. Model the capability the operation belongs to as a **Function** if you need the grouping: "filling" is a Function, "operation 0030, fill to 500 ml" is a Process within it. Use Composition from the Function to the Process when you need the containment and leave it out when you do not.

Assign a Role to the operation. The Role is the job as the industry names it — filling line operator, setter, shift supervisor — and the Assignment relationship is what says that person does that work. Assign the **Equipment** to the same Process, because in manufacturing the machine is an active structure element doing the work, not a passive resource the work happens to.

The work centre is the awkward one. It is not Equipment, because it is a scheduling and costing construct. It is not a Facility, because it may be a bench or a crew. **Resource** — a strategy element — is the most defensible choice, with Aggregation to the Equipment it contains, and you should say in the view's note that this is a modelling decision and not a reading of the standard. If your repository and your audience would prefer a Grouping, use a Grouping. Either way, declare it once in your modelling conventions and do not change it between views.

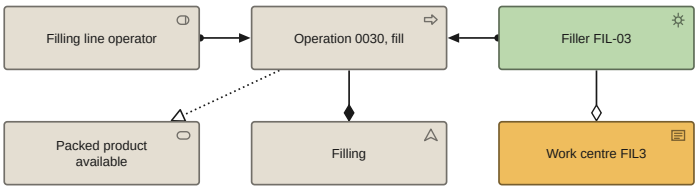


Figure 17.4 A routing operation with its Role and its Equipment both assigned to it, the Function that groups it, and the work centre modelled as a Resource — a declared modelling decision, not a reading of the standard.

Batch, lot and genealogy

Here the answer is that it does not work at all, and saying so plainly is the value.

A **batch** is a quantity of material produced under uniform conditions and treated as homogeneous for quality purposes. A **serial number** is the identity of one physical unit. **Genealogy** is the recorded relationship between material consumed and material produced, at batch or serial grain, through every production and movement event.

Genealogy is a directed graph of *instances*. One resin lot feeds four hundred production orders; one finished batch consumed fourteen component lots; a silo blends incoming lots continuously so an output lot's parentage is a set with proportions; a mother lot is split into daughter lots on repack; rework reintroduces output as input and creates a cycle unless each pass is a distinct node. ArchiMate models **types**, not instances. There is no instance layer in the notation, and there should not be — a genealogy view with four hundred nodes on it is not a view.

So do not draw it. The correct move in an ArchiMate view is to name the *type* — a Business Object called "material lot", a Business Object called "genealogy link", with an Association between them and a note — and point at the artefact that holds the real structure: the traceability data model, and behind it the recall simulation. Then make the architectural statement that actually matters, which is that **the graph exists in no single system**. The MES holds the production links, the WMS the movement links, the ERP the procurement and sales links, and the subcontractor holds the links across the fence. That statement belongs in an application co-operation view, drawn as four components each accessing a part of one Data Object, and it is far more useful than any attempt to draw the lots.

> Where-used tells you which products *could* contain the contaminated lot. Only genealogy tells you which *do*. That is the difference between recalling a product line and recalling four pallets, and it is the single most valuable sentence an architect can bring to a traceability workshop.

The tag

A historian tag is a named, typed, continuously sampled signal with its own master data: engineering unit, data type, source address, scan rate, deadband and compression settings, archival policy, valid range, alarm limits, asset binding, quality handling. A mid-sized plant has tens of thousands of them. A large process plant has hundreds of thousands.

A tag is not a Data Object in any useful sense. A Data Object represents a structured unit of information that an application component acts on; a tag is one column of one signal, and the thing applications act on is the series, the aggregate or the asset model above it. More practically: a view with tags in it is a view nobody can read, and a view with a representative sample of tags in it is a lie about scale.

The rule: model the tag namespace, not the tags. Draw one **Artifact** called `PAARL.FIL3.*` — the namespace, with its convention stated — deployed on the historian Node, and one Data Object for each *decision-grain aggregate* that the analytical layer actually publishes: downtime event, shift production summary, batch parameter profile. Those are the objects the business uses and the objects a report depends on. The tags behind them belong in the tag register, which is a spreadsheet or a historian export, not an architecture model.

Two things to say in the note on that view, because both will be news to an IT audience. First, a reading is not a number; it is a timestamp, a value and a **quality** flag, and analytics that ignore quality produce confidently wrong averages, because a frozen sensor reads a plausible constant. Second, the archive does not contain what the instrument measured. Historians apply exception reporting at the interface and swinging-door compression at the archive; a status tag over forty-eight hours at one-second scans is 172,800 events uncompressed and can be fifteen events with minimal compression [12]. Every query against a time grid is therefore interpolating, and interpolation is a modelling choice. **Two reports of the same number can both be right and differ.** Put the exception and compression settings, the step flag and the required aggregation method into the model as a Requirement on the reporting component, and put them under change control.

The recipe

ISA-88 gives a recipe hierarchy — general, site, master, control — and a procedural hierarchy — procedure, unit procedure, operation, phase [5]. The current United States edition of Part 1 is **ANSI/ISA-88.00.01-2010**, not the 1995 original; IEC 61512-1 is 1997 [4]. The dates of Parts 2 to 5 should be checked at the time of writing rather than quoted from memory, because the best public summary of the part landscape is a 2006 document with stale status columns.

The mapping onto ArchiMate's behaviour elements works better than you would expect, with one gap.

ISA-88 concept	ArchiMate element	Why
General or site recipe	Business Object	Equipment-independent product definition the business asserts
Master recipe	Business Object, realized by a Data Object	Bound to a process cell, version-controlled, held in a named system
Procedure, unit procedure	Process	Behaviour with a start, an end and a transformation
Operation	Process	A major sequence within one unit
Phase	Process realized by an Artifact	The phase is the unit of code — it is written in the controller
Control recipe	Nothing	It is an *instance*, created for one batch

The gap is the control recipe, and it is the same gap as genealogy: the instance layer. A control recipe is a copy of the master recipe for one specific batch with the actual formula values, the actual units allocated and the actual batch identifier, and with its execution record it becomes the batch production record. It is not drawable and it is also, in a regulated plant, the most important record in the building. Name it as a Business Object type, show the Process that creates it, and point at the batch record specification.

The modelling point that earns its keep here: **in a process plant the structure and the process conditions are inseparable.** A master recipe carries temperature, hold time, addition rate and mixing sequence as part of the specification. A BOM with a routing beside it cannot express "add B over twenty minutes while holding 65 °C". So a discrete pattern that versions the BOM and the routing separately produces invalid combinations the moment it is applied to a batch plant, and an ArchiMate view that shows a Business Object "BOM" and a Business Object "routing" side by side has already made that error visible — or hidden it, depending on whether you label the recipe as one versioned object.

Measures and key performance indicators

There is no metric element in ArchiMate, and Value is not a number. Value is a motivation element representing worth to a stakeholder. It does not hold 68.4 per cent. Neither does Goal, Outcome or Driver, though all three are useful for saying what the business is trying to achieve and why.

So OEE, yield, first-pass yield, rolled throughput yield, on-time-in-full and energy per tonne have no home in the notation. That is not a defect in ArchiMate; it is a category difference. A measure is a definition plus a formula plus a time scope plus a denominator convention plus a data lineage plus an owner, and none of those is an architecture element.

The measure definition register is a separate artefact and it is a first-class deliverable. State that in the model by drawing a **Deliverable** called "measure definition register" realized by the work package that produces it, and a Requirement on every reporting component that says the measure version must be recorded on the number. Then put the register itself in a table.

Why this matters more in manufacturing than anywhere else:

- **There is no OEE standard.** The research behind this book found at least seven incompatible availability denominators in use: all calendar time; rostered shift time; shift time less planned maintenance; shift time less maintenance and less breaks; shift time less no-demand periods; shift time less external causes; and the variations introduced by where the micro-stop threshold sits [13][14].
- **The micro-stop threshold shifts loss between availability and performance without changing OEE** [14]. So OEE can be comparable between two plants where its components are not — a genuinely useful and widely missed point.
- **OEE is documented as unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level** [14]. A line's OEE is not the mean of its stations' OEEs. Compute on the constraint, or on the line as one resource.
- **ISO 22400-2's middle term is Effectiveness, not Performance** [8]. The widely used Nakajima-derived formulation calls it the performance rate. An architect writing "performance rate" while citing the ISO standard is implementing one thing and referencing another. The published edition of Part 2 is **2014**; a second edition was still at draft stage in October 2026 [6][7].
- **The 85 per cent benchmark is an invention**, and the site that publishes it says so, calling it a convenient, compelling and completely artificial benchmark [13]. Never put it in a client deliverable as a target.

> When you are asked to build a group manufacturing dashboard, the first deliverable is not a dashboard. It is a measure definition register with an owner per measure, a formula, a denominator convention and a declared comparability scope. Ship the dashboard first and it will be rebuilt.

The control layer

This is where ArchiMate is at its best and where architects most often produce a network diagram by accident.

Thing on the plant floor	Element	Note
PLC or PAC	Device	The controller is a physical node that runs logic
DCS controller pair	Device	Draw the pair as one Device unless redundancy is the point
SCADA server	Node	A server with system software on it
HMI or panel	Device	And say in the note that it is a view, not a system of record
OPC UA server, historian engine, batch engine	System Software	Software deployed on a Node
Historian as a platform	Node, with Artifact for the archive	The series is an Artifact, not a Data Object
Plant control network, cell network	Communication Network	One per zone, not one per cable
Tag namespace, archive, controller program	Artifact	Deployed on a Node by Assignment

The discipline that keeps it from becoming a network diagram is four rules. Draw the Communication Network once per zone and never per link. Do not draw switches, firewalls or cabling unless the view's question is about them — a firewall pair is a technology service realized by a Node, and it earns a box only in a security view. Collapse redundant pairs. And decide before you start whether the view's question is *what runs where* (an implementation and deployment view) or *what talks to what* (a technology view); answering both in one picture produces neither.

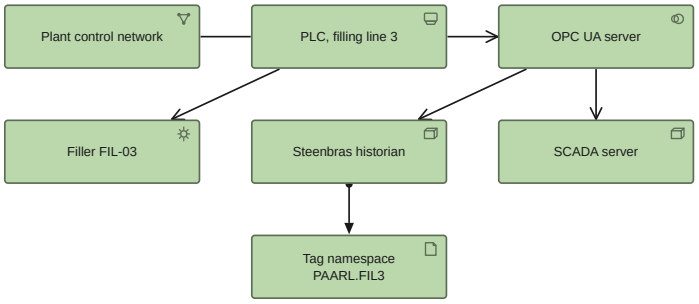


Figure 17.5 The control layer without becoming a network diagram: Device for controllers, Node for servers, System Software for the OPC UA server, Artifact for the tag namespace, and the Communication Network drawn once.

Zones and conduits

IEC 62443's zones group assets that share security requirements, with a boundary separating them from everything outside, and may be divided into sub-zones at different security levels to support defence in depth. Conduits group the elements carrying communication between two zones and provide the security functions protecting it, which is what lets zones at different security levels coexist [9]. Security levels — zero to four, describing resistance to a class of attacker — are evaluated per technical requirement, not used to classify whole products [9].

In ArchiMate: Grouping for a zone, Path for a conduit, Requirement for the target security level. Grouping is the only element that means "these things belong together for a reason that is not composition", which is exactly what a zone is. Path is the shared element for a logical communication route, which is close enough to a conduit to be useful.

And the honest note: the notation was not designed for this. Grouping carries no boundary semantics, so nothing in the model stops you putting an element in two zones. Path carries no security function, so the protection the conduit provides has to be expressed as a separate Requirement or a technology service. Zone-to-zone trust relationships have no element at all. If the engagement is primarily an OT security programme, the zone-and-conduit diagram should be produced in the 62443 convention, and your ArchiMate view should reference it rather than replace it.

Three facts to carry into that conversation, because getting them wrong loses you the control systems engineer in the first ten minutes.

The Purdue model is not a security model. Williams's 1992 technical guide and the Purdue Reference Model contain no cybersecurity content at all [11]. The Level 3.5 industrial DMZ, Level 5 and the whole zone language came later, from the ISA-99 and IEC 62443 community and from vendor reference architectures. Attributing the industrial DMZ to Williams is wrong and an OT engineer will notice.

The editions are uneven and the famous parts are the oldest. 62443-2-1 is Edition 2.0 of August 2024 and added a maturity model; 62443-2-4 is Edition 2.0 of December 2023; but 62443-3-3 is still Edition 1.0 of August 2013 and 62443-4-2 Edition 1.0 of February 2019 [9].

62443 has a conformance scheme and ISA-95 does not. ISASecure, run by the ISA Security Compliance Institute, certifies against 62443 through named schemes [10]. So a vendor claiming 62443 compliance can be asked for a certificate, a scheme and a level. There is no ISA-95 conformance scheme at all, which makes **"ISA-95 compliant" an uncertifiable claim** — treat it as a conversation opener and ask instead which ISA-95 objects the product exchanges, at what attribute level, over what transport, and whether you can see the profile.

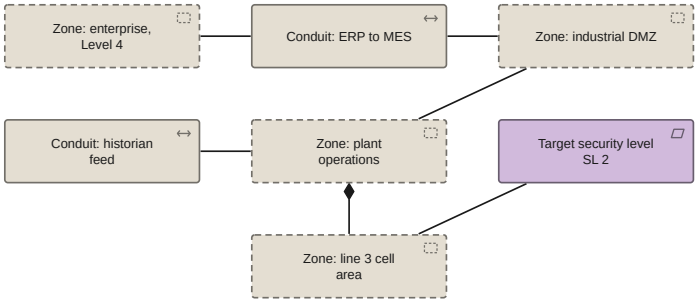


Figure 17.6 The IT/OT zone view: Grouping for an IEC 62443 zone, Path for a conduit, Requirement for the target security level. The notation was not designed for this, and the view should say so.

The viewpoints that actually get used

Eleven standard viewpoints carry almost all of a manufacturing engagement, and two more have to be invented because no standard viewpoint covers them. For each one, the four questions that matter are what it is for, what it may contain, what feeds it, and — the question architects skip — **what it cannot show**.

Capability map

For: agreeing what the business does, independent of who does it or which system supports it. The first artefact an executive committee can argue about productively. *May contain:* Capability, Resource, Value Stream, Course of Action, and Grouping for levels. *Fed by:* the organisation chart only as a cross-

check, never as the source; the ISA-95 functional enterprise-control model, which names business functions and shows which side of the Level 3/4 line each sits on, and is more directly useful for a manufacturing capability assessment than the level pyramid [1][2]. *Cannot show:* maturity, cost, ownership or sequence. A heat-map colour on a capability is an assessment result pinned to a picture, and the assessment, its scale and its date belong in a table beside it.

Value stream

For: the end-to-end flow that delivers value, used to frame scope before any system appears. *May contain:* Value Stream composed of Value Streams, with Capability realizing each stage and Role assigned. *Fed by:* the plant's own process documentation, and the twelve end-to-end processes a manufacturer actually runs — ideate to launch, plan to produce, procure to pay, order to cash, make-to-stock replenishment, quality hold to disposition, maintenance work management, engineering change, physical inventory, sales and operations planning, month-end costing close, recall. *Cannot show:* hand-offs, decisions, exceptions or timing. If the question is "who hands what to whom", you need a level-2 process model, and that is BPMN.

Layered

For: the single most useful picture in a manufacturing engagement: one business question, traced from the Role who asks it down to the Device the number comes from. *May contain:* anything, which is the danger. *Fed by:* everything. *Cannot show:* more than one question at a time. The layered view is where element budgets die. Pick one thread — "how does the plant manager know line 3's OEE" — and draw that thread only.

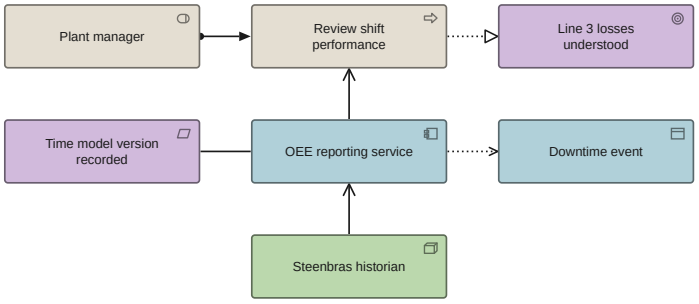


Figure 17.7 A layered view answering exactly one question — how the plant manager knows line 3's OEE — with the time-model version carried as a Requirement, because the number is meaningless without it.

Business process co-operation

For: which processes depend on which, and where the organisational boundaries fall. *May contain:* Process, Function, Event, Role, Collaboration, Business Object, junctions. *Fed by:* interviews, and the shift handover. *Cannot show:* swimlanes, and therefore cannot show a hand-off clearly. See the BPMN exception below.

Information structure

For: the twelve to twenty things the business talks about, and what asserts each. *May contain:* Business Object, Data Object, Artifact, Product, Representation of structure through Aggregation, Composition, Association and Specialization. *Fed by:* the ERP's master data, the PLM, the MES, and the one spreadsheet that turns out to be authoritative. *Cannot show:* attributes, keys, cardinality or effectivity. For a conceptual data model, draw Business Objects and relationships and nothing else; where cardinality matters, draw it in crow's foot notation as a plain diagram and say that is what you have done.

Application co-operation

For: which applications exchange which objects, and in which direction. In manufacturing this is also where you make the genealogy-spans-systems statement. *May contain:* Application Component, Application Interface, Data Object, Serving, Flow, Access. *Fed by:* the integration inventory, and the firewall rule base, which is usually more honest than the integration inventory. *Cannot show:* volume, latency, batch window, protocol or failure behaviour. Those are columns in an interface register, and in manufacturing the failure behaviour is the important one, because a plant must keep producing when the corporate network is down.

Application usage

For: which Role uses which component for which process. The view that exposes the fact that the operator is the primary data-entry population in the MES, and that every field you add costs cycle time. *May contain:* Role, Process, Application Component, Application Service, Serving. *Cannot show:* how often, how long it takes, or whether it is done at the end of the shift from memory on a paper note. That last one is a finding, not a model element, and it belongs in the text.

Technology

For: what runs where, on what, over which networks. *May contain:* Node, Device, System Software, Technology Interface, Communication Network, Path, Artifact, Equipment, Facility. *Cannot show:* IP addressing, VLANs, firewall rules or protocol versions. If those are the question, the answer is a network diagram produced by the people who own the switches.

Implementation and deployment

For: which artefact is deployed on which node, which is the only view that will reconcile against the plant's own records. *May contain:* Artifact, Node, Device, System Software, Assignment, and Application Component realized by Artifact. *Cannot show:* firmware revisions, validated baselines or patch state. In a plant those are the important facts, and they live in the asset register and the validation documentation.

Migration

For: the sequence. Plateau, gap, and the transition states between as-is and to-be. *May contain:* Plateau, Deliverable, Work Package, Gap expressed as an Assessment, Triggering between plateaus. *Fed by:* the shutdown calendar, the trading calendar, the wage-agreement dates and the month-end close, none of which are in any system you will be given access to. *Cannot show:* dependencies with lags, resource contention or risk. Those belong in a plan.

Requirements realization

For: tracing a requirement to the thing that satisfies it, which is how you survive a steering committee. *May contain:* Requirement, Principle, Goal, Outcome, Driver, Assessment, Stakeholder, and whatever realizes them. *Cannot show:* priority, cost or acceptance criteria. And remember that this is also where every constraint in your architecture must land, because there is no Constraint element.

The two you have to invent: the ERP/MES boundary view

No standard viewpoint covers the single most contested boundary in manufacturing architecture. Production scheduling, detailed scheduling, dispatch, labour and material confirmation, quality disposition, inventory within the plant and performance analysis can all sit in ERP, all sit in MES, or — in most real estates — sit in both and disagree.

Draw it as an application co-operation view with one addition: **a line down the middle, labelled Level 3 / Level 4, and a Requirement on each side stating what that side owns.** Put the ISA-95 object names on the Data Objects that cross it, because those are the entities the interface actually carries: an operations schedule with segment requirements going down, an operations performance with segment responses coming up, operations definition and operations capability alongside [1]. That is the standard's real content, and it is more useful than the pyramid.

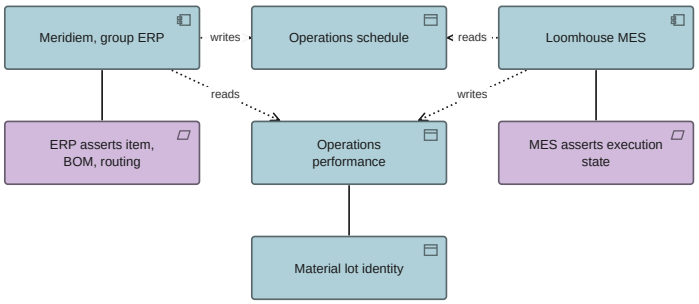


Figure 17.8 The ERP/MES boundary view, which no standard viewpoint covers: ISA-95 object names on the Data Objects that cross the Level 3 and Level 4 line, and a Requirement on each side stating what that side owns.

Three cautions. **B2MML is MESA's, not ISA's** — it is an XML implementation of the ISA-95 data models, royalty-free with attribution. Its current release is reported as V7.00.00 of 24 November 2020, built to the 2018 and 2019 editions — a version number the research for this book could not confirm from the repository itself, so check it before you print it. On either reading **it predates the 2025 Part 1**, so do not write that B2MML implements the current standard. **ANSI/ISA-95.00.01 is the 2025 edition**, announced in April 2025, revising the 2010 edition, and worked from IEC 62264-1 Edition 2.0 of 2013 [1][2] — so the ISA and IEC lineages are out of step and are different documents, and your view must say which one it means. And **ISA-95 has eight parts plus a technical report, not five**, though the designations and dates of Parts 2 to 8 should be checked rather than printed from memory.

And the IT/OT zone view

The other invented view. It answers one question: **what survives the loss of everything above Level 2?**

Draw the zones as Groupings, the conduits as Paths, and then overlay what the plant needs locally to keep making conforming, traceable product with no connectivity upward: the active recipes or routings at the correct revision, the specifications and work instructions, the open order queue, the label templates, the operator qualification data, the serial-number blocks. Mark each as a Data Object or Artifact with a Requirement stating its staleness tolerance. Then mark what the plant *captures* locally and queues upward: confirmations, parameter data, genealogy, scans, weights, inspection verdicts, rejects, downtime.

Add a Requirement that names the agreed period. "Indefinitely" is not an answer. "Seventy-two hours at full rate" is, and it sizes the local storage and the serial-number block. And add an Assessment recording what is *not* available in degraded mode and what the business has accepted losing: real-time customer visibility, automatic available-to-promise, cross-plant stock visibility, dynamic scheduling.

> The test of this view is not a review. It is pulling the uplink on a Tuesday. Most manufacturers have never done that deliberately and all of them have done it accidentally, and the incident report from the accident is the best requirements document available.

The BPMN exception, stated once

Level-2 process models are BPMN 2.0, not ArchiMate. ArchiMate has no swimlanes, and in a process model the hand-off is the point. An engineering change arriving from a customer, crossing the design engineer, the manufacturing engineer and two systems, with a gateway on whether tooling is affected, is a picture about who hands what to whom. ArchiMate's business process co-operation view can show that the processes depend on each other; it cannot show the boundary being crossed, and the boundary is where the work goes wrong.

The practical limits: **a maximum of five lanes and seven columns.** Beyond that it will not fit a page, and a process model that does not fit a page is not read. If your process needs nine lanes, you have two processes and a hand-off between them, and modelling it as one is the reason nobody can agree who owns the middle.

ArchiMate stays responsible for everything above and below: the capability the process realizes, the applications that support it, the objects it touches, the requirements it satisfies. The two notations meet at the process, and the discipline is that the same process has the same name in both.

The practical discipline

How many elements before a view stops communicating. The honest numbers, from producing these for a living: a view of **eight to twelve** elements is read and remembered; **twenty** is read with effort; **thirty-five** is looked at and not read; beyond that it is wallpaper that proves work was done. When a view exceeds about twenty elements, the fix is almost never a bigger page. It is that the view is answering two questions, and splitting it produces two views people will actually use.

Three grid columns is the practical width for a printed figure. Four columns renders small enough to be unreadable after downscaling, and a label beyond about sixty characters is silently truncated — which is how sibling volumes in this series shipped with half-named elements. **Validate, then look at the render.**

Naming conventions. Six rules that prevent most of the arguments:

1. **Name a Role as the industry names the job**, and say in one line what the person does all day. "Maintenance planner", not "asset care resource". An architect who cannot name the jobs cannot run a workshop.
2. **Name a Process with a verb and an object**: "release production order", "disposition quality hold". Not "order management".
3. **Name a Function with a noun**: "filling", "heat treatment", "quality operations".
4. **Name an application by what it does, with the product name in brackets** on the as-is only: "group ERP (Meridiem)". The to-be view carries capability names, because the product has not been chosen and naming it decides the procurement by accident.
5. **Encode physical structure in technology names, never organisational structure**, because organisations reorganise far more often than pipework moves. This is the same rule that governs historian tag naming, and for the same reason.
6. **Agree the naming convention before you integrate anything.** It is the single most valuable sentence in the ISA-95 literature and it is free.

Keeping a view and the repository in step. The failure is always the same: the repository holds the elements, the views are drawn over them, somebody edits a view to make it legible for a steering pack, and the repository and the pack diverge. Four things keep them together. Every element exists once, in the repository, with one name. Every view is generated from the repository and annotated, never drawn independently. Every view carries its own metadata — the question it answers, the audience, the date, the source of each assertion. And there is a standing reconciliation: the plant-structure registry, the application inventory and the interface register are compared to the model on a schedule, and the differences are worked, not explained.

And the rule that matters most. A view exists for an audience and a decision. If you cannot name the person who will use the view and the decision they will use it for, do not draw it, because nobody will maintain it. The manufacturing version of this is sharper than most: the maintenance planner will maintain the asset hierarchy view because they need it; nobody will maintain the enterprise-wide layered view, and in eighteen months it will be the most confidently wrong artefact in the repository.

THE CHAPTER IN TEN LINES

- ArchiMate 4 gives you 42 elements across seven domains, 11 relationships and junctions; the element is Role, and there is no Business Role.
- There is no Constraint element: express every constraint as a Requirement associated with the thing it governs.
- Never reproduce specification text, tables or figures. Describe the concept and draw your own picture.
- Model the physical plant in Location, Facility and Equipment, and put the ISA-95 classification and the ERP plant alongside it as Groupings with an explicit dated mapping.
- ISA-95's equipment enumeration has fifteen values, several of them alternatives selected by process type, and equipment module and control module are ISA-88 concepts it references.
- ArchiMate cannot express effectivity, revision or a scrap factor on a BOM line; approximate with Requirements and say where the real data model lives.
- A genealogy is a graph of instances, so it is not drawable: name the type, and make the point that the graph exists in no single system.
- Model the tag namespace and the decision-grain aggregates, never the tags.
- There is no metric element and Value is not a number, so the measure definition register is a separate deliverable and it comes before the dashboard.
- Level-2 process models are BPMN 2.0, at most five lanes and seven columns, because in a process model the hand-off is the point.
- A view of eight to twelve elements is read and remembered; one with no named audience and no decision will not be maintained.

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CHAPTER 18

Karoo Industrial Group, end to end

One invented South African manufacturing group, one quarter, one architect. Everything this book has taught, applied once, so that you see a whole engagement rather than nineteen topics — including the three things that turn out to be true instead of the brief you were given.

READ THIS FIRST

Nothing about Karoo Industrial Group is real. The group, its two divisions, its three plants, its eight named systems and its eight named people are all invented for this book. No figure in this chapter is sourced to Karoo Industrial Group, because there is nothing to source it to. Every *general* fact used to reason about the group — a standard, a regulation, a retention period, a measurement convention — is sourced and dated in the normal way. The group exists so that you can watch one engagement end to end instead of reading nineteen topics, and because almost every real South African plant is several archetypes at once, which a single-archetype example could not show.

What the architect walks into

The brief, as Dawid Scholtz wrote it, ran to one paragraph. *We need a target application architecture for manufacturing. The ERP is eighteen years old and the board has asked whether we replace it. We would like a recommendation and a roadmap by the end of the quarter.*

Nomthandazo Mahlangu, the group chief operating officer, named three symptoms in the first meeting, in this order.

1. **"I cannot get one number for output."** The monthly pack shows production three ways and the divisions argue about which is right.
2. **"Highveld keeps missing the rebate deadline."** The quarterly customs submission is late, and when it is not late it is queried.
3. **"The Paarl plant says it is running at 72 per cent and the Gqeberha plant says 81 per cent, and I do not believe either of them."**

Ingrid Lourens, the group's lead enterprise architect, spent the first two weeks on the floor rather than in the meeting room. Three things turned out to be true instead, and none of them was the thing she was asked about.

First, the ERP's age is not the problem. Meridiem is eighteen years old and has been extended heavily, and it does what an eighteen-year-old ERP does: it holds the item master, the bill of material, the routing, the work centre, the production order, the cost and the general ledger, and it holds them adequately. The problem is that there are **two instances of it and they do not share an item master**. Replacing the ERP without settling that question replaces one estate with two copies of the same estate.

Second, the output number is not a reporting problem. It is a measurement-definition problem wearing reporting clothes. The three numbers in the pack are produced from three different time models with three different availability denominators, and all three are arithmetically correct. No reporting tool fixes that.

Third, the rebate deadline is not a finance problem. The automotive production rebate certificate requires a bill-of-material-level country-of-origin and cost-component split, propagated through several tiers of supply. Highveld's costing does not preserve the component split through multi-level roll-up, so the claim is reconstructed by hand each quarter by one costing analyst. The deadline is missed because the data does not exist, not because somebody is slow.

> The brief you are given is a statement of the symptom the executive can see. The first deliverable is almost never what was asked for. It is a description of what is actually true, written so that the person who gave you the brief can recognise their own business in it and change their mind without losing face.

The group as it is

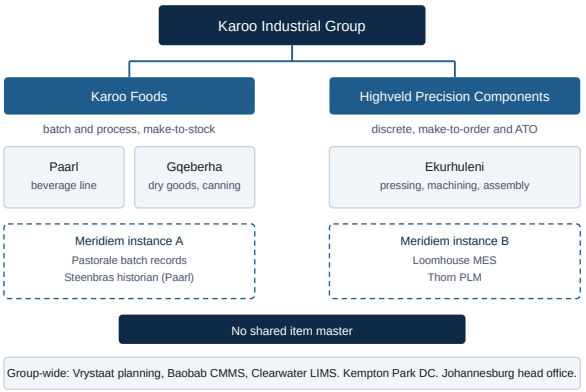


Figure 18.1 Karoo Industrial Group as the architect found it. Two divisions, three plants, two ERP instances and no shared item master. Nothing here is real.

Karoo Foods makes food and beverage, batch and process, make-to-stock. Two plants: a beverage line at Paarl and a dry-goods and canning site at Gqeberha. Recipes rather than bills of material. Co-products. Allergen segregation with a validated cleaning requirement between runs. Shelf life on

every batch. Catch weight on one line at Gqeberha, where the product is sold by declared net content and filled by actual mass. And a pasteuriser at Paarl that cannot be stopped and restarted cheaply — the sort of asset whose cool-down and re-heat is measured in hours, which is why an interruption there is a production event and not an inconvenience.

Highveld Precision Components makes machined and pressed components in Ekurhuleni, discrete, make-to-order and assemble-to-order. Sequenced just-in-time delivery to one vehicle manufacturer, with aftermarket make-to-stock running on the same presses alongside it. A five-level bill of material from finished assembly down to coil. Engineering change arriving from the customer rather than originating inside the business. And exposure to the automotive rebate regime, which turns a costing convenience into a customs obligation.

A Kempton Park distribution centre serves both. Head office is in Johannesburg.

This is the point of the group, and it is the point of the book. A design that fits Karoo Foods fails at Highveld, and a design that fits Highveld fails at Karoo Foods. Not in detail — in structure.

Decision	Karoo Foods	Highveld
Production definition	Master recipe: formula and process conditions in one versioned object	BOM plus routing, two objects joined at the production order
Quantities	Continuous, scaled, some lines fixed-quantity; yield variable by physics	Integers of identifiable things; scrap is an exception
Genealogy	Many-to-many lot graph; silos blend, so parentage is a set with proportions	Serial-number tree, mostly; lot behind the serial
Quality disposition	The batch is released or rejected whole; no despatch before disposition	Per piece, per operation, against a dimensional specification
Planning control mode	Campaign wheel at Paarl; daily schedule at Gqeberha; wheel length sets cover	Three modes on one press: sequenced firm, forecast MTS, quoted MTO
The binding constraint	Changeover and validated allergen cleaning; the pasteuriser cannot stop	Setter availability and die changeover; the customer release
Statutory pressure	R638 traceability and recall; R146 labelling and the eight allergen classes	Rebate substantiation at BOM level; origin and cost component split

Figure 18.2 The two archetypes side by side. A design that fits one column fails in the other, and not in detail — in structure.

Work through what that means for four decisions an architect would otherwise take once.

The production definition. At Karoo Foods the structure and the process conditions are inseparable: a master recipe carries temperature, hold time, addition rate and mixing sequence as part of the specification, so material and sequence cannot be versioned independently without producing invalid combinations. At Highveld the bill of material and the routing are genuinely separate objects, joined at the production order through the component-to-operation assignment. A group-wide "single product definition" programme that versions a BOM and a routing separately is correct at Ekurhuleni and wrong at Paarl.

Genealogy. At Highveld it is a serial-number tree, mostly. At Karoo Foods it is a many-to-many lot graph, because the Gqeberha silos blend incoming lots continuously, so an output lot's parentage is a set with proportions and "which input lot caused this" has no single answer. The same traceability design cannot serve both.

Quality disposition. At Karoo Foods the batch is the unit of release: it is released or rejected as a whole, and it cannot ship until disposition is recorded. At Highveld disposition is per piece, per operation, against a dimensional specification, with a first-article approval behind it. The status-of-stock model has to carry both.

Planning control mode. Highveld runs three control modes on one press line: a firm sequenced release from the vehicle manufacturer, a statistical make-to-stock forecast for aftermarket, and quoted-lead-time make-to-order for industrial customers. Karoo Foods runs a campaign wheel at Paarl and daily scheduling at Gqeberha, and the wheel's length sets the finished-goods cover for every product on it. A pull system and a material-requirements-planning push system make genuinely incompatible demands on the same data, so the deliverable is a control-mode map per material segment, not a tool choice.

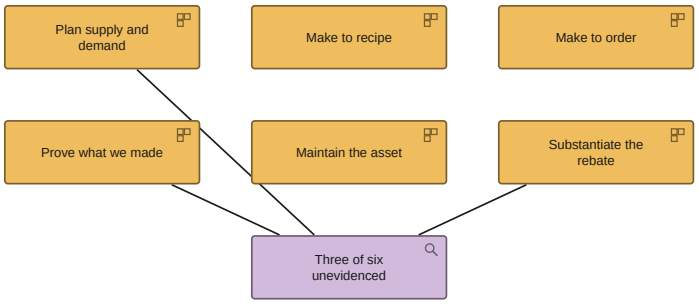


Figure 18.3 The as-is capability view. The assessment is the finding; the colour on a capability is not. Three of the six capabilities could not be evidenced at the time of the review.

The as-is, drawn

The group's eight systems, in the order they matter.

System	What it is	Where	What it asserts
Meridiem	Group ERP, two instances	Both divisions	Item, BOM, routing, order, cost, ledger
Loomhouse	MES	Ekurhuleni	Execution state, confirmation, genealogy
Pastorale	Spreadsheet and Access estate	Paarl, Gqeberha	Batch records, in-process results
Steenbras	Process historian	Paarl	The tag series, and nothing else
Clearwater	LIMS	Both divisions	Sample, test result, specification
Thorn	PLM	Ekurhuleni	Part, drawing, engineering BOM, change

Plus **Vrystaat**, the group planning tool, and **Baobab**, the computerised maintenance management system. Both run group-wide and both resolve plant structure differently from Meridiem.

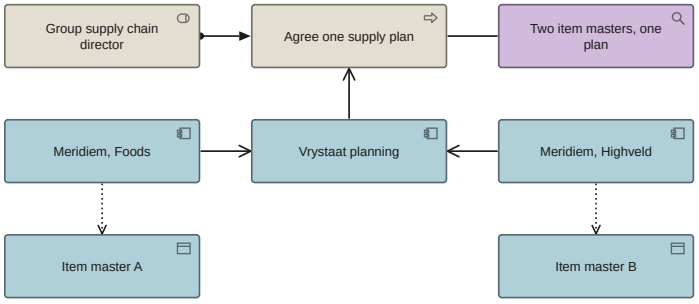


Figure 18.4 The as-is, traced from the one question the group chief operating officer asked. Two item masters, two ERP instances, one plan, and an assessment that says so.

The two weeks on the floor, which produced most of this chapter

The interview order matters, and the most useful first interview is not the one on the org chart. Ingrid started with a shift supervisor at Gqeberha, on shift, including a night shift, and walked the line. That produced the real process, the real data capture and the workarounds: production figures entered at the end of the shift from a paper note, an allergen changeover matrix laminated and taped to a wall rather than held in any system, and a planner's spreadsheet that is the actual schedule because Meridiem cannot honour the changeover rules.

Then Karel du Toit, the Highveld maintenance planner. He is the single most useful first interview in most plants, because the maintenance planner is the person who knows where everything actually is. He converts maintenance requirements into executable work — scopes the job, secures the spares, books the window with production, closes out the history — and he is therefore the primary user of Baobab and the only person in the group who can say whether the asset hierarchy is usable. His answers settled three things in an hour. The asset hierarchy in Baobab is 200-odd functional locations on the press line, against four work centres in Meridiem over the same steel, with no mapping — so a downtime event recorded in Baobab cannot be attributed to a work centre's loss. Failure history is free text, which means it is unanalysable, which means the group has a work-order backlog rather than a maintenance strategy. And equipment and functional location are merged in the data, so either the position's failure history or the object's rebuild history has been lost.

Then Clearwater, with Sipho Khumalo. The lab's turnaround is a planning constraint and nobody models it. At Gqeberha a release test takes most of a working day, so finished goods sit in quarantine for that long, and that is inventory, lead time and cash sitting in a building with nobody's name against it. **Quality assurance holds the veto over despatch**, and in a food plant that veto must exist as a status on the stock rather than as an email from the quality manager. At Karoo Industrial Group it is an email.

And then, last, Lerato Mokoena and Johan Terblanche. Plant managers are the most important interviews and they should not be the first, because you want to arrive with specific questions rather than general ones. Lerato's most useful sentence was about which numbers in the corporate pack she does not believe, and why.

The findings

Eight findings. Each one is traced to evidence that a reader could go and check in a real plant within an afternoon, because a finding an architect cannot evidence is an opinion with a template round it.

Finding	The check that evidences it
1 Two ERP instances, no shared item master	Extract both masters; compare on description, UoM, supplier part number
2 Engineering and manufacturing BOM gap	Trace the last five engineering changes from Thorn into Meridiem
3 Rebate cannot be substantiated	Ask for origin and cost split at every level of one five-level part
4 Batch record in a spreadsheet estate	Ask who can edit one batch record, and what the audit trail shows
5 Three availability denominators	Ask each plant separately, in writing, what its denominator is
6 Reported number is not the recorded number	Ask for compression settings on the tags behind OEE, and who changed them
7 Resilience tiers never tested	Ask what happened the last time the uplink dropped for real
8 Biometric clocking on contractual consent	Ask for the alternative for an employee who refuses, and the refusal record

Figure 18.5 The eight findings on one page, each with the check that evidences it. A finding an architect cannot evidence is an opinion with a template round it.

Finding 1 — Two ERP instances with no shared item master

Evidence: extract the item master from both Meridiem instances and compare on description, purchasing unit of measure, base unit of measure and supplier part number. At Karoo Industrial Group the same packaging film, bought from the same converter, exists twice with two codes, two base units and two conversion factors. Nobody can state group spend on film without a manual mapping spreadsheet, and the spreadsheet is maintained by one person in group procurement.

Why it is the first finding: the item master is the hardest master data object in any industry, and it is harder in manufacturing for a structural reason — it is co-owned by six functions who each need the record to exist before they can supply their part of it. That circularity is the whole governance problem, and no amount of tooling resolves it. Nobody owns the record; different functions own different fields of it, and what is owned centrally is the process, the rules and the arbitration.

Architectural consequence: every other finding depends on this one. You cannot build a group measure, a group spend view, a group plan or a group recall capability over two item masters.

Finding 2 — The engineering BOM and the manufacturing BOM do not meet

Evidence: at Highveld, Thorn holds the engineering BOM and Meridiem holds the manufacturing BOM, and the transformation between them is done by a manufacturing engineer in a spreadsheet. Ask for the last five engineering changes and trace each one into Meridiem. Two will not be there; one will be there with the wrong effectivity date; one will have been applied without the change number, so the question "why is this component here" has no answer.

The structural fact: these are not two views of one BOM. They are separate structures with their own identity, lifecycle, approval and effectivity, which happen to reference the same items. The engineering BOM follows functional decomposition as the drawing decomposes the product; the manufacturing BOM adds consumables, packaging, phantoms, scrap and yield, operation assignment and alternates, and frequently re-levels the engineering structure entirely. **A programme that tries to make them the same structure will fail and will be re-run under a different name in four years.** The achievable goal is a governed, traceable, auditable mapping.

The aggravating factor at Highveld: the change does not originate inside the business. It arrives from the customer, with a date attached, and the disposition of existing stock and work in progress — use as is, rework, scrap, return, segregate — is a property of the change, not of the system. That disposition field is missing, which is why the plant decides it informally on the line.

Finding 3 — The rebate claim cannot be substantiated

This is the finding with the hardest edge, because it is the one with a statutory clock on it.

The automotive production and development programme is a **customs duty-rebate-certificate regime, not a tax credit** [1], and that distinction changes who in the business owns it. Under the amended regulations: the production incentive is based on local value addition, adjusted by a factor and then by the applicable customs duty; the component value test requires at least 25 per cent of the ex-factory selling price to be labour, SACU-origin material and factory overhead incurred in South Africa; a production rebate certificate is applied for only after full payment is received and **within 12 calendar months of the eligible product's invoice date**, is valid for 12 months from the first day of the quarter in which the claim was submitted, is usable only by the original holder and may be transferred once at the commission's sole discretion [1]. And the sentence that decides the architecture: **if no Form C2 is obtained from a supplier, the purchase price is treated entirely as imported content** [1].

Evidence: ask Highveld to produce, for one current-production part, the country of origin and the cost-component split of every component at every level of its five-level structure, with the Form C2 reference for each purchased input. At Karoo Industrial Group this exists for about sixty parts because one costing analyst maintains it, and it does not exist for the other two thousand.

Architectural consequence: this is a **bill-of-material-level country-of-origin and cost-component-split requirement, propagated through multiple tiers of supply**, with document retention and a twelve-month window. A plant whose costing does not preserve the component split through multi-level roll-up physically cannot substantiate the rebate. Origin and cost-component split must be governed master-data attributes on the item and on the BOM position, not a quarterly reconstruction.

> The phase-two detail of the programme could not be verified from primary sources in the research behind this book, and secondary commentary on it disagrees with itself. The structural requirement above is stable and sourced; any phase-two figure should be read from the gazetted regulations and the commission's own administrative requirements before it goes in front of a client.

Finding 4 — The batch record is in a spreadsheet estate, against a traceability obligation

Evidence: Pastorale. Ask to see the batch record for a specific production date at Gqeberha, then ask who can edit it, then ask what the audit trail shows. At Karoo Industrial Group the answer is a shared folder, four people, and nothing.

The obligation: South Africa has no equivalent of the United States food traceability rule, and importing that rule's structure by default over-specifies a domestic manufacturer. The obligation here is **R638**, which replaced R962 of 2012: a certificate of acceptability per food premises, a traceability system, a recall procedure, records kept for six months beyond the product's shelf life, and notification to the local inspector and to the national food control directorate [2][3]. **There is no food traceability list, no prescribed critical tracking events and no prescribed key data elements.** The requirement is that you can do it, not that you do it in a prescribed shape.

And the harder half: the Gqeberha silos blend. The recall question is not "which products could contain this lot" — that is where-used, and the system can answer it. It is "which products *do*", which is genealogy. The difference between those two answers is the difference between recalling a product line and recalling four pallets, and at Karoo Industrial Group the genealogy is complete at Paarl and broken at the Gqeberha silos, where lots co-mingle.

The test that settles it: a recall simulation on a randomly chosen input lot, timed, with the amplification factor measured — the ratio of product implicated to product actually affected. A traceability design that has never been tested by simulation is a claim, not a capability. And Sipho Khumalo, the group quality manager, should run it, not the architect.

Finding 5 — Three plants, three availability denominators, no measure definition register

Evidence: ask each of the three plants, separately, in writing, what their overall equipment effectiveness denominator is. At Karoo Industrial Group: Paarl excludes planned maintenance and breaks; Gqeberha uses rostered shift time including breaks; Ekurhuleni excludes the hours when the vehicle manufacturer's release leaves the press with nothing to run. All three are defensible. All three are different. None is written down.

Why this is not pedantry. **There is no OEE standard.** The research behind this book found at six incompatible availability denominators in circulation, and a seventh divergence in where the micro-stop threshold sits [4][5]. That threshold — typically one to five minutes — shifts loss between availability and performance **without changing OEE** [5], so OEE can be comparable between plants where its components are not. And OEE is documented as unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level [5]: a line's OEE is not the mean of its stations'. Compute on the constraint, or on the line as one resource.

Two further precision points, because an architect who gets these wrong is corrected by the continuous-improvement lead in week three. **ISO 22400-2's middle term is Effectiveness, not Performance** [6] — the published edition is 2014 and the second edition was still at draft stage in October 2026 [7]. And **the 85 per cent benchmark is an invention**: the site that publishes it calls it a convenient, compelling and completely artificial benchmark [4]. It must never appear in a client deliverable as a target.

Architectural consequence: **the first deliverable is a measure definition register, not a dashboard.** One row per measure, with the formula, the time scope, the denominator convention, the data lineage, the owner and the declared comparability scope. Then the time model behind it: a partition of all calendar time into mutually exclusive, exhaustive, typed states, versioned, with the version stamped on every number produced.

Finding 6 — The reported number and the recorded number are not the same number

This one surprises everybody with an information-technology background and nobody with a control systems background.

Evidence: ask for the exception and compression settings on the twenty tags that feed the Paarl OEE calculation, and ask when they were last changed and by whom. At Karoo Industrial Group nobody knows, and the defaults are non-zero, which means compression is on unless somebody turned it off [8].

The mechanism. A historian of the class Steenbras belongs to applies two separate reductions at two points. **Exception reporting** at the interface sends only values differing enough from the last value sent. **Compression** at the archive uses a swinging-door algorithm that removes data which can be recreated by interpolation [8]. The magnitudes are not marginal: a nought-or-one status tag over forty-eight hours at one-second scans is **172,800 events with no compression and fifteen events with minimal compression** [8]. Out-of-order data bypasses compression, so late-arriving data is stored at full fidelity while live data is compressed, and the same tag has two fidelities in the same archive depending on how the data arrived [8].

The consequence. The archive does not contain what the instrument measured; it contains the smallest set of points from which the measured curve can be reconstructed to within the compression deviation. Every query against a time grid is therefore interpolating, and the tag's step flag decides whether an analogue interpolation or a last-known value is returned — so a discrete state mis-flagged as analogue returns a valve 37 per cent open when it has two positions. A time-weighted average over archived points is not the arithmetic mean of the original scans, and the two diverge most during the transients that matter. **Two reports of the same thing can both be right and differ.**

Architectural consequence: a statement, per tag class, of the exception and compression settings, the step flag, the retention period and the aggregation method reports must use — under change control. It is unusual, it is cheap, and it is the difference between a data platform that can support a yield investigation and one that cannot.

Finding 7 — The resilience tiers have never been tested

Evidence: ask what happens at Paarl when the link to Johannesburg drops. At Karoo Industrial Group there is a documented answer, written in 2019, which nobody has exercised. Then ask what happened the last time it dropped for real. There is an incident report, and it is a better requirements document than the design.

The design rule the group has not met: the plant must be able to continue producing conforming, traceable product for a **defined period** with no connectivity to any system above Level 2, and must be able to reconcile on recovery without loss of record. That requires local residency of the active recipes at the correct revision, the specifications, the work instructions, the open order queue, the label templates, the operator qualification data and the serial-number blocks; local durable capture of confirmations, parameter data, genealogy, scans, weights, verdicts, rejects and downtime; a declared and rehearsed degraded mode; a reconciliation procedure with a named owner; and an explicit list of what is *not* available and what the business has accepted losing.

"Indefinitely" is not an answer. "Seventy-two hours at full rate" is, and it sizes the local storage and the serial-number block.

And the South African specific, which is the best-evidenced design input in this whole book. Load-shedding as a daily event has ended — Eskom recorded 504 consecutive days without it as at 2 October 2026, with energy availability at 68.11 per cent financial-year-to-date against 62.59 per cent the prior year [9]. But **the real power issue for operational technology is voltage quality, not availability, and there is a national standard for it.** NRS 048-2:2025 Edition 5, approved by the regulator on 6 May 2025, classifies voltage dips into windows and **explicitly assigns window-Y immunity to the customer**; it names the windows in which dips may trip drives and those in which they trip contactors; and it publishes characteristic unplanned interruption counts — for one medium-voltage network category, 18 a year at the fiftieth percentile of sites and 75 at the ninety-fifth — **with no compliance criteria** [10].

Read as a design input, that says: dips in one window are yours to be immune to, dips in two others will trip your drives, and a site on overhead supply should expect tens of sustained interruptions a year with no recourse. The controller usually survives a dip because it is a small load on an uninterruptible supply — **and that is precisely the problem**, because it stays up, retains its state, and now believes a pump is running that has stopped. An interlock written on a run *command* rather than a run *feedback* does not notice.

Finding 8 — Biometric clocking, with consent taken in the employment contract

Evidence: the time-and-attendance terminals at all three plants read fingerprints. The lawful basis is a clause in the employment contract. There is no alternative mechanism for an employee who refuses, and no record of refusals. Templates are not deleted on termination.

The provision. **Biometric information is special personal information under section 26 of POPIA**, so processing is prohibited unless a section 27 ground applies [11][12]. The Information Regulator's guidance note of 28 June 2021 lists those grounds, and the first of them is consent [12]. **"We got consent in the employment contract" runs straight into the power-imbalance objection**, and if there is no alternative for an employee who refuses, consent was not freely given.

What the record has to show, regardless of how the open questions resolve: the lawful basis as a dated determination with an author; whether a template or an image is stored and whether the template is irreversible; where the template lives and in which country; the alternative mechanism and the record of refusals; deletion on termination with evidence; access control and logging on the biometric store; the purpose-limitation boundary, because a template collected for access control and used for productivity measurement is a new purpose; the operator agreement with the device vendor; and the entry in the processing register.

> A manufacturer's privacy exposure is its employee file, not its customer file. The longest-retained special personal information in the building is the occupational health file: **30 years of chemical exposure data and 40 years of noise exposure and audiometry** [13]. Any scoping exercise that starts with the customer database has started in the wrong place.

Whether the Information Regulator has issued guidance specific to biometric time-and-attendance in employment, and what its position is on consent in the employment relationship, could not be verified in the research behind this book. Ingrid's finding is therefore written as an exposure with a named open question and a date, not as a compliance conclusion — which is also the only honest way to write it.

What it costs to be wrong

This section is the one the executive committee reads, and it is the one where architects lose credibility by inventing numbers. Three figures circulate in South African manufacturing material — a rand cost of inventory shrinkage, a rand cost of power interruption, and a percentage of manufacturing jobs lost — and none of the three could be sourced in the research behind this book. **So none of them appears here.** What appears instead is the *shape* of each exposure, in terms an executive committee already uses, with the sourced constraint that gives it force.

An unsubstantiated rebate is a hard-edged, dated customs exposure. The claim window is twelve calendar months from the invoice date, the certificate is valid twelve months from the first day of its quarter, and a missing Form C2 means the whole purchase price counts as imported content [1]. The exposure is therefore not "some margin" — it is a specific, quantifiable amount per quarter that Highveld's own finance function can calculate in an afternoon, and the architect's job is to ask them to, not to guess it.

An unbounded recall is a working-capital event, not a quality event. The cost is the product implicated, not the product affected, and the ratio between those two is a design output — the amplification factor. Four levers reduce it: smaller batches, lot-specific rather than backflushed consumption, physically rather than nominally enforced first-in-first-out, and avoiding blending where a quality-critical input is involved. Each has a cost. Making the trade explicit before a recall is the architecture's contribution; discovering it during one is the alternative.

Incomparable measures misallocate capital. When Paarl reports 72 per cent and Gqeberha 81 per cent on different denominators, the capital that should go to Paarl goes to Gqeberha, for three years, and nobody can see why the returns did not arrive. This is the most expensive consequence in the list and the hardest to put a number on, which is exactly why it survives.

Backflushing hides the usage variance that explains the shrinkage. Backflush consumes standard BOM quantities by definition, so over-issues and unreported scrap cannot appear as a variance; they surface later as stock-count differences. **If a plant has unexplained inventory shrinkage and backflushes, you have found the mechanism** — and where substitution occurs, backflush produces a confidently false genealogy, which is worse than a gap.

And one exposure that is genuinely unrecoverable. Gqeberha does tolling work: it packs a customer's own material on a customer's own account. In a tolling arrangement the material belongs to the customer, stock in one silo may be owned by three parties, and the same tonne may be the plant's asset on Monday and a customer's on Tuesday. **Ownership is a first-class attribute of inventory, and omitting it is one of the few genuinely unrecoverable design errors**, because you cannot retrofit ownership onto historic stock movements. Everything else in this chapter can be fixed late and expensively. That one cannot be fixed at all.

> Resist the request for a benefits case in month one. The honest position is that four of the eight findings have exposures the business itself can quantify from its own records within a week, and that an architect's estimate of them would be less accurate and less credible than the finance function's. Ask for the calculation; do not manufacture it.

The target and the transition

Three transition states. Each has written exit criteria, and each exit criterion is a thing somebody can verify rather than a thing somebody can assert.

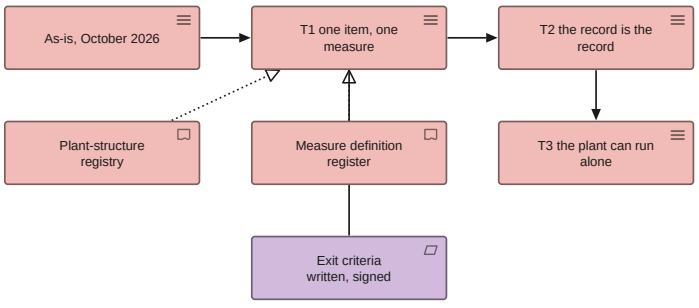


Figure 18.6 Three transition states with verifiable exit criteria. The first year buys agreement, not software: no new system class appears before transition state two.

Transition state one — one item, one measure. Scope: the item master reconciled across the two Meridiem instances with an arbitration process; the plant-structure registry built, holding both the ISA-95 equipment tree and the ERP plant hierarchy as separate typed trees with an explicit dated mapping; the measure definition register published with an owner per measure; the time model versioned.

Exit criteria. A query returns one item for one physical thing across both instances. A named person arbitrates a disputed item field within two working days, demonstrated three times. A downtime event recorded at Paarl resolves to a work centre and a cost centre without a manual rule. Each of the three plants' OEE carries a time-model version, and the group pack states the comparability scope on the page.

Transition state two — the record is the record. Scope: the batch record off Pastorale and onto a system with an audit trail and controlled access; genealogy closed at the Gqeberha silos to the extent blending allows, with the residual uncertainty *stated* rather than hidden; origin and cost-component split carried as governed attributes on the Highveld item and BOM position; the engineering-to-manufacturing BOM mapping governed, with change number in and change number out on every position.

Exit criteria. A recall simulation on a randomly chosen input lot completes inside the window Sipho Khumalo sets, with the amplification factor measured and the completeness gaps named. A rebate claim is produced from the system with no spreadsheet step. The last five engineering changes are each traceable from the customer's notice to the manufacturing BOM position, with effectivity and stock disposition recorded.

Transition state three — the plant can run alone. Scope: the autonomous-running period agreed per plant and *tested*; local residency and local capture in place; the zone and conduit model enforced in the switch configuration and not only on the drawing; historian exception, compression, step-flag and retention settings documented per tag class and under change control; the biometric processing either re-based on a defensible ground with an alternative mechanism, or replaced.

Exit criteria. The uplink is pulled on a Tuesday, deliberately, with the business informed, and the plant produces conforming traceable product for the agreed period and reconciles on recovery with no loss of record. A conduit not in the model does not pass traffic. A tag's compression setting cannot be changed without a change record.

The decisions that must be taken before anything can start

An architect's most useful quarter-end artefact is not a roadmap. It is a short list of decisions that only the executive committee can take, written so that each one can be taken in a single meeting.

Decision	Who	Why it blocks everything
One group item master, or two with a governed mapping	Nomthandazo Mahlangu	Findings 1, 3 and 5 all sit on it
Which Meridiem instance survives, if either	Dawid Scholtz	Determines whether this is consolidation or replacement
Is ownership an attribute of inventory	Priya Naidoo	Tolling at Gqeberha; unrecoverable if deferred
Who owns the measure definition register	Nomthandazo Mahlangu	Without an owner it is a document, not a control
Is the Highveld manufacturing BOM authored in Thorn or Meridiem	Johan Terblanche	Decides the engineering change path
The autonomous-running period, per plant	Lerato Mokoena	Sizes local storage and the serial-number block

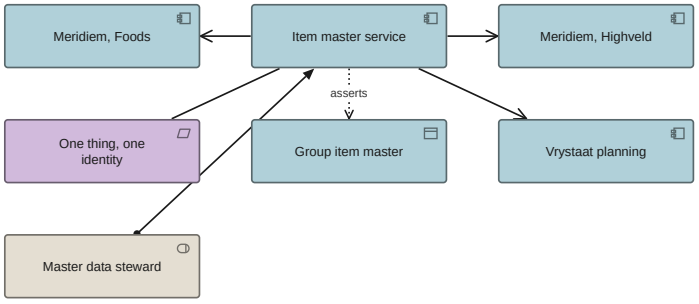


Figure 18.7 The target for transition state one. Deliberately unexciting: one item master service, one group item master, a named steward, and the rule that makes it worth building.

The target application picture is deliberately unexciting. One item master service asserting one group item master and serving both Meridiem instances and Vrystaat; a master data steward assigned to the service; and a Requirement on the service stating the rule that makes it worth building. No new system class appears in transition state one. That is the point: **the first year buys agreement, not software.**

The roadmap and what the year allows

Here is where most manufacturing roadmaps are written wrong, because they are written against twelve months.

The planning and trading rhythm. A monthly sales and operations planning cycle occupies the whole month: data gathering and product review in week one, demand review in week two, supply review in weeks two and three, integrated reconciliation in week three, the management business review in week four. The people you need for an architecture workshop are the people in those meetings, and the reconciliation step — the one most often skipped — is the one you must not be the reason for skipping.

Month-end close. Karoo Foods takes about ten working days to a reported number, of which several are inventory and intercompany. Highveld takes about eight, and its rebate submission is a separate parallel close that nobody else in finance understands. The plant controller is unavailable for the first two weeks of every month. That is twenty-four weeks of the year in which your most important finance ally has no time.

Shutdown and maintenance windows. Karoo Foods' Paarl wet line goes down every weekend for clean-in-place and sanitation, which is a hygiene requirement rather than a demand decision — so Monday's first batches carry a start-up quality risk and a start-up scrap allowance. Gqeberha takes the December industry shutdown. Highveld's presses take a planned Saturday maintenance slot weekly and a longer window at the vehicle manufacturer's model-year changeover. **Any change that touches the control layer happens in one of those windows, and there are not many of them.** Operational technology patches when the plant is down and the vendor has validated the patch against that firmware revision — not on Tuesday.

The wage and cost-roll dates. Manufacturing wages in the metals and engineering sector are set by the bargaining council rather than by the firm, and the increase lands on 1 July. Highveld rolls standard cost on 1 July to align with it, which means the cost roll and the wage increase arrive together and the margin step is unattributable without care. Karoo Foods rolls on 1 March. **Two cost rolls, two blackout periods, and neither is negotiable.**

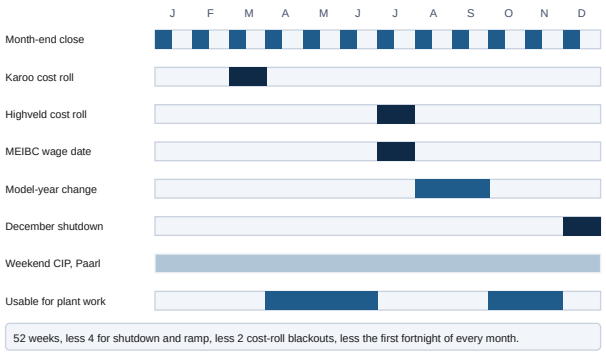


Figure 18.8 The roadmap against the real calendar. A delivery year that needs plant access is about thirty-two to thirty-five usable weeks, not fifty-two.

The honest arithmetic. Fifty-two weeks, less four for the December shutdown and its ramp down and up, less the two cost-roll blackouts, less the first fortnight of each month for anything needing the plant controller, less the weeks around the model-year changeover at Highveld. For work that needs plant people and plant access, a delivery year is about **thirty-two to thirty-five usable weeks**. For work that needs only head office, it is about forty-four. A plan that assumes forty-eight will be eight weeks late by March and will be described as a delivery failure rather than as an arithmetic error.

One more constraint, and it is structural rather than calendrical. **The scarcest person in this programme is not an architect.** South Africa's preliminary 2026 critical-skills results put engineers first at 29 per cent of surveyed employers, with information and communications technology specialists and artisans joint second at 25 per cent; the engineering category includes automation and industrial engineers, and the artisan shortage includes millwrights and electricians [14]. Those are exactly the people who execute everything below Level 2. A roadmap that assumes a control systems engineer is available in the quarter you need them has assumed the thing least likely to be true.

What the architect hands over

At the end of the first quarter, Ingrid handed over seven things.

1. **The two plant structures and the mapping between them** — the plant-structure registry, as a governed artefact with an owner, not as a diagram.
2. **The measure definition register**, with twenty-two measures, an owner each, and a comparability scope stated for every one. Four measures are marked "not comparable across plants" on the face of the register.
3. **The item master reconciliation**, with the duplicate population counted rather than estimated, and the arbitration process written and agreed.
4. **The eight findings**, each with the evidence, the exposure shape, and the name of the person who can quantify the exposure from the business's own records.
5. **The six decisions**, each framed so that it can be taken in one meeting, with the consequence of deferring each one written beside it.
6. **Three transition states with verifiable exit criteria**, and a roadmap drawn against the real calendar.
7. **A one-page statement of what is not known**, dated, with what would settle each item. Including the phase-two rebate detail, the regulator's position on employment-context biometric consent, and the mycotoxin regulation whose scope has to be read from the gazette rather than from a secondary list.

And four things she deliberately did not do.

She did not recommend replacing the ERP. The question she was asked. The honest answer is that the decision cannot be taken until the item master question is settled, because the answer changes depending on it, and a recommendation made before the decision would be an expensive guess dressed as advice.

She did not build a dashboard. Asked for one twice. The measure definition register came first, which made the second request easy to decline and the third request unnecessary.

She did not propose to replace Loomhouse. The Highveld MES is adequate and it is the only system holding a complete production genealogy. Replacing it before the item master is settled would break the one trace that works.

She did not model the documented process. The documented process at Gqeberha is three years old and describes a line that was re-laid in 2024. She modelled what the shift supervisor does, which differs, and put the difference in the findings — because the gap between the documented process and the actual one is not a defect to be corrected on paper. It is the requirement.

THE CHAPTER IN TEN LINES

- Nothing about Karoo Industrial Group is real; every general fact used to reason about it is sourced and dated.
- The brief you are given names the symptom the executive can see. The first deliverable is a description of what is actually true.
- Two archetypes under one roof is the normal case, and a design that fits one fails in the other structurally, not in detail.
- Every finding needs evidence a reader could check in an afternoon, or it is an opinion with a template round it.
- The rebate regime is a customs duty-rebate-certificate scheme that demands country-of-origin and cost-component split at bill-of-material level, propagated through several supply tiers.
- Three plants with three availability denominators are all arithmetically correct, which is why the first deliverable is a measure definition register and not a dashboard.
- Historian compression means the archive does not contain what the instrument measured, so the reported number and the recorded number differ legitimately.
- Ownership of inventory is the one design error that cannot be fixed later, because you cannot retrofit ownership onto historic stock movements.
- Write transition states with exit criteria somebody can verify, and put the decisions only the executive can take on a single page.
- A delivery year that needs plant access is about thirty-two to thirty-five usable weeks once shutdowns, cost rolls and month-end close are taken out.
- Say what you deliberately did not do, and why, because that list is where your judgement is visible.

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CHAPTER 19

Your first ninety days

The chapter you will come back to. What to look at in week one, what to count before you model anything, what to put in front of the executive committee and when — and the four things not to do, each of which has cost somebody a year.

Week one: see the plant

Not the meeting room. The floor.

An architect's instinct on day one is to ask for the application inventory, the integration list and an hour with the chief information officer. In manufacturing that instinct is wrong, and it is wrong in a way that costs you the whole engagement, because the documents you will be handed describe a plant that does not exist. The schedule in the system is not the schedule the plant runs. The process in the quality manual is not the process on the line. The asset hierarchy in the maintenance system is not the steel on the floor.

So go and look. Arrange a walk on shift, with a shift supervisor, in the clothes you are told to wear, and include a night shift. Night shift is where the data gets entered from memory, where the stand-in setter runs the changeover, and where the workarounds are least disguised.

What to look at, in roughly this order.

Walk the flow, not the org chart. Follow one product from goods receipt to despatch. Where does material physically stop? Every stop is a buffer somebody chose or a constraint nobody admitted. Where does it change identity — from coil to blank, from bulk to pack, from lot to serial? Every identity change is a data interface, and in most plants at least one of them is a clipboard.

Look for paper. Paper on a plant floor is not backwardness; it is a map of the gaps. A laminated sheet on a wall is master data that no system holds. A printed schedule with handwriting on it is the real schedule. A tally sheet by a machine is a measurement the system does not take. Photograph them, with permission, and bring them to the workshops. They are the most honest requirements document in the building.

Look at the screens. How many are in use, how slow are they, and how many fields does an operator fill per unit? **Operators are the primary data-entry population in any manufacturing execution system**, which means every field you add costs cycle time and will be degraded or circumvented if it costs too much. An operator entering a reason code from a dropdown of ninety options will pick the first one, forever.

Look at the changeover. Watch one. Time it. Notice who does it — in a high-mix plant the setter is a scarce, high-skill bottleneck role, and nobody knows more about changeover loss. Ask what the sequence rules are. If the answer is "Jannie knows", the changeover matrix is in a person.

Look at the board. Most plants have a visual management board. Read it, and then compare it with the corporate pack. Where they disagree, the board is usually right and the pack is usually what gets reported.

And ask three questions of everyone you meet on the floor. *What goes wrong most often? What do you have to do that the system does not know about? What would you fix first?* The answers are specific, they are consistent across shifts, and they are the requirements.

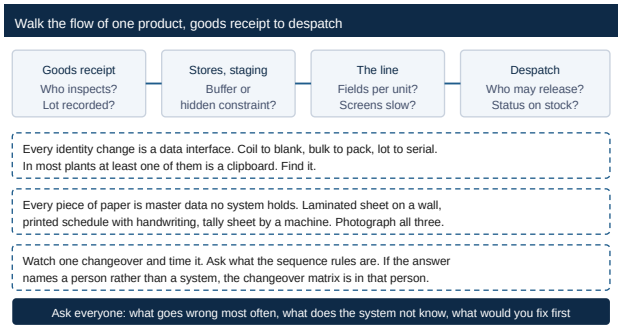


Figure 19.1 The plant walk, in order. Follow one product, and treat every stop, every identity change and every piece of paper as a finding.

> Do not interview the plant manager first. They are the most important interview and they should be one of the last, because you want to arrive with specific questions and a floor walk behind you. Arriving with general questions wastes the one hour you get, and it tells them you have not been outside.

The interview order

Eleven interviews, in this order, each chosen for what it reveals rather than for seniority.

Order	Role	What you are actually getting
1	Shift supervisor, plus the floor walk, on shift	The real process, the real data capture, the workarounds
2	Production planner or scheduler	Where the plan breaks, and which spreadsheet is the real system
3	**Maintenance planner**	Asset hierarchy quality, failure coding quality, and where everything actually is
4	Industrial engineer	Whether the routing and standard times are current, which is the foundation of costing
5	Plant controller	The costing model, the absorption base, the variance population
6	Quality manager and the lab	The release process, the status-of-stock model, lab turnaround as lead time
7	Demand planner	Forecast grain, horizon, bias, and the override culture
8	Master data owners, plural	The governance gap, because there is no single owner
9	Safety, health, environment and quality officer	The statutory reporting obligations nobody put in scope
10	Plant manager	Validation, priorities and sponsorship
11	Shop steward forum	What will stop the project

The maintenance planner is the single most useful first interview in most plants. Not the most senior, not the most political — the most useful. They convert maintenance requirements into executable work: scope the job, secure the spares, book the window with production, close out the history. To do that they must know where every asset is, what state it is in, what it is called in three systems, and which of those names is right. They are the primary user of the maintenance system and therefore the only person who can tell you whether its asset hierarchy is usable or decorative.

Four things they will tell you in the first hour. Whether failure history is coded or free text — and it is usually free text, which means it is unanalysable, which means the plant has a work-order backlog rather than a maintenance strategy. Whether equipment and functional location are kept apart, because collapsing them loses either the position's failure history or the object's rebuild history. What the real spares availability is, as against what the system says. And which assets are nursed rather than reliable, which is the answer to the capacity question nobody else will give you straight.

Two more notes on the list. The **shop steward forum** is not stakeholder management; elected worker representatives have statutory rights, and a system that measures individual operator performance, monitors location or changes manning is a consultation matter. Raise it early, in their own forum, and be straightforward about what the system will and will not measure. The alternative is discovering the objection at go-live. And the **safety, health, environment and quality officer** is now a significant data consumer and producer — statutory appointments, incident records, exposure monitoring, emission licence conditions, water use licence reporting, waste manifests, carbon activity data — and is almost never in scope of a manufacturing systems programme. They should be.

What to count before you model anything

Ten checks. Each takes an afternoon or less. Together they change the engagement, because each one either finds a defect that invalidates a design decision or removes a risk you would otherwise carry for a year. **Count, do not ask.** "Yes, we have one item master" is an assertion; two extracts and a comparison is a count.

1. Does the item master exist once, or more than once — and do the instances agree? Extract each instance and compare on description, base unit of measure, purchasing unit of measure and supplier part number. Count the duplicates rather than estimating them. The item master is the hardest master data object in any industry, and it is harder in manufacturing because it is co-owned by six functions who each need the record to exist before they can supply their part of it. Nobody owns the record; different functions own different fields, and what must be owned centrally is the process, the rules and the arbitration.

2. Is there a measure definition register? Ask for it by name. If there is not one, **that is the first deliverable, not a dashboard.** The reason is specific: there is no OEE standard, the research behind this book found six incompatible availability denominators in circulation and a seventh divergence in where the micro-stop threshold sits [1][2], that threshold shifts loss between availability and performance without changing OEE [2], and OEE is explicitly documented as unsuitable for comparing machines, for benchmarking and for aggregating to department or plant level [2]. A line's OEE is not the mean of its stations'. Also get the vocabulary right: ISO 22400-2's middle term is **Effectiveness**, not Performance [3], the published edition is 2014 and the second edition was still at draft stage in October 2026 [4]. And never put the 85 per cent figure in a deliverable as a benchmark — the site that publishes it calls it a convenient, compelling and completely artificial benchmark [1].

3. Is the time fence shorter than the cumulative material lead time? Take one representative finished item, explode its structure, and add up the longest purchase lead time plus the manufacturing lead time at each level. Compare that to the frozen and firm fence lengths in the planning configuration. **If the fence is shorter than the cumulative lead time it is arithmetically unenforceable** — it is the commonest configuration error in this area and it is checkable in an afternoon. Fence lengths are usually set from a template at implementation and never validated against actual lead times and changeover constraints.

4. Does the system overwrite the promised date on reschedule? Open one order line that has been rescheduled and look for the original promise. If the field has been overwritten, **on-time delivery cannot be measured at all** — and this is extremely common. An order line needs, at minimum, the customer request date, the first promise date, the current promise date, the despatch date, the delivery date, the quantities ordered, confirmed, despatched and received, and a reason code for every variance, all retained rather than overwritten.

5. Does the plant backflush? Ask the confirmation strategy. Backflushing consumes the standard bill-of-material quantities on reporting the finished quantity, which means it **systematically hides material usage variance** — over-issues and unreported scrap cannot appear as a variance and surface later

as stock-count differences. **If the plant has unexplained inventory shrinkage and backflushes, you have found the mechanism.** And where substitution occurs, backflush produces a confidently false genealogy, which is worse than a gap.

6. Is ownership an attribute of inventory? Ask whether there is any tolling, contract manufacturing, consignment or vendor-managed stock. If there is, and ownership is not a field on the stock position, stop and raise it immediately. In a tolling plant the material belongs to the customer; stock in one silo may be owned by three parties; the same tonne may be the plant's asset on Monday and a customer's on Tuesday. **This is one of the very few genuinely unrecoverable design errors**, because you cannot retrofit ownership onto historic stock movements.

7. Are scrap factors applied as $1 \div (1 - s)$, or as $(1 + s)$? The second is wrong. To get 100 good units at 10 per cent scrap you must start $100 \div (1 - 0.10) = \mathbf{111.1}$, not 110. At one level the error is about 1 per cent; compounded at 3 per cent over five levels the correct uplift is about 15.9 per cent against the naive 15 per cent. If the standard cost is built on one formula and the plant consumes according to the other, the result is a permanent, structural material usage variance that will be investigated every month for years and never explained. While you are in there, check two more things: whether component scrap, assembly scrap, operation scrap and yield are kept as four separate numbers, and whether the phantom indicator sits on the bill-of-material **line** or on the item. **Phantom is a property of the line, not of the item** — the same item is legitimately a phantom in one parent and a stocked assembly in another, and an item-level flag makes that impossible.

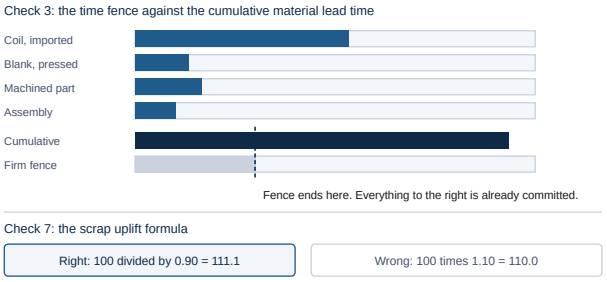


Figure 19.2 Two arithmetic checks that take an afternoon. A fence shorter than the cumulative material lead time cannot be enforced, and the scrap uplift formula is wrong more often than it is right.

8. Do the ISA-95 equipment model and the ERP plant hierarchy reconcile? Put the two trees side by side and try to map the nodes. Expect them not to match, because they are built for different purposes: the ERP's structure follows accounting and legal boundaries, and ISA-95's follows physical and operational decomposition. Two corrections to carry into that conversation. **ISA-95's equipment-element level enumeration has fifteen values**, and the middle branches — process cell and unit; production line and work cell; production unit; storage zone and storage unit — are **alternatives selected by process type, not levels stacked on each other** [5]. And **equipment module and control module are ISA-88 concepts that ISA-95 references**, not ISA-95's own [5]. The deliverable is a dated many-to-many mapping held in a governed plant-structure registry, not an attempt to make one tree the parent of the other. Remember there is a third tree, the asset and maintenance hierarchy, and a fourth, the fixed-asset register.

9. What are the historian's exception and compression settings, and does anyone know? Ask for them on the twenty tags that feed the most-quoted number in the plant. Then ask when they were last changed and by whom. Historians apply exception reporting at the interface and swinging-door compression at the archive; the defaults are non-zero, which means compression is on unless somebody turned it off, and out-of-order data bypasses compression so the same tag has two fidelities in one archive [6]. The magnitudes are not marginal: a nought-or-one status tag over forty-eight hours at one-second scans is **172,800 events uncompressed and can be fifteen events with minimal compression** [6]. The consequence is that the archive does not contain what the instrument measured, every query against a time grid is interpolating, and two reports of the same thing can both be right and differ. If nobody knows the settings, the number on the executive dashboard has no stated fidelity.

10. What does the retention policy say, and which clocks does it breach? Ask for the corporate retention policy. It will give a single period. **A single corporate retention period is always wrong**, because a manufacturer sits under at least eight different statutory clocks:

Record	Period	Instrument
Noise exposure, audiometry and medical surveillance	**40 years**	Noise-Induced Hearing Loss Regulations [7]
Chemical exposure assessments and air monitoring	**30 years minimum**	Hazardous Chemical Agents Regulations 2021 [8]
Pressure equipment manufacturing records	**12 years minimum**	Pressure Equipment Regulations [9]
Lifting equipment full service history	**At least 10 years**	Driven Machinery Regulations 2015 [10]
Company records and accounting records	**7 years**	Companies Act 71 of 2008, s 24 [11]
Tax and VAT records	**5 years from submission**	Tax Administration Act; VAT Act [11]
Employment records, collective agreements	**3 years**	BCEA s 31; LRA s 205 [11]
Incident records and committee records	**At least 3 years**	General Administrative Regulations [11]

So a five-year policy breaches the Companies Act, and a seven-year policy breaches four occupational-health instruments. And the two longest clocks sit on data held in occupational health systems that are replaced every seven to ten years, which means **the retention obligation outlives the application by a factor of four or five.** A retention design that assumes the system will still exist is not a design. The requirement is an export format, with its reference data, readable without the source application — cheap at the start, impossible to reconstruct later.

And one question to ask before you leave the first plant

Who inspects you, and what do they ask to see? Nobody in information technology will know, and almost nobody will have been asked. The safety, health, environment and quality officer will know, and the answer reorders your scope.

A South African manufacturer is inspected by, at least: a labour department inspector on machinery, chemicals, noise, pressure equipment, lifting equipment and incidents; a separate inspection regime if the site is a major hazard installation; the Compensation Fund on registration, assessments and accident reporting, with contractor liability now in scope; an environmental management inspector on the authorisation and **every one of its conditions**; an air quality officer on the emission licence, including whether continuous monitoring yielded at least 80 per cent valid hourly values and whether the audit by an accredited laboratory happened; the water authority on abstraction volumes and effluent quality; the compulsory specifications regulator on regulated products, with the right to require samples at the supplier's cost and with a test result generally valid for the whole batch; the legal metrology regulator on measuring instruments and prepackage net content; a local environmental health practitioner on food premises; the customs authority on classification, origin and duty relief; a bargaining council agent on grades and rates; the labour department on employment equity reporting; and the Information Regulator on lawful processing.

Three architectural consequences fall straight out of that list. **Batch size is an exposure decision, not a production convenience**, because a test result is generally valid for the whole batch. **The emissions compliance record belongs on the historian, not on a relational extract**, because the obligation is expressed as a percentage of valid hourly values. And **two obligations turn out to be bounded by the item revision rather than by the item** — the regulated-product register and the label — which means the product regulatory register must bind to a revision or it cannot answer the inspector's question.

None of this is legal advice and your deliverable must say so. What it is, is a list of record sets that must exist, with owners and clocks, and that list is architecture.

The ninety-day shape

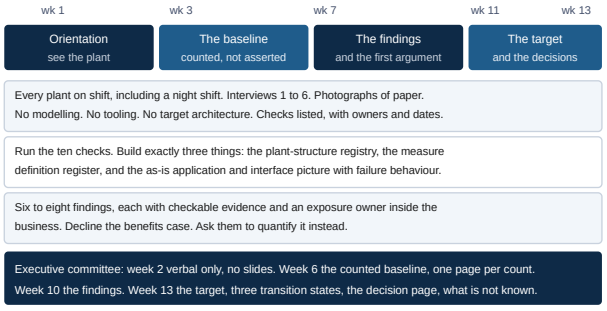


Figure 19.3 The ninety-day shape. Two weeks on the floor, four weeks counting, four weeks arguing, three weeks designing — and four executive conversations, each sized to what you can honestly carry.

Weeks 1 and 2 — orientation and the plant walk. Every plant, on shift, including a night shift. The first six interviews. No modelling, no tooling, no target architecture. The output is a set of notes, a set of photographs of paper, and a list of the ten checks with owners and dates. If somebody asks for an architecture in week two, the answer is that you have not yet seen the second plant.

Weeks 3 to 6 — the baseline, counted not asserted. Run the ten checks. Build three things, and only three: the **plant-structure registry** holding both hierarchies with an explicit dated mapping; the **measure definition register**; and the **as-is application and interface picture**, with the interface register's most important column being failure behaviour, because the plant must keep producing when the corporate network is down. Everything in the baseline carries a count and a date. Nothing carries an estimate.

Weeks 7 to 10 — the findings and the first argument. Six to eight findings, each with the evidence, each with the exposure shape, and each with the name of the person inside the business who can quantify the exposure from the business's own records. Do not quantify it yourself. This is also the window in which you will be most tempted to accept a benefits-case request, and declining it is the single most credibility-preserving decision of the quarter.

Weeks 11 to 13 — the target, the transition states and the decisions. The target architecture at the level the decisions require and no further. Three transition states with **written, verifiable exit criteria** — something somebody can check, not something somebody can assert. And a single page of decisions that only the executive committee can take, each framed so it can be taken in one meeting, with the consequence of deferral written beside it.

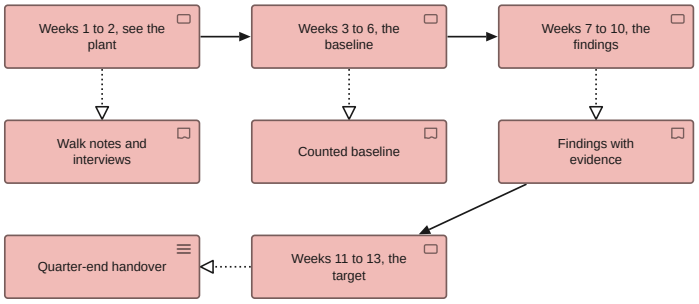


Figure 19.4 The quarter as work packages and deliverables. Only two of the twenty checklist items are modelling, and the plateau at the end is a handover, not a solution.

What goes in front of the executive committee, and when.

- **End of week 2:** nothing written. A verbal readout of what you saw, five minutes, no slides. It buys you the next four weeks.
- **End of week 6:** the counted baseline. One page per count, with the number and the method. This is the meeting where the engagement is either believed or not, and it is believed because everything in it is a count.
- **End of week 10:** the findings, with exposures the business will quantify. Ask for the quantification in that meeting, with owners and dates.
- **End of week 13:** the target, the three transition states with exit criteria, the roadmap drawn against the real calendar, the decision page, and a one-page statement of what is not known.

And plan all four of those around the plant's own rhythm. A monthly sales and operations planning cycle occupies the whole month; the plant controller is unavailable for the first fortnight of every month because of the close; changes touching the control layer happen only in shutdown windows, and there are not many of them. **Operational technology patches when the plant is down and the vendor has validated the patch against that firmware revision — not on Tuesday.**

What to say when you do not know

Manufacturing's standards and regulatory landscape is genuinely uncertain in places. The research behind this book carries more than two hundred items marked unverified, and that is not a failure of diligence — it is what an honest search of this subject returns. Standards move between editions without secondary sources noticing. Commencement notices for repealing regulations cannot always be confirmed as issued. Regulation numbers are reused across different instruments and years, so a number alone is not an identifier: it takes a number, a year and a subject.

So learn to say this, out loud, in front of a client: **"I do not know. I will find out, and I will record the date I found it."**

Then make the dated, sourced answer the deliverable. A line in a register that reads *"ISO 22400-2: published edition is 2014; second edition at draft stage; checked 16 October 2026"* is worth more than a confident sentence, because it can be re-checked by somebody else in two years and because it tells the reader exactly how much weight to put on it. Every assertion in an architecture that rests on a standard, a regulation or a vendor's ownership should carry the date it was checked. Vendor ownership in this market changes two or three times a decade, and an undated ownership claim in a deliverable is a liability.

And one specific instruction, because it saves a month. **Do not look for the manufacturing BIAN. There is not one.** Unlike banking and insurance, manufacturing has no single maintained, vendor-neutral industry reference model covering the business end to end. What it has is a control-systems hierarchy standard, a batch-control standard, three competing interoperability protocol families from three different bodies with no reconciliation between them, an identification system, engineering exchange formats, and a stack of sector quality schemes — all maintained by different organisations and covering different ground. An architect who spends the first month of a manufacturing engagement looking for the equivalent of a banking reference model has wasted the month.

The constructive answer is to build the house reference model from four sourced inputs: **ISA-95 object names for the ERP and manufacturing-execution boundary** [12], **the applicable quality scheme's record obligations for the data-ownership matrix**, **ISO 22400 for measure definitions** [3][4], and **ISA and IEC 62443 zones and conduits for technology and security** [13]. Then declare in the architecture principles that no single industry reference model governs. That declaration is itself a deliverable, and it stops the question being re-asked every quarter.

What not to do

Do not promise a dashboard in month one. You will be asked, twice. The measure definition register comes first, and once it exists the second request becomes easy to decline and the third becomes unnecessary. A dashboard built over undefined measures is rebuilt, and you pay for it twice.

Do not propose to replace the manufacturing execution system before the item master is settled. The execution system is frequently the only place holding a complete production genealogy. Replacing it while the master data question is open breaks the one trace that works, and it is the most common sequencing error in manufacturing programmes.

Do not model the documented process. The documented process describes the line as it was when somebody last had time to write it down. Model what the shift supervisor does, and put the difference between the two in the findings. **The gap between the documented process and the actual one is not a defect to be corrected on paper. It is the requirement.**

Do not assume the people below Level 2 are available. South Africa's preliminary 2026 critical-skills results put engineers first at **29 per cent** of surveyed employers, with information and communications technology specialists and artisans joint second at **25 per cent**; the engineering category

includes automation and industrial engineers, and the **artisan shortage is millwrights and electricians** [14] — exactly the people who execute everything you design below Level 2. A roadmap that assumes a control systems engineer will be free in the quarter you need them has assumed the least likely thing in it.

Do not quote a number you cannot source. Three figures circulate in South African manufacturing material — a rand cost of inventory shrinkage, a rand cost of power interruption, and a count of manufacturing jobs lost — and none of them survives a check. Name the survey whenever you give an employment number, because the two official series disagree with each other and both disagree with the circulating claims.

And do not write load-shedding in the present tense. It ended: 504 consecutive days without it as at 2 October 2026 [15]. But *load reduction* is a separate, continuing programme, and the real power issue for operational technology is voltage quality rather than availability — for which there is a national standard that assigns immunity in one dip window to the customer, names the windows in which dips trip drives and contactors, and publishes characteristic unplanned interruption counts with no compliance criteria attached [16]. Conflating the two will date your document in a sentence.

The ninety-day checklist

Print this. One page.

#	Do this	By when	Evidence it is done
1	Walk every plant on shift, including a night shift	Week 2	Walk notes; photographs of every laminated sheet
2	Interview the shift supervisor, planner and maintenance planner	Week 2	Three interview records with named workarounds
3	Count the item master instances and their disagreements	Week 4	A duplicate count, with the method stated
4	Ask for the measure definition register	Week 3	Either the register, or it is now deliverable one
5	Add the cumulative material lead time and compare to the fence	Week 4	One worked example per division
6	Check whether the promised date is overwritten on reschedule	Week 4	One rescheduled order line, inspected
7	Establish the confirmation strategy and whether the plant backflushes	Week 4	Named strategy per plant, with the shrinkage position
8	Establish whether ownership is an attribute of inventory	Week 3	Yes or no, in writing, with the tolling scope
9	Check the scrap formula, the four scrap numbers, and the phantom flag	Week 5	The formula actually in use, with a worked uplift
10	Map the ISA-95 tree to the ERP plant hierarchy	Week 6	A dated node-to-node mapping, with the gaps named
11	Get the historian exception, compression, step and retention settings	Week 5	Settings per tag class, and who may change them
12	Compare the retention policy to the eight statutory clocks	Week 5	A breach list, instrument by instrument
13	Build the plant-structure registry	Week 6	Both trees, one mapping, one named owner
14	Publish the counted baseline to the executive committee	Week 6	One page per count, no estimates
15	Write six to eight findings, each with checkable evidence	Week 10	Each with an exposure owner inside the business
16	Ask the business to quantify the exposures	Week 10	Owners and dates agreed in the meeting
17	Agree the autonomous-running period per plant, and test it	Week 12	A number of hours, and an uplink pulled deliberately
18	Write three transition states with verifiable exit criteria	Week 13	Criteria somebody can check, not assert
19	Put the decisions only the executive can take on one page	Week 13	Each takeable in one meeting, with deferral cost
20	Write the one page of what is not known, dated	Week 13	What would settle each item, item by item

> Twenty items, and only two of them are modelling. That ratio is right. In manufacturing the model is cheap and the truth is expensive, and the quarter is spent buying the truth.

THE CHAPTER IN TEN LINES

- Week one is the floor, on shift, including a night shift. The documents you are handed describe a plant that does not exist.
- The maintenance planner is the single most useful first interview in most plants, because they know where everything actually is.
- Interview the plant manager late, with a floor walk behind you and specific questions in hand.
- Count, do not ask. Ten checks, each an afternoon, each capable of invalidating a design decision.
- If there is no measure definition register, that is deliverable one, because there is no OEE standard and six incompatible availability denominators plus a micro-stop threshold divergence are in circulation.
- A time fence shorter than the cumulative material lead time is arithmetically unenforceable, and a system that overwrites the promised date destroys on-time measurement entirely.
- Backflushing plus unexplained shrinkage means you have found the mechanism; missing inventory ownership in a tolling plant is the one error that cannot be fixed later.
- Scrap uplift is 1 divided by 1 minus s, never 1 plus s, and the error compounds down the structure.
- A single corporate retention period is always wrong: a manufacturer sits under at least eight statutory clocks, the longest being 40 years for audiometry.
- Say I do not know, then make the dated sourced answer the deliverable. There is no manufacturing BIAN, and looking for one wastes a month.
- Do not promise a dashboard in month one, do not replace the execution system before the item master is settled, and do not model the documented process.
- The artisan shortage is millwrights and electricians, who are exactly the people who execute what you design below Level 2.

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APPENDIX A

The cross-maps

Ten tables that let you translate between the vocabularies a factory will throw at you. None of them is a standard, each is wrong at the edges, and every one is worth more than a glossary entry.

Ten translation tables. None is a standard. Each is the mapping an architect ends up drawing on a whiteboard in the second week, so every map says what it is for and where it breaks down. Use them forward, to place a name you have just heard; and backward, to check a design uses the word the plant uses, in the plant's sense. The second use earns its keep.

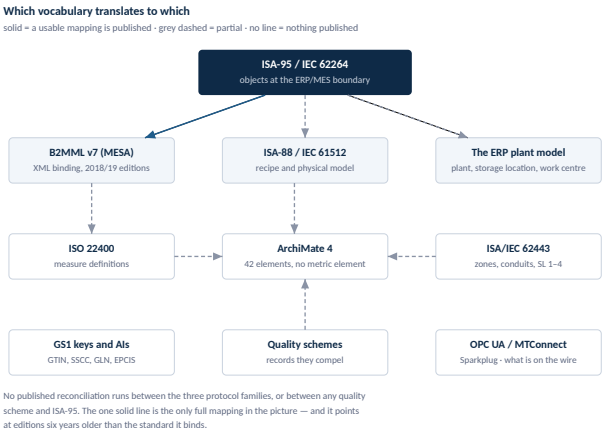


Figure 20.1 Which manufacturing vocabulary translates to which, and where nothing is published

Map 1 — ISA-95 objects to ArchiMate 4 elements

For: modelling a manufacturing boundary in ArchiMate while the client talks ISA-95. **Breaks down because:** ISA-95 objects are information structures; ArchiMate elements are architecture concepts. Some split across two elements, three have no home, one is not drawable.

ISA-95 object	ArchiMate 4 element	Fit, and what to watch
Personnel; personnel class	Business Actor; Role	Good. "Operator" is a class, so a Role. Name the job as the industry names it
Equipment; physical asset	Equipment	Good, with a caveat: ISA-95 separates the serial-numbered machine from its hierarchy position; ArchiMate does not
Site, area	Facility and Location	Good. Facility for the fence, Location for the geography
Operations capability	Capability plus Resource	The cleanest alignment in the set
Process segment	Process plus Capability	Work *and* what it needs: Realization between them
Material definition; material lot; material class	Business Object; Material; Grouping	Loose throughout. The definition is information and the stuff is Material; Material is typeless, so a lot and a tonne look identical; and Grouping carries no semantics
Product definition	Business Object	**Not Product.** ArchiMate's Product is a bundle of services and contracts sold to a customer. The commonest ArchiMate error in manufacturing models
Operations definition, work master	Business Object	Loose. Version, approval and effectivity are attributes no diagram shows
Operations schedule and performance	Business Object plus Event	The schedule is an object, releasing it an event. Draw both, in both directions, or you cannot show why planner and plant disagree
Equipment hierarchy node	Equipment, Composition	Right until lines branch and merge, which Composition cannot express
A measure or KPI	**none**	**No metric element exists, and Value is not a number** — Value models worth to a stakeholder. Use a Business Object for the definition, a Requirement for the target, attributes for the formula
Genealogy	**none**	A graph over *instances* — this lot, that serial, those four pallets. ArchiMate models types. **Not drawable.** Model the capability; draw an example as a plain figure
A time model, an effectivity window	**none**	Neither a typed partition of calendar time nor a date on a relationship is expressible. Use a versioned Business Object, and accept that two revisions cannot share a diagram

> Put this in the model's documentation: **the notation carries the structure and not the arithmetic.** Every measure, effectivity date and genealogy link is an attribute, a register or a separate figure — never an element.

Map 2 — The ISA-95 equipment hierarchy to the ERP plant hierarchy

For: the reconciliation project, whose first deliverable is this table filled in for the client's plant. **Breaks down:** everywhere — ISA-95's hierarchy exists for physical decomposition, the ERP's for valuation.

Two corrections, because almost every published version gets both wrong. **It is not a seven-level stack ending in control module:** the equipment-element level enumeration has **fifteen** values [1], and below area four branches are **alternatives selected by process type, not levels stacked on each other** [1]. `WORKCENTER` and `WORKUNIT` are the generic terms, which is why "work centre" means related but non-identical things here and in an ERP. And **equipment module and control module are ISA-88 concepts referenced by ISA-95, not ISA-95's own** [1].

ISA-95 element	For	Nearest ERP concept	Where it does not reconcile
Enterprise	all	Client, group	Group reporting nets out inter-company transfers
Site	all	Company code, plant	A company code is a set of books, a site is a fence; either holds several of the other
Area	all	Plant, production or MRP area	One ERP plant usually covers several areas, so plant measures cannot be decomposed
Process cell	batch	nothing	A recipe boundary with no ERP object; faked with a work-centre group
Unit	batch	Work centre, resource	One unit maps to several work centres, so downtime needs an attribution rule
Production line	discrete	Work centre, planning line	A planning line is not the routing work centre's grain
Work cell	discrete	Work centre	The nearest thing to a genuine match
Production unit	continuous	Work centre, resource	No discrete operations exist, so the routing is a fiction kept to make the ERP post
Storage zone	logistics	Warehouse number, storage type	The WMS structure sits below the ERP's, with its own identity
Storage unit	logistics	Storage location, bin	**Storage location has no ISA-95 equivalent** — an accounting subdivision, so a bin has no accounting home
Work centre, generic	all	Work centre	Same word: a grouping of equipment against a costing construct with a rate
Work unit, generic	all	nothing	No ERP object at this grain
Equipment module; control module	batch, ISA-88	nothing	They live in the control system and the PLC, so tag-to-asset binding has no ERP anchor at all
Other	all	nothing	The escape hatch, where awkward assets end up

Two structures are missing from both columns: the **cost centre**, which cuts across the physical tree because cost responsibility follows people, and the maintenance **functional-location tree**, a third tree diverging the first time a machine is moved. What works is one governed registry holding all three as typed trees plus a dated, many-to-many mapping between their nodes.

Map 3 — The four bills of material, side by side

For: working out which BOM someone means and who to ask. **Breaks down:** two structures people call BOMs are not BOMs, and the conflation is the most damaging error in the subject.

BOM	The question it answers	Owner	What it holds that the others do not
Engineering (eBOM)	What is this product, as designed?	Engineering, in PLM	Functional decomposition, drawing references, specifications, revision. Often no quantities in the manufacturing sense
Manufacturing (mBOM)	What does the plant consume, in what sequence, at which operation?	Manufacturing engineering	Consumables, packaging, phantoms, scrap and yield factors, operation assignment, alternates. Often re-levels the eBOM entirely
Planning (pBOM)	What do we plan for before the variant is known?	Demand planning	Option take-rates summing to 100%, not necessarily integers. **Never used to issue material**
Service (sBOM)	What can be ordered, replaced and repaired in the field?	Service and aftermarket	Kits that are not manufacturing items, supersession chains, interchangeability. The longest retention horizon

Each has its own grain: the eBOM decomposes as the drawing does, the mBOM as the process builds, the pBOM nominally, the sBOM as a technician strips the unit down. The **costing BOM** is the mBOM with a nominated alternative; the **master recipe** is the process industry's mBOM, with parameters inseparable from quantities that scale non-linearly with batch size.

> **As-built is not a kind of BOM — it is genealogy.** A BOM is a statement about a *kind* of thing; as-built is a statement about *one* thing, with the actual lots and serials consumed. Different grain, different volume class — millions of transactional rows against thousands of master rows — different retention driver, different consumers. **As-maintained** is a third thing again.

A "single source of truth BOM" programme that tries to make eBOM and mBOM one structure will fail and be re-run under another name in four years; the achievable goal is a governed, auditable mapping. And **phantom is a property of the BOM line, not of the item** — the same item is legitimately a phantom in one parent and a stocked assembly in another.

Map 4 — System class to what it masters

For: the data-ownership argument, which consumes more manufacturing architecture time than anything else. A default to argue with, not to apply. **Breaks down:** in a mixed-mode group the answer differs by plant.

Object	System of record	Also claims it
Item master	ERP	PLM, MES, WMS
Engineering BOM, revision, effectivity; manufacturing BOM and routing	PLM; ERP	ERP, PLM, MES
Master recipe; control recipe	PLM or a recipe manager; MES or DCS	MES, DCS
Work centre, resource, capacity	ERP	APS, MES
Production order header and state	ERP	MES
Operation state, confirmation	MES	ERP
Material lot, batch, serial	MES, or ERP without one	WMS, LIMS
Genealogy	MES	—
Stock by bin, pallet, handling unit	WMS	ERP, MES
Stock by stage, status and value	ERP	WMS, MES
Quality status of stock	ERP, set by LIMS or QMS	MES
Functional location, equipment, PM plan, maintenance work order	EAM or CMMS	ERP, MES, historian
Sample, result, specification, CoA; non-conformance, CAPA, deviation	LIMS; QMS	ERP, MES
Tag time series; live process state and alarm	Historian; SCADA or DCS	IIoT platform
Finite-capacity schedule and sequence; consensus demand plan	APS; the planning tool	ERP, MES, a spreadsheet
Serial, aggregation, EPCIS event	Traceability layer	MES, WMS
Measure definitions	**Nobody**	Everybody computes them

Every row has a settling question, always of the form *who may assert this, and who may only cache it?* Three objects are habitually mastered in more than one place and a fourth nowhere. **The item master** — ERP creates the code, PLM the part, MES whatever it confirms against; in a two-instance ERP estate it is not unique within the group, so two codes exist for one physical thing with different units of measure. **The bill of material** — PLM owns the eBOM, the ERP the mBOM, the MES a third structure at the line, and the manual transformation surfaces at a recall. **Equipment** — three unmapped trees over one set of machines, so energy, downtime and maintenance cost report against three denominators. And **the measure definition is mastered nowhere**: OEE is computed in three places with three denominators, which is why the first deliverable on a group reporting programme is a measure definition register.

Map 5 — Measure to definition to source

For: filling in with the client's own names and their own systems. **Breaks down:** the "diverges on" column is not a footnote — almost nothing here is comparable between plants, and several measures are not comparable between months in one plant.

Two corrections. **ISO 22400-2's middle term is Effectiveness, not Performance**: the standard decomposes OEE as availability times effectiveness times quality ratio [2], while the Nakajima and total-productive-maintenance formulation calls the middle term the performance rate and computes it against a different denominator. And **ISO 22400-2's current published edition is 2014** [3] — the second edition was still at DIS stage 40.99 in October 2026 [4], and Part 1 was *confirmed*, not revised, in August 2025 [5].

Measure	Formula	ISO 22400 term	Diverges on	Needs
OEE	Availability x Effectiveness x Quality ratio	OEE	all three factors, compounded	one governed time model
Availability	Actual production time / planned busy time	Availability	**seven denominators in use**, from calendar time to rostered-less-external-causes, plus the micro-stop threshold	a typed, versioned, reason-coded partition of all calendar time
Effectiveness, usually called performance	(Quantity x ideal cycle time) / actual production time	Effectiveness	ideal cycle time: nameplate, design, demonstrated best, or routing standard	ideal cycle time per item and resource, dated
Quality ratio	Good quantity / produced quantity	Quality ratio	whether rework is a first-pass reject or re-enters as new material	both quantities at one stage boundary
Yield, final	Good units out / units started	—	permits rework, so a repaired unit counts good	a start boundary and a defect definition
First-pass yield	Units passing with no rework / units started	—	98% yield and 92% first-pass yield can be the same plant	per-unit pass and fail at every step
Rolled throughput yield	Product of every step's first-pass yield	—	fixed step count only; a new step mechanically lowers it. Ten steps at 98% gives about 81.7%	first-pass yield per step
Takt time	Available time / demand in that time	—	**not a performance measure** — a demand-derived rhythm that moves when demand moves	a demand figure, a declared available time
Cycle time	three uses	—	operation time, one over the output rate, and elapsed time including waiting, used interchangeably	declared stage boundaries per use
Lead time	four uses	—	manufacturing, customer, cumulative, replenishment. **Cumulative lead time is the real constraint and nobody calculates it**	routing times, BOM depth, supplier lead times, the port
Throughput	Good units or value per unit time	—	**Theory of Constraints means revenue minus totally variable cost.** Both sit in one room	output quantity, or revenue and a variable-cost rule
Capacity	four figures, one usable	—	theoretical, rated, effective, demonstrated. **Only demonstrated capacity enters a plan**, and it is mix-specific	a sustained-output history by mix
On-time-in-full	Orders or lines complete and on time / total	—	four date bases, order or line level, four clock stops. 96% and 78% can be identical performance	five dates per line, never overwritten
Inventory turns	Cost of goods sold / average inventory at cost	—	sales in the numerator inflates turns by the margin; period-end inventory makes turns depend on month-end timing	COGS and average inventory, by stage
Days of supply	Average inventory / (annual COGS / 365)	—	supply chain counts demand days, finance counts cost days; both are called days of supply	inventory split by stage

OEE: two things never to put in a client deliverable. The **85% figure everybody quotes as the global benchmark is an invention, and the site that publishes it says so**, calling it "a convenient, compelling and completely artificial benchmark" [6]; the component values behind it are unverified. And **OEE is documented as unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level** [7] — a line's OEE is not the average of its stations'. Compute it on the constraint, or on the line as one resource. The micro-stop threshold, typically one to five minutes, shifts loss between Availability and Performance **without changing OEE**, so OEE can be comparable where its components are not, which is the opposite of what everyone assumes.

Map 6 — Quality scheme to the records it compels

For: turning a compliance topic into a data-ownership matrix. A scheme that compels specific records on demand, with traceable lineage, constrains the architecture exactly as a regulation does; absent an industry reference model, **the applicable scheme's record obligations are the nearest thing a manufacturer has to a mandated information architecture. Breaks down:** the last column is a practitioner summary, not a clause citation, and numbering moved in 2026.

Scheme	What it is	Who it binds	The records it compels
ISO 9001:2026 — sixth edition, 16 September 2026; the 2015 edition withdrawn [8]	Quality management system	Voluntary, routinely a customer requirement	Documented information under version, approval and retention control; competence per person per role; audit findings and closure; corrective action with root cause; calibration evidence
ISO 14001:2026 — 15 April 2026 [9]	Environmental management system	Voluntary, often a tender requirement	As above, plus measured performance against objectives, with stronger treatment of climate, biodiversity and resource efficiency
ISO 45001:2018 — still the published edition [10]	Occupational health and safety management system	Voluntary, and distinct from the statutory duties	Hazard identification, consultation and participation, incident investigation
IATF 16949:2016 — first edition, still current; second planned mid-2027 [11] [12]	Automotive quality management, supplementing ISO 9001	Any supplier into an OEM chain, so effectively mandatory	Per part number and run: design records, process flow, process failure mode and effects analysis, control plan, measurement system analysis, dimensional and material results, capability studies, samples, submission warrant
AS9100, becoming IA 9100 — series expected early 2027 [13]	Aerospace, space and defence quality management	Aerospace suppliers	Configuration management; traceability including qualified special processes; counterfeit-part prevention; first-article inspection; **retention for the life of the product**, which outlives the systems
ISO 13485	Medical device quality management, built for regulatory use	Device manufacturers; referenced by regulators	Device master and device history record; design history file; risk management; unique device identification; complaints routed back to a batch; **validation of every computerised system in the chain**
FSSC 22000 v6 — published 31 March 2023, mandatory 1 April 2024, recognised 23 August 2024; **v7 dates unverified** [14]	A scheme on ISO 22000 plus prerequisites plus scheme requirements	Food manufacturers whose customers require a benchmarked scheme	Traceability one-up and one-down, retrievable inside a *timed* mock recall; control point monitoring with measurement, limit, person, time, corrective action; allergen and changeover verification; label approval; calibration at every control point
GMP, via the PIC/S guide	Good manufacturing practice, enforced by inspection, not certification	Pharmaceutical manufacturers	The batch record, reviewed and released; computer-generated audit trails independent of the operator; electronic signatures with meaning; validation per system; data integrity; deviation and change control linked to batches

> **Do not cite the South African medicines regulator for GMP detail** — its guideline adopts and adapts the PIC/S guide, and batch-record retention, batch numbering, deviation and change control, product quality review, data integrity and the computerised-systems annex are in the PIC/S guide, not in the local guideline. And **South Africa has no Qualified Person**: the regulator states the PIC/S "Responsible Person" is equivalent to the **Responsible Pharmacist**, who is responsible for batch release. Import the European title and you put a role in the model the organisation does not have.

Map 7 — Regulator to evidence

For: an inspection-readiness workshop. **Breaks down:** this is condensed, every instrument has conditions the table cannot carry, and none of it is legal advice.

Body	What it inspects	What it asks to see
Labour inspector, general	Machinery, chemicals, noise, pressure equipment, lifting, incidents	Legal appointments; incidents investigated in seven days; two-yearly risk assessments; air and noise monitoring by approved authorities; medical surveillance; equipment registers
Labour inspector, major hazard installation	Major hazard installations	Substance inventory with safety data sheets; registration certificate; risk assessment not older than five years by an approved authority; emergency plans, tested annually
Mine health inspectorate	Mine and mine-adjacent facilities	Codes of practice; risk assessments; medical surveillance and exit certificates; hygiene measurements — a different record set, hence site jurisdiction as an attribute
Compensation Fund	Registration, assessments, accident reporting, contractors	Return of earnings; letter of good standing; accident register with seven-day reporting; **sub-contractors' registration and assessment evidence**
Environmental management inspector	Authorisations and the duty of care	The authorisation and compliance with **every condition**; incident and contamination records; audit reports
Air quality officer	Atmospheric emission licence	Licence conditions; the minimum emission standard with any postponement; **data availability against the 80% valid-hourly-values rule**; the two-yearly accredited audit
Revenue service and environment department	Greenhouse gas inventory, carbon tax	Activity data with source measurements; versioned emission factors; allowance evidence — trade intensity, mitigation plan, retired offset serials, budget position
Waste authority; water department	Classification, licensed activities, manifests; water use authorisation	Classification per stream with the standard and date; consignment chain of custody; volumes abstracted and discharged; effluent quality with method and laboratory
Compulsory specifications regulator	Regulated products, and instruments used for prescribed purposes	The specification and version; letter of authority with scope and expiry; accredited test reports; mark and labelling on the pack; **samples at the supplier's cost, with a test result generally valid for the whole batch.** For metrology: type-approval certificates, verification history and net-content control against the prescribed error limits
Local environmental health practitioner	Food premises and foodstuffs	Certificate of acceptability; hazard analysis plan and control point monitoring; handler training; labels with allergens, batch identification and date marking; recall capability
Medicines regulator	Licensed manufacturing sites	Manufacturing licence; master and executed batch records; deviations and change control; out-of-specification results; the Responsible Pharmacist's release decision; audit trails
Revenue service, customs	Declarations, duty relief, licensed premises	Classification with its basis; origin with supplier declarations and the rule-of-origin computation; duty-relief registers reconciling imports to exports; **drawback within six months of export**
Bargaining council agent	Collective agreement compliance	Grade per employee under the agreement's structure; rates against the gazetted schedule; hours by category; allowances; fund contributions
Information Regulator	Lawful processing	The lawful basis for **biometric** processing; the processing register; operator agreements; retention and deletion evidence; **the alternative offered to an employee who refuses**
Empowerment verification agency	Scorecard evidence	Supplier status levels **as at each transaction date**, with certificates and expiries

> The three most-missed rows. **Contractor liability** — the compensation regime now reaches a manufacturer for its sub-contractors' employees, so the dated contractor credential model is a compliance asset. **The batch rule** — a test result on a sample is generally valid for the whole batch, so batch size is an exposure decision. And **biometric time and attendance is a privacy question**: processing special personal information is prohibited unless a specific ground applies, and consent in an employment contract runs into the power-imbalance objection.

Map 8 — Retention clock by record class

For: setting a retention policy that does not breach something. **Breaks down:** only verified periods appear — anything absent was not verified, not absent from the law. Re-check every row before it reaches a client.

Record class	Period	Instrument
Noise exposure assessment and monitoring; audiometric and medical surveillance records including baseline audiograms	**40 years**	Noise-Induced Hearing Loss Regulations
Chemical exposure assessments and air monitoring	**Minimum 30 years**	Hazardous Chemical Agents Regulations 2021
Pressure equipment manufacturing records, held by the manufacturer	**Minimum 12 years**	Pressure Equipment Regulations
Lifting equipment full service history	**At least 10 years**	Driven Machinery Regulations 2015
Company records, financial statements, accounting records	**Seven years**	Companies Act, section 24
Tax and value-added tax records, zero-rating proof, input-tax support	**Five years from submission**, extended while under audit or appeal	Tax Administration Act; Value-Added Tax Act
Compensation-fund employer records	**Five years**, producible on demand	Compensation for Occupational Injuries and Diseases Act
Employment equity workforce records, plan, compliance records	**Five years after the plan expires**	Employment Equity Act, section 26
Major hazard installation registration and risk assessment interval	**Five years** each	Major Hazard Installation Regulations 2022
Safety data sheet review — an interval, not a retention	**At least every five years**	Hazardous Chemical Agents Regulations 2021
Incident records; health and safety committee records	**At least three years** each	General Administrative Regulations
Engineering-control examination and test records, chemicals	**At least three years**	Hazardous Chemical Agents Regulations 2021
Employee name, occupation, time worked, remuneration	**Three years from the last entry**	Basic Conditions of Employment Act
Written particulars of employment; employer records of collective agreements, awards and determinations	**Three years** after termination, or from the event	Basic Conditions of Employment Act; Labour Relations Act
Disciplinary records	**Indefinite**	Labour Relations Act, Code of Good Practice
Lifting equipment leased-item test records on the premises	Latest, **not older than 12 months**	Driven Machinery Regulations 2015
Drawback lodgement — a deadline, not a retention	**Within six months of export**	Customs machinery

> **A single corporate retention period is always wrong.** A five-year policy breaches the Companies Act's seven; a seven-year policy breaches four occupational-health instruments. And the two longest clocks sit in occupational-health systems replaced every seven to ten years, so **the obligation outlives the application by a factor of four or five.** The real requirement is an export format, with its reference data, readable without the source application: cheap at the start, impossible to reconstruct later.

Map 9 — Protocol to layer to what it carries

For: reading a plant network diagram and knowing which question to ask about each line. **Breaks down:** the level column says what a protocol is *for*, not where it is deployed — real plants run Modbus from a corporate server and OPC UA between two controllers.

Protocol or format	Level	What it carries	Standing
OPC UA core	2–3, 3–4	A typed address space and information model, not a flat tag list	IEC 62541, core at 1.05.x. Specifications free behind a terms gate; reference code free to members, GPL 2.0 otherwise
OPC UA PubSub	2–3	The same model over a broker or UDP multicast, **including an MQTT mapping** — so "OPC UA or MQTT" is a false choice	Part 14, at 1.05.x
UAFX, field exchange	1–2	Field-level exchange, the territory the fieldbuses held	Parts 80 to 84, **at 1.00.x against 1.05 for the core.** That gap is the honest maturity indicator
OPC UA companion specifications	2–3	Standardised node sets per class of machine, so two vendors' machines present the same semantics	**"Over 430" at the Foundation's April 2026 release**; the repeated "about 60" counts equipment types, not documents
MTConnect	2–3	Machine-tool and CNC monitoring, read-only over HTTP: a self-describing device model, current values, a sample stream	ANSI/MTC1.4-2018 on the vendor-facing designation, documentation a major version ahead. **Which version holds ANSI approval is unverified**
MQTT	2–3	A topic string and an opaque payload — nothing about content, naming or current state, which in a plant is the whole problem	ISO/IEC 20922 for 3.1.1; 5.0 from OASIS
Sparkplug	2–3	What MQTT leaves out: a topic namespace, a payload encoding, and birth and death certificates so a new subscriber gets a full definition and current state	**Sparkplug 3.0 is ISO/IEC 20237:2023, the Eclipse Foundation's** — not the OPC Foundation's
Modbus TCP and RTU	1–2	Register reads and writes. No types, no model, **no security whatsoever**	The lingua franca of third-party skids and weighbridges
PROFIBUS and PROFINET; EtherNet/IP and DeviceNet	0–2	Cyclic I/O and device parameters — the first pair on Siemens-centric estates, with PROFIBUS PA on process instruments; the second on Rockwell-centric ones	Vendor-ecosystem fieldbuses, each with its own consortium
HART	0–1	Instrument diagnostics over the 4–20 mA loop, which in most plants nothing reads	Widely installed, rarely harvested
B2MML	3–4	The ISA-95 objects as XML Schema. **A schema set, not a protocol**	**V7.00.00, 24 November 2020, MESA's, not ISA's.** Royalty-free with attribution, built to the 2018/19 editions, so it **predates the 2025 Part 1**
EDI: UN/EDIFACT	4	ORDERS, ORDRSP, DELFOR, DELJIT, DESADV, RECADV, INVOIC, APERAK	The cross-industry family in Europe, widely used locally
EDI: ANSI ASC X12	4	850, 855, 860, 830, 862, 856, 861, 810, 820, 997. **830 and 862 are the automotive pair** — 830 the planning horizon, 862 the firm shipping schedule	North American practice, met by any exporter
EDI: VDA, Odette	4	Delivery instruction, despatch advice, invoice, just-in-time and sequenced call-off. VDA's fixed-length formats are retiring for EDIFACT subsets	German and European automotive. Plan for OFTP2 specifically

> The sentence that matters most: **three standardised interoperability stacks exist, from three bodies, with no reconciliation between them.** A plant with machine tools and process equipment runs at least two, and the integration layer reconciles them. Not a failure of anyone's architecture — the state of the industry.

Map 10 — Archetype to design consequence

For: predicting most of what will go wrong from the cell a client sits in. **Breaks down:** almost every real plant is several cells at once, because **the archetype is a property of the order, not only of the plant.** A group with a beverage line, a canning site and an automotive component plant sits in five of these nine cells at once.

	Make-to-stock	Make-to-order	Engineer-to-order
Discrete	Forecast accuracy and finished-goods deployment; a shallow fixed BOM under change control. *Trap: a configurator nobody needs*	Capable-to-promise against capacity, not stock; finite scheduling with a moving bottleneck; job costing. *Trap: promising dates from infinite capacity*	The project is the cost object, the BOM a deliverable released while building, change the business model. *Trap: freezing a business that lives on change*
Batch	Lot genealogy and quality release; the allergen and changeover matrix; shelf life on ageing stock; a product wheel whose length sets the inventory requirement for everything on it. *Trap: promising quarantined stock*	Customer-specific grades and artwork against a fixed vessel, so batch quantity is granular — one, two or three batches, not 1.37. *Trap: a planning engine that cannot consume a changeover matrix*	Rare: a pilot plant or a toll campaign, where each run is its own recipe development, validation and cost object. *Trap: no master recipe exists to cost against*
Process	Mass balance, grade and specification management, shutdown planning on a multi-year cycle, and **transition material that is neither product** — real, costed, modelled by no discrete system. *Trap: counting pieces where there are none*	Tonnage of a grade on a contract with a price formula, where the saleable grade is set *after* production by the laboratory. *Trap: assuming the product is known at order entry*	Does not occur. It occurs as a capital project that *builds* a process plant, which is a different business

Three consequences the matrix cannot show. **The non-stoppable assets in the bottom row are the structural reason power interruption hit manufacturing categorically harder than retail** — a kiln cool-down and re-heat runs to days, a blast-furnace hearth can freeze and the asset be lost, a batch interrupted mid-process may not be quality-releasable at all. **A pull system and a material-requirements-planning push system make genuinely incompatible demands on the same data**, so the deliverable is a control-mode map per material segment, not a tool choice. And **the order-penetration point is the single most important fact about a manufacturing business, and it is almost never written down.** Find it, draw it, and most of the architecture follows.

THE CHAPTER IN TEN LINES

- ArchiMate 4 has no metric element and Value is not a number, so a manufacturing measure is a Business Object plus a Requirement plus a register, never an element.
- A genealogy is a graph over instances, so it is not drawable in a notation that models types; model the capability and draw the example separately.
- The ISA-95 equipment-element enumeration has fifteen values, four branches are alternatives selected by process type, and equipment module and control module are ISA-88 concepts it references.
- The ERP storage location has no ISA-95 equivalent and the ISA-88 control module has no ERP equivalent, so neither hierarchy can be derived from the other.
- Engineering, manufacturing, planning and service bills of material are four separate structures over the same items; as-built is not a bill of material at all, it is genealogy.
- The item master, the bill of material and the equipment hierarchy are habitually mastered in three places each, and the measure definition is mastered nowhere.
- ISO 22400-2's middle term is Effectiveness, not Performance, and its current published edition is 2014.
- Never print 85% as a global OEE benchmark: the site that publishes it calls it completely artificial, and OEE is documented as unsuitable for benchmarking or plant-level aggregation.
- A single corporate retention period is always wrong, because the clocks run from three years to forty and the longest outlive the systems holding the data.
- Three standardised interoperability stacks exist, from three bodies, with no reconciliation between them, and a mixed plant runs at least two.

SOURCES FOR THIS CHAPTER

- **1.** OPC Foundation, ISA-95 companion specification v1.00, section 7.4.1, ISA95EquipmentElementLevelEnum https://reference.opcfoundation.org/ISA-95/v100/docs/7.4.1
- **2.** MachineWise, ISO 22400-2 OEE definitions reference (states it does not reproduce the standard's text) https://machinewise.in/knowledge/iso-22400-2-oee-definitions/index.html
- **3.** ISO 22400-2:2014, Key performance indicators for manufacturing operations management, Part 2: definitions and descriptions (catalogue entry) https://standards.iteh.ai/catalog/standards/iso/8a9efc01-6c74-42a2-ad8f-ec19e84b48f0/iso-22400-2-2014
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APPENDIX B

Glossary

The barrier to entry in manufacturing is not the long words, it is the short ones. Yield, capacity, batch, order, on time: each sounds like a word you already know, and each means something narrower, or several things at once.

Manufacturing's vocabulary is the barrier to entry, and the barrier is not the long words. It is the short ones. Yield, capacity, batch, lot, order, routing, on time, throughput, work centre: each sounds like a word you already know, and each means something narrower, or several things at once, or something different two metres away on the same site.

This appendix is for lookup, so it runs alphabetically; the vocabulary chapter earlier in the book organises the same ground by domain, which is the better way to learn it the first time. Entries are short on purpose, and each says what the thing is and why it matters rather than only what it is called. Where the industry genuinely disagrees, the entry says so instead of picking a side.

Three terms get a warning rather than a definition, because the definition is not the problem.

> **OEE**
>
> **Overall equipment effectiveness: availability times effectiveness times quality ratio.** That is the only uncontroversial sentence about it.
>
> There is **no OEE standard**, and the research behind this book found **seven incompatible availability denominators** in use. The micro-stop threshold shifts loss between availability and performance **without changing OEE**, so OEE can be comparable in a group where its components are not — the opposite of what everyone assumes.
>
> **The 85% figure quoted as the global benchmark is an invention, and the site that publishes it says so**, calling it "a convenient, compelling and completely artificial benchmark" [1]. The component values usually attributed to total-productive-maintenance literature could not be verified. **Never put 85% in a client deliverable as a benchmark.**
>
> OEE is also documented as **unsuitable for comparing machines, for benchmarking, and for aggregating to department or plant level** [2]. A line's OEE is not the average of its stations': compute it on the constraint, or on the line as one resource. The first deliverable on any group dashboard is a **measure definition register**.

> **"ISA-95 COMPLIANT"**
>
> **Uncertifiable, because there is no ISA-95 conformance scheme.** Draw the contrast: the industrial-security series does have one — ISASecure, with named schemes for the development lifecycle, for components and for systems — and the batch-control standard's Part 1 carries conformance criteria in its own clause 9. ISA-95 has neither.
>
> What it usually means, in descending order of substance: the product ships an import or export in the XML binding; its internal model has an enterprise-site-area-line hierarchy; or somebody in marketing read the level diagram. **Ask instead:** which objects do you exchange, at which attribute level, over which transport, and can you show me the profile?

> **"SABS APPROVAL IS REQUIRED"**
>
> Wrong, in a way that will be corrected in front of a client. **The standards body publishes; the compulsory-specifications regulator regulates.** A national standard is voluntary unless an instrument makes it compulsory, and then **the instrument, not the standard, is the citation.** There is also **no general list of products needing a letter of authority** — the requirement arises from the individual compulsory specification, so the list of applicable specifications is the artefact to maintain, not a blanket rule.

A

Absorption costing. Assigning all manufacturing costs, fixed overhead included, to the units produced. The required basis for reported inventory, and the reason unit cost falls when volume rises without the product changing.

Activity-based costing. Replacing one or two plant-wide overhead drivers with many activity-level ones — setups, inspections, material moves. It usually shows low-volume complex products to be unprofitable. **If a client wants it, the real project is instrumentation**, because the activity data is not captured.

Allergen matrix. A from-product by to-product table stating what cleaning and verification each changeover requires. In food it makes a sequence legal rather than merely efficient, and most planning engines cannot consume it.

Alternate. A component the bill of material permits in place of the primary, as a planned and approved option. Distinct from a substitute, which is an unplanned swap made under pressure.

APDP. The Automotive Production and Development Programme — **a customs-duty-rebate-certificate regime, not a tax credit.** It demands bill-of-material-level country of origin and cost-component split through several supply tiers, so a plant whose costing loses component split through multi-level roll-up cannot substantiate a claim.

APQP. Advanced Product Quality Planning: the automotive industry's phased new-product introduction discipline, published as a manual and obliged by the automotive quality scheme.

APS. Advanced planning and scheduling: the engine that produces a finite-capacity schedule with a sequence, where material planning assumes infinite capacity. The schedule it publishes is never the schedule the plant runs.

As-built. What actually went into one physical unit — these lots, these serials, this date. **Not a kind of bill of material:** a bill describes a kind of thing, as-built one thing. Different grain, volume and retention driver.

As-maintained. What is inside a unit now, after years of field replacement. It diverges from as-built on the first repair, and it is what a recall or a warranty claim actually needs.

Assemble-to-order. Subassemblies made to forecast and held; final assembly waits for the order. Inventory sits at module level — you hold variety's common ancestors, not its leaves.

Asset Administration Shell. A standardised digital representation of an asset carrying submodels — nameplate, technical data, carbon footprint — so information travels with the asset. **Two version lines:** an international structural standard and a separate metamodel. Locally you will meet it as a file a European supplier sends, with nowhere to put it.

Availability. Run time over a planned production time. **The numerator is uncontested and the denominator is not:** seven conventions were found in use for this book. Two plants behaving identically report different availability.

B

B2MML. The XML Schema rendering of the ISA-95 object models, and the only freely available concrete form of them. **MESA International's, not ISA's**, royalty-free with attribution, at V7.00.00 of 24 November 2020 — built to the 2018/19 editions, so it predates the 2025 Part 1.

Backflush. Consuming components automatically at standard bill quantity when output is confirmed. It **systematically hides material usage variance**, and where substitution occurs it produces a confidently false genealogy, which is worse than a gap.

Batch. Two senses. In batch manufacturing, the quantity processed together through a sequence of steps, constrained by the vessel — one, two or three batches, not 1.37. In identification, the quantity treated as homogeneous for quality. **The trap:** batch and lot get used interchangeably, and nobody can then say whether the identifier describes what was made or what is held.

Batch record. The complete manufacturing record for one batch: recipe version, equipment, every parameter, deviation and signature. In regulated manufacture it is the artefact the whole landscape exists to produce.

B-BBEE. Broad-Based Black Economic Empowerment. For a manufacturer it bites through procurement and through enterprise and supplier development — the largest single element at 40 points, with sub-minimums and a level-dropping penalty. **Development targets are a percentage of net profit after tax, not of procurement spend.**

BCEA. The Basic Conditions of Employment Act: hours, overtime, leave, and the reason employee name, occupation, hours worked and remuneration are retained three years from the last entry.

Bill of material. The structure stating what a parent is made of, in what quantity, with what loss. **There is never one:** engineering, manufacturing, planning and service bills are separate structures with their own identity, lifecycle, approval and effectivity.

Breakeven volume. Fixed costs over contribution per unit. **Systematically misleading in South Africa**, because retrenchment procedure and collectively agreed shift arrangements make direct labour substantially not variable in the short run.

C

Campaign. A planned run of one grade before changing over. Process plants often cycle through grades in a fixed sequence, and **the cycle's length sets the finished-goods inventory requirement for everything on it.**

CAPA. Corrective and preventive action: the record linking a problem to its root cause, the action, the verification, and the batches affected. Every quality scheme compels it; most estates hold it in three systems.

Capable-to-promise. Answering "when can you have it?" by checking material *and* capacity *and* the order book. An order-driven plant needs this; promising against stock returns zero, the sales team overrides it, and the order book loses credibility.

Capacity. Four figures, one usable. **Theoretical** (design rate, no stoppages — a fiction), **rated** (adjusted for a shift pattern usually left unstated), **effective** (less planned maintenance and changeovers), **demonstrated** (highest sustained output actually achieved at acceptable quality in the current mix). **Only demonstrated capacity belongs in a plan, and it moves when the mix moves.**

Catch weight. Where an item is traded in one unit and priced or stocked in another that varies per unit — chicken portions sold by the kilogram, handled by the carton. It breaks any model assuming a constant conversion factor.

CFIHOS. What a contractor must hand over at the end of a capital project, in what structure, with a reference data library. **Version 2.0, November 2025, managed by the oil and gas producers' association, free of charge** — the most directly reusable free artefact in the standards landscape.

Changeover. Last good part of A to first good part of B, including run-up and quality approval. **Not one number but a matrix**, from-product by to-product: white to black paint is quick, black to white needs a flush.

COIDA. The compensation statute for occupational injuries and diseases. Its amendment took effect **23 January 2026**, extending claim prescription from twelve months to **three years**, requiring accidents to be reported within seven days, and adding liability for a sub-contractor's employees.

Companion specification. A standardised OPC UA information model for a class of machine, so every machine of that class exposes the same nodes. **Over 430 exist** per the Foundation's April 2026 release; the repeated "about 60" counts equipment types, not documents.

Compulsory specification. A technical specification an instrument has made legally binding, administered by the national regulator. A voluntary standard becomes compulsory only when such an instrument says so, and **the instrument, not the standard, is the citation.**

Condition-based maintenance. Intervening when a measured condition crosses a threshold rather than on a calendar. It needs a measurement point, a history and a threshold under change control — three objects most maintenance systems hold badly.

Configure-to-order. The customer specifies a configuration from an option set with rules, and the bill is generated at order entry. There may be millions of valid configurations and no structure for the one ordered. **The product master is the rule engine.**

Contribution margin. Selling price less variable cost per unit. The right frame for a short-run decision inside a fixed capacity envelope, and the wrong frame for pricing, capital allocation and exit.

Control module. The lowest level of the batch-control physical model — a valve, a motor, one actuated element. **An ISA-88 concept referenced by ISA-95, not ISA-95's own**, and it exists in the controller and nowhere in the enterprise system.

Control recipe. The instance of a recipe for one batch on specific equipment, with parameters resolved. Distinct from the master recipe, which is the approved template. Which system owns which settles most process-plant integration disputes.

Cumulative lead time. The longest path through the bill of material, from the lowest purchased part to the finished good. The real constraint on responsiveness, and **almost nobody calculates it** — which is why time fences are set shorter than it.

Cycle time. Three things. **Operation** cycle time is one operation on one unit; **line** cycle time is the reciprocal of the output rate; **total** cycle time is elapsed time through the whole process, waiting included. Used interchangeably, so the term is dangerous without a declared stage boundary.

D

Days of supply. Average inventory over average daily cost of sales. Supply chain measures forward demand days against the forecast, finance measures cost days against the ledger. Both are called days of supply, and only the by-stage form says anything.

DCS. Distributed control system: the integrated control, operator-station and engineering environment of a continuous plant, usually one vendor's, usually on its own network, usually not negotiable.

Deviation. A documented departure from an approved procedure or specification, with its justification, approval and link to the batches affected. In regulated manufacture the deviation record matters as much as the thing it departs from.

Digital twin. A model of an asset or process kept in step with it by live data. The research behind this book found **no credible independent adoption data at all** — every statistics source located was search-optimised content citing other such content. Treat any specific percentage as fabricated.

Discrete manufacturing. Output is countable, separable units with identity. Material is assembled, the bill is a tree with integer quantities, and the dominant architectural problem is engineering change control.

dtic. The department that owns industrial policy, the sector master plans and the automotive programme. **The master-plan status document publicly available is dated 27 October 2020**, so its figures are 2020 targets, not achievements.

E

Effectivity. The window in which a bill-of-material line or routing step is valid — by date, serial, lot or unit number. It is why two revisions cannot share a diagram, and why a system with one bill per product will be rebuilt.

Engineering change order. The authorised instruction to change a product's definition. The architecture question is never the document but the fan-out: orders in flight, stock of the superseded part, customer approvals, regulatory certificates.

Engineer-to-order. The order arrives as a specification or a drawing, and the product is designed after it. The bill is created during execution and changes while manufacturing runs. **The project is the cost object, the bill is a deliverable, and change is the business model.**

EPCIS. The traceability event standard: the what, when, where and why of an object, aggregation, transaction or transformation event. **The transformation event is the one that matters in manufacturing,** because it records inputs consumed and outputs produced. The vocabulary exists; do not invent one.

EPR. Extended producer responsibility: the obligation to fund and report end-of-life handling of packaging or product. For a manufacturer it is a data obligation on material placed on the market, by mass and material type, which the item master rarely carries.

Equipment module. A functional grouping of control modules performing a defined processing activity — a dosing skid, a weigh hopper. **An ISA-88 concept referenced by ISA-95, not ISA-95's own.**

F

Fieldbus. A digital network carrying cyclic input and output between controllers and field devices. Which family a plant runs follows from whose controllers it bought a decade ago, and is not an architecture choice available now.

First-pass yield. Units passing with no rework, over units started. Always at or below yield. **A plant quoting 98% yield and one quoting 92% first-pass yield may be the same plant.**

FSSC 22000. A food-safety certification scheme on the management standard plus sector prerequisites plus its own requirements, recognised by the retailer-facing benchmarking body. Version 6 published 31 March 2023, mandatory 1 April 2024. **Version 7 dates could not be verified for this book.**

Functional location. A position in the maintenance hierarchy independent of whatever equipment occupies it — "pump station 2, duty pump" rather than a serial number. Separating position from machine is what lets history survive a replacement.

G

Genealogy. The graph linking every input lot and serial to every output lot and serial, both ways. **Where-used tells you which products *could* contain a contaminated lot; only genealogy tells you which *do*** — the difference between recalling a line and recalling four pallets.

GFSI. The Global Food Safety Initiative. **It certifies nobody.** It benchmarks schemes against its own requirements, and retailers then accept the recognised ones. Confusing recognition with certification is common and consequential in a tender.

GLN. Global Location Number: a GS1 key identifying a party or a location. Plant, warehouse, legal entity, ship-to, bill-to and remit-to are distinct numbers, and conflating them surfaces at the first invoice dispute.

GMP. Good manufacturing practice, **enforced by inspection rather than certification** — a manufacturer is inspected, not certified. The detail lives in the inspection scheme's guide that national regulators adopt and adapt, not in the local guideline.

Grade. A quality classification of otherwise identical material, determined by assay. **The saleable product is therefore determined after production**, and an order model assuming the product is known at order entry will not fit.

GTIN. Global Trade Item Number, identifying a trade item at a packaging level. **A consumer unit, an inner pack and a case are three different numbers with a hierarchy between them**, so an item master with one row per product cannot reconcile a despatch against a customer receipt.

H

HACCP. Hazard Analysis and Critical Control Points: the method inside the food schemes, rarely certifiable by itself. **"It is mandatory in South Africa" is unsafe as written** — an instrument exists, but which classes of manufacturer it compels could not be verified.

Heijunka. Levelling: deliberately smoothing the production sequence so the plant sees steady demand. **It holds finished goods on purpose**, to absorb variability rather than transmit it into the factory — which is why "lean means no inventory" is wrong.

Hidden factory. The capacity consumed by rework, scrap, unrecorded micro-stops and unscheduled time, which never appears in a production report. Rolled throughput yield exposes part of it.

Historian. A time-series store built to write millions of tag values a second and read one at a time. **Not a relational database, and a system of record for nothing but tag values.** It compresses, so a past value may be interpolated rather than measured.

I

IATF 16949. The automotive quality standard, a supplement to the general one. **Still the 2016 first edition**, with the second planned mid-2027. The churn is in the auditor guide, the recognition rules and the sanctioned interpretations — so the edition date is not the measure of change.

Industrial DMZ. The brokered network segment between plant and enterprise, with no protocol passing straight through. **Not Williams' invention:** the Purdue reference model contains no cybersecurity content at all, and the zone language came later.

Interchangeability. Whether one part may replace another. **It is not symmetric** — "B may replace A" does not imply the converse, and one-way interchange is the normal supersession case.

Inventory turns. Cost of sales over average inventory at cost. Sales in the numerator inflates turns by the gross margin; period-end rather than average inventory makes turns a function of where month-end fell relative to a campaign.

ISA-88. The batch-control standard: recipe separated from equipment, the procedural model, the physical model, the batch production record. **Part 1's current American edition is ANSI/ISA-88.00.01-2010**, the international Part 1 1997. The best public account of the parts is a 2006 document with stale status columns, so **do not state Parts 2 to 5 dates from memory.**

ISA-95. The enterprise-control integration standard: the levels 0 to 4 taxonomy, the equipment hierarchy, the functional model, and the object models that are the nearest thing to a canonical information model at the enterprise-to-execution boundary. **Part 1 is the 2025 edition**, and there are **eight parts plus a technical report, not five.**

ITAC. The trade administration body: tariffs, rebates, anti-dumping duties, scrap export restraints and the automotive production rebate certificates. Its decisions move a landed price by double digits.

Item master. The register of everything bought, made, stocked or sold, with units of measure, classifications and plant-specific attributes. **The hardest master data object in manufacturing**, because nearly every attribute is really a dated, plant-specific, function-owned assertion.

J

JIS. Just in sequence: the customer specifies the order in which parts must arrive to match its build sequence, in a window measured in hours. It makes the customer's build schedule a live input to the supplier's plant.

Job shop. Many products in small quantities, each routed through shared general-purpose machines grouped by process. **Queue time dominates elapsed time**, so quoted lead times bear little relation to the sum of operation times, and the bottleneck moves with the mix.

K

Kaizen. Continuous incremental improvement by the people doing the work, as a habit rather than a project. Its architectural implication: the data to see a small problem must reach the person who can fix it within the shift.

Kanban. A signal authorising replenishment of a fixed quantity when a container is consumed. **A pull system and a push planning system make genuinely incompatible demands on the same data**, so the deliverable is a control-mode map per material segment, not a tool choice.

L

Lead time. Four measures wearing one word: **manufacturing** (release to completion), **customer** (placement to receipt), **cumulative** (the longest path through the bill), and **replenishment** for a purchased item — which includes receiving, inspection, put-away and, locally, the port.

Lean. The tradition arguing that lead time, not cost, is the primary variable, and that waste is what lies between value-adding steps. Its claim is not that inventory is evil but that inventory is a symptom of long lead time and large batches.

LIMS. Laboratory information management system: system of record for samples, tests, methods, specifications, results and certificates. Its results set the quality status of stock, so it holds the authority to make inventory sellable.

Little's Law. Work in progress equals throughput times cycle time. Rearranged, cycle time equals work in progress over throughput — the arithmetic behind lean's obsession with cutting work in progress, and the most useful formula in the subject.

Lot. An identified quantity of material treated as homogeneous, with its own quality status, expiry and genealogy. **The trap:** a lot is what you hold, a batch is what you made — and in a continuous plant lots are cut out of a flow by convention, so the convention is part of the data model.

M

Mass balance. Inputs equal outputs plus losses plus inventory change, within tolerance. In a process plant it is **not a report you add later, it is the transaction model** — and it will not close unless transition material is represented.

Master recipe. The approved, plant-localised template from which control recipes are instantiated: formula, procedure, parameters and ranges. Which system owns it, and where site localisation happens, is the architecture question.

mBOM. The manufacturing bill of material: what the plant consumes, in what sequence, at which operation, with consumables, packaging, phantoms, scrap factors and alternates. **The transformation from the engineering bill is the most expensive data transformation in a discrete manufacturer**, and almost never fully automated.

Measurement point. A named place on an asset where a reading is taken, with a unit, an expected range and a history. It is the join between the maintenance model and the tag world, and it is usually missing.

MEIBC. The bargaining council that sets wages across metals and engineering rather than firms doing so. The circulated agreement for 1 July 2026 gives 5.00% to 6.00% by grade with an **inverted differential** — 5.00% at the lowest grade, 6.00% at the highest — and names several employer bodies as **not covered.**

MES. Manufacturing execution system: system of record for work order execution state, operation confirmation, genealogy and the electronic batch record. Per plant, not per group — and **where there is no execution system there is no genealogy.**

MIBCO. The motor industry bargaining council. The two-word distinction that costs the most money in South African wage agreements: **component manufacturers' increases apply to *actual* rates of pay, not to the prescribed minimum.**

Micro-stop. A stoppage shorter than a company-set threshold, conventionally one to five minutes, counted under Performance rather than Availability. **Two plants behaving identically with different thresholds report different availability and performance, with the same OEE.**

Mixed-mode. A plant or group running more than one process type or fulfilment strategy at once. Almost every real South African plant is mixed-mode, and the handover between modes is a named integration problem.

MRO. Maintenance, repair and operations inventory — the spares held against failure. Often larger in value than raw material, governed by reliability engineering rather than planning, and deliberately turning over badly: you hold a gearbox for five years so you never need it urgently.

MRP. Material requirements planning: exploding a plan through the bill and offsetting by lead time, assuming infinite capacity. **Updating its planned lead times from observed performance creates a positive feedback loop** — longer observed, longer planned, more released work in progress, longer still — and it is mostly undiagnosed.

MTBF and MTTR. Mean time between failures and mean time to repair. Both require a failure definition and a clock start, and both are quietly incomparable between sites that define "failure" differently.

MTConnect. A read-only, HTTP-based standard for getting data out of machine tools, with a self-describing device model. **Its American designation, ANSI/MTC1.4-2018, is a full major version behind its own documentation, and which version holds that approval could not be verified.**

N

NEMA. The national environmental management frame under which authorisations, the duty of care and the air, waste and water instruments sit. An authorisation's **conditions** are the architecture requirement, not the authorisation.

NERSA. The energy regulator: it approves electricity tariffs and approved the current power-quality standard. It approved 8.76% from 1 April 2026 for direct customers and 9.01% from 1 July 2026 for municipal bulk purchasers, with 8.83% from April 2027.

NRCS. The national regulator for compulsory specifications and legal metrology. **It regulates; the standards body publishes.** Its section 20 is most often missed: on written request a supplier must provide samples at its own cost, and **a test result is generally valid for the whole batch unless shown otherwise.**

NRS 048-2. The South African power-quality standard, **Edition 5, approved by the regulator on 6 May 2025.** It classifies voltage dips into windows, **assigns the shallowest to the customer**, names those that trip drives and those that trip contactors, and publishes characteristic interruption counts with **no compliance criteria.**

O

OHSA. The occupational health and safety statute and its regulations: machinery, chemicals, noise, pressure equipment, lifting, major hazard installations, incidents. The source of the longest retention clocks a manufacturer carries.

On-time-in-full. The customer-facing service measure, and **the most frequently mis-specified metric in manufacturing architecture.** Four defensible date bases, order-level or line-level scoring, four places the clock can stop. **Two plants reporting 96% and 78% may be performing identically**, and early delivery is often a failure.

OPC UA. The industrial interoperability architecture: an address space, an information model, services, profiles, and two patterns — client-server, and a publish-subscribe part with an MQTT mapping, so "OPC UA or MQTT" is a false choice. **Maturity is uneven:** core parts at 1.05.x, field-exchange parts only at 1.00.x.

Operation. One step of a routing at a work centre, with a setup time, a run time per unit and a standard. The grain at which the floor confirms and at which labour and overhead are absorbed — so the join between execution and the ledger.

Order. The word splits. A **customer order** commits to supply; a **production order** (process order, work order) authorises making, with its own quantity, dates, components, operations and cost collection; a **purchase order** commits to buy. **The trap:** "the order" means the production order in planning and the customer order in sales.

Order-penetration point. Where the customer order enters the material flow. Everything upstream runs on forecast, everything downstream on the order. **The single most important fact about a manufacturing business, and almost never written down.**

Overhead absorption variance. Overhead absorbed less budgeted overhead at budget volume — the money consequence of running at a volume other than the rate assumed. **Owned by nobody in the plant, and that is the point:** it follows sales volume, and it is the variance most often misread as a manufacturing failure.

P

Phantom assembly. A bill-of-material level that exists for structure but is never stocked. **Phantom is a property of the bill-of-material *line*, not of the item** — the same item is legitimately a phantom in one parent and a stocked assembly in another, so an item-level flag cannot represent the truth.

Planning bill. A nominal structure expressing option take-rates as percentages, used to plan long-lead components before anybody knows which configuration will be ordered. Quantities need not be integers and must never issue material.

PLC. Programmable logic controller: the deterministic computer running the control logic and holding the live input and output image on a millisecond scan. Patching one is not like patching a server, because the plant is running and the vendor must have validated the patch against that firmware.

PLM. Product lifecycle management: system of record for the part, the design document, the engineering bill, the revision, the effectivity and the change order. Where the engineering truth lives, as distinct from the manufacturing truth.

POPIA. The personal information statute. **Its bite on a manufacturer is employee and biometric data, not customer marketing.** Biometric information is special personal information, so processing is prohibited unless a specific ground applies — and consent in an employment contract runs into the power-imbalance objection.

PPAP. Production Part Approval Process: the package proving a part can be made repeatably to specification. Assembling it for one part number spans engineering, execution, laboratory, metrology, quality and document management, and **no single system holds all of it.**

Process manufacturing. Output is a flowing, measured substance, transformed irreversibly, with the plant running for weeks between shutdowns. The bill is a formula in ratios with co-products, quantity truth is a meter with a tolerance rather than a count, and the problems are mass balance and grade.

Purdue model. The levels 0 to 4 hierarchy from Williams' 1992 reference architecture, used today as the backbone of operational-technology zoning. **Williams put no security in it** — no demilitarised zone, no Level 5, no zone language. Attributing those to him is wrong, and an engineer will notice.

Q

Quality hold. A state in which stock is physically present and not available. **Stock is not available merely because it is present**, and a planning engine that cannot see quality status will promise quarantined material.

Quality ratio. Good quantity over produced quantity, the third factor of OEE and the standard's own term. It diverges on whether rework is a first-pass reject, and on whether laboratory results arriving days later are included.

R

RAMI 4.0. A three-axis classification cube: hierarchy levels, lifecycle and value stream, and six interoperability layers. **A pre-standard** — a German specification and an internationally Publicly Available Specification, not a full standard. Good for showing a client their programme sits in one corner;

you cannot build from it.

Recipe. The process industry's bill and routing in one object: a formula, a procedure and the process conditions, where parameters are inseparable from quantities and quantities scale non-linearly with batch size.

Responsible Pharmacist. The person responsible for batch release at a South African pharmaceutical manufacturer. **South Africa has no Qualified Person** — the regulator's guideline states the inspection scheme's "Responsible Person" is equivalent to the Responsible Pharmacist.

Rolled throughput yield. The product of every step's first-pass yield. Ten steps at 98% gives about 81.7%, a figure most plants would not recognise as their own. Comparable only within a fixed step count, because adding a step mechanically lowers it.

Routing. The ordered sequence of operations turning components into a parent, with work centre, setup time, run time and standard for each. **The trap:** a process plant has no routing in this sense, and the one in the system exists to make the ledger post.

S

SABS. The national standards body: it **develops and publishes** standards and operates as a certification body. It does not regulate. **"SABS approval is required" is wrong** — the regulator for compulsory specifications is a different body.

SAHPRA. The health products regulator: it licenses pharmaceutical and device manufacturing sites and inspects them. **Do not cite it for good-manufacturing-practice detail** — its guideline adopts and adapts the international inspection scheme's guide, where that detail actually lives.

SANAS. The national accreditation system for laboratories and inspection bodies. Several obligations read "by an accredited laboratory" or "by an approved inspection authority", which makes accreditation status a dated attribute on a supplier record.

SANS. A South African National Standard. **Voluntary by default.** It becomes compulsory only when an instrument says so, and then the instrument is the citation. Regulations referencing particular standards are how a voluntary document acquires legal force without changing its own status.

SARS. The revenue service: tax, value-added tax and customs. Its customs side is the system of record a manufacturer does not know it has — classification, origin, duty relief and drawback, all reconciling against production genealogy.

Scheduling agreement. A longer-term agreement carrying a rolling forecast horizon and a firm release horizon, against which call-offs are made. The quantity is expressed cumulatively, and **a cumulative-quantity discrepancy is the commonest automotive data dispute.**

Scrap factor. An allowance on a bill-of-material line for material expected to be lost. **"10% scrap means uplift by 10%" is arithmetically wrong:** 100 divided by 0.9 is 111.1, not 110. Over five levels at 3% a scrap factor compounds to $1 \div 0.97^5$, an uplift of **16.4%** — not the 15.9% that 1.03^5 gives, which repeats the same error one level up. Which figure is right depends on whether the client's system treats the number as a loss rate or as an instruction to add; ask before you reconcile anything. The gap is a permanent, unexplainable usage variance. See the data chapter for the worked table.

SEP. Single Exit Price: the regulated price at which a medicine manufacturer or importer may sell to **any** buyer — a manufacturer-side price constraint, not a retail mechanic. The 2.88% adjustment took effect 1 October 2026. **It does not apply to the State.**

Serial number. An identifier for one individual unit, as distinct from a lot which identifies many. It is what makes as-built genealogy possible at unit grain, and what a warranty, a recall and a configuration record hang off.

SEZ. Special Economic Zone: a designated area whose qualifying companies are taxed at 15%. One of the incentives actually open — unlike the section 12I investment allowance, **closed since 31 March 2020** and still offered in presentations.

SMED. Single-minute exchange of dies: converting setup work that needs the machine stopped into work that can be done while it runs. The lever that shortens a product wheel, which is the single biggest inventory intervention available in a process plant.

Sparkplug. What makes MQTT usable on a plant floor: a topic namespace, a payload encoding, and birth and death certificates so a new subscriber gets a full definition and current state. **Sparkplug 3.0 is ISO/IEC 20237:2023, and it is the Eclipse Foundation's, not the OPC Foundation's.**

SPC. Statistical process control: monitoring a capable process to detect special-cause variation before it produces defects. It presumes a proven measurement system, which is why quality data must be a structured asset rather than a scan of a printout.

Specification. The characteristics, methods and limits a material or product must meet. Versioned, referenced by inspection lots and certificates, and **the version in force at the time of manufacture is the one that matters** — so it is never a current-state attribute.

Standard cost. A pre-determined cost per unit, rolled up from standard quantities and times at standard prices and rates. Stale by construction, and **the annual roll steps reported margin with no operational cause.** "Standard costing is not compliant with international reporting" is wrong: the inventories standard expressly permits it.

Standard time. The planned setup and run time for an operation. Always wrong in a known direction: it was set once, the method has changed, and nobody re-times it because the re-timing is a negotiation.

Store and forward. Buffering transactions locally when the link upward is down, and replaying them when it returns. **Not an optional resilience feature in manufacturing**, because the physical process does not stop when the software does.

T

Tag. A named, typed, continuously sampled signal. **A tag is not a data entity** — no identity beyond its name, no referential integrity, no transaction and no business meaning until something binds it to an asset and a context.

Takt time. Available production time over customer demand in that time: a demand-derived target rhythm, not a measurement of the plant. **It changes when demand changes, without the plant changing**, which is why it is not a performance measure however often it is reported as one.

TEEP. Total effective equipment performance: OEE multiplied by utilisation, where utilisation is planned production time over all calendar time. It exposes capacity you own and never schedule, and its calendar-time denominator makes it the least convention-dependent of the family.

Theory of Constraints. The tradition that output is set by one constraint, so improvement elsewhere is illusory. **Its "throughput" is revenue minus totally variable cost, not units**, and it refuses to absorb fixed cost into units — so an adopter runs two measurement systems permanently.

Throughput. Good output per unit time — or, in Theory of Constraints, revenue minus totally variable cost. **Both definitions sit in the same room and nobody says which they mean.** Establish it in the first conversation.

Time fence. The horizon inside which the plan is frozen or changes need authority. **A time fence shorter than the cumulative material lead time is arithmetically unenforceable** — the commonest configuration error in this area, and checkable in an afternoon.

Tolling. Processing material the plant does not own, for a fee. **Ownership is a first-class attribute of inventory, and omitting it is one of the few genuinely unrecoverable design errors** — you cannot retrofit ownership onto historic stock movements.

TPM. Total productive maintenance: operator-led care of equipment, with OEE as its scoreboard and the six big losses as its taxonomy. Its arithmetic is where the widely quoted OEE formulation comes from — and that is not the international standard's.

Transition material. What a plant produces while moving from one grade to another, which is neither product. It is real, it has a cost and it has a disposition. **No discrete-manufacturing system models it, and the mass balance will not close without it.**

U

Unified namespace. A pattern in which every system publishes to and subscribes from one broker-mediated, self-describing namespace. The topology argument is strong. **"Single source of truth" is false** — it holds current state only, with no history, transactions, referential integrity or audit trail, and **no specification and no standards body exist for it.**

Unit of measure. The dimension a quantity is expressed in — and manufacturing routinely carries an item in two or three at once. Conversion factors are item-specific, sometimes lot-specific because density varies by grade, sometimes not constant. **Every quantity must declare its unit and whether it was converted.**

V

Value stream mapping. Drawing the material and information flow for one product family with times attached, to show how little of the elapsed time adds value. The only common artefact that puts information flow and material flow on the same page.

Variance. The gap between what the standard said production should have cost and what it did, decomposed so each piece has one cause and one owner: purchase price, material usage, mix and yield, labour rate, labour efficiency, overhead spending, and overhead absorption — which has no owner. **Each needs a drill-down to a transaction.**

Voltage dip. A sudden reduction in supply voltage lasting more than 20 milliseconds and up to three seconds. **The real power problem for operational technology is dip immunity, not availability**, and the national standard classifies dips into windows and assigns the shallowest to the customer.

W

Where-used. The query asking which parents contain a component, directly or at any level. A statement about *kinds* of thing — which is why it tells you which products could contain a contaminated lot, and never which do.

Work centre. The word that means two things in one conversation. In the enterprise system, a costing and scheduling construct with a rate, a cost centre and a capacity calendar. In ISA-95, the generic term for a grouping of equipment scheduled from above. Neither is derivable from the other.

Work in progress. Material released to the floor, partly transformed, not yet sellable — valued at material consumed plus labour and overhead absorbed to the operation reached. **The hardest inventory in business to value correctly**, because its value depends on how far through the routing it is, which is a shop-floor fact.

Y

Yield. Good units out over units started, permitting rework. **The most dangerous innocent-looking word in the subject**, because it is defined against a stage boundary and a defect definition that differ between plants. Ask: started where, good according to whom, and does rework count?

THE CHAPTER IN TEN LINES

- The dangerous terms are the ones that look obvious: yield, throughput, capacity, work centre, on time, availability, batch, lot, routing and order each carry more than one defensible meaning.
- Work centre means a costing and scheduling construct in the enterprise system and a grouping of equipment in ISA-95, and neither is derivable from the other.
- Lot is what you hold and batch is what you made; in a continuous plant lots are cut out of a flow by convention, which makes the convention part of the data model.
- Order splits into customer order, production order and purchase order, with different states and different owners, and the word alone never says which.
- There is no OEE standard, seven availability denominators are in use, and the 85% benchmark is an invention the publishing site itself calls artificial.
- ISA-95 compliance is uncertifiable because no conformance scheme exists, while the industrial-security series has one and the batch-control standard carries conformance criteria.
- A South African national standard is voluntary unless an instrument makes it compulsory, and then the instrument is the citation, not the standard.
- Phantom is a property of the bill-of-material line and not of the item, and interchangeability is not symmetric.
- Ten per cent scrap means dividing by 0.9, not multiplying by 1.1, and the difference compounds through every bill-of-material level.
- Ownership is a first-class attribute of inventory, and it cannot be retrofitted onto historic stock movements.

SOURCES FOR THIS CHAPTER

- **1.** Vorne Industries, OEE FAQ (oee.com) — source of the "completely artificial benchmark" quotation https://www.oee.com/faq/
- **2.** Wikipedia, Overall equipment effectiveness — used for the catalogue of calculation conventions and documented criticisms, not for benchmark values https://en.wikipedia.org/wiki/Overall_equipment_effectiveness

APPENDIX C

Standards, bodies and where to find them

Who publishes what, at which version, on what terms, and whether you will genuinely meet it. Four of these editions moved inside this book's own research window, so the last section is the one that matters: how to check a version yourself.

Everything in this appendix carries a version and a date, and every date is as at **16 October 2026**. That is not pedantry. Four standards editions moved inside the research window for this book, and seven ownership or product-name changes landed in the same two years. **Never state a standard's version from memory** — including the ones you are confident about, because the ones you are confident about are the ones that have moved.

Read each entry for three things: whether you will genuinely meet it, what it costs to get, and what goes stale first. The last section tells you how to check a version yourself, which is the only part of this appendix that cannot go out of date.

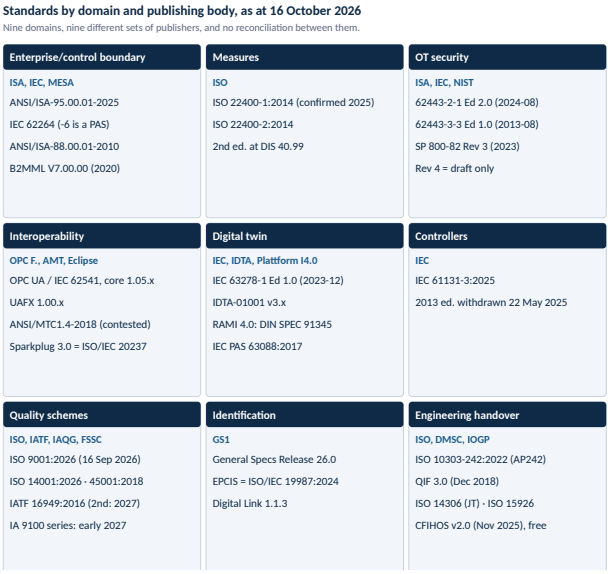


Figure 22.1 Nine domains, nine sets of publishers, and no reconciliation between them

The bodies

The standards bodies

Body	What it is, and what it publishes	Terms	Will an architect meet it?
ISA — International Society of Automation	A United States professional association founded in 1945, Research Triangle Park. Publishes through sector committees — ISA88, ISA95, ISA99, ISA101, ISA106 — as American National Standards, several with simultaneous IEC adoption [1]	Sold, not published. No free normative text [1]	Constantly. ISA-95 is the vocabulary the industry argues in
IEC	The international electrotechnical body. Publishes the IEC 62264, 61512, 62443, 62541, 61131 and 63278 series, in parallel lineages with ISA's	Sold by IEC and national bodies	Yes, and **its lineage is out of step with ISA's** — say which document you mean
ISO	Publishes ISO 22400 for measures, the management-system standards, ISO 10303, 14306, 15926, and the ISO/IEC adoptions of Sparkplug and EPCIS	Sold; catalogue pages are free and carry the status	Yes, mostly through the quality schemes
MESA International	A manufacturing-operations association. Maintains **B2MML** and BatchML through its XML Committee [2]	**Royalty-free with attribution**; download behind a MESA account, schemas on a public repository [2][3]	Yes, if you build an enterprise-to-execution canonical message set
OPC Foundation	Publishes OPC UA and the companion specification programme [4]	Specifications **free of charge behind a terms gate**; reference implementations free to members, GPL 2.0 otherwise	Yes. The single most consequential body in the protocol layer
AMT and the MTConnect Institute	AMT hosts the committee activity; the Institute publishes MTConnect [5][6]	The site does not state licensing [5]	In metalworking, yes; elsewhere rarely
Eclipse Foundation	Publishes **Sparkplug**, which became an ISO/IEC standard [7][8]	Eclipse specification process; the ISO/IEC adoption is sold	Yes, wherever a unified-namespace project has landed
IDTA — Industrial Digital Twin Association	Publishes the Asset Administration Shell metamodel and the submodel-template catalogue [9]	Specifications published on its site	Through imported European machinery, before anything local publishes one
Plattform Industrie 4.0	The German originator of RAMI 4.0, published as a pre-standard [10][11]	A specification, not a standard	As a slide, not as a design artefact
Industry IoT Consortium and Digital Twin Consortium	Formerly the Industrial Internet Consortium. **On 18 January 2024 OMG announced it was integrating the Consortium within the Digital Twin Consortium** [12]. The reference architecture is at **v1.9** on the Consortium's own page, undated [13]	Published free on their sites	Rarely, and as a viewpoint method rather than a reference model
OMG — Object Management Group	The parent body of the two above [12]	—	Indirectly

The scheme owners, registries and regulators

Body	What it is, and what it publishes	Terms	Will an architect meet it?
GS1	The identification system: keys, application identifiers, data carriers, EPCIS, Digital Link. **General Specifications at Release 26.0 (2026)** [14][15]	Member organisation licence; locally the member organisation trades as GS1 South Africa	Yes, in traceability, labelling and customer integration
DMSC — Digital Metrology Standards Consortium	Publishes QIF, the metrology and quality data schema library [16]	An American National Standard; download terms unverified	Only in model-based aerospace and automotive chains
IOGP	The oil and gas producers' association. Manages **CFIHOS** for capital-project information handover [17]	**Free of charge** [17]	Worth seeking out for any capital project, in any sector
IATF — International Automotive Task Force	Publishes IATF 16949 and its surrounding rules and interpretations [18]	Sold through regional bodies	Mandatory in any automotive chain
IAQG — International Aerospace Quality Group	Governs the 9100 series, now moving to a single **IA** designation [19]	Sold through the sector bodies	In aerospace only, and unavoidably there
GFSI — Global Food Safety Initiative	**Benchmarks food-safety schemes; certifies nobody.** Published Benchmarking Requirements version 2024 in two handbooks [20]	Free	Indirectly, through whichever scheme your client holds
PIC/S	The pharmaceutical inspection co-operation scheme. Maintains the good-manufacturing-practice guide that national regulators adopt and adapt	Published on its site	**Yes, and this is where the detail is** — not in the local regulator's guideline
INCIT	Publishes and governs SIRI, the industrial maturity assessment: 3 blocks, 8 pillars, 16 dimensions, bands 0 to 5, assessed on site by a certified assessor [21]. **Not a reference architecture** — it prescribes no systems	Assessor-mediated, commercial	Possibly, as the business case that precedes the architecture
NIST	Publishes SP 800-82, the operational-technology security guide [22]	**Free**	Yes, as the document to put in front of a security officer who knows IT
SABS	The South African Bureau of Standards: **develops and publishes** SANS standards, and certifies	Sold; copyright	Yes, by reference from regulations
NRCS	The National Regulator for Compulsory Specifications, and the legal-metrology regulator. **It regulates; SABS publishes**	Compulsory specifications are gazetted	Yes, in any regulated product category
SANAS	The national accreditation system for laboratories and inspection bodies	Accreditation is a service	Yes — several obligations read "by an accredited laboratory"

The enterprise-to-control boundary

ANSI/ISA-95.00.01-2025, (IEC 62264-1 Mod), *Models and Terminology* — announced 10 April 2025, revising the 2010 edition [23][24]. Almost all circulating material cites 2010, so this is the single most likely stale citation in a manufacturing architecture document.

The series has **eight parts plus ISA-TR95.01-2018, not five** — Parts 6 (messaging service model), 7 (alias service model) and 8 (information exchange profiles) exist and matter [1][25]. But **the designations and dates of Parts 2 to 8 are unverified** for this book: the only detailed public part list found is community-edited and internally inconsistent [25]. Print the shape, not a part table, and tell the reader to buy the current list from ISA [1].

Two more facts that get mangled. **The ISA and IEC lineages are out of step and are different documents:** ANSI/ISA-95.00.01-2025 was worked from IEC 62264-1 Edition 2.0, dated 2013-05, and "(IEC 62264-1 Mod)" means the ANSI text is modified — **ISA's release does not say how** [23][24]. Citing "ISA-95 Part 1" and "IEC 62264-1" interchangeably with one year is wrong. And **IEC 62264-6 is a Publicly Available Specification**, a pre-standard, not a full standard [26].

> **There is no ISA-95 conformance scheme, so "ISA-95 compliant" is uncertifiable.** Draw the contrast in a procurement document: ISA/IEC 62443 has ISASecure, with named schemes for the development lifecycle, for components and for systems [27]; and ANSI/ISA-88.00.01-2010 carries conformance criteria in its clause 9 [28]. ISA-95 has neither.

B2MML V7.00.00, 24 November 2020 [2]. **It is MESA International's, not ISA's**, royalty-free with attribution, and it was built to the **2018/19 editions** of ISA-95 and IEC 62264 — so it **predates the 2025 Part 1** [2][3]. Do not write that B2MML implements the current ISA-95. An architect who builds a canonical-model strategy on it must own that gap as a design decision rather than discover it later. Whether a version 8 is in progress is unverified [2][3].

ANSI/ISA-88.00.01-2010 is Part 1's current American edition; **IEC 61512-1 is 1997** [28]. **Parts 2 to 5 are unverified** — ISA's own committee page lists no parts, designations or dates, and the best public status account is a **2006** conference paper whose status columns are stale [1][29]. Do not print ISA-88 part dates from memory. Part 4, the batch production record, is the architecture-relevant one: it defines what a batch record *is* as a data object, and any discussion of electronic batch records that omits it is incomplete.

Measures

ISO 22400-1:2014 — edition 1, published October 2014, **"last reviewed and confirmed in 2025"**, with a 2023 review stage marked "to be revised" that ended in **confirmation in August 2025 instead** [30]. **ISO 22400-2:2014** is the part practitioners actually buy, and **it is still the current published edition**: the second edition was at **DIS stage 40.99** as at October 2026 [31]. Any text citing a 2024 or 2025 edition of Part 2 is wrong.

The standard does not stop the argument about denominators. It makes it resolvable, because both sides can be asked which definition they are using. Its real use is as the authority behind one measure definition in the repository, owned by one function, computed in one place.

Security

ISA/IEC 62443 editions are uneven, and the famous parts are the oldest.

Part	Subject	Edition and date
62443-1-5	Scheme for security profiles	Technical Specification 1.0, 2023-09
62443-2-1	Security programme requirements for asset owners	**Edition 2.0, 2024-08** — reorganised into programme elements and **added a maturity model**
62443-2-4	Requirements for service providers	**Edition 2.0, 2023-12**
62443-3-3	System security requirements and security levels	**Edition 1.0, 2013-08**
62443-4-2	Technical security requirements for components	**Edition 1.0, 2019-02**
62443-6-1	Security evaluation methodology for 62443-2-4	Technical Specification 1.0, 2024-03

Sources [32][33]. An architect writing an operational-technology target state in 2026 should be using the maturity model from the 2024 edition of 2-1, not the 2010 structure — and should notice that the requirement sets everybody quotes, 3-3 and 4-2, are the oldest live documents in the family. The most-cited public status summary of the series is dated **25 May 2021** and describes as "in development" documents that have since published [34]. **Never cite a 62443 status summary without its date.**

62443 is also the one item in this appendix that routinely becomes **contractually binding**, and the only one with a conformance scheme: **ISASecure**, with SDLA for the development lifecycle, CSA for components and SSA for systems [27]. A vendor claiming compliance can be asked for a certificate, a scheme name and a security level; if there is none, the claim is self-assessed.

NIST SP 800-82 Revision 3, *Guide to Operational Technology* (*OT*) *Security*, September 2023, is current and free [35]. **Revision 4 exists only as an Initial Public Draft dated 21 September 2026** [35][36]. Much secondary material still calls Revision 2 current. Revision 3 is where the title changed from industrial control systems to operational technology, and the scope broadened with it.

Controllers

IEC 61131-3:2025 is the current edition. The 2013 third edition was withdrawn on 22 May 2025 [37][38]. Nearly every source still cites 2013. The familiar "five languages" claim is also contestable: the 2013 edition specifies instruction list, structured text, ladder diagram and function block diagram, with sequential function chart as a *structuring element* rather than a fifth language.

Interoperability

OPC UA is IEC 62541, and its maturity is not uniform. The core parts are at **1.05.x**; **UAFX — field exchange, parts 80 to 84 — is only at 1.00.x** [4]. That version gap is the honest indicator: the core is mature, controller-to-controller exchange is young. PubSub, part 14, includes an MQTT mapping, which is why "OPC UA or MQTT" is a false choice.

Companion specifications: the OPC Foundation's own release of 20 April 2026 says there are "over 430" [39]. A secondary catalogue describes them as covering "over 60 types of industrial equipment" [62]. Both can be true — equipment types against documents — but quoting "about 60 companion specs" understates the programme by an order of magnitude. Note also that many are published by **VDMA**, a trade association, rather than by the Foundation, and **VDMA's licensing terms are unverified** [40]. Do not assume a VDMA companion specification is free on the same basis as a Foundation one.

MTConnect is a trap. The site's own designation is **ANSI/MTC1.4-2018** [5], while the documentation set is published at version 2.5.0 and the Institute announced a version 2.0 of the standard through AMT [6][41]. **Which version currently holds ANSI approval is unverified**, and the documentation is a full major version ahead of the vendor-facing designation. Do not write "ANSI/MTC1.4-2018 is the current MTConnect standard" as though that were the whole truth.

Sparkplug 3.0 is ISO/IEC 20237:2023, and it is the Eclipse Foundation's [7][8] — not the OPC Foundation's, which is where architects routinely file it. The practical consequence is that Sparkplug can be named with an international standard number in a procurement document, which is more than can be said for the unified namespace it underpins: **that pattern has no specification and no standards body at all.**

Digital twin and the Industry 4.0 layer

The Asset Administration Shell is two version lines, and citing one misrepresents the landscape. IEC 63278-1 Edition 1.0, 2023-12 is the structural standard, adopted in Europe and in Germany as DIN EN IEC 63278-1 (2024-09) [42][43]. The **IDTA publishes the implementable metamodel, IDTA-01001 at v3.x**, with version 3.0.2 posted in March 2025 [9]. The submodel templates are where the substance is, and the digital nameplate is the one with real traction.

RAMI 4.0 is a pre-standard, twice over: DIN SPEC 91345 (2016-04), a German specification rather than a DIN standard, and **IEC PAS 63088:2017**, a Publicly Available Specification rather than an International Standard [10][44]. Do not describe it as an IEC or ISO standard. It is pedagogically useful and operationally rare, and this research found no maintained conformance programme or tooling for it.

> **No credible independent adoption data exists for digital twin or for the Asset Administration Shell**, in any geography. Every statistics source located in this research was search-optimised content aggregating other search-optimised content. Any specific percentage an author finds should be assumed fabricated. State the absence as the finding — it is more useful than a number.

The quality schemes

Scheme	Status as at October 2026
ISO 9001	**ISO 9001:2026, sixth edition, published 16 September 2026**, at stage 60.60; **ISO 9001:2015 and its 2024 amendment are withdrawn** [45][46]. The certification transition period is set by accreditation-forum resolution, not by ISO, and is **unverified**
ISO 14001	**ISO 14001:2026, published 15 April 2026** [47]
ISO 45001	**ISO 45001:2018 remains the published edition.** A draft was circulating in 2026; the stage and projected date are **unverified** [48]
IATF 16949	**Still the 2016 first edition, 1 October 2016**, with the **second edition planned for mid-2027** [18][49]. The churn is in the Auditor Guide (5th edition, January 2025), the Rules (6th edition, 1 January 2025) and the Sanctioned Interpretations (November 2025) [18]
AS9100	**Being renamed, not just revised.** IAQG is moving AS, EN, SJAC and JISQ designations to a single **IA** designation at revision, with the **IA 9100 series expected early 2027** [19]. "Rev E" is third-party vocabulary and **unconfirmed**
ISO 13485	Current edition and revision status **unverified**; settled by the ISO catalogue entry
FSSC 22000	**v6 published 31 March 2023, mandatory 1 April 2024, recognised by GFSI 23 August 2024** against Benchmarking Requirements v2020 [50]. A **v7 page exists and its dates are unverified** [51] — do not print one
GMP	Enforced by inspection, not certification. The **PIC/S guide** is where the detail lives; its current revision number and date are **unverified**. EU Annex 11 on computerised systems is **under revision**, which is the most important thing to watch for any pharmaceutical client

Sources as cited. The pattern to teach from this table is not the versions but the question: **what records does this scheme compel, and which system is the authoritative source of each?** Answer that and the scheme has generated a data-ownership matrix, which is a real architecture artefact.

Identification

GS1 General Specifications, Release 26.0 (2026) [14][15], in nine sections. For a manufacturer the sections that bite are **data carriers and barcode placement**, because they decide whether a customer's receiving scanner can read the label your printer applied, at line speed, on a shrink-wrapped pallet. **EPCIS 2.0 is ISO/IEC 19987:2024** [52]; **GS1 Digital Link is at 1.1.3** on GS1's own reference site [53].

> **GS1 does not require anyone to adopt a 2D barcode by 2027.** GS1 US's own page calls Sunrise 2027 "a milestone in the transition to 2D barcodes", states that **"it is not necessary to implement all the capabilities of 2D by that date"**, names one retailer-side capability, and frames adoption as crawl, walk, run [54]. **The correct sentence:** Sunrise 2027 is a voluntary industry capability ambition coordinated by GS1, with no day or month, not a regulation and not a GS1 mandate. The manufacturer-side consequence is real even though the mandate is not — putting batch, expiry and serial data into a 2D carrier on a line is a printing, verification and data-availability project, and the data has to come from somewhere.

Engineering and plant-lifecycle data

ISO 10303-242:2022 is the STEP application protocol for managed model-based 3D engineering, and what you will actually meet is AP242 as a *contractual deliverable format* in aerospace, automotive and defence chains [55]. There is a newer architectural direction in **ISO/TS 10303-4000:2024, the core model** [56]. **QIF has not moved since December 2018 (ANSI QIF 3.0)** [16] — a stable, slow-cadence metrology schema, narrow and deep rather than broad. **JT** is ISO 14306, with Part 3 (2025) and Part 4 (2026) in the catalogue [57]; in a product-lifecycle estate it is a *derived* artefact, so the question is whether derivation is automated and versioned against the master.

ISO 15926 is hard, rigorous and sparsely implemented outside oil, gas, chemicals and large capital projects — and even there it is usually met through CFIHOS rather than directly [58]. **CFIHOS is at version 2.0, November 2025, managed by IOGP, free of charge** [17]. Two things about it. It is **the most directly reusable free artefact in this entire appendix**: a manufacturer commissioning a new line needs exactly a list of what data the contractor must deliver, in what form, before retention is released. And **its own standards page does not mention ISO 15926**, naming ISO 14224, the Energistics unit-of-measure standard and JIP33's reference data library instead [17] — so the familiar claim that CFIHOS is built on ISO 15926 is at best incomplete at version 2.0.

South Africa

NRS 048-2:2025, Edition 5, *Electricity supply: Quality of supply*, NERSA-approved 6 May 2025, superseding the 2015 fourth edition [59]. This is the best-evidenced South African design input in the whole research set behind this book, and it is free to download. It defines a dip as a sudden reduction in voltage lasting more than 20 milliseconds and up to three seconds, classifies dips into windows Y, X1, X2, S, T1, T2, Z1 and Z2, **explicitly assigns window-Y immunity to the customer**, names X2 and S as dips that may trip drives and T as dips that trip contactors, and publishes characteristic unplanned interruption counts — for a medium-voltage overhead network category, 18 a year at the 50th percentile of sites and 75 at the 95th — **with no**

compliance criteria [59]. Read from an architect's seat, the national standard says: window-Y immunity is your problem, X and S will trip your drives, and a rural overhead-fed site should expect tens of sustained interruptions a year with no recourse. That is a design input, and it is written down.

SANS and compulsory specifications. SABS develops and publishes SANS standards; the NRCS regulates compulsory specifications [60][61]. A SANS standard is **voluntary unless an instrument makes it compulsory**, and **the instrument, not the standard, is the citation.** Two sentences a competent author will write and should not: "SABS approval is required", and "the product must be SANS-compliant". There is also **no general list of products needing a letter of authority** — the requirement arises from the individual compulsory specification, so the register to maintain is the list of applicable specifications per product and version, not a blanket rule [61].

How to check a standard's version

This is the one procedural skill that keeps this appendix from going stale, and it takes about four minutes per standard.

1. **Go to the publisher's catalogue, not to a search engine summary.** The ISO catalogue page for a standard shows the edition, the publication date, the current **stage code** and whether previous editions are withdrawn. The IEC webstore shows the edition number and date. ANSI's webstore shows the ANSI designation. A blog post is not a catalogue.
2. **Read the stage code.** On the ISO side, **60.60 means published and current**; a number in the 30s or 40s means a draft in ballot, which is why **"stage 40.99" against ISO 22400-2 means the second edition has not published.** A standard at an early stage has no authority at all.
3. **Check whether the previous edition is marked withdrawn.** This is how you catch a supersession nobody announced — it is how IEC 61131-3:2013's withdrawal on 22 May 2025 is visible [37].
4. **Distinguish the document type.** A Technical Specification, a Technical Report, a Publicly Available Specification and a DIN SPEC are **not standards**. IEC 62264-6 and IEC PAS 63088 are pre-standards; several 62443 parts are Technical Specifications or Reports [26][32][44]. A requirement written against a pre-standard is weaker than it looks.
5. **Separate the lineages.** For anything with both an ANSI/ISA and an IEC number, check both, because they move independently and the texts are not guaranteed identical [23][24].
6. **Write down the date you checked, next to the version.** Then the next person knows how much to trust it, and you know when to look again.
7. **For the local instruments, go to the gazette.** A compulsory specification's authority is its gazette notice, with its number, year and subject — and a South African regulation number alone is not an identifier, because the same number is reused across years.

The questions to put to a vendor follow directly. *Which parts, at which editions, in which binding, and can you show me the profile?* **Any vendor who answers "we support ISA-95" with a picture of the level pyramid has answered nothing.**

> **More than 200 items in the research behind this book are recorded as unverified**, and that is a feature of honest research rather than a defect in it. Several of them are things a reader would reasonably expect this appendix to assert: the designations and dates of ISA-95 Parts 2 to 8, the whole ISA-88 part structure beyond Part 1, ISO 13485's current edition, FSSC 22000 version 7's dates, the PIC/S guide's current revision, which MTConnect version holds ANSI approval, and VDMA's companion-specification licensing.
>
> The honest practice is three steps, in order. **Find out** — most of these take one catalogue lookup. **Record the date** you found out. **Cite it**, with that date. What is not acceptable is the fourth option, which is to launder an unverified item into a confident sentence because the document needed a number in that sentence. A client who later discovers you invented a version has learned something about everything else in the document.

THE CHAPTER IN TEN LINES

- ANSI/ISA-95.00.01 is the 2025 edition, the series has eight parts plus a technical report, and the ANSI and IEC lineages are different documents that move independently.
- B2MML V7.00.00 of November 2020 is MESA's, royalty-free with attribution, and built to the 2018/19 editions, so it predates the 2025 Part 1.
- There is no ISA-95 conformance scheme, while IEC 62443 has ISASecure and ISA-88 Part 1 carries conformance criteria in its clause 9.
- IEC 62443 editions are uneven and the famous parts are the oldest: 2-1 is Edition 2.0 of 2024 with a maturity model, while 3-3 is still Edition 1.0 of 2013.
- NIST SP 800-82 Revision 3 of September 2023 is current and free; Revision 4 exists only as an initial public draft dated 21 September 2026.
- IEC 61131-3:2013 was withdrawn on 22 May 2025 and superseded by the 2025 edition, which almost no secondary source reflects.
- OPC UA core parts are at 1.05.x while UAFX is only at 1.00.x, and the Foundation's own April 2026 figure is over 430 companion specifications.
- Sparkplug 3.0 is ISO/IEC 20237:2023 and belongs to the Eclipse Foundation, so it can be named with a standard number in a procurement document.
- GS1 does not require a 2D barcode by 2027: Sunrise 2027 is a voluntary industry capability ambition with no day or month and no mandate language.
- SABS publishes SANS standards and the NRCS regulates compulsory specifications, so a SANS standard is voluntary unless an instrument makes it compulsory, and the instrument is the citation.
- CFIHOS version 2.0 of November 2025 is free of charge and directly reusable as a capital-project handover specification, and its own standards page does not mention ISO 15926.
- More than 200 items in the research behind this book are unverified, and the honest practice is to find out, record the date, and cite it.

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APPENDIX D

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The consolidated source list for the volume, grouped by domain, with the date each source was retrieved. The last section names the seven worth re-checking before any of this reaches a client.

How to read this list

Every source below was retrieved on **16 October 2026** unless the entry says otherwise, and that date is the most important thing on the page. Manufacturing statistics go stale monthly, port and utility figures change weekly, tariff and bargaining-council figures are gazette-dated and annual, and four standards editions moved inside this book's own research window. **A source consulted in October 2026 may have moved, and some of these certainly have.**

Three conventions. **Figures carry their own as-at dates** — a number in the chapters is dated at the point of use, because a production index or a port waiting time is an observation of a period, not a standing fact. **A dagger marks a source seen only as a secondary report or a search-result headline**, from which only the headline-level fact is claimed; anything load-bearing resting on one should be re-sourced to the primary document first. And **this list is deduplicated and grouped, not exhaustive**: the four research files behind the book carry roughly 330 numbered sources between them with substantial overlap, and where a primary and a daggered secondary covered the same fact the primary is listed and the secondary dropped. Background-only entries, and secondary claims the research rejected, are not reproduced.

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Re-check these before the material goes in front of a client

Seven items, in the order a client is most likely to notice the error.

1. **The ISA store listing for ISA-95.** Only Part 1's 2025 edition is primary-sourced. **Parts 2 to 8's designations and dates are unverified** and the public part lists are inconsistent. One lookup settles it.
2. **The ISO catalogue entry for ISO 22400-2.** The second edition was at DIS stage 40.99 in October 2026. If it has published, every measure-definition passage in this book needs its citation changed.
3. **FSSC 22000 version 7's publication date, mandatory-audit date and recognition status.** A version 7 page exists; its dates do not. Third-party transition commentary circulates freely and is not the scheme owner.
4. **IATF Global Oversight, on the second edition of IATF 16949** — and on how the automotive scheme responds to ISO 9001:2026, since it is built on a standard that has just been replaced. This is the largest open question in the quality layer.
5. **Vendor ownership, for every product name you print.** Specifically: **whether the former Infor EAM product moved into Octave** with the rest of the asset-lifecycle division, **who owns Alfabet** now, and **whether the Honeywell Automation and Aerospace separation has completed.** All three were unresolved at retrieval.
6. **Which MTConnect version holds ANSI approval**, and the current designation. The documentation is a full major version ahead of the vendor-facing designation, and the discrepancy is on the publisher's own site.
7. **The Stats SA national-accounts rebase to 2022.** Publication was expected in October 2026, and Stats SA said the figures may change. **Every share-of-GDP figure in this book is hedged for that reason**, and the hedge becomes a correction the moment the rebased series appears.

And four categories, rather than items, that will have moved by the time you read this. **Standards versions** — five moved inside the research window, so assume the next five move inside your project. **Vendor ownership and product names** — seven changes landed in industrial software in two years, so a name quoted from a three-year-old deck is often a name that no longer exists. **Tax rates and incentive status** — the carbon-tax rate, the tariff path and the bargaining-council increases are all dated and annual, and one investment allowance closed in 2020 is still presented as available. **Anything about the ports, the railway or the grid** — weekly facts reported as standing ones, with the direction of travel reversing more than once.

Two circulating employment claims — that manufacturing lost 100,000 jobs, and that it lost 45,995 in two years — were examined and **rejected** as irreconcilable with both official series, so neither appears above. **Name the survey whenever you give an employment number**, and the problem does not arise.

THE CHAPTER IN TEN LINES

- Every source here was retrieved on 16 October 2026 unless stated, and that date matters more than the title.
- A dagger marks a source seen only as a secondary report or a search-result headline; only the headline-level fact is claimed from it.
- Where a primary and a daggered secondary covered the same fact, the primary is listed and the secondary dropped.
- Figures in the chapters carry their own as-at dates, because a production index or a port waiting time is an observation of a period, not a standing fact.
- Seven items need re-checking first: the ISA-95 part list, ISO 22400-2's edition, FSSC 22000 version 7, the IATF second edition, three vendor-ownership questions, MTConnect's ANSI designation, and the Stats SA rebase.
- Four categories will have moved by the time you read this: standards versions, vendor ownership and product names, tax rates and incentive status, and anything about the ports, the railway or the grid.
- Two circulating manufacturing employment claims were examined and rejected as irreconcilable with the official series, so neither appears in the list.
- Name the survey whenever you give a South African employment number, and the reconciliation problem does not arise.